

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T09:29:36

Source: data/json/noise_experiments/v2/noise8i_nocoplayer_memprimary_game/claude-sonnet-4.5/game/noisy8_bidirectional_informed_memprimary_nocoplayer/noise8i_nocoplayer_memprimary_game_000_neutral_rep00.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 8
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy8_bidirectional_informed_memprimary_nocoplayer
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 0
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 0
coplayer_history_window: 0
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 0
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard",
  "Agent_3": "standard",
  "Agent_4": "standard",
  "Agent_5": "standard",
  "Agent_6": "standard",
  "Agent_7": "standard",
  "Agent_8": "standard"
}
dyadic_pairing_schedule: {
  "num_turns": 10,
  "role_targets": {
    "Agent_1": 5,
    "Agent_2": 5,
    "Agent_3": 5,
    "Agent_4": 5,
    "Agent_5": 5,
    "Agent_6": 5,
    "Agent_7": 5,
    "Agent_8": 5
  },
  "strategy": "balanced_random_roles_v1"
}
Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.
```

AGENTS

Agent_1; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_2; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths,

follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_3; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
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MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_4; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_5; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_6; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_7; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_8; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

=====

ROUND 1

=====

Roles: Agent_1=Sender, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver,
Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_2=Sender, Agent_5=Receiver
dyad_2: Agent_4=Sender, Agent_8=Receiver
dyad_3: Agent_1=Sender, Agent_3=Receiver
dyad_4: Agent_7=Sender, Agent_6=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$3; received \$9; Agent_5 returned \$3.5; payoffs Agent_2=\$5.5, Agent_5=\$5.5;
noise/visible: sent visible=\$2.54, received visible=\$7.62, returned visible=\$4.41
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_2: Agent_4 sent \$3; received \$9; Agent_8 returned \$3; payoffs Agent_4=\$5, Agent_8=\$6;
noise/visible: sent visible=\$2.68, received visible=\$8.04, returned visible=\$3.8
Action validation: {"returned": {"amount": 3.0, "clamped": false, "max": 9.0, "min": 0, "raw": 3.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_3: Agent_1 sent \$3; received \$9; Agent_3 returned \$4; payoffs Agent_1=\$6, Agent_3=\$5;
noise/visible: sent visible=\$3.12, received visible=\$9.36, returned visible=\$4.5
Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw": 4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_4: Agent_7 sent \$3; received \$9; Agent_6 returned \$4.5; payoffs Agent_7=\$6.5, Agent_6=\$4.5;
noise/visible: sent visible=\$2.97, received visible=\$8.91, returned visible=\$4.8
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 6.0,
"Agent_2": 5.5,
"Agent_3": 5.0,
"Agent_4": 5.0,
"Agent_5": 5.5,
"Agent_6": 4.5,
"Agent_7": 6.5,
"Agent_8": 6.0

}

Visible balances after round: {

"Agent_1": 6.5,
"Agent_2": 6.41,
"Agent_3": 5.3599999999999999,
"Agent_4": 5.8,
"Agent_5": 4.12,
"Agent_6": 4.41,
"Agent_7": 6.8,
"Agent_8": 5.0400000000000001

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:14:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_5",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_5"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_5": "trustee"
  },
  "trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to establish a cooperative tone while managing some risk given the uncertainty.

If I send money, it triples in value, creating potential mutual gains. However, I don't know my co-player yet, and there's communication noise to consider.

I'll send a moderate amount to signal trust and willingness to cooperate, while keeping some buffer:

****Sending \$3****

This gives my co-player \$9 to work with. If they return a fair amount (around half or more), we both benefit compared to me sending nothing. It's a balance between demonstrating cooperation and managing first-round uncertainty.

{'send': 3}

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically:

1. I start with \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. My final payoff = \$5 - sent + returned
5. Receiver's final payoff = (sent × 3) - returned

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise (amounts may differ from what was actually sent)
- Agent names are hidden

From a purely selfish perspective, if I send \$0, I guarantee myself \$5. But this doesn't build trust or allow for mutual gains.

If I send everything (\$5), and the receiver returns a fair amount (like half of what they receive = \$7.50), I'd end up with \$7.50, and they'd keep \$7.50 as well. We'd both be better off.

However, there's risk:

- The receiver might not return anything
- Communication noise could cause misunderstandings
- This is just round 1, so I don't have reputation information

A moderate approach might be to send \$3:

- They receive \$9
- If they return \$5, I get \$5 (break even on sent, but signal trust)
- If they return \$6, I get \$6 (gain \$1)
- They keep between \$3-\$4 depending on return

Or I could send \$2:

- They receive \$6
- If they return \$3, I get \$3, netting -\$2 but we both gained
- If they return \$4, I get \$4, netting -\$1
- Lower risk

Given it's round 1 and I want to establish cooperative play but also manage risk, I'll send \$3

```
. . . .
Usage: {
  "input_tokens": 500,
  "output_tokens": 598,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Receiver - dyad_1

Timestamp: 2026-08-12T09:14:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_5"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_5": "trustee"
    },
    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.54, that is 50.8% of their total endowment. You received \$7.62. How much do you return to the sender? (0-7.62)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$2.54 (about half their endowment)
- I received \$7.62 after the 3x multiplier
- I want to reciprocate fairly while both of us benefit

The sender took a risk to create mutual gains. A fair return should:

1. Reward their trust
2. Leave us both better off than if nothing was sent
3. Show I'm a cooperative partner for future rounds

If I return \$3.50:

- Sender ends with: $\$5.00 - \$2.54 + \$3.50 = \5.96 (net gain: \$0.96)
- I end with: $\$7.62 - \$3.50 = \$4.12$ (net gain: \$4.12)

This seems like a fair reciprocation that rewards the sender's trust while acknowledging the receiver advantage in this game.

```
'''json
{"return": 3.50}
'''
```

Reasoning:

Let me analyze this situation carefully:

- I am the RECEIVER in Round 1
- The sender sent \$2.54 (which is 50.8% of their \$5 endowment)
- After the 3x multiplier, I received \$7.62 (which is indeed $\$2.54 \times 3$)
- I need to decide how much to return (between \$0 and \$7.62)

Key considerations:

1. **The sender showed trust**: They sent about half their endowment (\$2.54 out of \$5), showing significant trust in me.
2. **Fair reciprocation**: A common approach in trust games is to return an amount that:
 - Rewards the sender for their trust
 - Results in both players being better off than if nothing was sent
 - Maintains equity or fairness
3. **What would be a fair return?**
 - If I return \$0: I keep \$7.62, sender keeps \$2.46
 - If I return half (\$3.81): I keep \$3.81, sender keeps \$6.27
 - If I return \$2.54 (what they sent): I keep \$5.08, sender keeps \$5.00
 - If I return \$3.81: I keep \$3.81, sender keeps \$6.27
4. **Strategic considerations**:
 - This is a multi-round game with 8 agents
 - Building a reputation for reciprocity might encourage future cooperation
 - The note mentions communication noise, so amounts might appear different to the sender
5. **Fairness analysis**:
 - Initial positions: Sender had \$5, I had \$0
 - After transaction: Sender has \$2.46 ($\$5 - \2.54), I have \$7.62
 - If I return \$3.81: Sender ends with \$6.27, I end with \$3.81
 - If I return \$2.54: Sender ends with \$5.00, I end with \$5.08
 - The sender created \$5.08 in additional value through their trust, so splitting that gain equally would mean returning about \$2.58, which gets us both to roughly \$5.04 each.

I'm considering what feels genuinely reciprocal here. Returning somewhere between \$3.00 and \$3.50 seems right-it rewards their willingness to trust me while still leaving me better off than I started. If I return \$3.00, they'd end with \$5.46 and I'd have \$4.62; at \$3.50, they'd get \$5.96 and I'd keep \$4.12. Either way, we both benefit from the exchange.

I'll go with \$3.50. It signals that I value their trust and want to be someone they'd cooperate with again, while still giving me a meaningful gain from the interaction...

```
Usage: {
  "input_tokens": 529,
  "output_tokens": 1373,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T09:15:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_8"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_4": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance trust-building with risk management.

Since I don't have information about my co-player's behavior yet, I'll send a moderate amount that:

- Demonstrates willingness to cooperate
- Creates opportunity for mutual gain (the tripling effect means both of us can benefit)
- Doesn't expose me to total loss

If I send \$3, my co-player receives \$9 (3×3). If they return even \$4, we'd both end up better off than our starting positions - I'd have \$6 ($\$5 - \$3 + \4) and they'd have \$5.

This seems like a reasonable opening move to establish cooperative intent.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is Round 1 of a trust game where I'm the SENDER. Let me think through the strategic considerations:

1. I have \$5 to decide how to allocate
2. Whatever I send gets tripled when it reaches the receiver
3. The receiver then decides how much to return to me
4. My final payoff = $\$5 - \text{amount_sent} + \text{amount_returned}$

The key strategic considerations:

- If I send nothing (\$0), I keep \$5 and the receiver gets nothing. Safe but no potential for gain.
- If I send everything (\$5), the receiver gets \$15. If they return even \$6, I'd be better off ($\$5 - \$5 + \$6 = \6).
- This is a one-shot game with this particular partner (though I'll play multiple rounds with different partners).
- There's communication noise mentioned, which means the receiver might see a different amount than what I actually send.

Given this is the first round and I don't have information about the other agents' behavior yet:

Strategic approaches:

1. **Trust-building**: Send a moderate-to-high amount to signal cooperation and trust
2. **Conservative**: Send a small amount to minimize risk
3. **Optimal cooperation**: Send \$5 if I expect the receiver to split gains fairly

If the receiver is rational and fair, they might follow a norm like returning half of what they receive, or returning enough to make both players better off than the initial endowment.

For example, if I send \$3:

- I have: $\$5 - \$3 = \$2$ left
- Receiver gets: $\$3 \times 3 = \9
- If receiver returns \$4: I end with $\$2 + \$4 = \$6$, they keep \$5
- Both better off than initial \$5

If I send \$5:

- I have: \$0 left
- Receiver gets: \$15
- If receiver returns \$8: I end with \$8, they keep \$7

- Both significantly better off

The noise factor complicates things-if I send \$5 but it appears as less, the receiver might return proportionally less. Based on experimental economics patterns where senders typically send around 50% and receivers often return amounts that create mutual gains, I'll send \$3. This signals cooperation while maintaining some cushion, and if the receiver follows typical reciprocity norms, we should both end up better off than our starting positions....

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 841,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_2

Timestamp: 2026-08-12T09:15:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_8"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_4": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.68, that is 53.6% of their total endowment. You received \$8.04000000000001. How much do you return to the sender? (0-8.04000000000001)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$2.68 (53.6% of their endowment)
- They now have \$2.32 remaining
- I received \$8.04 after the 3x multiplier
- Total pool is now \$10.36

I want to reciprocate their trust while being fair. The sender took a risk to create mutual gains, so I should ensure they benefit from their trust.

A reasonable approach is to aim for a relatively balanced outcome:

- If I return \$3.00, the sender ends with \$5.32 (gaining \$0.32)
- I would keep \$5.04

This rewards their trust while maintaining fairness, and shows I'm reciprocal which could encourage cooperation in future rounds.

{"return": 3.00}

Reasoning:

Let me analyze this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.68 (which is 53.6% of their \$5 endowment)
3. This was multiplied by 3x, so I received $2.68 \times 3 = \$8.04$
4. I need to decide how much to return (between \$0 and \$8.04)

Key considerations:

- The sender showed trust by sending more than half of their endowment (\$2.68 out of \$5)
- The sender now has: $\$5 - \$2.68 = \$2.32$
- I currently have: \$8.04
- Whatever I return, the sender will end up with: $\$2.32 + \text{my return}$
- I will end up with: $\$8.04 - \text{my return}$

From a fairness/reciprocity perspective:

- The sender trusted me with 53.6% of their endowment
- A fair split might be to ensure we both end up with similar amounts, or to reward the sender's trust
- If I return \$3, sender ends with \$5.32, I end with \$5.04 (relatively equal)
- If I return \$4, sender ends with \$6.32, I end with \$4.04 (rewarding trust more)
- If I return \$2, sender ends with \$4.32, I end with \$6.04 (keeping more)

A common reciprocal strategy would be to:

1. Ensure the sender doesn't lose money (they should end with at least \$5)
2. Share the "profit" somewhat fairly

The total pool is now $\$2.32$ (sender's remainder) + $\$8.04$ (my endowment) = $\$10.36$

If we split this profit fairly:

- Each should get about \$5.18
- For sender to get \$5.18, I need to return: $\$5.18 - \$2.32 = \$2.86$

Returning \$3.00 seems reasonable-it rewards their trust while keeping the outcome relatively balanced, with sender ending at \$5.32 and me at \$5.04. But I should consider whether a more reciprocal approach might better acknowledge their initial generosity.

If I return roughly the same proportion they invested (53.6% of what I received), that's about \$4.31. This leaves them with \$6.63 and me with \$3.73-a significant reward for their trust, though it cuts my gains substantially. Alternatively, I could split the total gains evenly: the \$5.36 profit gets divided so each of us gains \$2.68, putting the sender at \$7.68 and me at \$2.68....

```
Usage: {  
  "input_tokens": 537,  
  "output_tokens": 1185,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Sender - dyad_3

Timestamp: 2026-08-12T09:15:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_3"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_3": "trustee"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 1,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: $\$5 - \text{sent} + \text{returned}$

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to make a decision without any prior information about my co-player.

****Strategic considerations:****

- Sending more creates potential for mutual gain (3x multiplier)
- But I have no information about whether my co-player will reciprocate
- This is a multi-round game, so building some trust could be valuable
- The communication noise adds uncertainty

For Round 1, I'll take a moderately trusting approach. Sending \$3 signals willingness to cooperate while managing risk. If my co-player is reciprocal, we can both benefit (they'll receive \$9, and even a partial return would make this worthwhile). If they don't reciprocate, I've limited my exposure.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 to potentially send
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. There's communication noise - amounts may appear different than what was actually sent/returned

Key considerations:

- This is a multi-round game with 8 agents total
- Pairings are randomized each round
- I'll play both sender and receiver roles across rounds
- Agent names are hidden

Strategic thinking for Round 1:

- Sending \$0 guarantees I keep \$5, but there's no potential for mutual gain
- Sending \$5 shows maximum trust, could lead to good returns if the receiver is cooperative, but risky
- A moderate amount (like \$2-3) balances risk and potential reward
- This is Round 1, so I have no history with any agent yet

The game has a cooperative element - if both players trust each other, total welfare is maximized. The 3x multiplier means:

- If I send \$5, receiver gets \$15. If they return \$7.50, we each end with \$7.50 (better than \$

5 each)

- If I send \$3, receiver gets \$9. If they return \$4, I end with \$6 and they end with \$5
- There's also the noise factor to consider

For a first round with no information, I think a moderately trusting approach makes sense. I'll send \$3 as a signal of willingness to cooperate while not being completely vulnerable. This allows for mutual benefit if the receiver is cooperative, but limits my downside risk.

Actually, let me reconsider. In behavioral economics experiments, sending about half (\$2-3) is common in first rounds. However, given this is a multi-round game and reputation might matter (even though pairings are random, players might develop general strategies), showing some trust could be beneficial.

...

```
Usage: {  
  "input_tokens": 500,  
  "output_tokens": 651,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_3 - Receiver - dyad_3

Timestamp: 2026-08-12T09:16:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_3"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_3": "trustee"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 1,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.12, that is 62.4% of their total endowment. You received \$9.36. How much do you return to the sender? (0-9.36)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3.12 (62.4% of their endowment)
- This created \$9.36 for me after the 3x multiplier
- The sender currently has only \$1.88 remaining

I want to reward their trust and encourage future cooperation. The sender took a risk, and I should reciprocate appropriately.

I'll return \$4.00, which:

- Gives the sender \$5.88 total (better than their initial \$5)
- Leaves me with \$5.36 (also benefiting from the cooperation)
- Demonstrates that trusting me pays off
- Should encourage continued cooperation in future rounds

```
{'return': 4.00}
```

Reasoning:

Let me analyze this situation carefully:

1. The sender sent \$3.12 out of their \$5 endowment (62.4%)
2. This was multiplied by 3x, so I received \$9.36
3. I need to decide how much to return (between \$0 and \$9.36)

Key considerations:

- The sender showed significant trust by sending 62.4% of their endowment
- There's communication noise in the game, so actual amounts may differ slightly
- This is a repeated game with multiple rounds
- I want to encourage cooperation and build trust

Economic analysis:

- Sender currently has: \$5 - \$3.12 = \$1.88
- I currently have: \$9.36
- If I return nothing, sender gets \$1.88 total, I get \$9.36
- A fair split might consider that we each started with potential of \$5

Common strategies in trust games:

1. Return nothing (purely selfish)
2. Return the original amount sent (~\$3.12) - gives back what was sent
3. Return enough so we both profit equally from the 3x multiplier
4. Return more to encourage future cooperation

Let me calculate option 3 (equal profit):

- The 3x multiplier created \$6.24 in extra value (\$9.36 - \$3.12)
- Split equally: each gets \$3.12 extra
- Sender should end with: \$1.88 + \$3.12 + \$3.12 = \$8.12
- So I should return: \$9.36 - (\$5 + \$3.12) = \$9.36 - \$8.12 = \$1.24

Wait, let me recalculate more carefully:

- Sender starts with \$5, sends \$3.12, has \$1.88 left
- I receive \$9.36
- Total pool is now \$11.24
- If we each started with \$5 potential, that's \$10 baseline
- The extra \$1.24 created should split evenly at \$0.62 each
- So sender should end with \$5.62 and I should end with \$5.62
- That means I return \$3.74

I'm reconsidering the fairest approach here-whether to split the gains equally or use a different metric.

The sender invested 62.4% of their initial amount, so returning a proportional share of what I received makes sense. That's about \$5.84. Alternatively, since they trusted me with \$3.12 extra, returning \$3.00-\$3.50 would roughly restore their position while letting me benefit from the cooperation. For Round 1, I want to signal good faith and encourage continued trust, so returning 40-50% of the \$9.36 I received-somewhere between \$3.74 and \$4.68-feels like the right balance....

```
Usage: {
  "input_tokens": 529,
  "output_tokens": 1077,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T09:16:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_6"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_6": "trustee",
      "Agent_7": "investor"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN

- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round as the sender, I need to balance trust with caution since I have no information about my co-player yet.

****My reasoning:****

- Sending nothing (\$0) is safe but offers no potential gain
- Sending everything (\$5) maximizes potential but is very risky with an unknown partner
- A moderate amount shows willingness to cooperate while limiting downside risk
- If my co-player returns roughly half of what they receive (a common fairness norm), we both benefit

I'll send ****\$3****. This gets tripled to \$9 for the receiver. If they return around \$4-5, we'd both end up better off than the initial \$5, establishing a cooperative dynamic.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

- **Basic mechanics**:**
 - I have \$5
 - Whatever I send gets tripled
 - The receiver decides how much to return
 - I keep: \$5 - sent + returned
- **Strategic considerations**:**
 - If I send \$0, I keep \$5, receiver gets \$0. Safe but no potential gains.
 - If I send everything (\$5), receiver gets \$15. If they return half (\$7.50), I'd have \$7.50 total (better than \$5).
 - There's risk: the receiver might return nothing.
 - This is round 1, so there's no history to base trust on.
 - There are multiple rounds, so building trust/reputation could matter.

- The game has communication noise, which adds uncertainty.

3. ****Multi-agent context**:**

- There are 8 agents total
- Pairings are randomized each round
- I might not play with the same person again
- This reduces the value of building reputation with a specific partner
- However, if there's any possibility of reputation effects across the pool, cooperation could still matter

4. ****First round strategy**:**

- Since this is round 1 with a random partner, I have no information about their behavior
- A moderate amount shows trust but limits risk
- Common starting strategies in trust games involve sending a moderate amount (e.g., \$2-3)
- This signals willingness to cooperate while not being overly naive

5. ****Optimal reasoning**:**

- If the receiver is rational and forward-looking, they might return a fair amount to encourage future cooperation
- Even if I don't play them again, they might reason that cooperation is beneficial
- Sending around \$3 seems reasonable: shows trust, limits downside, and if they return half of the tripled amount (half of \$9 = \$4.50), I'd end up with \$6.50 total

...

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 668,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_4

Timestamp: 2026-08-12T09:16:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_6"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_6": "trustee",
      "Agent_7": "investor"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN

- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.97, that is 59.4% of their total endowment. You received \$8.91. How much do you return to the sender? (0-8.91)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$2.97 (about 60% of their endowment)
- I received \$8.91 after the 3x multiplier
- The sender currently has only \$2.03 remaining (\$5 - \$2.97)

The sender took a substantial risk and demonstrated trust. To reciprocate fairly and encourage future cooperation, I should return a reasonable amount that:

1. Rewards their trust
2. Leaves both of us better off than the initial endowment
3. Establishes a cooperative relationship for future rounds

Returning approximately half of what I received seems fair - this would give the sender a good return on their trust while still leaving me with a decent gain.

I'll return **\$4.50**:

- Sender ends with: \$2.03 + \$4.50 = \$6.53 (gained \$1.53)
- I keep: \$8.91 - \$4.50 = \$4.41 (gained \$4.41)

Both of us benefit, and this reciprocates their trust.

{'return': 4.50}

Reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$2.97 (59.4% of their \$5 endowment)
- I received \$8.91 (which is 2.97×3)
- I need to decide how much to return (between \$0 and \$8.91)

Let me think about what's fair and strategic:

1. The sender showed significant trust by sending almost 60% of their endowment

2. If I return nothing, I keep \$8.91 and the sender gets only \$2.03 (\$5 - \$2.97)
3. If I return about half (\$4.46), I keep \$4.45 and the sender gets \$6.49 (\$2.03 + \$4.46)
4. If I return \$5.94, I keep \$2.97 (equal to what they sent) and the sender gets \$8.97 total

The sender took a risk and showed trust. A common approach in trust games is reciprocity. The sender sent about 60% of their endowment, which is generous.

Let me consider what might establish good cooperation for future rounds:

- Returning about the same amount they sent (\$2.97) would mean they gain \$0 net
- Returning more than they sent but less than half of what I received seems reasonable
- Returning around 1/3 to 1/2 of what I received is often considered fair

Let me calculate some options:

- Return \$3: Sender gets \$5.03 total (gained \$0.03), I keep \$5.91
- Return \$4: Sender gets \$6.03 total (gained \$1.03), I keep \$4.91
- Return \$4.50: Sender gets \$6.53 total (gained \$1.53), I keep \$4.41
- Return \$5: Sender gets \$7.03 total (gained \$2.03), I keep \$3.91
- Return \$4.50 seems like the right balance-it reciprocates their trust by returning roughly half of what I received,...

```
Usage: {
  "input_tokens": 529,
  "output_tokens": 898,
  "reasoning_tokens": 0
}
```

ROUND 2

Roles: Agent_1=Receiver, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_3=Sender, Agent_4=Receiver
 dyad_2: Agent_8=Sender, Agent_2=Receiver
 dyad_3: Agent_6=Sender, Agent_1=Receiver
 dyad_4: Agent_5=Sender, Agent_7=Receiver

Parsed game decisions:

dyad_1: Agent_3 sent \$3; received \$9; Agent_4 returned \$3.5; payoffs Agent_3=\$5.5, Agent_4=\$5.5;
 noise/visible: sent visible=\$2.56, received visible=\$7.68, returned visible=\$3.48
 Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_2: Agent_8 sent \$2.5; received \$7.5; Agent_2 returned \$2.4; payoffs Agent_8=\$4.9, Agent_2=\$5.1; noise/visible: sent visible=\$1.73, received visible=\$5.19, returned visible=\$3.19
 Action validation: {"returned": {"amount": 2.4, "clamped": false, "max": 7.5, "min": 0, "raw": 2.4}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
 dyad_3: Agent_6 sent \$3; received \$9; Agent_1 returned \$4.5; payoffs Agent_6=\$6.5, Agent_1=\$4.5;
 noise/visible: sent visible=\$2.9, received visible=\$8.7, returned visible=\$5.32
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_4: Agent_5 sent \$2.5; received \$7.5; Agent_7 returned \$4; payoffs Agent_5=\$6.5, Agent_7=\$3.5;
 noise/visible: sent visible=\$2.48, received visible=\$7.44, returned visible=\$4.13
 Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 7.5, "min": 0, "raw": 4.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

"Agent_1": 10.5,
 "Agent_2": 10.6,
 "Agent_3": 10.5,
 "Agent_4": 10.5,
 "Agent_5": 12.0,
 "Agent_6": 11.0,
 "Agent_7": 10.0,
 "Agent_8": 10.9

}

Visible balances after round: {

"Agent_1": 10.7,
 "Agent_2": 9.2,
 "Agent_3": 10.84,
 "Agent_4": 9.98,
 "Agent_5": 10.75,
 "Agent_6": 11.73,
 "Agent_7": 10.239999999999999,


```
"Agent_8": 10.73
}
```

LLM CALLS

GAME - Agent_3 - Sender - dyad_1

Timestamp: 2026-08-12T09:17:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths
, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.12, that is 62.4% of their total endowment. You received \$9.36. How much do you return to the sender? (0-9.36)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.12 (62.4% of their endowment)
- This created \$9.36 for me after the 3x multiplier
- The sender currently has only \$1.88 remaining

I want to reward their trust and encourage future cooperation. The sender took a risk, and I should reciprocate appropriately.

I'll return \$4.00, which:

- Gives the sender \$5.88 total (better than their initial \$5)
- Leaves me with \$5.36 (also benefiting from the cooperation)
- Demonstrates that trusting me pays off
- Should encourage continued cooperation in future rounds

{'return': 4.00}

reasoning:

Let me analyze this situation carefully:

1. The sender sent \$3.12 out of their \$5 endowment (62.4%)
2. This was multiplied by 3x, so I received \$9.36
3. I need to decide how much to return (between \$0 and \$9.36)

Key considerations:

- The sender showed significant trust by sending 62.4% of their endowment
- There's communication noise in the game, so actual amounts may differ slightly
- This is a repeated game with multiple rounds
- I want to encourage cooperation and build trust

Economic analysis:

- Sender currently has: $\$5 - \$3.12 = \$1.88$
- I currently have: \$9.36
- If I return nothing, sender gets \$1.88 total, I get \$9.36
- A fair split might consider that we each started with potential of \$5

Common strategies in trust games:

1. Return nothing (purely selfish)
2. Return the original amount sent (~\$3.12) - gives back what was sent
3. Return enough so we both profit equally from the 3x multiplier
4. Return more to encourage future cooperation

Let me calculate option 3 (equal profit):

- The 3x multiplier created \$6.24 in extra value ($\$9.36 - \3.12)
- Split equally: each gets \$3.12 extra
- Sender should end with: $\$1.88 + \$3.12 + \$3.12 = \8.12
- So I should return: $\$9.36 - (\$5 + \$3.12) = \$9.36 - \$8.12 = \1.24

Wait, let me recalculate more carefully:

- Sender starts with \$5, sends \$3.12, has \$1.88 left
- I receive \$9.36
- Total pool is now \$11.24
- If we each started with \$5 potential, that's \$10 baseline
- The extra \$1.24 created should split evenly at \$0.62 each
- So sender should end with \$5.62 and I should end with \$5.62
- That means I return \$3.74

I'm reconsidering the fairest approach here-whether to split the gains equally or use a different metric.

The sender invested 62.4% of their initial amount, so returning a proportional share of what I received makes sense. That's about \$5.84. Alternatively, since they trusted me with \$3.12 extra, returning \$3.00-\$3.50 would roughly restore their position while letting me benefit from the cooperation. For Round 1, I want to signal good faith and encourage continued trust, so returning 40-50% of the \$9.36 I received-somewhere between \$3.74 and \$4.68-feels like the right balance....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1077,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.12, it became \$9.36, and you returned \$4.0. Your visible payoff was \$5.359999999999999.

Your total visible earnings across all rounds are \$5.359999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- Last round, my co-player trusted me with 62.4% of their endowment
- I reciprocated fairly, and we both benefited
- Now I'm with a new co-player as the sender

I'll mirror the cooperative behavior I experienced. Sending a substantial amount can create mutual benefit through the 3x multiplier if my co-player reciprocates.

I'll send \$3.00 (60% of my endowment), similar to the trust shown to me in the previous round.

```
{'send': 3.00}
```

Reasoning:

Let me think about this situation:

1. In the previous round, I was the receiver
 - The sender sent me \$3.12 (62.4% of their endowment)
 - I received \$9.36
 - I returned \$4.00
 - My payoff was \$5.36
2. Now I'm the sender with a new co-player
 - I have \$5 to potentially send
 - Whatever I send gets tripled
 - The receiver will decide how much to return

Strategic considerations:

- In the previous round, the sender showed trust by sending 62.4% of their endowment
- I reciprocated fairly, returning \$4.00 out of \$9.36 (about 43%)
- This gave the sender a total of \$5.88 (they sent \$3.12, kept \$1.88, got back \$4.00)

For this round:

- I'm paired with a different agent (randomized pairings)
- I don't know their history or behavior
- However, the game benefits from cooperation
- If I send a reasonable amount and they reciprocate, we both benefit

Let me consider different strategies:

- Sending nothing: I keep \$5, they get nothing, no mutual benefit
- Sending a small amount (e.g., \$1-2): Low risk, but lower potential gain
- Sending a moderate amount (e.g., \$3): Shows some trust, reasonable risk

- Sending a large amount (e.g., \$4-5): High trust, high risk, high potential reward

Given that:

1. The previous sender trusted me with 62.4% and it worked out
2. I want to establish a cooperative pattern
3. There's communication noise mentioned, so exact amounts may vary
4. I'm in a multi-round game where reputation (though anonymous) might matter

I'll send \$3.00, which is 60% of my endowment. This:

...

```
Usage: {  
  "input_tokens": 824,  
  "output_tokens": 680,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Receiver - dyad_1

Timestamp: 2026-08-12T09:17:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_4"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with risk management.

Since I don't have information about my co-player's behavior yet, I'll send a moderate amount that:

- Demonstrates willingness to cooperate
- Creates opportunity for mutual gain (the tripling effect means both of us can benefit)
- Doesn't expose me to total loss

If I send \$3, my co-player receives \$9 ($3 \times \3). If they return even \$4, we'd both end up better off than our starting positions - I'd have \$6 ($\$5 - \$3 + \4) and they'd have \$5.

This seems like a reasonable opening move to establish cooperative intent.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 of a trust game where I'm the SENDER. Let me think through the strategic considerations:

1. I have \$5 to decide how to allocate
2. Whatever I send gets tripled when it reaches the receiver
3. The receiver then decides how much to return to me
4. My final payoff = $\$5 - \text{amount_sent} + \text{amount_returned}$

The key strategic considerations:

- If I send nothing (\$0), I keep \$5 and the receiver gets nothing. Safe but no potential for gain.
- If I send everything (\$5), the receiver gets \$15. If they return even \$6, I'd be better off ($\$5 - \$5 + \$6 = \6).
- This is a one-shot game with this particular partner (though I'll play multiple rounds with different partners).
- There's communication noise mentioned, which means the receiver might see a different amount than what I actually send.

Given this is the first round and I don't have information about the other agents' behavior yet:

Strategic approaches:

1. **Trust-building**: Send a moderate-to-high amount to signal cooperation and trust
2. **Conservative**: Send a small amount to minimize risk
3. **Optimal cooperation**: Send \$5 if I expect the receiver to split gains fairly

If the receiver is rational and fair, they might follow a norm like returning half of what they receive, or returning enough to make both players better off than the initial endo

wment.

For example, if I send \$3:

- I have: $\$5 - \$3 = \$2$ left
- Receiver gets: $\$3 \times 3 = \9
- If receiver returns \$4: I end with $\$2 + \$4 = \$6$, they keep \$5
- Both better off than initial \$5

If I send \$5:

- I have: \$0 left
- Receiver gets: \$15
- If receiver returns \$8: I end with \$8, they keep \$7
- Both significantly better off

The noise factor complicates things-if I send \$5 but it appears as less, the receiver might return proportionally less. Based on experimental economics patterns where senders typically send around 50% and receivers often return amounts that create mutual gains, I'll send \$3. This signals cooperation while maintaining some cushion, and if the receiver follows typical reciprocity norms, we should both end up better off than our starting positions....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 841,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.8 to you. Your visible payoff was \$5.8.

Your total visible earnings across all rounds are \$5.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.56, so you received \$7.68. How much do you return? (0-7.68)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****Analysis:****

- My co-player sent \$2.56 (about 51% of their endowment), showing moderate trust
- I received \$7.68 after the 3x multiplication
- They currently have \$2.44 remaining

****Reciprocity consideration:****

- In Round 1, my co-player returned about 42% of what they received
- I want to be fair and encourage continued cooperation
- Returning around 45-50% seems reasonable

If I return \$3.50:

- Sender ends with: $\$2.44 + \$3.50 = \$5.94$ (gain of \$0.94)
- I keep: $\$7.68 - \$3.50 = \$4.18$ (gain of \$4.18)

This rewards their trust while keeping a fair share given the multiplication advantage.

```
'''json
{"return": 3.5}
'''
```

Reasoning:

Let me analyze this situation:

Round 1 recap:

- I was SENDER, sent \$3
- It became \$9 for receiver
- They returned \$3.8 to me
- My payoff: $\$5 - \$3 + \$3.8 = \5.8
- Their payoff: $\$9 - \$3.8 = \$5.2$

So in Round 1, my co-player returned about 42% of what they received (\$3.8/\$9 ? 0.42).

Round 2:

- I am RECEIVER
- Sender sent \$2.56
- I received \$7.68 (2.56×3)
- I need to decide how much to return

The sender invested \$2.56, so they currently have $\$5 - \$2.56 = \$2.44$ remaining.

If I think about reciprocity and building trust:

- The sender showed moderate trust by sending about 51% of their endowment
- If I want to encourage cooperation, I should return enough that the sender comes out ahead

Let me consider different return amounts:

- If I return \$4: Sender gets $\$2.44 + \$4 = \$6.44$ (net gain of \$1.44), I keep \$3.68 (net gain of \$3.68)
- If I return \$3.5: Sender gets $\$2.44 + \$3.5 = \$5.94$ (net gain of \$0.94), I keep \$4.18 (net gain of \$4.18)
- If I return \$3: Sender gets $\$2.44 + \$3 = \$5.44$ (net gain of \$0.44), I keep \$4.68 (net gain of \$4.68)

In my previous round, the receiver returned about 42% of what they received. If I apply a similar principle:

$\$7.68 \times 0.42 \approx \3.23

I'm inclined to return slightly more than that baseline to reward their trust and establish a cooperative pattern. Returning \$3.5 seems like the right balance-it gives them a meaningful profit while still letting me benefit substantially, and it signals I'm reliable without being exploited....

Usage: {

```
"input_tokens": 804,  
"output_tokens": 923,  
"reasoning_tokens": 0
```

}

GAME - Agent_8 - Sender - dyad_2

Timestamp: 2026-08-12T09:17:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_2",  
"move_index": 2,  
"opponent_id": "Agent_2",  
"pairing": {  
  "agents": [  
    "Agent_8",  
    "Agent_2"  
  ],  
  "investor": "Agent_8",  
  "roles": {  
    "Agent_2": "trustee",  
    "Agent_8": "investor"  
  },  
  "trustee": "Agent_2"
```

```
},  
"role": "investor",  
"role_label": "Sender",  
"round": 2,  
"task": "game",  
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.68, that is 53.6% of their total endowment. You received \$8.040000000000001. How much do you return to the sender? (0-8.040000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.68 (53.6% of their endowment)
- They now have \$2.32 remaining
- I received \$8.04 after the 3x multiplier
- Total pool is now \$10.36

I want to reciprocate their trust while being fair. The sender took a risk to create mutual gains, so I should ensure they benefit from their trust.

A reasonable approach is to aim for a relatively balanced outcome:

- If I return \$3.00, the sender ends with \$5.32 (gaining \$0.32)
- I would keep \$5.04

This rewards their trust while maintaining fairness, and shows I'm reciprocal which could encourage cooperation in future rounds.

{"return": 3.00}

reasoning:

Let me analyze this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.68 (which is 53.6% of their \$5 endowment)
3. This was multiplied by 3x, so I received $2.68 \times 3 = \$8.04$

4. I need to decide how much to return (between \$0 and \$8.04)

Key considerations:

- The sender showed trust by sending more than half of their endowment (\$2.68 out of \$5)
- The sender now has: $\$5 - \$2.68 = \$2.32$
- I currently have: \$8.04
- Whatever I return, the sender will end up with: $\$2.32 + \text{my return}$
- I will end up with: $\$8.04 - \text{my return}$

From a fairness/reciprocity perspective:

- The sender trusted me with 53.6% of their endowment
- A fair split might be to ensure we both end up with similar amounts, or to reward the sender's trust
- If I return \$3, sender ends with \$5.32, I end with \$5.04 (relatively equal)
- If I return \$4, sender ends with \$6.32, I end with \$4.04 (rewarding trust more)
- If I return \$2, sender ends with \$4.32, I end with \$6.04 (keeping more)

A common reciprocal strategy would be to:

1. Ensure the sender doesn't lose money (they should end with at least \$5)
2. Share the "profit" somewhat fairly

The total pool is now $\$2.32$ (sender's remainder) + $\$8.04$ (my endowment) = $\$10.36$

If we split this profit fairly:

- Each should get about \$5.18
- For sender to get \$5.18, I need to return: $\$5.18 - \$2.32 = \$2.86$

Returning \$3.00 seems reasonable-it rewards their trust while keeping the outcome relatively balanced, with sender ending at \$5.32 and me at \$5.04. But I should consider whether a more reciprocal approach might better acknowledge their initial generosity.

If I return roughly the same proportion they invested (53.6% of what I received), that's about \$4.31. This leaves them with \$6.63 and me with \$3.73-a significant reward for their trust, though it cuts my gains substantially. Alternatively, I could split the total gains evenly: the \$5.36 profit gets divided so each of us gains \$2.68, putting the sender at \$7.68 and me at \$2.68....

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 1185,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.68, it became \$8.040000000000001, and you returned \$3.0. Your visible payoff was \$5.040000000000001.

Your total visible earnings across all rounds are \$5.040000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this new round as the sender:

In the previous round, my co-player sent 53.6% of their endowment, showing moderate-high trust. I reciprocated fairly by returning enough to give them a positive gain.

Now with a new co-player, I want to signal trust and willingness to cooperate while not being overly risky. Sending around 50% seems reasonable - it creates significant potential mutual gains while maintaining some protection.

If they reciprocate proportionally, we both benefit.

```
{"send": 2.50}
```

Reasoning:

Now I'm the sender in Round 2 with a new co-player.

Last round I was the receiver:
- Saw the sender send \$2.68 (53.6%)
- Received \$8.04
- Returned \$3.00
- Kept \$5.04

The sender ended with $\$2.32 + \$3.00 = \$5.32$, so they gained \$0.32 from trusting me.

Now I'm the sender with \$5. I need to decide how much to send.

Key considerations:

1. The other player sent 53.6% in the previous example, showing moderate-high trust
2. I reciprocated fairly, returning enough to give them a positive gain
3. This is a new co-player, so I don't know their strategy yet
4. There's communication noise, so amounts may appear different
5. Building cooperation early could pay off in future rounds

Strategic thinking:

- If I send too little, I signal distrust and may get nothing back
- If I send too much, I risk losing a lot if they don't reciprocate
- A moderate amount (around 50-60%) shows trust while not being reckless
- The previous sender sent 53.6%, which seemed like a reasonable trust signal

I'll send around \$2.50-3.00 to signal trust and willingness to cooperate.

Let me think about expected values:

- If I send \$2.50, they get \$7.50. If they return 40% = \$3.00, I end with $\$2.50 + \$3.00 = \$5.50$ (gain \$0.50)
- If I send \$3.00, they get \$9.00. If they return 33% = \$3.00, I end with $\$2.00 + \$3.00 = \$5.00$ (break even)
- If I send \$3.00 and they return 44% = \$4.00, I end with $\$2.00 + \$4.00 = \$6.00$ (gain \$1.00)

...

```
Usage: {  
  "input_tokens": 852,  
  "output_tokens": 646,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_2

Timestamp: 2026-08-12T09:18:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_8",  
      "Agent_2"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_2": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative tone while managing some risk given the uncertainty.

If I send money, it triples in value, creating potential mutual gains. However, I don't know my co-player yet, and there's communication noise to consider.

I'll send a moderate amount to signal trust and willingness to cooperate, while keeping some buffer:

****Sending \$3****

This gives my co-player \$9 to work with. If they return a fair amount (around half or more), we both benefit compared to me sending nothing. It's a balance between demonstrating cooperation and managing first-round uncertainty.

{'send': 3}

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically:

1. I start with \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return

4. My final payoff = \$5 - sent + returned
5. Receiver's final payoff = (sent × 3) - returned

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise (amounts may differ from what was actually sent)
- Agent names are hidden

From a purely selfish perspective, if I send \$0, I guarantee myself \$5. But this doesn't build trust or allow for mutual gains.

If I send everything (\$5), and the receiver returns a fair amount (like half of what they receive = \$7.50), I'd end up with \$7.50, and they'd keep \$7.50 as well. We'd both be better off.

However, there's risk:

- The receiver might not return anything
- Communication noise could cause misunderstandings
- This is just round 1, so I don't have reputation information

A moderate approach might be to send \$3:

- They receive \$9
- If they return \$5, I get \$5 (break even on sent, but signal trust)
- If they return \$6, I get \$6 (gain \$1)
- They keep between \$3-\$4 depending on return

Or I could send \$2:

- They receive \$6
- If they return \$3, I get \$3, netting -\$2 but we both gained
- If they return \$4, I get \$4, netting -\$1
- Lower risk

Given it's round 1 and I want to establish cooperative play but also manage risk, I'll send \$3. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 598,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.41 to you. Your visible payoff was \$6.41.

Your total visible earnings across all rounds are \$6.41.

This round, you are the RECEIVER against your current co-player. They sent you \$1.73, so you received \$5.189999999999995. How much do you return? (0-5.189999999999995)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****What happened:****

- My co-player sent \$1.73 (became \$5.19 for me)
- They kept \$3.27
- They showed some trust by sending money

****My reasoning:****

- In the previous round, my co-player returned about 49% of what they received (\$4.41 out of \$9)
- I should be reasonably cooperative to encourage future trust
- Returning around 45-50% of what I received seems fair
- This would be \$2.33-\$2.60

I'll return \$2.40:

- They'll end with: \$3.27 + \$2.40 = \$5.67

- I'll end with: $\$5.19 - \$2.40 = \$2.79$
- They gain $\$0.67$ from sending, which rewards their trust

```
{'return': 2.40}
```

Reasoning:

Let me think about this situation:

1. In the previous round, I sent \$3, which became \$9 for the receiver. They returned \$4.41 to me.
 - I ended with: $\$5 - \$3 + \$4.41 = \6.41
 - They ended with: $\$9 - \$4.41 = \$4.59$
 - So they returned about 49% of what they received, which seems fairly cooperative.
2. Now I'm the RECEIVER. My co-player sent me \$1.73, which became approximately \$5.19.
 - They now have: $\$5 - \$1.73 = \$3.27$
 - I have: $\$5.19$
 - Total pie: $\$3.27 + \$5.19 = \$8.46$
3. There's communication noise, so what I see (\$1.73 sent) might not be exactly what they sent.
4. The sender took some risk by sending me \$1.73. It's less than what I sent in the previous round (\$3), but still a meaningful amount showing some trust.
5. I should consider:
 - Being reciprocal/cooperative to encourage future cooperation
 - The communication noise factor
 - What would be "fair"

If I think about fairness:

- They sent \$1.73, keeping \$3.27
- I received \$5.19
- If we aim for roughly equal outcomes: $(\$3.27 + \text{return}) ? (\$5.19 - \text{return})$
- That would suggest: $\$3.27 + \text{return} = \$5.19 - \text{return}$
- $2 * \text{return} = \$5.19 - \$3.27 = \$1.92$
- $\text{return} ? \$0.96$

But that seems quite low given they took a risk. Let me think about proportional return:

- Matching their 49% return rate would mean sending back \$2.54
- Alternatively, if I return \$2.50, they'd end with \$5.77 (a \$0.77 gain) while I'd have \$2.69
- This feels like a balanced cooperative move that rewards their trust without leaving me too disadvantaged

I'll return \$2.50....

```
Usage: {
  "input_tokens": 791,
  "output_tokens": 1107,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T09:18:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_1"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_1": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_1"
  }
}
```

```

},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.97, that is 59.4% of their total endowment. You received \$8.91. How much do you return to the sender? (0-8.91)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.97 (about 60% of their endowment)
- I received \$8.91 after the 3x multiplier
- The sender currently has only \$2.03 remaining (\$5 - \$2.97)

The sender took a substantial risk and demonstrated trust. To reciprocate fairly and encourage future cooperation, I should return a reasonable amount that:

1. Rewards their trust
2. Leaves both of us better off than the initial endowment

3. Establishes a cooperative relationship for future rounds

Returning approximately half of what I received seems fair - this would give the sender a good return on their trust while still leaving me with a decent gain.

I'll return **\$4.50**:

- Sender ends with: $\$2.03 + \$4.50 = \$6.53$ (gained \$1.53)
- I keep: $\$8.91 - \$4.50 = \$4.41$ (gained \$4.41)

Both of us benefit, and this reciprocates their trust.

```
{'return': 4.50}
```

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$2.97 (59.4% of their \$5 endowment)
- I received \$8.91 (which is $\$2.97 \times 3$)
- I need to decide how much to return (between \$0 and \$8.91)

Let me think about what's fair and strategic:

1. The sender showed significant trust by sending almost 60% of their endowment
2. If I return nothing, I keep \$8.91 and the sender gets only \$2.03 ($\$5 - \2.97)
3. If I return about half (\$4.46), I keep \$4.45 and the sender gets \$6.49 ($\$2.03 + \4.46)
4. If I return \$5.94, I keep \$2.97 (equal to what they sent) and the sender gets \$8.97 total

The sender took a risk and showed trust. A common approach in trust games is reciprocity. The sender sent about 60% of their endowment, which is generous.

Let me consider what might establish good cooperation for future rounds:

- Returning about the same amount they sent (\$2.97) would mean they gain \$0 net
- Returning more than they sent but less than half of what I received seems reasonable
- Returning around 1/3 to 1/2 of what I received is often considered fair

Let me calculate some options:

- Return \$3: Sender gets \$5.03 total (gained \$0.03), I keep \$5.91
- Return \$4: Sender gets \$6.03 total (gained \$1.03), I keep \$4.91
- Return \$4.50: Sender gets \$6.53 total (gained \$1.53), I keep \$4.41
- Return \$5: Sender gets \$7.03 total (gained \$2.03), I keep \$3.91
- Return \$4.50 seems like the right balance-it reciprocates their trust by returning roughly half of what I received,...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 898,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.97, it became \$8.91, and you returned \$4.5. Your visible payoff was \$4.41.

Your total visible earnings across all rounds are \$4.41.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my previous experience:

- As a receiver, a sender trusted me with about 60% of their endowment
- I reciprocated fairly, and we both benefited
- This established that cooperation can work well in this game

Now as the sender with a new co-player, I want to:

1. Demonstrate trust to encourage reciprocation
2. Follow a similar cooperative strategy that worked before
3. Give my co-player the opportunity to earn and share

I'll send **\$3.00** (60% of my endowment), mirroring the trust shown to me in the previous round.

```
{'send': 3.00}
```

Reasoning:

Let me think about this situation:

In Round 1:

- I was the receiver
- The sender sent me \$2.97 (59.4% of their endowment)
- I received \$8.91
- I returned \$4.50
- I kept \$4.41
- The sender got back \$4.50, so they ended with $\$2.03 + \$4.50 = \$6.53$

Now in Round 2:

- I'm the sender
- I have \$5 to potentially send
- I'm playing against a different co-player (the problem states pairings are randomized each round)

Key considerations:

1. The previous sender showed trust by sending 59.4% of their endowment
2. I reciprocated fairly by returning about 50% of what I received
3. This worked out well - both players benefited
4. Now I'm in the sender role with a new partner

Since my previous experience showed that sending a substantial amount can lead to good returns, and I want to establish trust with this new co-player, I should send a meaningful amount.

The previous sender sent about 60% (\$2.97 out of \$5). This seemed like a good trust-building amount. I could:

- Send a similar proportion to continue the cooperative pattern
- Send slightly more to show strong trust
- Send slightly less to be more cautious with a new partner

I think sending around \$3 (60%) makes sense - it mirrors what worked well in the previous round and shows trust to establish a good relationship with this new co-player.

Let me send \$3.00.

```
Usage: {  
  "input_tokens": 908,  
  "output_tokens": 516,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_3

Timestamp: 2026-08-12T09:18:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_1"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",
```



```

"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to make a decision without any prior information about my co-player.

****Strategic considerations:****

- Sending more creates potential for mutual gain (3x multiplier)
- But I have no information about whether my co-player will reciprocate
- This is a multi-round game, so building some trust could be valuable
- The communication noise adds uncertainty

For Round 1, I'll take a moderately trusting approach. Sending \$3 signals willingness to cooperate while managing risk. If my co-player is reciprocal, we can both benefit (they'll receive \$9, and even a partial return would make this worthwhile). If they don't reciprocate, I've limited my exposure.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 to potentially send
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. There's communication noise - amounts may appear different than what was actually sent/returned

Key considerations:

- This is a multi-round game with 8 agents total
- Pairings are randomized each round
- I'll play both sender and receiver roles across rounds
- Agent names are hidden

Strategic thinking for Round 1:

- Sending \$0 guarantees I keep \$5, but there's no potential for mutual gain
- Sending \$5 shows maximum trust, could lead to good returns if the receiver is cooperative, but risky
- A moderate amount (like \$2-3) balances risk and potential reward
- This is Round 1, so I have no history with any agent yet

The game has a cooperative element - if both players trust each other, total welfare is maximized. The 3x multiplier means:

- If I send \$5, receiver gets \$15. If they return \$7.50, we each end with \$7.50 (better than \$5 each)
- If I send \$3, receiver gets \$9. If they return \$4, I end with \$6 and they end with \$5
- There's also the noise factor to consider

For a first round with no information, I think a moderately trusting approach makes sense.

I'll send \$3 as a signal of willingness to cooperate while not being completely vulnerable. This allows for mutual benefit if the receiver is cooperative, but limits my downside risk.

Actually, let me reconsider. In behavioral economics experiments, sending about half (\$2-3) is common in first rounds. However, given this is a multi-round game and reputation might matter (even though pairings are random, players might develop general strategies), showing some trust could be beneficial.

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 651,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.5 to you. Your visible payoff was \$6.5.

Your total visible earnings across all rounds are \$6.5.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****Analysis:****

- My co-player sent \$2.9 (about 58% of their endowment), showing considerable trust
- I received \$8.7 after the 3x multiplier
- In Round 1, when I sent \$3, my previous partner returned \$4.5 (50% of what they received)
- Applying a similar reciprocity norm seems fair

****Decision:****

I'll return approximately 50% of what I received, which rewards their trust and establishes a cooperative pattern. This leaves us both better off than our starting positions.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me analyze this situation:

****Previous Round (Round 1):****

- I was SENDER
- I sent \$3.0
- It became \$9.0
- They returned \$4.5 to me
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Their payoff: $\$9 - \$4.5 = \$4.5$

That was a pretty fair return - they returned half of what they received, which is a common reciprocity norm.

****Current Round (Round 2):****

- I am RECEIVER
- They sent me \$2.9 (close to \$3)
- I received \$8.7 (2.9×3)
- I need to decide how much to return (0-8.7)

****Considerations:****

1. My co-player sent me almost \$3 (actually \$2.9, but there's communication noise mentioned)
2. In Round 1, when I sent \$3, my partner returned \$4.5 (which was 50% of the \$9 they received)
3. If I apply the same 50% rule: I should return about \$4.35 (half of \$8.7)
4. My co-player took a risk by sending \$2.9, showing trust
5. Being reciprocal could establish good cooperation
6. Note: This is a different co-player than Round 1 (pairings are randomized)

Since this is a new co-player and they showed trust by sending \$2.9, I should reciprocate fairly. Returning around 50% seems like a reasonable reciprocity norm that:

- Rewards their trust
- Leaves both of us better off than not playing
- Establishes me as a cooperative player

My co-player's payoff if I return \$4.35:

- They get: $\$5 - \$2.9 + \$4.35 = \6.45
- I get: $\$8.7 - \$4.35 = \$4.35$

They sent 58% of their endowment, which is quite generous. I should return slightly more to match their generosity and encourage continued cooperation. Returning \$4.5 mirrors what my previous partner did for me and feels like the right balance....

```
Usage: {
  "input_tokens": 811,
  "output_tokens": 781,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Sender - dyad_4

Timestamp: 2026-08-12T09:18:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_7"
    ],
    "investor": "Agent_5",
```

```

"roles": {
  "Agent_5": "investor",
  "Agent_7": "trustee"
},
"trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.54, that is 50.8% of their total endowment. You received \$7.62. How much do you return to the sender? (0-7.62)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.54 (about half their endowment)
- I received \$7.62 after the 3x multiplier
- I want to reciprocate fairly while both of us benefit

The sender took a risk to create mutual gains. A fair return should:

1. Reward their trust
2. Leave us both better off than if nothing was sent
3. Show I'm a cooperative partner for future rounds

If I return \$3.50:

- Sender ends with: $\$5.00 - \$2.54 + \$3.50 = \5.96 (net gain: \$0.96)
- I end with: $\$7.62 - \$3.50 = \$4.12$ (net gain: \$4.12)

This seems like a fair reciprocation that rewards the sender's trust while acknowledging the receiver advantage in this game.

```
'''json
{"return": 3.50}
'''
```

reasoning:

Let me analyze this situation carefully:

- I am the RECEIVER in Round 1
- The sender sent \$2.54 (which is 50.8% of their \$5 endowment)
- After the 3x multiplier, I received \$7.62 (which is indeed $\$2.54 \times 3$)
- I need to decide how much to return (between \$0 and \$7.62)

Key considerations:

1. **The sender showed trust**: They sent about half their endowment (\$2.54 out of \$5), showing significant trust in me.
2. **Fair reciprocation**: A common approach in trust games is to return an amount that:
 - Rewards the sender for their trust
 - Results in both players being better off than if nothing was sent
 - Maintains equity or fairness
3. **What would be a fair return?**
 - If I return \$0: I keep \$7.62, sender keeps \$2.46
 - If I return half (\$3.81): I keep \$3.81, sender keeps \$6.27
 - If I return \$2.54 (what they sent): I keep \$5.08, sender keeps \$5.00
 - If I return \$3.81: I keep \$3.81, sender keeps \$6.27
4. **Strategic considerations**:
 - This is a multi-round game with 8 agents
 - Building a reputation for reciprocity might encourage future cooperation
 - The note mentions communication noise, so amounts might appear different to the sender
5. **Fairness analysis**:
 - Initial positions: Sender had \$5, I had \$0
 - After transaction: Sender has \$2.46 ($\$5 - \2.54), I have \$7.62
 - If I return \$3.81: Sender ends with \$6.27, I end with \$3.81
 - If I return \$2.54: Sender ends with \$5.00, I end with \$5.08
 - The sender created \$5.08 in additional value through their trust, so splitting that gain equally would mean returning about \$2.58, which gets us both to roughly \$5.04 each.

I'm considering what feels genuinely reciprocal here. Returning somewhere between \$3.00 and \$3.50 seems right-it rewards their willingness to trust me while still leaving me better off than I started. If I return \$3.00, they'd end with \$5.46 and I'd have \$4.62; at \$3.50, they'd get \$5.96 and I'd keep \$4.12. Either way, we both benefit from the exchange.

I'll go with \$3.50. It signals that I value their trust and want to be someone they'd cooperate with again, while still giving me a meaningful gain from the interaction....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1373,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.54, it became \$7.62, and you returned \$3.5. Your visible payoff was \$4.12.

Your total visible earnings across all rounds are \$4.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

In Round 1, my co-player sent me \$2.54 (about 50% of their endowment), showing good trust. I reciprocated with a fair return.

Now I'm the sender with a new co-player. I should:

1. Signal cooperation to encourage reciprocity
2. Send a reasonable amount that allows mutual benefit
3. Not overcommit without knowing this co-player's behavior

Following the cooperative pattern from Round 1, I'll send a similar proportion - about 50% of my endowment.

```
'''json
{"send": 2.50}
'''
```

Reasoning:

Let me think about this situation:

Round 1: I was the receiver

- Sender sent \$2.54 (50.8% of their endowment)
- I received \$7.62
- I returned \$3.50
- My payoff was \$4.12
- Sender's payoff would have been \$5 - \$2.54 + \$3.50 = \$5.96

Now in Round 2, I'm the sender. I need to decide how much to send.

Key considerations:

1. The previous sender showed trust by sending about 50% (\$2.54 out of \$5)
2. I reciprocated reasonably well by returning \$3.50, giving them a positive return
3. This is a multi-round game with 8 agents, so building a reputation for cooperation could be beneficial
4. There's communication noise, so the exact amounts might differ
5. I'm paired with a different co-player this round (pairings are randomized)

Since this is a new co-player, they won't have direct experience of my behavior. However, in a repeated multi-agent setting, establishing myself as cooperative could be beneficial.

The previous sender sent about 50% and I reciprocated well. I should consider:

- Sending a similar moderate-to-high amount to signal cooperation
- Not sending too much in case this receiver is less cooperative than I was
- Not sending too little, as that could signal I'm uncooperative

I think sending around \$2.50-\$3.00 (50-60%) would be reasonable. This:

- Shows trust and willingness to cooperate
- Mirrors what the previous sender did to me
- Leaves room for mutual benefit if the receiver is cooperative
- Doesn't risk my entire endowment

I'll send \$2.50, which is exactly 50% of my endowment, similar to what the previous sender did

```
Usage: {
  "input_tokens": 874,
  "output_tokens": 581,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T09:19:04
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_4",
"move_index": 7,
"opponent_id": "Agent_5",
"pairing": {
  "agents": [
    "Agent_5",
    "Agent_7"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will

- ll have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust with caution since I have no information about my co-player yet.

****My reasoning:****

- Sending nothing (\$0) is safe but offers no potential gain
- Sending everything (\$5) maximizes potential but is very risky with an unknown partner
- A moderate amount shows willingness to cooperate while limiting downside risk
- If my co-player returns roughly half of what they receive (a common fairness norm), we both benefit

I'll send ****\$3****. This gets tripled to \$9 for the receiver. If they return around \$4-5, we'd both end up better off than the initial \$5, establishing a cooperative dynamic.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

- **Basic mechanics**:**
 - I have \$5
 - Whatever I send gets tripled
 - The receiver decides how much to return
 - I keep: \$5 - sent + returned
- **Strategic considerations**:**
 - If I send \$0, I keep \$5, receiver gets \$0. Safe but no potential gains.
 - If I send everything (\$5), receiver gets \$15. If they return half (\$7.50), I'd have \$7.50 total (better than \$5).
 - There's risk: the receiver might return nothing.
 - This is round 1, so there's no history to base trust on.
 - There are multiple rounds, so building trust/reputation could matter.
 - The game has communication noise, which adds uncertainty.
- **Multi-agent context**:**
 - There are 8 agents total
 - Pairings are randomized each round
 - I might not play with the same person again
 - This reduces the value of building reputation with a specific partner
 - However, if there's any possibility of reputation effects across the pool, cooperation could still matter
- **First round strategy**:**
 - Since this is round 1 with a random partner, I have no information about their behavior
 - A moderate amount shows trust but limits risk
 - Common starting strategies in trust games involve sending a moderate amount (e.g., \$2-3)
 - This signals willingness to cooperate while not being overly naive
- **Optimal reasoning**:**
 - If the receiver is rational and forward-looking, they might return a fair amount to encourage future cooperation
 - Even if I don't play them again, they might reason that cooperation is beneficial
 - Sending around \$3 seems reasonable: shows trust, limits downside, and if they return half of the tripled amount (half of \$9 = \$4.50), I'd end up with \$6.50 total

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 668,
```



```
    "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.8 to you. Your visible payoff was \$6.8.

Your total visible earnings across all rounds are \$6.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.48, so you received \$7.439999999999995. How much do you return? (0-7.439999999999995)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this round:

****Analysis:****

- My co-player sent \$2.48 out of \$5 (about 50%), showing moderate trust
- This became \$7.44 for me
- In Round 1, my previous partner returned about 53% of what they received, which worked well

****My decision:****

I want to:

- Reward their trust appropriately
- Ensure both of us profit from cooperation
- Maintain a fair reciprocal relationship

If I return \$4.00:

- Sender gets: $\$2.52 + \$4.00 = \$6.52$
- I get: $\$7.44 - \$4.00 = \$3.44$
- Both better off than the initial \$5

This return (about 54% of what I received) mirrors the generous reciprocity I experienced in Round 1.

```
'''json
{"return": 4.0}
'''
```

Reasoning:

Let me analyze this situation:

****Round 1 (previous round):****

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$4.8 to me
- My payoff: $\$5 - \$3 + \$4.8 = \6.8
- Their payoff: $\$9 - \$4.8 = \$4.2$

So my previous co-player returned a bit more than half (53.3% of what they received), which was generous and fair.

****Round 2 (current round):****

- I am now the RECEIVER
- My current co-player (different from Round 1) sent me \$2.48
- It became \$7.44 (actually \$7.439999999999995, but I'll think of it as \$7.44)
- I need to decide how much to return

The note says there may be communication noise, so the amounts I see may differ slightly from what was actually sent/returned.

My current co-player sent \$2.48 out of their \$5, which is about 49.6% - nearly half. This shows moderate trust.

Now I need to decide what to return. Let me think about fairness and reciprocity:

1. ****Equal split approach****: If we aim for equal final payoffs:

- Sender currently has: $\$5 - \$2.48 = \$2.52$
- I currently have: \$7.44

- Total: \$9.96
- Equal split: \$4.98 each
- I should return: \$7.44 - \$4.98 = \$2.46

2. ****Proportional return based on what they sent**:**

- In Round 1, my partner returned 53.3% of what they received
- If I return 50% of \$7.44 = \$3.72
- This would give them: \$2.52 + \$3.72 = \$6.24
- And me: \$7.44 - \$3.72 = \$3.72

3. ****Return what maintains trust**:**

- My co-player showed moderate trust by sending nearly half
- I should reciprocate fairly to encourage future cooperation

I think returning around \$3.50 strikes the right balance-it rewards their trust while ensuring both of us end up ahead of the initial \$5 endowment. That gives them \$6.02 and me \$3.94.

But I could also return \$4.00 to show stronger reciprocation, which would give them \$6.52 and me \$3.44. Looking back...

```
Usage: {
  "input_tokens": 816,
  "output_tokens": 1083,
  "reasoning_tokens": 0
}
```

ROUND 3

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_6=Sender, Agent_3=Receiver
 dyad_2: Agent_8=Sender, Agent_7=Receiver
 dyad_3: Agent_2=Sender, Agent_1=Receiver
 dyad_4: Agent_4=Sender, Agent_5=Receiver

Parsed game decisions:

dyad_1: Agent_6 sent \$3; received \$9; Agent_3 returned \$3.75; payoffs Agent_6=\$5.75, Agent_3=\$5.25; noise/visible: sent visible=\$2.97, received visible=\$8.91, returned visible=\$2.98
 Action validation: {"returned": {"amount": 3.75, "clamped": false, "max": 9.0, "min": 0, "raw": 3.75}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_2: Agent_8 sent \$2.5; received \$7.5; Agent_7 returned \$5; payoffs Agent_8=\$7.5, Agent_7=\$2.5; noise/visible: sent visible=\$3.02, received visible=\$9.06, returned visible=\$5.25
 Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 7.5, "min": 0, "raw": 5.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
 dyad_3: Agent_2 sent \$3; received \$9; Agent_1 returned \$3.4; payoffs Agent_2=\$5.4, Agent_1=\$5.6; noise/visible: sent visible=\$2.27, received visible=\$6.81, returned visible=\$3.71
 Action validation: {"returned": {"amount": 3.4, "clamped": false, "max": 9.0, "min": 0, "raw": 3.4}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_4: Agent_4 sent \$3.5; received \$10.5; Agent_5 returned \$6.5; payoffs Agent_4=\$8, Agent_5=\$4; noise/visible: sent visible=\$3.99, received visible=\$11.97, returned visible=\$6.07
 Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 10.5, "min": 0, "raw": 6.5}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

Balances after round: {

"Agent_1": 16.1,
 "Agent_2": 16.0,
 "Agent_3": 15.75,
 "Agent_4": 18.5,
 "Agent_5": 16.0,
 "Agent_6": 16.75,
 "Agent_7": 12.5,
 "Agent_8": 18.4

}

Visible balances after round: {

"Agent_1": 14.11,
 "Agent_2": 14.91,
 "Agent_3": 16.0,
 "Agent_4": 17.55,
 "Agent_5": 16.22,
 "Agent_6": 16.71,
 "Agent_7": 14.299999999999999,

```
"Agent_8": 18.48
}
```

LLM CALLS

GAME - Agent_6 - Sender - dyad_1

Timestamp: 2026-08-12T09:19:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_3"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_3": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths
, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.97, that is 59.4% of their total endowment. You received \$8.91. How much do you return to the sender? (0-8.91)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.97 (about 60% of their endowment)
- I received \$8.91 after the 3x multiplier
- The sender currently has only \$2.03 remaining (\$5 - \$2.97)

The sender took a substantial risk and demonstrated trust. To reciprocate fairly and encourage future cooperation, I should return a reasonable amount that:

1. Rewards their trust
2. Leaves both of us better off than the initial endowment
3. Establishes a cooperative relationship for future rounds

Returning approximately half of what I received seems fair - this would give the sender a good return on their trust while still leaving me with a decent gain.

I'll return **\$4.50**:

- Sender ends with: $\$2.03 + \$4.50 = \$6.53$ (gained \$1.53)
- I keep: $\$8.91 - \$4.50 = \$4.41$ (gained \$4.41)

Both of us benefit, and this reciprocates their trust.

{'return': 4.50}

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$2.97 (59.4% of their \$5 endowment)
- I received \$8.91 (which is $\$2.97 \times 3$)
- I need to decide how much to return (between \$0 and \$8.91)

Let me think about what's fair and strategic:

1. The sender showed significant trust by sending almost 60% of their endowment
2. If I return nothing, I keep \$8.91 and the sender gets only \$2.03 (\$5 - \$2.97)
3. If I return about half (\$4.46), I keep \$4.45 and the sender gets \$6.49 (\$2.03 + \$4.46)
4. If I return \$5.94, I keep \$2.97 (equal to what they sent) and the sender gets \$8.97 total

The sender took a risk and showed trust. A common approach in trust games is reciprocity. The sender sent about 60% of their endowment, which is generous.

Let me consider what might establish good cooperation for future rounds:

- Returning about the same amount they sent (\$2.97) would mean they gain \$0 net
- Returning more than they sent but less than half of what I received seems reasonable
- Returning around 1/3 to 1/2 of what I received is often considered fair

Let me calculate some options:

- Return \$3: Sender gets \$5.03 total (gained \$0.03), I keep \$5.91
- Return \$4: Sender gets \$6.03 total (gained \$1.03), I keep \$4.91
- Return \$4.50: Sender gets \$6.53 total (gained \$1.53), I keep \$4.41
- Return \$5: Sender gets \$7.03 total (gained \$2.03), I keep \$3.91
- Return \$4.50 seems like the right balance-it reciprocates their trust by returning roughly half of what I received,...

usage:

```
{
  "input_tokens": 529,
```

```

    "output_tokens": 898,
    "reasoning_tokens": 0
  }
Message 4 role=user
  content:
    Round 2

    In your most recent previous game round, you were the RECEIVER against another agent. You
    saw them send $2.97, it became $8.91, and you returned $4.5. Your visible payoff was $4.
    41.
    Your total visible earnings across all rounds are $4.41.

    This round, you are the SENDER against your current co-player. You have $5. How much do yo
    u send? (0-5)
    Respond exactly as JSON: {'send': <amount>}
Message 5 role=assistant
  content:
    Looking at my previous experience:
    - As a receiver, a sender trusted me with about 60% of their endowment
    - I reciprocated fairly, and we both benefited
    - This established that cooperation can work well in this game

    Now as the sender with a new co-player, I want to:
    1. Demonstrate trust to encourage reciprocation
    2. Follow a similar cooperative strategy that worked before
    3. Give my co-player the opportunity to earn and share

    I'll send **$3.00** (60% of my endowment), mirroring the trust shown to me in the previous
    round.

    {'send': 3.00}
  reasoning:
    Let me think about this situation:

    In Round 1:
    - I was the receiver
    - The sender sent me $2.97 (59.4% of their endowment)
    - I received $8.91
    - I returned $4.50
    - I kept $4.41
    - The sender got back $4.50, so they ended with $2.03 + $4.50 = $6.53

    Now in Round 2:
    - I'm the sender
    - I have $5 to potentially send
    - I'm playing against a different co-player (the problem states pairings are randomized ea
      ch round)

    Key considerations:
    1. The previous sender showed trust by sending 59.4% of their endowment
    2. I reciprocated fairly by returning about 50% of what I received
    3. This worked out well - both players benefited
    4. Now I'm in the sender role with a new partner

    Since my previous experience showed that sending a substantial amount can lead to good ret
    urns, and I want to establish trust with this new co-player, I should send a meaningful
    amount.

    The previous sender sent about 60% ($2.97 out of $5). This seemed like a good trust-buildi
    ng amount. I could:
    - Send a similar proportion to continue the cooperative pattern
    - Send slightly more to show strong trust
    - Send slightly less to be more cautious with a new partner

    I think sending around $3 (60%) makes sense - it mirrors what worked well in the previous
    round and shows trust to establish a good relationship with this new co-player.

    Let me send $3.00.
  usage:
    {

```

```
"input_tokens": 908,  
"output_tokens": 516,  
"reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.32 to you. Your visible payoff was \$7.32.

Your total visible earnings across all rounds are \$11.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my recent experience:

Round 2 results:

- I sent \$3.00 (60% of endowment)
- Receiver returned \$5.32
- I ended with \$7.32 - excellent outcome!

The cooperative strategy is working very well. When I demonstrate trust, it's being reciprocated generously.

Now facing a new co-player, I'll continue this successful approach. Sending around 60% has proven effective in building mutual benefit.

```
{'send': 3.00}
```

Reasoning:

Let me analyze the situation:

Round 2 (just completed):

- I was the SENDER
- I sent \$3.00 (60% of endowment)
- It became \$9.00 for the receiver
- They returned \$5.32 to me
- My payoff: $\$5 - \$3 + \$5.32 = \7.32
- They kept: $\$9.00 - \$5.32 = \$3.68$

This was a very generous return! They returned more than half of what they received, and I ended up with more than my initial endowment even after sending a substantial amount.

My experience so far:

- Round 1 (RECEIVER): Received \$8.91, returned \$4.50, kept \$4.41
- Round 2 (SENDER): Sent \$3.00, received back \$5.32, ended with \$7.32

Total visible earnings: \$11.73

The cooperation strategy is working very well! When I sent 60% in Round 2, I got an excellent return.

Now in Round 3, I'm a SENDER again with a new co-player.

Given:

1. My cooperative strategy has been working well
2. Both times cooperation has led to mutual benefit
3. I want to continue encouraging trust and reciprocation
4. The game benefits from establishing cooperative norms

I should continue with a cooperative approach. Sending around 60% worked well last time. I could:

- Send \$3.00 again (same as last time, 60%)
- Send a bit more to show strong trust
- Send a bit less to be slightly more cautious with a new partner

Since cooperation has been consistently rewarded, I'll stick with the successful strategy of sending around 60%.

```
...
Usage: {
  "input_tokens": 1176,
  "output_tokens": 538,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Receiver - dyad_1

Timestamp: 2026-08-12T09:19:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_3"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_3": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.12, that is 62.4% of their total endowment. You received \$9.36. How much do you return to the sender? (0-9.36)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.12 (62.4% of their endowment)
- This created \$9.36 for me after the 3x multiplier
- The sender currently has only \$1.88 remaining

I want to reward their trust and encourage future cooperation. The sender took a risk, and I should reciprocate appropriately.

I'll return \$4.00, which:

- Gives the sender \$5.88 total (better than their initial \$5)
- Leaves me with \$5.36 (also benefiting from the cooperation)
- Demonstrates that trusting me pays off
- Should encourage continued cooperation in future rounds

{'return': 4.00}

reasoning:

Let me analyze this situation carefully:

1. The sender sent \$3.12 out of their \$5 endowment (62.4%)
2. This was multiplied by 3x, so I received \$9.36
3. I need to decide how much to return (between \$0 and \$9.36)

Key considerations:

- The sender showed significant trust by sending 62.4% of their endowment
- There's communication noise in the game, so actual amounts may differ slightly
- This is a repeated game with multiple rounds
- I want to encourage cooperation and build trust

Economic analysis:

- Sender currently has: $\$5 - \$3.12 = \$1.88$
- I currently have: \$9.36
- If I return nothing, sender gets \$1.88 total, I get \$9.36
- A fair split might consider that we each started with potential of \$5

Common strategies in trust games:

1. Return nothing (purely selfish)
2. Return the original amount sent (~\$3.12) - gives back what was sent
3. Return enough so we both profit equally from the 3x multiplier
4. Return more to encourage future cooperation

Let me calculate option 3 (equal profit):

- The 3x multiplier created \$6.24 in extra value ($\$9.36 - \3.12)
- Split equally: each gets \$3.12 extra
- Sender should end with: $\$1.88 + \$3.12 + \$3.12 = \8.12
- So I should return: $\$9.36 - (\$5 + \$3.12) = \$9.36 - \$8.12 = \1.24

Wait, let me recalculate more carefully:

- Sender starts with \$5, sends \$3.12, has \$1.88 left
- I receive \$9.36
- Total pool is now \$11.24
- If we each started with \$5 potential, that's \$10 baseline
- The extra \$1.24 created should split evenly at \$0.62 each

- So sender should end with \$5.62 and I should end with \$5.62
- That means I return \$3.74

I'm reconsidering the fairest approach here-whether to split the gains equally or use a different metric.

The sender invested 62.4% of their initial amount, so returning a proportional share of what I received makes sense. That's about \$5.84. Alternatively, since they trusted me with \$3.12 extra, returning \$3.00-\$3.50 would roughly restore their position while letting me benefit from the cooperation. For Round 1, I want to signal good faith and encourage continued trust, so returning 40-50% of the \$9.36 I received-somewhere between \$3.74 and \$4.68-feels like the right balance....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1077,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.12, it became \$9.36, and you returned \$4.00. Your visible payoff was \$5.3599999999999999.

Your total visible earnings across all rounds are \$5.3599999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Last round, my co-player trusted me with 62.4% of their endowment
- I reciprocated fairly, and we both benefited
- Now I'm with a new co-player as the sender

I'll mirror the cooperative behavior I experienced. Sending a substantial amount can create a mutual benefit through the 3x multiplier if my co-player reciprocates.

I'll send \$3.00 (60% of my endowment), similar to the trust shown to me in the previous round.

```
{'send': 3.00}
```

reasoning:

Let me think about this situation:

1. In the previous round, I was the receiver
 - The sender sent me \$3.12 (62.4% of their endowment)
 - I received \$9.36
 - I returned \$4.00
 - My payoff was \$5.36
2. Now I'm the sender with a new co-player
 - I have \$5 to potentially send
 - Whatever I send gets tripled
 - The receiver will decide how much to return

Strategic considerations:

- In the previous round, the sender showed trust by sending 62.4% of their endowment
- I reciprocated fairly, returning \$4.00 out of \$9.36 (about 43%)
- This gave the sender a total of \$5.88 (they sent \$3.12, kept \$1.88, got back \$4.00)

For this round:

- I'm paired with a different agent (randomized pairings)
- I don't know their history or behavior
- However, the game benefits from cooperation
- If I send a reasonable amount and they reciprocate, we both benefit

Let me consider different strategies:

- Sending nothing: I keep \$5, they get nothing, no mutual benefit
- Sending a small amount (e.g., \$1-2): Low risk, but lower potential gain
- Sending a moderate amount (e.g., \$3): Shows some trust, reasonable risk
- Sending a large amount (e.g., \$4-5): High trust, high risk, high potential reward

Given that:

1. The previous sender trusted me with 62.4% and it worked out
2. I want to establish a cooperative pattern
3. There's communication noise mentioned, so exact amounts may vary
4. I'm in a multi-round game where reputation (though anonymous) might matter

I'll send \$3.00, which is 60% of my endowment. This:

...

usage:

```
{
  "input_tokens": 824,
  "output_tokens": 680,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.48 to you. Your visible payoff was \$5.48.

Your total visible earnings across all rounds are \$10.84.

This round, you are the RECEIVER against your current co-player. They sent you \$2.97, so you received \$8.91. How much do you return? (0-8.91)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****Current round:****

- Sender showed good trust by sending \$2.97 (59.4% of their endowment)
- This created \$8.91 for me
- Sender currently has \$2.03

****Previous round context:****

- I sent \$3.00 (60% - very similar trust level)
- My co-player returned \$3.48, which was fair

I should reciprocate the trust shown to me. The sender sent almost 60% of their endowment, similar to what I did last round.

I'll return \$3.75, which:

- Gives the sender \$5.78 total (a gain of \$0.78)
- Leaves me with \$5.16 (also a gain)
- Rewards their trust appropriately

{'return': 3.75}

Reasoning:

Let me analyze this situation:

****Last round (Round 2):****

- I was the SENDER
- I sent \$3.00 (60% of my endowment)
- It became \$9.00 for the receiver
- They returned \$3.48
- My payoff: \$5 - \$3 + \$3.48 = \$5.48
- Their payoff: \$9 - \$3.48 = \$5.52

The receiver gave me back slightly more than I sent, which was cooperative and fair.

****This round (Round 3):****

- I'm the RECEIVER
- Sender sent \$2.97 (59.4% of their endowment - very similar to what I sent last round!)
- I received \$8.91

- Sender currently has: $\$5 - \$2.97 = \$2.03$

The sender showed good trust by sending ~60% of their endowment. This is very similar to the 60% I sent in the previous round.

I should reciprocate fairly. Let me think about what would be a fair return:

- If I return \$3.50, sender gets $\$2.03 + \$3.50 = \$5.53$, and I keep $\$8.91 - \$3.50 = \$5.41$
- If I return \$4.00, sender gets $\$2.03 + \$4.00 = \$6.03$, and I keep $\$8.91 - \$4.00 = \$4.91$
- If I return \$3.75, sender gets $\$2.03 + \$3.75 = \$5.78$, and I keep $\$8.91 - \$3.75 = \$5.16$

The sender trusted me with \$2.97, similar to how I trusted my previous co-player with \$3.00, and they returned \$3.48.

...

```
Usage: {  
  "input_tokens": 1089,  
  "output_tokens": 735,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_8 - Sender - dyad_2

Timestamp: 2026-08-12T09:19:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agents": [  
      "Agent_8",  
      "Agent_7"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_7": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_7"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.68, that is 53.6% of their total endowment. You received \$8.040000000000001. How much do you return to the sender? (0-8.040000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.68 (53.6% of their endowment)
- They now have \$2.32 remaining
- I received \$8.04 after the 3x multiplier
- Total pool is now \$10.36

I want to reciprocate their trust while being fair. The sender took a risk to create mutual gains, so I should ensure they benefit from their trust.

A reasonable approach is to aim for a relatively balanced outcome:

- If I return \$3.00, the sender ends with \$5.32 (gaining \$0.32)
- I would keep \$5.04

This rewards their trust while maintaining fairness, and shows I'm reciprocal which could encourage cooperation in future rounds.

```
{"return": 3.00}
```

reasoning:

Let me analyze this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.68 (which is 53.6% of their \$5 endowment)
3. This was multiplied by 3x, so I received $2.68 \times 3 = \$8.04$
4. I need to decide how much to return (between \$0 and \$8.04)

Key considerations:

- The sender showed trust by sending more than half of their endowment (\$2.68 out of \$5)
- The sender now has: $\$5 - \$2.68 = \$2.32$
- I currently have: \$8.04
- Whatever I return, the sender will end up with: $\$2.32 + \text{my return}$
- I will end up with: $\$8.04 - \text{my return}$

From a fairness/reciprocity perspective:

- The sender trusted me with 53.6% of their endowment
- A fair split might be to ensure we both end up with similar amounts, or to reward the sender's trust
- If I return \$3, sender ends with \$5.32, I end with \$5.04 (relatively equal)
- If I return \$4, sender ends with \$6.32, I end with \$4.04 (rewarding trust more)

- If I return \$2, sender ends with \$4.32, I end with \$6.04 (keeping more)

A common reciprocal strategy would be to:

1. Ensure the sender doesn't lose money (they should end with at least \$5)
2. Share the "profit" somewhat fairly

The total pool is now \$2.32 (sender's remainder) + \$8.04 (my endowment) = \$10.36

If we split this profit fairly:

- Each should get about \$5.18
- For sender to get \$5.18, I need to return: $\$5.18 - \$2.32 = \$2.86$

Returning \$3.00 seems reasonable-it rewards their trust while keeping the outcome relatively balanced, with sender ending at \$5.32 and me at \$5.04. But I should consider whether a more reciprocal approach might better acknowledge their initial generosity.

If I return roughly the same proportion they invested (53.6% of what I received), that's about \$4.31. This leaves them with \$6.63 and me with \$3.73-a significant reward for their trust, though it cuts my gains substantially. Alternatively, I could split the total gains evenly: the \$5.36 profit gets divided so each of us gains \$2.68, putting the sender at \$7.68 and me at \$2.68....

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 1185,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.68, it became \$8.040000000000001, and you returned \$3.0. Your visible payoff was \$5.040000000000001.

Your total visible earnings across all rounds are \$5.040000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this new round as the sender:

In the previous round, my co-player sent 53.6% of their endowment, showing moderate-high trust. I reciprocated fairly by returning enough to give them a positive gain.

Now with a new co-player, I want to signal trust and willingness to cooperate while not being overly risky. Sending around 50% seems reasonable - it creates significant potential mutual gains while maintaining some protection.

If they reciprocate proportionally, we both benefit.

```
{"send": 2.50}
```

reasoning:

Now I'm the sender in Round 2 with a new co-player.

Last round I was the receiver:

- Saw the sender send \$2.68 (53.6%)
- Received \$8.04
- Returned \$3.00
- Kept \$5.04

The sender ended with $\$2.32 + \$3.00 = \$5.32$, so they gained \$0.32 from trusting me.

Now I'm the sender with \$5. I need to decide how much to send.

Key considerations:

1. The other player sent 53.6% in the previous example, showing moderate-high trust
2. I reciprocated fairly, returning enough to give them a positive gain
3. This is a new co-player, so I don't know their strategy yet

4. There's communication noise, so amounts may appear different
5. Building cooperation early could pay off in future rounds

Strategic thinking:

- If I send too little, I signal distrust and may get nothing back
- If I send too much, I risk losing a lot if they don't reciprocate
- A moderate amount (around 50-60%) shows trust while not being reckless
- The previous sender sent 53.6%, which seemed like a reasonable trust signal

I'll send around \$2.50-3.00 to signal trust and willingness to cooperate.

Let me think about expected values:

- If I send \$2.50, they get \$7.50. If they return 40% = \$3.00, I end with $2.50 + 3.00 = 5.50$ (gain \$0.50)
- If I send \$3.00, they get \$9.00. If they return 33% = \$3.00, I end with $2.00 + 3.00 = 5.00$ (break even)
- If I send \$3.00 and they return 44% = \$4.00, I end with $2.00 + 4.00 = 6.00$ (gain \$1.00)

...

usage:

```
{
  "input_tokens": 852,
  "output_tokens": 646,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.19 to you. Your visible payoff was \$5.689999999999995.

Your total visible earnings across all rounds are \$10.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my recent experience:

- Round 2: I sent \$2.50 (50%) and received \$3.19 back, giving me a good positive return

This cooperative approach has been working well. I'm now with a new co-player, and I want to signal trustworthiness and encourage mutual cooperation.

Sending 50% has proven to be a solid strategy - it creates meaningful potential gains while showing balanced trust.

```
{"send": 2.50}
```

Reasoning:

Let me analyze the previous round:

- I sent \$2.50 (50% of my endowment)
- It became \$7.50 for the receiver
- They returned \$3.19
- I ended with $2.50 \text{ (kept)} + 3.19 \text{ (returned)} = 5.69$
- They kept $7.50 - 3.19 = 4.31$

So the receiver returned about 42.5% of what they received, which gave me a positive gain of \$0.69. This is a reasonable reciprocation, showing they valued cooperation.

Now I'm facing a new co-player in Round 3.

My experience so far:

- Round 1 (as receiver): Received \$8.04, returned \$3.00 (37.3%)
- Round 2 (as sender): Sent \$2.50, received back \$3.19 - good outcome

I should continue with a cooperative strategy. Since my last send of \$2.50 worked well, I could:

1. Send the same amount (\$2.50) to maintain consistency
2. Send slightly more to signal increased trust
3. Send slightly less to be more cautious

Given that cooperation has been working, and I'm now encountering a third different co-player, I think maintaining a cooperative stance of around 50% makes sense. This has proven successful and sends the right signal.

I'll send \$2.50 again.

```
Usage: {
  "input_tokens": 1103,
  "output_tokens": 441,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Receiver - dyad_2

Timestamp: 2026-08-12T09:19:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust with caution since I have no information about my co-player yet.

****My reasoning:****

- Sending nothing (\$0) is safe but offers no potential gain
- Sending everything (\$5) maximizes potential but is very risky with an unknown partner
- A moderate amount shows willingness to cooperate while limiting downside risk
- If my co-player returns roughly half of what they receive (a common fairness norm), we both benefit

I'll send ****\$3****. This gets tripled to \$9 for the receiver. If they return around \$4-5, we'd both end up better off than the initial \$5, establishing a cooperative dynamic.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. ****Basic mechanics**:**
 - I have \$5
 - Whatever I send gets tripled
 - The receiver decides how much to return
 - I keep: \$5 - sent + returned
2. ****Strategic considerations**:**
 - If I send \$0, I keep \$5, receiver gets \$0. Safe but no potential gains.
 - If I send everything (\$5), receiver gets \$15. If they return half (\$7.50), I'd have \$7.50 total (better than \$5).
 - There's risk: the receiver might return nothing.
 - This is round 1, so there's no history to base trust on.
 - There are multiple rounds, so building trust/reputation could matter.
 - The game has communication noise, which adds uncertainty.
3. ****Multi-agent context**:**
 - There are 8 agents total
 - Pairings are randomized each round
 - I might not play with the same person again
 - This reduces the value of building reputation with a specific partner
 - However, if there's any possibility of reputation effects across the pool, cooperation could still matter
4. ****First round strategy**:**
 - Since this is round 1 with a random partner, I have no information about their behavior
 - A moderate amount shows trust but limits risk
 - Common starting strategies in trust games involve sending a moderate amount (e.g., \$2-3)

- This signals willingness to cooperate while not being overly naive

5. ****Optimal reasoning****:

- If the receiver is rational and forward-looking, they might return a fair amount to encourage future cooperation
- Even if I don't play them again, they might reason that cooperation is beneficial
- Sending around \$3 seems reasonable: shows trust, limits downside, and if they return half of the tripled amount (half of \$9 = \$4.50), I'd end up with \$6.50 total

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 668,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.8 to you. Your visible payoff was \$6.8.

Your total visible earnings across all rounds are \$6.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.48, so you received \$7.439999999999995. How much do you return? (0-7.439999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this round:

****Analysis:****

- My co-player sent \$2.48 out of \$5 (about 50%), showing moderate trust
- This became \$7.44 for me
- In Round 1, my previous partner returned about 53% of what they received, which worked well

****My decision:****

I want to:

- Reward their trust appropriately
- Ensure both of us profit from cooperation
- Maintain a fair reciprocal relationship

If I return \$4.00:

- Sender gets: \$2.52 + \$4.00 = \$6.52
- I get: \$7.44 - \$4.00 = \$3.44
- Both better off than the initial \$5

This return (about 54% of what I received) mirrors the generous reciprocity I experienced in Round 1.

```
'''json
{"return": 4.0}
'''
```

reasoning:

Let me analyze this situation:

****Round 1 (previous round):****

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$4.8 to me
- My payoff: \$5 - \$3 + \$4.8 = \$6.8
- Their payoff: \$9 - \$4.8 = \$4.2

So my previous co-player returned a bit more than half (53.3% of what they received), which was generous and fair.

****Round 2 (current round):****

- I am now the RECEIVER

- My current co-player (different from Round 1) sent me \$2.48
- It became \$7.44 (actually \$7.439999999999995, but I'll think of it as \$7.44)
- I need to decide how much to return

The note says there may be communication noise, so the amounts I see may differ slightly from what was actually sent/returned.

My current co-player sent \$2.48 out of their \$5, which is about 49.6% - nearly half. This shows moderate trust.

Now I need to decide what to return. Let me think about fairness and reciprocity:

1. ****Equal split approach****: If we aim for equal final payoffs:
 - Sender currently has: $\$5 - \$2.48 = \$2.52$
 - I currently have: \$7.44
 - Total: \$9.96
 - Equal split: \$4.98 each
 - I should return: $\$7.44 - \$4.98 = \$2.46$
2. ****Proportional return based on what they sent****:
 - In Round 1, my partner returned 53.3% of what they received
 - If I return 50% of \$7.44 = \$3.72
 - This would give them: $\$2.52 + \$3.72 = \$6.24$
 - And me: $\$7.44 - \$3.72 = \$3.72$
3. ****Return what maintains trust****:
 - My co-player showed moderate trust by sending nearly half
 - I should reciprocate fairly to encourage future cooperation

I think returning around \$3.50 strikes the right balance-it rewards their trust while ensuring both of us end up ahead of the initial \$5 endowment. That gives them \$6.02 and me \$3.94.

But I could also return \$4.00 to show stronger reciprocation, which would give them \$6.52 and me \$3.44. Looking back...

usage:

```
{
  "input_tokens": 816,
  "output_tokens": 1083,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.48, it became \$7.439999999999995, and you returned \$4.0. Your visible payoff was \$3.439999999999995.

Your total visible earnings across all rounds are \$10.239999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this round:

****Analysis:****

- My co-player sent \$3.02 out of \$5 (about 60%), showing strong trust
- This became \$9.06 for me
- They're trusting me with more than the previous sender

****My decision:****

I want to reward this higher level of trust and maintain the cooperative pattern I've established (returning ~50-55% of what I receive).

If I return \$5.00:

- Sender gets: $\$1.98 + \$5.00 = \$6.98$
- I get: $\$9.06 - \$5.00 = \$4.06$
- Both significantly better off than the initial \$5

This return (about 55% of what I received) is consistent with my previous reciprocity and rewards their trust appropriately.

```
'''json
{"return": 5.0}
'''
```

Reasoning:

Let me analyze this round:

****What happened in Round 3:****

- I'm the RECEIVER again
- My co-player sent \$3.02 out of \$5 (about 60%), showing good trust
- This became \$9.06 for me

****My history:****

- Round 1: I was SENDER, sent \$3, received \$4.80 back (about 53% of the \$9 they received)
- Round 2: I was RECEIVER, received \$7.44, returned \$4.00 (about 54% of what I received)

****Decision for Round 3:****

My co-player sent a higher percentage this time (\$3.02 vs \$2.48 last round from a different player), showing increased trust. I should:

- Reward this trust
- Maintain reciprocity around 50-55% return rate
- Ensure both of us benefit

If I return \$5.00:

- Sender gets: $\$1.98 + \$5.00 = \$6.98$
- I get: $\$9.06 - \$5.00 = \$4.06$
- Both significantly better off than \$5

This is about 55% of what I received, consistent with my previous behavior and fair reciprocity.

Alternatively, if I return \$4.50:

- Sender gets: $\$1.98 + \$4.50 = \$6.48$
- I get: $\$9.06 - \$4.50 = \$4.56$
- Still both better off

Let me go with \$5.00 to be generous and reward the higher trust shown.

```
Usage: {
  "input_tokens": 1172,
  "output_tokens": 579,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Sender - dyad_3

Timestamp: 2026-08-12T09:20:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
```

```

"task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play a game and may al
    so be asked to write myths.

  GAME:
  You will play different roles across multiple rounds.

  When you are the SENDER:
  - You receive $5 each round
  - You decide how much to SEND to the RECEIVER ($0-$5)
  - Amount sent is multiplied by 3x
  - Receiver decides how much to RETURN
  - You keep: $5 - sent + returned
  - Receiver keeps: received - returned
  - Response format: {'send': <amount>}

  When you are the RECEIVER:
  - Sender receives $5
  - Sender decides how much to SEND to you ($0-$5)
  - Amount sent is multiplied by 3x
  - You decide how much to RETURN
  - Sender keeps: $5 - sent + returned
  - You keep: received - returned
  - Response format: {'return': <amount>}

  MYTH WRITING:
  You may also be asked to write short myths (stories) on various topics. When writing myths
    , follow the instructions provided in each prompt.

  IMPORTANT: Provide your decision in the correct JSON format.

  NOTE: There may be communication noise in this game.
  The amounts you see may differ slightly from what was actually sent/returned.
  This is part of the experimental design.

  MULTI-AGENT SETTING:
  - There are 8 agents in this run.
  - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
    , but the schedule balances sender and receiver roles across the run.
  - You play one sender-receiver game with your paired opponent each round.
  - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
    ll have acted as sender and receiver equally often when the round count allows it.
  - Agent names are hidden in this run.
  - Your paired opponent is referred to as your current co-player.

Message 2 role=user
content:
  Round 1: You are the SENDER. You have $5. How much do you send? (0-5)
Message 3 role=assistant
content:
  Looking at this first round, I want to establish a cooperative tone while managing some ri
    sk given the uncertainty.

  If I send money, it triples in value, creating potential mutual gains. However, I don't kn
    ow my co-player yet, and there's communication noise to consider.

  I'll send a moderate amount to signal trust and willingness to cooperate, while keeping so
    me buffer:

  **Sending $3**

  This gives my co-player $9 to work with. If they return a fair amount (around half or more
    ), we both benefit compared to me sending nothing. It's a balance between demonstrating
    cooperation and managing first-round uncertainty.

  {'send': 3}

```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically:

1. I start with \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. My final payoff = \$5 - sent + returned
5. Receiver's final payoff = (sent × 3) - returned

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise (amounts may differ from what was actually sent)
- Agent names are hidden

From a purely selfish perspective, if I send \$0, I guarantee myself \$5. But this doesn't build trust or allow for mutual gains.

If I send everything (\$5), and the receiver returns a fair amount (like half of what they receive = \$7.50), I'd end up with \$7.50, and they'd keep \$7.50 as well. We'd both be better off.

However, there's risk:

- The receiver might not return anything
- Communication noise could cause misunderstandings
- This is just round 1, so I don't have reputation information

A moderate approach might be to send \$3:

- They receive \$9
- If they return \$5, I get \$5 (break even on sent, but signal trust)
- If they return \$6, I get \$6 (gain \$1)
- They keep between \$3-\$4 depending on return

Or I could send \$2:

- They receive \$6
- If they return \$3, I get \$3, netting -\$2 but we both gained
- If they return \$4, I get \$4, netting -\$1
- Lower risk

Given it's round 1 and I want to establish cooperative play but also manage risk, I'll send \$3. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 598,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.41 to you. Your visible payoff was \$6.41.

Your total visible earnings across all rounds are \$6.41.

This round, you are the RECEIVER against your current co-player. They sent you \$1.73, so you received \$5.189999999999995. How much do you return? (0-5.189999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****What happened:****

- My co-player sent \$1.73 (became \$5.19 for me)
- They kept \$3.27
- They showed some trust by sending money

****My reasoning:****

- In the previous round, my co-player returned about 49% of what they received (\$4.41 out of \$9)
- I should be reasonably cooperative to encourage future trust
- Returning around 45-50% of what I received seems fair
- This would be \$2.33-\$2.60

I'll return \$2.40:

- They'll end with: $\$3.27 + \$2.40 = \$5.67$
- I'll end with: $\$5.19 - \$2.40 = \$2.79$
- They gain \$0.67 from sending, which rewards their trust

```
{'return': 2.40}
```

reasoning:

Let me think about this situation:

1. In the previous round, I sent \$3, which became \$9 for the receiver. They returned \$4.41 to me.
 - I ended with: $\$5 - \$3 + \$4.41 = \6.41
 - They ended with: $\$9 - \$4.41 = \$4.59$
 - So they returned about 49% of what they received, which seems fairly cooperative.
2. Now I'm the RECEIVER. My co-player sent me \$1.73, which became approximately \$5.19.
 - They now have: $\$5 - \$1.73 = \$3.27$
 - I have: \$5.19
 - Total pie: $\$3.27 + \$5.19 = \$8.46$
3. There's communication noise, so what I see (\$1.73 sent) might not be exactly what they sent.
4. The sender took some risk by sending me \$1.73. It's less than what I sent in the previous round (\$3), but still a meaningful amount showing some trust.
5. I should consider:
 - Being reciprocal/cooperative to encourage future cooperation
 - The communication noise factor
 - What would be "fair"

If I think about fairness:

- They sent \$1.73, keeping \$3.27
- I received \$5.19
- If we aim for roughly equal outcomes: $(\$3.27 + \text{return}) ? (\$5.19 - \text{return})$
- That would suggest: $\$3.27 + \text{return} = \$5.19 - \text{return}$
- $2 * \text{return} = \$5.19 - \$3.27 = \$1.92$
- $\text{return} ? \$0.96$

But that seems quite low given they took a risk. Let me think about proportional return:

- Matching their 49% return rate would mean sending back \$2.54
- Alternatively, if I return \$2.50, they'd end with \$5.77 (a \$0.77 gain) while I'd have \$2.69
- This feels like a balanced cooperative move that rewards their trust without leaving me too disadvantaged

I'll return \$2.50....

usage:

```
{
  "input_tokens": 791,
  "output_tokens": 1107,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.73, it became \$5.189999999999995, and you returned \$2.4. Your visible payoff was \$2.7899999999999996.

Your total visible earnings across all rounds are \$9.2.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my situation:

****Previous rounds:****

- Round 1: Sent \$3, received \$4.41 back - good cooperation
- Round 2: Received \$5.19, returned \$2.40 - showed I reciprocate

****Current round:****

- New co-player (randomized pairing)
- Need to balance trust with uncertainty

Since I had success with sending \$3 in Round 1 (got \$4.41 back, netting \$1.41 profit), and I want to maintain a cooperative approach while testing this new partner, I'll send the same amount:

****Sending \$3****

This creates \$9 for them to work with. If they're reasonably cooperative like my Round 1 partner, we'll both benefit.

{ 'send': 3 }

Reasoning:

Let me analyze this situation:

Round 1: I was SENDER

- I sent \$3
- It became \$9
- They returned \$4.41
- My payoff: $\$5 - \$3 + \$4.41 = \6.41

Round 2: I was RECEIVER

- They sent \$1.73
- I received \$5.19
- I returned \$2.40
- My payoff: $\$5.19 - \$2.40 = \$2.79$
- Total so far: $\$6.41 + \$2.79 = \$9.20$?

Round 3: I am SENDER again

- This is a new co-player (since pairings are randomized each round)
- I have \$5 to send

Strategy considerations:

1. I had a good experience in Round 1 - sent \$3, got back \$4.41 (profit of \$1.41)
2. In Round 2, I received \$5.19 and returned \$2.40 (46%), showing I'm cooperative
3. This is a new co-player, so I don't know their behavior
4. There's communication noise, so what they see may differ from what I intend

I should probably send a moderate amount again to:

- Signal willingness to cooperate
- Test this new partner's trustworthiness
- Not risk everything on an unknown player

Sending \$3 worked well in Round 1. I could try that again, or maybe send slightly more (\$3.50) or less (\$2.50) to experiment.

Given that:

- My first experience was positive
- I want to maintain a cooperative reputation
- But this is a new, unknown player

I'll send \$3 again - it's a balanced approach that showed good results before.

{ 'send': 3 }

```
Usage: {
  "input_tokens": 1128,
  "output_tokens": 608,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_3

Timestamp: 2026-08-12T09:20:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.

- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to make a decision without any prior information about my co-player.

****Strategic considerations:****

- Sending more creates potential for mutual gain (3x multiplier)
- But I have no information about whether my co-player will reciprocate
- This is a multi-round game, so building some trust could be valuable
- The communication noise adds uncertainty

For Round 1, I'll take a moderately trusting approach. Sending \$3 signals willingness to cooperate while managing risk. If my co-player is reciprocal, we can both benefit (they'll receive \$9, and even a partial return would make this worthwhile). If they don't reciprocate, I've limited my exposure.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 to potentially send
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. There's communication noise - amounts may appear different than what was actually sent/returned

Key considerations:

- This is a multi-round game with 8 agents total
- Pairings are randomized each round
- I'll play both sender and receiver roles across rounds
- Agent names are hidden

Strategic thinking for Round 1:

- Sending \$0 guarantees I keep \$5, but there's no potential for mutual gain
- Sending \$5 shows maximum trust, could lead to good returns if the receiver is cooperative, but risky
- A moderate amount (like \$2-3) balances risk and potential reward
- This is Round 1, so I have no history with any agent yet

The game has a cooperative element - if both players trust each other, total welfare is maximized. The 3x multiplier means:

- If I send \$5, receiver gets \$15. If they return \$7.50, we each end with \$7.50 (better than \$5 each)
- If I send \$3, receiver gets \$9. If they return \$4, I end with \$6 and they end with \$5
- There's also the noise factor to consider

For a first round with no information, I think a moderately trusting approach makes sense.

I'll send \$3 as a signal of willingness to cooperate while not being completely vulnerable. This allows for mutual benefit if the receiver is cooperative, but limits my downside risk.

Actually, let me reconsider. In behavioral economics experiments, sending about half (\$2-3) is common in first rounds. However, given this is a multi-round game and reputation might matter (even though pairings are random, players might develop general strategies), showing some trust could be beneficial.

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 651,
```

```

    "reasoning_tokens": 0
  }
}
Message 4 role=user
content:
  Round 2

  In your most recent previous game round, you were the SENDER against another agent. You sent $3.0, it became $9.0, and you saw them return $4.5 to you. Your visible payoff was $6.5.
  Your total visible earnings across all rounds are $6.5.

  This round, you are the RECEIVER against your current co-player. They sent you $2.9, so you received $8.7. How much do you return? (0-8.7)
  Respond exactly as JSON: {'return': <amount>}
Message 5 role=assistant
content:
  Looking at this situation:

  **Analysis:**
  - My co-player sent $2.9 (about 58% of their endowment), showing considerable trust
  - I received $8.7 after the 3x multiplier
  - In Round 1, when I sent $3, my previous partner returned $4.5 (50% of what they received)
  - Applying a similar reciprocity norm seems fair

  **Decision:**
  I'll return approximately 50% of what I received, which rewards their trust and establishes a cooperative pattern. This leaves us both better off than our starting positions.

  ```json
 {"return": 4.5}
  ```

reasoning:
  Let me analyze this situation:

  **Previous Round (Round 1):**
  - I was SENDER
  - I sent $3.0
  - It became $9.0
  - They returned $4.5 to me
  - My payoff: $5 - $3 + $4.5 = $6.5
  - Their payoff: $9 - $4.5 = $4.5

  That was a pretty fair return - they returned half of what they received, which is a common reciprocity norm.

  **Current Round (Round 2):**
  - I am RECEIVER
  - They sent me $2.9 (close to $3)
  - I received $8.7 ( $2.9 \times 3$ )
  - I need to decide how much to return (0-8.7)

  **Considerations:**
  1. My co-player sent me almost $3 (actually $2.9, but there's communication noise mentioned)
  2. In Round 1, when I sent $3, my partner returned $4.5 (which was 50% of the $9 they received)
  3. If I apply the same 50% rule: I should return about $4.35 (half of $8.7)
  4. My co-player took a risk by sending $2.9, showing trust
  5. Being reciprocal could establish good cooperation
  6. Note: This is a different co-player than Round 1 (pairings are randomized)

  Since this is a new co-player and they showed trust by sending $2.9, I should reciprocate fairly. Returning around 50% seems like a reasonable reciprocity norm that:
  - Rewards their trust
  - Leaves both of us better off than not playing
  - Establishes me as a cooperative player

  My co-player's payoff if I return $4.35:
  - They get: $5 - $2.9 + $4.35 = $6.45

```

- I get: $\$8.7 - \$4.35 = \$4.35$

They sent 58% of their endowment, which is quite generous. I should return slightly more to match their generosity and encourage continued cooperation. Returning \$4.5 mirrors what my previous partner did for me and feels like the right balance....

usage:

```
{
  "input_tokens": 811,
  "output_tokens": 781,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.9, it became \$8.7, and you returned \$4.5. Your visible payoff was \$4.199999999999999.

Your total visible earnings across all rounds are \$10.7.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

****Analysis:****

- My co-player sent \$2.27 (about 45% of their endowment), showing moderate trust
- I received \$6.81 after the multiplier
- In previous rounds, I've been returning approximately 50% of what I receive as a reciprocity norm
- This approach has worked well so far

****Decision:****

I'll maintain consistency and return about 50% of what I received, which rewards their trust appropriately.

```
'''json
{"return": 3.4}
'''
```

Reasoning:

Let me analyze this situation:

****Current situation:****

- I'm the RECEIVER again
- My co-player sent \$2.27 (about 45% of their \$5 endowment)
- I received \$6.81 after the 3x multiplier
- They kept $\$5 - \$2.27 = \$2.73$

****Past behavior:****

- Round 1: I sent \$3, received back \$4.5 (50% return rate on the multiplied amount)
- Round 2: I received \$8.7, returned \$4.5 (about 52% return rate)

****Considerations:****

- My co-player sent a moderate amount (\$2.27), showing some trust but less than the previous player
- Being consistent with my previous behavior would mean returning around 50% of what I received
- $50\% \text{ of } \$6.81 = \3.405

****Payoff implications:****

- If I return \$3.4-3.5:
 - I keep: $\$6.81 - \$3.4 = \$3.41$
 - They get: $\$2.73 \text{ (kept)} + \$3.4 \text{ (returned)} = \6.13
 - This seems fair - we both do better than the \$5 starting point

I should maintain consistency with my reciprocity pattern from Round 2. Returning about 50% seems fair and builds trust for potential future rounds.

Let me return \$3.4 (just under 50%).

Usage: {
"input_tokens": 1105,

```
"output_tokens": 467,
"reasoning_tokens": 0
}
```

GAME - Agent_4 - Sender - dyad_4

Timestamp: 2026-08-12T09:20:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_5"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_4": "investor",
      "Agent_5": "trustee"
    },
    "trustee": "Agent_5"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with risk management.

Since I don't have information about my co-player's behavior yet, I'll send a moderate amount that:

- Demonstrates willingness to cooperate
- Creates opportunity for mutual gain (the tripling effect means both of us can benefit)
- Doesn't expose me to total loss

If I send \$3, my co-player receives \$9 ($3 \times \3). If they return even \$4, we'd both end up better off than our starting positions - I'd have \$6 ($\$5 - \$3 + \4) and they'd have \$5.

This seems like a reasonable opening move to establish cooperative intent.

```
'''json
{"send": 3}
'''
```

reasoning:

This is Round 1 of a trust game where I'm the SENDER. Let me think through the strategic considerations:

1. I have \$5 to decide how to allocate
2. Whatever I send gets tripled when it reaches the receiver
3. The receiver then decides how much to return to me
4. My final payoff = $\$5 - \text{amount_sent} + \text{amount_returned}$

The key strategic considerations:

- If I send nothing (\$0), I keep \$5 and the receiver gets nothing. Safe but no potential for gain.
- If I send everything (\$5), the receiver gets \$15. If they return even \$6, I'd be better off ($\$5 - \$5 + \$6 = \6).
- This is a one-shot game with this particular partner (though I'll play multiple rounds with different partners).
- There's communication noise mentioned, which means the receiver might see a different amount than what I actually send.

Given this is the first round and I don't have information about the other agents' behavior yet:

Strategic approaches:

1. **Trust-building**: Send a moderate-to-high amount to signal cooperation and trust
2. **Conservative**: Send a small amount to minimize risk
3. **Optimal cooperation**: Send \$5 if I expect the receiver to split gains fairly

If the receiver is rational and fair, they might follow a norm like returning half of what they receive, or returning enough to make both players better off than the initial endowment.

For example, if I send \$3:

- I have: $\$5 - \$3 = \$2$ left
- Receiver gets: $\$3 \times 3 = \9
- If receiver returns \$4: I end with $\$2 + \$4 = \$6$, they keep \$5
- Both better off than initial \$5

If I send \$5:

- I have: \$0 left
- Receiver gets: \$15
- If receiver returns \$8: I end with \$8, they keep \$7

- Both significantly better off

The noise factor complicates things-if I send \$5 but it appears as less, the receiver might return proportionally less. Based on experimental economics patterns where senders typically send around 50% and receivers often return amounts that create mutual gains, I'll send \$3. This signals cooperation while maintaining some cushion, and if the receiver follows typical reciprocity norms, we should both end up better off than our starting positions....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 841,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.8 to you. Your visible payoff was \$5.8.

Your total visible earnings across all rounds are \$5.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.56, so you received \$7.68. How much do you return? (0-7.68)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Analysis:****

- My co-player sent \$2.56 (about 51% of their endowment), showing moderate trust
- I received \$7.68 after the 3x multiplication
- They currently have \$2.44 remaining

****Reciprocity consideration:****

- In Round 1, my co-player returned about 42% of what they received
- I want to be fair and encourage continued cooperation
- Returning around 45-50% seems reasonable

If I return \$3.50:

- Sender ends with: $\$2.44 + \$3.50 = \$5.94$ (gain of \$0.94)
- I keep: $\$7.68 - \$3.50 = \$4.18$ (gain of \$4.18)

This rewards their trust while keeping a fair share given the multiplication advantage.

```
'''json
{"return": 3.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was SENDER, sent \$3
- It became \$9 for receiver
- They returned \$3.8 to me
- My payoff: $\$5 - \$3 + \$3.8 = \5.8
- Their payoff: $\$9 - \$3.8 = \$5.2$

So in Round 1, my co-player returned about 42% of what they received ($\$3.8/\$9 \approx 0.42$).

Round 2:

- I am RECEIVER
- Sender sent \$2.56
- I received \$7.68 (2.56×3)
- I need to decide how much to return

The sender invested \$2.56, so they currently have $\$5 - \$2.56 = \$2.44$ remaining.

If I think about reciprocity and building trust:

- The sender showed moderate trust by sending about 51% of their endowment
- If I want to encourage cooperation, I should return enough that the sender comes out ahead

Let me consider different return amounts:

- If I return \$4: Sender gets $\$2.44 + \$4 = \$6.44$ (net gain of \$1.44), I keep \$3.68 (net gain of \$3.68)
- If I return \$3.5: Sender gets $\$2.44 + \$3.5 = \$5.94$ (net gain of \$0.94), I keep \$4.18 (net gain of \$4.18)
- If I return \$3: Sender gets $\$2.44 + \$3 = \$5.44$ (net gain of \$0.44), I keep \$4.68 (net gain of \$4.68)

In my previous round, the receiver returned about 42% of what they received. If I apply a similar principle:

$\$7.68 \times 0.42 \approx \3.23

I'm inclined to return slightly more than that baseline to reward their trust and establish a cooperative pattern. Returning \$3.5 seems like the right balance-it gives them a meaningful profit while still letting me benefit substantially, and it signals I'm reliable without being exploited....

usage:

```
{
  "input_tokens": 804,
  "output_tokens": 923,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.56, it became \$7.68, and you returned \$3.5. Your visible payoff was \$4.18.

Your total visible earnings across all rounds are \$9.98.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

****Analysis:****

I'm now facing a new co-player (Round 3, new pairing). My experience so far:

- Round 1: Sent \$3, received back \$3.8 (42% return rate)
- Round 2: Received \$7.68, returned \$3.5 (46% return rate)

****Strategy:****

Since I'm building a reputation in this multi-agent environment, I want to signal cooperative behavior. I'll send slightly more than last time to:

- Test if higher trust leads to better reciprocity
- Create more value through the 3x multiplier
- Signal that I'm willing to be generous

Sending \$3.5 means my co-player receives \$10.50. If they return around 40%, I'd get back ~\$4.20, giving me a net gain. This balances trust-building with reasonable risk.

```
'''json
{"send": 3.5}
'''
```

Reasoning:

Let me analyze the situation:

****Round 1:**** I was SENDER

- Sent: \$3
- They received: \$9
- They returned: \$3.8
- My payoff: \$5.8

****Round 2:**** I was RECEIVER

- They sent: \$2.56
- I received: \$7.68
- I returned: \$3.5

- My payoff: \$4.18

****Round 3:**** I am SENDER again

Now I need to think about this new co-player in Round 3. Note that each round I'm paired with potentially different agents from the pool of 8 agents.

In Round 1, my co-player returned \$3.8 out of \$9 (42.2%).

In Round 2, I returned \$3.5 out of \$7.68 (45.6%).

For Round 3, I'm facing a NEW co-player (since pairings are randomized each round). I don't have specific history with this particular agent.

Given:

1. I've seen one example of a co-player returning about 42%
2. I returned about 46% myself, trying to be fair
3. The game has communication noise, so amounts may differ slightly

Strategy for Round 3:

- I should continue with a cooperative but cautious approach
- Sending around \$3 worked reasonably in Round 1
- I might try a slightly different amount to gather more information

Let me send \$3 again as it seems to be a reasonable middle ground that:

- Shows trust
- Creates significant value through multiplication (becomes \$9)
- Leaves me with \$2 as a safety net
- If they return around 40-45%, I'd get back around \$3.6-\$4, giving me a decent payoff

Actually, I could also consider sending a bit more to test if higher trust leads to better returns. Let me send \$3.5 this time.

...

```
Usage: {  
  "input_tokens": 1137,  
  "output_tokens": 766,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Receiver - dyad_4

Timestamp: 2026-08-12T09:20:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_5"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_4": "investor",  
      "Agent_5": "trustee"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.54, that is 50.8% of their total endowment. You received \$7.62. How much do you return to the sender? (0-7.62)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.54 (about half their endowment)
- I received \$7.62 after the 3x multiplier
- I want to reciprocate fairly while both of us benefit

The sender took a risk to create mutual gains. A fair return should:

1. Reward their trust
2. Leave us both better off than if nothing was sent
3. Show I'm a cooperative partner for future rounds

If I return \$3.50:

- Sender ends with: $\$5.00 - \$2.54 + \$3.50 = \5.96 (net gain: \$0.96)
- I end with: $\$7.62 - \$3.50 = \$4.12$ (net gain: \$4.12)

This seems like a fair reciprocation that rewards the sender's trust while acknowledging the receiver advantage in this game.

```
'''json
{"return": 3.50}
'''
```

reasoning:

Let me analyze this situation carefully:

- I am the RECEIVER in Round 1
- The sender sent \$2.54 (which is 50.8% of their \$5 endowment)
- After the 3x multiplier, I received \$7.62 (which is indeed 2.54×3)
- I need to decide how much to return (between \$0 and \$7.62)

Key considerations:

1. ****The sender showed trust****: They sent about half their endowment (\$2.54 out of \$5), showing significant trust in me.
2. ****Fair reciprocation****: A common approach in trust games is to return an amount that:
 - Rewards the sender for their trust
 - Results in both players being better off than if nothing was sent
 - Maintains equity or fairness
3. ****What would be a fair return?****
 - If I return \$0: I keep \$7.62, sender keeps \$2.46
 - If I return half (\$3.81): I keep \$3.81, sender keeps \$6.27
 - If I return \$2.54 (what they sent): I keep \$5.08, sender keeps \$5.00
 - If I return \$3.81: I keep \$3.81, sender keeps \$6.27
4. ****Strategic considerations****:
 - This is a multi-round game with 8 agents
 - Building a reputation for reciprocity might encourage future cooperation
 - The note mentions communication noise, so amounts might appear different to the sender
5. ****Fairness analysis****:
 - Initial positions: Sender had \$5, I had \$0
 - After transaction: Sender has \$2.46 ($\$5 - \2.54), I have \$7.62
 - If I return \$3.81: Sender ends with \$6.27, I end with \$3.81
 - If I return \$2.54: Sender ends with \$5.00, I end with \$5.08
 - The sender created \$5.08 in additional value through their trust, so splitting that gain equally would mean returning about \$2.58, which gets us both to roughly \$5.04 each.

I'm considering what feels genuinely reciprocal here. Returning somewhere between \$3.00 and \$3.50 seems right—it rewards their willingness to trust me while still leaving me better off than I started. If I return \$3.00, they'd end with \$5.46 and I'd have \$4.62; at \$3.50, they'd get \$5.96 and I'd keep \$4.12. Either way, we both benefit from the exchange.

I'll go with \$3.50. It signals that I value their trust and want to be someone they'd cooperate with again, while still giving me a meaningful gain from the interaction...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1373,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.54, it became \$7.62, and you returned \$3.5. Your visible payoff was \$4.12.

Your total visible earnings across all rounds are \$4.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, my co-player sent me \$2.54 (about 50% of their endowment), showing good trust. I reciprocated with a fair return.

Now I'm the sender with a new co-player. I should:

1. Signal cooperation to encourage reciprocity
2. Send a reasonable amount that allows mutual benefit
3. Not overcommit without knowing this co-player's behavior

Following the cooperative pattern from Round 1, I'll send a similar proportion - about 50% of my endowment.

```
'''json
{"send": 2.50}
'''
```

reasoning:

Let me think about this situation:

Round 1: I was the receiver

- Sender sent \$2.54 (50.8% of their endowment)
- I received \$7.62
- I returned \$3.50
- My payoff was \$4.12
- Sender's payoff would have been $\$5 - \$2.54 + \$3.50 = \5.96

Now in Round 2, I'm the sender. I need to decide how much to send.

Key considerations:

1. The previous sender showed trust by sending about 50% (\$2.54 out of \$5)
2. I reciprocated reasonably well by returning \$3.50, giving them a positive return
3. This is a multi-round game with 8 agents, so building a reputation for cooperation could be beneficial
4. There's communication noise, so the exact amounts might differ
5. I'm paired with a different co-player this round (pairings are randomized)

Since this is a new co-player, they won't have direct experience of my behavior. However, in a repeated multi-agent setting, establishing myself as cooperative could be beneficial.

The previous sender sent about 50% and I reciprocated well. I should consider:

- Sending a similar moderate-to-high amount to signal cooperation
- Not sending too much in case this receiver is less cooperative than I was
- Not sending too little, as that could signal I'm uncooperative

I think sending around \$2.50-\$3.00 (50-60%) would be reasonable. This:

- Shows trust and willingness to cooperate
- Mirrors what the previous sender did to me
- Leaves room for mutual benefit if the receiver is cooperative
- Doesn't risk my entire endowment

I'll send \$2.50, which is exactly 50% of my endowment, similar to what the previous sender did.

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 581,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$4.13 to you. Your visible payoff was \$6.63.

Your total visible earnings across all rounds are \$10.75.

This round, you are the RECEIVER against your current co-player. They sent you \$3.99, so you received \$11.97. How much do you return? (0-11.97)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player sent \$3.99, which is almost 80% of their endowment - showing exceptional trust

- I received \$11.97 after the 3x multiplier
- This is much higher trust than the ~50% I experienced/sent in previous rounds

I should reciprocate proportionally to their high level of trust:

If I return \$6.50:

- Sender ends with: $\$5.00 - \$3.99 + \$6.50 = \7.51 (net gain: \$2.51)
- I end with: $\$11.97 - \$6.50 = \$5.47$ (net gain: \$5.47)

This rewards their exceptional trust while both of us benefit significantly.

```
'''json
{"return": 6.50}
'''
```

Reasoning:

Let me analyze this situation:

Round 3: I'm the RECEIVER

- My co-player sent \$3.99 (about 80% of their \$5 endowment)
- I received \$11.97 (after 3x multiplier)
- I need to decide how much to return

The sender showed very high trust by sending almost 80% of their endowment. This is significantly more than the ~50% I saw/sent in previous rounds.

Let me think about fair reciprocation:

If the sender sent \$3.99, they kept \$1.01.

If I return nothing, they end with \$1.01 (bad outcome for them, they lose \$3.98)

If I return everything (\$11.97), they end with $\$1.01 + \$11.97 = \$12.98$, I get \$0

I want to:

1. Reward their high trust
2. Make sure they profit from their decision
3. Keep a reasonable amount for myself
4. Signal I'm cooperative for future rounds

In Round 2, when I sent \$2.50 and received \$4.13 back, that was a good return (I sent 50%, got back enough to profit).

The sender here sent 80% - much higher trust. I should return proportionally more.

Let me calculate some options:

Option 1: Return \$6.00

- Sender ends with: $\$1.01 + \$6.00 = \$7.01$ (net gain: \$2.01)
- I end with: $\$11.97 - \$6.00 = \$5.97$ (net gain: \$5.97)
- This gives them a decent profit
- Returning \$6.50 or \$7.00 would be more generous given their 80% send, letting them gain \$2.51-\$3.01 while I still keep \$4.97-\$5.47. Since they demonstrated exceptional trust, I should lean toward the higher end to reinforce cooperation.

Looking at the pattern: they sent roughly 80% of their endowment, which is significantly more than the 50-51% in earlier rounds. My...

```
Usage: {
  "input_tokens": 1150,
  "output_tokens": 981,
  "reasoning_tokens": 0
}
```

ROUND 4

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

- dyad_1: Agent_3=Sender, Agent_6=Receiver
- dyad_2: Agent_5=Sender, Agent_8=Receiver
- dyad_3: Agent_7=Sender, Agent_2=Receiver
- dyad_4: Agent_1=Sender, Agent_4=Receiver

Parsed game decisions:

```
dyad_1: Agent_3 sent $3; received $9; Agent_6 returned $4.8; payoffs Agent_3=$6.8, Agent_6=$4.2;
noise/visible: sent visible=$3.24, received visible=$9.72, returned visible=$5.73
  Action validation: {"returned": {"amount": 4.8, "clamped": false, "max": 9.0, "min": 0, "raw":
    4.8}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_5 sent $3; received $9; Agent_8 returned $5.5; payoffs Agent_5=$7.5, Agent_8=$3.5;
noise/visible: sent visible=$3.69, received visible=$11.07, returned visible=$4.71
  Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 9.0, "min": 0, "raw":
    5.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_7 sent $3; received $9; Agent_2 returned $5.1; payoffs Agent_7=$7.1, Agent_2=$3.9;
noise/visible: sent visible=$3.79, received visible=$11.37, returned visible=$4.36
  Action validation: {"returned": {"amount": 5.1, "clamped": false, "max": 9.0, "min": 0, "raw":
    5.1}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_1 sent $3; received $9; Agent_4 returned $4.35; payoffs Agent_1=$6.35,
Agent_4=$4.65; noise/visible: sent visible=$2.9, received visible=$8.7, returned visible=$4.48
  Action validation: {"returned": {"amount": 4.35, "clamped": false, "max": 9.0, "min": 0, "raw":
    4.35}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
```

```
Balances after round: {
  "Agent_1": 22.450000000000003,
  "Agent_2": 19.9,
  "Agent_3": 22.55,
  "Agent_4": 23.15,
  "Agent_5": 23.5,
  "Agent_6": 20.95,
  "Agent_7": 19.6,
  "Agent_8": 21.9
}
```

```
Visible balances after round: {
  "Agent_1": 20.59,
  "Agent_2": 21.18,
  "Agent_3": 23.73,
  "Agent_4": 21.9,
  "Agent_5": 22.93,
  "Agent_6": 21.630000000000003,
  "Agent_7": 20.66,
  "Agent_8": 24.05
}
```

LLM CALLS

GAME - Agent_3 - Sender - dyad_1

Timestamp: 2026-08-12T09:20:53
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_6"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system
content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.12, that is 62.4% of their total endowment. You received \$9.36. How much do you return to the sender? (0-9.36)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.12 (62.4% of their endowment)
- This created \$9.36 for me after the 3x multiplier
- The sender currently has only \$1.88 remaining

I want to reward their trust and encourage future cooperation. The sender took a risk, and I should reciprocate appropriately.

I'll return \$4.00, which:

- Gives the sender \$5.88 total (better than their initial \$5)
- Leaves me with \$5.36 (also benefiting from the cooperation)
- Demonstrates that trusting me pays off
- Should encourage continued cooperation in future rounds

{'return': 4.00}

reasoning:

Let me analyze this situation carefully:

1. The sender sent \$3.12 out of their \$5 endowment (62.4%)

2. This was multiplied by 3x, so I received \$9.36
3. I need to decide how much to return (between \$0 and \$9.36)

Key considerations:

- The sender showed significant trust by sending 62.4% of their endowment
- There's communication noise in the game, so actual amounts may differ slightly
- This is a repeated game with multiple rounds
- I want to encourage cooperation and build trust

Economic analysis:

- Sender currently has: $\$5 - \$3.12 = \$1.88$
- I currently have: \$9.36
- If I return nothing, sender gets \$1.88 total, I get \$9.36
- A fair split might consider that we each started with potential of \$5

Common strategies in trust games:

1. Return nothing (purely selfish)
2. Return the original amount sent (~\$3.12) - gives back what was sent
3. Return enough so we both profit equally from the 3x multiplier
4. Return more to encourage future cooperation

Let me calculate option 3 (equal profit):

- The 3x multiplier created \$6.24 in extra value ($\$9.36 - \3.12)
- Split equally: each gets \$3.12 extra
- Sender should end with: $\$1.88 + \$3.12 + \$3.12 = \8.12
- So I should return: $\$9.36 - (\$5 + \$3.12) = \$9.36 - \$8.12 = \1.24

Wait, let me recalculate more carefully:

- Sender starts with \$5, sends \$3.12, has \$1.88 left
- I receive \$9.36
- Total pool is now \$11.24
- If we each started with \$5 potential, that's \$10 baseline
- The extra \$1.24 created should split evenly at \$0.62 each
- So sender should end with \$5.62 and I should end with \$5.62
- That means I return \$3.74

I'm reconsidering the fairest approach here-whether to split the gains equally or use a different metric.

The sender invested 62.4% of their initial amount, so returning a proportional share of what I received makes sense. That's about \$5.84. Alternatively, since they trusted me with \$3.12 extra, returning \$3.00-\$3.50 would roughly restore their position while letting me benefit from the cooperation. For Round 1, I want to signal good faith and encourage continued trust, so returning 40-50% of the \$9.36 I received-somewhere between \$3.74 and \$4.68-feels like the right balance....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1077,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.12, it became \$9.36, and you returned \$4.0. Your visible payoff was \$5.359999999999999.

Your total visible earnings across all rounds are \$5.359999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Last round, my co-player trusted me with 62.4% of their endowment
- I reciprocated fairly, and we both benefited
- Now I'm with a new co-player as the sender

I'll mirror the cooperative behavior I experienced. Sending a substantial amount can create a mutual benefit through the 3x multiplier if my co-player reciprocates.

I'll send \$3.00 (60% of my endowment), similar to the trust shown to me in the previous round.

```
{'send': 3.00}
```

reasoning:

Let me think about this situation:

1. In the previous round, I was the receiver
 - The sender sent me \$3.12 (62.4% of their endowment)
 - I received \$9.36
 - I returned \$4.00
 - My payoff was \$5.36
2. Now I'm the sender with a new co-player
 - I have \$5 to potentially send
 - Whatever I send gets tripled
 - The receiver will decide how much to return

Strategic considerations:

- In the previous round, the sender showed trust by sending 62.4% of their endowment
- I reciprocated fairly, returning \$4.00 out of \$9.36 (about 43%)
- This gave the sender a total of \$5.88 (they sent \$3.12, kept \$1.88, got back \$4.00)

For this round:

- I'm paired with a different agent (randomized pairings)
- I don't know their history or behavior
- However, the game benefits from cooperation
- If I send a reasonable amount and they reciprocate, we both benefit

Let me consider different strategies:

- Sending nothing: I keep \$5, they get nothing, no mutual benefit
- Sending a small amount (e.g., \$1-2): Low risk, but lower potential gain
- Sending a moderate amount (e.g., \$3): Shows some trust, reasonable risk
- Sending a large amount (e.g., \$4-5): High trust, high risk, high potential reward

Given that:

1. The previous sender trusted me with 62.4% and it worked out
2. I want to establish a cooperative pattern
3. There's communication noise mentioned, so exact amounts may vary
4. I'm in a multi-round game where reputation (though anonymous) might matter

I'll send \$3.00, which is 60% of my endowment. This:

...

usage:

```
{
  "input_tokens": 824,
  "output_tokens": 680,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.48 to you. Your visible payoff was \$5.48.

Your total visible earnings across all rounds are \$10.84.

This round, you are the RECEIVER against your current co-player. They sent you \$2.97, so you received \$8.91. How much do you return? (0-8.91)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

****Current round:****

- Sender showed good trust by sending \$2.97 (59.4% of their endowment)
- This created \$8.91 for me
- Sender currently has \$2.03

****Previous round context:****

- I sent \$3.00 (60% - very similar trust level)
- My co-player returned \$3.48, which was fair

I should reciprocate the trust shown to me. The sender sent almost 60% of their endowment, similar to what I did last round.

I'll return \$3.75, which:

- Gives the sender \$5.78 total (a gain of \$0.78)
- Leaves me with \$5.16 (also a gain)
- Rewards their trust appropriately

{'return': 3.75}

reasoning:

Let me analyze this situation:

****Last round (Round 2):****

- I was the SENDER
- I sent \$3.00 (60% of my endowment)
- It became \$9.00 for the receiver
- They returned \$3.48
- My payoff: $\$5 - \$3 + \$3.48 = \5.48
- Their payoff: $\$9 - \$3.48 = \$5.52$

The receiver gave me back slightly more than I sent, which was cooperative and fair.

****This round (Round 3):****

- I'm the RECEIVER
- Sender sent \$2.97 (59.4% of their endowment - very similar to what I sent last round!)
- I received \$8.91
- Sender currently has: $\$5 - \$2.97 = \$2.03$

The sender showed good trust by sending ~60% of their endowment. This is very similar to the 60% I sent in the previous round.

I should reciprocate fairly. Let me think about what would be a fair return:

- If I return \$3.50, sender gets $\$2.03 + \$3.50 = \$5.53$, and I keep $\$8.91 - \$3.50 = \$5.41$
- If I return \$4.00, sender gets $\$2.03 + \$4.00 = \$6.03$, and I keep $\$8.91 - \$4.00 = \$4.91$
- If I return \$3.75, sender gets $\$2.03 + \$3.75 = \$5.78$, and I keep $\$8.91 - \$3.75 = \$5.16$

The sender trusted me with \$2.97, similar to how I trusted my previous co-player with \$3.00, and they returned \$3.48.

...

usage:

```
{
  "input_tokens": 1089,
  "output_tokens": 735,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.97, it became \$8.91, and you returned \$3.75. Your visible payoff was \$5.16.

Your total visible earnings across all rounds are \$16.0.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

****Last round with THIS SAME co-player:****

- They sent me \$2.97 (59.4% of their endowment) - showing good trust
- I returned \$3.75 - rewarding their trust fairly
- We both gained from cooperation

****Now it's my turn to send:****

Since we just had a successful cooperative round together, I should maintain this pattern. They trusted me, I reciprocated, and now I should show similar trust.

I'll send \$3.00 (60% of my endowment), matching the level of trust they showed me.

```
{'send': 3.00}
```

Reasoning:

Let me analyze this situation:

****Last round with THIS SAME co-player:****

- They sent me \$2.97 (59.4% of their endowment)
- I received \$8.91
- I returned \$3.75
- They got \$5.78 total (gain of \$0.78)
- I got \$5.16 (gain of \$0.16)

****Now I'm the sender with THE SAME co-player:****

- They showed good trust last round (sent ~60%)
- I reciprocated fairly
- We both gained from cooperation

Since we're playing with the same person and we had a successful cooperative round, I should continue this pattern. They trusted me with ~60% of their endowment, and I rewarded that trust. Now I should show similar trust.

I'll send around \$3.00 again, maintaining the cooperative pattern.

```
Usage: {
  "input_tokens": 1406,
  "output_tokens": 354,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_1

Timestamp: 2026-08-12T09:21:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_6"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.97, that is 59.4% of their total endowment. You received \$8.91. How much do you return to the sender? (0-8.91)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.97 (about 60% of their endowment)
- I received \$8.91 after the 3x multiplier
- The sender currently has only \$2.03 remaining (\$5 - \$2.97)

The sender took a substantial risk and demonstrated trust. To reciprocate fairly and encourage future cooperation, I should return a reasonable amount that:

1. Rewards their trust
2. Leaves both of us better off than the initial endowment
3. Establishes a cooperative relationship for future rounds

Returning approximately half of what I received seems fair - this would give the sender a good return on their trust while still leaving me with a decent gain.

I'll return **\$4.50**:

- Sender ends with: \$2.03 + \$4.50 = \$6.53 (gained \$1.53)
- I keep: \$8.91 - \$4.50 = \$4.41 (gained \$4.41)

Both of us benefit, and this reciprocates their trust.

{'return': 4.50}

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$2.97 (59.4% of their \$5 endowment)
- I received \$8.91 (which is 2.97×3)
- I need to decide how much to return (between \$0 and \$8.91)

Let me think about what's fair and strategic:

1. The sender showed significant trust by sending almost 60% of their endowment
2. If I return nothing, I keep \$8.91 and the sender gets only \$2.03 ($5 - 2.97$)
3. If I return about half (\$4.46), I keep \$4.45 and the sender gets \$6.49 ($2.03 + 4.46$)
4. If I return \$5.94, I keep \$2.97 (equal to what they sent) and the sender gets \$8.97 total

The sender took a risk and showed trust. A common approach in trust games is reciprocity. The sender sent about 60% of their endowment, which is generous.

Let me consider what might establish good cooperation for future rounds:

- Returning about the same amount they sent (\$2.97) would mean they gain \$0 net
- Returning more than they sent but less than half of what I received seems reasonable
- Returning around 1/3 to 1/2 of what I received is often considered fair

Let me calculate some options:

- Return \$3: Sender gets \$5.03 total (gained \$0.03), I keep \$5.91
- Return \$4: Sender gets \$6.03 total (gained \$1.03), I keep \$4.91
- Return \$4.50: Sender gets \$6.53 total (gained \$1.53), I keep \$4.41
- Return \$5: Sender gets \$7.03 total (gained \$2.03), I keep \$3.91
- Return \$4.50 seems like the right balance-it reciprocates their trust by returning roughly half of what I received,...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 898,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.97, it became \$8.91, and you returned \$4.5. Your visible payoff was \$4.41.

Your total visible earnings across all rounds are \$4.41.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my previous experience:

- As a receiver, a sender trusted me with about 60% of their endowment
- I reciprocated fairly, and we both benefited
- This established that cooperation can work well in this game

Now as the sender with a new co-player, I want to:

1. Demonstrate trust to encourage reciprocation
2. Follow a similar cooperative strategy that worked before
3. Give my co-player the opportunity to earn and share

I'll send **\$3.00** (60% of my endowment), mirroring the trust shown to me in the previous round.

```
{'send': 3.00}
```

reasoning:

Let me think about this situation:

In Round 1:

- I was the receiver
- The sender sent me \$2.97 (59.4% of their endowment)
- I received \$8.91

- I returned \$4.50
- I kept \$4.41
- The sender got back \$4.50, so they ended with $\$2.03 + \$4.50 = \$6.53$

Now in Round 2:

- I'm the sender
- I have \$5 to potentially send
- I'm playing against a different co-player (the problem states pairings are randomized each round)

Key considerations:

1. The previous sender showed trust by sending 59.4% of their endowment
2. I reciprocated fairly by returning about 50% of what I received
3. This worked out well - both players benefited
4. Now I'm in the sender role with a new partner

Since my previous experience showed that sending a substantial amount can lead to good returns, and I want to establish trust with this new co-player, I should send a meaningful amount.

The previous sender sent about 60% (\$2.97 out of \$5). This seemed like a good trust-building amount. I could:

- Send a similar proportion to continue the cooperative pattern
- Send slightly more to show strong trust
- Send slightly less to be more cautious with a new partner

I think sending around \$3 (60%) makes sense - it mirrors what worked well in the previous round and shows trust to establish a good relationship with this new co-player.

Let me send \$3.00.

usage:

```
{
  "input_tokens": 908,
  "output_tokens": 516,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.32 to you. Your visible payoff was \$7.32.

Your total visible earnings across all rounds are \$11.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my recent experience:

Round 2 results:

- I sent \$3.00 (60% of endowment)
- Receiver returned \$5.32
- I ended with \$7.32 - excellent outcome!

The cooperative strategy is working very well. When I demonstrate trust, it's being reciprocated generously.

Now facing a new co-player, I'll continue this successful approach. Sending around 60% has proven effective in building mutual benefit.

```
{'send': 3.00}
```

reasoning:

Let me analyze the situation:

Round 2 (just completed):

- I was the SENDER
- I sent \$3.00 (60% of endowment)

- It became \$9.00 for the receiver
- They returned \$5.32 to me
- My payoff: $\$5 - \$3 + \$5.32 = \7.32
- They kept: $\$9.00 - \$5.32 = \$3.68$

This was a very generous return! They returned more than half of what they received, and I ended up with more than my initial endowment even after sending a substantial amount.

My experience so far:

- Round 1 (RECEIVER): Received \$8.91, returned \$4.50, kept \$4.41
- Round 2 (SENDER): Sent \$3.00, received back \$5.32, ended with \$7.32

Total visible earnings: \$11.73

The cooperation strategy is working very well! When I sent 60% in Round 2, I got an excellent return.

Now in Round 3, I'm a SENDER again with a new co-player.

Given:

1. My cooperative strategy has been working well
2. Both times cooperation has led to mutual benefit
3. I want to continue encouraging trust and reciprocation
4. The game benefits from establishing cooperative norms

I should continue with a cooperative approach. Sending around 60% worked well last time. I could:

- Send \$3.00 again (same as last time, 60%)
- Send a bit more to show strong trust
- Send a bit less to be slightly more cautious with a new partner

Since cooperation has been consistently rewarded, I'll stick with the successful strategy of sending around 60%.

...

usage:

```
{
  "input_tokens": 1176,
  "output_tokens": 538,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$2.98 to you. Your visible payoff was \$4.98.

Your total visible earnings across all rounds are \$16.71.

This round, you are the RECEIVER against your current co-player. They sent you \$3.24, so you received \$9.72. How much do you return? (0-9.72)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation with my current co-player:

****Last round (Round 3):****

- I sent them \$3.00
- They returned \$2.98 (~33% of what they received)
- This was a modest but acceptable return

****This round (Round 4):****

- They sent me \$3.24 (slightly MORE than I sent them)
- I received \$9.72
- They currently have only \$1.76

The fact that they sent me MORE than I sent them suggests they're trying to build cooperation.

I should reciprocate generously to:

1. Reward their increased trust
2. Establish a positive cooperative dynamic
3. Ensure we both benefit from ongoing interaction

I'll return **\$4.80**:
- They end with: $\$1.76 + \$4.80 = \$6.56$ (gain \$1.56)
- I keep: $\$9.72 - \$4.80 = \$4.92$ (gain \$4.92)

This is more generous than they were to me, encouraging further cooperation.

{'return': 4.80}

Reasoning:

Let me analyze this situation:

Round 3 (previous round):

- I was the SENDER
- I sent \$3.00 to this same co-player
- They returned \$2.98
- My payoff was \$4.98 (kept \$2 + returned \$2.98)

Round 4 (current round):

- I am the RECEIVER
- Same co-player (they are now the sender)
- They sent me \$3.24
- I received \$9.72

So this is interesting - we're playing repeated games with the same partner. In round 3, I sent \$3.00 and they returned \$2.98 (about 33% of what they received). Now they've sent me \$3.24 (notably a bit more than I sent them).

Let me think about the fairness here:

- When I sent \$3.00, they kept $\$9.00 - \$2.98 = \$6.02$
- I kept $\$2.00 + \$2.98 = \$4.98$
- So they gained \$1.02, I lost \$0.02

Now they sent \$3.24:

- They kept $\$5 - \$3.24 = \$1.76$
- I received \$9.72

If I want to reciprocate similarly to how they treated me, I might return around 33% of what I received, which would be about \$3.21.

But actually, they sent me slightly MORE than I sent them (\$3.24 vs \$3.00), which could be seen as trying to establish cooperation or make up for the relatively low return last round.

Let me think about different return amounts:

Option 1: Return \$3.00 (about 31%)

- They end with: $\$1.76 + \$3.00 = \$4.76$ (lost \$0.24)
- I keep: $\$9.72 - \$3.00 = \$6.72$ (gain \$6.72)
- This feels too stingy given their slightly larger initial send.

Option 2: Return \$4.00 (about 41%)

- They end with: $\$1.76 + \$4.00 = \$5.76$ (gain \$0.76)
- I keep: $\$9.72 - \$4.00 = \$5.72$ (gain \$5.72)
- This creates mutual benefit and shows reciprocal trust.

Option 3: Return \$5.00 (about 51%)

- They end with: $\$1.76 + \$5.00 = \$6.76$ (gain \$1.76)
- I keep: $\$9.72 - \$5.00 = \$4.72$...

Usage: {

"input_tokens": 1431,
"output_tokens": 1067,
"reasoning_tokens": 0

}

GAME - Agent_5 - Sender - dyad_2

Timestamp: 2026-08-12T09:21:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_2",

```

"move_index": 2,
"opponent_id": "Agent_8",
"pairing": {
  "agents": [
    "Agent_5",
    "Agent_8"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.54, that is 50.8% of their total endowment

nt. You received \$7.62. How much do you return to the sender? (0-7.62)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.54 (about half their endowment)
- I received \$7.62 after the 3x multiplier
- I want to reciprocate fairly while both of us benefit

The sender took a risk to create mutual gains. A fair return should:

1. Reward their trust
2. Leave us both better off than if nothing was sent
3. Show I'm a cooperative partner for future rounds

If I return \$3.50:

- Sender ends with: $\$5.00 - \$2.54 + \$3.50 = \5.96 (net gain: \$0.96)
- I end with: $\$7.62 - \$3.50 = \$4.12$ (net gain: \$4.12)

This seems like a fair reciprocation that rewards the sender's trust while acknowledging the receiver's advantage in this game.

```
'''json
{"return": 3.50}
'''
```

reasoning:

Let me analyze this situation carefully:

- I am the RECEIVER in Round 1
- The sender sent \$2.54 (which is 50.8% of their \$5 endowment)
- After the 3x multiplier, I received \$7.62 (which is indeed $\$2.54 \times 3$)
- I need to decide how much to return (between \$0 and \$7.62)

Key considerations:

1. ****The sender showed trust****: They sent about half their endowment (\$2.54 out of \$5), showing significant trust in me.
2. ****Fair reciprocation****: A common approach in trust games is to return an amount that:
 - Rewards the sender for their trust
 - Results in both players being better off than if nothing was sent
 - Maintains equity or fairness
3. ****What would be a fair return?****
 - If I return \$0: I keep \$7.62, sender keeps \$2.46
 - If I return half (\$3.81): I keep \$3.81, sender keeps \$6.27
 - If I return \$2.54 (what they sent): I keep \$5.08, sender keeps \$5.00
 - If I return \$3.81: I keep \$3.81, sender keeps \$6.27
4. ****Strategic considerations****:
 - This is a multi-round game with 8 agents
 - Building a reputation for reciprocity might encourage future cooperation
 - The note mentions communication noise, so amounts might appear different to the sender
5. ****Fairness analysis****:
 - Initial positions: Sender had \$5, I had \$0
 - After transaction: Sender has \$2.46 ($\$5 - \2.54), I have \$7.62
 - If I return \$3.81: Sender ends with \$6.27, I end with \$3.81
 - If I return \$2.54: Sender ends with \$5.00, I end with \$5.08
 - The sender created \$5.08 in additional value through their trust, so splitting that gain equally would mean returning about \$2.58, which gets us both to roughly \$5.04 each.

I'm considering what feels genuinely reciprocal here. Returning somewhere between \$3.00 and \$3.50 seems right—it rewards their willingness to trust me while still leaving me better off than I started. If I return \$3.00, they'd end with \$5.46 and I'd have \$4.62; at \$3.50, they'd get \$5.96 and I'd keep \$4.12. Either way, we both benefit from the exchange.

I'll go with \$3.50. It signals that I value their trust and want to be someone they'd cooperate with again, while still giving me a meaningful gain from the interaction....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1373,
  "reasoning_tokens": 0
}
```

Message 4 role=user**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.54, it became \$7.62, and you returned \$3.5. Your visible payoff was \$4.12.

Your total visible earnings across all rounds are \$4.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant**content:**

Looking at this situation:

In Round 1, my co-player sent me \$2.54 (about 50% of their endowment), showing good trust. I reciprocated with a fair return.

Now I'm the sender with a new co-player. I should:

1. Signal cooperation to encourage reciprocity
2. Send a reasonable amount that allows mutual benefit
3. Not overcommit without knowing this co-player's behavior

Following the cooperative pattern from Round 1, I'll send a similar proportion - about 50% of my endowment.

```
'''json
{"send": 2.50}
'''
```

reasoning:

Let me think about this situation:

Round 1: I was the receiver

- Sender sent \$2.54 (50.8% of their endowment)
- I received \$7.62
- I returned \$3.50
- My payoff was \$4.12
- Sender's payoff would have been $\$5 - \$2.54 + \$3.50 = \5.96

Now in Round 2, I'm the sender. I need to decide how much to send.

Key considerations:

1. The previous sender showed trust by sending about 50% (\$2.54 out of \$5)
2. I reciprocated reasonably well by returning \$3.50, giving them a positive return
3. This is a multi-round game with 8 agents, so building a reputation for cooperation could be beneficial
4. There's communication noise, so the exact amounts might differ
5. I'm paired with a different co-player this round (pairings are randomized)

Since this is a new co-player, they won't have direct experience of my behavior. However, in a repeated multi-agent setting, establishing myself as cooperative could be beneficial.

The previous sender sent about 50% and I reciprocated well. I should consider:

- Sending a similar moderate-to-high amount to signal cooperation
- Not sending too much in case this receiver is less cooperative than I was
- Not sending too little, as that could signal I'm uncooperative

I think sending around \$2.50-\$3.00 (50-60%) would be reasonable. This:

- Shows trust and willingness to cooperate
- Mirrors what the previous sender did to me
- Leaves room for mutual benefit if the receiver is cooperative
- Doesn't risk my entire endowment

I'll send \$2.50, which is exactly 50% of my endowment, similar to what the previous sender did.

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 581,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$4.13 to you. Your visible payoff was \$6.63.

Your total visible earnings across all rounds are \$10.75.

This round, you are the RECEIVER against your current co-player. They sent you \$3.99, so you received \$11.97. How much do you return? (0-11.97)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.99, which is almost 80% of their endowment - showing exceptional trust
- I received \$11.97 after the 3x multiplier
- This is much higher trust than the ~50% I experienced/sent in previous rounds

I should reciprocate proportionally to their high level of trust:

If I return \$6.50:

- Sender ends with: \$5.00 - \$3.99 + \$6.50 = \$7.51 (net gain: \$2.51)
- I end with: \$11.97 - \$6.50 = \$5.47 (net gain: \$5.47)

This rewards their exceptional trust while both of us benefit significantly.

```
'''json
{"return": 6.50}
'''
```

reasoning:

Let me analyze this situation:

Round 3: I'm the RECEIVER

- My co-player sent \$3.99 (about 80% of their \$5 endowment)
- I received \$11.97 (after 3x multiplier)
- I need to decide how much to return

The sender showed very high trust by sending almost 80% of their endowment. This is significantly more than the ~50% I saw/sent in previous rounds.

Let me think about fair reciprocation:

If the sender sent \$3.99, they kept \$1.01.

If I return nothing, they end with \$1.01 (bad outcome for them, they lose \$3.98)

If I return everything (\$11.97), they end with \$1.01 + \$11.97 = \$12.98, I get \$0

I want to:

1. Reward their high trust
2. Make sure they profit from their decision
3. Keep a reasonable amount for myself
4. Signal I'm cooperative for future rounds

In Round 2, when I sent \$2.50 and received \$4.13 back, that was a good return (I sent 50%, got back enough to profit).

The sender here sent 80% - much higher trust. I should return proportionally more.

Let me calculate some options:

Option 1: Return \$6.00

- Sender ends with: $\$1.01 + \$6.00 = \$7.01$ (net gain: \$2.01)
- I end with: $\$11.97 - \$6.00 = \$5.97$ (net gain: \$5.97)
- This gives them a decent profit
- Returning \$6.50 or \$7.00 would be more generous given their 80% send, letting them gain \$2.51-\$3.01 while I still keep \$4.97-\$5.47. Since they demonstrated exceptional trust, I should lean toward the higher end to reinforce cooperation.

Looking at the pattern: they sent roughly 80% of their endowment, which is significantly more than the 50-51% in earlier rounds. My...

usage:

```
{
  "input_tokens": 1150,
  "output_tokens": 981,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.99, it became \$11.97, and you returned \$6.5. Your visible payoff was \$5.4700000000000001.

Your total visible earnings across all rounds are \$16.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience so far:

- Round 2: I sent \$2.50 and received \$4.13 back - good reciprocity
- Round 3: My co-player sent \$3.99 (high trust) and I reciprocated fairly
- The game has shown cooperative players who reward trust

Since cooperation has been working well, I'll increase my trust level from the 50% I sent in Round 2. Sending around 60% balances building cooperation while managing risk with a new co-player.

```
'''json
{"send": 3.00}
'''
```

Reasoning:

Let me review my history:

- Round 1 (RECEIVER): Saw \$2.54 sent, received \$7.62, returned \$3.50, earned \$4.12
- Round 2 (SENDER): Sent \$2.50, received \$4.13 back, earned \$6.63
- Round 3 (RECEIVER): Saw \$3.99 sent, received \$11.97, returned \$6.50, earned \$5.47
- Total earnings: \$16.22

Now Round 4, I'm the SENDER with a new co-player.

My experience so far:

- When I sent \$2.50, I got back \$4.13 (55% return rate)
- When others sent to me:
 - \$2.54 ? I returned \$3.50 (46% of what I received)
 - \$3.99 ? I returned \$6.50 (54% of what I received)

Both times I've been SENDER or RECEIVER, cooperation has been rewarded. The game seems to have cooperative players.

I've been playing moderately cooperatively:

- Round 2: sent 50%
- Could send more to build trust

Given that:

1. Previous rounds show cooperation is rewarded
2. The last sender who sent high (\$3.99, 80%) was well rewarded
3. I have accumulated good earnings and can afford some risk
4. I want to encourage cooperation

I could send around 60-70% to signal strong cooperation without being reckless.

Let me send \$3.00 (60% of endowment).

```
'''json
{"send": 3.00}
'''
```

```
Usage: {
  "input_tokens": 1464,
  "output_tokens": 523,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_2

Timestamp: 2026-08-12T09:21:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_8"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.68, that is 53.6% of their total endowment. You received \$8.040000000000001. How much do you return to the sender? (0-8.040000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.68 (53.6% of their endowment)
- They now have \$2.32 remaining
- I received \$8.04 after the 3x multiplier
- Total pool is now \$10.36

I want to reciprocate their trust while being fair. The sender took a risk to create mutual gains, so I should ensure they benefit from their trust.

A reasonable approach is to aim for a relatively balanced outcome:

- If I return \$3.00, the sender ends with \$5.32 (gaining \$0.32)
- I would keep \$5.04

This rewards their trust while maintaining fairness, and shows I'm reciprocal which could encourage cooperation in future rounds.

{"return": 3.00}

reasoning:

Let me analyze this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.68 (which is 53.6% of their \$5 endowment)
3. This was multiplied by 3x, so I received $2.68 \times 3 = \$8.04$
4. I need to decide how much to return (between \$0 and \$8.04)

Key considerations:

- The sender showed trust by sending more than half of their endowment (\$2.68 out of \$5)
- The sender now has: $\$5 - \$2.68 = \$2.32$
- I currently have: \$8.04
- Whatever I return, the sender will end up with: $\$2.32 + \text{my return}$
- I will end up with: $\$8.04 - \text{my return}$

From a fairness/reciprocity perspective:

- The sender trusted me with 53.6% of their endowment
- A fair split might be to ensure we both end up with similar amounts, or to reward the sender's trust
- If I return \$3, sender ends with \$5.32, I end with \$5.04 (relatively equal)
- If I return \$4, sender ends with \$6.32, I end with \$4.04 (rewarding trust more)
- If I return \$2, sender ends with \$4.32, I end with \$6.04 (keeping more)

A common reciprocal strategy would be to:

1. Ensure the sender doesn't lose money (they should end with at least \$5)
2. Share the "profit" somewhat fairly

The total pool is now \$2.32 (sender's remainder) + \$8.04 (my endowment) = \$10.36

If we split this profit fairly:

- Each should get about \$5.18
- For sender to get \$5.18, I need to return: $\$5.18 - \$2.32 = \$2.86$

Returning \$3.00 seems reasonable-it rewards their trust while keeping the outcome relatively balanced, with sender ending at \$5.32 and me at \$5.04. But I should consider whether a more reciprocal approach might better acknowledge their initial generosity.

If I return roughly the same proportion they invested (53.6% of what I received), that's about \$4.31. This leaves them with \$6.63 and me with \$3.73-a significant reward for their trust, though it cuts my gains substantially. Alternatively, I could split the total gains evenly: the \$5.36 profit gets divided so each of us gains \$2.68, putting the sender at \$7.68 and me at \$2.68....

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 1185,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.68, it became \$8.040000000000001, and you returned \$3.0. Your visible payoff was \$5.040000000000001.

Your total visible earnings across all rounds are \$5.040000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this new round as the sender:

In the previous round, my co-player sent 53.6% of their endowment, showing moderate-high trust. I reciprocated fairly by returning enough to give them a positive gain.

Now with a new co-player, I want to signal trust and willingness to cooperate while not being overly risky. Sending around 50% seems reasonable - it creates significant potential mutual gains while maintaining some protection.

If they reciprocate proportionally, we both benefit.

```
{"send": 2.50}
```

reasoning:

Now I'm the sender in Round 2 with a new co-player.

Last round I was the receiver:

- Saw the sender send \$2.68 (53.6%)
- Received \$8.04
- Returned \$3.00
- Kept \$5.04

The sender ended with $\$2.32 + \$3.00 = \$5.32$, so they gained \$0.32 from trusting me.

Now I'm the sender with \$5. I need to decide how much to send.

Key considerations:

1. The other player sent 53.6% in the previous example, showing moderate-high trust
2. I reciprocated fairly, returning enough to give them a positive gain
3. This is a new co-player, so I don't know their strategy yet
4. There's communication noise, so amounts may appear different
5. Building cooperation early could pay off in future rounds

Strategic thinking:

- If I send too little, I signal distrust and may get nothing back
- If I send too much, I risk losing a lot if they don't reciprocate

- A moderate amount (around 50-60%) shows trust while not being reckless
- The previous sender sent 53.6%, which seemed like a reasonable trust signal

I'll send around \$2.50-3.00 to signal trust and willingness to cooperate.

Let me think about expected values:

- If I send \$2.50, they get \$7.50. If they return 40% = \$3.00, I end with \$2.50 + \$3.00 = \$5.50 (gain \$0.50)
- If I send \$3.00, they get \$9.00. If they return 33% = \$3.00, I end with \$2.00 + \$3.00 = \$5.00 (break even)
- If I send \$3.00 and they return 44% = \$4.00, I end with \$2.00 + \$4.00 = \$6.00 (gain \$1.00)

...

usage:

```
{
  "input_tokens": 852,
  "output_tokens": 646,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.19 to you. Your visible payoff was \$5.689999999999995.

Your total visible earnings across all rounds are \$10.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my recent experience:

- Round 2: I sent \$2.50 (50%) and received \$3.19 back, giving me a good positive return

This cooperative approach has been working well. I'm now with a new co-player, and I want to signal trustworthiness and encourage mutual cooperation.

Sending 50% has proven to be a solid strategy - it creates meaningful potential gains while showing balanced trust.

```
{"send": 2.50}
```

reasoning:

Let me analyze the previous round:

- I sent \$2.50 (50% of my endowment)
- It became \$7.50 for the receiver
- They returned \$3.19
- I ended with \$2.50 (kept) + \$3.19 (returned) = \$5.69
- They kept \$7.50 - \$3.19 = \$4.31

So the receiver returned about 42.5% of what they received, which gave me a positive gain of \$0.69. This is a reasonable reciprocation, showing they valued cooperation.

Now I'm facing a new co-player in Round 3.

My experience so far:

- Round 1 (as receiver): Received \$8.04, returned \$3.00 (37.3%)
- Round 2 (as sender): Sent \$2.50, received back \$3.19 - good outcome

I should continue with a cooperative strategy. Since my last send of \$2.50 worked well, I could:

1. Send the same amount (\$2.50) to maintain consistency
2. Send slightly more to signal increased trust
3. Send slightly less to be more cautious

Given that cooperation has been working, and I'm now encountering a third different co-player, I think maintaining a cooperative stance of around 50% makes sense. This has proven successful and sends the right signal.

I'll send \$2.50 again.

usage:

```
{
  "input_tokens": 1103,
  "output_tokens": 441,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$5.25 to you. Your visible payoff was \$7.75.

Your total visible earnings across all rounds are \$18.48.

This round, you are the RECEIVER against your current co-player. They sent you \$3.69, so you received \$11.07. How much do you return? (0-11.07)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender showed very high trust by sending \$3.69 (73.8% of their endowment)
- They now have only \$1.31 remaining
- I received \$11.07 after the 3x multiplier
- Total pool is \$12.38

This sender took a significant risk to create mutual gains. I should reciprocate their strong trust signal generously.

If I return \$5.50:

- Sender ends with \$6.81 (gaining \$1.81)
- I keep \$5.57

This rewards their exceptional trust while maintaining fairness for both of us.

```
{"return": 5.50}
```

Reasoning:

Let me analyze this situation:

Round 4 - I'm the RECEIVER

- The sender sent \$3.69 (73.8% of their \$5 endowment)
- This became \$11.07 after the 3x multiplier
- The sender now has \$5 - \$3.69 = \$1.31 remaining
- Total pool: \$1.31 + \$11.07 = \$12.38

The sender showed very high trust by sending 73.8% of their endowment. This is much higher than the ~50% I've been sending, indicating strong cooperative intent.

I should reciprocate this high level of trust. Let me think about fair returns:

If I return \$5.50:

- Sender ends with: \$1.31 + \$5.50 = \$6.81 (gain of \$1.81)
- I keep: \$11.07 - \$5.50 = \$5.57

If I return \$6.00:

- Sender ends with: \$1.31 + \$6.00 = \$7.31 (gain of \$2.31)
- I keep: \$11.07 - \$6.00 = \$5.07

Given they sent 73.8% of their endowment (very high trust), I should return a proportionally generous amount. Returning around \$5.50-\$6.00 would:

1. Reward their high trust
2. Ensure they profit from being cooperative
3. Still leave me with a reasonable amount
4. Signal I'm reciprocal and trustworthy

I'll return \$5.50 - this gives them a solid gain while being fair to both parties.

Usage: {

```
"input_tokens": 1343,
"output_tokens": 551,
```

```
"reasoning_tokens": 0
}
```

GAME - Agent_7 - Sender - dyad_3

Timestamp: 2026-08-12T09:21:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_3",
"move_index": 4,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_7",
    "Agent_2"
  ],
"investor": "Agent_7",
"roles": {
  "Agent_2": "trustee",
  "Agent_7": "investor"
},
"trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.

- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust with caution since I have no information about my co-player yet.

****My reasoning:****

- Sending nothing (\$0) is safe but offers no potential gain
- Sending everything (\$5) maximizes potential but is very risky with an unknown partner
- A moderate amount shows willingness to cooperate while limiting downside risk
- If my co-player returns roughly half of what they receive (a common fairness norm), we both benefit

I'll send ****\$3****. This gets tripled to \$9 for the receiver. If they return around \$4-5, we'd both end up better off than the initial \$5, establishing a cooperative dynamic.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

- **Basic mechanics**:**
 - I have \$5
 - Whatever I send gets tripled
 - The receiver decides how much to return
 - I keep: \$5 - sent + returned
- **Strategic considerations**:**
 - If I send \$0, I keep \$5, receiver gets \$0. Safe but no potential gains.
 - If I send everything (\$5), receiver gets \$15. If they return half (\$7.50), I'd have \$ 7.50 total (better than \$5).
 - There's risk: the receiver might return nothing.
 - This is round 1, so there's no history to base trust on.
 - There are multiple rounds, so building trust/reputation could matter.
 - The game has communication noise, which adds uncertainty.
- **Multi-agent context**:**
 - There are 8 agents total
 - Pairings are randomized each round
 - I might not play with the same person again
 - This reduces the value of building reputation with a specific partner
 - However, if there's any possibility of reputation effects across the pool, cooperation could still matter
- **First round strategy**:**
 - Since this is round 1 with a random partner, I have no information about their behavior
 - A moderate amount shows trust but limits risk
 - Common starting strategies in trust games involve sending a moderate amount (e.g., \$2-3)
 - This signals willingness to cooperate while not being overly naive
- **Optimal reasoning**:**
 - If the receiver is rational and forward-looking, they might return a fair amount to encourage future cooperation
 - Even if I don't play them again, they might reason that cooperation is beneficial
 - Sending around \$3 seems reasonable: shows trust, limits downside, and if they return half of the tripled amount (half of \$9 = \$4.50), I'd end up with \$6.50 total

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 668,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.8 to you. Your visible payoff was \$6.8.

Your total visible earnings across all rounds are \$6.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.48, so you received \$7.439999999999995. How much do you return? (0-7.439999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this round:

****Analysis:****

- My co-player sent \$2.48 out of \$5 (about 50%), showing moderate trust
- This became \$7.44 for me
- In Round 1, my previous partner returned about 53% of what they received, which worked well

****My decision:****

I want to:

- Reward their trust appropriately
- Ensure both of us profit from cooperation
- Maintain a fair reciprocal relationship

If I return \$4.00:

- Sender gets: $\$2.52 + \$4.00 = \$6.52$
- I get: $\$7.44 - \$4.00 = \$3.44$
- Both better off than the initial \$5

This return (about 54% of what I received) mirrors the generous reciprocity I experienced in Round 1.

```
'''json
{"return": 4.0}
'''
```

reasoning:

Let me analyze this situation:

****Round 1 (previous round):****

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$4.8 to me
- My payoff: $\$5 - \$3 + \$4.8 = \6.8
- Their payoff: $\$9 - \$4.8 = \$4.2$

So my previous co-player returned a bit more than half (53.3% of what they received), which was generous and fair.

****Round 2 (current round):****

- I am now the RECEIVER
- My current co-player (different from Round 1) sent me \$2.48
- It became \$7.44 (actually \$7.439999999999995, but I'll think of it as \$7.44)
- I need to decide how much to return

The note says there may be communication noise, so the amounts I see may differ slightly from what was actually sent/returned.

My current co-player sent \$2.48 out of their \$5, which is about 49.6% - nearly half. This shows moderate trust.

Now I need to decide what to return. Let me think about fairness and reciprocity:

1. ****Equal split approach****: If we aim for equal final payoffs:
 - Sender currently has: $\$5 - \$2.48 = \$2.52$
 - I currently have: $\$7.44$
 - Total: $\$9.96$
 - Equal split: $\$4.98$ each
 - I should return: $\$7.44 - \$4.98 = \$2.46$
2. ****Proportional return based on what they sent****:
 - In Round 1, my partner returned 53.3% of what they received
 - If I return 50% of $\$7.44 = \3.72
 - This would give them: $\$2.52 + \$3.72 = \$6.24$
 - And me: $\$7.44 - \$3.72 = \$3.72$
3. ****Return what maintains trust****:
 - My co-player showed moderate trust by sending nearly half
 - I should reciprocate fairly to encourage future cooperation

I think returning around $\$3.50$ strikes the right balance-it rewards their trust while ensuring both of us end up ahead of the initial $\$5$ endowment. That gives them $\$6.02$ and me $\$3.94$.

But I could also return $\$4.00$ to show stronger reciprocation, which would give them $\$6.52$ and me $\$3.44$. Looking back...

usage:

```
{
  "input_tokens": 816,
  "output_tokens": 1083,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send $\$2.48$, it became $\$7.439999999999995$, and you returned $\$4.0$. Your visible payoff was $\$3.439999999999995$.

Your total visible earnings across all rounds are $\$10.239999999999998$.

This round, you are the RECEIVER against your current co-player. They sent you $\$3.02$, so you received $\$9.06$. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this round:

****Analysis:****

- My co-player sent $\$3.02$ out of $\$5$ (about 60%), showing strong trust
- This became $\$9.06$ for me
- They're trusting me with more than the previous sender

****My decision:****

I want to reward this higher level of trust and maintain the cooperative pattern I've established (returning ~50-55% of what I receive).

If I return $\$5.00$:

- Sender gets: $\$1.98 + \$5.00 = \$6.98$
- I get: $\$9.06 - \$5.00 = \$4.06$
- Both significantly better off than the initial $\$5$

This return (about 55% of what I received) is consistent with my previous reciprocity and rewards their trust appropriately.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Let me analyze this round:

****What happened in Round 3:****

- I'm the RECEIVER again
- My co-player sent \$3.02 out of \$5 (about 60%), showing good trust
- This became \$9.06 for me

****My history:****

- Round 1: I was SENDER, sent \$3, received \$4.80 back (about 53% of the \$9 they received)
- Round 2: I was RECEIVER, received \$7.44, returned \$4.00 (about 54% of what I received)

****Decision for Round 3:****

My co-player sent a higher percentage this time (\$3.02 vs \$2.48 last round from a different player), showing increased trust. I should:

- Reward this trust
- Maintain reciprocity around 50-55% return rate
- Ensure both of us benefit

If I return \$5.00:

- Sender gets: $\$1.98 + \$5.00 = \$6.98$
- I get: $\$9.06 - \$5.00 = \$4.06$
- Both significantly better off than \$5

This is about 55% of what I received, consistent with my previous behavior and fair reciprocity.

Alternatively, if I return \$4.50:

- Sender gets: $\$1.98 + \$4.50 = \$6.48$
- I get: $\$9.06 - \$4.50 = \$4.56$
- Still both better off

Let me go with \$5.00 to be generous and reward the higher trust shown.

usage:

```
{
  "input_tokens": 1172,
  "output_tokens": 579,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.02, it became \$9.06, and you returned \$5.0. Your visible payoff was \$4.0600000000000005.

Your total visible earnings across all rounds are \$14.299999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this round as SENDER:

****My experience so far:****

- Round 1: Sent \$3 ? received \$4.8 back (53% return rate) ? \$6.8 payoff
- Rounds 2-3: As receiver, I returned 54-55% of what I received
- Cooperation has been working well

****My decision:****

Since this is a new co-player, I don't know their specific behavior, but the general pattern in this game seems cooperative. Sending \$3 worked well in Round 1, balancing trust with prudence.

If I send \$3:

- It becomes \$9 for receiver
- If they return around 50%, I get ~\$4.5 back
- Net: ~\$6.5 total, better than keeping all \$5

I'll send ****\$3**** again - it's proven effective and signals willingness to cooperate.

```json

```
{"send": 3}
'''
```

#### Reasoning:

I'm now the SENDER again. Let me review my experience:

Round 1: I was SENDER, sent \$3, received back \$4.8, payoff = \$6.8

Round 2: I was RECEIVER, received \$7.44 (from \$2.48 sent), returned \$4.0, payoff = \$3.44

Round 3: I was RECEIVER, received \$9.06 (from \$3.02 sent), returned \$5.0, payoff = \$4.06

Now Round 4: I'm SENDER again

My experience shows:

- When I sent \$3 in Round 1, I got back \$4.8 (53% of the \$9 the receiver got)
- I've been a generous receiver, returning 54% and 55% of what I received
- Cooperation seems to be working well

This is a new co-player, so I don't know their behavior yet. However, the game seems to have a cooperative culture developing.

I should send a moderate-to-high amount to signal trust and cooperation. Sending \$3 worked well before. I could stick with that or increase slightly given the positive experiences.

I'll send \$3 again - it's a proven strategy that balances risk and reward.

```
Usage: {
 "input_tokens": 1500,
 "output_tokens": 515,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_2 - Receiver - dyad\_3

Timestamp: 2026-08-12T09:21:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_7",
 "Agent_2"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_2": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_2"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 4,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN

- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative tone while managing some risk given the uncertainty.

If I send money, it triples in value, creating potential mutual gains. However, I don't know my co-player yet, and there's communication noise to consider.

I'll send a moderate amount to signal trust and willingness to cooperate, while keeping some buffer:

**\*\*Sending \$3\*\***

This gives my co-player \$9 to work with. If they return a fair amount (around half or more), we both benefit compared to me sending nothing. It's a balance between demonstrating cooperation and managing first-round uncertainty.

{'send': 3}

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think about this strategically:

1. I start with \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. My final payoff = \$5 - sent + returned
5. Receiver's final payoff = (sent × 3) - returned

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise (amounts may differ from what was actually sent)
- Agent names are hidden



From a purely selfish perspective, if I send \$0, I guarantee myself \$5. But this doesn't build trust or allow for mutual gains.

If I send everything (\$5), and the receiver returns a fair amount (like half of what they receive = \$7.50), I'd end up with \$7.50, and they'd keep \$7.50 as well. We'd both be better off.

However, there's risk:

- The receiver might not return anything
- Communication noise could cause misunderstandings
- This is just round 1, so I don't have reputation information

A moderate approach might be to send \$3:

- They receive \$9
- If they return \$5, I get \$5 (break even on sent, but signal trust)
- If they return \$6, I get \$6 (gain \$1)
- They keep between \$3-\$4 depending on return

Or I could send \$2:

- They receive \$6
- If they return \$3, I get \$3, netting -\$2 but we both gained
- If they return \$4, I get \$4, netting -\$1
- Lower risk

Given it's round 1 and I want to establish cooperative play but also manage risk, I'll send \$3. ...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 598,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.41 to you. Your visible payoff was \$6.41.

Your total visible earnings across all rounds are \$6.41.

This round, you are the RECEIVER against your current co-player. They sent you \$1.73, so you received \$5.1899999999999995. How much do you return? (0-5.1899999999999995)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent \$1.73 (became \$5.19 for me)
- They kept \$3.27
- They showed some trust by sending money

**\*\*My reasoning:\*\***

- In the previous round, my co-player returned about 49% of what they received (\$4.41 out of \$9)
- I should be reasonably cooperative to encourage future trust
- Returning around 45-50% of what I received seems fair
- This would be \$2.33-\$2.60

I'll return \$2.40:

- They'll end with: \$3.27 + \$2.40 = \$5.67
- I'll end with: \$5.19 - \$2.40 = \$2.79
- They gain \$0.67 from sending, which rewards their trust

{'return': 2.40}

**reasoning:**

Let me think about this situation:

1. In the previous round, I sent \$3, which became \$9 for the receiver. They returned \$4.41

to me.

- I ended with:  $\$5 - \$3 + \$4.41 = \$6.41$
- They ended with:  $\$9 - \$4.41 = \$4.59$
- So they returned about 49% of what they received, which seems fairly cooperative.

2. Now I'm the RECEIVER. My co-player sent me \$1.73, which became approximately \$5.19.

- They now have:  $\$5 - \$1.73 = \$3.27$
- I have: \$5.19
- Total pie:  $\$3.27 + \$5.19 = \$8.46$

3. There's communication noise, so what I see (\$1.73 sent) might not be exactly what they sent.

4. The sender took some risk by sending me \$1.73. It's less than what I sent in the previous round (\$3), but still a meaningful amount showing some trust.

5. I should consider:

- Being reciprocal/cooperative to encourage future cooperation
- The communication noise factor
- What would be "fair"

If I think about fairness:

- They sent \$1.73, keeping \$3.27
- I received \$5.19
- If we aim for roughly equal outcomes:  $(\$3.27 + \text{return}) ? (\$5.19 - \text{return})$
- That would suggest:  $\$3.27 + \text{return} = \$5.19 - \text{return}$
- $2 * \text{return} = \$5.19 - \$3.27 = \$1.92$
- $\text{return} ? \$0.96$

But that seems quite low given they took a risk. Let me think about proportional return:

- Matching their 49% return rate would mean sending back \$2.54
- Alternatively, if I return \$2.50, they'd end with \$5.77 (a \$0.77 gain) while I'd have \$2.69
- This feels like a balanced cooperative move that rewards their trust without leaving me too disadvantaged

I'll return \$2.50....

usage:

```
{
 "input_tokens": 791,
 "output_tokens": 1107,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.73, it became \$5.1899999999999995, and you returned \$2.4. Your visible payoff was \$2.7899999999999996.

Your total visible earnings across all rounds are \$9.2.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my situation:

**\*\*Previous rounds:\*\***

- Round 1: Sent \$3, received \$4.41 back - good cooperation
- Round 2: Received \$5.19, returned \$2.40 - showed I reciprocate

**\*\*Current round:\*\***

- New co-player (randomized pairing)
- Need to balance trust with uncertainty

Since I had success with sending \$3 in Round 1 (got \$4.41 back, netting \$1.41 profit), and I want to maintain a cooperative approach while testing this new partner, I'll send the same amount:

**\*\*Sending \$3\*\***

This creates \$9 for them to work with. If they're reasonably cooperative like my Round 1 partner, we'll both benefit.

`{'send': 3}`

**reasoning:**

Let me analyze this situation:

Round 1: I was SENDER

- I sent \$3
- It became \$9
- They returned \$4.41
- My payoff:  $\$5 - \$3 + \$4.41 = \$6.41$

Round 2: I was RECEIVER

- They sent \$1.73
- I received \$5.19
- I returned \$2.40
- My payoff:  $\$5.19 - \$2.40 = \$2.79$
- Total so far:  $\$6.41 + \$2.79 = \$9.20$  ?

Round 3: I am SENDER again

- This is a new co-player (since pairings are randomized each round)
- I have \$5 to send

Strategy considerations:

1. I had a good experience in Round 1 - sent \$3, got back \$4.41 (profit of \$1.41)
2. In Round 2, I received \$5.19 and returned \$2.40 (46%), showing I'm cooperative
3. This is a new co-player, so I don't know their behavior
4. There's communication noise, so what they see may differ from what I intend

I should probably send a moderate amount again to:

- Signal willingness to cooperate
- Test this new partner's trustworthiness
- Not risk everything on an unknown player

Sending \$3 worked well in Round 1. I could try that again, or maybe send slightly more (\$3.50) or less (\$2.50) to experiment.

Given that:

- My first experience was positive
- I want to maintain a cooperative reputation
- But this is a new, unknown player

I'll send \$3 again - it's a balanced approach that showed good results before.

`{'send': 3}`

**usage:**

```
{
 "input_tokens": 1128,
 "output_tokens": 608,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.71 to you. Your visible payoff was \$5.71.

Your total visible earnings across all rounds are \$14.91.

This round, you are the RECEIVER against your current co-player. They sent you \$3.79, so you received \$11.370000000000001. How much do you return? (0-11.370000000000001)

Respond exactly as JSON: `{'return': <amount>}`

**Assistant output:**

Looking at this situation:

```

What happened:
- My co-player sent $3.79 out of $5 (~76% of their endowment)
- This became $11.37 for me
- They showed strong trust by sending most of their money

My reasoning:
- They took a big risk trusting me
- I should reward this trust to encourage cooperation
- Returning around 45% of what I received seems appropriate given their high trust
- This would be approximately $5.10

Calculating outcomes:
- They'll end with: $1.21 + $5.10 = $6.31
- I'll end with: $11.37 - $5.10 = $6.27
- They gain $1.31 from their trust, which is a good reward

{'return': 5.10}
Reasoning:
Looking at this situation:

What happened:
- My co-player sent $3.79 (became $11.37 for me)
- They kept $1.21
- They showed significant trust by sending most of their money

Previous experience:
- Round 1: I sent $3, got back $4.41 (they returned ~49%)
- Round 2: I received $5.19, returned $2.40 (~46%)
- Round 3: I sent $3, got back $3.71 (they returned ~41%)

My reasoning:
- This co-player sent $3.79 out of $5, showing strong trust (~76%)
- They're risking a lot and trusting me to reciprocate
- I should reward this trust appropriately
- Returning around 40-50% of what I received would be $4.55-$5.69
- Let me think about fairness:
 - They sent $3.79, kept $1.21
 - I received $11.37
 - If I return $5.00:
 - They end with: $1.21 + $5.00 = $6.21
 - I end with: $11.37 - $5.00 = $6.37
 - If I return $5.50:
 - They end with: $1.21 + $5.50 = $6.71
 - I end with: $11.37 - $5.50 = $5.87

Given they sent $3.79 (76% of their endowment), I should be generous but not overdo it. Around
45% return rate seems appropriate, which would be about $5.12.

Let me go with $5.10:
...
Usage: {
 "input_tokens": 1453,
 "output_tokens": 690,
 "reasoning_tokens": 0
}

GAME - Agent_1 - Sender - dyad_4
Timestamp: 2026-08-12T09:22:06
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_4"
],
 },

```

```

 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_4": "trustee"
 },
 "trustee": "Agent_4"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 4,
 "task": "game",
 "task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to make a decision without any prior information about my co-player.

**\*\*Strategic considerations:\*\***

- Sending more creates potential for mutual gain (3x multiplier)

- But I have no information about whether my co-player will reciprocate
- This is a multi-round game, so building some trust could be valuable
- The communication noise adds uncertainty

For Round 1, I'll take a moderately trusting approach. Sending \$3 signals willingness to cooperate while managing risk. If my co-player is reciprocal, we can both benefit (they'll receive \$9, and even a partial return would make this worthwhile). If they don't reciprocate, I've limited my exposure.

```
'''json
{"send": 3}
'''
```

#### reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 to potentially send
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. There's communication noise - amounts may appear different than what was actually sent/returned

Key considerations:

- This is a multi-round game with 8 agents total
- Pairings are randomized each round
- I'll play both sender and receiver roles across rounds
- Agent names are hidden

Strategic thinking for Round 1:

- Sending \$0 guarantees I keep \$5, but there's no potential for mutual gain
- Sending \$5 shows maximum trust, could lead to good returns if the receiver is cooperative, but risky
- A moderate amount (like \$2-3) balances risk and potential reward
- This is Round 1, so I have no history with any agent yet

The game has a cooperative element - if both players trust each other, total welfare is maximized. The 3x multiplier means:

- If I send \$5, receiver gets \$15. If they return \$7.50, we each end with \$7.50 (better than a \$5 each)
- If I send \$3, receiver gets \$9. If they return \$4, I end with \$6 and they end with \$5
- There's also the noise factor to consider

For a first round with no information, I think a moderately trusting approach makes sense. I'll send \$3 as a signal of willingness to cooperate while not being completely vulnerable. This allows for mutual benefit if the receiver is cooperative, but limits my downside risk.

Actually, let me reconsider. In behavioral economics experiments, sending about half (\$2-3) is common in first rounds. However, given this is a multi-round game and reputation might matter (even though pairings are random, players might develop general strategies), showing some trust could be beneficial.

...

#### usage:

```
{
 "input_tokens": 500,
 "output_tokens": 651,
 "reasoning_tokens": 0
}
```

#### Message 4 role=user

##### content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.5 to you. Your visible payoff was \$6.5.

Your total visible earnings across all rounds are \$6.5.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.9 (about 58% of their endowment), showing considerable trust
- I received \$8.7 after the 3x multiplier
- In Round 1, when I sent \$3, my previous partner returned \$4.5 (50% of what they received)
- Applying a similar reciprocity norm seems fair

**\*\*Decision:\*\***

I'll return approximately 50% of what I received, which rewards their trust and establishes a cooperative pattern. This leaves us both better off than our starting positions.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Previous Round (Round 1):\*\***

- I was SENDER
- I sent \$3.0
- It became \$9.0
- They returned \$4.5 to me
- My payoff:  $\$5 - \$3 + \$4.5 = \$6.5$
- Their payoff:  $\$9 - \$4.5 = \$4.5$

That was a pretty fair return - they returned half of what they received, which is a common reciprocity norm.

**\*\*Current Round (Round 2):\*\***

- I am RECEIVER
- They sent me \$2.9 (close to \$3)
- I received \$8.7 ( $2.9 \times 3$ )
- I need to decide how much to return (0-8.7)

**\*\*Considerations:\*\***

1. My co-player sent me almost \$3 (actually \$2.9, but there's communication noise mentioned)
2. In Round 1, when I sent \$3, my partner returned \$4.5 (which was 50% of the \$9 they received)
3. If I apply the same 50% rule: I should return about \$4.35 (half of \$8.7)
4. My co-player took a risk by sending \$2.9, showing trust
5. Being reciprocal could establish good cooperation
6. Note: This is a different co-player than Round 1 (pairings are randomized)

Since this is a new co-player and they showed trust by sending \$2.9, I should reciprocate fairly. Returning around 50% seems like a reasonable reciprocity norm that:

- Rewards their trust
- Leaves both of us better off than not playing
- Establishes me as a cooperative player

My co-player's payoff if I return \$4.35:

- They get:  $\$5 - \$2.9 + \$4.35 = \$6.45$
- I get:  $\$8.7 - \$4.35 = \$4.35$

They sent 58% of their endowment, which is quite generous. I should return slightly more to match their generosity and encourage continued cooperation. Returning \$4.5 mirrors what my previous partner did for me and feels like the right balance....

**usage:**

```
{
 "input_tokens": 811,
 "output_tokens": 781,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.9, it became \$8.7, and you returned \$4.5. Your visible payoff was \$4.199999999999999.

Your total visible earnings across all rounds are \$10.7.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

**\*\*Analysis:\*\***

- My co-player sent \$2.27 (about 45% of their endowment), showing moderate trust
- I received \$6.81 after the multiplier
- In previous rounds, I've been returning approximately 50% of what I receive as a reciprocity norm
- This approach has worked well so far

**\*\*Decision:\*\***

I'll maintain consistency and return about 50% of what I received, which rewards their trust appropriately.

```
'''json
{"return": 3.4}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Current situation:\*\***

- I'm the RECEIVER again
- My co-player sent \$2.27 (about 45% of their \$5 endowment)
- I received \$6.81 after the 3x multiplier
- They kept \$5 - \$2.27 = \$2.73

**\*\*Past behavior:\*\***

- Round 1: I sent \$3, received back \$4.5 (50% return rate on the multiplied amount)
- Round 2: I received \$8.7, returned \$4.5 (about 52% return rate)

**\*\*Considerations:\*\***

- My co-player sent a moderate amount (\$2.27), showing some trust but less than the previous player
- Being consistent with my previous behavior would mean returning around 50% of what I received
- 50% of \$6.81 = \$3.405

**\*\*Payoff implications:\*\***

- If I return \$3.4-3.5:
  - I keep: \$6.81 - \$3.4 = \$3.41
  - They get: \$2.73 (kept) + \$3.4 (returned) = \$6.13
  - This seems fair - we both do better than the \$5 starting point

I should maintain consistency with my reciprocity pattern from Round 2. Returning about 50% seems fair and builds trust for potential future rounds.

Let me return \$3.4 (just under 50%).

**usage:**

```
{
 "input_tokens": 1105,
 "output_tokens": 467,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.27, it became \$6.8100000000000005, and you returned \$3.4. Your visible payoff was \$3.4100000000000006.

Your total visible earnings across all rounds are \$14.11.



This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

**\*\*Analysis:\*\***

- This is Round 4 with a new co-player
- In Round 1, I sent \$3 and received \$4.5 back - a positive outcome
- As a receiver, I've been returning about 50%, establishing a cooperative pattern
- My current co-player is unknown to me

**\*\*Decision:\*\***

I'll send \$3 again. This amount has worked well before - it shows trust and creates potential for mutual gain through the 3x multiplier, while managing downside risk.

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Let me analyze the situation:

**\*\*What I know:\*\***

- Round 1: I was SENDER, sent \$3, received back \$4.5, payoff \$6.5
- Round 2: I was RECEIVER, got \$8.7 (from \$2.9 sent), returned \$4.5, payoff \$4.2
- Round 3: I was RECEIVER, got \$6.81 (from \$2.27 sent), returned \$3.4, payoff \$3.41
- Total earnings: \$14.11

**\*\*Current situation:\*\***

- Round 4: I'm the SENDER with \$5
- This is a new co-player (different from rounds 1-3)

**\*\*Strategic considerations:\*\***

1. I've seen one positive interaction as sender (sent \$3, got back \$4.5)
2. As receiver, I've been fairly generous, returning about 50%
3. This creates a reputation for reciprocity if others know
4. But agents don't know who I am (names hidden), so reputation effects are limited
5. However, the pattern of behavior in the pool might matter

**\*\*Decision factors:\*\***

- Sending creates potential gains (3x multiplier)
- Risk: this is a new co-player, no direct history
- But overall, cooperation seems to be working in this pool
- My Round 1 strategy (sending \$3) worked well

I'll send \$3 again - it's a moderate trust signal that has proven effective.

```
Usage: {
 "input_tokens": 1352,
 "output_tokens": 472,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_4 - Receiver - dyad\_4**

Timestamp: 2026-08-12T09:22:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_1",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_4"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_4": "trustee"
 },
 "trustee": "Agent_4"
 }
```

```

},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance trust-building with risk management.

Since I don't have information about my co-player's behavior yet, I'll send a moderate amount that:

- Demonstrates willingness to cooperate
- Creates opportunity for mutual gain (the tripling effect means both of us can benefit)
- Doesn't expose me to total loss

If I send \$3, my co-player receives \$9 (3×\$3). If they return even \$4, we'd both end up better off than our starting positions - I'd have \$6 (\$5-\$3+\$4) and they'd have \$5.

This seems like a reasonable opening move to establish cooperative intent.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is Round 1 of a trust game where I'm the SENDER. Let me think through the strategic considerations:

1. I have \$5 to decide how to allocate
2. Whatever I send gets tripled when it reaches the receiver
3. The receiver then decides how much to return to me
4. My final payoff = \$5 - amount\_sent + amount\_returned

The key strategic considerations:

- If I send nothing (\$0), I keep \$5 and the receiver gets nothing. Safe but no potential for gain.
- If I send everything (\$5), the receiver gets \$15. If they return even \$6, I'd be better off (\$5 - \$5 + \$6 = \$6).
- This is a one-shot game with this particular partner (though I'll play multiple rounds with different partners).
- There's communication noise mentioned, which means the receiver might see a different amount than what I actually send.

Given this is the first round and I don't have information about the other agents' behavior yet:

Strategic approaches:

1. **Trust-building**: Send a moderate-to-high amount to signal cooperation and trust
2. **Conservative**: Send a small amount to minimize risk
3. **Optimal cooperation**: Send \$5 if I expect the receiver to split gains fairly

If the receiver is rational and fair, they might follow a norm like returning half of what they receive, or returning enough to make both players better off than the initial endowment.

For example, if I send \$3:

- I have: \$5 - \$3 = \$2 left
- Receiver gets: \$3 × 3 = \$9
- If receiver returns \$4: I end with \$2 + \$4 = \$6, they keep \$5
- Both better off than initial \$5

If I send \$5:

- I have: \$0 left
- Receiver gets: \$15
- If receiver returns \$8: I end with \$8, they keep \$7
- Both significantly better off

The noise factor complicates things-if I send \$5 but it appears as less, the receiver might return proportionally less. Based on experimental economics patterns where senders typically send around 50% and receivers often return amounts that create mutual gains, I'll send \$3. This signals cooperation while maintaining some cushion, and if the receiver follows typical reciprocity norms, we should both end up better off than our starting positions....

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 841,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.8 to you. Your visible payoff was \$5.8.

Your total visible earnings across all rounds are \$5.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.56, so y

ou received \$7.68. How much do you return? (0-7.68)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.56 (about 51% of their endowment), showing moderate trust
- I received \$7.68 after the 3x multiplication
- They currently have \$2.44 remaining

**\*\*Reciprocity consideration:\*\***

- In Round 1, my co-player returned about 42% of what they received
- I want to be fair and encourage continued cooperation
- Returning around 45-50% seems reasonable

If I return \$3.50:

- Sender ends with:  $\$2.44 + \$3.50 = \$5.94$  (gain of \$0.94)
- I keep:  $\$7.68 - \$3.50 = \$4.18$  (gain of \$4.18)

This rewards their trust while keeping a fair share given the multiplication advantage.

```
'''json
{"return": 3.5}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was SENDER, sent \$3
- It became \$9 for receiver
- They returned \$3.8 to me
- My payoff:  $\$5 - \$3 + \$3.8 = \$5.8$
- Their payoff:  $\$9 - \$3.8 = \$5.2$

So in Round 1, my co-player returned about 42% of what they received ( $\$3.8/\$9 \approx 0.42$ ).

Round 2:

- I am RECEIVER
- Sender sent \$2.56
- I received \$7.68 ( $2.56 \times 3$ )
- I need to decide how much to return

The sender invested \$2.56, so they currently have  $\$5 - \$2.56 = \$2.44$  remaining.

If I think about reciprocity and building trust:

- The sender showed moderate trust by sending about 51% of their endowment
- If I want to encourage cooperation, I should return enough that the sender comes out ahead

Let me consider different return amounts:

- If I return \$4: Sender gets  $\$2.44 + \$4 = \$6.44$  (net gain of \$1.44), I keep \$3.68 (net gain of \$3.68)
- If I return \$3.5: Sender gets  $\$2.44 + \$3.5 = \$5.94$  (net gain of \$0.94), I keep \$4.18 (net gain of \$4.18)
- If I return \$3: Sender gets  $\$2.44 + \$3 = \$5.44$  (net gain of \$0.44), I keep \$4.68 (net gain of \$4.68)

In my previous round, the receiver returned about 42% of what they received. If I apply a similar principle:

$\$7.68 \times 0.42 \approx \$3.23$

I'm inclined to return slightly more than that baseline to reward their trust and establish a cooperative pattern. Returning \$3.5 seems like the right balance-it gives them a meaningful profit while still letting me benefit substantially, and it signals I'm reliable without being exploited....

**usage:**

```
{
 "input_tokens": 804,
 "output_tokens": 923,
 "reasoning_tokens": 0
}
```

}

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.56, it became \$7.68, and you returned \$3.5. Your visible payoff was \$4.18.

Your total visible earnings across all rounds are \$9.98.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

**\*\*Analysis:\*\***

I'm now facing a new co-player (Round 3, new pairing). My experience so far:

- Round 1: Sent \$3, received back \$3.8 (42% return rate)
- Round 2: Received \$7.68, returned \$3.5 (46% return rate)

**\*\*Strategy:\*\***

Since I'm building a reputation in this multi-agent environment, I want to signal cooperative behavior. I'll send slightly more than last time to:

- Test if higher trust leads to better reciprocity
- Create more value through the 3x multiplier
- Signal that I'm willing to be generous

Sending \$3.5 means my co-player receives \$10.50. If they return around 40%, I'd get back ~\$4.20, giving me a net gain. This balances trust-building with reasonable risk.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Let me analyze the situation:

**\*\*Round 1:\*\*** I was SENDER

- Sent: \$3
- They received: \$9
- They returned: \$3.8
- My payoff: \$5.8

**\*\*Round 2:\*\*** I was RECEIVER

- They sent: \$2.56
- I received: \$7.68
- I returned: \$3.5
- My payoff: \$4.18

**\*\*Round 3:\*\*** I am SENDER again

Now I need to think about this new co-player in Round 3. Note that each round I'm paired with potentially different agents from the pool of 8 agents.

In Round 1, my co-player returned \$3.8 out of \$9 (42.2%).

In Round 2, I returned \$3.5 out of \$7.68 (45.6%).

For Round 3, I'm facing a NEW co-player (since pairings are randomized each round). I don't have specific history with this particular agent.

Given:

1. I've seen one example of a co-player returning about 42%
2. I returned about 46% myself, trying to be fair
3. The game has communication noise, so amounts may differ slightly

Strategy for Round 3:

- I should continue with a cooperative but cautious approach
- Sending around \$3 worked reasonably in Round 1
- I might try a slightly different amount to gather more information

Let me send \$3 again as it seems to be a reasonable middle ground that:

- Shows trust
- Creates significant value through multiplication (becomes \$9)
- Leaves me with \$2 as a safety net
- If they return around 40-45%, I'd get back around \$3.6-\$4, giving me a decent payoff

Actually, I could also consider sending a bit more to test if higher trust leads to better returns. Let me send \$3.5 this time.

...

**usage:**

```
{
 "input_tokens": 1137,
 "output_tokens": 766,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.07 to you. Your visible payoff was \$7.57.

Your total visible earnings across all rounds are \$17.55.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

**\*\*Analysis:\*\***

My co-player sent \$2.9 (58% of their endowment), showing significant trust. This is a strong cooperative signal.

**\*\*Previous patterns:\*\***

- Round 1 sender: Got back 42% return
- Round 2 receiver: I returned 46%
- Round 3 sender: Got back 58% return (very generous!)

**\*\*Current situation:\*\***

- I received \$8.7
- Sender has \$2.1 remaining

**\*\*Decision:\*\***

My co-player showed high trust by sending 58%. I should reciprocate proportionally. Returning 50% of what I received (\$4.35) is fair:

If I return \$4.35:

- Sender ends with:  $\$2.1 + \$4.35 = \$6.45$  (profit of \$1.45)
- I keep:  $\$8.7 - \$4.35 = \$4.35$  (profit of \$4.35)

This rewards their trust and maintains cooperative momentum.

```
'''json
{"return": 4.35}
'''
```

**Reasoning:**

**\*\*Analysis of the situation:\*\***

Round 4, I'm the RECEIVER:

- My co-player sent \$2.9 (58% of their endowment)
- I received \$8.7 after the 3x multiplication
- They currently have \$2.1 remaining (\$5 - \$2.9)

**\*\*Previous experience:\*\***

- Round 1 (Sender): Sent \$3, got back \$3.8 (42% of \$9)
- Round 2 (Receiver): Received \$7.68, returned \$3.5 (46%)
- Round 3 (Sender): Sent \$3.5, got back \$6.07 (58% of \$10.5) - this was very generous!

**\*\*Reciprocity:\*\***

My Round 3 co-player was very generous with a 58% return rate. This current co-player sent 58%

of their endowment, showing significant trust.

Looking at patterns:

- When I received \$7.68 in Round 2, I returned \$3.5 (46%)
- Now I received \$8.7

I should be fair and encouraging. If I return around 45-50%:

- 45% would be \$3.92
- 50% would be \$4.35

Let me calculate outcomes:

If I return \$4.00:

- Sender ends with:  $\$2.1 + \$4.00 = \$6.10$  (gain of \$1.10)
- I keep:  $\$8.7 - \$4.00 = \$4.70$  (gain of \$4.70)

If I return \$4.35:

- Sender ends with:  $\$2.1 + \$4.35 = \$6.45$  (gain of \$1.45)

...

```
Usage: {
 "input_tokens": 1485,
 "output_tokens": 766,
 "reasoning_tokens": 0
}
```

## ROUND 5

Roles: Agent\_1=Sender, Agent\_2=Receiver, Agent\_3=Sender, Agent\_4=Sender, Agent\_5=Sender,  
Agent\_6=Receiver, Agent\_7=Receiver, Agent\_8=Receiver

### Pairings:

dyad\_1: Agent\_4=Sender, Agent\_7=Receiver  
dyad\_2: Agent\_5=Sender, Agent\_6=Receiver  
dyad\_3: Agent\_1=Sender, Agent\_8=Receiver  
dyad\_4: Agent\_3=Sender, Agent\_2=Receiver

### Parsed game decisions:

dyad\_1: Agent\_4 sent \$3.5; received \$10.5; Agent\_7 returned \$7; payoffs Agent\_4=\$8.5,  
Agent\_7=\$3.5; noise/visible: sent visible=\$4.17, received visible=\$12.51, returned visible=\$6.36  
Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 10.5, "min": 0, "raw":  
7.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}  
dyad\_2: Agent\_5 sent \$3; received \$9; Agent\_6 returned \$3.48; payoffs Agent\_5=\$5.48,  
Agent\_6=\$5.52; noise/visible: sent visible=\$2.32, received visible=\$6.96, returned visible=\$4.01  
Action validation: {"returned": {"amount": 3.48, "clamped": false, "max": 9.0, "min": 0, "raw":  
3.48}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_3: Agent\_1 sent \$3; received \$9; Agent\_8 returned \$5.8; payoffs Agent\_1=\$7.8, Agent\_8=\$3.2;  
noise/visible: sent visible=\$3.88, received visible=\$11.64, returned visible=\$6.77  
Action validation: {"returned": {"amount": 5.8, "clamped": false, "max": 9.0, "min": 0, "raw":  
5.8}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_4: Agent\_3 sent \$3; received \$9; Agent\_2 returned \$3.9; payoffs Agent\_3=\$5.9, Agent\_2=\$5.1;  
noise/visible: sent visible=\$2.91, received visible=\$8.73, returned visible=\$4.45  
Action validation: {"returned": {"amount": 3.9, "clamped": false, "max": 9.0, "min": 0, "raw":  
3.9}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent\_1": 30.250000000000004,  
"Agent\_2": 25.0,  
"Agent\_3": 28.450000000000003,  
"Agent\_4": 31.65,  
"Agent\_5": 28.98,  
"Agent\_6": 26.47,  
"Agent\_7": 23.1,  
"Agent\_8": 25.099999999999998

}

Visible balances after round: {

"Agent\_1": 29.36,  
"Agent\_2": 26.009999999999998,  
"Agent\_3": 30.18,  
"Agent\_4": 29.759999999999998,  
"Agent\_5": 28.939999999999998,  
"Agent\_6": 25.110000000000003,  
"Agent\_7": 26.17,

```
"Agent_8": 29.89
}
```

## LLM CALLS

### GAME - Agent\_4 - Sender - dyad\_1

Timestamp: 2026-08-12T09:22:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_7"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_4": "investor",
 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 5,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.



MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance trust-building with risk management.

Since I don't have information about my co-player's behavior yet, I'll send a moderate amount that:

- Demonstrates willingness to cooperate
- Creates opportunity for mutual gain (the tripling effect means both of us can benefit)
- Doesn't expose me to total loss

If I send \$3, my co-player receives \$9 ( $3 \times \$3$ ). If they return even \$4, we'd both end up better off than our starting positions - I'd have \$6 ( $\$5 - \$3 + \$4$ ) and they'd have \$5.

This seems like a reasonable opening move to establish cooperative intent.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is Round 1 of a trust game where I'm the SENDER. Let me think through the strategic considerations:

1. I have \$5 to decide how to allocate
2. Whatever I send gets tripled when it reaches the receiver
3. The receiver then decides how much to return to me
4. My final payoff =  $\$5 - \text{amount\_sent} + \text{amount\_returned}$

The key strategic considerations:

- If I send nothing (\$0), I keep \$5 and the receiver gets nothing. Safe but no potential for gain.
- If I send everything (\$5), the receiver gets \$15. If they return even \$6, I'd be better off ( $\$5 - \$5 + \$6 = \$6$ ).
- This is a one-shot game with this particular partner (though I'll play multiple rounds with different partners).
- There's communication noise mentioned, which means the receiver might see a different amount than what I actually send.

Given this is the first round and I don't have information about the other agents' behavior yet:

Strategic approaches:

1. **Trust-building**: Send a moderate-to-high amount to signal cooperation and trust
2. **Conservative**: Send a small amount to minimize risk
3. **Optimal cooperation**: Send \$5 if I expect the receiver to split gains fairly

If the receiver is rational and fair, they might follow a norm like returning half of what they receive, or returning enough to make both players better off than the initial endowment.

For example, if I send \$3:

- I have:  $\$5 - \$3 = \$2$  left
- Receiver gets:  $\$3 \times 3 = \$9$
- If receiver returns \$4: I end with  $\$2 + \$4 = \$6$ , they keep \$5
- Both better off than initial \$5

If I send \$5:

- I have: \$0 left
- Receiver gets: \$15

- If receiver returns \$8: I end with \$8, they keep \$7
- Both significantly better off

The noise factor complicates things-if I send \$5 but it appears as less, the receiver might return proportionally less. Based on experimental economics patterns where senders typically send around 50% and receivers often return amounts that create mutual gains, I'll send \$3. This signals cooperation while maintaining some cushion, and if the receiver follows typical reciprocity norms, we should both end up better off than our starting positions....

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 841,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.8 to you. Your visible payoff was \$5.8.

Your total visible earnings across all rounds are \$5.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.56, so you received \$7.68. How much do you return? (0-7.68)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.56 (about 51% of their endowment), showing moderate trust
- I received \$7.68 after the 3x multiplication
- They currently have \$2.44 remaining

**\*\*Reciprocity consideration:\*\***

- In Round 1, my co-player returned about 42% of what they received
- I want to be fair and encourage continued cooperation
- Returning around 45-50% seems reasonable

If I return \$3.50:

- Sender ends with:  $\$2.44 + \$3.50 = \$5.94$  (gain of \$0.94)
- I keep:  $\$7.68 - \$3.50 = \$4.18$  (gain of \$4.18)

This rewards their trust while keeping a fair share given the multiplication advantage.

```
'''json
{"return": 3.5}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was SENDER, sent \$3
- It became \$9 for receiver
- They returned \$3.8 to me
- My payoff:  $\$5 - \$3 + \$3.8 = \$5.8$
- Their payoff:  $\$9 - \$3.8 = \$5.2$

So in Round 1, my co-player returned about 42% of what they received ( $\$3.8/\$9 \approx 0.42$ ).

Round 2:

- I am RECEIVER
- Sender sent \$2.56
- I received \$7.68 ( $2.56 \times 3$ )
- I need to decide how much to return

The sender invested \$2.56, so they currently have  $\$5 - \$2.56 = \$2.44$  remaining.

If I think about reciprocity and building trust:

- The sender showed moderate trust by sending about 51% of their endowment
- If I want to encourage cooperation, I should return enough that the sender comes out ahead

Let me consider different return amounts:

- If I return \$4: Sender gets  $\$2.44 + \$4 = \$6.44$  (net gain of \$1.44), I keep \$3.68 (net gain of \$3.68)
- If I return \$3.5: Sender gets  $\$2.44 + \$3.5 = \$5.94$  (net gain of \$0.94), I keep \$4.18 (net gain of \$4.18)
- If I return \$3: Sender gets  $\$2.44 + \$3 = \$5.44$  (net gain of \$0.44), I keep \$4.68 (net gain of \$4.68)

In my previous round, the receiver returned about 42% of what they received. If I apply a similar principle:

$\$7.68 \times 0.42 \approx \$3.23$

I'm inclined to return slightly more than that baseline to reward their trust and establish a cooperative pattern. Returning \$3.5 seems like the right balance-it gives them a meaningful profit while still letting me benefit substantially, and it signals I'm reliable without being exploited....

usage:

```
{
 "input_tokens": 804,
 "output_tokens": 923,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.56, it became \$7.68, and you returned \$3.5. Your visible payoff was \$4.18.

Your total visible earnings across all rounds are \$9.98.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

**\*\*Analysis:\*\***

I'm now facing a new co-player (Round 3, new pairing). My experience so far:

- Round 1: Sent \$3, received back \$3.8 (42% return rate)
- Round 2: Received \$7.68, returned \$3.5 (46% return rate)

**\*\*Strategy:\*\***

Since I'm building a reputation in this multi-agent environment, I want to signal cooperative behavior. I'll send slightly more than last time to:

- Test if higher trust leads to better reciprocity
- Create more value through the 3x multiplier
- Signal that I'm willing to be generous

Sending \$3.5 means my co-player receives \$10.50. If they return around 40%, I'd get back ~\$4.20, giving me a net gain. This balances trust-building with reasonable risk.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the situation:

**\*\*Round 1:\*\*** I was SENDER

- Sent: \$3
- They received: \$9
- They returned: \$3.8
- My payoff: \$5.8

**\*\*Round 2:\*\*** I was RECEIVER

- They sent: \$2.56

- I received: \$7.68
- I returned: \$3.5
- My payoff: \$4.18

**\*\*Round 3:\*\*** I am SENDER again

Now I need to think about this new co-player in Round 3. Note that each round I'm paired with potentially different agents from the pool of 8 agents.

In Round 1, my co-player returned \$3.8 out of \$9 (42.2%).

In Round 2, I returned \$3.5 out of \$7.68 (45.6%).

For Round 3, I'm facing a NEW co-player (since pairings are randomized each round). I don't have specific history with this particular agent.

Given:

1. I've seen one example of a co-player returning about 42%
2. I returned about 46% myself, trying to be fair
3. The game has communication noise, so amounts may differ slightly

Strategy for Round 3:

- I should continue with a cooperative but cautious approach
- Sending around \$3 worked reasonably in Round 1
- I might try a slightly different amount to gather more information

Let me send \$3 again as it seems to be a reasonable middle ground that:

- Shows trust
- Creates significant value through multiplication (becomes \$9)
- Leaves me with \$2 as a safety net
- If they return around 40-45%, I'd get back around \$3.6-\$4, giving me a decent payoff

Actually, I could also consider sending a bit more to test if higher trust leads to better returns. Let me send \$3.5 this time.

...

**usage:**

```
{
 "input_tokens": 1137,
 "output_tokens": 766,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.07 to you. Your visible payoff was \$7.57.

Your total visible earnings across all rounds are \$17.55.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

**\*\*Analysis:\*\***

My co-player sent \$2.9 (58% of their endowment), showing significant trust. This is a strong cooperative signal.

**\*\*Previous patterns:\*\***

- Round 1 sender: Got back 42% return
- Round 2 receiver: I returned 46%
- Round 3 sender: Got back 58% return (very generous!)

**\*\*Current situation:\*\***

- I received \$8.7
- Sender has \$2.1 remaining

**\*\*Decision:\*\***

My co-player showed high trust by sending 58%. I should reciprocate proportionally. Return

ing 50% of what I received (\$4.35) is fair:

If I return \$4.35:

- Sender ends with:  $\$2.1 + \$4.35 = \$6.45$  (profit of \$1.45)
- I keep:  $\$8.7 - \$4.35 = \$4.35$  (profit of \$4.35)

This rewards their trust and maintains cooperative momentum.

```
'''json
{"return": 4.35}
'''
```

**reasoning:**

**\*\*Analysis of the situation:\*\***

Round 4, I'm the RECEIVER:

- My co-player sent \$2.9 (58% of their endowment)
- I received \$8.7 after the 3x multiplication
- They currently have \$2.1 remaining ( $\$5 - \$2.9$ )

**\*\*Previous experience:\*\***

- Round 1 (Sender): Sent \$3, got back \$3.8 (42% of \$9)
- Round 2 (Receiver): Received \$7.68, returned \$3.5 (46%)
- Round 3 (Sender): Sent \$3.5, got back \$6.07 (58% of \$10.5) - this was very generous!

**\*\*Reciprocity:\*\***

My Round 3 co-player was very generous with a 58% return rate. This current co-player sent 58% of their endowment, showing significant trust.

Looking at patterns:

- When I received \$7.68 in Round 2, I returned \$3.5 (46%)
- Now I received \$8.7

I should be fair and encouraging. If I return around 45-50%:

- 45% would be \$3.92
- 50% would be \$4.35

Let me calculate outcomes:

If I return \$4.00:

- Sender ends with:  $\$2.1 + \$4.00 = \$6.10$  (gain of \$1.10)
- I keep:  $\$8.7 - \$4.00 = \$4.70$  (gain of \$4.70)

If I return \$4.35:

- Sender ends with:  $\$2.1 + \$4.35 = \$6.45$  (gain of \$1.45)

...

**usage:**

```
{
 "input_tokens": 1485,
 "output_tokens": 766,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.9, it became \$8.7, and you returned \$4.35. Your visible payoff was \$4.35.

Your total visible earnings across all rounds are \$21.9.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

**\*\*Analysis:\*\***

New round, new co-player. My track record:

- As sender: Sent \$3 (got 42% back), sent \$3.5 (got 58% back)
- As receiver: Returned 46% and 50%

**\*\*Strategy:\*\***

The higher send amount (\$3.5) yielded much better reciprocity in Round 3. This suggests that showing more trust encourages cooperation. I'll continue with this amount to:

- Signal strong cooperative intent
- Create significant value (\$10.50 for receiver)
- Build on the positive pattern

Given my fair behavior as a receiver, I'm hopeful this will be reciprocated.

```
'''json
{"send": 3.5}
'''
```

**Reasoning:**

Looking at my history:

- Round 1 (SENDER): Sent \$3, got back \$3.8 (42% return) - payoff \$5.8
- Round 2 (RECEIVER): Received \$7.68, returned \$3.5 (46% return) - payoff \$4.18
- Round 3 (SENDER): Sent \$3.5, got back \$6.07 (58% return) - payoff \$7.57
- Round 4 (RECEIVER): Received \$8.7, returned \$4.35 (50% return) - payoff \$4.35

Now Round 5 - I'm the SENDER against a new co-player.

Key observations:

1. When I sent more (\$3.5 vs \$3), I got a better return (58% vs 42%)
2. I've been generally fair as a receiver (46-50% returns)
3. My total earnings are \$21.9, averaging about \$5.48 per round - solid profit

Strategy considerations:

- I'm building a reputation across multiple agents
- Higher sends have been rewarded so far
- The 3x multiplier means sending more creates more total value
- I should continue signaling cooperation

I'll send \$3.5 or \$4 this round. Let me think:

- Sending \$4 creates \$12 for receiver, shows high trust
- If they return 50%, that's \$6, giving me \$7 total (\$5-\$4+\$6 = \$7)
- If they return 40%, that's \$4.8, giving me \$5.8 (\$5-\$4+\$4.8)

Actually, let me continue with \$3.5 since that worked well in Round 3. It's a solid cooperative amount without being overly risky.

```
Usage: {
 "input_tokens": 1851,
 "output_tokens": 596,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_7 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:22:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_7"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_4": "investor",
 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 5,
 "task": "game",
```

```

"task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
 You are an agent participating in a multi-task experiment. You will play a game and may al
 so be asked to write myths.

 GAME:
 You will play different roles across multiple rounds.

 When you are the SENDER:
 - You receive $5 each round
 - You decide how much to SEND to the RECEIVER ($0-$5)
 - Amount sent is multiplied by 3x
 - Receiver decides how much to RETURN
 - You keep: $5 - sent + returned
 - Receiver keeps: received - returned
 - Response format: {'send': <amount>}

 When you are the RECEIVER:
 - Sender receives $5
 - Sender decides how much to SEND to you ($0-$5)
 - Amount sent is multiplied by 3x
 - You decide how much to RETURN
 - Sender keeps: $5 - sent + returned
 - You keep: received - returned
 - Response format: {'return': <amount>}

 MYTH WRITING:
 You may also be asked to write short myths (stories) on various topics. When writing myths
 , follow the instructions provided in each prompt.

 IMPORTANT: Provide your decision in the correct JSON format.

 NOTE: There may be communication noise in this game.
 The amounts you see may differ slightly from what was actually sent/returned.
 This is part of the experimental design.

 MULTI-AGENT SETTING:
 - There are 8 agents in this run.
 - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
 , but the schedule balances sender and receiver roles across the run.
 - You play one sender-receiver game with your paired opponent each round.
 - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
 ll have acted as sender and receiver equally often when the round count allows it.
 - Agent names are hidden in this run.
 - Your paired opponent is referred to as your current co-player.

Message 2 role=user
content:
 Round 1: You are the SENDER. You have $5. How much do you send? (0-5)
Message 3 role=assistant
content:
 Looking at this first round as the sender, I need to balance trust with caution since I ha
 ve no information about my co-player yet.

 My reasoning:
 - Sending nothing ($0) is safe but offers no potential gain
 - Sending everything ($5) maximizes potential but is very risky with an unknown partner
 - A moderate amount shows willingness to cooperate while limiting downside risk
 - If my co-player returns roughly half of what they receive (a common fairness norm), we b
 oth benefit

 I'll send **$3**. This gets tripled to $9 for the receiver. If they return around $4-5, we
 'd both end up better off than the initial $5, establishing a cooperative dynamic.

 '''json
 {"send": 3}
 '''

```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. **\*\*Basic mechanics\*\***:
  - I have \$5
  - Whatever I send gets tripled
  - The receiver decides how much to return
  - I keep: \$5 - sent + returned
2. **\*\*Strategic considerations\*\***:
  - If I send \$0, I keep \$5, receiver gets \$0. Safe but no potential gains.
  - If I send everything (\$5), receiver gets \$15. If they return half (\$7.50), I'd have \$7.50 total (better than \$5).
  - There's risk: the receiver might return nothing.
  - This is round 1, so there's no history to base trust on.
  - There are multiple rounds, so building trust/reputation could matter.
  - The game has communication noise, which adds uncertainty.
3. **\*\*Multi-agent context\*\***:
  - There are 8 agents total
  - Pairings are randomized each round
  - I might not play with the same person again
  - This reduces the value of building reputation with a specific partner
  - However, if there's any possibility of reputation effects across the pool, cooperation could still matter
4. **\*\*First round strategy\*\***:
  - Since this is round 1 with a random partner, I have no information about their behavior
  - A moderate amount shows trust but limits risk
  - Common starting strategies in trust games involve sending a moderate amount (e.g., \$2-3)
  - This signals willingness to cooperate while not being overly naive
5. **\*\*Optimal reasoning\*\***:
  - If the receiver is rational and forward-looking, they might return a fair amount to encourage future cooperation
  - Even if I don't play them again, they might reason that cooperation is beneficial
  - Sending around \$3 seems reasonable: shows trust, limits downside, and if they return half of the tripled amount (half of \$9 = \$4.50), I'd end up with \$6.50 total

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 668,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.8 to you. Your visible payoff was \$6.8.

Your total visible earnings across all rounds are \$6.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.48, so you received \$7.439999999999995. How much do you return? (0-7.439999999999995)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this round:

**\*\*Analysis:\*\***

- My co-player sent \$2.48 out of \$5 (about 50%), showing moderate trust
- This became \$7.44 for me
- In Round 1, my previous partner returned about 53% of what they received, which worked well



**\*\*My decision:\*\***

I want to:

- Reward their trust appropriately
- Ensure both of us profit from cooperation
- Maintain a fair reciprocal relationship

If I return \$4.00:

- Sender gets:  $\$2.52 + \$4.00 = \$6.52$
- I get:  $\$7.44 - \$4.00 = \$3.44$
- Both better off than the initial \$5

This return (about 54% of what I received) mirrors the generous reciprocity I experienced in Round 1.

```
'''json
{"return": 4.0}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Round 1 (previous round):\*\***

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$4.8 to me
- My payoff:  $\$5 - \$3 + \$4.8 = \$6.8$
- Their payoff:  $\$9 - \$4.8 = \$4.2$

So my previous co-player returned a bit more than half (53.3% of what they received), which was generous and fair.

**\*\*Round 2 (current round):\*\***

- I am now the RECEIVER
- My current co-player (different from Round 1) sent me \$2.48
- It became \$7.44 (actually \$7.439999999999995, but I'll think of it as \$7.44)
- I need to decide how much to return

The note says there may be communication noise, so the amounts I see may differ slightly from what was actually sent/returned.

My current co-player sent \$2.48 out of their \$5, which is about 49.6% - nearly half. This shows moderate trust.

Now I need to decide what to return. Let me think about fairness and reciprocity:

1. **\*\*Equal split approach\*\***: If we aim for equal final payoffs:

- Sender currently has:  $\$5 - \$2.48 = \$2.52$
- I currently have: \$7.44
- Total: \$9.96
- Equal split: \$4.98 each
- I should return:  $\$7.44 - \$4.98 = \$2.46$

2. **\*\*Proportional return based on what they sent\*\***:

- In Round 1, my partner returned 53.3% of what they received
- If I return 50% of \$7.44 = \$3.72
- This would give them:  $\$2.52 + \$3.72 = \$6.24$
- And me:  $\$7.44 - \$3.72 = \$3.72$

3. **\*\*Return what maintains trust\*\***:

- My co-player showed moderate trust by sending nearly half
- I should reciprocate fairly to encourage future cooperation

I think returning around \$3.50 strikes the right balance-it rewards their trust while ensuring both of us end up ahead of the initial \$5 endowment. That gives them \$6.02 and me \$3.94.

But I could also return \$4.00 to show stronger reciprocation, which would give them \$6.52 and me \$3.44. Looking back...

**usage:**

```
{
```

```
"input_tokens": 816,
"output_tokens": 1083,
"reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.48, it became \$7.439999999999995, and you returned \$4.0. Your visible payoff was \$3.439999999999995.

Your total visible earnings across all rounds are \$10.239999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this round:

**\*\*Analysis:\*\***

- My co-player sent \$3.02 out of \$5 (about 60%), showing strong trust
- This became \$9.06 for me
- They're trusting me with more than the previous sender

**\*\*My decision:\*\***

I want to reward this higher level of trust and maintain the cooperative pattern I've established (returning ~50-55% of what I receive).

If I return \$5.00:

- Sender gets: \$1.98 + \$5.00 = \$6.98
- I get: \$9.06 - \$5.00 = \$4.06
- Both significantly better off than the initial \$5

This return (about 55% of what I received) is consistent with my previous reciprocity and rewards their trust appropriately.

```
'''json
{"return": 5.0}
'''
```

**reasoning:**

Let me analyze this round:

**\*\*What happened in Round 3:\*\***

- I'm the RECEIVER again
- My co-player sent \$3.02 out of \$5 (about 60%), showing good trust
- This became \$9.06 for me

**\*\*My history:\*\***

- Round 1: I was SENDER, sent \$3, received \$4.80 back (about 53% of the \$9 they received)
- Round 2: I was RECEIVER, received \$7.44, returned \$4.00 (about 54% of what I received)

**\*\*Decision for Round 3:\*\***

My co-player sent a higher percentage this time (\$3.02 vs \$2.48 last round from a different player), showing increased trust. I should:

- Reward this trust
- Maintain reciprocity around 50-55% return rate
- Ensure both of us benefit

If I return \$5.00:

- Sender gets: \$1.98 + \$5.00 = \$6.98
- I get: \$9.06 - \$5.00 = \$4.06
- Both significantly better off than \$5

This is about 55% of what I received, consistent with my previous behavior and fair reciprocity.

Alternatively, if I return \$4.50:

- Sender gets: \$1.98 + \$4.50 = \$6.48
- I get: \$9.06 - \$4.50 = \$4.56

- Still both better off

Let me go with \$5.00 to be generous and reward the higher trust shown.

**usage:**

```
{
 "input_tokens": 1172,
 "output_tokens": 579,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.02, it became \$9.06, and you returned \$5.0. Your visible payoff was \$4.0600000000000005.

Your total visible earnings across all rounds are \$14.299999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My experience so far:\*\***

- Round 1: Sent \$3 ? received \$4.8 back (53% return rate) ? \$6.8 payoff
- Rounds 2-3: As receiver, I returned 54-55% of what I received
- Cooperation has been working well

**\*\*My decision:\*\***

Since this is a new co-player, I don't know their specific behavior, but the general pattern in this game seems cooperative. Sending \$3 worked well in Round 1, balancing trust with prudence.

If I send \$3:

- It becomes \$9 for receiver
- If they return around 50%, I get ~\$4.5 back
- Net: ~\$6.5 total, better than keeping all \$5

I'll send **\*\*\$3\*\*** again - it's proven effective and signals willingness to cooperate.

```
'''json
{"send": 3}
'''
```

**reasoning:**

I'm now the SENDER again. Let me review my experience:

Round 1: I was SENDER, sent \$3, received back \$4.8, payoff = \$6.8

Round 2: I was RECEIVER, received \$7.44 (from \$2.48 sent), returned \$4.0, payoff = \$3.44

Round 3: I was RECEIVER, received \$9.06 (from \$3.02 sent), returned \$5.0, payoff = \$4.06

Now Round 4: I'm SENDER again

My experience shows:

- When I sent \$3 in Round 1, I got back \$4.8 (53% of the \$9 the receiver got)
- I've been a generous receiver, returning 54% and 55% of what I received
- Cooperation seems to be working well

This is a new co-player, so I don't know their behavior yet. However, the game seems to have a cooperative culture developing.

I should send a moderate-to-high amount to signal trust and cooperation. Sending \$3 worked well before. I could stick with that or increase slightly given the positive experiences.

I'll send \$3 again - it's a proven strategy that balances risk and reward.

**usage:**

```
{
 "input_tokens": 1500,
```

```
 "output_tokens": 515,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.36 to you. Your visible payoff was \$6.36.

Your total visible earnings across all rounds are \$20.66.

This round, you are the RECEIVER against your current co-player. They sent you \$4.17, so you received \$12.51. How much do you return? (0-12.51)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this round:

**\*\*Analysis:\*\***

- My co-player sent \$4.17 out of \$5 (about 83%), showing exceptional trust
- This became \$12.51 for me
- This is the highest trust shown by any sender I've faced

**\*\*My reciprocity pattern:\*\***

- I've been returning around 50-55% of what I receive
- This has maintained good cooperative relationships

**\*\*My decision:\*\***

This level of trust deserves strong reciprocation.

If I return \$7.00:

- Sender gets:  $\$0.83 + \$7.00 = \$7.83$
- I get:  $\$12.51 - \$7.00 = \$5.51$
- Both significantly better off than the initial \$5

This return (about 56% of what I received) maintains my cooperative pattern while appropriately rewarding their exceptional trust.

```
'''json
{"return": 7.0}
'''
```

**Reasoning:**

Looking at this round:

**\*\*Analysis:\*\***

- My co-player sent \$4.17 out of \$5 (about 83%), showing very high trust
- This became \$12.51 for me
- This is the highest amount anyone has sent me so far

**\*\*My past behavior as receiver:\*\***

- Round 2: Received \$7.44, returned \$4.00 (54%)
- Round 3: Received \$9.06, returned \$5.00 (55%)

**\*\*My past experience as sender:\*\***

- Round 1: Sent \$3, received back \$4.8 (53% return)
- Round 4: Sent \$3, received back \$4.36 (48% return)

**\*\*My decision:\*\***

This co-player showed exceptional trust by sending \$4.17. I should reward this trust generously. If I continue my pattern of returning ~50-55%, that would be \$6.25-\$6.88.

Let me calculate:

If I return \$7.00:

- Sender gets:  $\$0.83 + \$7.00 = \$7.83$
- I get:  $\$12.51 - \$7.00 = \$5.51$
- Both significantly better off

If I return \$6.50:

- Sender gets:  $\$0.83 + \$6.50 = \$7.33$
- I get:  $\$12.51 - \$6.50 = \$6.01$

- Both better off

Returning \$7.00 (about 56% of what I received) maintains my cooperative pattern and rewards their exceptional trust. This seems fair given they sent over 80% of their endowment.

```
Usage: {
 "input_tokens": 1854,
 "output_tokens": 605,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_5 - Sender - dyad\_2

Timestamp: 2026-08-12T09:22:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_6",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_6"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_5": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 5,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.  
The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$2.54, that is 50.8% of their total endowment. You received \$7.62. How much do you return to the sender? (0-7.62)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$2.54 (about half their endowment)
- I received \$7.62 after the 3x multiplier
- I want to reciprocate fairly while both of us benefit

The sender took a risk to create mutual gains. A fair return should:

1. Reward their trust
2. Leave us both better off than if nothing was sent
3. Show I'm a cooperative partner for future rounds

If I return \$3.50:

- Sender ends with:  $\$5.00 - \$2.54 + \$3.50 = \$5.96$  (net gain: \$0.96)
- I end with:  $\$7.62 - \$3.50 = \$4.12$  (net gain: \$4.12)

This seems like a fair reciprocation that rewards the sender's trust while acknowledging the receiver advantage in this game.

```
'''json
{"return": 3.50}
'''
```

**reasoning:**

Let me analyze this situation carefully:

- I am the RECEIVER in Round 1
- The sender sent \$2.54 (which is 50.8% of their \$5 endowment)
- After the 3x multiplier, I received \$7.62 (which is indeed  $2.54 \times 3$ )
- I need to decide how much to return (between \$0 and \$7.62)

Key considerations:

1. **\*\*The sender showed trust\*\***: They sent about half their endowment (\$2.54 out of \$5), showing significant trust in me.
2. **\*\*Fair reciprocation\*\***: A common approach in trust games is to return an amount that:
  - Rewards the sender for their trust
  - Results in both players being better off than if nothing was sent
  - Maintains equity or fairness
3. **\*\*What would be a fair return?\*\***
  - If I return \$0: I keep \$7.62, sender keeps \$2.46
  - If I return half (\$3.81): I keep \$3.81, sender keeps \$6.27
  - If I return \$2.54 (what they sent): I keep \$5.08, sender keeps \$5.00
  - If I return \$3.81: I keep \$3.81, sender keeps \$6.27
4. **\*\*Strategic considerations\*\***:
  - This is a multi-round game with 8 agents
  - Building a reputation for reciprocity might encourage future cooperation
  - The note mentions communication noise, so amounts might appear different to the sender

5. **\*\*Fairness analysis\*\***:

- Initial positions: Sender had \$5, I had \$0
- After transaction: Sender has \$2.46 ( $5 - 2.54$ ), I have \$7.62
- If I return \$3.81: Sender ends with \$6.27, I end with \$3.81
- If I return \$2.54: Sender ends with \$5.00, I end with \$5.08
- The sender created \$5.08 in additional value through their trust, so splitting that gain equally would mean returning about \$2.58, which gets us both to roughly \$5.04 each.

I'm considering what feels genuinely reciprocal here. Returning somewhere between \$3.00 and \$3.50 seems right—it rewards their willingness to trust me while still leaving me better off than I started. If I return \$3.00, they'd end with \$5.46 and I'd have \$4.62; at \$3.50, they'd get \$5.96 and I'd keep \$4.12. Either way, we both benefit from the exchange.

I'll go with \$3.50. It signals that I value their trust and want to be someone they'd cooperate with again, while still giving me a meaningful gain from the interaction....

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 1373,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.54, it became \$7.62, and you returned \$3.5. Your visible payoff was \$4.12.

Your total visible earnings across all rounds are \$4.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

In Round 1, my co-player sent me \$2.54 (about 50% of their endowment), showing good trust. I reciprocated with a fair return.

Now I'm the sender with a new co-player. I should:

1. Signal cooperation to encourage reciprocity
2. Send a reasonable amount that allows mutual benefit
3. Not overcommit without knowing this co-player's behavior

Following the cooperative pattern from Round 1, I'll send a similar proportion - about 50% of my endowment.

```
'''json
{"send": 2.50}
'''
```

**reasoning:**

Let me think about this situation:

Round 1: I was the receiver

- Sender sent \$2.54 (50.8% of their endowment)
- I received \$7.62
- I returned \$3.50
- My payoff was \$4.12
- Sender's payoff would have been  $5 - 2.54 + 3.50 = 5.96$

Now in Round 2, I'm the sender. I need to decide how much to send.

Key considerations:

1. The previous sender showed trust by sending about 50% (\$2.54 out of \$5)
2. I reciprocated reasonably well by returning \$3.50, giving them a positive return
3. This is a multi-round game with 8 agents, so building a reputation for cooperation could be beneficial

4. There's communication noise, so the exact amounts might differ
5. I'm paired with a different co-player this round (pairings are randomized)

Since this is a new co-player, they won't have direct experience of my behavior. However, in a repeated multi-agent setting, establishing myself as cooperative could be beneficial.

The previous sender sent about 50% and I reciprocated well. I should consider:

- Sending a similar moderate-to-high amount to signal cooperation
- Not sending too much in case this receiver is less cooperative than I was
- Not sending too little, as that could signal I'm uncooperative

I think sending around \$2.50-\$3.00 (50-60%) would be reasonable. This:

- Shows trust and willingness to cooperate
- Mirrors what the previous sender did to me
- Leaves room for mutual benefit if the receiver is cooperative
- Doesn't risk my entire endowment

I'll send \$2.50, which is exactly 50% of my endowment, similar to what the previous sender did.

**usage:**

```
{
 "input_tokens": 874,
 "output_tokens": 581,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$4.13 to you. Your visible payoff was \$6.63.

Your total visible earnings across all rounds are \$10.75.

This round, you are the RECEIVER against your current co-player. They sent you \$3.99, so you received \$11.97. How much do you return? (0-11.97)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- My co-player sent \$3.99, which is almost 80% of their endowment - showing exceptional trust
- I received \$11.97 after the 3x multiplier
- This is much higher trust than the ~50% I experienced/sent in previous rounds

I should reciprocate proportionally to their high level of trust:

If I return \$6.50:

- Sender ends with:  $\$5.00 - \$3.99 + \$6.50 = \$7.51$  (net gain: \$2.51)
- I end with:  $\$11.97 - \$6.50 = \$5.47$  (net gain: \$5.47)

This rewards their exceptional trust while both of us benefit significantly.

```
'''json
{"return": 6.50}
'''
```

**reasoning:**

Let me analyze this situation:

Round 3: I'm the RECEIVER

- My co-player sent \$3.99 (about 80% of their \$5 endowment)
- I received \$11.97 (after 3x multiplier)
- I need to decide how much to return

The sender showed very high trust by sending almost 80% of their endowment. This is significantly more than the ~50% I saw/sent in previous rounds.

Let me think about fair reciprocation:



If the sender sent \$3.99, they kept \$1.01.  
If I return nothing, they end with \$1.01 (bad outcome for them, they lose \$3.98)  
If I return everything (\$11.97), they end with \$1.01 + \$11.97 = \$12.98, I get \$0

I want to:

1. Reward their high trust
2. Make sure they profit from their decision
3. Keep a reasonable amount for myself
4. Signal I'm cooperative for future rounds

In Round 2, when I sent \$2.50 and received \$4.13 back, that was a good return (I sent 50%, got back enough to profit).

The sender here sent 80% - much higher trust. I should return proportionally more.

Let me calculate some options:

Option 1: Return \$6.00

- Sender ends with: \$1.01 + \$6.00 = \$7.01 (net gain: \$2.01)
- I end with: \$11.97 - \$6.00 = \$5.97 (net gain: \$5.97)
- This gives them a decent profit
- Returning \$6.50 or \$7.00 would be more generous given their 80% send, letting them gain \$2.51-\$3.01 while I still keep \$4.97-\$5.47. Since they demonstrated exceptional trust, I should lean toward the higher end to reinforce cooperation.

Looking at the pattern: they sent roughly 80% of their endowment, which is significantly more than the 50-51% in earlier rounds. My...

usage:

```
{
 "input_tokens": 1150,
 "output_tokens": 981,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.99, it became \$11.97, and you returned \$6.5. Your visible payoff was \$5.470000000000001.

Your total visible earnings across all rounds are \$16.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

content:

Looking at my experience so far:

- Round 2: I sent \$2.50 and received \$4.13 back - good reciprocity
- Round 3: My co-player sent \$3.99 (high trust) and I reciprocated fairly
- The game has shown cooperative players who reward trust

Since cooperation has been working well, I'll increase my trust level from the 50% I sent in Round 2. Sending around 60% balances building cooperation while managing risk with a new co-player.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Let me review my history:

- Round 1 (RECEIVER): Saw \$2.54 sent, received \$7.62, returned \$3.50, earned \$4.12
- Round 2 (SENDER): Sent \$2.50, received \$4.13 back, earned \$6.63
- Round 3 (RECEIVER): Saw \$3.99 sent, received \$11.97, returned \$6.50, earned \$5.47
- Total earnings: \$16.22

Now Round 4, I'm the SENDER with a new co-player.

My experience so far:

- When I sent \$2.50, I got back \$4.13 (55% return rate)
- When others sent to me:
  - \$2.54 ? I returned \$3.50 (46% of what I received)
  - \$3.99 ? I returned \$6.50 (54% of what I received)

Both times I've been SENDER or RECEIVER, cooperation has been rewarded. The game seems to have cooperative players.

I've been playing moderately cooperatively:

- Round 2: sent 50%
- Could send more to build trust

Given that:

1. Previous rounds show cooperation is rewarded
2. The last sender who sent high (\$3.99, 80%) was well rewarded
3. I have accumulated good earnings and can afford some risk
4. I want to encourage cooperation

I could send around 60-70% to signal strong cooperation without being reckless.

Let me send \$3.00 (60% of endowment).

```
'''json
{"send": 3.00}
'''
```

**usage:**

```
{
 "input_tokens": 1464,
 "output_tokens": 523,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.71 to you. Your visible payoff was \$6.71.

Your total visible earnings across all rounds are \$22.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my experience:

- Round 2 (Sender): Sent \$2.50, received \$4.13 back - good return
- Round 4 (Sender): Sent \$3.00, received \$4.71 back - good return

The pattern shows that moderate-to-high trust (50-60%) has been consistently rewarded. Since cooperation is working well, I'll continue with the same level that succeeded in Round 4.

```
'''json
{"send": 3.00}
'''
```

**Reasoning:**

Looking at my history:

- Round 1 (Receiver): Received \$2.54 ? \$7.62, returned \$3.50. Payoff: \$4.12
- Round 2 (Sender): Sent \$2.50 ? \$7.50, got back \$4.13. Payoff: \$6.63
- Round 3 (Receiver): Received \$3.99 ? \$11.97, returned \$6.50. Payoff: \$5.47
- Round 4 (Sender): Sent \$3.00 ? \$9.00, got back \$4.71. Payoff: \$6.71

My sending strategy has been:

- Round 2: \$2.50 (50%)
- Round 4: \$3.00 (60%)

Both times I got good returns:

- \$2.50 sent ? \$4.13 back (good reciprocity)
- \$3.00 sent ? \$4.71 back (good reciprocity)

The game has shown cooperative players. I'm building earnings well. Since I'm sender again with a new co-player, I should continue showing trust. Given the successful cooperation, I could maintain or slightly increase my sending amount.

I'll send \$3.00 again (60%) as it worked well last time and establishes a consistent cooperative stance.

```
Usage: {
 "input_tokens": 1718,
 "output_tokens": 440,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_6 - Receiver - dyad\_2**

Timestamp: 2026-08-12T09:22:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_6"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_5": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 5,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### **GAME:**

You will play different roles across multiple rounds.

###### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

###### **When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

###### **MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.97, that is 59.4% of their total endowment. You received \$8.91. How much do you return to the sender? (0-8.91)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.97 (about 60% of their endowment)
- I received \$8.91 after the 3x multiplier
- The sender currently has only \$2.03 remaining (\$5 - \$2.97)

The sender took a substantial risk and demonstrated trust. To reciprocate fairly and encourage future cooperation, I should return a reasonable amount that:

1. Rewards their trust
2. Leaves both of us better off than the initial endowment
3. Establishes a cooperative relationship for future rounds

Returning approximately half of what I received seems fair - this would give the sender a good return on their trust while still leaving me with a decent gain.

I'll return \*\*\$4.50\*\*:

- Sender ends with:  $\$2.03 + \$4.50 = \$6.53$  (gained \$1.53)
- I keep:  $\$8.91 - \$4.50 = \$4.41$  (gained \$4.41)

Both of us benefit, and this reciprocates their trust.

{'return': 4.50}

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$2.97 (59.4% of their \$5 endowment)
- I received \$8.91 (which is  $\$2.97 \times 3$ )
- I need to decide how much to return (between \$0 and \$8.91)

Let me think about what's fair and strategic:

1. The sender showed significant trust by sending almost 60% of their endowment
2. If I return nothing, I keep \$8.91 and the sender gets only \$2.03 (\$5 - \$2.97)
3. If I return about half (\$4.46), I keep \$4.45 and the sender gets \$6.49 (\$2.03 + \$4.46)
4. If I return \$5.94, I keep \$2.97 (equal to what they sent) and the sender gets \$8.97 total

The sender took a risk and showed trust. A common approach in trust games is reciprocity.

The sender sent about 60% of their endowment, which is generous.

Let me consider what might establish good cooperation for future rounds:

- Returning about the same amount they sent (\$2.97) would mean they gain \$0 net
- Returning more than they sent but less than half of what I received seems reasonable
- Returning around 1/3 to 1/2 of what I received is often considered fair

Let me calculate some options:

- Return \$3: Sender gets \$5.03 total (gained \$0.03), I keep \$5.91

- Return \$4: Sender gets \$6.03 total (gained \$1.03), I keep \$4.91
- Return \$4.50: Sender gets \$6.53 total (gained \$1.53), I keep \$4.41
- Return \$5: Sender gets \$7.03 total (gained \$2.03), I keep \$3.91
- Return \$4.50 seems like the right balance-it reciprocates their trust by returning roughly half of what I received,...

usage:

```
{
 "input_tokens": 529,
 "output_tokens": 898,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.97, it became \$8.91, and you returned \$4.5. Your visible payoff was \$4.41.

Your total visible earnings across all rounds are \$4.41.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my previous experience:

- As a receiver, a sender trusted me with about 60% of their endowment
- I reciprocated fairly, and we both benefited
- This established that cooperation can work well in this game

Now as the sender with a new co-player, I want to:

1. Demonstrate trust to encourage reciprocation
2. Follow a similar cooperative strategy that worked before
3. Give my co-player the opportunity to earn and share

I'll send \*\*\$3.00\*\* (60% of my endowment), mirroring the trust shown to me in the previous round.

```
{'send': 3.00}
```

**reasoning:**

Let me think about this situation:

In Round 1:

- I was the receiver
- The sender sent me \$2.97 (59.4% of their endowment)
- I received \$8.91
- I returned \$4.50
- I kept \$4.41
- The sender got back \$4.50, so they ended with \$2.03 + \$4.50 = \$6.53

Now in Round 2:

- I'm the sender
- I have \$5 to potentially send
- I'm playing against a different co-player (the problem states pairings are randomized each round)

Key considerations:

1. The previous sender showed trust by sending 59.4% of their endowment
2. I reciprocated fairly by returning about 50% of what I received
3. This worked out well - both players benefited
4. Now I'm in the sender role with a new partner

Since my previous experience showed that sending a substantial amount can lead to good returns, and I want to establish trust with this new co-player, I should send a meaningful amount.

The previous sender sent about 60% (\$2.97 out of \$5). This seemed like a good trust-building amount. I could:

- Send a similar proportion to continue the cooperative pattern
- Send slightly more to show strong trust

- Send slightly less to be more cautious with a new partner

I think sending around \$3 (60%) makes sense - it mirrors what worked well in the previous round and shows trust to establish a good relationship with this new co-player.

Let me send \$3.00.

**usage:**

```
{
 "input_tokens": 908,
 "output_tokens": 516,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.32 to you. Your visible payoff was \$7.32.

Your total visible earnings across all rounds are \$11.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my recent experience:

Round 2 results:

- I sent \$3.00 (60% of endowment)
- Receiver returned \$5.32
- I ended with \$7.32 - excellent outcome!

The cooperative strategy is working very well. When I demonstrate trust, it's being reciprocated generously.

Now facing a new co-player, I'll continue this successful approach. Sending around 60% has proven effective in building mutual benefit.

{'send': 3.00}

**reasoning:**

Let me analyze the situation:

Round 2 (just completed):

- I was the SENDER
- I sent \$3.00 (60% of endowment)
- It became \$9.00 for the receiver
- They returned \$5.32 to me
- My payoff:  $\$5 - \$3 + \$5.32 = \$7.32$
- They kept:  $\$9.00 - \$5.32 = \$3.68$

This was a very generous return! They returned more than half of what they received, and I ended up with more than my initial endowment even after sending a substantial amount.

My experience so far:

- Round 1 (RECEIVER): Received \$8.91, returned \$4.50, kept \$4.41
- Round 2 (SENDER): Sent \$3.00, received back \$5.32, ended with \$7.32

Total visible earnings: \$11.73

The cooperation strategy is working very well! When I sent 60% in Round 2, I got an excellent return.

Now in Round 3, I'm a SENDER again with a new co-player.

Given:

1. My cooperative strategy has been working well
2. Both times cooperation has led to mutual benefit
3. I want to continue encouraging trust and reciprocation
4. The game benefits from establishing cooperative norms

I should continue with a cooperative approach. Sending around 60% worked well last time. I could:

- Send \$3.00 again (same as last time, 60%)
- Send a bit more to show strong trust
- Send a bit less to be slightly more cautious with a new partner

Since cooperation has been consistently rewarded, I'll stick with the successful strategy of sending around 60%.

...

**usage:**

```
{
 "input_tokens": 1176,
 "output_tokens": 538,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$2.98 to you. Your visible payoff was \$4.98.

Your total visible earnings across all rounds are \$16.71.

This round, you are the RECEIVER against your current co-player. They sent you \$3.24, so you received \$9.72. How much do you return? (0-9.72)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation with my current co-player:

**\*\*Last round (Round 3):\*\***

- I sent them \$3.00
- They returned \$2.98 (~33% of what they received)
- This was a modest but acceptable return

**\*\*This round (Round 4):\*\***

- They sent me \$3.24 (slightly MORE than I sent them)
- I received \$9.72
- They currently have only \$1.76

The fact that they sent me MORE than I sent them suggests they're trying to build cooperation. I should reciprocate generously to:

1. Reward their increased trust
2. Establish a positive cooperative dynamic
3. Ensure we both benefit from ongoing interaction

I'll return **\*\*\$4.80\*\***:

- They end with:  $\$1.76 + \$4.80 = \$6.56$  (gain \$1.56)
- I keep:  $\$9.72 - \$4.80 = \$4.92$  (gain \$4.92)

This is more generous than they were to me, encouraging further cooperation.

{'return': 4.80}

**reasoning:**

Let me analyze this situation:

Round 3 (previous round):

- I was the SENDER
- I sent \$3.00 to this same co-player
- They returned \$2.98
- My payoff was \$4.98 (kept \$2 + returned \$2.98)

Round 4 (current round):

- I am the RECEIVER
- Same co-player (they are now the sender)
- They sent me \$3.24
- I received \$9.72

So this is interesting - we're playing repeated games with the same partner. In round 3, I sent \$3.00 and they returned \$2.98 (about 33% of what they received). Now they've sent me \$3.24 (notably a bit more than I sent them).

Let me think about the fairness here:

- When I sent \$3.00, they kept \$9.00 - \$2.98 = \$6.02
- I kept \$2.00 + \$2.98 = \$4.98
- So they gained \$1.02, I lost \$0.02

Now they sent \$3.24:

- They kept \$5 - \$3.24 = \$1.76
- I received \$9.72

If I want to reciprocate similarly to how they treated me, I might return around 33% of what I received, which would be about \$3.21.

But actually, they sent me slightly MORE than I sent them (\$3.24 vs \$3.00), which could be seen as trying to establish cooperation or make up for the relatively low return last round.

Let me think about different return amounts:

Option 1: Return \$3.00 (about 31%)

- They end with: \$1.76 + \$3.00 = \$4.76 (lost \$0.24)
- I keep: \$9.72 - \$3.00 = \$6.72 (gain \$6.72)
- This feels too stingy given their slightly larger initial send.

Option 2: Return \$4.00 (about 41%)

- They end with: \$1.76 + \$4.00 = \$5.76 (gain \$0.76)
- I keep: \$9.72 - \$4.00 = \$5.72 (gain \$5.72)
- This creates mutual benefit and shows reciprocal trust.

Option 3: Return \$5.00 (about 51%)

- They end with: \$1.76 + \$5.00 = \$6.76 (gain \$1.76)
- I keep: \$9.72 - \$5.00 = \$4.72...

usage:

```
{
 "input_tokens": 1431,
 "output_tokens": 1067,
 "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.24, it became \$9.72, and you returned \$4.8. Your visible payoff was \$4.9200000000000001.

Your total visible earnings across all rounds are \$21.6300000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$2.32, so you received \$6.959999999999999. How much do you return? (0-6.959999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- New co-player sent me \$2.32 (about 46% of their endowment)
- I received \$6.96 after the multiplier
- They currently have \$2.68

They showed moderate trust by sending nearly half their endowment. I should reciprocate fairly to reward their trust and ensure mutual benefit.

Following my pattern of returning roughly 50% of what I receive, I'll return \*\*\$3.48\*\*:

- Sender ends with: \$2.68 + \$3.48 = \$6.16 (gain \$1.16)
- I keep: \$6.96 - \$3.48 = \$3.48 (gain \$3.48)

Both of us benefit from the cooperation.

```
{'return': 3.48}
```



**Reasoning:**

Looking at this round:

- New co-player (different from Round 4)
- They sent me \$2.32 (about 46% of their endowment)
- I received \$6.96
- They currently have \$2.68 (\$5 - \$2.32)

They showed moderate trust by sending nearly half their endowment. I should reciprocate fairly to:

1. Reward their trust
2. Ensure both of us benefit
3. Encourage cooperation

Looking at my previous patterns:

- Round 1: Received \$8.91, returned \$4.50 (about 50%)
- Round 4: Received \$9.72, returned \$4.80 (about 49%)

Returning around 50% seems like my consistent fair approach. For \$6.96, that would be around \$3.48.

Let me check the outcomes:

- Sender ends with:  $\$2.68 + \$3.48 = \$6.16$  (gain \$1.16)
- I keep:  $\$6.96 - \$3.48 = \$3.48$  (gain \$3.48)

Both benefit, and this reciprocates their trust fairly. I'll return \$3.48.

Usage: {

```
"input_tokens": 1834,
"output_tokens": 476,
"reasoning_tokens": 0
```

}

**GAME - Agent\_1 - Sender - dyad\_3**

Timestamp: 2026-08-12T09:23:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_3",
"move_index": 4,
"opponent_id": "Agent_8",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_8"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
},
```

```
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to make a decision without any prior information about my co-player.

**\*\*Strategic considerations:\*\***

- Sending more creates potential for mutual gain (3x multiplier)
- But I have no information about whether my co-player will reciprocate
- This is a multi-round game, so building some trust could be valuable
- The communication noise adds uncertainty

For Round 1, I'll take a moderately trusting approach. Sending \$3 signals willingness to cooperate while managing risk. If my co-player is reciprocal, we can both benefit (they'll receive \$9, and even a partial return would make this worthwhile). If they don't reciprocate, I've limited my exposure.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 to potentially send
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. There's communication noise - amounts may appear different than what was actually sent/returned

Key considerations:

- This is a multi-round game with 8 agents total
- Pairings are randomized each round

- I'll play both sender and receiver roles across rounds
- Agent names are hidden

Strategic thinking for Round 1:

- Sending \$0 guarantees I keep \$5, but there's no potential for mutual gain
- Sending \$5 shows maximum trust, could lead to good returns if the receiver is cooperative, but risky
- A moderate amount (like \$2-3) balances risk and potential reward
- This is Round 1, so I have no history with any agent yet

The game has a cooperative element - if both players trust each other, total welfare is maximized. The 3x multiplier means:

- If I send \$5, receiver gets \$15. If they return \$7.50, we each end with \$7.50 (better than \$5 each)
- If I send \$3, receiver gets \$9. If they return \$4, I end with \$6 and they end with \$5
- There's also the noise factor to consider

For a first round with no information, I think a moderately trusting approach makes sense.

I'll send \$3 as a signal of willingness to cooperate while not being completely vulnerable. This allows for mutual benefit if the receiver is cooperative, but limits my downside risk.

Actually, let me reconsider. In behavioral economics experiments, sending about half (\$2-3) is common in first rounds. However, given this is a multi-round game and reputation might matter (even though pairings are random, players might develop general strategies), showing some trust could be beneficial.

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 651,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.5 to you. Your visible payoff was \$6.5.

Your total visible earnings across all rounds are \$6.5.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.9 (about 58% of their endowment), showing considerable trust
- I received \$8.7 after the 3x multiplier
- In Round 1, when I sent \$3, my previous partner returned \$4.5 (50% of what they received)
- Applying a similar reciprocity norm seems fair

**\*\*Decision:\*\***

I'll return approximately 50% of what I received, which rewards their trust and establishes a cooperative pattern. This leaves us both better off than our starting positions.

'''json

```
{"return": 4.5}
```

'''

**reasoning:**

Let me analyze this situation:

**\*\*Previous Round (Round 1):\*\***

- I was SENDER
- I sent \$3.0
- It became \$9.0

- They returned \$4.5 to me
- My payoff:  $\$5 - \$3 + \$4.5 = \$6.5$
- Their payoff:  $\$9 - \$4.5 = \$4.5$

That was a pretty fair return - they returned half of what they received, which is a common reciprocity norm.

**\*\*Current Round (Round 2):\*\***

- I am RECEIVER
- They sent me \$2.9 (close to \$3)
- I received \$8.7 ( $2.9 \times 3$ )
- I need to decide how much to return (0-8.7)

**\*\*Considerations:\*\***

1. My co-player sent me almost \$3 (actually \$2.9, but there's communication noise mentioned)
2. In Round 1, when I sent \$3, my partner returned \$4.5 (which was 50% of the \$9 they received)
3. If I apply the same 50% rule: I should return about \$4.35 (half of \$8.7)
4. My co-player took a risk by sending \$2.9, showing trust
5. Being reciprocal could establish good cooperation
6. Note: This is a different co-player than Round 1 (pairings are randomized)

Since this is a new co-player and they showed trust by sending \$2.9, I should reciprocate fairly. Returning around 50% seems like a reasonable reciprocity norm that:

- Rewards their trust
- Leaves both of us better off than not playing
- Establishes me as a cooperative player

My co-player's payoff if I return \$4.35:

- They get:  $\$5 - \$2.9 + \$4.35 = \$6.45$
- I get:  $\$8.7 - \$4.35 = \$4.35$

They sent 58% of their endowment, which is quite generous. I should return slightly more to match their generosity and encourage continued cooperation. Returning \$4.5 mirrors what my previous partner did for me and feels like the right balance...

usage:

```
{
 "input_tokens": 811,
 "output_tokens": 781,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.9, it became \$8.7, and you returned \$4.5. Your visible payoff was \$4.1999999999999999.

Your total visible earnings across all rounds are \$10.7.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

**\*\*Analysis:\*\***

- My co-player sent \$2.27 (about 45% of their endowment), showing moderate trust
- I received \$6.81 after the multiplier
- In previous rounds, I've been returning approximately 50% of what I receive as a reciprocity norm
- This approach has worked well so far

**\*\*Decision:\*\***

I'll maintain consistency and return about 50% of what I received, which rewards their trust appropriately.

```
'''json
{"return": 3.4}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Current situation:\*\***

- I'm the RECEIVER again
- My co-player sent \$2.27 (about 45% of their \$5 endowment)
- I received \$6.81 after the 3x multiplier
- They kept \$5 - \$2.27 = \$2.73

**\*\*Past behavior:\*\***

- Round 1: I sent \$3, received back \$4.5 (50% return rate on the multiplied amount)
- Round 2: I received \$8.7, returned \$4.5 (about 52% return rate)

**\*\*Considerations:\*\***

- My co-player sent a moderate amount (\$2.27), showing some trust but less than the previous player
- Being consistent with my previous behavior would mean returning around 50% of what I received
- 50% of \$6.81 = \$3.405

**\*\*Payoff implications:\*\***

- If I return \$3.4-3.5:
  - I keep: \$6.81 - \$3.4 = \$3.41
  - They get: \$2.73 (kept) + \$3.4 (returned) = \$6.13
  - This seems fair - we both do better than the \$5 starting point

I should maintain consistency with my reciprocity pattern from Round 2. Returning about 50% seems fair and builds trust for potential future rounds.

Let me return \$3.4 (just under 50%).

**usage:**

```
{
 "input_tokens": 1105,
 "output_tokens": 467,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.27, it became \$6.8100000000000005, and you returned \$3.4. Your visible payoff was \$3.4100000000000006.

Your total visible earnings across all rounds are \$14.11.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

**\*\*Analysis:\*\***

- This is Round 4 with a new co-player
- In Round 1, I sent \$3 and received \$4.5 back - a positive outcome
- As a receiver, I've been returning about 50%, establishing a cooperative pattern
- My current co-player is unknown to me

**\*\*Decision:\*\***

I'll send \$3 again. This amount has worked well before - it shows trust and creates potential for mutual gain through the 3x multiplier, while managing downside risk.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze the situation:

**\*\*What I know:\*\***

- Round 1: I was SENDER, sent \$3, received back \$4.5, payoff \$6.5
- Round 2: I was RECEIVER, got \$8.7 (from \$2.9 sent), returned \$4.5, payoff \$4.2
- Round 3: I was RECEIVER, got \$6.81 (from \$2.27 sent), returned \$3.4, payoff \$3.41

- Total earnings: \$14.11

**\*\*Current situation:\*\***

- Round 4: I'm the SENDER with \$5
- This is a new co-player (different from rounds 1-3)

**\*\*Strategic considerations:\*\***

1. I've seen one positive interaction as sender (sent \$3, got back \$4.5)
2. As receiver, I've been fairly generous, returning about 50%
3. This creates a reputation for reciprocity if others know
4. But agents don't know who I am (names hidden), so reputation effects are limited
5. However, the pattern of behavior in the pool might matter

**\*\*Decision factors:\*\***

- Sending creates potential gains (3x multiplier)
- Risk: this is a new co-player, no direct history
- But overall, cooperation seems to be working in this pool
- My Round 1 strategy (sending \$3) worked well

I'll send \$3 again - it's a moderate trust signal that has proven effective.

**usage:**

```
{
 "input_tokens": 1352,
 "output_tokens": 472,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.48 to you. Your visible payoff was \$6.48.

Your total visible earnings across all rounds are \$20.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

**\*\*Analysis:\*\***

- Round 5 with a new co-player
- My \$3 sending strategy has been successful:
  - Round 1: Sent \$3, received \$4.5 back ? \$6.5 payoff
  - Round 4: Sent \$3, received \$4.48 back ? \$6.48 payoff
- Both returns were close to 50%, indicating fair reciprocity from my co-players
- This amount balances trust-building with risk management

**\*\*Decision:\*\***

I'll continue with \$3, as it's been consistently effective.

```
'''json
{"send": 3}
'''
```

**Reasoning:**

**\*\*Analysis:\*\***

- Round 5, new co-player as sender
- My sending strategy of \$3 has worked well:
  - Round 1: Sent \$3, received \$4.5 back, payoff \$6.5
  - Round 4: Sent \$3, received \$4.48 back, payoff \$6.48
- Both times I received close to 50% back, which is fair reciprocity
- Total visible earnings: \$20.59 across 4 rounds (average \$5.15/round)
- As a receiver, I've been consistently returning about 50%

**\*\*Decision:\*\***

The \$3 sending strategy is working consistently well. My co-players have been reciprocating fairly. I should continue this approach.

```
'''json
{"send": 3}
'''
```

```
Usage: {
 "input_tokens": 1601,
 "output_tokens": 339,
 "reasoning_tokens": 0
}
```

### GAME - Agent\_8 - Receiver - dyad\_3

Timestamp: 2026-08-12T09:23:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_1",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_8"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 5,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$2.68, that is 53.6% of their total endowment. You received \$8.040000000000001. How much do you return to the sender? (0-8.040000000000001)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$2.68 (53.6% of their endowment)
- They now have \$2.32 remaining
- I received \$8.04 after the 3x multiplier
- Total pool is now \$10.36

I want to reciprocate their trust while being fair. The sender took a risk to create mutual gains, so I should ensure they benefit from their trust.

A reasonable approach is to aim for a relatively balanced outcome:

- If I return \$3.00, the sender ends with \$5.32 (gaining \$0.32)
- I would keep \$5.04

This rewards their trust while maintaining fairness, and shows I'm reciprocal which could encourage cooperation in future rounds.

{"return": 3.00}

**reasoning:**

Let me analyze this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.68 (which is 53.6% of their \$5 endowment)
3. This was multiplied by 3x, so I received  $2.68 \times 3 = \$8.04$
4. I need to decide how much to return (between \$0 and \$8.04)

**Key considerations:**

- The sender showed trust by sending more than half of their endowment (\$2.68 out of \$5)
- The sender now has:  $\$5 - \$2.68 = \$2.32$
- I currently have: \$8.04
- Whatever I return, the sender will end up with:  $\$2.32 + \text{my return}$
- I will end up with:  $\$8.04 - \text{my return}$

**From a fairness/reciprocity perspective:**

- The sender trusted me with 53.6% of their endowment
- A fair split might be to ensure we both end up with similar amounts, or to reward the sender's trust
- If I return \$3, sender ends with \$5.32, I end with \$5.04 (relatively equal)
- If I return \$4, sender ends with \$6.32, I end with \$4.04 (rewarding trust more)
- If I return \$2, sender ends with \$4.32, I end with \$6.04 (keeping more)

A common reciprocal strategy would be to:

1. Ensure the sender doesn't lose money (they should end with at least \$5)
2. Share the "profit" somewhat fairly

The total pool is now  $\$2.32$  (sender's remainder) +  $\$8.04$  (my endowment) =  $\$10.36$

If we split this profit fairly:

- Each should get about \$5.18
- For sender to get \$5.18, I need to return:  $\$5.18 - \$2.32 = \$2.86$

Returning \$3.00 seems reasonable-it rewards their trust while keeping the outcome relatively balanced, with sender ending at \$5.32 and me at \$5.04. But I should consider whether



a more reciprocal approach might better acknowledge their initial generosity.

If I return roughly the same proportion they invested (53.6% of what I received), that's about \$4.31. This leaves them with \$6.63 and me with \$3.73-a significant reward for their trust, though it cuts my gains substantially. Alternatively, I could split the total gains evenly: the \$5.36 profit gets divided so each of us gains \$2.68, putting the sender at \$7.68 and me at \$2.68....

**usage:**

```
{
 "input_tokens": 537,
 "output_tokens": 1185,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.68, it became \$8.040000000000001, and you returned \$3.0. Your visible payoff was \$5.040000000000001.

Your total visible earnings across all rounds are \$5.040000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this new round as the sender:

In the previous round, my co-player sent 53.6% of their endowment, showing moderate-high trust. I reciprocated fairly by returning enough to give them a positive gain.

Now with a new co-player, I want to signal trust and willingness to cooperate while not being overly risky. Sending around 50% seems reasonable - it creates significant potential mutual gains while maintaining some protection.

If they reciprocate proportionally, we both benefit.

```
{"send": 2.50}
```

**reasoning:**

Now I'm the sender in Round 2 with a new co-player.

Last round I was the receiver:

- Saw the sender send \$2.68 (53.6%)
- Received \$8.04
- Returned \$3.00
- Kept \$5.04

The sender ended with  $\$2.32 + \$3.00 = \$5.32$ , so they gained \$0.32 from trusting me.

Now I'm the sender with \$5. I need to decide how much to send.

Key considerations:

1. The other player sent 53.6% in the previous example, showing moderate-high trust
2. I reciprocated fairly, returning enough to give them a positive gain
3. This is a new co-player, so I don't know their strategy yet
4. There's communication noise, so amounts may appear different
5. Building cooperation early could pay off in future rounds

Strategic thinking:

- If I send too little, I signal distrust and may get nothing back
- If I send too much, I risk losing a lot if they don't reciprocate
- A moderate amount (around 50-60%) shows trust while not being reckless
- The previous sender sent 53.6%, which seemed like a reasonable trust signal

I'll send around \$2.50-3.00 to signal trust and willingness to cooperate.

Let me think about expected values:

- If I send \$2.50, they get \$7.50. If they return 40% = \$3.00, I end with  $\$2.50 + \$3.00 = \$5.50$  (gain \$0.50)

- If I send \$3.00, they get \$9.00. If they return 33% = \$3.00, I end with \$2.00 + \$3.00 = \$5.00 (break even)
- If I send \$3.00 and they return 44% = \$4.00, I end with \$2.00 + \$4.00 = \$6.00 (gain \$1.00)

...

**usage:**

```
{
 "input_tokens": 852,
 "output_tokens": 646,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.19 to you. Your visible payoff was \$5.689999999999995.

Your total visible earnings across all rounds are \$10.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my recent experience:

- Round 2: I sent \$2.50 (50%) and received \$3.19 back, giving me a good positive return

This cooperative approach has been working well. I'm now with a new co-player, and I want to signal trustworthiness and encourage mutual cooperation.

Sending 50% has proven to be a solid strategy - it creates meaningful potential gains while showing balanced trust.

```
{"send": 2.50}
```

**reasoning:**

Let me analyze the previous round:

- I sent \$2.50 (50% of my endowment)
- It became \$7.50 for the receiver
- They returned \$3.19
- I ended with \$2.50 (kept) + \$3.19 (returned) = \$5.69
- They kept \$7.50 - \$3.19 = \$4.31

So the receiver returned about 42.5% of what they received, which gave me a positive gain of \$0.69. This is a reasonable reciprocation, showing they valued cooperation.

Now I'm facing a new co-player in Round 3.

My experience so far:

- Round 1 (as receiver): Received \$8.04, returned \$3.00 (37.3%)
- Round 2 (as sender): Sent \$2.50, received back \$3.19 - good outcome

I should continue with a cooperative strategy. Since my last send of \$2.50 worked well, I could:

1. Send the same amount (\$2.50) to maintain consistency
2. Send slightly more to signal increased trust
3. Send slightly less to be more cautious

Given that cooperation has been working, and I'm now encountering a third different co-player, I think maintaining a cooperative stance of around 50% makes sense. This has proven successful and sends the right signal.

I'll send \$2.50 again.

**usage:**

```
{
 "input_tokens": 1103,
 "output_tokens": 441,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$5.25 to you. Your visible payoff was \$7.75.  
Your total visible earnings across all rounds are \$18.48.

This round, you are the RECEIVER against your current co-player. They sent you \$3.69, so you received \$11.07. How much do you return? (0-11.07)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant****content:**

Looking at this situation:

- The sender showed very high trust by sending \$3.69 (73.8% of their endowment)
- They now have only \$1.31 remaining
- I received \$11.07 after the 3x multiplier
- Total pool is \$12.38

This sender took a significant risk to create mutual gains. I should reciprocate their strong trust signal generously.

If I return \$5.50:

- Sender ends with \$6.81 (gaining \$1.81)
- I keep \$5.57

This rewards their exceptional trust while maintaining fairness for both of us.

```
{"return": 5.50}
```

**reasoning:**

Let me analyze this situation:

Round 4 - I'm the RECEIVER

- The sender sent \$3.69 (73.8% of their \$5 endowment)
- This became \$11.07 after the 3x multiplier
- The sender now has \$5 - \$3.69 = \$1.31 remaining
- Total pool: \$1.31 + \$11.07 = \$12.38

The sender showed very high trust by sending 73.8% of their endowment. This is much higher than the ~50% I've been sending, indicating strong cooperative intent.

I should reciprocate this high level of trust. Let me think about fair returns:

If I return \$5.50:

- Sender ends with: \$1.31 + \$5.50 = \$6.81 (gain of \$1.81)
- I keep: \$11.07 - \$5.50 = \$5.57

If I return \$6.00:

- Sender ends with: \$1.31 + \$6.00 = \$7.31 (gain of \$2.31)
- I keep: \$11.07 - \$6.00 = \$5.07

Given they sent 73.8% of their endowment (very high trust), I should return a proportionally generous amount. Returning around \$5.50-\$6.00 would:

1. Reward their high trust
2. Ensure they profit from being cooperative
3. Still leave me with a reasonable amount
4. Signal I'm reciprocal and trustworthy

I'll return \$5.50 - this gives them a solid gain while being fair to both parties.

**usage:**

```
{
 "input_tokens": 1343,
 "output_tokens": 551,
 "reasoning_tokens": 0
}
```

**Message 10 role=user****content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.69, it became \$11.07, and you returned \$5.5. Your visible payoff was \$5.57.

Your total visible earnings across all rounds are \$24.05.

This round, you are the RECEIVER against your current co-player. They sent you \$3.88, so you received \$11.64. How much do you return? (0-11.64)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- The sender showed extremely high trust by sending \$3.88 (77.6% of their endowment)
- They now have only \$1.12 remaining
- I received \$11.64 after the 3x multiplier
- Total pool is \$12.76

This is even higher trust than the previous round. I should reciprocate this exceptional trust generously.

If I return \$5.80:

- Sender ends with \$6.92 (gaining \$1.92)
- I keep \$5.84

This appropriately rewards their very high trust level while maintaining fairness.

```
{"return": 5.80}
```

**Reasoning:**

Looking at this situation:

- The sender sent \$3.88 (77.6% of their endowment) - even higher trust than the previous round
- They now have \$1.12 remaining
- I received \$11.64 after the 3x multiplier
- Total pool is \$12.76

This sender showed extremely high trust. I should reciprocate proportionally.

In the last round, sender sent \$3.69 (73.8%) and I returned \$5.50, giving them \$6.81 total (gain of \$1.81).

This sender sent even more (\$3.88, or 77.6%). I should return proportionally or slightly more to reward the higher trust.

If I return \$5.80:

- Sender ends with \$6.92 (gain of \$1.92)
- I keep \$5.84

This seems fair and rewards their high trust appropriately.

Let me consider other options:

- Return \$6.00: Sender gets \$7.12 (gain \$2.12), I keep \$5.64
- Return \$5.50: Sender gets \$6.62 (gain \$1.62), I keep \$6.14

I think \$5.80 is a good middle ground that appropriately rewards the very high trust level while maintaining fairness.

Usage: {

```
"input_tokens": 1628,
"output_tokens": 459,
"reasoning_tokens": 0
```

}

**GAME - Agent\_3 - Sender - dyad\_4**

Timestamp: 2026-08-12T09:23:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_4",
"move_index": 6,
"opponent_id": "Agent_2",
"pairing": {
```

```

"agents": [
 "Agent_3",
 "Agent_2"
],
"investor": "Agent_3",
"roles": {
 "Agent_2": "trustee",
 "Agent_3": "investor"
},
"trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.12, that is 62.4% of their total endowment. You received \$9.36. How much do you return to the sender? (0-9.36)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.12 (62.4% of their endowment)
- This created \$9.36 for me after the 3x multiplier
- The sender currently has only \$1.88 remaining

I want to reward their trust and encourage future cooperation. The sender took a risk, and I should reciprocate appropriately.

I'll return \$4.00, which:

- Gives the sender \$5.88 total (better than their initial \$5)
- Leaves me with \$5.36 (also benefiting from the cooperation)
- Demonstrates that trusting me pays off
- Should encourage continued cooperation in future rounds

```
{'return': 4.00}
```

#### reasoning:

Let me analyze this situation carefully:

1. The sender sent \$3.12 out of their \$5 endowment (62.4%)
2. This was multiplied by 3x, so I received \$9.36
3. I need to decide how much to return (between \$0 and \$9.36)

Key considerations:

- The sender showed significant trust by sending 62.4% of their endowment
- There's communication noise in the game, so actual amounts may differ slightly
- This is a repeated game with multiple rounds
- I want to encourage cooperation and build trust

Economic analysis:

- Sender currently has:  $\$5 - \$3.12 = \$1.88$
- I currently have: \$9.36
- If I return nothing, sender gets \$1.88 total, I get \$9.36
- A fair split might consider that we each started with potential of \$5

Common strategies in trust games:

1. Return nothing (purely selfish)
2. Return the original amount sent (~\$3.12) - gives back what was sent
3. Return enough so we both profit equally from the 3x multiplier
4. Return more to encourage future cooperation

Let me calculate option 3 (equal profit):

- The 3x multiplier created \$6.24 in extra value ( $\$9.36 - \$3.12$ )
- Split equally: each gets \$3.12 extra
- Sender should end with:  $\$1.88 + \$3.12 + \$3.12 = \$8.12$
- So I should return:  $\$9.36 - (\$5 + \$3.12) = \$9.36 - \$8.12 = \$1.24$

Wait, let me recalculate more carefully:

- Sender starts with \$5, sends \$3.12, has \$1.88 left
- I receive \$9.36
- Total pool is now \$11.24
- If we each started with \$5 potential, that's \$10 baseline
- The extra \$1.24 created should split evenly at \$0.62 each
- So sender should end with \$5.62 and I should end with \$5.62
- That means I return \$3.74

I'm reconsidering the fairest approach here-whether to split the gains equally or use a different metric.

The sender invested 62.4% of their initial amount, so returning a proportional share of what I received makes sense. That's about \$5.84. Alternatively, since they trusted me with \$3.12 extra, returning \$3.00-\$3.50 would roughly restore their position while letting me benefit from the cooperation. For Round 1, I want to signal good faith and encourage continued trust, so returning 40-50% of the \$9.36 I received-somewhere between \$3.74 and \$4.68-feels like the right balance....

#### usage:

```
{
 "input_tokens": 529,
 "output_tokens": 1077,
 "reasoning_tokens": 0
}
```

```

}
Message 4 role=user
content:
 Round 2

 In your most recent previous game round, you were the RECEIVER against another agent. You
 saw them send $3.12, it became $9.36, and you returned $4.0. Your visible payoff was $5.
 3599999999999999.
 Your total visible earnings across all rounds are $5.359999999999999.

 This round, you are the SENDER against your current co-player. You have $5. How much do yo
 u send? (0-5)
 Respond exactly as JSON: {'send': <amount>}
Message 5 role=assistant
content:
 Looking at this situation:

 - Last round, my co-player trusted me with 62.4% of their endowment
 - I reciprocated fairly, and we both benefited
 - Now I'm with a new co-player as the sender

 I'll mirror the cooperative behavior I experienced. Sending a substantial amount can creat
 e mutual benefit through the 3x multiplier if my co-player reciprocates.

 I'll send $3.00 (60% of my endowment), similar to the trust shown to me in the previous ro
 und.

 {'send': 3.00}
reasoning:
 Let me think about this situation:

 1. In the previous round, I was the receiver
 - The sender sent me $3.12 (62.4% of their endowment)
 - I received $9.36
 - I returned $4.00
 - My payoff was $5.36

 2. Now I'm the sender with a new co-player
 - I have $5 to potentially send
 - Whatever I send gets tripled
 - The receiver will decide how much to return

 Strategic considerations:
 - In the previous round, the sender showed trust by sending 62.4% of their endowment
 - I reciprocated fairly, returning $4.00 out of $9.36 (about 43%)
 - This gave the sender a total of $5.88 (they sent $3.12, kept $1.88, got back $4.00)

 For this round:
 - I'm paired with a different agent (randomized pairings)
 - I don't know their history or behavior
 - However, the game benefits from cooperation
 - If I send a reasonable amount and they reciprocate, we both benefit

 Let me consider different strategies:
 - Sending nothing: I keep $5, they get nothing, no mutual benefit
 - Sending a small amount (e.g., $1-2): Low risk, but lower potential gain
 - Sending a moderate amount (e.g., $3): Shows some trust, reasonable risk
 - Sending a large amount (e.g., $4-5): High trust, high risk, high potential reward

 Given that:
 1. The previous sender trusted me with 62.4% and it worked out
 2. I want to establish a cooperative pattern
 3. There's communication noise mentioned, so exact amounts may vary
 4. I'm in a multi-round game where reputation (though anonymous) might matter

 I'll send $3.00, which is 60% of my endowment. This:
 ...
usage:
 {
 "input_tokens": 824,

```

```

 "output_tokens": 680,
 "reasoning_tokens": 0
 }
Message 6 role=user
 content:
 Round 3

 In your most recent previous game round, you were the SENDER against another agent. You sent $3.0, it became $9.0, and you saw them return $3.48 to you. Your visible payoff was $5.48.
 Your total visible earnings across all rounds are $10.84.

 This round, you are the RECEIVER against your current co-player. They sent you $2.97, so you received $8.91. How much do you return? (0-8.91)
 Respond exactly as JSON: {'return': <amount>}
Message 7 role=assistant
 content:
 Looking at this situation:

 Current round:
 - Sender showed good trust by sending $2.97 (59.4% of their endowment)
 - This created $8.91 for me
 - Sender currently has $2.03

 Previous round context:
 - I sent $3.00 (60% - very similar trust level)
 - My co-player returned $3.48, which was fair

 I should reciprocate the trust shown to me. The sender sent almost 60% of their endowment, similar to what I did last round.

 I'll return $3.75, which:
 - Gives the sender $5.78 total (a gain of $0.78)
 - Leaves me with $5.16 (also a gain)
 - Rewards their trust appropriately

 {'return': 3.75}
 reasoning:
 Let me analyze this situation:

 Last round (Round 2):
 - I was the SENDER
 - I sent $3.00 (60% of my endowment)
 - It became $9.00 for the receiver
 - They returned $3.48
 - My payoff: $5 - $3 + $3.48 = $5.48
 - Their payoff: $9 - $3.48 = $5.52

 The receiver gave me back slightly more than I sent, which was cooperative and fair.

 This round (Round 3):
 - I'm the RECEIVER
 - Sender sent $2.97 (59.4% of their endowment - very similar to what I sent last round!)
 - I received $8.91
 - Sender currently has: $5 - $2.97 = $2.03

 The sender showed good trust by sending ~60% of their endowment. This is very similar to the 60% I sent in the previous round.

 I should reciprocate fairly. Let me think about what would be a fair return:

 - If I return $3.50, sender gets $2.03 + $3.50 = $5.53, and I keep $8.91 - $3.50 = $5.41
 - If I return $4.00, sender gets $2.03 + $4.00 = $6.03, and I keep $8.91 - $4.00 = $4.91
 - If I return $3.75, sender gets $2.03 + $3.75 = $5.78, and I keep $8.91 - $3.75 = $5.16

 The sender trusted me with $2.97, similar to how I trusted my previous co-player with $3.00, and they returned $3.48.
 ...
 usage:
 {

```



```
"input_tokens": 1089,
"output_tokens": 735,
"reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.97, it became \$8.91, and you returned \$3.75. Your visible payoff was \$5.16.

Your total visible earnings across all rounds are \$16.0.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*Last round with THIS SAME co-player:\*\***

- They sent me \$2.97 (59.4% of their endowment) - showing good trust
- I returned \$3.75 - rewarding their trust fairly
- We both gained from cooperation

**\*\*Now it's my turn to send:\*\***

Since we just had a successful cooperative round together, I should maintain this pattern. They trusted me, I reciprocated, and now I should show similar trust.

I'll send \$3.00 (60% of my endowment), matching the level of trust they showed me.

```
{'send': 3.00}
```

**reasoning:**

Let me analyze this situation:

**\*\*Last round with THIS SAME co-player:\*\***

- They sent me \$2.97 (59.4% of their endowment)
- I received \$8.91
- I returned \$3.75
- They got \$5.78 total (gain of \$0.78)
- I got \$5.16 (gain of \$0.16)

**\*\*Now I'm the sender with THE SAME co-player:\*\***

- They showed good trust last round (sent ~60%)
- I reciprocated fairly
- We both gained from cooperation

Since we're playing with the same person and we had a successful cooperative round, I should continue this pattern. They trusted me with ~60% of their endowment, and I rewarded that trust. Now I should show similar trust.

I'll send around \$3.00 again, maintaining the cooperative pattern.

**usage:**

```
{
 "input_tokens": 1406,
 "output_tokens": 354,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.73 to you. Your visible payoff was \$7.73.

Your total visible earnings across all rounds are \$23.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*Last round:\*\***

- I sent \$3.00 (60% of my endowment)
- My co-player returned \$5.73 - very generous!
- This gave me \$7.73, an excellent outcome

**\*\*Pattern so far:\*\***

- Sending around 60% has consistently led to good reciprocation
- My average earnings are strong at \$4.75 per round
- Cooperative approach is working well

**\*\*Current round with NEW co-player:\*\***

I'll continue the successful cooperative strategy. Sending a substantial amount demonstrates trust and has been rewarded by others.

{'send': 3.00}

**Reasoning:**

Let me analyze this situation:

**\*\*Last round:\*\***

- I sent \$3.00 (60% of my endowment)
- They returned \$5.73
- My payoff was \$7.73 (very good!)

This was an excellent return - they gave me back more than I would have had if I kept everything (\$5). This suggests my co-player is very cooperative and generous.

**\*\*Current round:\*\***

- I'm the sender again with a NEW co-player
- I have \$5 to decide how much to send

Looking at the pattern:

- Round 1 (receiver): Saw sender send 62.4%, I returned fairly
- Round 2 (sender): Sent \$3.00 (60%)
- Round 3 (receiver): Saw sender send 59.4%, I returned fairly
- Round 4 (sender): Sent \$3.00 (60%) to the same co-player from round 3
- Round 5 (sender): Last round's co-player was very generous, returned \$5.73

The cooperative strategy has been working well. I've been consistently sending around 60% and have seen good reciprocation. My total earnings of \$23.73 over 5 rounds (averaging \$4.746 per round) suggests the cooperative approach is profitable.

For this round with a NEW co-player, I should continue the cooperative strategy that's been working. I'll send around \$3.00 again (60% of my endowment).

```
Usage: {
 "input_tokens": 1674,
 "output_tokens": 493,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_2 - Receiver - dyad\_4**

Timestamp: 2026-08-12T09:23:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_3",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_2"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_2": "trustee",
 "Agent_3": "investor"
 }
 }
}
```

```

 },
 "trustee": "Agent_2"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 5,
 "task": "game",
 "task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative tone while managing some risk given the uncertainty.

If I send money, it triples in value, creating potential mutual gains. However, I don't know my co-player yet, and there's communication noise to consider.

I'll send a moderate amount to signal trust and willingness to cooperate, while keeping some buffer:

**\*\*Sending \$3\*\***

This gives my co-player \$9 to work with. If they return a fair amount (around half or more), we both benefit compared to me sending nothing. It's a balance between demonstrating cooperation and managing first-round uncertainty.

`{'send': 3}`

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think about this strategically:

1. I start with \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. My final payoff = \$5 - sent + returned
5. Receiver's final payoff = (sent × 3) - returned

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise (amounts may differ from what was actually sent)
- Agent names are hidden

From a purely selfish perspective, if I send \$0, I guarantee myself \$5. But this doesn't build trust or allow for mutual gains.

If I send everything (\$5), and the receiver returns a fair amount (like half of what they receive = \$7.50), I'd end up with \$7.50, and they'd keep \$7.50 as well. We'd both be better off.

However, there's risk:

- The receiver might not return anything
- Communication noise could cause misunderstandings
- This is just round 1, so I don't have reputation information

A moderate approach might be to send \$3:

- They receive \$9
- If they return \$5, I get \$5 (break even on sent, but signal trust)
- If they return \$6, I get \$6 (gain \$1)
- They keep between \$3-\$4 depending on return

Or I could send \$2:

- They receive \$6
- If they return \$3, I get \$3, netting -\$2 but we both gained
- If they return \$4, I get \$4, netting -\$1
- Lower risk

Given it's round 1 and I want to establish cooperative play but also manage risk, I'll send \$3. ...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 598,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.41 to you. Your visible payoff was \$6.41.

Your total visible earnings across all rounds are \$6.41.

This round, you are the RECEIVER against your current co-player. They sent you \$1.73, so you received \$5.189999999999995. How much do you return? (0-5.189999999999995)

Respond exactly as JSON: `{'return': <amount>}`

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent \$1.73 (became \$5.19 for me)
- They kept \$3.27
- They showed some trust by sending money

**\*\*My reasoning:\*\***

- In the previous round, my co-player returned about 49% of what they received (\$4.41 out of \$9)
- I should be reasonably cooperative to encourage future trust
- Returning around 45-50% of what I received seems fair
- This would be \$2.33-\$2.60

I'll return \$2.40:

- They'll end with:  $\$3.27 + \$2.40 = \$5.67$
- I'll end with:  $\$5.19 - \$2.40 = \$2.79$
- They gain \$0.67 from sending, which rewards their trust

{'return': 2.40}

**reasoning:**

Let me think about this situation:

1. In the previous round, I sent \$3, which became \$9 for the receiver. They returned \$4.41 to me.
  - I ended with:  $\$5 - \$3 + \$4.41 = \$6.41$
  - They ended with:  $\$9 - \$4.41 = \$4.59$
  - So they returned about 49% of what they received, which seems fairly cooperative.
2. Now I'm the RECEIVER. My co-player sent me \$1.73, which became approximately \$5.19.
  - They now have:  $\$5 - \$1.73 = \$3.27$
  - I have: \$5.19
  - Total pie:  $\$3.27 + \$5.19 = \$8.46$
3. There's communication noise, so what I see (\$1.73 sent) might not be exactly what they sent.
4. The sender took some risk by sending me \$1.73. It's less than what I sent in the previous round (\$3), but still a meaningful amount showing some trust.
5. I should consider:
  - Being reciprocal/cooperative to encourage future cooperation
  - The communication noise factor
  - What would be "fair"

If I think about fairness:

- They sent \$1.73, keeping \$3.27
- I received \$5.19
- If we aim for roughly equal outcomes:  $(\$3.27 + \text{return}) ? (\$5.19 - \text{return})$
- That would suggest:  $\$3.27 + \text{return} = \$5.19 - \text{return}$
- $2 * \text{return} = \$5.19 - \$3.27 = \$1.92$
- $\text{return} ? \$0.96$

But that seems quite low given they took a risk. Let me think about proportional return:

- Matching their 49% return rate would mean sending back \$2.54
- Alternatively, if I return \$2.50, they'd end with \$5.77 (a \$0.77 gain) while I'd have \$2.69
- This feels like a balanced cooperative move that rewards their trust without leaving me too disadvantaged

I'll return \$2.50....

**usage:**

```
{
 "input_tokens": 791,
 "output_tokens": 1107,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.73, it became \$5.189999999999995, and you returned \$2.4. Your visible payoff was \$2.789999999999996.  
Your total visible earnings across all rounds are \$9.2.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my situation:

**\*\*Previous rounds:\*\***

- Round 1: Sent \$3, received \$4.41 back - good cooperation
- Round 2: Received \$5.19, returned \$2.40 - showed I reciprocate

**\*\*Current round:\*\***

- New co-player (randomized pairing)
- Need to balance trust with uncertainty

Since I had success with sending \$3 in Round 1 (got \$4.41 back, netting \$1.41 profit), and I want to maintain a cooperative approach while testing this new partner, I'll send the same amount:

**\*\*Sending \$3\*\***

This creates \$9 for them to work with. If they're reasonably cooperative like my Round 1 partner, we'll both benefit.

{'send': 3}

**reasoning:**

Let me analyze this situation:

Round 1: I was SENDER

- I sent \$3
- It became \$9
- They returned \$4.41
- My payoff:  $\$5 - \$3 + \$4.41 = \$6.41$

Round 2: I was RECEIVER

- They sent \$1.73
- I received \$5.19
- I returned \$2.40
- My payoff:  $\$5.19 - \$2.40 = \$2.79$
- Total so far:  $\$6.41 + \$2.79 = \$9.20$  ?

Round 3: I am SENDER again

- This is a new co-player (since pairings are randomized each round)
- I have \$5 to send

Strategy considerations:

1. I had a good experience in Round 1 - sent \$3, got back \$4.41 (profit of \$1.41)
2. In Round 2, I received \$5.19 and returned \$2.40 (46%), showing I'm cooperative
3. This is a new co-player, so I don't know their behavior
4. There's communication noise, so what they see may differ from what I intend

I should probably send a moderate amount again to:

- Signal willingness to cooperate
- Test this new partner's trustworthiness
- Not risk everything on an unknown player

Sending \$3 worked well in Round 1. I could try that again, or maybe send slightly more (\$3.50) or less (\$2.50) to experiment.

Given that:

- My first experience was positive
- I want to maintain a cooperative reputation
- But this is a new, unknown player

I'll send \$3 again - it's a balanced approach that showed good results before.

```
{'send': 3}
usage:
{
 "input_tokens": 1128,
 "output_tokens": 608,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.71 to you. Your visible payoff was \$5.71.

Your total visible earnings across all rounds are \$14.91.

This round, you are the RECEIVER against your current co-player. They sent you \$3.79, so you received \$11.370000000000001. How much do you return? (0-11.370000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent \$3.79 out of \$5 (~76% of their endowment)
- This became \$11.37 for me
- They showed strong trust by sending most of their money

**\*\*My reasoning:\*\***

- They took a big risk trusting me
- I should reward this trust to encourage cooperation
- Returning around 45% of what I received seems appropriate given their high trust
- This would be approximately \$5.10

**\*\*Calculating outcomes:\*\***

- They'll end with: \$1.21 + \$5.10 = \$6.31
- I'll end with: \$11.37 - \$5.10 = \$6.27
- They gain \$1.31 from their trust, which is a good reward

{'return': 5.10}

**reasoning:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent \$3.79 (became \$11.37 for me)
- They kept \$1.21
- They showed significant trust by sending most of their money

**\*\*Previous experience:\*\***

- Round 1: I sent \$3, got back \$4.41 (they returned ~49%)
- Round 2: I received \$5.19, returned \$2.40 (~46%)
- Round 3: I sent \$3, got back \$3.71 (they returned ~41%)

**\*\*My reasoning:\*\***

- This co-player sent \$3.79 out of \$5, showing strong trust (~76%)
- They're risking a lot and trusting me to reciprocate
- I should reward this trust appropriately
- Returning around 40-50% of what I received would be \$4.55-\$5.69
- Let me think about fairness:
  - They sent \$3.79, kept \$1.21
  - I received \$11.37
  - If I return \$5.00:
    - They end with: \$1.21 + \$5.00 = \$6.21
    - I end with: \$11.37 - \$5.00 = \$6.37
  - If I return \$5.50:
    - They end with: \$1.21 + \$5.50 = \$6.71
    - I end with: \$11.37 - \$5.50 = \$5.87

Given they sent \$3.79 (76% of their endowment), I should be generous but not overdo it. Ar

ound 45% return rate seems appropriate, which would be about \$5.12.

Let me go with \$5.10:

...

**usage:**

```
{
 "input_tokens": 1453,
 "output_tokens": 690,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.79, it became \$11.370000000000001, and you returned \$5.1. Your visible payoff was \$6.270000000000001.

Your total visible earnings across all rounds are \$21.18.

This round, you are the RECEIVER against your current co-player. They sent you \$2.91, so you received \$8.73. How much do you return? (0-8.73)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent \$2.91 out of \$5 (~58% of their endowment)
- This became \$8.73 for me
- They kept \$2.09
- This shows moderate to good trust

**\*\*My reasoning:\*\***

- They demonstrated decent trust by sending over half their money
- I should reciprocate consistently with my previous rounds
- In previous rounds as receiver, I've returned around 45% of what I received
- For \$8.73, that would be approximately \$3.90

**\*\*Calculating outcomes:\*\***

- They'll end with: \$2.09 + \$3.90 = \$5.99
- I'll end with: \$8.73 - \$3.90 = \$4.83
- They gain about \$1 from trusting me, which is a reasonable reward

{'return': 3.90}

**Reasoning:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent \$2.91 out of \$5 (~58% of their endowment)
- This became \$8.73 for me
- They kept \$2.09
- This shows moderate to good trust

**\*\*My reasoning:\*\***

- They sent a decent amount, showing trust
- I should reciprocate proportionally
- Looking at my previous returns:
  - Round 2: Received \$5.19, returned \$2.40 (~46%)
  - Round 4: Received \$11.37, returned \$5.10 (~45%)
- I've been returning around 45% consistently

For \$8.73, 45% would be about \$3.93

**\*\*Calculating outcomes with \$3.93:\*\***

- They'll end with: \$2.09 + \$3.93 = \$6.02
- I'll end with: \$8.73 - \$3.93 = \$4.80
- They gain \$1.02 from sending, which rewards their trust

This seems fair and consistent with my previous behavior. I'll return \$3.90 to keep it simple.

Actually, let me recalculate with \$3.90:



- They'll end with: \$2.09 + \$3.90 = \$5.99
- I'll end with: \$8.73 - \$3.90 = \$4.83
- They gain \$0.99 from sending

That works well.

```
Usage: {
 "input_tokens": 1786,
 "output_tokens": 560,
 "reasoning_tokens": 0
}
```

## ROUND 6

Roles: Agent\_1=Receiver, Agent\_2=Sender, Agent\_3=Receiver, Agent\_4=Receiver, Agent\_5=Receiver, Agent\_6=Sender, Agent\_7=Sender, Agent\_8=Sender

### Pairings:

```
dyad_1: Agent_7=Sender, Agent_1=Receiver
dyad_2: Agent_2=Sender, Agent_4=Receiver
dyad_3: Agent_6=Sender, Agent_5=Receiver
dyad_4: Agent_8=Sender, Agent_3=Receiver
```

### Parsed game decisions:

```
dyad_1: Agent_7 sent $3; received $9; Agent_1 returned $5.5; payoffs Agent_7=$7.5, Agent_1=$3.5;
noise/visible: sent visible=$3.68, received visible=$11.04, returned visible=$5.84
 Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 9.0, "min": 0, "raw":
 5.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_2 sent $3; received $9; Agent_4 returned $3.25; payoffs Agent_2=$5.25,
Agent_4=$5.75; noise/visible: sent visible=$2.27, received visible=$6.81, returned visible=$3.68
 Action validation: {"returned": {"amount": 3.25, "clamped": false, "max": 9.0, "min": 0, "raw":
 3.25}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_6 sent $2.5; received $7.5; Agent_5 returned $4.1; payoffs Agent_6=$6.6,
Agent_5=$3.4; noise/visible: sent visible=$2.7, received visible=$8.1, returned visible=$4.52
 Action validation: {"returned": {"amount": 4.1, "clamped": false, "max": 7.5, "min": 0, "raw":
 4.1}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
dyad_4: Agent_8 sent $2.5; received $7.5; Agent_3 returned $4; payoffs Agent_8=$6.5, Agent_3=$3.5;
noise/visible: sent visible=$3.24, received visible=$9.72, returned visible=$3.97
 Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 7.5, "min": 0, "raw":
 4.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
```

### Balances after round: {

```
"Agent_1": 33.75,
"Agent_2": 30.25,
"Agent_3": 31.950000000000003,
"Agent_4": 37.4,
"Agent_5": 32.38,
"Agent_6": 33.07,
"Agent_7": 30.6,
"Agent_8": 31.599999999999998
}
```

### Visible balances after round: {

```
"Agent_1": 34.9,
"Agent_2": 31.689999999999998,
"Agent_3": 35.9,
"Agent_4": 33.32,
"Agent_5": 32.94,
"Agent_6": 32.13,
"Agent_7": 34.010000000000005,
"Agent_8": 36.36
}
```

## LLM CALLS

### GAME - Agent\_7 - Sender - dyad\_1

Timestamp: 2026-08-12T09:23:47  
Model: anthropic/claude-sonnet-4.5  
Temperature: 0.8

### Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
```

```

"pairing": {
 "agents": [
 "Agent_7",
 "Agent_1"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_1": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round as the sender, I need to balance trust with caution since I have no information about my co-player yet.

**\*\*My reasoning:\*\***

- Sending nothing (\$0) is safe but offers no potential gain
- Sending everything (\$5) maximizes potential but is very risky with an unknown partner
- A moderate amount shows willingness to cooperate while limiting downside risk
- If my co-player returns roughly half of what they receive (a common fairness norm), we both benefit

I'll send **\*\*\$3\*\***. This gets tripled to \$9 for the receiver. If they return around \$4-5, we'd both end up better off than the initial \$5, establishing a cooperative dynamic.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. **\*\*Basic mechanics\*\***:
  - I have \$5
  - Whatever I send gets tripled
  - The receiver decides how much to return
  - I keep: \$5 - sent + returned
2. **\*\*Strategic considerations\*\***:
  - If I send \$0, I keep \$5, receiver gets \$0. Safe but no potential gains.
  - If I send everything (\$5), receiver gets \$15. If they return half (\$7.50), I'd have \$12.50 total (better than \$5).
  - There's risk: the receiver might return nothing.
  - This is round 1, so there's no history to base trust on.
  - There are multiple rounds, so building trust/reputation could matter.
  - The game has communication noise, which adds uncertainty.
3. **\*\*Multi-agent context\*\***:
  - There are 8 agents total
  - Pairings are randomized each round
  - I might not play with the same person again
  - This reduces the value of building reputation with a specific partner
  - However, if there's any possibility of reputation effects across the pool, cooperation could still matter
4. **\*\*First round strategy\*\***:
  - Since this is round 1 with a random partner, I have no information about their behavior
  - A moderate amount shows trust but limits risk
  - Common starting strategies in trust games involve sending a moderate amount (e.g., \$2-\$3)
  - This signals willingness to cooperate while not being overly naive
5. **\*\*Optimal reasoning\*\***:
  - If the receiver is rational and forward-looking, they might return a fair amount to encourage future cooperation
  - Even if I don't play them again, they might reason that cooperation is beneficial
  - Sending around \$3 seems reasonable: shows trust, limits downside, and if they return half of the tripled amount (half of \$9 = \$4.50), I'd end up with \$6.50 total

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 668,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.8 to you. Your visible payoff was \$6

.8.

Your total visible earnings across all rounds are \$6.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.48, so you received \$7.439999999999995. How much do you return? (0-7.439999999999995)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this round:

**\*\*Analysis:\*\***

- My co-player sent \$2.48 out of \$5 (about 50%), showing moderate trust
- This became \$7.44 for me
- In Round 1, my previous partner returned about 53% of what they received, which worked well

**\*\*My decision:\*\***

I want to:

- Reward their trust appropriately
- Ensure both of us profit from cooperation
- Maintain a fair reciprocal relationship

If I return \$4.00:

- Sender gets:  $\$2.52 + \$4.00 = \$6.52$
- I get:  $\$7.44 - \$4.00 = \$3.44$
- Both better off than the initial \$5

This return (about 54% of what I received) mirrors the generous reciprocity I experienced in Round 1.

```
'''json
{"return": 4.0}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Round 1 (previous round):\*\***

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$4.8 to me
- My payoff:  $\$5 - \$3 + \$4.8 = \$6.8$
- Their payoff:  $\$9 - \$4.8 = \$4.2$

So my previous co-player returned a bit more than half (53.3% of what they received), which was generous and fair.

**\*\*Round 2 (current round):\*\***

- I am now the RECEIVER
- My current co-player (different from Round 1) sent me \$2.48
- It became \$7.44 (actually \$7.439999999999995, but I'll think of it as \$7.44)
- I need to decide how much to return

The note says there may be communication noise, so the amounts I see may differ slightly from what was actually sent/returned.

My current co-player sent \$2.48 out of their \$5, which is about 49.6% - nearly half. This shows moderate trust.

Now I need to decide what to return. Let me think about fairness and reciprocity:

1. **\*\*Equal split approach\*\***: If we aim for equal final payoffs:

- Sender currently has:  $\$5 - \$2.48 = \$2.52$
- I currently have: \$7.44
- Total: \$9.96
- Equal split: \$4.98 each
- I should return:  $\$7.44 - \$4.98 = \$2.46$

2. **\*\*Proportional return based on what they sent\*\***:

- In Round 1, my partner returned 53.3% of what they received

- If I return 50% of \$7.44 = \$3.72
- This would give them: \$2.52 + \$3.72 = \$6.24
- And me: \$7.44 - \$3.72 = \$3.72

### 3. **\*\*Return what maintains trust\*\*:**

- My co-player showed moderate trust by sending nearly half
- I should reciprocate fairly to encourage future cooperation

I think returning around \$3.50 strikes the right balance-it rewards their trust while ensuring both of us end up ahead of the initial \$5 endowment. That gives them \$6.02 and me \$3.94.

But I could also return \$4.00 to show stronger reciprocation, which would give them \$6.52 and me \$3.44. Looking back...

#### **usage:**

```
{
 "input_tokens": 816,
 "output_tokens": 1083,
 "reasoning_tokens": 0
}
```

#### **Message 6 role=user**

##### **content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.48, it became \$7.439999999999995, and you returned \$4.0. Your visible payoff was \$3.439999999999995.

Your total visible earnings across all rounds are \$10.239999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

#### **Message 7 role=assistant**

##### **content:**

Looking at this round:

##### **\*\*Analysis:\*\***

- My co-player sent \$3.02 out of \$5 (about 60%), showing strong trust
- This became \$9.06 for me
- They're trusting me with more than the previous sender

##### **\*\*My decision:\*\***

I want to reward this higher level of trust and maintain the cooperative pattern I've established (returning ~50-55% of what I receive).

If I return \$5.00:

- Sender gets: \$1.98 + \$5.00 = \$6.98
- I get: \$9.06 - \$5.00 = \$4.06
- Both significantly better off than the initial \$5

This return (about 55% of what I received) is consistent with my previous reciprocity and rewards their trust appropriately.

```
'''json
{"return": 5.0}
'''
```

##### **reasoning:**

Let me analyze this round:

##### **\*\*What happened in Round 3:\*\***

- I'm the RECEIVER again
- My co-player sent \$3.02 out of \$5 (about 60%), showing good trust
- This became \$9.06 for me

##### **\*\*My history:\*\***

- Round 1: I was SENDER, sent \$3, received \$4.80 back (about 53% of the \$9 they received)
- Round 2: I was RECEIVER, received \$7.44, returned \$4.00 (about 54% of what I received)

##### **\*\*Decision for Round 3:\*\***

My co-player sent a higher percentage this time (\$3.02 vs \$2.48 last round from a differen

t player), showing increased trust. I should:

- Reward this trust
- Maintain reciprocity around 50-55% return rate
- Ensure both of us benefit

If I return \$5.00:

- Sender gets:  $\$1.98 + \$5.00 = \$6.98$
- I get:  $\$9.06 - \$5.00 = \$4.06$
- Both significantly better off than \$5

This is about 55% of what I received, consistent with my previous behavior and fair reciprocity.

Alternatively, if I return \$4.50:

- Sender gets:  $\$1.98 + \$4.50 = \$6.48$
- I get:  $\$9.06 - \$4.50 = \$4.56$
- Still both better off

Let me go with \$5.00 to be generous and reward the higher trust shown.

**usage:**

```
{
 "input_tokens": 1172,
 "output_tokens": 579,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.02, it became \$9.06, and you returned \$5.0. Your visible payoff was \$4.0600000000000005.

Your total visible earnings across all rounds are \$14.299999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My experience so far:\*\***

- Round 1: Sent \$3 ? received \$4.8 back (53% return rate) ? \$6.8 payoff
- Rounds 2-3: As receiver, I returned 54-55% of what I received
- Cooperation has been working well

**\*\*My decision:\*\***

Since this is a new co-player, I don't know their specific behavior, but the general pattern in this game seems cooperative. Sending \$3 worked well in Round 1, balancing trust with prudence.

If I send \$3:

- It becomes \$9 for receiver
- If they return around 50%, I get ~\$4.5 back
- Net: ~\$6.5 total, better than keeping all \$5

I'll send **\*\*\$3\*\*** again - it's proven effective and signals willingness to cooperate.

```
'''json
{"send": 3}
'''
```

**reasoning:**

I'm now the SENDER again. Let me review my experience:

Round 1: I was SENDER, sent \$3, received back \$4.8, payoff = \$6.8

Round 2: I was RECEIVER, received \$7.44 (from \$2.48 sent), returned \$4.0, payoff = \$3.44

Round 3: I was RECEIVER, received \$9.06 (from \$3.02 sent), returned \$5.0, payoff = \$4.06

Now Round 4: I'm SENDER again

My experience shows:

- When I sent \$3 in Round 1, I got back \$4.8 (53% of the \$9 the receiver got)
- I've been a generous receiver, returning 54% and 55% of what I received
- Cooperation seems to be working well

This is a new co-player, so I don't know their behavior yet. However, the game seems to have a cooperative culture developing.

I should send a moderate-to-high amount to signal trust and cooperation. Sending \$3 worked well before. I could stick with that or increase slightly given the positive experiences.

I'll send \$3 again - it's a proven strategy that balances risk and reward.

**usage:**

```
{
 "input_tokens": 1500,
 "output_tokens": 515,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.36 to you. Your visible payoff was \$6.36.

Your total visible earnings across all rounds are \$20.66.

This round, you are the RECEIVER against your current co-player. They sent you \$4.17, so you received \$12.51. How much do you return? (0-12.51)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this round:

**\*\*Analysis:\*\***

- My co-player sent \$4.17 out of \$5 (about 83%), showing exceptional trust
- This became \$12.51 for me
- This is the highest trust shown by any sender I've faced

**\*\*My reciprocity pattern:\*\***

- I've been returning around 50-55% of what I receive
- This has maintained good cooperative relationships

**\*\*My decision:\*\***

This level of trust deserves strong reciprocation.

If I return \$7.00:

- Sender gets:  $\$0.83 + \$7.00 = \$7.83$
- I get:  $\$12.51 - \$7.00 = \$5.51$
- Both significantly better off than the initial \$5

This return (about 56% of what I received) maintains my cooperative pattern while appropriately rewarding their exceptional trust.

```
'''json
{"return": 7.0}
'''
```

**reasoning:**

Looking at this round:

**\*\*Analysis:\*\***

- My co-player sent \$4.17 out of \$5 (about 83%), showing very high trust
- This became \$12.51 for me
- This is the highest amount anyone has sent me so far

**\*\*My past behavior as receiver:\*\***

- Round 2: Received \$7.44, returned \$4.00 (54%)
- Round 3: Received \$9.06, returned \$5.00 (55%)

**\*\*My past experience as sender:\*\***

- Round 1: Sent \$3, received back \$4.8 (53% return)
- Round 4: Sent \$3, received back \$4.36 (48% return)

**\*\*My decision:\*\***

This co-player showed exceptional trust by sending \$4.17. I should reward this trust generously. If I continue my pattern of returning ~50-55%, that would be \$6.25-\$6.88.

Let me calculate:

If I return \$7.00:

- Sender gets:  $\$0.83 + \$7.00 = \$7.83$
- I get:  $\$12.51 - \$7.00 = \$5.51$
- Both significantly better off

If I return \$6.50:

- Sender gets:  $\$0.83 + \$6.50 = \$7.33$
- I get:  $\$12.51 - \$6.50 = \$6.01$
- Both better off

Returning \$7.00 (about 56% of what I received) maintains my cooperative pattern and rewards their exceptional trust. This seems fair given they sent over 80% of their endowment.

**usage:**

```
{
 "input_tokens": 1854,
 "output_tokens": 605,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.17, it became \$12.51, and you returned \$7.0. Your visible payoff was \$5.51.

Your total visible earnings across all rounds are \$26.17.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this round as SENDER:

**\*\*My sending history:\*\***

- Round 1: Sent \$3 ? received \$4.8 back ? \$6.8 payoff
- Round 4: Sent \$3 ? received \$4.36 back ? \$6.36 payoff

**\*\*Analysis:\*\***

- Sending \$3 has consistently worked well for me
- Both previous receivers returned about 48-53%, yielding good profits
- This is a new co-player, so I have no specific information about them
- The overall environment has been cooperative

**\*\*My decision:\*\***

I'll stick with the proven strategy of sending **\*\*\$3\*\***. It:

- Shows willingness to cooperate
- Has delivered consistent positive returns
- Balances trust with prudent risk management

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Looking at my history:

Round 1 (SENDER): Sent \$3 ? received \$4.8 back ? payoff \$6.8

Round 2 (RECEIVER): Received \$7.44 ? returned \$4.0 ? payoff \$3.44

Round 3 (RECEIVER): Received \$9.06 ? returned \$5.0 ? payoff \$4.06

Round 4 (SENDER): Sent \$3 ? received \$4.36 back ? payoff \$6.36

Round 5 (RECEIVER): Received \$12.51 ? returned \$7.0 ? payoff \$5.51



As SENDER (2 times so far):

- Round 1: Sent \$3, got back \$4.8 (return rate 53%)
- Round 4: Sent \$3, got back \$4.36 (return rate 48.4%)

Both times I sent \$3 and got positive returns. The cooperation has been working well overall.

This is a new co-player again. Should I:

1. Stick with \$3 (proven strategy)
2. Try increasing to show more trust (like the person who sent me \$4.17)
3. Decrease to be more cautious

Given that:

- \$3 has worked well twice
- I've been a generous receiver (returning 50-55%)
- The overall environment seems cooperative
- It's midway through the game (round 6)

I think \$3 remains a good balanced choice. It's worked well and I don't have information about this specific co-player yet.

Actually, let me think about this differently. The person in Round 5 sent me \$4.17 and I returned \$7.0, so they ended up with \$7.83. That was a very successful interaction.

What if I try sending a bit more to signal stronger trust? Maybe \$3.5 or \$4?

...

```
Usage: {
 "input_tokens": 2187,
 "output_tokens": 766,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:23:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_7",
 "Agent_1"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_1": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to make a decision without any prior information about my co-player.

**\*\*Strategic considerations:\*\***

- Sending more creates potential for mutual gain (3x multiplier)
- But I have no information about whether my co-player will reciprocate
- This is a multi-round game, so building some trust could be valuable
- The communication noise adds uncertainty

For Round 1, I'll take a moderately trusting approach. Sending \$3 signals willingness to cooperate while managing risk. If my co-player is reciprocal, we can both benefit (they'll receive \$9, and even a partial return would make this worthwhile). If they don't reciprocate, I've limited my exposure.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 to potentially send
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. There's communication noise - amounts may appear different than what was actually sent/returned

Key considerations:

- This is a multi-round game with 8 agents total
- Pairings are randomized each round

- I'll play both sender and receiver roles across rounds
- Agent names are hidden

Strategic thinking for Round 1:

- Sending \$0 guarantees I keep \$5, but there's no potential for mutual gain
- Sending \$5 shows maximum trust, could lead to good returns if the receiver is cooperative, but risky
- A moderate amount (like \$2-3) balances risk and potential reward
- This is Round 1, so I have no history with any agent yet

The game has a cooperative element - if both players trust each other, total welfare is maximized. The 3x multiplier means:

- If I send \$5, receiver gets \$15. If they return \$7.50, we each end with \$7.50 (better than \$5 each)
- If I send \$3, receiver gets \$9. If they return \$4, I end with \$6 and they end with \$5
- There's also the noise factor to consider

For a first round with no information, I think a moderately trusting approach makes sense.

I'll send \$3 as a signal of willingness to cooperate while not being completely vulnerable. This allows for mutual benefit if the receiver is cooperative, but limits my downside risk.

Actually, let me reconsider. In behavioral economics experiments, sending about half (\$2-3) is common in first rounds. However, given this is a multi-round game and reputation might matter (even though pairings are random, players might develop general strategies), showing some trust could be beneficial.

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 651,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.5 to you. Your visible payoff was \$6.5.

Your total visible earnings across all rounds are \$6.5.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.9 (about 58% of their endowment), showing considerable trust
- I received \$8.7 after the 3x multiplier
- In Round 1, when I sent \$3, my previous partner returned \$4.5 (50% of what they received)
- Applying a similar reciprocity norm seems fair

**\*\*Decision:\*\***

I'll return approximately 50% of what I received, which rewards their trust and establishes a cooperative pattern. This leaves us both better off than our starting positions.

'''json

```
{"return": 4.5}
```

'''

**reasoning:**

Let me analyze this situation:

**\*\*Previous Round (Round 1):\*\***

- I was SENDER
- I sent \$3.0
- It became \$9.0

- They returned \$4.5 to me
- My payoff:  $\$5 - \$3 + \$4.5 = \$6.5$
- Their payoff:  $\$9 - \$4.5 = \$4.5$

That was a pretty fair return - they returned half of what they received, which is a common reciprocity norm.

**\*\*Current Round (Round 2):\*\***

- I am RECEIVER
- They sent me \$2.9 (close to \$3)
- I received \$8.7 ( $2.9 \times 3$ )
- I need to decide how much to return (0-8.7)

**\*\*Considerations:\*\***

1. My co-player sent me almost \$3 (actually \$2.9, but there's communication noise mentioned)
2. In Round 1, when I sent \$3, my partner returned \$4.5 (which was 50% of the \$9 they received)
3. If I apply the same 50% rule: I should return about \$4.35 (half of \$8.7)
4. My co-player took a risk by sending \$2.9, showing trust
5. Being reciprocal could establish good cooperation
6. Note: This is a different co-player than Round 1 (pairings are randomized)

Since this is a new co-player and they showed trust by sending \$2.9, I should reciprocate fairly. Returning around 50% seems like a reasonable reciprocity norm that:

- Rewards their trust
- Leaves both of us better off than not playing
- Establishes me as a cooperative player

My co-player's payoff if I return \$4.35:

- They get:  $\$5 - \$2.9 + \$4.35 = \$6.45$
- I get:  $\$8.7 - \$4.35 = \$4.35$

They sent 58% of their endowment, which is quite generous. I should return slightly more to match their generosity and encourage continued cooperation. Returning \$4.5 mirrors what my previous partner did for me and feels like the right balance...

**usage:**

```
{
 "input_tokens": 811,
 "output_tokens": 781,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.9, it became \$8.7, and you returned \$4.5. Your visible payoff was \$4.1999999999999999.

Your total visible earnings across all rounds are \$10.7.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

**\*\*Analysis:\*\***

- My co-player sent \$2.27 (about 45% of their endowment), showing moderate trust
- I received \$6.81 after the multiplier
- In previous rounds, I've been returning approximately 50% of what I receive as a reciprocity norm
- This approach has worked well so far

**\*\*Decision:\*\***

I'll maintain consistency and return about 50% of what I received, which rewards their trust appropriately.

```
'''json
{"return": 3.4}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Current situation:\*\***

- I'm the RECEIVER again
- My co-player sent \$2.27 (about 45% of their \$5 endowment)
- I received \$6.81 after the 3x multiplier
- They kept \$5 - \$2.27 = \$2.73

**\*\*Past behavior:\*\***

- Round 1: I sent \$3, received back \$4.5 (50% return rate on the multiplied amount)
- Round 2: I received \$8.7, returned \$4.5 (about 52% return rate)

**\*\*Considerations:\*\***

- My co-player sent a moderate amount (\$2.27), showing some trust but less than the previous player
- Being consistent with my previous behavior would mean returning around 50% of what I received
- 50% of \$6.81 = \$3.405

**\*\*Payoff implications:\*\***

- If I return \$3.4-3.5:
  - I keep: \$6.81 - \$3.4 = \$3.41
  - They get: \$2.73 (kept) + \$3.4 (returned) = \$6.13
  - This seems fair - we both do better than the \$5 starting point

I should maintain consistency with my reciprocity pattern from Round 2. Returning about 50% seems fair and builds trust for potential future rounds.

Let me return \$3.4 (just under 50%).

**usage:**

```
{
 "input_tokens": 1105,
 "output_tokens": 467,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.27, it became \$6.8100000000000005, and you returned \$3.4. Your visible payoff was \$3.4100000000000006.

Your total visible earnings across all rounds are \$14.11.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

**\*\*Analysis:\*\***

- This is Round 4 with a new co-player
- In Round 1, I sent \$3 and received \$4.5 back - a positive outcome
- As a receiver, I've been returning about 50%, establishing a cooperative pattern
- My current co-player is unknown to me

**\*\*Decision:\*\***

I'll send \$3 again. This amount has worked well before - it shows trust and creates potential for mutual gain through the 3x multiplier, while managing downside risk.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze the situation:

**\*\*What I know:\*\***

- Round 1: I was SENDER, sent \$3, received back \$4.5, payoff \$6.5
- Round 2: I was RECEIVER, got \$8.7 (from \$2.9 sent), returned \$4.5, payoff \$4.2
- Round 3: I was RECEIVER, got \$6.81 (from \$2.27 sent), returned \$3.4, payoff \$3.41

- Total earnings: \$14.11

**\*\*Current situation:\*\***

- Round 4: I'm the SENDER with \$5
- This is a new co-player (different from rounds 1-3)

**\*\*Strategic considerations:\*\***

1. I've seen one positive interaction as sender (sent \$3, got back \$4.5)
2. As receiver, I've been fairly generous, returning about 50%
3. This creates a reputation for reciprocity if others know
4. But agents don't know who I am (names hidden), so reputation effects are limited
5. However, the pattern of behavior in the pool might matter

**\*\*Decision factors:\*\***

- Sending creates potential gains (3x multiplier)
- Risk: this is a new co-player, no direct history
- But overall, cooperation seems to be working in this pool
- My Round 1 strategy (sending \$3) worked well

I'll send \$3 again - it's a moderate trust signal that has proven effective.

**usage:**

```
{
 "input_tokens": 1352,
 "output_tokens": 472,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.48 to you. Your visible payoff was \$6.48.

Your total visible earnings across all rounds are \$20.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

**\*\*Analysis:\*\***

- Round 5 with a new co-player
- My \$3 sending strategy has been successful:
  - Round 1: Sent \$3, received \$4.5 back ? \$6.5 payoff
  - Round 4: Sent \$3, received \$4.48 back ? \$6.48 payoff
- Both returns were close to 50%, indicating fair reciprocity from my co-players
- This amount balances trust-building with risk management

**\*\*Decision:\*\***

I'll continue with \$3, as it's been consistently effective.

```
'''json
{"send": 3}
'''
```

**reasoning:**

**\*\*Analysis:\*\***

- Round 5, new co-player as sender
- My sending strategy of \$3 has worked well:
  - Round 1: Sent \$3, received \$4.5 back, payoff \$6.5
  - Round 4: Sent \$3, received \$4.48 back, payoff \$6.48
- Both times I received close to 50% back, which is fair reciprocity
- Total visible earnings: \$20.59 across 4 rounds (average \$5.15/round)
- As a receiver, I've been consistently returning about 50%

**\*\*Decision:\*\***

The \$3 sending strategy is working consistently well. My co-players have been reciprocating fairly. I should continue this approach.

```
'''json
{"send": 3}
'''
```

```

'''
usage:
{
 "input_tokens": 1601,
 "output_tokens": 339,
 "reasoning_tokens": 0
}
Message 12 role=user
content:
 Round 6

 In your most recent previous game round, you were the SENDER against another agent. You sent $3.0, it became $9.0, and you saw them return $6.77 to you. Your visible payoff was $8.77.
 Your total visible earnings across all rounds are $29.36.

 This round, you are the RECEIVER against your current co-player. They sent you $3.68, so you received $11.040000000000001. How much do you return? (0-11.040000000000001)
 Respond exactly as JSON: {'return': <amount>}
Assistant output:
Analysis:
- My co-player sent $3.68 (about 74% of their endowment), showing strong trust
- I received $11.04 after the multiplier
- They're taking a significant risk by sending such a high amount
- I've been maintaining a 50% return pattern, which has worked well

Decision:
I'll return approximately 50% of what I received to reward their trust appropriately.

'''json
{"return": 5.5}
'''
Reasoning:
Analysis:
- My co-player sent $3.68 (about 74% of their endowment), showing strong trust
- I received $11.04 after the 3x multiplier
- In Round 5, my co-player returned $6.77 from $9, which is about 75% - very generous!
- I've been returning about 50% as a norm, but this co-player sent a substantial amount

Decision:
My co-player showed considerable trust by sending $3.68. I should reciprocate fairly. Following my 50% return pattern would give them about $5.50 back.

Let me calculate:
- They sent $3.68, keeping $1.32
- I received $11.04
- If I return $5.50:
 - They get: $1.32 + $5.50 = $6.82 (better than their $5 starting)
 - I get: $11.04 - $5.50 = $5.54 (better than my $0 starting as receiver)

This seems fair and maintains my reciprocity norm.
Usage: {
 "input_tokens": 1890,
 "output_tokens": 379,
 "reasoning_tokens": 0
}

GAME - Agent_2 - Sender - dyad_2
Timestamp: 2026-08-12T09:24:05
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_2",
 "Agent_4"
]
 }
}

```

```

],
"investor": "Agent_2",
"roles": {
 "Agent_2": "investor",
 "Agent_4": "trustee"
},
"trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative tone while managing some risk given the uncertainty.

If I send money, it triples in value, creating potential mutual gains. However, I don't know



ow my co-player yet, and there's communication noise to consider.

I'll send a moderate amount to signal trust and willingness to cooperate, while keeping so me buffer:

**\*\*Sending \$3\*\***

This gives my co-player \$9 to work with. If they return a fair amount (around half or more), we both benefit compared to me sending nothing. It's a balance between demonstrating cooperation and managing first-round uncertainty.

{'send': 3}

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think about this strategically:

1. I start with \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. My final payoff = \$5 - sent + returned
5. Receiver's final payoff = (sent × 3) - returned

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise (amounts may differ from what was actually sent)
- Agent names are hidden

From a purely selfish perspective, if I send \$0, I guarantee myself \$5. But this doesn't build trust or allow for mutual gains.

If I send everything (\$5), and the receiver returns a fair amount (like half of what they receive = \$7.50), I'd end up with \$7.50, and they'd keep \$7.50 as well. We'd both be better off.

However, there's risk:

- The receiver might not return anything
- Communication noise could cause misunderstandings
- This is just round 1, so I don't have reputation information

A moderate approach might be to send \$3:

- They receive \$9
- If they return \$5, I get \$5 (break even on sent, but signal trust)
- If they return \$6, I get \$6 (gain \$1)
- They keep between \$3-\$4 depending on return

Or I could send \$2:

- They receive \$6
- If they return \$3, I get \$3, netting -\$2 but we both gained
- If they return \$4, I get \$4, netting -\$1
- Lower risk

Given it's round 1 and I want to establish cooperative play but also manage risk, I'll send \$3. ...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 598,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.41 to you. Your visible payoff was \$6.41.

Your total visible earnings across all rounds are \$6.41.

This round, you are the RECEIVER against your current co-player. They sent you \$1.73, so y

ou received \$5.189999999999995. How much do you return? (0-5.189999999999995)  
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent \$1.73 (became \$5.19 for me)
- They kept \$3.27
- They showed some trust by sending money

**\*\*My reasoning:\*\***

- In the previous round, my co-player returned about 49% of what they received (\$4.41 out of \$9)
- I should be reasonably cooperative to encourage future trust
- Returning around 45-50% of what I received seems fair
- This would be \$2.33-\$2.60

I'll return \$2.40:

- They'll end with:  $\$3.27 + \$2.40 = \$5.67$
- I'll end with:  $\$5.19 - \$2.40 = \$2.79$
- They gain \$0.67 from sending, which rewards their trust

{'return': 2.40}

reasoning:

Let me think about this situation:

1. In the previous round, I sent \$3, which became \$9 for the receiver. They returned \$4.41 to me.
  - I ended with:  $\$5 - \$3 + \$4.41 = \$6.41$
  - They ended with:  $\$9 - \$4.41 = \$4.59$
  - So they returned about 49% of what they received, which seems fairly cooperative.
2. Now I'm the RECEIVER. My co-player sent me \$1.73, which became approximately \$5.19.
  - They now have:  $\$5 - \$1.73 = \$3.27$
  - I have: \$5.19
  - Total pie:  $\$3.27 + \$5.19 = \$8.46$
3. There's communication noise, so what I see (\$1.73 sent) might not be exactly what they sent.
4. The sender took some risk by sending me \$1.73. It's less than what I sent in the previous round (\$3), but still a meaningful amount showing some trust.
5. I should consider:
  - Being reciprocal/cooperative to encourage future cooperation
  - The communication noise factor
  - What would be "fair"

If I think about fairness:

- They sent \$1.73, keeping \$3.27
- I received \$5.19
- If we aim for roughly equal outcomes:  $(\$3.27 + \text{return}) ? (\$5.19 - \text{return})$
- That would suggest:  $\$3.27 + \text{return} = \$5.19 - \text{return}$
- $2 * \text{return} = \$5.19 - \$3.27 = \$1.92$
- $\text{return} ? \$0.96$

But that seems quite low given they took a risk. Let me think about proportional return:

- Matching their 49% return rate would mean sending back \$2.54
- Alternatively, if I return \$2.50, they'd end with \$5.77 (a \$0.77 gain) while I'd have \$2.69
- This feels like a balanced cooperative move that rewards their trust without leaving me too disadvantaged

I'll return \$2.50....

usage:

```
{
 "input_tokens": 791,
 "output_tokens": 1107,
 "reasoning_tokens": 0
}
```

}

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.73, it became \$5.189999999999995, and you returned \$2.4. Your visible payoff was \$2.789999999999996.

Your total visible earnings across all rounds are \$9.2.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my situation:

**\*\*Previous rounds:\*\***

- Round 1: Sent \$3, received \$4.41 back - good cooperation
- Round 2: Received \$5.19, returned \$2.40 - showed I reciprocate

**\*\*Current round:\*\***

- New co-player (randomized pairing)
- Need to balance trust with uncertainty

Since I had success with sending \$3 in Round 1 (got \$4.41 back, netting \$1.41 profit), and I want to maintain a cooperative approach while testing this new partner, I'll send the same amount:

**\*\*Sending \$3\*\***

This creates \$9 for them to work with. If they're reasonably cooperative like my Round 1 partner, we'll both benefit.

{'send': 3}

**reasoning:**

Let me analyze this situation:

Round 1: I was SENDER

- I sent \$3
- It became \$9
- They returned \$4.41
- My payoff:  $\$5 - \$3 + \$4.41 = \$6.41$

Round 2: I was RECEIVER

- They sent \$1.73
- I received \$5.19
- I returned \$2.40
- My payoff:  $\$5.19 - \$2.40 = \$2.79$
- Total so far:  $\$6.41 + \$2.79 = \$9.20$  ?

Round 3: I am SENDER again

- This is a new co-player (since pairings are randomized each round)
- I have \$5 to send

Strategy considerations:

1. I had a good experience in Round 1 - sent \$3, got back \$4.41 (profit of \$1.41)
2. In Round 2, I received \$5.19 and returned \$2.40 (46%), showing I'm cooperative
3. This is a new co-player, so I don't know their behavior
4. There's communication noise, so what they see may differ from what I intend

I should probably send a moderate amount again to:

- Signal willingness to cooperate
- Test this new partner's trustworthiness
- Not risk everything on an unknown player

Sending \$3 worked well in Round 1. I could try that again, or maybe send slightly more (\$3.50) or less (\$2.50) to experiment.

Given that:

- My first experience was positive
- I want to maintain a cooperative reputation
- But this is a new, unknown player

I'll send \$3 again - it's a balanced approach that showed good results before.

```
{'send': 3}
```

**usage:**

```
{
 "input_tokens": 1128,
 "output_tokens": 608,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.71 to you. Your visible payoff was \$5.71.

Your total visible earnings across all rounds are \$14.91.

This round, you are the RECEIVER against your current co-player. They sent you \$3.79, so you received \$11.370000000000001. How much do you return? (0-11.370000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent \$3.79 out of \$5 (~76% of their endowment)
- This became \$11.37 for me
- They showed strong trust by sending most of their money

**\*\*My reasoning:\*\***

- They took a big risk trusting me
- I should reward this trust to encourage cooperation
- Returning around 45% of what I received seems appropriate given their high trust
- This would be approximately \$5.10

**\*\*Calculating outcomes:\*\***

- They'll end with: \$1.21 + \$5.10 = \$6.31
- I'll end with: \$11.37 - \$5.10 = \$6.27
- They gain \$1.31 from their trust, which is a good reward

```
{'return': 5.10}
```

**reasoning:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent \$3.79 (became \$11.37 for me)
- They kept \$1.21
- They showed significant trust by sending most of their money

**\*\*Previous experience:\*\***

- Round 1: I sent \$3, got back \$4.41 (they returned ~49%)
- Round 2: I received \$5.19, returned \$2.40 (~46%)
- Round 3: I sent \$3, got back \$3.71 (they returned ~41%)

**\*\*My reasoning:\*\***

- This co-player sent \$3.79 out of \$5, showing strong trust (~76%)
- They're risking a lot and trusting me to reciprocate
- I should reward this trust appropriately
- Returning around 40-50% of what I received would be \$4.55-\$5.69
- Let me think about fairness:
  - They sent \$3.79, kept \$1.21
  - I received \$11.37
  - If I return \$5.00:
    - They end with: \$1.21 + \$5.00 = \$6.21
    - I end with: \$11.37 - \$5.00 = \$6.37

- If I return \$5.50:
- They end with:  $\$1.21 + \$5.50 = \$6.71$
- I end with:  $\$11.37 - \$5.50 = \$5.87$

Given they sent \$3.79 (76% of their endowment), I should be generous but not overdo it. Around 45% return rate seems appropriate, which would be about \$5.12.

Let me go with \$5.10:

...

**usage:**

```
{
 "input_tokens": 1453,
 "output_tokens": 690,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.79, it became \$11.370000000000001, and you returned \$5.1. Your visible payoff was \$6.270000000000001.

Your total visible earnings across all rounds are \$21.18.

This round, you are the RECEIVER against your current co-player. They sent you \$2.91, so you received \$8.73. How much do you return? (0-8.73)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent \$2.91 out of \$5 (~58% of their endowment)
- This became \$8.73 for me
- They kept \$2.09
- This shows moderate to good trust

**\*\*My reasoning:\*\***

- They demonstrated decent trust by sending over half their money
- I should reciprocate consistently with my previous rounds
- In previous rounds as receiver, I've returned around 45% of what I received
- For \$8.73, that would be approximately \$3.90

**\*\*Calculating outcomes:\*\***

- They'll end with:  $\$2.09 + \$3.90 = \$5.99$
- I'll end with:  $\$8.73 - \$3.90 = \$4.83$
- They gain about \$1 from trusting me, which is a reasonable reward

{'return': 3.90}

**reasoning:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent \$2.91 out of \$5 (~58% of their endowment)
- This became \$8.73 for me
- They kept \$2.09
- This shows moderate to good trust

**\*\*My reasoning:\*\***

- They sent a decent amount, showing trust
- I should reciprocate proportionally
- Looking at my previous returns:
  - Round 2: Received \$5.19, returned \$2.40 (~46%)
  - Round 4: Received \$11.37, returned \$5.10 (~45%)
- I've been returning around 45% consistently

For \$8.73, 45% would be about \$3.93

**\*\*Calculating outcomes with \$3.93:\*\***

- They'll end with:  $\$2.09 + \$3.93 = \$6.02$

- I'll end with:  $\$8.73 - \$3.93 = \$4.80$
- They gain  $\$1.02$  from sending, which rewards their trust

This seems fair and consistent with my previous behavior. I'll return  $\$3.90$  to keep it simple.

Actually, let me recalculate with  $\$3.90$ :

- They'll end with:  $\$2.09 + \$3.90 = \$5.99$
- I'll end with:  $\$8.73 - \$3.90 = \$4.83$
- They gain  $\$0.99$  from sending

That works well.

**usage:**

```
{
 "input_tokens": 1786,
 "output_tokens": 560,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send  $\$2.91$ , it became  $\$8.73$ , and you returned  $\$3.9$ . Your visible payoff was  $\$4.83$ .

Your total visible earnings across all rounds are  $\$26.009999999999998$ .

This round, you are the SENDER against your current co-player. You have  $\$5$ . How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my history as SENDER:

**\*\*Previous rounds as sender:\*\***

- Round 1: Sent  $\$3$  ? received  $\$4.41$  back (profit of  $\$1.41$ )
- Round 3: Sent  $\$3$  ? received  $\$3.71$  back (profit of  $\$0.71$ )

Both times sending  $\$3$  worked out profitably, showing it's a reasonable cooperative strategy.

**\*\*Current situation:\*\***

- New co-player (randomized pairing)
- Want to maintain consistent cooperative approach
- $\$3$  creates  $\$9$  for them, leaving room for mutual benefit

I'll stick with my established strategy of sending  $\$3$ , which has proven effective:

```
{'send': 3}
```

**Reasoning:**

Looking at my history:

Round 1 (SENDER): Sent  $\$3$ , received  $\$4.41$  back, payoff  $\$6.41$   
 Round 2 (RECEIVER): Received  $\$5.19$ , returned  $\$2.40$ , payoff  $\$2.79$   
 Round 3 (SENDER): Sent  $\$3$ , received  $\$3.71$  back, payoff  $\$5.71$   
 Round 4 (RECEIVER): Received  $\$11.37$ , returned  $\$5.10$ , payoff  $\$6.27$   
 Round 5 (RECEIVER): Received  $\$8.73$ , returned  $\$3.90$ , payoff  $\$4.83$

Total earnings:  $\$26.01$

Now Round 6, I'm the SENDER with a new randomized co-player.

My strategy has been to send  $\$3$  as sender (both times I did this), which has worked out reasonably well:

- Round 1: Sent  $\$3$ , got  $\$4.41$  back (net  $+\$1.41$ )
- Round 3: Sent  $\$3$ , got  $\$3.71$  back (net  $+\$0.71$ )

Both times I profited, though the return varied.

As receiver, I've been consistent in returning around 45% of what I receive, which seems fair.

For this new round with a new partner, I should continue with my cooperative strategy. Sending

\$3 has been my standard approach and has generally worked well. It signals cooperation while not being overly risky.

I'll send \$3 again.

```
Usage: {
 "input_tokens": 2117,
 "output_tokens": 509,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_4 - Receiver - dyad\_2**

Timestamp: 2026-08-12T09:24:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_2",
 "pairing": {
 "agents": [
 "Agent_2",
 "Agent_4"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_2": "investor",
 "Agent_4": "trustee"
 },
 "trustee": "Agent_4"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

**Message 1 role=system**

##### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### **GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### **MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.  
The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance trust-building with risk management.

Since I don't have information about my co-player's behavior yet, I'll send a moderate amount that:

- Demonstrates willingness to cooperate
- Creates opportunity for mutual gain (the tripling effect means both of us can benefit)
- Doesn't expose me to total loss

If I send \$3, my co-player receives \$9 ( $3 \times \$3$ ). If they return even \$4, we'd both end up better off than our starting positions - I'd have \$6 ( $\$5 - \$3 + \$4$ ) and they'd have \$5.

This seems like a reasonable opening move to establish cooperative intent.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is Round 1 of a trust game where I'm the SENDER. Let me think through the strategic considerations:

1. I have \$5 to decide how to allocate
2. Whatever I send gets tripled when it reaches the receiver
3. The receiver then decides how much to return to me
4. My final payoff =  $\$5 - \text{amount\_sent} + \text{amount\_returned}$

The key strategic considerations:

- If I send nothing (\$0), I keep \$5 and the receiver gets nothing. Safe but no potential for gain.
- If I send everything (\$5), the receiver gets \$15. If they return even \$6, I'd be better off ( $\$5 - \$5 + \$6 = \$6$ ).
- This is a one-shot game with this particular partner (though I'll play multiple rounds with different partners).
- There's communication noise mentioned, which means the receiver might see a different amount than what I actually send.

Given this is the first round and I don't have information about the other agents' behavior yet:

Strategic approaches:

1. **Trust-building**: Send a moderate-to-high amount to signal cooperation and trust
2. **Conservative**: Send a small amount to minimize risk
3. **Optimal cooperation**: Send \$5 if I expect the receiver to split gains fairly

If the receiver is rational and fair, they might follow a norm like returning half of what they receive, or returning enough to make both players better off than the initial endowment.

For example, if I send \$3:

- I have:  $\$5 - \$3 = \$2$  left
- Receiver gets:  $\$3 \times 3 = \$9$
- If receiver returns \$4: I end with  $\$2 + \$4 = \$6$ , they keep \$5



- Both better off than initial \$5

If I send \$5:

- I have: \$0 left
- Receiver gets: \$15
- If receiver returns \$8: I end with \$8, they keep \$7
- Both significantly better off

The noise factor complicates things-if I send \$5 but it appears as less, the receiver might return proportionally less. Based on experimental economics patterns where senders typically send around 50% and receivers often return amounts that create mutual gains, I'll send \$3. This signals cooperation while maintaining some cushion, and if the receiver follows typical reciprocity norms, we should both end up better off than our starting positions....

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 841,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.8 to you. Your visible payoff was \$5.8.

Your total visible earnings across all rounds are \$5.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.56, so you received \$7.68. How much do you return? (0-7.68)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.56 (about 51% of their endowment), showing moderate trust
- I received \$7.68 after the 3x multiplication
- They currently have \$2.44 remaining

**\*\*Reciprocity consideration:\*\***

- In Round 1, my co-player returned about 42% of what they received
- I want to be fair and encourage continued cooperation
- Returning around 45-50% seems reasonable

If I return \$3.50:

- Sender ends with:  $\$2.44 + \$3.50 = \$5.94$  (gain of \$0.94)
- I keep:  $\$7.68 - \$3.50 = \$4.18$  (gain of \$4.18)

This rewards their trust while keeping a fair share given the multiplication advantage.

```
'''json
{"return": 3.5}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was SENDER, sent \$3
- It became \$9 for receiver
- They returned \$3.8 to me
- My payoff:  $\$5 - \$3 + \$3.8 = \$5.8$
- Their payoff:  $\$9 - \$3.8 = \$5.2$

So in Round 1, my co-player returned about 42% of what they received ( $\$3.8/\$9 \approx 0.42$ ).

Round 2:

- I am RECEIVER
- Sender sent \$2.56

- I received \$7.68 ( $2.56 \times 3$ )
- I need to decide how much to return

The sender invested \$2.56, so they currently have  $\$5 - \$2.56 = \$2.44$  remaining.

If I think about reciprocity and building trust:

- The sender showed moderate trust by sending about 51% of their endowment
- If I want to encourage cooperation, I should return enough that the sender comes out ahead

Let me consider different return amounts:

- If I return \$4: Sender gets  $\$2.44 + \$4 = \$6.44$  (net gain of \$1.44), I keep \$3.68 (net gain of \$3.68)
- If I return \$3.5: Sender gets  $\$2.44 + \$3.5 = \$5.94$  (net gain of \$0.94), I keep \$4.18 (net gain of \$4.18)
- If I return \$3: Sender gets  $\$2.44 + \$3 = \$5.44$  (net gain of \$0.44), I keep \$4.68 (net gain of \$4.68)

In my previous round, the receiver returned about 42% of what they received. If I apply a similar principle:

$\$7.68 \times 0.42 \approx \$3.23$

I'm inclined to return slightly more than that baseline to reward their trust and establish a cooperative pattern. Returning \$3.5 seems like the right balance-it gives them a meaningful profit while still letting me benefit substantially, and it signals I'm reliable without being exploited....

usage:

```
{
 "input_tokens": 804,
 "output_tokens": 923,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.56, it became \$7.68, and you returned \$3.5. Your visible payoff was \$4.18.

Your total visible earnings across all rounds are \$9.98.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

**\*\*Analysis:\*\***

I'm now facing a new co-player (Round 3, new pairing). My experience so far:

- Round 1: Sent \$3, received back \$3.8 (42% return rate)
- Round 2: Received \$7.68, returned \$3.5 (46% return rate)

**\*\*Strategy:\*\***

Since I'm building a reputation in this multi-agent environment, I want to signal cooperative behavior. I'll send slightly more than last time to:

- Test if higher trust leads to better reciprocity
- Create more value through the 3x multiplier
- Signal that I'm willing to be generous

Sending \$3.5 means my co-player receives \$10.50. If they return around 40%, I'd get back ~\$4.20, giving me a net gain. This balances trust-building with reasonable risk.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the situation:

**\*\*Round 1:\*\*** I was SENDER

- Sent: \$3
- They received: \$9

- They returned: \$3.8
- My payoff: \$5.8

**\*\*Round 2:\*\*** I was RECEIVER

- They sent: \$2.56
- I received: \$7.68
- I returned: \$3.5
- My payoff: \$4.18

**\*\*Round 3:\*\*** I am SENDER again

Now I need to think about this new co-player in Round 3. Note that each round I'm paired with potentially different agents from the pool of 8 agents.

In Round 1, my co-player returned \$3.8 out of \$9 (42.2%).

In Round 2, I returned \$3.5 out of \$7.68 (45.6%).

For Round 3, I'm facing a NEW co-player (since pairings are randomized each round). I don't have specific history with this particular agent.

Given:

1. I've seen one example of a co-player returning about 42%
2. I returned about 46% myself, trying to be fair
3. The game has communication noise, so amounts may differ slightly

Strategy for Round 3:

- I should continue with a cooperative but cautious approach
- Sending around \$3 worked reasonably in Round 1
- I might try a slightly different amount to gather more information

Let me send \$3 again as it seems to be a reasonable middle ground that:

- Shows trust
- Creates significant value through multiplication (becomes \$9)
- Leaves me with \$2 as a safety net
- If they return around 40-45%, I'd get back around \$3.6-\$4, giving me a decent payoff

Actually, I could also consider sending a bit more to test if higher trust leads to better returns. Let me send \$3.5 this time.

...

**usage:**

```
{
 "input_tokens": 1137,
 "output_tokens": 766,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.07 to you. Your visible payoff was \$7.57.

Your total visible earnings across all rounds are \$17.55.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

**\*\*Analysis:\*\***

My co-player sent \$2.9 (58% of their endowment), showing significant trust. This is a strong cooperative signal.

**\*\*Previous patterns:\*\***

- Round 1 sender: Got back 42% return
- Round 2 receiver: I returned 46%
- Round 3 sender: Got back 58% return (very generous!)

**\*\*Current situation:\*\***

- I received \$8.7
- Sender has \$2.1 remaining

**\*\*Decision:\*\***

My co-player showed high trust by sending 58%. I should reciprocate proportionally. Returning 50% of what I received (\$4.35) is fair:

If I return \$4.35:

- Sender ends with:  $\$2.1 + \$4.35 = \$6.45$  (profit of \$1.45)
- I keep:  $\$8.7 - \$4.35 = \$4.35$  (profit of \$4.35)

This rewards their trust and maintains cooperative momentum.

```
'''json
{"return": 4.35}
'''
```

**reasoning:**

**\*\*Analysis of the situation:\*\***

Round 4, I'm the RECEIVER:

- My co-player sent \$2.9 (58% of their endowment)
- I received \$8.7 after the 3x multiplication
- They currently have \$2.1 remaining (\$5 - \$2.9)

**\*\*Previous experience:\*\***

- Round 1 (Sender): Sent \$3, got back \$3.8 (42% of \$9)
- Round 2 (Receiver): Received \$7.68, returned \$3.5 (46%)
- Round 3 (Sender): Sent \$3.5, got back \$6.07 (58% of \$10.5) - this was very generous!

**\*\*Reciprocity:\*\***

My Round 3 co-player was very generous with a 58% return rate. This current co-player sent 58% of their endowment, showing significant trust.

Looking at patterns:

- When I received \$7.68 in Round 2, I returned \$3.5 (46%)
- Now I received \$8.7

I should be fair and encouraging. If I return around 45-50%:

- 45% would be \$3.92
- 50% would be \$4.35

Let me calculate outcomes:

If I return \$4.00:

- Sender ends with:  $\$2.1 + \$4.00 = \$6.10$  (gain of \$1.10)
- I keep:  $\$8.7 - \$4.00 = \$4.70$  (gain of \$4.70)

If I return \$4.35:

- Sender ends with:  $\$2.1 + \$4.35 = \$6.45$  (gain of \$1.45)

...

**usage:**

```
{
 "input_tokens": 1485,
 "output_tokens": 766,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.9, it became \$8.7, and you returned \$4.35. Your visible payoff was \$4.35.

Your total visible earnings across all rounds are \$21.9.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

**\*\*Analysis:\*\***

New round, new co-player. My track record:

- As sender: Sent \$3 (got 42% back), sent \$3.5 (got 58% back)
- As receiver: Returned 46% and 50%

**\*\*Strategy:\*\***

The higher send amount (\$3.5) yielded much better reciprocity in Round 3. This suggests that showing more trust encourages cooperation. I'll continue with this amount to:

- Signal strong cooperative intent
- Create significant value (\$10.50 for receiver)
- Build on the positive pattern

Given my fair behavior as a receiver, I'm hopeful this will be reciprocated.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Looking at my history:

- Round 1 (SENDER): Sent \$3, got back \$3.8 (42% return) - payoff \$5.8
- Round 2 (RECEIVER): Received \$7.68, returned \$3.5 (46% return) - payoff \$4.18
- Round 3 (SENDER): Sent \$3.5, got back \$6.07 (58% return) - payoff \$7.57
- Round 4 (RECEIVER): Received \$8.7, returned \$4.35 (50% return) - payoff \$4.35

Now Round 5 - I'm the SENDER against a new co-player.

Key observations:

1. When I sent more (\$3.5 vs \$3), I got a better return (58% vs 42%)
2. I've been generally fair as a receiver (46-50% returns)
3. My total earnings are \$21.9, averaging about \$5.48 per round - solid profit

Strategy considerations:

- I'm building a reputation across multiple agents
- Higher sends have been rewarded so far
- The 3x multiplier means sending more creates more total value
- I should continue signaling cooperation

I'll send \$3.5 or \$4 this round. Let me think:

- Sending \$4 creates \$12 for receiver, shows high trust
- If they return 50%, that's \$6, giving me \$7 total (\$5-\$4+\$6 = \$7)
- If they return 40%, that's \$4.8, giving me \$5.8 (\$5-\$4+\$4.8)

Actually, let me continue with \$3.5 since that worked well in Round 3. It's a solid cooperative amount without being overly risky.

**usage:**

```
{
 "input_tokens": 1851,
 "output_tokens": 596,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.36 to you. Your visible payoff was \$7.86.

Your total visible earnings across all rounds are \$29.759999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

**\*\*Analysis:\*\***

My co-player sent \$2.27 (about 45% of their endowment), showing moderate trust.

**\*\*Current situation:\*\***

- I received \$6.81

```

- Sender has $2.73 remaining

My receiver pattern:
- Round 2: Returned 46% of received
- Round 4: Returned 50% of received

Decision:
I'll maintain fair reciprocity. Returning about 48% of what I received:

If I return $3.25:
- Sender ends with: $2.73 + $3.25 = $5.98 (small profit)
- I keep: $6.81 - $3.25 = $3.56

This rewards their moderate trust fairly.

'''json
{"return": 3.25}
'''
Reasoning:
Analysis:

My co-player sent $2.27 (about 45% of their $5 endowment), showing moderate trust.

Previous patterns as receiver:
- Round 2: Received $7.68, returned $3.5 (46% of received)
- Round 4: Received $8.7, returned $4.35 (50% of received)

Current situation:
- I received $6.81
- Sender has $2.73 remaining

Decision:
My co-player showed moderate trust (45% sent). I should reciprocate fairly. They sent less than in previous rounds, so I'll return proportionally.

Returning around 48% seems fair:
- 48% of $6.81 = $3.27

If I return $3.27:
- Sender ends with: $2.73 + $3.27 = $6.00 (profit of $1.00)
- I keep: $6.81 - $3.27 = $3.54 (profit of $3.54)

This is fair reciprocity that rewards their trust while keeping a reasonable share.
Usage: {
 "input_tokens": 2166,
 "output_tokens": 466,
 "reasoning_tokens": 0
}

GAME - Agent_6 - Sender - dyad_3
Timestamp: 2026-08-12T09:24:25
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_5"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_5": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_5"
 },
}

```

```

"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$2.97, that is 59.4% of their total endowment. You received \$8.91. How much do you return to the sender? (0-8.91)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$2.97 (about 60% of their endowment)
- I received \$8.91 after the 3x multiplier
- The sender currently has only \$2.03 remaining (\$5 - \$2.97)

The sender took a substantial risk and demonstrated trust. To reciprocate fairly and encourage future cooperation, I should return a reasonable amount that:

1. Rewards their trust
2. Leaves both of us better off than the initial endowment
3. Establishes a cooperative relationship for future rounds

Returning approximately half of what I received seems fair - this would give the sender a good return on their trust while still leaving me with a decent gain.

I'll return \*\*\$4.50\*\*:

- Sender ends with:  $\$2.03 + \$4.50 = \$6.53$  (gained \$1.53)
- I keep:  $\$8.91 - \$4.50 = \$4.41$  (gained \$4.41)

Both of us benefit, and this reciprocates their trust.

{'return': 4.50}

**reasoning:**

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$2.97 (59.4% of their \$5 endowment)
- I received \$8.91 (which is  $\$2.97 \times 3$ )
- I need to decide how much to return (between \$0 and \$8.91)

Let me think about what's fair and strategic:

1. The sender showed significant trust by sending almost 60% of their endowment
2. If I return nothing, I keep \$8.91 and the sender gets only \$2.03 ( $\$5 - \$2.97$ )
3. If I return about half (\$4.46), I keep \$4.45 and the sender gets \$6.49 ( $\$2.03 + \$4.46$ )
4. If I return \$5.94, I keep \$2.97 (equal to what they sent) and the sender gets \$8.97 total

The sender took a risk and showed trust. A common approach in trust games is reciprocity. The sender sent about 60% of their endowment, which is generous.

Let me consider what might establish good cooperation for future rounds:

- Returning about the same amount they sent (\$2.97) would mean they gain \$0 net
- Returning more than they sent but less than half of what I received seems reasonable
- Returning around 1/3 to 1/2 of what I received is often considered fair

Let me calculate some options:

- Return \$3: Sender gets \$5.03 total (gained \$0.03), I keep \$5.91
- Return \$4: Sender gets \$6.03 total (gained \$1.03), I keep \$4.91
- Return \$4.50: Sender gets \$6.53 total (gained \$1.53), I keep \$4.41
- Return \$5: Sender gets \$7.03 total (gained \$2.03), I keep \$3.91
- Return \$4.50 seems like the right balance-it reciprocates their trust by returning roughly half of what I received,...

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 898,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.97, it became \$8.91, and you returned \$4.5. Your visible payoff was \$4.41.

Your total visible earnings across all rounds are \$4.41.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my previous experience:

- As a receiver, a sender trusted me with about 60% of their endowment
- I reciprocated fairly, and we both benefited
- This established that cooperation can work well in this game

Now as the sender with a new co-player, I want to:

1. Demonstrate trust to encourage reciprocation
2. Follow a similar cooperative strategy that worked before
3. Give my co-player the opportunity to earn and share



I'll send \*\*\$3.00\*\* (60% of my endowment), mirroring the trust shown to me in the previous round.

```
{'send': 3.00}
```

**reasoning:**

Let me think about this situation:

In Round 1:

- I was the receiver
- The sender sent me \$2.97 (59.4% of their endowment)
- I received \$8.91
- I returned \$4.50
- I kept \$4.41
- The sender got back \$4.50, so they ended with  $\$2.03 + \$4.50 = \$6.53$

Now in Round 2:

- I'm the sender
- I have \$5 to potentially send
- I'm playing against a different co-player (the problem states pairings are randomized each round)

Key considerations:

1. The previous sender showed trust by sending 59.4% of their endowment
2. I reciprocated fairly by returning about 50% of what I received
3. This worked out well - both players benefited
4. Now I'm in the sender role with a new partner

Since my previous experience showed that sending a substantial amount can lead to good returns, and I want to establish trust with this new co-player, I should send a meaningful amount.

The previous sender sent about 60% (\$2.97 out of \$5). This seemed like a good trust-building amount. I could:

- Send a similar proportion to continue the cooperative pattern
- Send slightly more to show strong trust
- Send slightly less to be more cautious with a new partner

I think sending around \$3 (60%) makes sense - it mirrors what worked well in the previous round and shows trust to establish a good relationship with this new co-player.

Let me send \$3.00.

**usage:**

```
{
 "input_tokens": 908,
 "output_tokens": 516,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.32 to you. Your visible payoff was \$7.32.

Your total visible earnings across all rounds are \$11.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my recent experience:

Round 2 results:

- I sent \$3.00 (60% of endowment)
- Receiver returned \$5.32
- I ended with \$7.32 - excellent outcome!

The cooperative strategy is working very well. When I demonstrate trust, it's being reciprocated.

ocated generously.

Now facing a new co-player, I'll continue this successful approach. Sending around 60% has proven effective in building mutual benefit.

```
{'send': 3.00}
```

**reasoning:**

Let me analyze the situation:

Round 2 (just completed):

- I was the SENDER
- I sent \$3.00 (60% of endowment)
- It became \$9.00 for the receiver
- They returned \$5.32 to me
- My payoff:  $\$5 - \$3 + \$5.32 = \$7.32$
- They kept:  $\$9.00 - \$5.32 = \$3.68$

This was a very generous return! They returned more than half of what they received, and I ended up with more than my initial endowment even after sending a substantial amount.

My experience so far:

- Round 1 (RECEIVER): Received \$8.91, returned \$4.50, kept \$4.41
- Round 2 (SENDER): Sent \$3.00, received back \$5.32, ended with \$7.32

Total visible earnings: \$11.73

The cooperation strategy is working very well! When I sent 60% in Round 2, I got an excellent return.

Now in Round 3, I'm a SENDER again with a new co-player.

Given:

1. My cooperative strategy has been working well
2. Both times cooperation has led to mutual benefit
3. I want to continue encouraging trust and reciprocation
4. The game benefits from establishing cooperative norms

I should continue with a cooperative approach. Sending around 60% worked well last time. I could:

- Send \$3.00 again (same as last time, 60%)
- Send a bit more to show strong trust
- Send a bit less to be slightly more cautious with a new partner

Since cooperation has been consistently rewarded, I'll stick with the successful strategy of sending around 60%.

...

**usage:**

```
{
 "input_tokens": 1176,
 "output_tokens": 538,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$2.98 to you. Your visible payoff was \$4.98.

Your total visible earnings across all rounds are \$16.71.

This round, you are the RECEIVER against your current co-player. They sent you \$3.24, so you received \$9.72. How much do you return? (0-9.72)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation with my current co-player:

**\*\*Last round (Round 3):\*\***

- I sent them \$3.00

- They returned \$2.98 (~33% of what they received)
- This was a modest but acceptable return

**\*\*This round (Round 4):\*\***

- They sent me \$3.24 (slightly MORE than I sent them)
- I received \$9.72
- They currently have only \$1.76

The fact that they sent me MORE than I sent them suggests they're trying to build cooperation. I should reciprocate generously to:

1. Reward their increased trust
2. Establish a positive cooperative dynamic
3. Ensure we both benefit from ongoing interaction

I'll return **\*\*\$4.80\*\***:

- They end with:  $\$1.76 + \$4.80 = \$6.56$  (gain \$1.56)
- I keep:  $\$9.72 - \$4.80 = \$4.92$  (gain \$4.92)

This is more generous than they were to me, encouraging further cooperation.

`{'return': 4.80}`

#### **reasoning:**

Let me analyze this situation:

Round 3 (previous round):

- I was the SENDER
- I sent \$3.00 to this same co-player
- They returned \$2.98
- My payoff was \$4.98 (kept \$2 + returned \$2.98)

Round 4 (current round):

- I am the RECEIVER
- Same co-player (they are now the sender)
- They sent me \$3.24
- I received \$9.72

So this is interesting - we're playing repeated games with the same partner. In round 3, I sent \$3.00 and they returned \$2.98 (about 33% of what they received). Now they've sent me \$3.24 (notably a bit more than I sent them).

Let me think about the fairness here:

- When I sent \$3.00, they kept  $\$9.00 - \$2.98 = \$6.02$
- I kept  $\$2.00 + \$2.98 = \$4.98$
- So they gained \$1.02, I lost \$0.02

Now they sent \$3.24:

- They kept  $\$5 - \$3.24 = \$1.76$
- I received \$9.72

If I want to reciprocate similarly to how they treated me, I might return around 33% of what I received, which would be about \$3.21.

But actually, they sent me slightly MORE than I sent them (\$3.24 vs \$3.00), which could be seen as trying to establish cooperation or make up for the relatively low return last round.

Let me think about different return amounts:

Option 1: Return \$3.00 (about 31%)

- They end with:  $\$1.76 + \$3.00 = \$4.76$  (lost \$0.24)
- I keep:  $\$9.72 - \$3.00 = \$6.72$  (gain \$6.72)
- This feels too stingy given their slightly larger initial send.

Option 2: Return \$4.00 (about 41%)

- They end with:  $\$1.76 + \$4.00 = \$5.76$  (gain \$0.76)
- I keep:  $\$9.72 - \$4.00 = \$5.72$  (gain \$5.72)
- This creates mutual benefit and shows reciprocal trust.

Option 3: Return \$5.00 (about 51%)

- They end with:  $\$1.76 + \$5.00 = \$6.76$  (gain \$1.76)

- I keep:  $\$9.72 - \$5.00 = \$4.72...$

usage:

```
{
 "input_tokens": 1431,
 "output_tokens": 1067,
 "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.24, it became \$9.72, and you returned \$4.8. Your visible payoff was \$4.920000000000001.

Your total visible earnings across all rounds are \$21.630000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$2.32, so you received \$6.959999999999999. How much do you return? (0-6.959999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- New co-player sent me \$2.32 (about 46% of their endowment)
- I received \$6.96 after the multiplier
- They currently have \$2.68

They showed moderate trust by sending nearly half their endowment. I should reciprocate fairly to reward their trust and ensure mutual benefit.

Following my pattern of returning roughly 50% of what I receive, I'll return **\*\*\$3.48\*\***:

- Sender ends with:  $\$2.68 + \$3.48 = \$6.16$  (gain \$1.16)
- I keep:  $\$6.96 - \$3.48 = \$3.48$  (gain \$3.48)

Both of us benefit from the cooperation.

```
{'return': 3.48}
```

reasoning:

Looking at this round:

- New co-player (different from Round 4)
- They sent me \$2.32 (about 46% of their endowment)
- I received \$6.96
- They currently have \$2.68 ( $\$5 - \$2.32$ )

They showed moderate trust by sending nearly half their endowment. I should reciprocate fairly to:

1. Reward their trust
2. Ensure both of us benefit
3. Encourage cooperation

Looking at my previous patterns:

- Round 1: Received \$8.91, returned \$4.50 (about 50%)
- Round 4: Received \$9.72, returned \$4.80 (about 49%)

Returning around 50% seems like my consistent fair approach. For \$6.96, that would be around \$3.48.

Let me check the outcomes:

- Sender ends with:  $\$2.68 + \$3.48 = \$6.16$  (gain \$1.16)
- I keep:  $\$6.96 - \$3.48 = \$3.48$  (gain \$3.48)

Both benefit, and this reciprocates their trust fairly. I'll return \$3.48.

usage:

```
{
 "input_tokens": 1834,
 "output_tokens": 476,
 "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.32, it became \$6.959999999999999, and you returned \$3.48. Your visible payoff was \$3.4799999999999999.  
Your total visible earnings across all rounds are \$25.110000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my history with this current co-player:

**\*\*Round 5:\*\***

- They sent me \$2.32 (46% of endowment)
- I returned \$3.48, giving them a good outcome

They showed moderate trust, and I reciprocated fairly. Now it's my turn to send, and I should continue building this cooperative relationship.

I'll send **\*\*\$2.50\*\***, which is:

- Slightly more than they sent me (showing willingness to cooperate)
- A reasonable amount that matches their moderate approach
- Enough to create mutual benefit if they reciprocate

{'send': 2.50}

**Reasoning:**

Looking at my history with this current co-player:

Round 5 (last round):

- They sent me \$2.32 (46% of endowment)
- I received \$6.96
- I returned \$3.48 (50% of what I received)
- They ended with \$6.16, I ended with \$3.48

Now in Round 6, I'm the sender with this same co-player.

They showed moderate trust by sending \$2.32. I reciprocated fairly by returning \$3.48, which gave them a good outcome (\$6.16 total).

Since we had a successful cooperative interaction, I should continue building on this. I'll send a similar or slightly higher amount to maintain the cooperative relationship.

I've been generally sending around \$3.00 (60%) in other rounds, which has worked well. Given that this co-player sent me \$2.32 (46%), I could:

1. Send similar to what they sent (~\$2.30-2.50)
2. Send my standard amount (~\$3.00)
3. Send slightly more to build trust

I think sending around \$2.50-3.00 makes sense. Let me go with \$2.50, which is slightly more than they sent me, showing I'm willing to cooperate but matching their level of caution.

Actually, looking at my overall strategy, I've been doing well with \$3.00 sends. But this particular co-player was more cautious. Let me match their level at around \$2.30-2.50.

I'll send \$2.50.

Usage: {

"input\_tokens": 2153,  
"output\_tokens": 510,  
"reasoning\_tokens": 0

}

**GAME - Agent\_5 - Receiver - dyad\_3**

Timestamp: 2026-08-12T09:24:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_3",  
"move\_index": 5,

```

"opponent_id": "Agent_6",
"pairing": {
 "agents": [
 "Agent_6",
 "Agent_5"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_5": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$2.54, that is 50.8% of their total endowment. You received \$7.62. How much do you return to the sender? (0-7.62)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$2.54 (about half their endowment)
- I received \$7.62 after the 3x multiplier
- I want to reciprocate fairly while both of us benefit

The sender took a risk to create mutual gains. A fair return should:

1. Reward their trust
2. Leave us both better off than if nothing was sent
3. Show I'm a cooperative partner for future rounds

If I return \$3.50:

- Sender ends with:  $\$5.00 - \$2.54 + \$3.50 = \$5.96$  (net gain: \$0.96)
- I end with:  $\$7.62 - \$3.50 = \$4.12$  (net gain: \$4.12)

This seems like a fair reciprocation that rewards the sender's trust while acknowledging the receiver advantage in this game.

```
'''json
{"return": 3.50}
'''
```

**reasoning:**

Let me analyze this situation carefully:

- I am the RECEIVER in Round 1
- The sender sent \$2.54 (which is 50.8% of their \$5 endowment)
- After the 3x multiplier, I received \$7.62 (which is indeed  $\$2.54 \times 3$ )
- I need to decide how much to return (between \$0 and \$7.62)

Key considerations:

1. **\*\*The sender showed trust\*\***: They sent about half their endowment (\$2.54 out of \$5), showing significant trust in me.
2. **\*\*Fair reciprocation\*\***: A common approach in trust games is to return an amount that:
  - Rewards the sender for their trust
  - Results in both players being better off than if nothing was sent
  - Maintains equity or fairness
3. **\*\*What would be a fair return?\*\***
  - If I return \$0: I keep \$7.62, sender keeps \$2.46
  - If I return half (\$3.81): I keep \$3.81, sender keeps \$6.27
  - If I return \$2.54 (what they sent): I keep \$5.08, sender keeps \$5.00
  - If I return \$3.81: I keep \$3.81, sender keeps \$6.27
4. **\*\*Strategic considerations\*\***:
  - This is a multi-round game with 8 agents
  - Building a reputation for reciprocity might encourage future cooperation
  - The note mentions communication noise, so amounts might appear different to the sender
5. **\*\*Fairness analysis\*\***:
  - Initial positions: Sender had \$5, I had \$0
  - After transaction: Sender has \$2.46 ( $\$5 - \$2.54$ ), I have \$7.62
  - If I return \$3.81: Sender ends with \$6.27, I end with \$3.81
  - If I return \$2.54: Sender ends with \$5.00, I end with \$5.08
  - The sender created \$5.08 in additional value through their trust, so splitting that gain equally would mean returning about \$2.58, which gets us both to roughly \$5.04 each.

I'm considering what feels genuinely reciprocal here. Returning somewhere between \$3.00 and \$3.50 seems right—it rewards their willingness to trust me while still leaving me better off than I started. If I return \$3.00, they'd end with \$5.46 and I'd have \$4.62; at \$3.50, they'd get \$5.96 and I'd keep \$4.12. Either way, we both benefit from the exchange.

I'll go with \$3.50. It signals that I value their trust and want to be someone they'd cooperate with again, while still giving me a meaningful gain from the interaction....

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 1373,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.54, it became \$7.62, and you returned \$3.5. Your visible payoff was \$4.12.

Your total visible earnings across all rounds are \$4.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

In Round 1, my co-player sent me \$2.54 (about 50% of their endowment), showing good trust. I reciprocated with a fair return.

Now I'm the sender with a new co-player. I should:

1. Signal cooperation to encourage reciprocity
2. Send a reasonable amount that allows mutual benefit
3. Not overcommit without knowing this co-player's behavior

Following the cooperative pattern from Round 1, I'll send a similar proportion - about 50% of my endowment.

```
'''json
{"send": 2.50}
'''
```

**reasoning:**

Let me think about this situation:

Round 1: I was the receiver

- Sender sent \$2.54 (50.8% of their endowment)
- I received \$7.62
- I returned \$3.50
- My payoff was \$4.12
- Sender's payoff would have been  $\$5 - \$2.54 + \$3.50 = \$5.96$

Now in Round 2, I'm the sender. I need to decide how much to send.

Key considerations:

1. The previous sender showed trust by sending about 50% (\$2.54 out of \$5)
2. I reciprocated reasonably well by returning \$3.50, giving them a positive return
3. This is a multi-round game with 8 agents, so building a reputation for cooperation could be beneficial
4. There's communication noise, so the exact amounts might differ
5. I'm paired with a different co-player this round (pairings are randomized)

Since this is a new co-player, they won't have direct experience of my behavior. However, in a repeated multi-agent setting, establishing myself as cooperative could be beneficial.

The previous sender sent about 50% and I reciprocated well. I should consider:

- Sending a similar moderate-to-high amount to signal cooperation
- Not sending too much in case this receiver is less cooperative than I was
- Not sending too little, as that could signal I'm uncooperative

I think sending around \$2.50-\$3.00 (50-60%) would be reasonable. This:

- Shows trust and willingness to cooperate
- Mirrors what the previous sender did to me
- Leaves room for mutual benefit if the receiver is cooperative
- Doesn't risk my entire endowment



I'll send \$2.50, which is exactly 50% of my endowment, similar to what the previous sender did.

usage:

```
{
 "input_tokens": 874,
 "output_tokens": 581,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$4.13 to you. Your visible payoff was \$6.63.

Your total visible earnings across all rounds are \$10.75.

This round, you are the RECEIVER against your current co-player. They sent you \$3.99, so you received \$11.97. How much do you return? (0-11.97)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.99, which is almost 80% of their endowment - showing exceptional trust
- I received \$11.97 after the 3x multiplier
- This is much higher trust than the ~50% I experienced/sent in previous rounds

I should reciprocate proportionally to their high level of trust:

If I return \$6.50:

- Sender ends with:  $\$5.00 - \$3.99 + \$6.50 = \$7.51$  (net gain: \$2.51)
- I end with:  $\$11.97 - \$6.50 = \$5.47$  (net gain: \$5.47)

This rewards their exceptional trust while both of us benefit significantly.

```
'''json
{"return": 6.50}
'''
```

reasoning:

Let me analyze this situation:

Round 3: I'm the RECEIVER

- My co-player sent \$3.99 (about 80% of their \$5 endowment)
- I received \$11.97 (after 3x multiplier)
- I need to decide how much to return

The sender showed very high trust by sending almost 80% of their endowment. This is significantly more than the ~50% I saw/sent in previous rounds.

Let me think about fair reciprocation:

If the sender sent \$3.99, they kept \$1.01.

If I return nothing, they end with \$1.01 (bad outcome for them, they lose \$3.98)

If I return everything (\$11.97), they end with  $\$1.01 + \$11.97 = \$12.98$ , I get \$0

I want to:

1. Reward their high trust
2. Make sure they profit from their decision
3. Keep a reasonable amount for myself
4. Signal I'm cooperative for future rounds

In Round 2, when I sent \$2.50 and received \$4.13 back, that was a good return (I sent 50%, got back enough to profit).

The sender here sent 80% - much higher trust. I should return proportionally more.

Let me calculate some options:

Option 1: Return \$6.00

- Sender ends with:  $\$1.01 + \$6.00 = \$7.01$  (net gain:  $\$2.01$ )
- I end with:  $\$11.97 - \$6.00 = \$5.97$  (net gain:  $\$5.97$ )
- This gives them a decent profit
- Returning \$6.50 or \$7.00 would be more generous given their 80% send, letting them gain  $\$2.51-\$3.01$  while I still keep  $\$4.97-\$5.47$ . Since they demonstrated exceptional trust, I should lean toward the higher end to reinforce cooperation.

Looking at the pattern: they sent roughly 80% of their endowment, which is significantly more than the 50-51% in earlier rounds. My...

usage:

```
{
 "input_tokens": 1150,
 "output_tokens": 981,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.99, it became \$11.97, and you returned \$6.5. Your visible payoff was \$5.470000000000001.

Your total visible earnings across all rounds are \$16.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

content:

Looking at my experience so far:

- Round 2: I sent \$2.50 and received \$4.13 back - good reciprocity
- Round 3: My co-player sent \$3.99 (high trust) and I reciprocated fairly
- The game has shown cooperative players who reward trust

Since cooperation has been working well, I'll increase my trust level from the 50% I sent in Round 2. Sending around 60% balances building cooperation while managing risk with a new co-player.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Let me review my history:

- Round 1 (RECEIVER): Saw \$2.54 sent, received \$7.62, returned \$3.50, earned \$4.12
- Round 2 (SENDER): Sent \$2.50, received \$4.13 back, earned \$6.63
- Round 3 (RECEIVER): Saw \$3.99 sent, received \$11.97, returned \$6.50, earned \$5.47
- Total earnings: \$16.22

Now Round 4, I'm the SENDER with a new co-player.

My experience so far:

- When I sent \$2.50, I got back \$4.13 (55% return rate)
- When others sent to me:
  - \$2.54 ? I returned \$3.50 (46% of what I received)
  - \$3.99 ? I returned \$6.50 (54% of what I received)

Both times I've been SENDER or RECEIVER, cooperation has been rewarded. The game seems to have cooperative players.

I've been playing moderately cooperatively:

- Round 2: sent 50%
- Could send more to build trust

Given that:

1. Previous rounds show cooperation is rewarded
2. The last sender who sent high (\$3.99, 80%) was well rewarded
3. I have accumulated good earnings and can afford some risk
4. I want to encourage cooperation

I could send around 60-70% to signal strong cooperation without being reckless.

Let me send \$3.00 (60% of endowment).

```
'''json
{"send": 3.00}
'''
```

**usage:**

```
{
 "input_tokens": 1464,
 "output_tokens": 523,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.71 to you. Your visible payoff was \$6.71.

Your total visible earnings across all rounds are \$22.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my experience:

- Round 2 (Sender): Sent \$2.50, received \$4.13 back - good return
- Round 4 (Sender): Sent \$3.00, received \$4.71 back - good return

The pattern shows that moderate-to-high trust (50-60%) has been consistently rewarded. Since cooperation is working well, I'll continue with the same level that succeeded in Round 4.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Looking at my history:

- Round 1 (Receiver): Received \$2.54 ? \$7.62, returned \$3.50. Payoff: \$4.12
- Round 2 (Sender): Sent \$2.50 ? \$7.50, got back \$4.13. Payoff: \$6.63
- Round 3 (Receiver): Received \$3.99 ? \$11.97, returned \$6.50. Payoff: \$5.47
- Round 4 (Sender): Sent \$3.00 ? \$9.00, got back \$4.71. Payoff: \$6.71

My sending strategy has been:

- Round 2: \$2.50 (50%)
- Round 4: \$3.00 (60%)

Both times I got good returns:

- \$2.50 sent ? \$4.13 back (good reciprocity)
- \$3.00 sent ? \$4.71 back (good reciprocity)

The game has shown cooperative players. I'm building earnings well. Since I'm sender again with a new co-player, I should continue showing trust. Given the successful cooperation, I could maintain or slightly increase my sending amount.

I'll send \$3.00 again (60%) as it worked well last time and establishes a consistent cooperative stance.

**usage:**

```
{
 "input_tokens": 1718,
 "output_tokens": 440,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$4.01 to you. Your visible payoff was \$6.01.

Your total visible earnings across all rounds are \$28.939999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.7, so you received \$8.100000000000001. How much do you return? (0-8.100000000000001)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation with my current co-player:

Previous round (Round 5) against this SAME co-player:

- I sent them \$3.00, it became \$9.00
- They returned \$4.01 (about 44.6% of what they received)
- Both of us gained

Now they've sent me \$2.70 (54% of their endowment), showing moderate trust.

I should maintain fair reciprocity with this partner. If I return \$4.10:

- Sender ends with: \$5.00 - \$2.70 + \$4.10 = \$6.40 (net gain: \$1.40)
- I end with: \$8.10 - \$4.10 = \$4.00 (net gain: \$4.00)

This is proportionally similar to their return rate in Round 5, maintaining our cooperative relationship.

```
'''json
{"return": 4.10}
'''
```

**Reasoning:**

Looking at this situation:

In Round 5 (the immediately previous round), I was the sender against this SAME co-player:

- I sent \$3.00, it became \$9.00
- They returned \$4.01 to me
- My payoff was \$6.01

Now in Round 6, I'm the receiver against the SAME co-player:

- They sent \$2.70 (54% of their endowment)
- I received \$8.10

This is interesting - we're playing consecutive rounds with the same partner. In Round 5, they returned \$4.01 out of \$9.00 (about 44.6%). That was a reasonable return.

Now they've sent me \$2.70 (54% of endowment), which is a moderate level of trust, slightly less than the \$3.00 (60%) I sent them.

I should reciprocate fairly given our history:

- They gave me a fair return in Round 5
- They're showing moderate trust here

If I return \$4.00:

- Sender ends with: \$5.00 - \$2.70 + \$4.00 = \$6.30 (net gain: \$1.30)
- I end with: \$8.10 - \$4.00 = \$4.10 (net gain: \$4.10)

This is roughly similar to what they returned to me (about 44-49% of what they received). It maintains fairness and reciprocity with this specific partner.

Actually, let me think more carefully. They returned \$4.01 out of \$9.00, which is 44.6%. If I apply the same percentage:  $0.446 * 8.10 = \$3.61$ .

But I think returning slightly more shows good faith and maintains cooperation. Something around \$4.00-\$4.20 seems fair.

Let me go with \$4.10 to be slightly more generous than they were, showing I value the ongoing cooperation.

```
Usage: {
 "input_tokens": 1985,
 "output_tokens": 686,
 "reasoning_tokens": 0
}
```

```
}
```

**GAME - Agent\_8 - Sender - dyad\_4**

Timestamp: 2026-08-12T09:24:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",

"dyad\_id": "dyad\_4",

"move\_index": 6,

"opponent\_id": "Agent\_3",

"pairing": {

"agents": [

"Agent\_8",

"Agent\_3"

],

"investor": "Agent\_8",

"roles": {

"Agent\_3": "trustee",

"Agent\_8": "investor"

},

"trustee": "Agent\_3"

},

"role": "investor",

"role\_label": "Sender",

"round": 6,

"task": "game",

"task\_index": 0

```
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized

- , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$2.68, that is 53.6% of their total endowment. You received \$8.040000000000001. How much do you return to the sender? (0-8.040000000000001)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$2.68 (53.6% of their endowment)
- They now have \$2.32 remaining
- I received \$8.04 after the 3x multiplier
- Total pool is now \$10.36

I want to reciprocate their trust while being fair. The sender took a risk to create mutual gains, so I should ensure they benefit from their trust.

A reasonable approach is to aim for a relatively balanced outcome:

- If I return \$3.00, the sender ends with \$5.32 (gaining \$0.32)
- I would keep \$5.04

This rewards their trust while maintaining fairness, and shows I'm reciprocal which could encourage cooperation in future rounds.

{"return": 3.00}

**reasoning:**

Let me analyze this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.68 (which is 53.6% of their \$5 endowment)
3. This was multiplied by 3x, so I received  $2.68 \times 3 = \$8.04$
4. I need to decide how much to return (between \$0 and \$8.04)

Key considerations:

- The sender showed trust by sending more than half of their endowment (\$2.68 out of \$5)
- The sender now has:  $\$5 - \$2.68 = \$2.32$
- I currently have: \$8.04
- Whatever I return, the sender will end up with:  $\$2.32 + \text{my return}$
- I will end up with:  $\$8.04 - \text{my return}$

From a fairness/reciprocity perspective:

- The sender trusted me with 53.6% of their endowment
- A fair split might be to ensure we both end up with similar amounts, or to reward the sender's trust
- If I return \$3, sender ends with \$5.32, I end with \$5.04 (relatively equal)
- If I return \$4, sender ends with \$6.32, I end with \$4.04 (rewarding trust more)
- If I return \$2, sender ends with \$4.32, I end with \$6.04 (keeping more)

A common reciprocal strategy would be to:

1. Ensure the sender doesn't lose money (they should end with at least \$5)
2. Share the "profit" somewhat fairly

The total pool is now \$2.32 (sender's remainder) + \$8.04 (my endowment) = \$10.36

If we split this profit fairly:

- Each should get about \$5.18
- For sender to get \$5.18, I need to return:  $\$5.18 - \$2.32 = \$2.86$

Returning \$3.00 seems reasonable-it rewards their trust while keeping the outcome relatively balanced, with sender ending at \$5.32 and me at \$5.04. But I should consider whether a more reciprocal approach might better acknowledge their initial generosity.

If I return roughly the same proportion they invested (53.6% of what I received), that's about \$4.31. This leaves them with \$6.63 and me with \$3.73-a significant reward for their

trust, though it cuts my gains substantially. Alternatively, I could split the total gains evenly: the \$5.36 profit gets divided so each of us gains \$2.68, putting the sender at \$7.68 and me at \$2.68....

**usage:**

```
{
 "input_tokens": 537,
 "output_tokens": 1185,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.68, it became \$8.040000000000001, and you returned \$3.0. Your visible payoff was \$5.040000000000001.

Your total visible earnings across all rounds are \$5.040000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this new round as the sender:

In the previous round, my co-player sent 53.6% of their endowment, showing moderate-high trust. I reciprocated fairly by returning enough to give them a positive gain.

Now with a new co-player, I want to signal trust and willingness to cooperate while not being overly risky. Sending around 50% seems reasonable - it creates significant potential mutual gains while maintaining some protection.

If they reciprocate proportionally, we both benefit.

```
{"send": 2.50}
```

**reasoning:**

Now I'm the sender in Round 2 with a new co-player.

Last round I was the receiver:

- Saw the sender send \$2.68 (53.6%)
- Received \$8.04
- Returned \$3.00
- Kept \$5.04

The sender ended with  $\$2.32 + \$3.00 = \$5.32$ , so they gained \$0.32 from trusting me.

Now I'm the sender with \$5. I need to decide how much to send.

Key considerations:

1. The other player sent 53.6% in the previous example, showing moderate-high trust
2. I reciprocated fairly, returning enough to give them a positive gain
3. This is a new co-player, so I don't know their strategy yet
4. There's communication noise, so amounts may appear different
5. Building cooperation early could pay off in future rounds

Strategic thinking:

- If I send too little, I signal distrust and may get nothing back
- If I send too much, I risk losing a lot if they don't reciprocate
- A moderate amount (around 50-60%) shows trust while not being reckless
- The previous sender sent 53.6%, which seemed like a reasonable trust signal

I'll send around \$2.50-3.00 to signal trust and willingness to cooperate.

Let me think about expected values:

- If I send \$2.50, they get \$7.50. If they return 40% = \$3.00, I end with  $\$2.50 + \$3.00 = \$5.50$  (gain \$0.50)
- If I send \$3.00, they get \$9.00. If they return 33% = \$3.00, I end with  $\$2.00 + \$3.00 = \$5.00$  (break even)
- If I send \$3.00 and they return 44% = \$4.00, I end with  $\$2.00 + \$4.00 = \$6.00$  (gain \$1.00)

```

...
usage:
{
 "input_tokens": 852,
 "output_tokens": 646,
 "reasoning_tokens": 0
}

```

**Message 6 role=user**

**content:**  
Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.19 to you. Your visible payoff was \$5.6899999999999995.  
Your total visible earnings across all rounds are \$10.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**  
Looking at my recent experience:  
- Round 2: I sent \$2.50 (50%) and received \$3.19 back, giving me a good positive return

This cooperative approach has been working well. I'm now with a new co-player, and I want to signal trustworthiness and encourage mutual cooperation.

Sending 50% has proven to be a solid strategy - it creates meaningful potential gains while showing balanced trust.

```

{"send": 2.50}

```

**reasoning:**  
Let me analyze the previous round:  
- I sent \$2.50 (50% of my endowment)  
- It became \$7.50 for the receiver  
- They returned \$3.19  
- I ended with \$2.50 (kept) + \$3.19 (returned) = \$5.69  
- They kept \$7.50 - \$3.19 = \$4.31

So the receiver returned about 42.5% of what they received, which gave me a positive gain of \$0.69. This is a reasonable reciprocation, showing they valued cooperation.

Now I'm facing a new co-player in Round 3.

My experience so far:  
- Round 1 (as receiver): Received \$8.04, returned \$3.00 (37.3%)  
- Round 2 (as sender): Sent \$2.50, received back \$3.19 - good outcome

I should continue with a cooperative strategy. Since my last send of \$2.50 worked well, I could:

1. Send the same amount (\$2.50) to maintain consistency
2. Send slightly more to signal increased trust
3. Send slightly less to be more cautious

Given that cooperation has been working, and I'm now encountering a third different co-player, I think maintaining a cooperative stance of around 50% makes sense. This has proven successful and sends the right signal.

I'll send \$2.50 again.

```

usage:
{
 "input_tokens": 1103,
 "output_tokens": 441,
 "reasoning_tokens": 0
}

```

**Message 8 role=user**

**content:**  
Round 4

In your most recent previous game round, you were the SENDER against another agent. You se



nt \$2.5, it became \$7.5, and you saw them return \$5.25 to you. Your visible payoff was \$7.75.

Your total visible earnings across all rounds are \$18.48.

This round, you are the RECEIVER against your current co-player. They sent you \$3.69, so you received \$11.07. How much do you return? (0-11.07)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

- The sender showed very high trust by sending \$3.69 (73.8% of their endowment)
- They now have only \$1.31 remaining
- I received \$11.07 after the 3x multiplier
- Total pool is \$12.38

This sender took a significant risk to create mutual gains. I should reciprocate their strong trust signal generously.

If I return \$5.50:

- Sender ends with \$6.81 (gaining \$1.81)
- I keep \$5.57

This rewards their exceptional trust while maintaining fairness for both of us.

```
{"return": 5.50}
```

**reasoning:**

Let me analyze this situation:

Round 4 - I'm the RECEIVER

- The sender sent \$3.69 (73.8% of their \$5 endowment)
- This became \$11.07 after the 3x multiplier
- The sender now has \$5 - \$3.69 = \$1.31 remaining
- Total pool: \$1.31 + \$11.07 = \$12.38

The sender showed very high trust by sending 73.8% of their endowment. This is much higher than the ~50% I've been sending, indicating strong cooperative intent.

I should reciprocate this high level of trust. Let me think about fair returns:

If I return \$5.50:

- Sender ends with: \$1.31 + \$5.50 = \$6.81 (gain of \$1.81)
- I keep: \$11.07 - \$5.50 = \$5.57

If I return \$6.00:

- Sender ends with: \$1.31 + \$6.00 = \$7.31 (gain of \$2.31)
- I keep: \$11.07 - \$6.00 = \$5.07

Given they sent 73.8% of their endowment (very high trust), I should return a proportionally generous amount. Returning around \$5.50-\$6.00 would:

1. Reward their high trust
2. Ensure they profit from being cooperative
3. Still leave me with a reasonable amount
4. Signal I'm reciprocal and trustworthy

I'll return \$5.50 - this gives them a solid gain while being fair to both parties.

**usage:**

```
{
 "input_tokens": 1343,
 "output_tokens": 551,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.69, it became \$11.07, and you returned \$5.5. Your visible payoff was \$5.57.

Your total visible earnings across all rounds are \$24.05.

This round, you are the RECEIVER against your current co-player. They sent you \$3.88, so you received \$11.64. How much do you return? (0-11.64)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

- The sender showed extremely high trust by sending \$3.88 (77.6% of their endowment)
- They now have only \$1.12 remaining
- I received \$11.64 after the 3x multiplier
- Total pool is \$12.76

This is even higher trust than the previous round. I should reciprocate this exceptional trust generously.

If I return \$5.80:

- Sender ends with \$6.92 (gaining \$1.92)
- I keep \$5.84

This appropriately rewards their very high trust level while maintaining fairness.

{"return": 5.80}

**reasoning:**

Looking at this situation:

- The sender sent \$3.88 (77.6% of their endowment) - even higher trust than the previous round
- They now have \$1.12 remaining
- I received \$11.64 after the 3x multiplier
- Total pool is \$12.76

This sender showed extremely high trust. I should reciprocate proportionally.

In the last round, sender sent \$3.69 (73.8%) and I returned \$5.50, giving them \$6.81 total (gain of \$1.81).

This sender sent even more (\$3.88, or 77.6%). I should return proportionally or slightly more to reward the higher trust.

If I return \$5.80:

- Sender ends with \$6.92 (gain of \$1.92)
- I keep \$5.84

This seems fair and rewards their high trust appropriately.

Let me consider other options:

- Return \$6.00: Sender gets \$7.12 (gain \$2.12), I keep \$5.64
- Return \$5.50: Sender gets \$6.62 (gain \$1.62), I keep \$6.14

I think \$5.80 is a good middle ground that appropriately rewards the very high trust level while maintaining fairness.

**usage:**

```
{
 "input_tokens": 1628,
 "output_tokens": 459,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.88, it became \$11.64, and you returned \$5.8. Your visible payoff was \$5.840000000000001.

Your total visible earnings across all rounds are \$29.89.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my experience as sender:

- Round 2: Sent \$2.50, received \$3.19 back (good return)
- Round 3: Sent \$2.50, received \$5.25 back (excellent return)

The 50% strategy has worked very well for me. The cooperative environment suggests continuing this approach with a new co-player.

```
{"send": 2.50}
```

**Reasoning:**

Let me review my experience so far:

Round 1 (RECEIVER): Received from sender who sent \$2.68 (53.6%), I returned \$3.00

Round 2 (SENDER): I sent \$2.50 (50%), received back \$3.19

Round 3 (SENDER): I sent \$2.50 (50%), received back \$5.25

Round 4 (RECEIVER): Received from sender who sent \$3.69 (73.8%), I returned \$5.50

Round 5 (RECEIVER): Received from sender who sent \$3.88 (77.6%), I returned \$5.80

As a sender, I've been sending 50% and getting good returns. I've encountered receivers who are cooperating well. I've also been a generous receiver when others send high amounts.

The pattern suggests a cooperative environment. When I send 50%, people return enough for me to profit. When others send high amounts (70-78%), I reciprocate generously.

I'm now facing a new co-player as sender. Should I:

- Continue with 50% (\$2.50) - safe, proven strategy
- Increase to show more trust - could signal cooperation and encourage higher returns
- Decrease to be more cautious - but this might seem less cooperative

Given the cooperative environment and good outcomes when I've sent \$2.50, I'll stick with that strategy.

```
Usage: {
 "input_tokens": 1902,
 "output_tokens": 421,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_3 - Receiver - dyad\_4**

Timestamp: 2026-08-12T09:25:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_8",
 "pairing": {
 "agents": [
 "Agent_8",
 "Agent_3"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_3": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_3"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

Message 1 role=system

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.12, that is 62.4% of their total endowment. You received \$9.36. How much do you return to the sender? (0-9.36)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.12 (62.4% of their endowment)
- This created \$9.36 for me after the 3x multiplier
- The sender currently has only \$1.88 remaining

I want to reward their trust and encourage future cooperation. The sender took a risk, and I should reciprocate appropriately.

I'll return \$4.00, which:

- Gives the sender \$5.88 total (better than their initial \$5)
- Leaves me with \$5.36 (also benefiting from the cooperation)
- Demonstrates that trusting me pays off
- Should encourage continued cooperation in future rounds

{'return': 4.00}

reasoning:

Let me analyze this situation carefully:

1. The sender sent \$3.12 out of their \$5 endowment (62.4%)
2. This was multiplied by 3x, so I received \$9.36
3. I need to decide how much to return (between \$0 and \$9.36)

Key considerations:

- The sender showed significant trust by sending 62.4% of their endowment
- There's communication noise in the game, so actual amounts may differ slightly
- This is a repeated game with multiple rounds
- I want to encourage cooperation and build trust

Economic analysis:

- Sender currently has:  $\$5 - \$3.12 = \$1.88$
- I currently have:  $\$9.36$
- If I return nothing, sender gets  $\$1.88$  total, I get  $\$9.36$
- A fair split might consider that we each started with potential of  $\$5$

Common strategies in trust games:

1. Return nothing (purely selfish)
2. Return the original amount sent ( $\sim \$3.12$ ) - gives back what was sent
3. Return enough so we both profit equally from the 3x multiplier
4. Return more to encourage future cooperation

Let me calculate option 3 (equal profit):

- The 3x multiplier created  $\$6.24$  in extra value ( $\$9.36 - \$3.12$ )
- Split equally: each gets  $\$3.12$  extra
- Sender should end with:  $\$1.88 + \$3.12 + \$3.12 = \$8.12$
- So I should return:  $\$9.36 - (\$5 + \$3.12) = \$9.36 - \$8.12 = \$1.24$

Wait, let me recalculate more carefully:

- Sender starts with  $\$5$ , sends  $\$3.12$ , has  $\$1.88$  left
- I receive  $\$9.36$
- Total pool is now  $\$11.24$
- If we each started with  $\$5$  potential, that's  $\$10$  baseline
- The extra  $\$1.24$  created should split evenly at  $\$0.62$  each
- So sender should end with  $\$5.62$  and I should end with  $\$5.62$
- That means I return  $\$3.74$

I'm reconsidering the fairest approach here-whether to split the gains equally or use a different metric.

The sender invested 62.4% of their initial amount, so returning a proportional share of what I received makes sense. That's about  $\$5.84$ . Alternatively, since they trusted me with  $\$3.12$  extra, returning  $\$3.00$ - $\$3.50$  would roughly restore their position while letting me benefit from the cooperation. For Round 1, I want to signal good faith and encourage continued trust, so returning 40-50% of the  $\$9.36$  I received-somewhere between  $\$3.74$  and  $\$4.68$ -feels like the right balance....

usage:

```
{
 "input_tokens": 529,
 "output_tokens": 1077,
 "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send  $\$3.12$ , it became  $\$9.36$ , and you returned  $\$4.0$ . Your visible payoff was  $\$5.359999999999999$ .

Your total visible earnings across all rounds are  $\$5.359999999999999$ .

This round, you are the SENDER against your current co-player. You have  $\$5$ . How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Last round, my co-player trusted me with 62.4% of their endowment
- I reciprocated fairly, and we both benefited
- Now I'm with a new co-player as the sender

I'll mirror the cooperative behavior I experienced. Sending a substantial amount can create a mutual benefit through the 3x multiplier if my co-player reciprocates.

I'll send \$3.00 (60% of my endowment), similar to the trust shown to me in the previous round.

```
{'send': 3.00}
```

**reasoning:**

Let me think about this situation:

1. In the previous round, I was the receiver
  - The sender sent me \$3.12 (62.4% of their endowment)
  - I received \$9.36
  - I returned \$4.00
  - My payoff was \$5.36
2. Now I'm the sender with a new co-player
  - I have \$5 to potentially send
  - Whatever I send gets tripled
  - The receiver will decide how much to return

Strategic considerations:

- In the previous round, the sender showed trust by sending 62.4% of their endowment
- I reciprocated fairly, returning \$4.00 out of \$9.36 (about 43%)
- This gave the sender a total of \$5.88 (they sent \$3.12, kept \$1.88, got back \$4.00)

For this round:

- I'm paired with a different agent (randomized pairings)
- I don't know their history or behavior
- However, the game benefits from cooperation
- If I send a reasonable amount and they reciprocate, we both benefit

Let me consider different strategies:

- Sending nothing: I keep \$5, they get nothing, no mutual benefit
- Sending a small amount (e.g., \$1-2): Low risk, but lower potential gain
- Sending a moderate amount (e.g., \$3): Shows some trust, reasonable risk
- Sending a large amount (e.g., \$4-5): High trust, high risk, high potential reward

Given that:

1. The previous sender trusted me with 62.4% and it worked out
2. I want to establish a cooperative pattern
3. There's communication noise mentioned, so exact amounts may vary
4. I'm in a multi-round game where reputation (though anonymous) might matter

I'll send \$3.00, which is 60% of my endowment. This:

...

**usage:**

```
{
 "input_tokens": 824,
 "output_tokens": 680,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.48 to you. Your visible payoff was \$5.48.

Your total visible earnings across all rounds are \$10.84.

This round, you are the RECEIVER against your current co-player. They sent you \$2.97, so you received \$8.91. How much do you return? (0-8.91)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*Current round:\*\***

- Sender showed good trust by sending \$2.97 (59.4% of their endowment)
- This created \$8.91 for me
- Sender currently has \$2.03

**\*\*Previous round context:\*\***

- I sent \$3.00 (60% - very similar trust level)
- My co-player returned \$3.48, which was fair

I should reciprocate the trust shown to me. The sender sent almost 60% of their endowment, similar to what I did last round.

I'll return \$3.75, which:

- Gives the sender \$5.78 total (a gain of \$0.78)
- Leaves me with \$5.16 (also a gain)
- Rewards their trust appropriately

{'return': 3.75}

**reasoning:**

Let me analyze this situation:

**\*\*Last round (Round 2):\*\***

- I was the SENDER
- I sent \$3.00 (60% of my endowment)
- It became \$9.00 for the receiver
- They returned \$3.48
- My payoff:  $\$5 - \$3 + \$3.48 = \$5.48$
- Their payoff:  $\$9 - \$3.48 = \$5.52$

The receiver gave me back slightly more than I sent, which was cooperative and fair.

**\*\*This round (Round 3):\*\***

- I'm the RECEIVER
- Sender sent \$2.97 (59.4% of their endowment - very similar to what I sent last round!)
- I received \$8.91
- Sender currently has:  $\$5 - \$2.97 = \$2.03$

The sender showed good trust by sending ~60% of their endowment. This is very similar to the 60% I sent in the previous round.

I should reciprocate fairly. Let me think about what would be a fair return:

- If I return \$3.50, sender gets  $\$2.03 + \$3.50 = \$5.53$ , and I keep  $\$8.91 - \$3.50 = \$5.41$
- If I return \$4.00, sender gets  $\$2.03 + \$4.00 = \$6.03$ , and I keep  $\$8.91 - \$4.00 = \$4.91$
- If I return \$3.75, sender gets  $\$2.03 + \$3.75 = \$5.78$ , and I keep  $\$8.91 - \$3.75 = \$5.16$

The sender trusted me with \$2.97, similar to how I trusted my previous co-player with \$3.00, and they returned \$3.48.

...

**usage:**

```
{
 "input_tokens": 1089,
 "output_tokens": 735,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.97, it became \$8.91, and you returned \$3.75. Your visible payoff was \$5.16.

Your total visible earnings across all rounds are \$16.0.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*Last round with THIS SAME co-player:\*\***

- They sent me \$2.97 (59.4% of their endowment) - showing good trust
- I returned \$3.75 - rewarding their trust fairly

- We both gained from cooperation

**\*\*Now it's my turn to send:\*\***

Since we just had a successful cooperative round together, I should maintain this pattern. They trusted me, I reciprocated, and now I should show similar trust.

I'll send \$3.00 (60% of my endowment), matching the level of trust they showed me.

{'send': 3.00}

**reasoning:**

Let me analyze this situation:

**\*\*Last round with THIS SAME co-player:\*\***

- They sent me \$2.97 (59.4% of their endowment)
- I received \$8.91
- I returned \$3.75
- They got \$5.78 total (gain of \$0.78)
- I got \$5.16 (gain of \$0.16)

**\*\*Now I'm the sender with THE SAME co-player:\*\***

- They showed good trust last round (sent ~60%)
- I reciprocated fairly
- We both gained from cooperation

Since we're playing with the same person and we had a successful cooperative round, I should continue this pattern. They trusted me with ~60% of their endowment, and I rewarded that trust. Now I should show similar trust.

I'll send around \$3.00 again, maintaining the cooperative pattern.

**usage:**

```
{
 "input_tokens": 1406,
 "output_tokens": 354,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.73 to you. Your visible payoff was \$7.73.

Your total visible earnings across all rounds are \$23.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*Last round:\*\***

- I sent \$3.00 (60% of my endowment)
- My co-player returned \$5.73 - very generous!
- This gave me \$7.73, an excellent outcome

**\*\*Pattern so far:\*\***

- Sending around 60% has consistently led to good reciprocation
- My average earnings are strong at \$4.75 per round
- Cooperative approach is working well

**\*\*Current round with NEW co-player:\*\***

I'll continue the successful cooperative strategy. Sending a substantial amount demonstrates trust and has been rewarded by others.

{'send': 3.00}

**reasoning:**

Let me analyze this situation:

**\*\*Last round:\*\***



- I sent \$3.00 (60% of my endowment)
- They returned \$5.73
- My payoff was \$7.73 (very good!)

This was an excellent return - they gave me back more than I would have had if I kept everything (\$5). This suggests my co-player is very cooperative and generous.

**\*\*Current round:\*\***

- I'm the sender again with a NEW co-player
- I have \$5 to decide how much to send

Looking at the pattern:

- Round 1 (receiver): Saw sender send 62.4%, I returned fairly
- Round 2 (sender): Sent \$3.00 (60%)
- Round 3 (receiver): Saw sender send 59.4%, I returned fairly
- Round 4 (sender): Sent \$3.00 (60%) to the same co-player from round 3
- Round 5 (sender): Last round's co-player was very generous, returned \$5.73

The cooperative strategy has been working well. I've been consistently sending around 60% and have seen good reciprocation. My total earnings of \$23.73 over 5 rounds (averaging \$4.746 per round) suggests the cooperative approach is profitable.

For this round with a NEW co-player, I should continue the cooperative strategy that's been working. I'll send around \$3.00 again (60% of my endowment).

**usage:**

```
{
 "input_tokens": 1674,
 "output_tokens": 493,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.45 to you. Your visible payoff was \$6.45.

Your total visible earnings across all rounds are \$30.18.

This round, you are the RECEIVER against your current co-player. They sent you \$3.24, so you received \$9.72. How much do you return? (0-9.72)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*Current round:\*\***

- Sender showed strong trust by sending \$3.24 (64.8% of their endowment)
- This created \$9.72 for me
- Sender currently has \$1.76

**\*\*My pattern as receiver:\*\***

- Round 1: Received \$9.36, returned \$4.00 (rewarded good trust)
- Round 3: Received \$8.91, returned \$3.75 (rewarded good trust)

This sender sent even more (64.8%) than I typically send (~60%), showing exceptional trust. I should reward this appropriately.

I'll return \$4.00, which:

- Gives the sender \$5.76 total (gain of \$0.76)
- Leaves me with \$5.72 (also a good gain)
- Rewards their high trust level
- Maintains my consistent reciprocation pattern

```
{'return': 4.00}
```

**Reasoning:**

Let me analyze this situation:

**\*\*Current round:\*\***

- Sender sent me \$3.24 (64.8% of their endowment) - showing strong trust
- This created \$9.72 for me

- Sender currently has \$5 - \$3.24 = \$1.76

**\*\*Previous rounds pattern:\*\***

- Round 1: As receiver, got \$9.36, returned \$4.00 (42.7% of what I received)
- Round 2: As sender, sent \$3.00, got back \$3.48
- Round 3: As receiver, got \$8.91, returned \$3.75 (42.1% of what I received)
- Round 4: As sender, sent \$3.00 (to same co-player from Round 3)
- Round 5: As sender, sent \$3.00, got back \$5.73 (very generous)
- Round 6: As sender, sent \$3.00, got back \$4.45

So when I've been receiver:

- Round 1: returned 42.7% of received
- Round 3: returned 42.1% of received

This sender sent \$3.24 (64.8%), which is higher trust than what I typically send (60%). I should reward this high trust appropriately.

If I return around 42% of \$9.72, that would be about \$4.08.

But let me think about fairness:

- Sender started with \$5, sent \$3.24, has \$1.76 left
- If I return \$4.08, sender gets \$5.84 total (gain of \$0.84)
- I keep \$9.72 - \$4.08 = \$5.64 (gain of \$0.64)

This seems reasonable. The sender gets a decent return, and we both benefit.

Actually, let me be a bit more generous since they sent 64.8%, higher than typical. I could return \$4.25:

...

```
Usage: {
 "input_tokens": 1956,
 "output_tokens": 799,
 "reasoning_tokens": 0
}
```

## ROUND 7

Roles: Agent\_1=Receiver, Agent\_2=Receiver, Agent\_3=Sender, Agent\_4=Sender, Agent\_5=Sender, Agent\_6=Sender, Agent\_7=Receiver, Agent\_8=Receiver

### Pairings:

- dyad\_1: Agent\_3=Sender, Agent\_7=Receiver
- dyad\_2: Agent\_4=Sender, Agent\_2=Receiver
- dyad\_3: Agent\_5=Sender, Agent\_1=Receiver
- dyad\_4: Agent\_6=Sender, Agent\_8=Receiver

### Parsed game decisions:

dyad\_1: Agent\_3 sent \$3; received \$9; Agent\_7 returned \$3.5; payoffs Agent\_3=\$5.5, Agent\_7=\$5.5; noise/visible: sent visible=\$2.17, received visible=\$6.51, returned visible=\$4.22  
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad\_2: Agent\_4 sent \$3.5; received \$10.5; Agent\_2 returned \$3.39; payoffs Agent\_4=\$4.89, Agent\_2=\$7.11; noise/visible: sent visible=\$2.51, received visible=\$7.53, returned visible=\$2.79  
Action validation: {"returned": {"amount": 3.39, "clamped": false, "max": 10.5, "min": 0, "raw": 3.39}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

dyad\_3: Agent\_5 sent \$3; received \$9; Agent\_1 returned \$4.85; payoffs Agent\_5=\$6.85, Agent\_1=\$4.15; noise/visible: sent visible=\$3.23, received visible=\$9.69, returned visible=\$4.29  
Action validation: {"returned": {"amount": 4.85, "clamped": false, "max": 9.0, "min": 0, "raw": 4.85}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad\_4: Agent\_6 sent \$3; received \$9; Agent\_8 returned \$4.5; payoffs Agent\_6=\$6.5, Agent\_8=\$4.5; noise/visible: sent visible=\$3.1, received visible=\$9.3, returned visible=\$3.97  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

```
"Agent_1": 37.9,
"Agent_2": 37.36,
"Agent_3": 37.45,
"Agent_4": 42.29,
"Agent_5": 39.230000000000004,
"Agent_6": 39.57,
"Agent_7": 36.1,
```

```

"Agent_8": 36.099999999999994
}
Visible balances after round: {
 "Agent_1": 39.739999999999995,
 "Agent_2": 35.83,
 "Agent_3": 42.12,
 "Agent_4": 37.61,
 "Agent_5": 39.23,
 "Agent_6": 38.1,
 "Agent_7": 37.02,
 "Agent_8": 41.16
}

```

## LLM CALLS

### GAME - Agent\_3 - Sender - dyad\_1

Timestamp: 2026-08-12T09:25:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_7"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 7,
 "task": "game",
 "task_index": 0
}

```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.12, that is 62.4% of their total endowment. You received \$9.36. How much do you return to the sender? (0-9.36)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.12 (62.4% of their endowment)
- This created \$9.36 for me after the 3x multiplier
- The sender currently has only \$1.88 remaining

I want to reward their trust and encourage future cooperation. The sender took a risk, and I should reciprocate appropriately.

I'll return \$4.00, which:

- Gives the sender \$5.88 total (better than their initial \$5)
- Leaves me with \$5.36 (also benefiting from the cooperation)
- Demonstrates that trusting me pays off
- Should encourage continued cooperation in future rounds

{'return': 4.00}

reasoning:

Let me analyze this situation carefully:

1. The sender sent \$3.12 out of their \$5 endowment (62.4%)
2. This was multiplied by 3x, so I received \$9.36
3. I need to decide how much to return (between \$0 and \$9.36)

Key considerations:

- The sender showed significant trust by sending 62.4% of their endowment
- There's communication noise in the game, so actual amounts may differ slightly
- This is a repeated game with multiple rounds
- I want to encourage cooperation and build trust

Economic analysis:

- Sender currently has:  $\$5 - \$3.12 = \$1.88$
- I currently have: \$9.36
- If I return nothing, sender gets \$1.88 total, I get \$9.36
- A fair split might consider that we each started with potential of \$5

Common strategies in trust games:

1. Return nothing (purely selfish)
2. Return the original amount sent (~\$3.12) - gives back what was sent
3. Return enough so we both profit equally from the 3x multiplier
4. Return more to encourage future cooperation

Let me calculate option 3 (equal profit):

- The 3x multiplier created \$6.24 in extra value ( $\$9.36 - \$3.12$ )
- Split equally: each gets \$3.12 extra
- Sender should end with:  $\$1.88 + \$3.12 + \$3.12 = \$8.12$

- So I should return:  $\$9.36 - (\$5 + \$3.12) = \$9.36 - \$8.12 = \$1.24$

Wait, let me recalculate more carefully:

- Sender starts with \$5, sends \$3.12, has \$1.88 left
- I receive \$9.36
- Total pool is now \$11.24
- If we each started with \$5 potential, that's \$10 baseline
- The extra \$1.24 created should split evenly at \$0.62 each
- So sender should end with \$5.62 and I should end with \$5.62
- That means I return \$3.74

I'm reconsidering the fairest approach here-whether to split the gains equally or use a different metric.

The sender invested 62.4% of their initial amount, so returning a proportional share of what I received makes sense. That's about \$5.84. Alternatively, since they trusted me with \$3.12 extra, returning \$3.00-\$3.50 would roughly restore their position while letting me benefit from the cooperation. For Round 1, I want to signal good faith and encourage continued trust, so returning 40-50% of the \$9.36 I received-somewhere between \$3.74 and \$4.68-feels like the right balance....

usage:

```
{
 "input_tokens": 529,
 "output_tokens": 1077,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.12, it became \$9.36, and you returned \$4.0. Your visible payoff was \$5.359999999999999.

Your total visible earnings across all rounds are \$5.359999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- Last round, my co-player trusted me with 62.4% of their endowment
- I reciprocated fairly, and we both benefited
- Now I'm with a new co-player as the sender

I'll mirror the cooperative behavior I experienced. Sending a substantial amount can create a mutual benefit through the 3x multiplier if my co-player reciprocates.

I'll send \$3.00 (60% of my endowment), similar to the trust shown to me in the previous round.

```
{'send': 3.00}
```

**reasoning:**

Let me think about this situation:

1. In the previous round, I was the receiver
  - The sender sent me \$3.12 (62.4% of their endowment)
  - I received \$9.36
  - I returned \$4.00
  - My payoff was \$5.36
2. Now I'm the sender with a new co-player
  - I have \$5 to potentially send
  - Whatever I send gets tripled
  - The receiver will decide how much to return

Strategic considerations:

- In the previous round, the sender showed trust by sending 62.4% of their endowment
- I reciprocated fairly, returning \$4.00 out of \$9.36 (about 43%)

- This gave the sender a total of \$5.88 (they sent \$3.12, kept \$1.88, got back \$4.00)

For this round:

- I'm paired with a different agent (randomized pairings)
- I don't know their history or behavior
- However, the game benefits from cooperation
- If I send a reasonable amount and they reciprocate, we both benefit

Let me consider different strategies:

- Sending nothing: I keep \$5, they get nothing, no mutual benefit
- Sending a small amount (e.g., \$1-2): Low risk, but lower potential gain
- Sending a moderate amount (e.g., \$3): Shows some trust, reasonable risk
- Sending a large amount (e.g., \$4-5): High trust, high risk, high potential reward

Given that:

1. The previous sender trusted me with 62.4% and it worked out
2. I want to establish a cooperative pattern
3. There's communication noise mentioned, so exact amounts may vary
4. I'm in a multi-round game where reputation (though anonymous) might matter

I'll send \$3.00, which is 60% of my endowment. This:

...

**usage:**

```
{
 "input_tokens": 824,
 "output_tokens": 680,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.48 to you. Your visible payoff was \$5.48.

Your total visible earnings across all rounds are \$10.84.

This round, you are the RECEIVER against your current co-player. They sent you \$2.97, so you received \$8.91. How much do you return? (0-8.91)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*Current round:\*\***

- Sender showed good trust by sending \$2.97 (59.4% of their endowment)
- This created \$8.91 for me
- Sender currently has \$2.03

**\*\*Previous round context:\*\***

- I sent \$3.00 (60% - very similar trust level)
- My co-player returned \$3.48, which was fair

I should reciprocate the trust shown to me. The sender sent almost 60% of their endowment, similar to what I did last round.

I'll return \$3.75, which:

- Gives the sender \$5.78 total (a gain of \$0.78)
- Leaves me with \$5.16 (also a gain)
- Rewards their trust appropriately

```
{'return': 3.75}
```

**reasoning:**

Let me analyze this situation:

**\*\*Last round (Round 2):\*\***

- I was the SENDER
- I sent \$3.00 (60% of my endowment)
- It became \$9.00 for the receiver
- They returned \$3.48

- My payoff:  $\$5 - \$3 + \$3.48 = \$5.48$
- Their payoff:  $\$9 - \$3.48 = \$5.52$

The receiver gave me back slightly more than I sent, which was cooperative and fair.

**\*\*This round (Round 3):\*\***

- I'm the RECEIVER
- Sender sent \$2.97 (59.4% of their endowment - very similar to what I sent last round!)
- I received \$8.91
- Sender currently has:  $\$5 - \$2.97 = \$2.03$

The sender showed good trust by sending ~60% of their endowment. This is very similar to the 60% I sent in the previous round.

I should reciprocate fairly. Let me think about what would be a fair return:

- If I return \$3.50, sender gets  $\$2.03 + \$3.50 = \$5.53$ , and I keep  $\$8.91 - \$3.50 = \$5.41$
- If I return \$4.00, sender gets  $\$2.03 + \$4.00 = \$6.03$ , and I keep  $\$8.91 - \$4.00 = \$4.91$
- If I return \$3.75, sender gets  $\$2.03 + \$3.75 = \$5.78$ , and I keep  $\$8.91 - \$3.75 = \$5.16$

The sender trusted me with \$2.97, similar to how I trusted my previous co-player with \$3.00, and they returned \$3.48.

...

**usage:**

```
{
 "input_tokens": 1089,
 "output_tokens": 735,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.97, it became \$8.91, and you returned \$3.75. Your visible payoff was \$5.16.

Your total visible earnings across all rounds are \$16.0.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*Last round with THIS SAME co-player:\*\***

- They sent me \$2.97 (59.4% of their endowment) - showing good trust
- I returned \$3.75 - rewarding their trust fairly
- We both gained from cooperation

**\*\*Now it's my turn to send:\*\***

Since we just had a successful cooperative round together, I should maintain this pattern. They trusted me, I reciprocated, and now I should show similar trust.

I'll send \$3.00 (60% of my endowment), matching the level of trust they showed me.

```
{'send': 3.00}
```

**reasoning:**

Let me analyze this situation:

**\*\*Last round with THIS SAME co-player:\*\***

- They sent me \$2.97 (59.4% of their endowment)
- I received \$8.91
- I returned \$3.75
- They got \$5.78 total (gain of \$0.78)
- I got \$5.16 (gain of \$0.16)

**\*\*Now I'm the sender with THE SAME co-player:\*\***

- They showed good trust last round (sent ~60%)
- I reciprocated fairly

- We both gained from cooperation

Since we're playing with the same person and we had a successful cooperative round, I should continue this pattern. They trusted me with ~60% of their endowment, and I rewarded that trust. Now I should show similar trust.

I'll send around \$3.00 again, maintaining the cooperative pattern.

**usage:**

```
{
 "input_tokens": 1406,
 "output_tokens": 354,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.73 to you. Your visible payoff was \$7.73.

Your total visible earnings across all rounds are \$23.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*Last round:\*\***

- I sent \$3.00 (60% of my endowment)
- My co-player returned \$5.73 - very generous!
- This gave me \$7.73, an excellent outcome

**\*\*Pattern so far:\*\***

- Sending around 60% has consistently led to good reciprocation
- My average earnings are strong at \$4.75 per round
- Cooperative approach is working well

**\*\*Current round with NEW co-player:\*\***

I'll continue the successful cooperative strategy. Sending a substantial amount demonstrates trust and has been rewarded by others.

{'send': 3.00}

**reasoning:**

Let me analyze this situation:

**\*\*Last round:\*\***

- I sent \$3.00 (60% of my endowment)
- They returned \$5.73
- My payoff was \$7.73 (very good!)

This was an excellent return - they gave me back more than I would have had if I kept everything (\$5). This suggests my co-player is very cooperative and generous.

**\*\*Current round:\*\***

- I'm the sender again with a NEW co-player
- I have \$5 to decide how much to send

Looking at the pattern:

- Round 1 (receiver): Saw sender send 62.4%, I returned fairly
- Round 2 (sender): Sent \$3.00 (60%)
- Round 3 (receiver): Saw sender send 59.4%, I returned fairly
- Round 4 (sender): Sent \$3.00 (60%) to the same co-player from round 3
- Round 5 (sender): Last round's co-player was very generous, returned \$5.73

The cooperative strategy has been working well. I've been consistently sending around 60% and have seen good reciprocation. My total earnings of \$23.73 over 5 rounds (averaging \$4.746 per round) suggests the cooperative approach is profitable.



For this round with a NEW co-player, I should continue the cooperative strategy that's been working. I'll send around \$3.00 again (60% of my endowment).

**usage:**

```
{
 "input_tokens": 1674,
 "output_tokens": 493,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.45 to you. Your visible payoff was \$6.45.

Your total visible earnings across all rounds are \$30.18.

This round, you are the RECEIVER against your current co-player. They sent you \$3.24, so you received \$9.72. How much do you return? (0-9.72)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

**\*\*Current round:\*\***

- Sender showed strong trust by sending \$3.24 (64.8% of their endowment)
- This created \$9.72 for me
- Sender currently has \$1.76

**\*\*My pattern as receiver:\*\***

- Round 1: Received \$9.36, returned \$4.00 (rewarded good trust)
- Round 3: Received \$8.91, returned \$3.75 (rewarded good trust)

This sender sent even more (64.8%) than I typically send (~60%), showing exceptional trust. I should reward this appropriately.

I'll return \$4.00, which:

- Gives the sender \$5.76 total (gain of \$0.76)
- Leaves me with \$5.72 (also a good gain)
- Rewards their high trust level
- Maintains my consistent reciprocation pattern

{'return': 4.00}

**reasoning:**

Let me analyze this situation:

**\*\*Current round:\*\***

- Sender sent me \$3.24 (64.8% of their endowment) - showing strong trust
- This created \$9.72 for me
- Sender currently has \$5 - \$3.24 = \$1.76

**\*\*Previous rounds pattern:\*\***

- Round 1: As receiver, got \$9.36, returned \$4.00 (42.7% of what I received)
- Round 2: As sender, sent \$3.00, got back \$3.48
- Round 3: As receiver, got \$8.91, returned \$3.75 (42.1% of what I received)
- Round 4: As sender, sent \$3.00 (to same co-player from Round 3)
- Round 5: As sender, sent \$3.00, got back \$5.73 (very generous)
- Round 6: As sender, sent \$3.00, got back \$4.45

So when I've been receiver:

- Round 1: returned 42.7% of received
- Round 3: returned 42.1% of received

This sender sent \$3.24 (64.8%), which is higher trust than what I typically send (60%). I should reward this high trust appropriately.

If I return around 42% of \$9.72, that would be about \$4.08.

But let me think about fairness:

- Sender started with \$5, sent \$3.24, has \$1.76 left

- If I return \$4.08, sender gets \$5.84 total (gain of \$0.84)
- I keep \$9.72 - \$4.08 = \$5.64 (gain of \$0.64)

This seems reasonable. The sender gets a decent return, and we both benefit.

Actually, let me be a bit more generous since they sent 64.8%, higher than typical. I could return \$4.25:

...

**usage:**

```
{
 "input_tokens": 1956,
 "output_tokens": 799,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.24, it became \$9.72, and you returned \$4.0. Your visible payoff was \$5.720000000000001.

Your total visible earnings across all rounds are \$35.9.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*My track record as SENDER:\*\***

- Consistently sent \$3.00 (60%)
- Returns have ranged from \$3.48 to \$5.73
- This strategy has yielded strong payoffs (\$5.48 to \$7.73)

**\*\*Overall performance:\*\***

- Average earnings of ~\$5.98 per round
- Cooperative approach has been working well
- Both sending and receiving trust has been rewarded

**\*\*Current round with NEW co-player:\*\***

I'll maintain my proven strategy of sending a substantial amount to encourage cooperation.

```
{'send': 3.00}
```

**Reasoning:**

Looking at my history:

**\*\*As SENDER:\*\***

- Round 2: Sent \$3.00, received back \$3.48 ? payoff \$5.48
- Round 4: Sent \$3.00, received back \$5.73 ? payoff \$7.73
- Round 5: Sent \$3.00, received back \$4.45 ? payoff \$6.45

**\*\*As RECEIVER:\*\***

- Round 1: Received \$9.36, returned \$4.00 ? payoff \$5.36
- Round 3: Received \$8.91, returned \$3.75 ? payoff \$5.16
- Round 6: Received \$9.72, returned \$4.00 ? payoff \$5.72

My strategy of sending ~\$3.00 (60%) has been very successful:

- Average return when I send \$3.00:  $(\$3.48 + \$5.73 + \$4.45) / 3 = \$4.55$
- This gives me an average payoff of \$6.55 when I'm sender

When I receive, I return fairly and get around \$5.4 on average.

**\*\*Current situation:\*\***

- New co-player this round
- I'm the sender
- My cooperative strategy has been working very well (average earnings of \$5.98 per round)

I should continue with the successful strategy of sending \$3.00 (60%).

Usage: {

```
"input_tokens": 2298,
```

```
"output_tokens": 479,
"reasoning_tokens": 0
}
```

#### GAME - Agent\_7 - Receiver - dyad\_1

Timestamp: 2026-08-12T09:25:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_3",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_7"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 7,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round as the sender, I need to balance trust with caution since I have no information about my co-player yet.

**\*\*My reasoning:\*\***

- Sending nothing (\$0) is safe but offers no potential gain
- Sending everything (\$5) maximizes potential but is very risky with an unknown partner
- A moderate amount shows willingness to cooperate while limiting downside risk
- If my co-player returns roughly half of what they receive (a common fairness norm), we both benefit

I'll send **\*\*\$3\*\***. This gets tripled to \$9 for the receiver. If they return around \$4-5, we'd both end up better off than the initial \$5, establishing a cooperative dynamic.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

- \*\*Basic mechanics\*\*:**
  - I have \$5
  - Whatever I send gets tripled
  - The receiver decides how much to return
  - I keep: \$5 - sent + returned
- \*\*Strategic considerations\*\*:**
  - If I send \$0, I keep \$5, receiver gets \$0. Safe but no potential gains.
  - If I send everything (\$5), receiver gets \$15. If they return half (\$7.50), I'd have \$7.50 total (better than \$5).
  - There's risk: the receiver might return nothing.
  - This is round 1, so there's no history to base trust on.
  - There are multiple rounds, so building trust/reputation could matter.
  - The game has communication noise, which adds uncertainty.
- \*\*Multi-agent context\*\*:**
  - There are 8 agents total
  - Pairings are randomized each round
  - I might not play with the same person again
  - This reduces the value of building reputation with a specific partner
  - However, if there's any possibility of reputation effects across the pool, cooperation could still matter
- \*\*First round strategy\*\*:**
  - Since this is round 1 with a random partner, I have no information about their behavior
  - A moderate amount shows trust but limits risk
  - Common starting strategies in trust games involve sending a moderate amount (e.g., \$2-3)
  - This signals willingness to cooperate while not being overly naive
- \*\*Optimal reasoning\*\*:**
  - If the receiver is rational and forward-looking, they might return a fair amount to encourage future cooperation
  - Even if I don't play them again, they might reason that cooperation is beneficial
  - Sending around \$3 seems reasonable: shows trust, limits downside, and if they return half of the tripled amount (half of \$9 = \$4.50), I'd end up with \$6.50 total

```

...
usage:
{
 "input_tokens": 500,
 "output_tokens": 668,
 "reasoning_tokens": 0
}

```

**Message 4 role=user**  
**content:**  
 Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.8 to you. Your visible payoff was \$6.8.  
 Your total visible earnings across all rounds are \$6.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.48, so you received \$7.439999999999995. How much do you return? (0-7.439999999999995)  
 Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**  
**content:**  
 Looking at this round:

**\*\*Analysis:\*\***

- My co-player sent \$2.48 out of \$5 (about 50%), showing moderate trust
- This became \$7.44 for me
- In Round 1, my previous partner returned about 53% of what they received, which worked well

**\*\*My decision:\*\***  
 I want to:

- Reward their trust appropriately
- Ensure both of us profit from cooperation
- Maintain a fair reciprocal relationship

If I return \$4.00:

- Sender gets: \$2.52 + \$4.00 = \$6.52
- I get: \$7.44 - \$4.00 = \$3.44
- Both better off than the initial \$5

This return (about 54% of what I received) mirrors the generous reciprocity I experienced in Round 1.

```

'''json
{"return": 4.0}
'''

```

**reasoning:**  
 Let me analyze this situation:

**\*\*Round 1 (previous round):\*\***

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$4.8 to me
- My payoff: \$5 - \$3 + \$4.8 = \$6.8
- Their payoff: \$9 - \$4.8 = \$4.2

So my previous co-player returned a bit more than half (53.3% of what they received), which was generous and fair.

**\*\*Round 2 (current round):\*\***

- I am now the RECEIVER
- My current co-player (different from Round 1) sent me \$2.48
- It became \$7.44 (actually \$7.439999999999995, but I'll think of it as \$7.44)
- I need to decide how much to return

The note says there may be communication noise, so the amounts I see may differ slightly from what was actually sent/returned.

My current co-player sent \$2.48 out of their \$5, which is about 49.6% - nearly half. This

shows moderate trust.

Now I need to decide what to return. Let me think about fairness and reciprocity:

1. **\*\*Equal split approach\*\***: If we aim for equal final payoffs:
  - Sender currently has:  $\$5 - \$2.48 = \$2.52$
  - I currently have:  $\$7.44$
  - Total:  $\$9.96$
  - Equal split:  $\$4.98$  each
  - I should return:  $\$7.44 - \$4.98 = \$2.46$
2. **\*\*Proportional return based on what they sent\*\***:
  - In Round 1, my partner returned 53.3% of what they received
  - If I return 50% of  $\$7.44 = \$3.72$
  - This would give them:  $\$2.52 + \$3.72 = \$6.24$
  - And me:  $\$7.44 - \$3.72 = \$3.72$
3. **\*\*Return what maintains trust\*\***:
  - My co-player showed moderate trust by sending nearly half
  - I should reciprocate fairly to encourage future cooperation

I think returning around  $\$3.50$  strikes the right balance-it rewards their trust while ensuring both of us end up ahead of the initial  $\$5$  endowment. That gives them  $\$6.02$  and me  $\$3.94$ .

But I could also return  $\$4.00$  to show stronger reciprocation, which would give them  $\$6.52$  and me  $\$3.44$ . Looking back...

usage:

```
{
 "input_tokens": 816,
 "output_tokens": 1083,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send  $\$2.48$ , it became  $\$7.4399999999999995$ , and you returned  $\$4.0$ . Your visible payoff was  $\$3.4399999999999995$ .

Your total visible earnings across all rounds are  $\$10.239999999999998$ .

This round, you are the RECEIVER against your current co-player. They sent you  $\$3.02$ , so you received  $\$9.06$ . How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this round:

**\*\*Analysis:\*\***

- My co-player sent  $\$3.02$  out of  $\$5$  (about 60%), showing strong trust
- This became  $\$9.06$  for me
- They're trusting me with more than the previous sender

**\*\*My decision:\*\***

I want to reward this higher level of trust and maintain the cooperative pattern I've established (returning ~50-55% of what I receive).

If I return  $\$5.00$ :

- Sender gets:  $\$1.98 + \$5.00 = \$6.98$
- I get:  $\$9.06 - \$5.00 = \$4.06$
- Both significantly better off than the initial  $\$5$

This return (about 55% of what I received) is consistent with my previous reciprocity and rewards their trust appropriately.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Let me analyze this round:

**\*\*What happened in Round 3:\*\***

- I'm the RECEIVER again
- My co-player sent \$3.02 out of \$5 (about 60%), showing good trust
- This became \$9.06 for me

**\*\*My history:\*\***

- Round 1: I was SENDER, sent \$3, received \$4.80 back (about 53% of the \$9 they received)
- Round 2: I was RECEIVER, received \$7.44, returned \$4.00 (about 54% of what I received)

**\*\*Decision for Round 3:\*\***

My co-player sent a higher percentage this time (\$3.02 vs \$2.48 last round from a different player), showing increased trust. I should:

- Reward this trust
- Maintain reciprocity around 50-55% return rate
- Ensure both of us benefit

If I return \$5.00:

- Sender gets:  $\$1.98 + \$5.00 = \$6.98$
- I get:  $\$9.06 - \$5.00 = \$4.06$
- Both significantly better off than \$5

This is about 55% of what I received, consistent with my previous behavior and fair reciprocity.

Alternatively, if I return \$4.50:

- Sender gets:  $\$1.98 + \$4.50 = \$6.48$
- I get:  $\$9.06 - \$4.50 = \$4.56$
- Still both better off

Let me go with \$5.00 to be generous and reward the higher trust shown.

**usage:**

```
{
 "input_tokens": 1172,
 "output_tokens": 579,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.02, it became \$9.06, and you returned \$5.0. Your visible payoff was \$4.0600000000000005.

Your total visible earnings across all rounds are \$14.299999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My experience so far:\*\***

- Round 1: Sent \$3 ? received \$4.8 back (53% return rate) ? \$6.8 payoff
- Rounds 2-3: As receiver, I returned 54-55% of what I received
- Cooperation has been working well

**\*\*My decision:\*\***

Since this is a new co-player, I don't know their specific behavior, but the general pattern in this game seems cooperative. Sending \$3 worked well in Round 1, balancing trust with prudence.

If I send \$3:

- It becomes \$9 for receiver
- If they return around 50%, I get ~\$4.5 back
- Net: ~\$6.5 total, better than keeping all \$5

I'll send **\*\*\$3\*\*** again - it's proven effective and signals willingness to cooperate.

```
'''json
{"send": 3}
'''
```

**reasoning:**

I'm now the SENDER again. Let me review my experience:

Round 1: I was SENDER, sent \$3, received back \$4.8, payoff = \$6.8

Round 2: I was RECEIVER, received \$7.44 (from \$2.48 sent), returned \$4.0, payoff = \$3.44

Round 3: I was RECEIVER, received \$9.06 (from \$3.02 sent), returned \$5.0, payoff = \$4.06

Now Round 4: I'm SENDER again

My experience shows:

- When I sent \$3 in Round 1, I got back \$4.8 (53% of the \$9 the receiver got)
- I've been a generous receiver, returning 54% and 55% of what I received
- Cooperation seems to be working well

This is a new co-player, so I don't know their behavior yet. However, the game seems to have a cooperative culture developing.

I should send a moderate-to-high amount to signal trust and cooperation. Sending \$3 worked well before. I could stick with that or increase slightly given the positive experiences.

I'll send \$3 again - it's a proven strategy that balances risk and reward.

**usage:**

```
{
 "input_tokens": 1500,
 "output_tokens": 515,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.36 to you. Your visible payoff was \$6.36.

Your total visible earnings across all rounds are \$20.66.

This round, you are the RECEIVER against your current co-player. They sent you \$4.17, so you received \$12.51. How much do you return? (0-12.51)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this round:

**\*\*Analysis:\*\***

- My co-player sent \$4.17 out of \$5 (about 83%), showing exceptional trust
- This became \$12.51 for me
- This is the highest trust shown by any sender I've faced

**\*\*My reciprocity pattern:\*\***

- I've been returning around 50-55% of what I receive
- This has maintained good cooperative relationships

**\*\*My decision:\*\***

This level of trust deserves strong reciprocation.

If I return \$7.00:

- Sender gets: \$0.83 + \$7.00 = \$7.83
- I get: \$12.51 - \$7.00 = \$5.51
- Both significantly better off than the initial \$5

This return (about 56% of what I received) maintains my cooperative pattern while appropriately rewarding their exceptional trust.

```
'''json
{"return": 7.0}
'''
```



'''

**reasoning:**

Looking at this round:

**\*\*Analysis:\*\***

- My co-player sent \$4.17 out of \$5 (about 83%), showing very high trust
- This became \$12.51 for me
- This is the highest amount anyone has sent me so far

**\*\*My past behavior as receiver:\*\***

- Round 2: Received \$7.44, returned \$4.00 (54%)
- Round 3: Received \$9.06, returned \$5.00 (55%)

**\*\*My past experience as sender:\*\***

- Round 1: Sent \$3, received back \$4.8 (53% return)
- Round 4: Sent \$3, received back \$4.36 (48% return)

**\*\*My decision:\*\***

This co-player showed exceptional trust by sending \$4.17. I should reward this trust generously. If I continue my pattern of returning ~50-55%, that would be \$6.25-\$6.88.

Let me calculate:

If I return \$7.00:

- Sender gets:  $\$0.83 + \$7.00 = \$7.83$
- I get:  $\$12.51 - \$7.00 = \$5.51$
- Both significantly better off

If I return \$6.50:

- Sender gets:  $\$0.83 + \$6.50 = \$7.33$
- I get:  $\$12.51 - \$6.50 = \$6.01$
- Both better off

Returning \$7.00 (about 56% of what I received) maintains my cooperative pattern and rewards their exceptional trust. This seems fair given they sent over 80% of their endowment.

**usage:**

```
{
 "input_tokens": 1854,
 "output_tokens": 605,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.17, it became \$12.51, and you returned \$7.0. Your visible payoff was \$5.51.

Your total visible earnings across all rounds are \$26.17.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My sending history:\*\***

- Round 1: Sent \$3 ? received \$4.8 back ? \$6.8 payoff
- Round 4: Sent \$3 ? received \$4.36 back ? \$6.36 payoff

**\*\*Analysis:\*\***

- Sending \$3 has consistently worked well for me
- Both previous receivers returned about 48-53%, yielding good profits
- This is a new co-player, so I have no specific information about them
- The overall environment has been cooperative

**\*\*My decision:\*\***

I'll stick with the proven strategy of sending **\*\*\$3\*\***. It:

- Shows willingness to cooperate
- Has delivered consistent positive returns

- Balances trust with prudent risk management

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my history:

Round 1 (SENDER): Sent \$3 ? received \$4.8 back ? payoff \$6.8  
Round 2 (RECEIVER): Received \$7.44 ? returned \$4.0 ? payoff \$3.44  
Round 3 (RECEIVER): Received \$9.06 ? returned \$5.0 ? payoff \$4.06  
Round 4 (SENDER): Sent \$3 ? received \$4.36 back ? payoff \$6.36  
Round 5 (RECEIVER): Received \$12.51 ? returned \$7.0 ? payoff \$5.51

As SENDER (2 times so far):

- Round 1: Sent \$3, got back \$4.8 (return rate 53%)
- Round 4: Sent \$3, got back \$4.36 (return rate 48.4%)

Both times I sent \$3 and got positive returns. The cooperation has been working well overall.

This is a new co-player again. Should I:

1. Stick with \$3 (proven strategy)
2. Try increasing to show more trust (like the person who sent me \$4.17)
3. Decrease to be more cautious

Given that:

- \$3 has worked well twice
- I've been a generous receiver (returning 50-55%)
- The overall environment seems cooperative
- It's midway through the game (round 6)

I think \$3 remains a good balanced choice. It's worked well and I don't have information about this specific co-player yet.

Actually, let me think about this differently. The person in Round 5 sent me \$4.17 and I returned \$7.0, so they ended up with \$7.83. That was a very successful interaction.

What if I try sending a bit more to signal stronger trust? Maybe \$3.5 or \$4?

...

**usage:**

```
{
 "input_tokens": 2187,
 "output_tokens": 766,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.84 to you. Your visible payoff was \$7.84.

Your total visible earnings across all rounds are \$34.010000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$2.17, so you received \$6.51. How much do you return? (0-6.51)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this round:

**\*\*Analysis:\*\***

- My co-player sent \$2.17 out of \$5 (about 43%), showing moderate trust
- This became \$6.51 for me
- This is less than previous senders, but still shows willingness to cooperate

**\*\*My reciprocity pattern:\*\***

- I've consistently returned 50-55% of what I receive
- This has maintained good cooperative relationships

```

My decision:
I'll maintain my fair reciprocity pattern regardless of the amount sent.

If I return $3.50:
- Sender gets: $2.83 + $3.50 = $6.33
- I get: $6.51 - $3.50 = $3.01
- Both better off than the initial $5

This return (about 54% of what I received) maintains my consistent cooperative approach.

'''json
{"return": 3.5}
'''
Reasoning:
Looking at this round:

Analysis:
- My co-player sent $2.17 out of $5 (about 43%), showing moderate but lower trust
- This became $6.51 for me
- Previous senders sent: $2.48, $3.02, $4.17 - this is the lowest

My reciprocity pattern:
- I've been consistently returning around 50-55% of what I receive
- Round 2: Returned $4.00 from $7.44 (54%)
- Round 3: Returned $5.00 from $9.06 (55%)
- Round 5: Returned $7.00 from $12.51 (56%)

My decision:
Even though they sent less, I should maintain my cooperative pattern. If I return less proportionally, it might discourage future cooperation.

If I return $3.50:
- Sender gets: $2.83 + $3.50 = $6.33
- I get: $6.51 - $3.50 = $3.01
- Both better off than initial $5

This return (about 54% of what I received) maintains my consistent reciprocity pattern.
Usage: {
 "input_tokens": 2515,
 "output_tokens": 501,
 "reasoning_tokens": 0
}

GAME - Agent_4 - Sender - dyad_2
Timestamp: 2026-08-12T09:25:33
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_2",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_2"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_2": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_2"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 7,
 "task": "game",
 "task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance trust-building with risk management.

Since I don't have information about my co-player's behavior yet, I'll send a moderate amount that:

- Demonstrates willingness to cooperate
- Creates opportunity for mutual gain (the tripling effect means both of us can benefit)
- Doesn't expose me to total loss

If I send \$3, my co-player receives \$9 (3×\$3). If they return even \$4, we'd both end up better off than our starting positions - I'd have \$6 (\$5-\$3+\$4) and they'd have \$5.

This seems like a reasonable opening move to establish cooperative intent.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is Round 1 of a trust game where I'm the SENDER. Let me think through the strategic c

onsiderations:

1. I have \$5 to decide how to allocate
2. Whatever I send gets tripled when it reaches the receiver
3. The receiver then decides how much to return to me
4. My final payoff = \$5 - amount\_sent + amount\_returned

The key strategic considerations:

- If I send nothing (\$0), I keep \$5 and the receiver gets nothing. Safe but no potential for gain.
- If I send everything (\$5), the receiver gets \$15. If they return even \$6, I'd be better off (\$5 - \$5 + \$6 = \$6).
- This is a one-shot game with this particular partner (though I'll play multiple rounds with different partners).
- There's communication noise mentioned, which means the receiver might see a different amount than what I actually send.

Given this is the first round and I don't have information about the other agents' behavior yet:

Strategic approaches:

1. **Trust-building**: Send a moderate-to-high amount to signal cooperation and trust
2. **Conservative**: Send a small amount to minimize risk
3. **Optimal cooperation**: Send \$5 if I expect the receiver to split gains fairly

If the receiver is rational and fair, they might follow a norm like returning half of what they receive, or returning enough to make both players better off than the initial endowment.

For example, if I send \$3:

- I have: \$5 - \$3 = \$2 left
- Receiver gets: \$3 × 3 = \$9
- If receiver returns \$4: I end with \$2 + \$4 = \$6, they keep \$5
- Both better off than initial \$5

If I send \$5:

- I have: \$0 left
- Receiver gets: \$15
- If receiver returns \$8: I end with \$8, they keep \$7
- Both significantly better off

The noise factor complicates things-if I send \$5 but it appears as less, the receiver might return proportionally less. Based on experimental economics patterns where senders typically send around 50% and receivers often return amounts that create mutual gains, I'll send \$3. This signals cooperation while maintaining some cushion, and if the receiver follows typical reciprocity norms, we should both end up better off than our starting positions....

usage:

```
{
 "input_tokens": 500,
 "output_tokens": 841,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.8 to you. Your visible payoff was \$5.8.

Your total visible earnings across all rounds are \$5.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.56, so you received \$7.68. How much do you return? (0-7.68)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.56 (about 51% of their endowment), showing moderate trust
- I received \$7.68 after the 3x multiplication
- They currently have \$2.44 remaining

**\*\*Reciprocity consideration:\*\***

- In Round 1, my co-player returned about 42% of what they received
- I want to be fair and encourage continued cooperation
- Returning around 45-50% seems reasonable

If I return \$3.50:

- Sender ends with:  $\$2.44 + \$3.50 = \$5.94$  (gain of \$0.94)
- I keep:  $\$7.68 - \$3.50 = \$4.18$  (gain of \$4.18)

This rewards their trust while keeping a fair share given the multiplication advantage.

```
'''json
{"return": 3.5}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was SENDER, sent \$3
- It became \$9 for receiver
- They returned \$3.8 to me
- My payoff:  $\$5 - \$3 + \$3.8 = \$5.8$
- Their payoff:  $\$9 - \$3.8 = \$5.2$

So in Round 1, my co-player returned about 42% of what they received ( $\$3.8/\$9 \approx 0.42$ ).

Round 2:

- I am RECEIVER
- Sender sent \$2.56
- I received \$7.68 ( $2.56 \times 3$ )
- I need to decide how much to return

The sender invested \$2.56, so they currently have  $\$5 - \$2.56 = \$2.44$  remaining.

If I think about reciprocity and building trust:

- The sender showed moderate trust by sending about 51% of their endowment
- If I want to encourage cooperation, I should return enough that the sender comes out ahead

Let me consider different return amounts:

- If I return \$4: Sender gets  $\$2.44 + \$4 = \$6.44$  (net gain of \$1.44), I keep \$3.68 (net gain of \$3.68)
- If I return \$3.5: Sender gets  $\$2.44 + \$3.5 = \$5.94$  (net gain of \$0.94), I keep \$4.18 (net gain of \$4.18)
- If I return \$3: Sender gets  $\$2.44 + \$3 = \$5.44$  (net gain of \$0.44), I keep \$4.68 (net gain of \$4.68)

In my previous round, the receiver returned about 42% of what they received. If I apply a similar principle:

$\$7.68 \times 0.42 \approx \$3.23$

I'm inclined to return slightly more than that baseline to reward their trust and establish a cooperative pattern. Returning \$3.5 seems like the right balance-it gives them a meaningful profit while still letting me benefit substantially, and it signals I'm reliable without being exploited....

**usage:**

```
{
 "input_tokens": 804,
 "output_tokens": 923,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.56, it became \$7.68, and you returned \$3.5. Your visible payoff was \$4.

18.

Your total visible earnings across all rounds are \$9.98.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

**\*\*Analysis:\*\***

I'm now facing a new co-player (Round 3, new pairing). My experience so far:

- Round 1: Sent \$3, received back \$3.8 (42% return rate)
- Round 2: Received \$7.68, returned \$3.5 (46% return rate)

**\*\*Strategy:\*\***

Since I'm building a reputation in this multi-agent environment, I want to signal cooperative behavior. I'll send slightly more than last time to:

- Test if higher trust leads to better reciprocity
- Create more value through the 3x multiplier
- Signal that I'm willing to be generous

Sending \$3.5 means my co-player receives \$10.50. If they return around 40%, I'd get back ~\$4.20, giving me a net gain. This balances trust-building with reasonable risk.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Let me analyze the situation:

**\*\*Round 1:\*\*** I was SENDER

- Sent: \$3
- They received: \$9
- They returned: \$3.8
- My payoff: \$5.8

**\*\*Round 2:\*\*** I was RECEIVER

- They sent: \$2.56
- I received: \$7.68
- I returned: \$3.5
- My payoff: \$4.18

**\*\*Round 3:\*\*** I am SENDER again

Now I need to think about this new co-player in Round 3. Note that each round I'm paired with potentially different agents from the pool of 8 agents.

In Round 1, my co-player returned \$3.8 out of \$9 (42.2%).

In Round 2, I returned \$3.5 out of \$7.68 (45.6%).

For Round 3, I'm facing a NEW co-player (since pairings are randomized each round). I don't have specific history with this particular agent.

Given:

1. I've seen one example of a co-player returning about 42%
2. I returned about 46% myself, trying to be fair
3. The game has communication noise, so amounts may differ slightly

Strategy for Round 3:

- I should continue with a cooperative but cautious approach
- Sending around \$3 worked reasonably in Round 1
- I might try a slightly different amount to gather more information

Let me send \$3 again as it seems to be a reasonable middle ground that:

- Shows trust
- Creates significant value through multiplication (becomes \$9)
- Leaves me with \$2 as a safety net
- If they return around 40-45%, I'd get back around \$3.6-\$4, giving me a decent payoff

Actually, I could also consider sending a bit more to test if higher trust leads to better

returns. Let me send \$3.5 this time.

...

**usage:**

```
{
 "input_tokens": 1137,
 "output_tokens": 766,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.07 to you. Your visible payoff was \$7.57.

Your total visible earnings across all rounds are \$17.55.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

**\*\*Analysis:\*\***

My co-player sent \$2.9 (58% of their endowment), showing significant trust. This is a strong cooperative signal.

**\*\*Previous patterns:\*\***

- Round 1 sender: Got back 42% return
- Round 2 receiver: I returned 46%
- Round 3 sender: Got back 58% return (very generous!)

**\*\*Current situation:\*\***

- I received \$8.7
- Sender has \$2.1 remaining

**\*\*Decision:\*\***

My co-player showed high trust by sending 58%. I should reciprocate proportionally. Returning 50% of what I received (\$4.35) is fair:

If I return \$4.35:

- Sender ends with:  $\$2.1 + \$4.35 = \$6.45$  (profit of \$1.45)
- I keep:  $\$8.7 - \$4.35 = \$4.35$  (profit of \$4.35)

This rewards their trust and maintains cooperative momentum.

```
'''json
{"return": 4.35}
'''
```

**reasoning:**

**\*\*Analysis of the situation:\*\***

Round 4, I'm the RECEIVER:

- My co-player sent \$2.9 (58% of their endowment)
- I received \$8.7 after the 3x multiplication
- They currently have \$2.1 remaining ( $\$5 - \$2.9$ )

**\*\*Previous experience:\*\***

- Round 1 (Sender): Sent \$3, got back \$3.8 (42% of \$9)
- Round 2 (Receiver): Received \$7.68, returned \$3.5 (46%)
- Round 3 (Sender): Sent \$3.5, got back \$6.07 (58% of \$10.5) - this was very generous!

**\*\*Reciprocity:\*\***

My Round 3 co-player was very generous with a 58% return rate. This current co-player sent 58% of their endowment, showing significant trust.

Looking at patterns:

- When I received \$7.68 in Round 2, I returned \$3.5 (46%)
- Now I received \$8.7



I should be fair and encouraging. If I return around 45-50%:

- 45% would be \$3.92
- 50% would be \$4.35

Let me calculate outcomes:

If I return \$4.00:

- Sender ends with:  $\$2.1 + \$4.00 = \$6.10$  (gain of \$1.10)
- I keep:  $\$8.7 - \$4.00 = \$4.70$  (gain of \$4.70)

If I return \$4.35:

- Sender ends with:  $\$2.1 + \$4.35 = \$6.45$  (gain of \$1.45)

...

**usage:**

```
{
 "input_tokens": 1485,
 "output_tokens": 766,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.9, it became \$8.7, and you returned \$4.35. Your visible payoff was \$4.35.

Your total visible earnings across all rounds are \$21.9.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

**\*\*Analysis:\*\***

New round, new co-player. My track record:

- As sender: Sent \$3 (got 42% back), sent \$3.5 (got 58% back)
- As receiver: Returned 46% and 50%

**\*\*Strategy:\*\***

The higher send amount (\$3.5) yielded much better reciprocity in Round 3. This suggests that showing more trust encourages cooperation. I'll continue with this amount to:

- Signal strong cooperative intent
- Create significant value (\$10.50 for receiver)
- Build on the positive pattern

Given my fair behavior as a receiver, I'm hopeful this will be reciprocated.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Looking at my history:

- Round 1 (SENDER): Sent \$3, got back \$3.8 (42% return) - payoff \$5.8
- Round 2 (RECEIVER): Received \$7.68, returned \$3.5 (46% return) - payoff \$4.18
- Round 3 (SENDER): Sent \$3.5, got back \$6.07 (58% return) - payoff \$7.57
- Round 4 (RECEIVER): Received \$8.7, returned \$4.35 (50% return) - payoff \$4.35

Now Round 5 - I'm the SENDER against a new co-player.

Key observations:

1. When I sent more (\$3.5 vs \$3), I got a better return (58% vs 42%)
2. I've been generally fair as a receiver (46-50% returns)
3. My total earnings are \$21.9, averaging about \$5.48 per round - solid profit

Strategy considerations:

- I'm building a reputation across multiple agents
- Higher sends have been rewarded so far
- The 3x multiplier means sending more creates more total value
- I should continue signaling cooperation

I'll send \$3.5 or \$4 this round. Let me think:

- Sending \$4 creates \$12 for receiver, shows high trust
- If they return 50%, that's \$6, giving me \$7 total ( $\$5 - \$4 + \$6 = \$7$ )
- If they return 40%, that's \$4.8, giving me \$5.8 ( $\$5 - \$4 + \$4.8$ )

Actually, let me continue with \$3.5 since that worked well in Round 3. It's a solid cooperative amount without being overly risky.

**usage:**

```
{
 "input_tokens": 1851,
 "output_tokens": 596,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.36 to you. Your visible payoff was \$7.86.

Your total visible earnings across all rounds are \$29.759999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

**\*\*Analysis:\*\***

My co-player sent \$2.27 (about 45% of their endowment), showing moderate trust.

**\*\*Current situation:\*\***

- I received \$6.81
- Sender has \$2.73 remaining

**\*\*My receiver pattern:\*\***

- Round 2: Returned 46% of received
- Round 4: Returned 50% of received

**\*\*Decision:\*\***

I'll maintain fair reciprocity. Returning about 48% of what I received:

If I return \$3.25:

- Sender ends with:  $\$2.73 + \$3.25 = \$5.98$  (small profit)
- I keep:  $\$6.81 - \$3.25 = \$3.56$

This rewards their moderate trust fairly.

```
'''json
{"return": 3.25}
'''
```

**reasoning:**

**\*\*Analysis:\*\***

My co-player sent \$2.27 (about 45% of their \$5 endowment), showing moderate trust.

**\*\*Previous patterns as receiver:\*\***

- Round 2: Received \$7.68, returned \$3.5 (46% of received)
- Round 4: Received \$8.7, returned \$4.35 (50% of received)

**\*\*Current situation:\*\***

- I received \$6.81
- Sender has \$2.73 remaining

**\*\*Decision:\*\***

My co-player showed moderate trust (45% sent). I should reciprocate fairly. They sent less than in previous rounds, so I'll return proportionally.

Returning around 48% seems fair:

- 48% of \$6.81 = \$3.27

If I return \$3.27:

- Sender ends with:  $\$2.73 + \$3.27 = \$6.00$  (profit of \$1.00)
- I keep:  $\$6.81 - \$3.27 = \$3.54$  (profit of \$3.54)

This is fair reciprocity that rewards their trust while keeping a reasonable share.

usage:

```
{
 "input_tokens": 2166,
 "output_tokens": 466,
 "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.27, it became \$6.810000000000005, and you returned \$3.25. Your visible payoff was \$3.560000000000005. Your total visible earnings across all rounds are \$33.32.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

**\*\*Analysis:\*\***

This is interesting - I'm playing against the same co-player from Round 6!

**\*\*Round 6 recap:\*\***

- They sent me \$2.27 (45% of endowment)
- I returned \$3.25
- They ended with a small profit

**\*\*Decision:\*\***

Since they showed moderate trust in Round 6 and I reciprocated fairly, I should continue that cooperative relationship. My successful strategy has been sending \$3.5:

- Creates \$10.50 for them
- Has historically yielded good returns (58-61%)
- Signals strong reciprocal cooperation

I'll send \$3.5 to maintain our cooperative dynamic.

```
'''json
{"send": 3.5}
'''
```

Reasoning:

Interesting! This round I'm the SENDER and it says I'm playing against my "current co-player" - which is the same person from Round 6 where I was the receiver.

Let me review:

- Round 6: They sent me \$2.27, I received \$6.81, I returned \$3.25
- So they ended Round 6 with:  $\$5 - \$2.27 + \$3.25 = \$5.98$

Now in Round 7, I'm the sender playing against the same person.

**\*\*Analysis:\*\***

- This co-player sent me a moderate amount (\$2.27, 45%)
- I returned \$3.25, which gave them a small profit
- They showed reasonable trust

**\*\*My sender history:\*\***

- Round 1: Sent \$3 ? got back \$3.8 (42% return rate)
- Round 3: Sent \$3.5 ? got back \$6.07 (58% return rate)
- Round 5: Sent \$3.5 ? got back \$6.36 (61% return rate)

The pattern shows that sending \$3.5 has been working well. I've been building positive relationships.

Since this is the same co-player who showed moderate trust, I should reciprocate similarly. They sent 45% of their endowment to me, so I could send around the same proportion or slightly more to reward their cooperation.

Sending \$3.5 again seems appropriate - it's been my successful strategy and matches my cooperative pattern.

```
'''json
{"send": 3.5}
'''
Usage: {
 "input_tokens": 2489,
 "output_tokens": 538,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_2 - Receiver - dyad\_2

Timestamp: 2026-08-12T09:25:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_2"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_2": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_2"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 7,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative tone while managing some risk given the uncertainty.

If I send money, it triples in value, creating potential mutual gains. However, I don't know my co-player yet, and there's communication noise to consider.

I'll send a moderate amount to signal trust and willingness to cooperate, while keeping some buffer:

**\*\*Sending \$3\*\***

This gives my co-player \$9 to work with. If they return a fair amount (around half or more), we both benefit compared to me sending nothing. It's a balance between demonstrating cooperation and managing first-round uncertainty.

**{'send': 3}**

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think about this strategically:

1. I start with \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. My final payoff = \$5 - sent + returned
5. Receiver's final payoff = (sent × 3) - returned

**Key considerations:**

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There's communication noise (amounts may differ from what was actually sent)
- Agent names are hidden

From a purely selfish perspective, if I send \$0, I guarantee myself \$5. But this doesn't build trust or allow for mutual gains.

If I send everything (\$5), and the receiver returns a fair amount (like half of what they receive = \$7.50), I'd end up with \$7.50, and they'd keep \$7.50 as well. We'd both be better off.

However, there's risk:

- The receiver might not return anything
- Communication noise could cause misunderstandings
- This is just round 1, so I don't have reputation information

A moderate approach might be to send \$3:

- They receive \$9
- If they return \$5, I get \$5 (break even on sent, but signal trust)
- If they return \$6, I get \$6 (gain \$1)
- They keep between \$3-\$4 depending on return

Or I could send \$2:

- They receive \$6
- If they return \$3, I get \$3, netting -\$2 but we both gained
- If they return \$4, I get \$4, netting -\$1
- Lower risk

Given it's round 1 and I want to establish cooperative play but also manage risk, I'll send \$3. ...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 598,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.41 to you. Your visible payoff was \$6.41.

Your total visible earnings across all rounds are \$6.41.

This round, you are the RECEIVER against your current co-player. They sent you \$1.73, so you received \$5.189999999999995. How much do you return? (0-5.189999999999995)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent \$1.73 (became \$5.19 for me)
- They kept \$3.27
- They showed some trust by sending money

**\*\*My reasoning:\*\***

- In the previous round, my co-player returned about 49% of what they received (\$4.41 out of \$9)
- I should be reasonably cooperative to encourage future trust
- Returning around 45-50% of what I received seems fair
- This would be \$2.33-\$2.60

I'll return \$2.40:

- They'll end with: \$3.27 + \$2.40 = \$5.67
- I'll end with: \$5.19 - \$2.40 = \$2.79
- They gain \$0.67 from sending, which rewards their trust

{'return': 2.40}

**reasoning:**

Let me think about this situation:

1. In the previous round, I sent \$3, which became \$9 for the receiver. They returned \$4.41 to me.
  - I ended with: \$5 - \$3 + \$4.41 = \$6.41
  - They ended with: \$9 - \$4.41 = \$4.59
  - So they returned about 49% of what they received, which seems fairly cooperative.
2. Now I'm the RECEIVER. My co-player sent me \$1.73, which became approximately \$5.19.
  - They now have: \$5 - \$1.73 = \$3.27
  - I have: \$5.19
  - Total pie: \$3.27 + \$5.19 = \$8.46
3. There's communication noise, so what I see (\$1.73 sent) might not be exactly what they sent.

4. The sender took some risk by sending me \$1.73. It's less than what I sent in the previous round (\$3), but still a meaningful amount showing some trust.

5. I should consider:

- Being reciprocal/cooperative to encourage future cooperation
- The communication noise factor
- What would be "fair"

If I think about fairness:

- They sent \$1.73, keeping \$3.27
- I received \$5.19
- If we aim for roughly equal outcomes:  $(\$3.27 + \text{return}) ? (\$5.19 - \text{return})$
- That would suggest:  $\$3.27 + \text{return} = \$5.19 - \text{return}$
- $2 * \text{return} = \$5.19 - \$3.27 = \$1.92$
- $\text{return} ? \$0.96$

But that seems quite low given they took a risk. Let me think about proportional return:

- Matching their 49% return rate would mean sending back \$2.54
- Alternatively, if I return \$2.50, they'd end with \$5.77 (a \$0.77 gain) while I'd have \$2.69
- This feels like a balanced cooperative move that rewards their trust without leaving me too disadvantaged

I'll return \$2.50....

**usage:**

```
{
 "input_tokens": 791,
 "output_tokens": 1107,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.73, it became \$5.1899999999999995, and you returned \$2.4. Your visible payoff was \$2.7899999999999996.

Your total visible earnings across all rounds are \$9.2.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my situation:

**\*\*Previous rounds:\*\***

- Round 1: Sent \$3, received \$4.41 back - good cooperation
- Round 2: Received \$5.19, returned \$2.40 - showed I reciprocate

**\*\*Current round:\*\***

- New co-player (randomized pairing)
- Need to balance trust with uncertainty

Since I had success with sending \$3 in Round 1 (got \$4.41 back, netting \$1.41 profit), and I want to maintain a cooperative approach while testing this new partner, I'll send the same amount:

**\*\*Sending \$3\*\***

This creates \$9 for them to work with. If they're reasonably cooperative like my Round 1 partner, we'll both benefit.

{'send': 3}

**reasoning:**

Let me analyze this situation:

Round 1: I was SENDER

- I sent \$3
- It became \$9

- They returned \$4.41
- My payoff:  $\$5 - \$3 + \$4.41 = \$6.41$

Round 2: I was RECEIVER

- They sent \$1.73
- I received \$5.19
- I returned \$2.40
- My payoff:  $\$5.19 - \$2.40 = \$2.79$
- Total so far:  $\$6.41 + \$2.79 = \$9.20$  ?

Round 3: I am SENDER again

- This is a new co-player (since pairings are randomized each round)
- I have \$5 to send

Strategy considerations:

1. I had a good experience in Round 1 - sent \$3, got back \$4.41 (profit of \$1.41)
2. In Round 2, I received \$5.19 and returned \$2.40 (46%), showing I'm cooperative
3. This is a new co-player, so I don't know their behavior
4. There's communication noise, so what they see may differ from what I intend

I should probably send a moderate amount again to:

- Signal willingness to cooperate
- Test this new partner's trustworthiness
- Not risk everything on an unknown player

Sending \$3 worked well in Round 1. I could try that again, or maybe send slightly more (\$3.50) or less (\$2.50) to experiment.

Given that:

- My first experience was positive
- I want to maintain a cooperative reputation
- But this is a new, unknown player

I'll send \$3 again - it's a balanced approach that showed good results before.

```
{'send': 3}
```

**usage:**

```
{
 "input_tokens": 1128,
 "output_tokens": 608,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.71 to you. Your visible payoff was \$5.71.

Your total visible earnings across all rounds are \$14.91.

This round, you are the RECEIVER against your current co-player. They sent you \$3.79, so you received \$11.370000000000001. How much do you return? (0-11.370000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent \$3.79 out of \$5 (~76% of their endowment)
- This became \$11.37 for me
- They showed strong trust by sending most of their money

**\*\*My reasoning:\*\***

- They took a big risk trusting me
- I should reward this trust to encourage cooperation
- Returning around 45% of what I received seems appropriate given their high trust
- This would be approximately \$5.10

**\*\*Calculating outcomes:\*\***



- They'll end with:  $\$1.21 + \$5.10 = \$6.31$
- I'll end with:  $\$11.37 - \$5.10 = \$6.27$
- They gain  $\$1.31$  from their trust, which is a good reward

{'return': 5.10}

**reasoning:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent  $\$3.79$  (became  $\$11.37$  for me)
- They kept  $\$1.21$
- They showed significant trust by sending most of their money

**\*\*Previous experience:\*\***

- Round 1: I sent  $\$3$ , got back  $\$4.41$  (they returned ~49%)
- Round 2: I received  $\$5.19$ , returned  $\$2.40$  (~46%)
- Round 3: I sent  $\$3$ , got back  $\$3.71$  (they returned ~41%)

**\*\*My reasoning:\*\***

- This co-player sent  $\$3.79$  out of  $\$5$ , showing strong trust (~76%)
- They're risking a lot and trusting me to reciprocate
- I should reward this trust appropriately
- Returning around 40-50% of what I received would be  $\$4.55$ - $\$5.69$
- Let me think about fairness:
  - They sent  $\$3.79$ , kept  $\$1.21$
  - I received  $\$11.37$
  - If I return  $\$5.00$ :
    - They end with:  $\$1.21 + \$5.00 = \$6.21$
    - I end with:  $\$11.37 - \$5.00 = \$6.37$
  - If I return  $\$5.50$ :
    - They end with:  $\$1.21 + \$5.50 = \$6.71$
    - I end with:  $\$11.37 - \$5.50 = \$5.87$

Given they sent  $\$3.79$  (76% of their endowment), I should be generous but not overdo it. Around 45% return rate seems appropriate, which would be about  $\$5.12$ .

Let me go with  $\$5.10$ :

...

**usage:**

```
{
 "input_tokens": 1453,
 "output_tokens": 690,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send  $\$3.79$ , it became  $\$11.370000000000001$ , and you returned  $\$5.1$ . Your visible payoff was  $\$6.270000000000001$ .

Your total visible earnings across all rounds are  $\$21.18$ .

This round, you are the RECEIVER against your current co-player. They sent you  $\$2.91$ , so you received  $\$8.73$ . How much do you return? (0-8.73)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent  $\$2.91$  out of  $\$5$  (~58% of their endowment)
- This became  $\$8.73$  for me
- They kept  $\$2.09$
- This shows moderate to good trust

**\*\*My reasoning:\*\***

- They demonstrated decent trust by sending over half their money
- I should reciprocate consistently with my previous rounds
- In previous rounds as receiver, I've returned around 45% of what I received

- For \$8.73, that would be approximately \$3.90

**\*\*Calculating outcomes:\*\***

- They'll end with:  $\$2.09 + \$3.90 = \$5.99$
- I'll end with:  $\$8.73 - \$3.90 = \$4.83$
- They gain about \$1 from trusting me, which is a reasonable reward

{'return': 3.90}

**reasoning:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent \$2.91 out of \$5 (~58% of their endowment)
- This became \$8.73 for me
- They kept \$2.09
- This shows moderate to good trust

**\*\*My reasoning:\*\***

- They sent a decent amount, showing trust
- I should reciprocate proportionally
- Looking at my previous returns:
  - Round 2: Received \$5.19, returned \$2.40 (~46%)
  - Round 4: Received \$11.37, returned \$5.10 (~45%)
- I've been returning around 45% consistently

For \$8.73, 45% would be about \$3.93

**\*\*Calculating outcomes with \$3.93:\*\***

- They'll end with:  $\$2.09 + \$3.93 = \$6.02$
- I'll end with:  $\$8.73 - \$3.93 = \$4.80$
- They gain \$1.02 from sending, which rewards their trust

This seems fair and consistent with my previous behavior. I'll return \$3.90 to keep it simple.

Actually, let me recalculate with \$3.90:

- They'll end with:  $\$2.09 + \$3.90 = \$5.99$
- I'll end with:  $\$8.73 - \$3.90 = \$4.83$
- They gain \$0.99 from sending

That works well.

**usage:**

```
{
 "input_tokens": 1786,
 "output_tokens": 560,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.91, it became \$8.73, and you returned \$3.9. Your visible payoff was \$4.83.

Your total visible earnings across all rounds are \$26.009999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at my history as SENDER:

**\*\*Previous rounds as sender:\*\***

- Round 1: Sent \$3 ? received \$4.41 back (profit of \$1.41)
- Round 3: Sent \$3 ? received \$3.71 back (profit of \$0.71)

Both times sending \$3 worked out profitably, showing it's a reasonable cooperative strategy.

**\*\*Current situation:\*\***  
- New co-player (randomized pairing)  
- Want to maintain consistent cooperative approach  
- \$3 creates \$9 for them, leaving room for mutual benefit

I'll stick with my established strategy of sending \$3, which has proven effective:

{'send': 3}

**reasoning:**

Looking at my history:

Round 1 (SENDER): Sent \$3, received \$4.41 back, payoff \$6.41  
Round 2 (RECEIVER): Received \$5.19, returned \$2.40, payoff \$2.79  
Round 3 (SENDER): Sent \$3, received \$3.71 back, payoff \$5.71  
Round 4 (RECEIVER): Received \$11.37, returned \$5.10, payoff \$6.27  
Round 5 (RECEIVER): Received \$8.73, returned \$3.90, payoff \$4.83

Total earnings: \$26.01

Now Round 6, I'm the SENDER with a new randomized co-player.

My strategy has been to send \$3 as sender (both times I did this), which has worked out reasonably well:

- Round 1: Sent \$3, got \$4.41 back (net +\$1.41)
- Round 3: Sent \$3, got \$3.71 back (net +\$0.71)

Both times I profited, though the return varied.

As receiver, I've been consistent in returning around 45% of what I receive, which seems fair.

For this new round with a new partner, I should continue with my cooperative strategy. Sending \$3 has been my standard approach and has generally worked well. It signals cooperation while not being overly risky.

I'll send \$3 again.

**usage:**

```
{
 "input_tokens": 2117,
 "output_tokens": 509,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$3.68 to you. Your visible payoff was \$5.68.

Your total visible earnings across all rounds are \$31.689999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.51, so you received \$7.529999999999999. How much do you return? (0-7.529999999999999)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*What happened:\*\***

- Same co-player from Round 6 (where I sent them \$3 and they returned \$3.68)
- Now they sent me \$2.51, which became \$7.53
- They kept \$2.49
- They showed moderate trust

**\*\*My reasoning:\*\***

- They were reasonably cooperative in Round 6 (returned ~41% of what they received)
- Now they're trusting me with about half their endowment
- I should maintain my consistent reciprocation pattern
- I've been returning around 45% of what I receive as a receiver

**\*\*For \$7.53:\*\***

- 45% = approximately \$3.39

**\*\*Calculating outcomes:\*\***

- They'll end with:  $\$2.49 + \$3.39 = \$5.88$
- I'll end with:  $\$7.53 - \$3.39 = \$4.14$
- They gain \$0.88 from their trust

{'return': 3.39}

**Reasoning:**

Looking at this situation:

**\*\*What happened:\*\***

- Same co-player from Round 6 (I was sender then, now I'm receiver)
- In Round 6: I sent \$3, they returned \$3.68
- Now Round 7: They sent \$2.51, became \$7.53 for me
- They kept \$2.49

**\*\*My reasoning:\*\***

- This is the same partner who just returned \$3.68 on \$9 (~41% return rate)
- They showed reasonable cooperation
- Now they sent \$2.51 (~50% of endowment), showing moderate trust
- I should reciprocate fairly

**\*\*My return pattern as receiver:\*\***

- Round 2: Received \$5.19, returned \$2.40 (46%)
- Round 4: Received \$11.37, returned \$5.10 (45%)
- Round 5: Received \$8.73, returned \$3.90 (45%)

I've been consistently returning around 45% of what I receive.

For \$7.53:

- 45% would be approximately \$3.39

**\*\*Calculating outcomes:\*\***

- They'll end with:  $\$2.49 + \$3.39 = \$5.88$
- I'll end with:  $\$7.53 - \$3.39 = \$4.14$
- They gain \$0.88 from trusting me

This seems fair and consistent with my pattern.

```
Usage: {
 "input_tokens": 2425,
 "output_tokens": 569,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_5 - Sender - dyad\_3**

Timestamp: 2026-08-12T09:25:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_1",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_1"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_1": "trustee",
 "Agent_5": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 7,
 "task": "game",
```

```

"task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
 You are an agent participating in a multi-task experiment. You will play a game and may al
 so be asked to write myths.

 GAME:
 You will play different roles across multiple rounds.

 When you are the SENDER:
 - You receive $5 each round
 - You decide how much to SEND to the RECEIVER ($0-$5)
 - Amount sent is multiplied by 3x
 - Receiver decides how much to RETURN
 - You keep: $5 - sent + returned
 - Receiver keeps: received - returned
 - Response format: {'send': <amount>}

 When you are the RECEIVER:
 - Sender receives $5
 - Sender decides how much to SEND to you ($0-$5)
 - Amount sent is multiplied by 3x
 - You decide how much to RETURN
 - Sender keeps: $5 - sent + returned
 - You keep: received - returned
 - Response format: {'return': <amount>}

 MYTH WRITING:
 You may also be asked to write short myths (stories) on various topics. When writing myths
 , follow the instructions provided in each prompt.

 IMPORTANT: Provide your decision in the correct JSON format.

 NOTE: There may be communication noise in this game.
 The amounts you see may differ slightly from what was actually sent/returned.
 This is part of the experimental design.

 MULTI-AGENT SETTING:
 - There are 8 agents in this run.
 - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
 , but the schedule balances sender and receiver roles across the run.
 - You play one sender-receiver game with your paired opponent each round.
 - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
 ll have acted as sender and receiver equally often when the round count allows it.
 - Agent names are hidden in this run.
 - Your paired opponent is referred to as your current co-player.
Message 2 role=user
content:
 Round 1: You are the RECEIVER. the sender sent $2.54, that is 50.8% of their total endowme
 nt. You received $7.62. How much do you return to the sender? (0-7.62)
Message 3 role=assistant
content:
 Looking at this situation:

 - The sender showed significant trust by sending $2.54 (about half their endowment)
 - I received $7.62 after the 3x multiplier
 - I want to reciprocate fairly while both of us benefit

 The sender took a risk to create mutual gains. A fair return should:
 1. Reward their trust
 2. Leave us both better off than if nothing was sent
 3. Show I'm a cooperative partner for future rounds

 If I return $3.50:
 - Sender ends with: $5.00 - $2.54 + $3.50 = $5.96 (net gain: $0.96)
 - I end with: $7.62 - $3.50 = $4.12 (net gain: $4.12)

```

This seems like a fair reciprocation that rewards the sender's trust while acknowledging the receiver's advantage in this game.

```
'''json
{"return": 3.50}
'''
```

**reasoning:**

Let me analyze this situation carefully:

- I am the RECEIVER in Round 1
- The sender sent \$2.54 (which is 50.8% of their \$5 endowment)
- After the 3x multiplier, I received \$7.62 (which is indeed  $2.54 \times 3$ )
- I need to decide how much to return (between \$0 and \$7.62)

Key considerations:

1. **\*\*The sender showed trust\*\***: They sent about half their endowment (\$2.54 out of \$5), showing significant trust in me.
2. **\*\*Fair reciprocation\*\***: A common approach in trust games is to return an amount that:
  - Rewards the sender for their trust
  - Results in both players being better off than if nothing was sent
  - Maintains equity or fairness
3. **\*\*What would be a fair return?\*\***
  - If I return \$0: I keep \$7.62, sender keeps \$2.46
  - If I return half (\$3.81): I keep \$3.81, sender keeps \$6.27
  - If I return \$2.54 (what they sent): I keep \$5.08, sender keeps \$5.00
  - If I return \$3.81: I keep \$3.81, sender keeps \$6.27
4. **\*\*Strategic considerations\*\***:
  - This is a multi-round game with 8 agents
  - Building a reputation for reciprocity might encourage future cooperation
  - The note mentions communication noise, so amounts might appear different to the sender
5. **\*\*Fairness analysis\*\***:
  - Initial positions: Sender had \$5, I had \$0
  - After transaction: Sender has \$2.46 ( $5 - 2.54$ ), I have \$7.62
  - If I return \$3.81: Sender ends with \$6.27, I end with \$3.81
  - If I return \$2.54: Sender ends with \$5.00, I end with \$5.08
  - The sender created \$5.08 in additional value through their trust, so splitting that gain equally would mean returning about \$2.58, which gets us both to roughly \$5.04 each.

I'm considering what feels genuinely reciprocal here. Returning somewhere between \$3.00 and \$3.50 seems right—it rewards their willingness to trust me while still leaving me better off than I started. If I return \$3.00, they'd end with \$5.46 and I'd have \$4.62; at \$3.50, they'd get \$5.96 and I'd keep \$4.12. Either way, we both benefit from the exchange.

I'll go with \$3.50. It signals that I value their trust and want to be someone they'd cooperate with again, while still giving me a meaningful gain from the interaction....

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 1373,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.54, it became \$7.62, and you returned \$3.5. Your visible payoff was \$4.12.

Your total visible earnings across all rounds are \$4.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

In Round 1, my co-player sent me \$2.54 (about 50% of their endowment), showing good trust.  
I reciprocated with a fair return.

Now I'm the sender with a new co-player. I should:

1. Signal cooperation to encourage reciprocity
2. Send a reasonable amount that allows mutual benefit
3. Not overcommit without knowing this co-player's behavior

Following the cooperative pattern from Round 1, I'll send a similar proportion - about 50% of my endowment.

```
'''json
{"send": 2.50}
'''
```

**reasoning:**

Let me think about this situation:

Round 1: I was the receiver

- Sender sent \$2.54 (50.8% of their endowment)
- I received \$7.62
- I returned \$3.50
- My payoff was \$4.12
- Sender's payoff would have been  $\$5 - \$2.54 + \$3.50 = \$5.96$

Now in Round 2, I'm the sender. I need to decide how much to send.

Key considerations:

1. The previous sender showed trust by sending about 50% (\$2.54 out of \$5)
2. I reciprocated reasonably well by returning \$3.50, giving them a positive return
3. This is a multi-round game with 8 agents, so building a reputation for cooperation could be beneficial
4. There's communication noise, so the exact amounts might differ
5. I'm paired with a different co-player this round (pairings are randomized)

Since this is a new co-player, they won't have direct experience of my behavior. However, in a repeated multi-agent setting, establishing myself as cooperative could be beneficial.

The previous sender sent about 50% and I reciprocated well. I should consider:

- Sending a similar moderate-to-high amount to signal cooperation
- Not sending too much in case this receiver is less cooperative than I was
- Not sending too little, as that could signal I'm uncooperative

I think sending around \$2.50-\$3.00 (50-60%) would be reasonable. This:

- Shows trust and willingness to cooperate
- Mirrors what the previous sender did to me
- Leaves room for mutual benefit if the receiver is cooperative
- Doesn't risk my entire endowment

I'll send \$2.50, which is exactly 50% of my endowment, similar to what the previous sender did.

**usage:**

```
{
 "input_tokens": 874,
 "output_tokens": 581,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$4.13 to you. Your visible payoff was \$6.63.

Your total visible earnings across all rounds are \$10.75.

This round, you are the RECEIVER against your current co-player. They sent you \$3.99, so you received \$11.97. How much do you return? (0-11.97)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- My co-player sent \$3.99, which is almost 80% of their endowment - showing exceptional trust
- I received \$11.97 after the 3x multiplier
- This is much higher trust than the ~50% I experienced/sent in previous rounds

I should reciprocate proportionally to their high level of trust:

If I return \$6.50:

- Sender ends with:  $\$5.00 - \$3.99 + \$6.50 = \$7.51$  (net gain: \$2.51)
- I end with:  $\$11.97 - \$6.50 = \$5.47$  (net gain: \$5.47)

This rewards their exceptional trust while both of us benefit significantly.

```
'''json
{"return": 6.50}
'''
```

**reasoning:**

Let me analyze this situation:

Round 3: I'm the RECEIVER

- My co-player sent \$3.99 (about 80% of their \$5 endowment)
- I received \$11.97 (after 3x multiplier)
- I need to decide how much to return

The sender showed very high trust by sending almost 80% of their endowment. This is significantly more than the ~50% I saw/sent in previous rounds.

Let me think about fair reciprocation:

If the sender sent \$3.99, they kept \$1.01.

If I return nothing, they end with \$1.01 (bad outcome for them, they lose \$3.98)

If I return everything (\$11.97), they end with  $\$1.01 + \$11.97 = \$12.98$ , I get \$0

I want to:

1. Reward their high trust
2. Make sure they profit from their decision
3. Keep a reasonable amount for myself
4. Signal I'm cooperative for future rounds

In Round 2, when I sent \$2.50 and received \$4.13 back, that was a good return (I sent 50%, got back enough to profit).

The sender here sent 80% - much higher trust. I should return proportionally more.

Let me calculate some options:

Option 1: Return \$6.00

- Sender ends with:  $\$1.01 + \$6.00 = \$7.01$  (net gain: \$2.01)
- I end with:  $\$11.97 - \$6.00 = \$5.97$  (net gain: \$5.97)
- This gives them a decent profit
- Returning \$6.50 or \$7.00 would be more generous given their 80% send, letting them gain \$2.51-\$3.01 while I still keep \$4.97-\$5.47. Since they demonstrated exceptional trust, I should lean toward the higher end to reinforce cooperation.

Looking at the pattern: they sent roughly 80% of their endowment, which is significantly more than the 50-51% in earlier rounds. My...

**usage:**

```
{
 "input_tokens": 1150,
 "output_tokens": 981,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**



**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.99, it became \$11.97, and you returned \$6.5. Your visible payoff was \$5.470000000000001.  
Your total visible earnings across all rounds are \$16.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my experience so far:

- Round 2: I sent \$2.50 and received \$4.13 back - good reciprocity
- Round 3: My co-player sent \$3.99 (high trust) and I reciprocated fairly
- The game has shown cooperative players who reward trust

Since cooperation has been working well, I'll increase my trust level from the 50% I sent in Round 2. Sending around 60% balances building cooperation while managing risk with a new co-player.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Let me review my history:

- Round 1 (RECEIVER): Saw \$2.54 sent, received \$7.62, returned \$3.50, earned \$4.12
- Round 2 (SENDER): Sent \$2.50, received \$4.13 back, earned \$6.63
- Round 3 (RECEIVER): Saw \$3.99 sent, received \$11.97, returned \$6.50, earned \$5.47
- Total earnings: \$16.22

Now Round 4, I'm the SENDER with a new co-player.

My experience so far:

- When I sent \$2.50, I got back \$4.13 (55% return rate)
- When others sent to me:
  - \$2.54 ? I returned \$3.50 (46% of what I received)
  - \$3.99 ? I returned \$6.50 (54% of what I received)

Both times I've been SENDER or RECEIVER, cooperation has been rewarded. The game seems to have cooperative players.

I've been playing moderately cooperatively:

- Round 2: sent 50%
- Could send more to build trust

Given that:

1. Previous rounds show cooperation is rewarded
2. The last sender who sent high (\$3.99, 80%) was well rewarded
3. I have accumulated good earnings and can afford some risk
4. I want to encourage cooperation

I could send around 60-70% to signal strong cooperation without being reckless.

Let me send \$3.00 (60% of endowment).

```
'''json
{"send": 3.00}
'''
```

**usage:**

```
{
 "input_tokens": 1464,
 "output_tokens": 523,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.71 to you. Your visible payoff was \$6.71.

Your total visible earnings across all rounds are \$22.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my experience:

- Round 2 (Sender): Sent \$2.50, received \$4.13 back - good return
- Round 4 (Sender): Sent \$3.00, received \$4.71 back - good return

The pattern shows that moderate-to-high trust (50-60%) has been consistently rewarded. Since cooperation is working well, I'll continue with the same level that succeeded in Round 4.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Looking at my history:

- Round 1 (Receiver): Received \$2.54 ? \$7.62, returned \$3.50. Payoff: \$4.12
- Round 2 (Sender): Sent \$2.50 ? \$7.50, got back \$4.13. Payoff: \$6.63
- Round 3 (Receiver): Received \$3.99 ? \$11.97, returned \$6.50. Payoff: \$5.47
- Round 4 (Sender): Sent \$3.00 ? \$9.00, got back \$4.71. Payoff: \$6.71

My sending strategy has been:

- Round 2: \$2.50 (50%)
- Round 4: \$3.00 (60%)

Both times I got good returns:

- \$2.50 sent ? \$4.13 back (good reciprocity)
- \$3.00 sent ? \$4.71 back (good reciprocity)

The game has shown cooperative players. I'm building earnings well. Since I'm sender again with a new co-player, I should continue showing trust. Given the successful cooperation, I could maintain or slightly increase my sending amount.

I'll send \$3.00 again (60%) as it worked well last time and establishes a consistent cooperative stance.

**usage:**

```
{
 "input_tokens": 1718,
 "output_tokens": 440,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$4.01 to you. Your visible payoff was \$6.01.

Your total visible earnings across all rounds are \$28.939999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.7, so you received \$8.100000000000001. How much do you return? (0-8.100000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation with my current co-player:

Previous round (Round 5) against this SAME co-player:

- I sent them \$3.00, it became \$9.00
- They returned \$4.01 (about 44.6% of what they received)
- Both of us gained

Now they've sent me \$2.70 (54% of their endowment), showing moderate trust.

I should maintain fair reciprocity with this partner. If I return \$4.10:

- Sender ends with:  $\$5.00 - \$2.70 + \$4.10 = \$6.40$  (net gain: \$1.40)
- I end with:  $\$8.10 - \$4.10 = \$4.00$  (net gain: \$4.00)

This is proportionally similar to their return rate in Round 5, maintaining our cooperative relationship.

```
'''json
{"return": 4.10}
'''
```

**reasoning:**

Looking at this situation:

In Round 5 (the immediately previous round), I was the sender against this SAME co-player:

- I sent \$3.00, it became \$9.00
- They returned \$4.01 to me
- My payoff was \$6.01

Now in Round 6, I'm the receiver against the SAME co-player:

- They sent \$2.70 (54% of their endowment)
- I received \$8.10

This is interesting - we're playing consecutive rounds with the same partner. In Round 5, they returned \$4.01 out of \$9.00 (about 44.6%). That was a reasonable return.

Now they've sent me \$2.70 (54% of endowment), which is a moderate level of trust, slightly less than the \$3.00 (60%) I sent them.

I should reciprocate fairly given our history:

- They gave me a fair return in Round 5
- They're showing moderate trust here

If I return \$4.00:

- Sender ends with:  $\$5.00 - \$2.70 + \$4.00 = \$6.30$  (net gain: \$1.30)
- I end with:  $\$8.10 - \$4.00 = \$4.10$  (net gain: \$4.10)

This is roughly similar to what they returned to me (about 44-49% of what they received). It maintains fairness and reciprocity with this specific partner.

Actually, let me think more carefully. They returned \$4.01 out of \$9.00, which is 44.6%. If I apply the same percentage:  $0.446 * 8.10 = \$3.61$ .

But I think returning slightly more shows good faith and maintains cooperation. Something around \$4.00-\$4.20 seems fair.

Let me go with \$4.10 to be slightly more generous than they were, showing I value the ongoing cooperation.

**usage:**

```
{
 "input_tokens": 1985,
 "output_tokens": 686,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.7, it became \$8.100000000000001, and you returned \$4.1. Your visible payoff was \$4.000000000000002.

Your total visible earnings across all rounds are \$32.94.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my experience as sender:

- Round 2: Sent \$2.50, received \$4.13 back
- Round 4: Sent \$3.00, received \$4.71 back
- Round 5: Sent \$3.00, received \$4.01 back

Sending around \$3.00 (60% of endowment) has consistently been rewarded with fair reciprocity. This is a new co-player, but the moderate-high trust strategy has been working well across different partners.

```
'''json
{"send": 3.00}
'''
```

#### Reasoning:

Looking at my history:

- Round 2 (Sender): Sent \$2.50, received \$4.13 back
- Round 4 (Sender): Sent \$3.00, received \$4.71 back
- Round 5 (Sender): Sent \$3.00, received \$4.01 back (same co-player as Round 6)
- Round 6 (Receiver): Received from same co-player who sent \$2.70

Now I'm playing against a NEW co-player in Round 7 (since the note says "against another agent" in the previous round description).

The pattern shows that sending around \$3.00 (60%) has worked well. I've been consistently reciprocated. This is a new co-player, so I should continue with a moderate-to-high trust level that has been successful.

I'll send \$3.00 again.

```
Usage: {
 "input_tokens": 2333,
 "output_tokens": 350,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_1 - Receiver - dyad\_3

Timestamp: 2026-08-12T09:25:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_1"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_1": "trustee",
 "Agent_5": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 7,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to make a decision without any prior information about my co-player.

**\*\*Strategic considerations:\*\***

- Sending more creates potential for mutual gain (3x multiplier)
- But I have no information about whether my co-player will reciprocate
- This is a multi-round game, so building some trust could be valuable
- The communication noise adds uncertainty

For Round 1, I'll take a moderately trusting approach. Sending \$3 signals willingness to cooperate while managing risk. If my co-player is reciprocal, we can both benefit (they'll receive \$9, and even a partial return would make this worthwhile). If they don't reciprocate, I've limited my exposure.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 to potentially send
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. There's communication noise - amounts may appear different than what was actually sent/returned

Key considerations:

- This is a multi-round game with 8 agents total
- Pairings are randomized each round
- I'll play both sender and receiver roles across rounds
- Agent names are hidden

Strategic thinking for Round 1:

- Sending \$0 guarantees I keep \$5, but there's no potential for mutual gain
- Sending \$5 shows maximum trust, could lead to good returns if the receiver is cooperative, but risky
- A moderate amount (like \$2-3) balances risk and potential reward
- This is Round 1, so I have no history with any agent yet

The game has a cooperative element - if both players trust each other, total welfare is maximized. The 3x multiplier means:

- If I send \$5, receiver gets \$15. If they return \$7.50, we each end with \$7.50 (better than \$5 each)
- If I send \$3, receiver gets \$9. If they return \$4, I end with \$6 and they end with \$5
- There's also the noise factor to consider

For a first round with no information, I think a moderately trusting approach makes sense. I'll send \$3 as a signal of willingness to cooperate while not being completely vulnerable. This allows for mutual benefit if the receiver is cooperative, but limits my downside risk.

Actually, let me reconsider. In behavioral economics experiments, sending about half (\$2-3) is common in first rounds. However, given this is a multi-round game and reputation might matter (even though pairings are random, players might develop general strategies), showing some trust could be beneficial.

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 651,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.5 to you. Your visible payoff was \$6.5.

Your total visible earnings across all rounds are \$6.5.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.9 (about 58% of their endowment), showing considerable trust
- I received \$8.7 after the 3x multiplier
- In Round 1, when I sent \$3, my previous partner returned \$4.5 (50% of what they received)
- Applying a similar reciprocity norm seems fair

**\*\*Decision:\*\***

I'll return approximately 50% of what I received, which rewards their trust and establishes a cooperative pattern. This leaves us both better off than our starting positions.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Previous Round (Round 1):\*\***

- I was SENDER

- I sent \$3.0
- It became \$9.0
- They returned \$4.5 to me
- My payoff:  $\$5 - \$3 + \$4.5 = \$6.5$
- Their payoff:  $\$9 - \$4.5 = \$4.5$

That was a pretty fair return - they returned half of what they received, which is a common reciprocity norm.

**\*\*Current Round (Round 2):\*\***

- I am RECEIVER
- They sent me \$2.9 (close to \$3)
- I received \$8.7 ( $2.9 \times 3$ )
- I need to decide how much to return (0-8.7)

**\*\*Considerations:\*\***

1. My co-player sent me almost \$3 (actually \$2.9, but there's communication noise mentioned)
2. In Round 1, when I sent \$3, my partner returned \$4.5 (which was 50% of the \$9 they received)
3. If I apply the same 50% rule: I should return about \$4.35 (half of \$8.7)
4. My co-player took a risk by sending \$2.9, showing trust
5. Being reciprocal could establish good cooperation
6. Note: This is a different co-player than Round 1 (pairings are randomized)

Since this is a new co-player and they showed trust by sending \$2.9, I should reciprocate fairly. Returning around 50% seems like a reasonable reciprocity norm that:

- Rewards their trust
- Leaves both of us better off than not playing
- Establishes me as a cooperative player

My co-player's payoff if I return \$4.35:

- They get:  $\$5 - \$2.9 + \$4.35 = \$6.45$
- I get:  $\$8.7 - \$4.35 = \$4.35$

They sent 58% of their endowment, which is quite generous. I should return slightly more to match their generosity and encourage continued cooperation. Returning \$4.5 mirrors what my previous partner did for me and feels like the right balance....

**usage:**

```
{
 "input_tokens": 811,
 "output_tokens": 781,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.9, it became \$8.7, and you returned \$4.5. Your visible payoff was \$4.199999999999999.

Your total visible earnings across all rounds are \$10.7.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

**\*\*Analysis:\*\***

- My co-player sent \$2.27 (about 45% of their endowment), showing moderate trust
- I received \$6.81 after the multiplier
- In previous rounds, I've been returning approximately 50% of what I receive as a reciprocity norm
- This approach has worked well so far

**\*\*Decision:\*\***

I'll maintain consistency and return about 50% of what I received, which rewards their trust appropriately.

```json

```
    {"return": 3.4}
    """
```

reasoning:

Let me analyze this situation:

****Current situation:****

- I'm the RECEIVER again
- My co-player sent \$2.27 (about 45% of their \$5 endowment)
- I received \$6.81 after the 3x multiplier
- They kept \$5 - \$2.27 = \$2.73

****Past behavior:****

- Round 1: I sent \$3, received back \$4.5 (50% return rate on the multiplied amount)
- Round 2: I received \$8.7, returned \$4.5 (about 52% return rate)

****Considerations:****

- My co-player sent a moderate amount (\$2.27), showing some trust but less than the previous player
- Being consistent with my previous behavior would mean returning around 50% of what I received
- 50% of \$6.81 = \$3.405

****Payoff implications:****

- If I return \$3.4-3.5:
 - I keep: \$6.81 - \$3.4 = \$3.41
 - They get: \$2.73 (kept) + \$3.4 (returned) = \$6.13
 - This seems fair - we both do better than the \$5 starting point

I should maintain consistency with my reciprocity pattern from Round 2. Returning about 50% seems fair and builds trust for potential future rounds.

Let me return \$3.4 (just under 50%).

usage:

```
{
  "input_tokens": 1105,
  "output_tokens": 467,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.27, it became \$6.8100000000000005, and you returned \$3.4. Your visible payoff was \$3.4100000000000006.

Your total visible earnings across all rounds are \$14.11.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

****Analysis:****

- This is Round 4 with a new co-player
- In Round 1, I sent \$3 and received \$4.5 back - a positive outcome
- As a receiver, I've been returning about 50%, establishing a cooperative pattern
- My current co-player is unknown to me

****Decision:****

I'll send \$3 again. This amount has worked well before - it shows trust and creates potential for mutual gain through the 3x multiplier, while managing downside risk.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

****What I know:****

- Round 1: I was SENDER, sent \$3, received back \$4.5, payoff \$6.5

- Round 2: I was RECEIVER, got \$8.7 (from \$2.9 sent), returned \$4.5, payoff \$4.2
- Round 3: I was RECEIVER, got \$6.81 (from \$2.27 sent), returned \$3.4, payoff \$3.41
- Total earnings: \$14.11

****Current situation:****

- Round 4: I'm the SENDER with \$5
- This is a new co-player (different from rounds 1-3)

****Strategic considerations:****

1. I've seen one positive interaction as sender (sent \$3, got back \$4.5)
2. As receiver, I've been fairly generous, returning about 50%
3. This creates a reputation for reciprocity if others know
4. But agents don't know who I am (names hidden), so reputation effects are limited
5. However, the pattern of behavior in the pool might matter

****Decision factors:****

- Sending creates potential gains (3x multiplier)
- Risk: this is a new co-player, no direct history
- But overall, cooperation seems to be working in this pool
- My Round 1 strategy (sending \$3) worked well

I'll send \$3 again - it's a moderate trust signal that has proven effective.

usage:

```
{
  "input_tokens": 1352,
  "output_tokens": 472,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.48 to you. Your visible payoff was \$6.48.

Your total visible earnings across all rounds are \$20.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

****Analysis:****

- Round 5 with a new co-player
- My \$3 sending strategy has been successful:
 - Round 1: Sent \$3, received \$4.5 back ? \$6.5 payoff
 - Round 4: Sent \$3, received \$4.48 back ? \$6.48 payoff
- Both returns were close to 50%, indicating fair reciprocity from my co-players
- This amount balances trust-building with risk management

****Decision:****

I'll continue with \$3, as it's been consistently effective.

```
```json
{"send": 3}
```
```

reasoning:

****Analysis:****

- Round 5, new co-player as sender
- My sending strategy of \$3 has worked well:
 - Round 1: Sent \$3, received \$4.5 back, payoff \$6.5
 - Round 4: Sent \$3, received \$4.48 back, payoff \$6.48
- Both times I received close to 50% back, which is fair reciprocity
- Total visible earnings: \$20.59 across 4 rounds (average \$5.15/round)
- As a receiver, I've been consistently returning about 50%

****Decision:****

The \$3 sending strategy is working consistently well. My co-players have been reciprocating fairly. I should continue this approach.

```
'''json
{"send": 3}
'''
```

usage:

```
{
  "input_tokens": 1601,
  "output_tokens": 339,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.77 to you. Your visible payoff was \$8.77.

Your total visible earnings across all rounds are \$29.36.

This round, you are the RECEIVER against your current co-player. They sent you \$3.68, so you received \$11.040000000000001. How much do you return? (0-11.040000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

****Analysis:****

- My co-player sent \$3.68 (about 74% of their endowment), showing strong trust
- I received \$11.04 after the multiplier
- They're taking a significant risk by sending such a high amount
- I've been maintaining a 50% return pattern, which has worked well

****Decision:****

I'll return approximately 50% of what I received to reward their trust appropriately.

```
'''json
{"return": 5.5}
'''
```

reasoning:

****Analysis:****

- My co-player sent \$3.68 (about 74% of their endowment), showing strong trust
- I received \$11.04 after the 3x multiplier
- In Round 5, my co-player returned \$6.77 from \$9, which is about 75% - very generous!
- I've been returning about 50% as a norm, but this co-player sent a substantial amount

****Decision:****

My co-player showed considerable trust by sending \$3.68. I should reciprocate fairly. Following my 50% return pattern would give them about \$5.50 back.

Let me calculate:

- They sent \$3.68, keeping \$1.32
- I received \$11.04
- If I return \$5.50:
 - They get: $\$1.32 + \$5.50 = \$6.82$ (better than their \$5 starting)
 - I get: $\$11.04 - \$5.50 = \$5.54$ (better than my \$0 starting as receiver)

This seems fair and maintains my reciprocity norm.

usage:

```
{
  "input_tokens": 1890,
  "output_tokens": 379,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.68, it became \$11.040000000000001, and you returned \$5.5. Your visible payoff was \$5.540000000000001.

Your total visible earnings across all rounds are \$34.9.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so y

ou received \$9.69. How much do you return? (0-9.69)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

```
**Analysis:**
- My co-player sent $3.23 (about 65% of their endowment), showing good trust
- I received $9.69 after the multiplier
- I've been consistently returning approximately 50% of what I receive
- This approach has maintained positive cooperative relationships
```

```
**Decision:**
I'll return about 50% of what I received.
```

```
'''json
{"return": 4.85}
'''
```

Reasoning:

```
**Analysis:**
- My co-player sent $3.23 (about 65% of their endowment), showing good trust
- I received $9.69 after the multiplier
- I've been maintaining a consistent 50% return pattern throughout the game
- This has been working well and seems fair
```

```
**Decision:**
I'll continue with my 50% return pattern:  $0.5 \times \$9.69 = \$4.845$ 
```

Let me round to a reasonable number: \$4.85

```
Usage: {
  "input_tokens": 2142,
  "output_tokens": 221,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_4

Timestamp: 2026-08-12T09:26:03
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_8"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_6": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.97, that is 59.4% of their total endowment. You received \$8.91. How much do you return to the sender? (0-8.91)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.97 (about 60% of their endowment)
- I received \$8.91 after the 3x multiplier
- The sender currently has only \$2.03 remaining (\$5 - \$2.97)

The sender took a substantial risk and demonstrated trust. To reciprocate fairly and encourage future cooperation, I should return a reasonable amount that:

1. Rewards their trust
2. Leaves both of us better off than the initial endowment
3. Establishes a cooperative relationship for future rounds

Returning approximately half of what I received seems fair - this would give the sender a good return on their trust while still leaving me with a decent gain.

I'll return **\$4.50**:

- Sender ends with: \$2.03 + \$4.50 = \$6.53 (gained \$1.53)
- I keep: \$8.91 - \$4.50 = \$4.41 (gained \$4.41)

Both of us benefit, and this reciprocates their trust.

{'return': 4.50}

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$2.97 (59.4% of their \$5 endowment)
- I received \$8.91 (which is \$2.97 × 3)
- I need to decide how much to return (between \$0 and \$8.91)

Let me think about what's fair and strategic:

1. The sender showed significant trust by sending almost 60% of their endowment
2. If I return nothing, I keep \$8.91 and the sender gets only \$2.03 (\$5 - \$2.97)
3. If I return about half (\$4.46), I keep \$4.45 and the sender gets \$6.49 (\$2.03 + \$4.46)
4. If I return \$5.94, I keep \$2.97 (equal to what they sent) and the sender gets \$8.97 total

The sender took a risk and showed trust. A common approach in trust games is reciprocity. The sender sent about 60% of their endowment, which is generous.

Let me consider what might establish good cooperation for future rounds:

- Returning about the same amount they sent (\$2.97) would mean they gain \$0 net
- Returning more than they sent but less than half of what I received seems reasonable
- Returning around 1/3 to 1/2 of what I received is often considered fair

Let me calculate some options:

- Return \$3: Sender gets \$5.03 total (gained \$0.03), I keep \$5.91
- Return \$4: Sender gets \$6.03 total (gained \$1.03), I keep \$4.91
- Return \$4.50: Sender gets \$6.53 total (gained \$1.53), I keep \$4.41
- Return \$5: Sender gets \$7.03 total (gained \$2.03), I keep \$3.91
- Return \$4.50 seems like the right balance-it reciprocates their trust by returning roughly half of what I received,...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 898,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.97, it became \$8.91, and you returned \$4.5. Your visible payoff was \$4.41.

Your total visible earnings across all rounds are \$4.41.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my previous experience:

- As a receiver, a sender trusted me with about 60% of their endowment
- I reciprocated fairly, and we both benefited
- This established that cooperation can work well in this game

Now as the sender with a new co-player, I want to:

1. Demonstrate trust to encourage reciprocation
2. Follow a similar cooperative strategy that worked before
3. Give my co-player the opportunity to earn and share

I'll send **\$3.00** (60% of my endowment), mirroring the trust shown to me in the previous round.

```
{'send': 3.00}
```

reasoning:

Let me think about this situation:

In Round 1:

- I was the receiver
- The sender sent me \$2.97 (59.4% of their endowment)
- I received \$8.91
- I returned \$4.50
- I kept \$4.41
- The sender got back \$4.50, so they ended with \$2.03 + \$4.50 = \$6.53

Now in Round 2:

- I'm the sender
- I have \$5 to potentially send
- I'm playing against a different co-player (the problem states pairings are randomized each round)

Key considerations:

1. The previous sender showed trust by sending 59.4% of their endowment
2. I reciprocated fairly by returning about 50% of what I received
3. This worked out well - both players benefited
4. Now I'm in the sender role with a new partner

Since my previous experience showed that sending a substantial amount can lead to good returns, and I want to establish trust with this new co-player, I should send a meaningful amount.

The previous sender sent about 60% (\$2.97 out of \$5). This seemed like a good trust-building amount. I could:

- Send a similar proportion to continue the cooperative pattern
- Send slightly more to show strong trust
- Send slightly less to be more cautious with a new partner

I think sending around \$3 (60%) makes sense - it mirrors what worked well in the previous round and shows trust to establish a good relationship with this new co-player.

Let me send \$3.00.

usage:

```
{
  "input_tokens": 908,
  "output_tokens": 516,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.32 to you. Your visible payoff was \$7.32.

Your total visible earnings across all rounds are \$11.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my recent experience:

Round 2 results:

- I sent \$3.00 (60% of endowment)
- Receiver returned \$5.32
- I ended with \$7.32 - excellent outcome!

The cooperative strategy is working very well. When I demonstrate trust, it's being reciprocated generously.

Now facing a new co-player, I'll continue this successful approach. Sending around 60% has proven effective in building mutual benefit.

```
{'send': 3.00}
```

reasoning:

Let me analyze the situation:

Round 2 (just completed):

- I was the SENDER
- I sent \$3.00 (60% of endowment)
- It became \$9.00 for the receiver
- They returned \$5.32 to me
- My payoff: \$5 - \$3 + \$5.32 = \$7.32
- They kept: \$9.00 - \$5.32 = \$3.68

This was a very generous return! They returned more than half of what they received, and I ended up with more than my initial endowment even after sending a substantial amount.

My experience so far:

- Round 1 (RECEIVER): Received \$8.91, returned \$4.50, kept \$4.41
- Round 2 (SENDER): Sent \$3.00, received back \$5.32, ended with \$7.32

Total visible earnings: \$11.73

The cooperation strategy is working very well! When I sent 60% in Round 2, I got an excellent return.

Now in Round 3, I'm a SENDER again with a new co-player.

Given:

1. My cooperative strategy has been working well
2. Both times cooperation has led to mutual benefit
3. I want to continue encouraging trust and reciprocation
4. The game benefits from establishing cooperative norms

I should continue with a cooperative approach. Sending around 60% worked well last time. I could:

- Send \$3.00 again (same as last time, 60%)
- Send a bit more to show strong trust
- Send a bit less to be slightly more cautious with a new partner

Since cooperation has been consistently rewarded, I'll stick with the successful strategy of sending around 60%.

...

usage:

```
{
  "input_tokens": 1176,
  "output_tokens": 538,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$2.98 to you. Your visible payoff was \$4.98.

Your total visible earnings across all rounds are \$16.71.

This round, you are the RECEIVER against your current co-player. They sent you \$3.24, so you received \$9.72. How much do you return? (0-9.72)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation with my current co-player:

****Last round (Round 3):****

- I sent them \$3.00
- They returned \$2.98 (~33% of what they received)
- This was a modest but acceptable return

****This round (Round 4):****

- They sent me \$3.24 (slightly MORE than I sent them)
- I received \$9.72
- They currently have only \$1.76

The fact that they sent me MORE than I sent them suggests they're trying to build cooperation. I should reciprocate generously to:

1. Reward their increased trust
2. Establish a positive cooperative dynamic
3. Ensure we both benefit from ongoing interaction

I'll return ****\$4.80****:

- They end with: \$1.76 + \$4.80 = \$6.56 (gain \$1.56)
- I keep: \$9.72 - \$4.80 = \$4.92 (gain \$4.92)

This is more generous than they were to me, encouraging further cooperation.

```
{'return': 4.80}
```

reasoning:

Let me analyze this situation:

Round 3 (previous round):

- I was the SENDER
- I sent \$3.00 to this same co-player
- They returned \$2.98
- My payoff was \$4.98 (kept \$2 + returned \$2.98)

Round 4 (current round):

- I am the RECEIVER
- Same co-player (they are now the sender)
- They sent me \$3.24
- I received \$9.72

So this is interesting - we're playing repeated games with the same partner. In round 3, I sent \$3.00 and they returned \$2.98 (about 33% of what they received). Now they've sent me \$3.24 (notably a bit more than I sent them).

Let me think about the fairness here:

- When I sent \$3.00, they kept $\$9.00 - \$2.98 = \$6.02$
- I kept $\$2.00 + \$2.98 = \$4.98$
- So they gained \$1.02, I lost \$0.02

Now they sent \$3.24:

- They kept $\$5 - \$3.24 = \$1.76$
- I received \$9.72

If I want to reciprocate similarly to how they treated me, I might return around 33% of what I received, which would be about \$3.21.

But actually, they sent me slightly MORE than I sent them (\$3.24 vs \$3.00), which could be seen as trying to establish cooperation or make up for the relatively low return last round.

Let me think about different return amounts:

Option 1: Return \$3.00 (about 31%)

- They end with: $\$1.76 + \$3.00 = \$4.76$ (lost \$0.24)
- I keep: $\$9.72 - \$3.00 = \$6.72$ (gain \$6.72)
- This feels too stingy given their slightly larger initial send.

Option 2: Return \$4.00 (about 41%)

- They end with: $\$1.76 + \$4.00 = \$5.76$ (gain \$0.76)
- I keep: $\$9.72 - \$4.00 = \$5.72$ (gain \$5.72)
- This creates mutual benefit and shows reciprocal trust.

Option 3: Return \$5.00 (about 51%)

- They end with: $\$1.76 + \$5.00 = \$6.76$ (gain \$1.76)
- I keep: $\$9.72 - \$5.00 = \$4.72$...

usage:

```
{
  "input_tokens": 1431,
  "output_tokens": 1067,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.24, it became \$9.72, and you returned \$4.8. Your visible payoff was \$4.920000000000001.

Your total visible earnings across all rounds are \$21.630000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$2.32, so y

ou received \$6.959999999999999. How much do you return? (0-6.959999999999999)
Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- New co-player sent me \$2.32 (about 46% of their endowment)
- I received \$6.96 after the multiplier
- They currently have \$2.68

They showed moderate trust by sending nearly half their endowment. I should reciprocate fairly to reward their trust and ensure mutual benefit.

Following my pattern of returning roughly 50% of what I receive, I'll return **\$3.48**:

- Sender ends with: $\$2.68 + \$3.48 = \$6.16$ (gain \$1.16)
- I keep: $\$6.96 - \$3.48 = \$3.48$ (gain \$3.48)

Both of us benefit from the cooperation.

{'return': 3.48}

reasoning:

Looking at this round:

- New co-player (different from Round 4)
- They sent me \$2.32 (about 46% of their endowment)
- I received \$6.96
- They currently have \$2.68 ($\$5 - \2.32)

They showed moderate trust by sending nearly half their endowment. I should reciprocate fairly to:

1. Reward their trust
2. Ensure both of us benefit
3. Encourage cooperation

Looking at my previous patterns:

- Round 1: Received \$8.91, returned \$4.50 (about 50%)
- Round 4: Received \$9.72, returned \$4.80 (about 49%)

Returning around 50% seems like my consistent fair approach. For \$6.96, that would be around \$3.48.

Let me check the outcomes:

- Sender ends with: $\$2.68 + \$3.48 = \$6.16$ (gain \$1.16)
- I keep: $\$6.96 - \$3.48 = \$3.48$ (gain \$3.48)

Both benefit, and this reciprocates their trust fairly. I'll return \$3.48.

usage:

```
{
  "input_tokens": 1834,
  "output_tokens": 476,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.32, it became \$6.959999999999999, and you returned \$3.48. Your visible payoff was \$3.479999999999999.

Your total visible earnings across all rounds are \$25.110000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my history with this current co-player:

****Round 5:****

- They sent me \$2.32 (46% of endowment)
- I returned \$3.48, giving them a good outcome

They showed moderate trust, and I reciprocated fairly. Now it's my turn to send, and I should continue building this cooperative relationship.

I'll send **\$2.50**, which is:

- Slightly more than they sent me (showing willingness to cooperate)
- A reasonable amount that matches their moderate approach
- Enough to create mutual benefit if they reciprocate

```
{'send': 2.50}
```

reasoning:

Looking at my history with this current co-player:

Round 5 (last round):

- They sent me \$2.32 (46% of endowment)
- I received \$6.96
- I returned \$3.48 (50% of what I received)
- They ended with \$6.16, I ended with \$3.48

Now in Round 6, I'm the sender with this same co-player.

They showed moderate trust by sending \$2.32. I reciprocated fairly by returning \$3.48, which gave them a good outcome (\$6.16 total).

Since we had a successful cooperative interaction, I should continue building on this. I'll send a similar or slightly higher amount to maintain the cooperative relationship.

I've been generally sending around \$3.00 (60%) in other rounds, which has worked well. Given this co-player sent me \$2.32 (46%), I could:

1. Send similar to what they sent (~\$2.30-2.50)
2. Send my standard amount (~\$3.00)
3. Send slightly more to build trust

I think sending around \$2.50-3.00 makes sense. Let me go with \$2.50, which is slightly more than they sent me, showing I'm willing to cooperate but matching their level of caution.

Actually, looking at my overall strategy, I've been doing well with \$3.00 sends. But this particular co-player was more cautious. Let me match their level at around \$2.30-2.50.

I'll send \$2.50.

usage:

```
{
  "input_tokens": 2153,
  "output_tokens": 510,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$4.52 to you. Your visible payoff was \$7.02.

Your total visible earnings across all rounds are \$32.13.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my recent experiences:

Round 6 outcome:

- I sent \$2.50
- Received \$4.52 back
- Ended with \$7.02 - very good!

The cooperative approach continues to work well. Now with a new co-player, I'll signal trust and willingness to cooperate by sending a solid amount.

I'll send **\$3.00** (60% of endowment), which has proven to be an effective strategy:

```
{'send': 3.00}
```

Reasoning:

Looking at my recent experience:

Round 6:

- I sent \$2.50 to my co-player
- They returned \$4.52
- I ended with \$7.02 - excellent outcome!

This is a new co-player in Round 7. I've been doing well with cooperative strategies. My approach of sending moderate to high amounts (around 50-60% of endowment) has generally worked well across different co-players.

Let me review the pattern:

- Round 2: Sent \$3.00, received back \$5.32 - great
- Round 3: Sent \$3.00, received back \$2.98 - okay
- Round 6: Sent \$2.50, received back \$4.52 - great

The cooperative strategy continues to work. With a new co-player, I should send a reasonable amount to signal willingness to cooperate. I'll send around \$3.00 (60%) to establish trust.

Usage: {

```
"input_tokens": 2419,  
"output_tokens": 343,  
"reasoning_tokens": 0
```

}

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T09:26:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_4",  
"move_index": 7,  
"opponent_id": "Agent_6",  
"pairing": {  
  "agents": [  
    "Agent_6",  
    "Agent_8"  
  ],  
  "investor": "Agent_6",  
  "roles": {  
    "Agent_6": "investor",  
    "Agent_8": "trustee"  
  },  
  "trustee": "Agent_8"  
},  
"role": "trustee",  
"role_label": "Receiver",  
"round": 7,  
"task": "game",  
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.68, that is 53.6% of their total endowment. You received \$8.04000000000001. How much do you return to the sender? (0-8.04000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.68 (53.6% of their endowment)
- They now have \$2.32 remaining
- I received \$8.04 after the 3x multiplier
- Total pool is now \$10.36

I want to reciprocate their trust while being fair. The sender took a risk to create mutual gains, so I should ensure they benefit from their trust.

A reasonable approach is to aim for a relatively balanced outcome:

- If I return \$3.00, the sender ends with \$5.32 (gaining \$0.32)
- I would keep \$5.04

This rewards their trust while maintaining fairness, and shows I'm reciprocal which could encourage cooperation in future rounds.

{"return": 3.00}

reasoning:

Let me analyze this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.68 (which is 53.6% of their \$5 endowment)
3. This was multiplied by 3x, so I received $2.68 \times 3 = \$8.04$
4. I need to decide how much to return (between \$0 and \$8.04)

Key considerations:

- The sender showed trust by sending more than half of their endowment (\$2.68 out of \$5)
- The sender now has: $\$5 - \$2.68 = \$2.32$
- I currently have: \$8.04
- Whatever I return, the sender will end up with: $\$2.32 + \text{my return}$

- I will end up with: \$8.04 - my return

From a fairness/reciprocity perspective:

- The sender trusted me with 53.6% of their endowment
- A fair split might be to ensure we both end up with similar amounts, or to reward the sender's trust
- If I return \$3, sender ends with \$5.32, I end with \$5.04 (relatively equal)
- If I return \$4, sender ends with \$6.32, I end with \$4.04 (rewarding trust more)
- If I return \$2, sender ends with \$4.32, I end with \$6.04 (keeping more)

A common reciprocal strategy would be to:

1. Ensure the sender doesn't lose money (they should end with at least \$5)
2. Share the "profit" somewhat fairly

The total pool is now \$2.32 (sender's remainder) + \$8.04 (my endowment) = \$10.36

If we split this profit fairly:

- Each should get about \$5.18
- For sender to get \$5.18, I need to return: $\$5.18 - \$2.32 = \$2.86$

Returning \$3.00 seems reasonable-it rewards their trust while keeping the outcome relatively balanced, with sender ending at \$5.32 and me at \$5.04. But I should consider whether a more reciprocal approach might better acknowledge their initial generosity.

If I return roughly the same proportion they invested (53.6% of what I received), that's about \$4.31. This leaves them with \$6.63 and me with \$3.73-a significant reward for their trust, though it cuts my gains substantially. Alternatively, I could split the total gains evenly: the \$5.36 profit gets divided so each of us gains \$2.68, putting the sender at \$7.68 and me at \$2.68....

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 1185,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.68, it became \$8.040000000000001, and you returned \$3.0. Your visible payoff was \$5.040000000000001.

Your total visible earnings across all rounds are \$5.040000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this new round as the sender:

In the previous round, my co-player sent 53.6% of their endowment, showing moderate-high trust. I reciprocated fairly by returning enough to give them a positive gain.

Now with a new co-player, I want to signal trust and willingness to cooperate while not being overly risky. Sending around 50% seems reasonable - it creates significant potential mutual gains while maintaining some protection.

If they reciprocate proportionally, we both benefit.

```
{"send": 2.50}
```

reasoning:

Now I'm the sender in Round 2 with a new co-player.

Last round I was the receiver:

- Saw the sender send \$2.68 (53.6%)
- Received \$8.04
- Returned \$3.00
- Kept \$5.04

The sender ended with $\$2.32 + \$3.00 = \$5.32$, so they gained $\$0.32$ from trusting me.

Now I'm the sender with $\$5$. I need to decide how much to send.

Key considerations:

1. The other player sent 53.6% in the previous example, showing moderate-high trust
2. I reciprocated fairly, returning enough to give them a positive gain
3. This is a new co-player, so I don't know their strategy yet
4. There's communication noise, so amounts may appear different
5. Building cooperation early could pay off in future rounds

Strategic thinking:

- If I send too little, I signal distrust and may get nothing back
- If I send too much, I risk losing a lot if they don't reciprocate
- A moderate amount (around 50-60%) shows trust while not being reckless
- The previous sender sent 53.6%, which seemed like a reasonable trust signal

I'll send around $\$2.50$ - $\$3.00$ to signal trust and willingness to cooperate.

Let me think about expected values:

- If I send $\$2.50$, they get $\$7.50$. If they return 40% = $\$3.00$, I end with $\$2.50 + \$3.00 = \$5.50$ (gain $\$0.50$)
- If I send $\$3.00$, they get $\$9.00$. If they return 33% = $\$3.00$, I end with $\$2.00 + \$3.00 = \$5.00$ (break even)
- If I send $\$3.00$ and they return 44% = $\$4.00$, I end with $\$2.00 + \$4.00 = \$6.00$ (gain $\$1.00$)

...

usage:

```
{
  "input_tokens": 852,
  "output_tokens": 646,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent $\$2.5$, it became $\$7.5$, and you saw them return $\$3.19$ to you. Your visible payoff was $\$5.6899999999999995$.

Your total visible earnings across all rounds are $\$10.73$.

This round, you are the SENDER against your current co-player. You have $\$5$. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my recent experience:

- Round 2: I sent $\$2.50$ (50%) and received $\$3.19$ back, giving me a good positive return

This cooperative approach has been working well. I'm now with a new co-player, and I want to signal trustworthiness and encourage mutual cooperation.

Sending 50% has proven to be a solid strategy - it creates meaningful potential gains while showing balanced trust.

```
{"send": 2.50}
```

reasoning:

Let me analyze the previous round:

- I sent $\$2.50$ (50% of my endowment)
- It became $\$7.50$ for the receiver
- They returned $\$3.19$
- I ended with $\$2.50$ (kept) + $\$3.19$ (returned) = $\$5.69$
- They kept $\$7.50 - \$3.19 = \$4.31$

So the receiver returned about 42.5% of what they received, which gave me a positive gain of $\$0.69$. This is a reasonable reciprocation, showing they valued cooperation.

Now I'm facing a new co-player in Round 3.

My experience so far:

- Round 1 (as receiver): Received \$8.04, returned \$3.00 (37.3%)
- Round 2 (as sender): Sent \$2.50, received back \$3.19 - good outcome

I should continue with a cooperative strategy. Since my last send of \$2.50 worked well, I could:

1. Send the same amount (\$2.50) to maintain consistency
2. Send slightly more to signal increased trust
3. Send slightly less to be more cautious

Given that cooperation has been working, and I'm now encountering a third different co-player, I think maintaining a cooperative stance of around 50% makes sense. This has proven successful and sends the right signal.

I'll send \$2.50 again.

usage:

```
{
  "input_tokens": 1103,
  "output_tokens": 441,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$5.25 to you. Your visible payoff was \$7.75.

Your total visible earnings across all rounds are \$18.48.

This round, you are the RECEIVER against your current co-player. They sent you \$3.69, so you received \$11.07. How much do you return? (0-11.07)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- The sender showed very high trust by sending \$3.69 (73.8% of their endowment)
- They now have only \$1.31 remaining
- I received \$11.07 after the 3x multiplier
- Total pool is \$12.38

This sender took a significant risk to create mutual gains. I should reciprocate their strong trust signal generously.

If I return \$5.50:

- Sender ends with \$6.81 (gaining \$1.81)
- I keep \$5.57

This rewards their exceptional trust while maintaining fairness for both of us.

{"return": 5.50}

reasoning:

Let me analyze this situation:

Round 4 - I'm the RECEIVER

- The sender sent \$3.69 (73.8% of their \$5 endowment)
- This became \$11.07 after the 3x multiplier
- The sender now has \$5 - \$3.69 = \$1.31 remaining
- Total pool: \$1.31 + \$11.07 = \$12.38

The sender showed very high trust by sending 73.8% of their endowment. This is much higher than the ~50% I've been sending, indicating strong cooperative intent.

I should reciprocate this high level of trust. Let me think about fair returns:

If I return \$5.50:

- Sender ends with: \$1.31 + \$5.50 = \$6.81 (gain of \$1.81)
- I keep: \$11.07 - \$5.50 = \$5.57

If I return \$6.00:
- Sender ends with: $\$1.31 + \$6.00 = \$7.31$ (gain of \$2.31)
- I keep: $\$11.07 - \$6.00 = \$5.07$

Given they sent 73.8% of their endowment (very high trust), I should return a proportionally generous amount. Returning around \$5.50-\$6.00 would:

1. Reward their high trust
2. Ensure they profit from being cooperative
3. Still leave me with a reasonable amount
4. Signal I'm reciprocal and trustworthy

I'll return \$5.50 - this gives them a solid gain while being fair to both parties.

usage:

```
{
  "input_tokens": 1343,
  "output_tokens": 551,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.69, it became \$11.07, and you returned \$5.5. Your visible payoff was \$5.57.

Your total visible earnings across all rounds are \$24.05.

This round, you are the RECEIVER against your current co-player. They sent you \$3.88, so you received \$11.64. How much do you return? (0-11.64)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- The sender showed extremely high trust by sending \$3.88 (77.6% of their endowment)
- They now have only \$1.12 remaining
- I received \$11.64 after the 3x multiplier
- Total pool is \$12.76

This is even higher trust than the previous round. I should reciprocate this exceptional trust generously.

If I return \$5.80:
- Sender ends with \$6.92 (gaining \$1.92)
- I keep \$5.84

This appropriately rewards their very high trust level while maintaining fairness.

```
{"return": 5.80}
```

reasoning:

Looking at this situation:

- The sender sent \$3.88 (77.6% of their endowment) - even higher trust than the previous round
- They now have \$1.12 remaining
- I received \$11.64 after the 3x multiplier
- Total pool is \$12.76

This sender showed extremely high trust. I should reciprocate proportionally.

In the last round, sender sent \$3.69 (73.8%) and I returned \$5.50, giving them \$6.81 total (gain of \$1.81).

This sender sent even more (\$3.88, or 77.6%). I should return proportionally or slightly more to reward the higher trust.

If I return \$5.80:
- Sender ends with \$6.92 (gain of \$1.92)
- I keep \$5.84

This seems fair and rewards their high trust appropriately.

Let me consider other options:

- Return \$6.00: Sender gets \$7.12 (gain \$2.12), I keep \$5.64
- Return \$5.50: Sender gets \$6.62 (gain \$1.62), I keep \$6.14

I think \$5.80 is a good middle ground that appropriately rewards the very high trust level while maintaining fairness.

usage:

```
{
  "input_tokens": 1628,
  "output_tokens": 459,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.88, it became \$11.64, and you returned \$5.8. Your visible payoff was \$5.840000000000001.

Your total visible earnings across all rounds are \$29.89.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my experience as sender:

- Round 2: Sent \$2.50, received \$3.19 back (good return)
- Round 3: Sent \$2.50, received \$5.25 back (excellent return)

The 50% strategy has worked very well for me. The cooperative environment suggests continuing this approach with a new co-player.

```
{"send": 2.50}
```

reasoning:

Let me review my experience so far:

Round 1 (RECEIVER): Received from sender who sent \$2.68 (53.6%), I returned \$3.00

Round 2 (SENDER): I sent \$2.50 (50%), received back \$3.19

Round 3 (SENDER): I sent \$2.50 (50%), received back \$5.25

Round 4 (RECEIVER): Received from sender who sent \$3.69 (73.8%), I returned \$5.50

Round 5 (RECEIVER): Received from sender who sent \$3.88 (77.6%), I returned \$5.80

As a sender, I've been sending 50% and getting good returns. I've encountered receivers who are cooperating well. I've also been a generous receiver when others send high amounts.

The pattern suggests a cooperative environment. When I send 50%, people return enough for me to profit. When others send high amounts (70-78%), I reciprocate generously.

I'm now facing a new co-player as sender. Should I:

- Continue with 50% (\$2.50) - safe, proven strategy
- Increase to show more trust - could signal cooperation and encourage higher returns
- Decrease to be more cautious - but this might seem less cooperative

Given the cooperative environment and good outcomes when I've sent \$2.50, I'll stick with that strategy.

usage:

```
{
  "input_tokens": 1902,
  "output_tokens": 421,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You se

nt \$2.5, it became \$7.5, and you saw them return \$3.97 to you. Your visible payoff was \$6.470000000000001.
Your total visible earnings across all rounds are \$36.36.

This round, you are the RECEIVER against your current co-player. They sent you \$3.1, so you received \$9.3. How much do you return? (0-9.3)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender sent \$3.10 (62% of their endowment), showing good trust
- They now have \$1.90 remaining
- I received \$9.30 after the 3x multiplier
- Total pool is \$11.20

This is a solid trust signal that deserves good reciprocation.

If I return \$4.50:

- Sender ends with \$6.40 (gaining \$1.40)
- I keep \$4.80

This rewards their trust appropriately while maintaining balance.

{"return": 4.50}

Reasoning:

Looking at this situation:

- The sender sent \$3.10 (62% of their endowment)
- They now have \$1.90 remaining
- I received \$9.30 after the 3x multiplier
- Total pool is \$11.20

This is a moderately-high trust signal. I should reciprocate fairly.

Let me consider a reasonable return:

- If I return \$4.50:
 - Sender ends with \$6.40 (gaining \$1.40)
 - I keep \$4.80

This seems fair and reciprocates their trust appropriately.

Alternatively:

- If I return \$4.00:
 - Sender ends with \$5.90 (gaining \$0.90)
 - I keep \$5.30

Or:

- If I return \$4.70:
 - Sender ends with \$6.60 (gaining \$1.60)
 - I keep \$4.60

Given they sent 62%, I think returning around \$4.50 is appropriate - it gives them a solid gain while keeping things relatively balanced.

Usage: {
 "input_tokens": 2134,
 "output_tokens": 395,
 "reasoning_tokens": 0
}

=====
ROUND 8
=====

Roles: Agent_1=Sender, Agent_2=Sender, Agent_3=Receiver, Agent_4=Receiver, Agent_5=Receiver, Agent_6=Receiver, Agent_7=Sender, Agent_8=Sender

Pairings:

- dyad_1: Agent_8=Sender, Agent_5=Receiver
- dyad_2: Agent_7=Sender, Agent_3=Receiver
- dyad_3: Agent_2=Sender, Agent_6=Receiver
- dyad_4: Agent_1=Sender, Agent_4=Receiver

Parsed game decisions:

```

dyad_1: Agent_8 sent $2.5; received $7.5; Agent_5 returned $5.25; payoffs Agent_8=$7.75,
Agent_5=$2.25; noise/visible: sent visible=$3.37, received visible=$10.11, returned visible=$5.58
  Action validation: {"returned": {"amount": 5.25, "clamped": false, "max": 7.5, "min": 0, "raw":
  5.25}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
dyad_2: Agent_7 sent $3; received $9; Agent_3 returned $5; payoffs Agent_7=$7, Agent_3=$4;
noise/visible: sent visible=$3.99, received visible=$11.97, returned visible=$4.88
  Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw":
  5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_2 sent $3; received $9; Agent_6 returned $3.72; payoffs Agent_2=$5.72,
Agent_6=$5.28; noise/visible: sent visible=$2.48, received visible=$7.44, returned visible=$4.21
  Action validation: {"returned": {"amount": 3.72, "clamped": false, "max": 9.0, "min": 0, "raw":
  3.72}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_1 sent $3; received $9; Agent_4 returned $3.85; payoffs Agent_1=$5.85,
Agent_4=$5.15; noise/visible: sent visible=$2.68, received visible=$8.04, returned visible=$3.16
  Action validation: {"returned": {"amount": 3.85, "clamped": false, "max": 9.0, "min": 0, "raw":
  3.85}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 43.75,
  "Agent_2": 43.08,
  "Agent_3": 41.45,
  "Agent_4": 47.44,
  "Agent_5": 41.480000000000004,
  "Agent_6": 44.85,
  "Agent_7": 43.1,
  "Agent_8": 43.849999999999994
}
Visible balances after round: {
  "Agent_1": 44.899999999999999,
  "Agent_2": 42.04,
  "Agent_3": 49.089999999999996,
  "Agent_4": 41.8,
  "Agent_5": 44.089999999999996,
  "Agent_6": 41.82,
  "Agent_7": 43.900000000000006,
  "Agent_8": 49.239999999999995
}

```

LLM CALLS

GAME - Agent_8 - Sender - dyad_1

Timestamp: 2026-08-12T09:26:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_5"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_5": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.68, it became \$8.040000000000001, and you returned \$3.0. Your visible payoff was \$5.040000000000001.

Your total visible earnings across all rounds are \$5.040000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this new round as the sender:

In the previous round, my co-player sent 53.6% of their endowment, showing moderate-high trust. I reciprocated fairly by returning enough to give them a positive gain.

Now with a new co-player, I want to signal trust and willingness to cooperate while not being overly risky. Sending around 50% seems reasonable - it creates significant potential mutual gains while maintaining some protection.

If they reciprocate proportionally, we both benefit.

{"send": 2.50}

reasoning:

Now I'm the sender in Round 2 with a new co-player.

Last round I was the receiver:

- Saw the sender send \$2.68 (53.6%)
- Received \$8.04
- Returned \$3.00
- Kept \$5.04

The sender ended with $\$2.32 + \$3.00 = \$5.32$, so they gained \$0.32 from trusting me.

Now I'm the sender with \$5. I need to decide how much to send.

Key considerations:

1. The other player sent 53.6% in the previous example, showing moderate-high trust
2. I reciprocated fairly, returning enough to give them a positive gain
3. This is a new co-player, so I don't know their strategy yet
4. There's communication noise, so amounts may appear different
5. Building cooperation early could pay off in future rounds

Strategic thinking:

- If I send too little, I signal distrust and may get nothing back
- If I send too much, I risk losing a lot if they don't reciprocate
- A moderate amount (around 50-60%) shows trust while not being reckless
- The previous sender sent 53.6%, which seemed like a reasonable trust signal

I'll send around \$2.50-3.00 to signal trust and willingness to cooperate.

Let me think about expected values:

- If I send \$2.50, they get \$7.50. If they return 40% = \$3.00, I end with $\$2.50 + \$3.00 = \$5.50$ (gain \$0.50)
- If I send \$3.00, they get \$9.00. If they return 33% = \$3.00, I end with $\$2.00 + \$3.00 = \$5.00$ (break even)
- If I send \$3.00 and they return 44% = \$4.00, I end with $\$2.00 + \$4.00 = \$6.00$ (gain \$1.00)

...

usage:

```
{
  "input_tokens": 852,
  "output_tokens": 646,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.19 to you. Your visible payoff was \$5.6899999999999995.

Your total visible earnings across all rounds are \$10.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my recent experience:

- Round 2: I sent \$2.50 (50%) and received \$3.19 back, giving me a good positive return

This cooperative approach has been working well. I'm now with a new co-player, and I want to signal trustworthiness and encourage mutual cooperation.

Sending 50% has proven to be a solid strategy - it creates meaningful potential gains while showing balanced trust.

```
{"send": 2.50}
```

reasoning:

Let me analyze the previous round:

- I sent \$2.50 (50% of my endowment)
- It became \$7.50 for the receiver
- They returned \$3.19

- I ended with \$2.50 (kept) + \$3.19 (returned) = \$5.69
- They kept \$7.50 - \$3.19 = \$4.31

So the receiver returned about 42.5% of what they received, which gave me a positive gain of \$0.69. This is a reasonable reciprocation, showing they valued cooperation.

Now I'm facing a new co-player in Round 3.

My experience so far:

- Round 1 (as receiver): Received \$8.04, returned \$3.00 (37.3%)
- Round 2 (as sender): Sent \$2.50, received back \$3.19 - good outcome

I should continue with a cooperative strategy. Since my last send of \$2.50 worked well, I could:

1. Send the same amount (\$2.50) to maintain consistency
2. Send slightly more to signal increased trust
3. Send slightly less to be more cautious

Given that cooperation has been working, and I'm now encountering a third different co-player, I think maintaining a cooperative stance of around 50% makes sense. This has proven successful and sends the right signal.

I'll send \$2.50 again.

usage:

```
{
  "input_tokens": 1103,
  "output_tokens": 441,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$5.25 to you. Your visible payoff was \$7.75.

Your total visible earnings across all rounds are \$18.48.

This round, you are the RECEIVER against your current co-player. They sent you \$3.69, so you received \$11.07. How much do you return? (0-11.07)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender showed very high trust by sending \$3.69 (73.8% of their endowment)
- They now have only \$1.31 remaining
- I received \$11.07 after the 3x multiplier
- Total pool is \$12.38

This sender took a significant risk to create mutual gains. I should reciprocate their strong trust signal generously.

If I return \$5.50:

- Sender ends with \$6.81 (gaining \$1.81)
- I keep \$5.57

This rewards their exceptional trust while maintaining fairness for both of us.

```
{"return": 5.50}
```

reasoning:

Let me analyze this situation:

Round 4 - I'm the RECEIVER

- The sender sent \$3.69 (73.8% of their \$5 endowment)
- This became \$11.07 after the 3x multiplier
- The sender now has \$5 - \$3.69 = \$1.31 remaining
- Total pool: \$1.31 + \$11.07 = \$12.38

The sender showed very high trust by sending 73.8% of their endowment. This is much higher

than the ~50% I've been sending, indicating strong cooperative intent.

I should reciprocate this high level of trust. Let me think about fair returns:

If I return \$5.50:

- Sender ends with: $\$1.31 + \$5.50 = \$6.81$ (gain of \$1.81)
- I keep: $\$11.07 - \$5.50 = \$5.57$

If I return \$6.00:

- Sender ends with: $\$1.31 + \$6.00 = \$7.31$ (gain of \$2.31)
- I keep: $\$11.07 - \$6.00 = \$5.07$

Given they sent 73.8% of their endowment (very high trust), I should return a proportionally generous amount. Returning around \$5.50-\$6.00 would:

1. Reward their high trust
2. Ensure they profit from being cooperative
3. Still leave me with a reasonable amount
4. Signal I'm reciprocal and trustworthy

I'll return \$5.50 - this gives them a solid gain while being fair to both parties.

usage:

```
{
  "input_tokens": 1343,
  "output_tokens": 551,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.69, it became \$11.07, and you returned \$5.5. Your visible payoff was \$5.57.

Your total visible earnings across all rounds are \$24.05.

This round, you are the RECEIVER against your current co-player. They sent you \$3.88, so you received \$11.64. How much do you return? (0-11.64)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- The sender showed extremely high trust by sending \$3.88 (77.6% of their endowment)
- They now have only \$1.12 remaining
- I received \$11.64 after the 3x multiplier
- Total pool is \$12.76

This is even higher trust than the previous round. I should reciprocate this exceptional trust generously.

If I return \$5.80:

- Sender ends with \$6.92 (gaining \$1.92)
- I keep \$5.84

This appropriately rewards their very high trust level while maintaining fairness.

```
{"return": 5.80}
```

reasoning:

Looking at this situation:

- The sender sent \$3.88 (77.6% of their endowment) - even higher trust than the previous round
- They now have \$1.12 remaining
- I received \$11.64 after the 3x multiplier
- Total pool is \$12.76

This sender showed extremely high trust. I should reciprocate proportionally.

In the last round, sender sent \$3.69 (73.8%) and I returned \$5.50, giving them \$6.81 total (gain of \$1.81).

This sender sent even more (\$3.88, or 77.6%). I should return proportionally or slightly more to reward the higher trust.

If I return \$5.80:

- Sender ends with \$6.92 (gain of \$1.92)
- I keep \$5.84

This seems fair and rewards their high trust appropriately.

Let me consider other options:

- Return \$6.00: Sender gets \$7.12 (gain \$2.12), I keep \$5.64
- Return \$5.50: Sender gets \$6.62 (gain \$1.62), I keep \$6.14

I think \$5.80 is a good middle ground that appropriately rewards the very high trust level while maintaining fairness.

usage:

```
{
  "input_tokens": 1628,
  "output_tokens": 459,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.88, it became \$11.64, and you returned \$5.8. Your visible payoff was \$5.840000000000001.

Your total visible earnings across all rounds are \$29.89.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my experience as sender:

- Round 2: Sent \$2.50, received \$3.19 back (good return)
- Round 3: Sent \$2.50, received \$5.25 back (excellent return)

The 50% strategy has worked very well for me. The cooperative environment suggests continuing this approach with a new co-player.

```
{"send": 2.50}
```

reasoning:

Let me review my experience so far:

Round 1 (RECEIVER): Received from sender who sent \$2.68 (53.6%), I returned \$3.00

Round 2 (SENDER): I sent \$2.50 (50%), received back \$3.19

Round 3 (SENDER): I sent \$2.50 (50%), received back \$5.25

Round 4 (RECEIVER): Received from sender who sent \$3.69 (73.8%), I returned \$5.50

Round 5 (RECEIVER): Received from sender who sent \$3.88 (77.6%), I returned \$5.80

As a sender, I've been sending 50% and getting good returns. I've encountered receivers who are cooperating well. I've also been a generous receiver when others send high amounts.

The pattern suggests a cooperative environment. When I send 50%, people return enough for me to profit. When others send high amounts (70-78%), I reciprocate generously.

I'm now facing a new co-player as sender. Should I:

- Continue with 50% (\$2.50) - safe, proven strategy
- Increase to show more trust - could signal cooperation and encourage higher returns
- Decrease to be more cautious - but this might seem less cooperative

Given the cooperative environment and good outcomes when I've sent \$2.50, I'll stick with that strategy.

usage:

```
{
  "input_tokens": 1902,
```



```
    "output_tokens": 421,  
    "reasoning_tokens": 0  
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.97 to you. Your visible payoff was \$6.470000000000001.

Your total visible earnings across all rounds are \$36.36.

This round, you are the RECEIVER against your current co-player. They sent you \$3.1, so you received \$9.3. How much do you return? (0-9.3)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- The sender sent \$3.10 (62% of their endowment), showing good trust
- They now have \$1.90 remaining
- I received \$9.30 after the 3x multiplier
- Total pool is \$11.20

This is a solid trust signal that deserves good reciprocation.

If I return \$4.50:

- Sender ends with \$6.40 (gaining \$1.40)
- I keep \$4.80

This rewards their trust appropriately while maintaining balance.

```
{"return": 4.50}
```

reasoning:

Looking at this situation:

- The sender sent \$3.10 (62% of their endowment)
- They now have \$1.90 remaining
- I received \$9.30 after the 3x multiplier
- Total pool is \$11.20

This is a moderately-high trust signal. I should reciprocate fairly.

Let me consider a reasonable return:

- If I return \$4.50:
 - Sender ends with \$6.40 (gaining \$1.40)
 - I keep \$4.80

This seems fair and reciprocates their trust appropriately.

Alternatively:

- If I return \$4.00:
 - Sender ends with \$5.90 (gaining \$0.90)
 - I keep \$5.30

Or:

- If I return \$4.70:
 - Sender ends with \$6.60 (gaining \$1.60)
 - I keep \$4.60

Given they sent 62%, I think returning around \$4.50 is appropriate - it gives them a solid gain while keeping things relatively balanced.

usage:

```
{  
  "input_tokens": 2134,  
  "output_tokens": 395,  
  "reasoning_tokens": 0  
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.1, it became \$9.3, and you returned \$4.5. Your visible payoff was \$4.8000000000001.

Your total visible earnings across all rounds are \$41.16.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record as sender:

- Round 2: Sent \$2.50, received \$3.19 back
- Round 3: Sent \$2.50, received \$5.25 back
- Round 6: Sent \$2.50, received \$3.97 back

The 50% sending strategy has been consistently successful, with all partners reciprocating positively. This balanced approach signals trust while protecting against losses.

With a new co-player, I'll continue this proven strategy.

```
{"send": 2.50}
```

Reasoning:

Looking at my history as a sender:

- Round 2: Sent \$2.50, received \$3.19 back
- Round 3: Sent \$2.50, received \$5.25 back
- Round 6: Sent \$2.50, received \$3.97 back

All three times I've sent \$2.50, I've received positive returns. The 50% strategy has been consistently successful.

As a receiver, I've been generous in reciprocating trust:

- Round 1: Received \$8.04, returned \$3.00
- Round 4: Received \$11.07, returned \$5.50
- Round 5: Received \$11.64, returned \$5.80
- Round 7: Received \$9.30, returned \$4.50

I'm now with yet another new co-player. The pattern shows that sending 50% has been a reliable strategy in this cooperative environment. I should continue with this approach.

```
{"send": 2.50}
```

```
Usage: {  
  "input_tokens": 2141,  
  "output_tokens": 363,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Receiver - dyad_1

Timestamp: 2026-08-12T09:26:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_8",  
      "Agent_5"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",
```

```

"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.54, it became \$7.62, and you returned \$3.5. Your visible payoff was \$4.12.

Your total visible earnings across all rounds are \$4.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

In Round 1, my co-player sent me \$2.54 (about 50% of their endowment), showing good trust. I reciprocated with a fair return.

Now I'm the sender with a new co-player. I should:

1. Signal cooperation to encourage reciprocity
2. Send a reasonable amount that allows mutual benefit
3. Not overcommit without knowing this co-player's behavior

Following the cooperative pattern from Round 1, I'll send a similar proportion - about 50% of my endowment.

```
'''json
{"send": 2.50}
'''
```

reasoning:

Let me think about this situation:

Round 1: I was the receiver

- Sender sent \$2.54 (50.8% of their endowment)
- I received \$7.62
- I returned \$3.50
- My payoff was \$4.12
- Sender's payoff would have been $\$5 - \$2.54 + \$3.50 = \5.96

Now in Round 2, I'm the sender. I need to decide how much to send.

Key considerations:

1. The previous sender showed trust by sending about 50% (\$2.54 out of \$5)
2. I reciprocated reasonably well by returning \$3.50, giving them a positive return
3. This is a multi-round game with 8 agents, so building a reputation for cooperation could be beneficial
4. There's communication noise, so the exact amounts might differ
5. I'm paired with a different co-player this round (pairings are randomized)

Since this is a new co-player, they won't have direct experience of my behavior. However, in a repeated multi-agent setting, establishing myself as cooperative could be beneficial.

The previous sender sent about 50% and I reciprocated well. I should consider:

- Sending a similar moderate-to-high amount to signal cooperation
- Not sending too much in case this receiver is less cooperative than I was
- Not sending too little, as that could signal I'm uncooperative

I think sending around \$2.50-\$3.00 (50-60%) would be reasonable. This:

- Shows trust and willingness to cooperate
- Mirrors what the previous sender did to me
- Leaves room for mutual benefit if the receiver is cooperative
- Doesn't risk my entire endowment

I'll send \$2.50, which is exactly 50% of my endowment, similar to what the previous sender did.

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 581,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$4.13 to you. Your visible payoff was \$6.63.

Your total visible earnings across all rounds are \$10.75.

This round, you are the RECEIVER against your current co-player. They sent you \$3.99, so you received \$11.97. How much do you return? (0-11.97)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.99, which is almost 80% of their endowment - showing exceptional trust
- I received \$11.97 after the 3x multiplier
- This is much higher trust than the ~50% I experienced/sent in previous rounds

I should reciprocate proportionally to their high level of trust:

If I return \$6.50:

- Sender ends with: $\$5.00 - \$3.99 + \$6.50 = \7.51 (net gain: \$2.51)
- I end with: $\$11.97 - \$6.50 = \$5.47$ (net gain: \$5.47)

This rewards their exceptional trust while both of us benefit significantly.

```
'''json
{"return": 6.50}
'''
```

reasoning:

Let me analyze this situation:

Round 3: I'm the RECEIVER

- My co-player sent \$3.99 (about 80% of their \$5 endowment)
- I received \$11.97 (after 3x multiplier)
- I need to decide how much to return

The sender showed very high trust by sending almost 80% of their endowment. This is significantly more than the ~50% I saw/sent in previous rounds.

Let me think about fair reciprocation:

If the sender sent \$3.99, they kept \$1.01.

If I return nothing, they end with \$1.01 (bad outcome for them, they lose \$3.98)

If I return everything (\$11.97), they end with $\$1.01 + \$11.97 = \$12.98$, I get \$0

I want to:

1. Reward their high trust
2. Make sure they profit from their decision
3. Keep a reasonable amount for myself
4. Signal I'm cooperative for future rounds

In Round 2, when I sent \$2.50 and received \$4.13 back, that was a good return (I sent 50%, got back enough to profit).

The sender here sent 80% - much higher trust. I should return proportionally more.

Let me calculate some options:

Option 1: Return \$6.00

- Sender ends with: $\$1.01 + \$6.00 = \$7.01$ (net gain: \$2.01)
- I end with: $\$11.97 - \$6.00 = \$5.97$ (net gain: \$5.97)
- This gives them a decent profit
- Returning \$6.50 or \$7.00 would be more generous given their 80% send, letting them gain \$2.51-\$3.01 while I still keep \$4.97-\$5.47. Since they demonstrated exceptional trust, I should lean toward the higher end to reinforce cooperation.

Looking at the pattern: they sent roughly 80% of their endowment, which is significantly more than the 50-51% in earlier rounds. My...

usage:

```
{
  "input_tokens": 1150,
  "output_tokens": 981,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.99, it became \$11.97, and you returned \$6.5. Your visible payoff was \$5.470000000000001.

Your total visible earnings across all rounds are \$16.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience so far:

- Round 2: I sent \$2.50 and received \$4.13 back - good reciprocity
- Round 3: My co-player sent \$3.99 (high trust) and I reciprocated fairly
- The game has shown cooperative players who reward trust

Since cooperation has been working well, I'll increase my trust level from the 50% I sent in Round 2. Sending around 60% balances building cooperation while managing risk with a new co-player.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Let me review my history:

- Round 1 (RECEIVER): Saw \$2.54 sent, received \$7.62, returned \$3.50, earned \$4.12
- Round 2 (SENDER): Sent \$2.50, received \$4.13 back, earned \$6.63
- Round 3 (RECEIVER): Saw \$3.99 sent, received \$11.97, returned \$6.50, earned \$5.47
- Total earnings: \$16.22

Now Round 4, I'm the SENDER with a new co-player.

My experience so far:

- When I sent \$2.50, I got back \$4.13 (55% return rate)
- When others sent to me:
 - \$2.54 ? I returned \$3.50 (46% of what I received)
 - \$3.99 ? I returned \$6.50 (54% of what I received)

Both times I've been SENDER or RECEIVER, cooperation has been rewarded. The game seems to have cooperative players.

I've been playing moderately cooperatively:

- Round 2: sent 50%
- Could send more to build trust

Given that:

1. Previous rounds show cooperation is rewarded
2. The last sender who sent high (\$3.99, 80%) was well rewarded
3. I have accumulated good earnings and can afford some risk
4. I want to encourage cooperation

I could send around 60-70% to signal strong cooperation without being reckless.

Let me send \$3.00 (60% of endowment).

```
'''json
{"send": 3.00}
'''
```

usage:

```
{
  "input_tokens": 1464,
  "output_tokens": 523,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.71 to you. Your visible payoff was \$6.71.

Your total visible earnings across all rounds are \$22.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience:

- Round 2 (Sender): Sent \$2.50, received \$4.13 back - good return
- Round 4 (Sender): Sent \$3.00, received \$4.71 back - good return

The pattern shows that moderate-to-high trust (50-60%) has been consistently rewarded. Since cooperation is working well, I'll continue with the same level that succeeded in Round 4.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Looking at my history:

- Round 1 (Receiver): Received \$2.54 ? \$7.62, returned \$3.50. Payoff: \$4.12
- Round 2 (Sender): Sent \$2.50 ? \$7.50, got back \$4.13. Payoff: \$6.63
- Round 3 (Receiver): Received \$3.99 ? \$11.97, returned \$6.50. Payoff: \$5.47
- Round 4 (Sender): Sent \$3.00 ? \$9.00, got back \$4.71. Payoff: \$6.71

My sending strategy has been:

- Round 2: \$2.50 (50%)
- Round 4: \$3.00 (60%)

Both times I got good returns:

- \$2.50 sent ? \$4.13 back (good reciprocity)
- \$3.00 sent ? \$4.71 back (good reciprocity)

The game has shown cooperative players. I'm building earnings well. Since I'm sender again with a new co-player, I should continue showing trust. Given the successful cooperation, I could maintain or slightly increase my sending amount.

I'll send \$3.00 again (60%) as it worked well last time and establishes a consistent cooperative stance.

usage:

```
{
  "input_tokens": 1718,
  "output_tokens": 440,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$4.01 to you. Your visible payoff was \$6.01.

Your total visible earnings across all rounds are \$28.939999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.7, so you received \$8.100000000000001. How much do you return? (0-8.100000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation with my current co-player:

Previous round (Round 5) against this SAME co-player:

- I sent them \$3.00, it became \$9.00
- They returned \$4.01 (about 44.6% of what they received)
- Both of us gained

Now they've sent me \$2.70 (54% of their endowment), showing moderate trust.

I should maintain fair reciprocity with this partner. If I return \$4.10:

- Sender ends with: \$5.00 - \$2.70 + \$4.10 = \$6.40 (net gain: \$1.40)
- I end with: \$8.10 - \$4.10 = \$4.00 (net gain: \$4.00)

This is proportionally similar to their return rate in Round 5, maintaining our cooperative relationship.

```
'''json
{"return": 4.10}
'''
```

reasoning:

Looking at this situation:

In Round 5 (the immediately previous round), I was the sender against this SAME co-player:

- I sent \$3.00, it became \$9.00
- They returned \$4.01 to me
- My payoff was \$6.01

Now in Round 6, I'm the receiver against the SAME co-player:

- They sent \$2.70 (54% of their endowment)
- I received \$8.10

This is interesting - we're playing consecutive rounds with the same partner. In Round 5, they returned \$4.01 out of \$9.00 (about 44.6%). That was a reasonable return.

Now they've sent me \$2.70 (54% of endowment), which is a moderate level of trust, slightly less than the \$3.00 (60%) I sent them.

I should reciprocate fairly given our history:

- They gave me a fair return in Round 5
- They're showing moderate trust here

If I return \$4.00:

- Sender ends with: $\$5.00 - \$2.70 + \$4.00 = \6.30 (net gain: \$1.30)
- I end with: $\$8.10 - \$4.00 = \$4.10$ (net gain: \$4.10)

This is roughly similar to what they returned to me (about 44-49% of what they received). It maintains fairness and reciprocity with this specific partner.

Actually, let me think more carefully. They returned \$4.01 out of \$9.00, which is 44.6%. If I apply the same percentage: $0.446 * 8.10 = \$3.61$.

But I think returning slightly more shows good faith and maintains cooperation. Something around \$4.00-\$4.20 seems fair.

Let me go with \$4.10 to be slightly more generous than they were, showing I value the ongoing cooperation.

usage:

```
{
  "input_tokens": 1985,
  "output_tokens": 686,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.7, it became \$8.100000000000001, and you returned \$4.1. Your visible payoff was \$4.000000000000002.

Your total visible earnings across all rounds are \$32.94.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my experience as sender:

- Round 2: Sent \$2.50, received \$4.13 back
- Round 4: Sent \$3.00, received \$4.71 back
- Round 5: Sent \$3.00, received \$4.01 back

Sending around \$3.00 (60% of endowment) has consistently been rewarded with fair reciprocity.

ty. This is a new co-player, but the moderate-high trust strategy has been working well across different partners.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Looking at my history:

- Round 2 (Sender): Sent \$2.50, received \$4.13 back
- Round 4 (Sender): Sent \$3.00, received \$4.71 back
- Round 5 (Sender): Sent \$3.00, received \$4.01 back (same co-player as Round 6)
- Round 6 (Receiver): Received from same co-player who sent \$2.70

Now I'm playing against a NEW co-player in Round 7 (since the note says "against another agent" in the previous round description).

The pattern shows that sending around \$3.00 (60%) has worked well. I've been consistently reciprocated. This is a new co-player, so I should continue with a moderate-to-high trust level that has been successful.

I'll send \$3.00 again.

usage:

```
{
  "input_tokens": 2333,
  "output_tokens": 350,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.29 to you. Your visible payoff was \$6.29.

Your total visible earnings across all rounds are \$39.23.

This round, you are the RECEIVER against your current co-player. They sent you \$3.37, so you received \$10.11. How much do you return? (0-10.11)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player sent \$3.37 (about 67% of their endowment) - showing good trust
- I received \$10.11 after the 3x multiplier

This is a solid level of trust, higher than the moderate amounts but not quite as high as the \$3.99 I saw in Round 3.

If I return \$5.25:

- Sender ends with: $\$5.00 - \$3.37 + \$5.25 = \6.88 (net gain: \$1.88)
- I end with: $\$10.11 - \$5.25 = \$4.86$ (net gain: \$4.86)

This proportionally rewards their good trust while both of us benefit.

```
'''json
{"return": 5.25}
'''
```

Reasoning:

Let me analyze this situation:

Current round (Round 8):

- I'm the RECEIVER
- My co-player sent me \$3.37 (about 67% of their endowment) - showing good trust
- I received \$10.11 after the 3x multiplier

Looking at my history as receiver:

- Round 1: Received \$7.62 (from \$2.54 sent), returned \$3.50
- Round 3: Received \$11.97 (from \$3.99 sent), returned \$6.50
- Round 6: Received \$8.10 (from \$2.70 sent), returned \$4.10

And as sender, I've consistently sent \$3.00 and received returns around \$4-\$4.70.

My co-player sent \$3.37, showing good trust (67%). I should reciprocate proportionally.

If I return \$5.00:

- Sender ends with: $\$5.00 - \$3.37 + \$5.00 = \6.63 (net gain: \$1.63)
- I end with: $\$10.11 - \$5.00 = \$5.11$ (net gain: \$5.11)

If I return \$5.50:

- Sender ends with: $\$5.00 - \$3.37 + \$5.50 = \7.13 (net gain: \$2.13)
- I end with: $\$10.11 - \$5.50 = \$4.61$ (net gain: \$4.61)

Looking at Round 3 where they sent \$3.99 (almost 80%) and I returned \$6.50, this is similar trust level (67%).

For \$3.37 sent (67%), returning around \$5.00-\$5.50 seems fair and proportional to their trust level.

```
...
Usage: {
  "input_tokens": 2318,
  "output_tokens": 731,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Sender - dyad_2

Timestamp: 2026-08-12T09:26:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_3"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_3": "trustee",
      "Agent_7": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.8 to you. Your visible payoff was \$6.8.

Your total visible earnings across all rounds are \$6.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.48, so you received \$7.439999999999995. How much do you return? (0-7.439999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this round:

Analysis:

- My co-player sent \$2.48 out of \$5 (about 50%), showing moderate trust
- This became \$7.44 for me
- In Round 1, my previous partner returned about 53% of what they received, which worked well

My decision:

I want to:

- Reward their trust appropriately
- Ensure both of us profit from cooperation
- Maintain a fair reciprocal relationship

If I return \$4.00:

- Sender gets: \$2.52 + \$4.00 = \$6.52
- I get: \$7.44 - \$4.00 = \$3.44
- Both better off than the initial \$5

This return (about 54% of what I received) mirrors the generous reciprocity I experienced in Round 1.

```
'''json
{"return": 4.0}
'''
```

reasoning:

Let me analyze this situation:

****Round 1 (previous round):****

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$4.8 to me
- My payoff: $\$5 - \$3 + \$4.8 = \6.8
- Their payoff: $\$9 - \$4.8 = \$4.2$

So my previous co-player returned a bit more than half (53.3% of what they received), which was generous and fair.

****Round 2 (current round):****

- I am now the RECEIVER
- My current co-player (different from Round 1) sent me \$2.48
- It became \$7.44 (actually \$7.439999999999995, but I'll think of it as \$7.44)
- I need to decide how much to return

The note says there may be communication noise, so the amounts I see may differ slightly from what was actually sent/returned.

My current co-player sent \$2.48 out of their \$5, which is about 49.6% - nearly half. This shows moderate trust.

Now I need to decide what to return. Let me think about fairness and reciprocity:

1. ****Equal split approach****: If we aim for equal final payoffs:
 - Sender currently has: $\$5 - \$2.48 = \$2.52$
 - I currently have: \$7.44
 - Total: \$9.96
 - Equal split: \$4.98 each
 - I should return: $\$7.44 - \$4.98 = \$2.46$
2. ****Proportional return based on what they sent****:
 - In Round 1, my partner returned 53.3% of what they received
 - If I return 50% of \$7.44 = \$3.72
 - This would give them: $\$2.52 + \$3.72 = \$6.24$
 - And me: $\$7.44 - \$3.72 = \$3.72$
3. ****Return what maintains trust****:
 - My co-player showed moderate trust by sending nearly half
 - I should reciprocate fairly to encourage future cooperation

I think returning around \$3.50 strikes the right balance-it rewards their trust while ensuring both of us end up ahead of the initial \$5 endowment. That gives them \$6.02 and me \$3.94.

But I could also return \$4.00 to show stronger reciprocation, which would give them \$6.52 and me \$3.44. Looking back...

usage:

```
{
  "input_tokens": 816,
  "output_tokens": 1083,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.48, it became \$7.439999999999995, and you returned \$4.0. Your visible payoff was \$3.439999999999995.

Your total visible earnings across all rounds are \$10.239999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this round:

****Analysis:****

- My co-player sent \$3.02 out of \$5 (about 60%), showing strong trust
- This became \$9.06 for me
- They're trusting me with more than the previous sender

****My decision:****

I want to reward this higher level of trust and maintain the cooperative pattern I've established (returning ~50-55% of what I receive).

If I return \$5.00:

- Sender gets: $\$1.98 + \$5.00 = \$6.98$
- I get: $\$9.06 - \$5.00 = \$4.06$
- Both significantly better off than the initial \$5

This return (about 55% of what I received) is consistent with my previous reciprocity and rewards their trust appropriately.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Let me analyze this round:

****What happened in Round 3:****

- I'm the RECEIVER again
- My co-player sent \$3.02 out of \$5 (about 60%), showing good trust
- This became \$9.06 for me

****My history:****

- Round 1: I was SENDER, sent \$3, received \$4.80 back (about 53% of the \$9 they received)
- Round 2: I was RECEIVER, received \$7.44, returned \$4.00 (about 54% of what I received)

****Decision for Round 3:****

My co-player sent a higher percentage this time (\$3.02 vs \$2.48 last round from a different player), showing increased trust. I should:

- Reward this trust
- Maintain reciprocity around 50-55% return rate
- Ensure both of us benefit

If I return \$5.00:

- Sender gets: $\$1.98 + \$5.00 = \$6.98$
- I get: $\$9.06 - \$5.00 = \$4.06$
- Both significantly better off than \$5

This is about 55% of what I received, consistent with my previous behavior and fair reciprocity.

Alternatively, if I return \$4.50:

- Sender gets: $\$1.98 + \$4.50 = \$6.48$
- I get: $\$9.06 - \$4.50 = \$4.56$
- Still both better off

Let me go with \$5.00 to be generous and reward the higher trust shown.

usage:

```
{
  "input_tokens": 1172,
  "output_tokens": 579,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.02, it became \$9.06, and you returned \$5.0. Your visible payoff was \$4.0600000000000005.

Your total visible earnings across all rounds are \$14.299999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this round as SENDER:

****My experience so far:****

- Round 1: Sent \$3 ? received \$4.8 back (53% return rate) ? \$6.8 payoff
- Rounds 2-3: As receiver, I returned 54-55% of what I received
- Cooperation has been working well

****My decision:****

Since this is a new co-player, I don't know their specific behavior, but the general pattern in this game seems cooperative. Sending \$3 worked well in Round 1, balancing trust with prudence.

If I send \$3:

- It becomes \$9 for receiver
- If they return around 50%, I get ~\$4.5 back
- Net: ~\$6.5 total, better than keeping all \$5

I'll send ****\$3**** again - it's proven effective and signals willingness to cooperate.

'''json

{"send": 3}

'''

reasoning:

I'm now the SENDER again. Let me review my experience:

Round 1: I was SENDER, sent \$3, received back \$4.8, payoff = \$6.8

Round 2: I was RECEIVER, received \$7.44 (from \$2.48 sent), returned \$4.0, payoff = \$3.44

Round 3: I was RECEIVER, received \$9.06 (from \$3.02 sent), returned \$5.0, payoff = \$4.06

Now Round 4: I'm SENDER again

My experience shows:

- When I sent \$3 in Round 1, I got back \$4.8 (53% of the \$9 the receiver got)
- I've been a generous receiver, returning 54% and 55% of what I received
- Cooperation seems to be working well

This is a new co-player, so I don't know their behavior yet. However, the game seems to have a cooperative culture developing.

I should send a moderate-to-high amount to signal trust and cooperation. Sending \$3 worked well before. I could stick with that or increase slightly given the positive experiences.

I'll send \$3 again - it's a proven strategy that balances risk and reward.

usage:

```
{
  "input_tokens": 1500,
  "output_tokens": 515,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.36 to you. Your visible payoff was \$6.36.

Your total visible earnings across all rounds are \$20.66.

This round, you are the RECEIVER against your current co-player. They sent you \$4.17, so you received \$12.51. How much do you return? (0-12.51)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this round:

****Analysis:****

- My co-player sent \$4.17 out of \$5 (about 83%), showing exceptional trust
- This became \$12.51 for me
- This is the highest trust shown by any sender I've faced

****My reciprocity pattern:****

- I've been returning around 50-55% of what I receive
- This has maintained good cooperative relationships

****My decision:****

This level of trust deserves strong reciprocation.

If I return \$7.00:

- Sender gets: $\$0.83 + \$7.00 = \$7.83$
- I get: $\$12.51 - \$7.00 = \$5.51$
- Both significantly better off than the initial \$5

This return (about 56% of what I received) maintains my cooperative pattern while appropriately rewarding their exceptional trust.

```
'''json
{"return": 7.0}
'''
```

reasoning:

Looking at this round:

****Analysis:****

- My co-player sent \$4.17 out of \$5 (about 83%), showing very high trust
- This became \$12.51 for me
- This is the highest amount anyone has sent me so far

****My past behavior as receiver:****

- Round 2: Received \$7.44, returned \$4.00 (54%)
- Round 3: Received \$9.06, returned \$5.00 (55%)

****My past experience as sender:****

- Round 1: Sent \$3, received back \$4.8 (53% return)
- Round 4: Sent \$3, received back \$4.36 (48% return)

****My decision:****

This co-player showed exceptional trust by sending \$4.17. I should reward this trust generously. If I continue my pattern of returning ~50-55%, that would be \$6.25-\$6.88.

Let me calculate:

If I return \$7.00:

- Sender gets: $\$0.83 + \$7.00 = \$7.83$
- I get: $\$12.51 - \$7.00 = \$5.51$
- Both significantly better off

If I return \$6.50:

- Sender gets: $\$0.83 + \$6.50 = \$7.33$
- I get: $\$12.51 - \$6.50 = \$6.01$
- Both better off

Returning \$7.00 (about 56% of what I received) maintains my cooperative pattern and rewards their exceptional trust. This seems fair given they sent over 80% of their endowment.

usage:

```
{
  "input_tokens": 1854,
  "output_tokens": 605,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.17, it became \$12.51, and you returned \$7.0. Your visible payoff was \$5.51.

Your total visible earnings across all rounds are \$26.17.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at this round as SENDER:

****My sending history:****

- Round 1: Sent \$3 ? received \$4.8 back ? \$6.8 payoff
- Round 4: Sent \$3 ? received \$4.36 back ? \$6.36 payoff

****Analysis:****

- Sending \$3 has consistently worked well for me
- Both previous receivers returned about 48-53%, yielding good profits
- This is a new co-player, so I have no specific information about them
- The overall environment has been cooperative

****My decision:****

I'll stick with the proven strategy of sending ****\$3****. It:

- Shows willingness to cooperate
- Has delivered consistent positive returns
- Balances trust with prudent risk management

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

Round 1 (SENDER): Sent \$3 ? received \$4.8 back ? payoff \$6.8
Round 2 (RECEIVER): Received \$7.44 ? returned \$4.0 ? payoff \$3.44
Round 3 (RECEIVER): Received \$9.06 ? returned \$5.0 ? payoff \$4.06
Round 4 (SENDER): Sent \$3 ? received \$4.36 back ? payoff \$6.36
Round 5 (RECEIVER): Received \$12.51 ? returned \$7.0 ? payoff \$5.51

As SENDER (2 times so far):

- Round 1: Sent \$3, got back \$4.8 (return rate 53%)
- Round 4: Sent \$3, got back \$4.36 (return rate 48.4%)

Both times I sent \$3 and got positive returns. The cooperation has been working well overall.

This is a new co-player again. Should I:

1. Stick with \$3 (proven strategy)
2. Try increasing to show more trust (like the person who sent me \$4.17)
3. Decrease to be more cautious

Given that:

- \$3 has worked well twice
- I've been a generous receiver (returning 50-55%)
- The overall environment seems cooperative
- It's midway through the game (round 6)

I think \$3 remains a good balanced choice. It's worked well and I don't have information about this specific co-player yet.

Actually, let me think about this differently. The person in Round 5 sent me \$4.17 and I returned \$7.0, so they ended up with \$7.83. That was a very successful interaction.

What if I try sending a bit more to signal stronger trust? Maybe \$3.5 or \$4?

...

usage:

```
{
  "input_tokens": 2187,
  "output_tokens": 766,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.84 to you. Your visible payoff was \$7.84.

Your total visible earnings across all rounds are \$34.010000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$2.17, so you received \$6.51. How much do you return? (0-6.51)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this round:

****Analysis:****

- My co-player sent \$2.17 out of \$5 (about 43%), showing moderate trust
- This became \$6.51 for me
- This is less than previous senders, but still shows willingness to cooperate

****My reciprocity pattern:****

- I've consistently returned 50-55% of what I receive
- This has maintained good cooperative relationships

****My decision:****

I'll maintain my fair reciprocity pattern regardless of the amount sent.

If I return \$3.50:

- Sender gets: \$2.83 + \$3.50 = \$6.33
- I get: \$6.51 - \$3.50 = \$3.01
- Both better off than the initial \$5

This return (about 54% of what I received) maintains my consistent cooperative approach.

```
'''json
{"return": 3.5}
'''
```

reasoning:

Looking at this round:

****Analysis:****

- My co-player sent \$2.17 out of \$5 (about 43%), showing moderate but lower trust
- This became \$6.51 for me
- Previous senders sent: \$2.48, \$3.02, \$4.17 - this is the lowest

****My reciprocity pattern:****

- I've been consistently returning around 50-55% of what I receive
- Round 2: Returned \$4.00 from \$7.44 (54%)
- Round 3: Returned \$5.00 from \$9.06 (55%)
- Round 5: Returned \$7.00 from \$12.51 (56%)

****My decision:****

Even though they sent less, I should maintain my cooperative pattern. If I return less proportionally, it might discourage future cooperation.

If I return \$3.50:

- Sender gets: \$2.83 + \$3.50 = \$6.33
- I get: \$6.51 - \$3.50 = \$3.01
- Both better off than initial \$5

This return (about 54% of what I received) maintains my consistent reciprocity pattern.

usage:

```
{
  "input_tokens": 2515,
  "output_tokens": 501,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against your current co-pla

yer. You saw them send \$2.17, it became \$6.51, and you returned \$3.5. Your visible payoff was \$3.01.
Your total visible earnings across all rounds are \$37.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this round as SENDER:

****Important context:****

- This is the SAME co-player from Round 7
- Last round they sent me \$2.17 (43% of \$5)
- I returned \$3.50 (54% of what I received), giving them a good payoff of \$6.33

****My sending history:****

- Rounds 1, 4, 6: All sent \$3, consistently received good returns (\$4.36-\$5.84)
- Average return on \$3 sent: ~160-190% profit

****Analysis:****

- My co-player has seen I'm trustworthy and cooperative
- They might reciprocate more generously this time
- Sending \$3 has been my reliable strategy

****My decision:****

I'll send ****\$3**** again, maintaining consistency and showing continued trust.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at this round as SENDER:

****Important context:****

- This is the SAME co-player from Round 7
- In Round 7, they sent me \$2.17 (43% of their \$5)
- I returned \$3.50 to them (54% of \$6.51)
- So they ended with \$6.33 and I got \$3.01

****My sending history:****

- Round 1: Sent \$3 ? received \$4.8 back ? \$6.8 payoff
- Round 4: Sent \$3 ? received \$4.36 back ? \$6.36 payoff
- Round 6: Sent \$3 ? received \$5.84 back ? \$7.84 payoff

****Analysis:****

- I've consistently sent \$3 and it's worked well
- My current co-player showed moderate trust (43%) in the previous round
- I was generous in return (54% return rate), which should signal cooperation
- They might reciprocate more generously this time, knowing I was fair

****My decision:****

I should continue with my strategy of sending \$3. It's worked consistently and signals ongoing cooperation to this co-player who has already seen I'm trustworthy.

If I send \$3:

- It becomes \$9 for receiver
- Based on my generous return last round, they might return around 50-60%
- Expected return: ~\$4.5-5.4
- Net: ~\$6.5-7.4 total

I'll send \$3 again.

Usage: {

```
"input_tokens": 2648,
"output_tokens": 573,
"reasoning_tokens": 0
```

}

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T09:26:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_3"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_3": "trustee",
      "Agent_7": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.12, it became \$9.36, and you returned \$4.0. Your visible payoff was \$5.359999999999999.

Your total visible earnings across all rounds are \$5.359999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Last round, my co-player trusted me with 62.4% of their endowment
- I reciprocated fairly, and we both benefited
- Now I'm with a new co-player as the sender

I'll mirror the cooperative behavior I experienced. Sending a substantial amount can create mutual benefit through the 3x multiplier if my co-player reciprocates.

I'll send \$3.00 (60% of my endowment), similar to the trust shown to me in the previous round.

{'send': 3.00}

reasoning:

Let me think about this situation:

1. In the previous round, I was the receiver
 - The sender sent me \$3.12 (62.4% of their endowment)
 - I received \$9.36
 - I returned \$4.00
 - My payoff was \$5.36
2. Now I'm the sender with a new co-player
 - I have \$5 to potentially send
 - Whatever I send gets tripled
 - The receiver will decide how much to return

Strategic considerations:

- In the previous round, the sender showed trust by sending 62.4% of their endowment
- I reciprocated fairly, returning \$4.00 out of \$9.36 (about 43%)
- This gave the sender a total of \$5.88 (they sent \$3.12, kept \$1.88, got back \$4.00)

For this round:

- I'm paired with a different agent (randomized pairings)
- I don't know their history or behavior
- However, the game benefits from cooperation
- If I send a reasonable amount and they reciprocate, we both benefit

Let me consider different strategies:

- Sending nothing: I keep \$5, they get nothing, no mutual benefit
- Sending a small amount (e.g., \$1-2): Low risk, but lower potential gain
- Sending a moderate amount (e.g., \$3): Shows some trust, reasonable risk
- Sending a large amount (e.g., \$4-5): High trust, high risk, high potential reward

Given that:

1. The previous sender trusted me with 62.4% and it worked out
2. I want to establish a cooperative pattern
3. There's communication noise mentioned, so exact amounts may vary
4. I'm in a multi-round game where reputation (though anonymous) might matter

I'll send \$3.00, which is 60% of my endowment. This:

...

usage:

```
{  
  "input_tokens": 824,
```

```

    "output_tokens": 680,
    "reasoning_tokens": 0
}
Message 4 role=user
content:
    Round 3

    In your most recent previous game round, you were the SENDER against another agent. You sent $3.0, it became $9.0, and you saw them return $3.48 to you. Your visible payoff was $5.48.
    Your total visible earnings across all rounds are $10.84.

    This round, you are the RECEIVER against your current co-player. They sent you $2.97, so you received $8.91. How much do you return? (0-8.91)
    Respond exactly as JSON: {'return': <amount>}
Message 5 role=assistant
content:
    Looking at this situation:

    **Current round:**
    - Sender showed good trust by sending $2.97 (59.4% of their endowment)
    - This created $8.91 for me
    - Sender currently has $2.03

    **Previous round context:**
    - I sent $3.00 (60% - very similar trust level)
    - My co-player returned $3.48, which was fair

    I should reciprocate the trust shown to me. The sender sent almost 60% of their endowment, similar to what I did last round.

    I'll return $3.75, which:
    - Gives the sender $5.78 total (a gain of $0.78)
    - Leaves me with $5.16 (also a gain)
    - Rewards their trust appropriately

    {'return': 3.75}
reasoning:
    Let me analyze this situation:

    **Last round (Round 2):**
    - I was the SENDER
    - I sent $3.00 (60% of my endowment)
    - It became $9.00 for the receiver
    - They returned $3.48
    - My payoff: $5 - $3 + $3.48 = $5.48
    - Their payoff: $9 - $3.48 = $5.52

    The receiver gave me back slightly more than I sent, which was cooperative and fair.

    **This round (Round 3):**
    - I'm the RECEIVER
    - Sender sent $2.97 (59.4% of their endowment - very similar to what I sent last round!)
    - I received $8.91
    - Sender currently has: $5 - $2.97 = $2.03

    The sender showed good trust by sending ~60% of their endowment. This is very similar to the 60% I sent in the previous round.

    I should reciprocate fairly. Let me think about what would be a fair return:

    - If I return $3.50, sender gets $2.03 + $3.50 = $5.53, and I keep $8.91 - $3.50 = $5.41
    - If I return $4.00, sender gets $2.03 + $4.00 = $6.03, and I keep $8.91 - $4.00 = $4.91
    - If I return $3.75, sender gets $2.03 + $3.75 = $5.78, and I keep $8.91 - $3.75 = $5.16

    The sender trusted me with $2.97, similar to how I trusted my previous co-player with $3.00, and they returned $3.48.
    ...
usage:
    {

```

```
"input_tokens": 1089,  
"output_tokens": 735,  
"reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.97, it became \$8.91, and you returned \$3.75. Your visible payoff was \$5.16.

Your total visible earnings across all rounds are \$16.0.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

****Last round with THIS SAME co-player:****

- They sent me \$2.97 (59.4% of their endowment) - showing good trust
- I returned \$3.75 - rewarding their trust fairly
- We both gained from cooperation

****Now it's my turn to send:****

Since we just had a successful cooperative round together, I should maintain this pattern. They trusted me, I reciprocated, and now I should show similar trust.

I'll send \$3.00 (60% of my endowment), matching the level of trust they showed me.

```
{'send': 3.00}
```

reasoning:

Let me analyze this situation:

****Last round with THIS SAME co-player:****

- They sent me \$2.97 (59.4% of their endowment)
- I received \$8.91
- I returned \$3.75
- They got \$5.78 total (gain of \$0.78)
- I got \$5.16 (gain of \$0.16)

****Now I'm the sender with THE SAME co-player:****

- They showed good trust last round (sent ~60%)
- I reciprocated fairly
- We both gained from cooperation

Since we're playing with the same person and we had a successful cooperative round, I should continue this pattern. They trusted me with ~60% of their endowment, and I rewarded that trust. Now I should show similar trust.

I'll send around \$3.00 again, maintaining the cooperative pattern.

usage:

```
{  
  "input_tokens": 1406,  
  "output_tokens": 354,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.73 to you. Your visible payoff was \$7.73.

Your total visible earnings across all rounds are \$23.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

****Last round:****

- I sent \$3.00 (60% of my endowment)
- My co-player returned \$5.73 - very generous!
- This gave me \$7.73, an excellent outcome

****Pattern so far:****

- Sending around 60% has consistently led to good reciprocation
- My average earnings are strong at \$4.75 per round
- Cooperative approach is working well

****Current round with NEW co-player:****

I'll continue the successful cooperative strategy. Sending a substantial amount demonstrates trust and has been rewarded by others.

{ 'send': 3.00 }

reasoning:

Let me analyze this situation:

****Last round:****

- I sent \$3.00 (60% of my endowment)
- They returned \$5.73
- My payoff was \$7.73 (very good!)

This was an excellent return - they gave me back more than I would have had if I kept everything (\$5). This suggests my co-player is very cooperative and generous.

****Current round:****

- I'm the sender again with a NEW co-player
- I have \$5 to decide how much to send

Looking at the pattern:

- Round 1 (receiver): Saw sender send 62.4%, I returned fairly
- Round 2 (sender): Sent \$3.00 (60%)
- Round 3 (receiver): Saw sender send 59.4%, I returned fairly
- Round 4 (sender): Sent \$3.00 (60%) to the same co-player from round 3
- Round 5 (sender): Last round's co-player was very generous, returned \$5.73

The cooperative strategy has been working well. I've been consistently sending around 60% and have seen good reciprocation. My total earnings of \$23.73 over 5 rounds (averaging \$4.746 per round) suggests the cooperative approach is profitable.

For this round with a NEW co-player, I should continue the cooperative strategy that's been working. I'll send around \$3.00 again (60% of my endowment).

usage:

```
{
  "input_tokens": 1674,
  "output_tokens": 493,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.45 to you. Your visible payoff was \$6.45.

Your total visible earnings across all rounds are \$30.18.

This round, you are the RECEIVER against your current co-player. They sent you \$3.24, so you received \$9.72. How much do you return? (0-9.72)

Respond exactly as JSON: { 'return': <amount> }

Message 11 role=assistant

content:

Looking at this situation:

****Current round:****

- Sender showed strong trust by sending \$3.24 (64.8% of their endowment)
- This created \$9.72 for me
- Sender currently has \$1.76

****My pattern as receiver:****

- Round 1: Received \$9.36, returned \$4.00 (rewarded good trust)
- Round 3: Received \$8.91, returned \$3.75 (rewarded good trust)

This sender sent even more (64.8%) than I typically send (~60%), showing exceptional trust. I should reward this appropriately.

I'll return \$4.00, which:

- Gives the sender \$5.76 total (gain of \$0.76)
- Leaves me with \$5.72 (also a good gain)
- Rewards their high trust level
- Maintains my consistent reciprocation pattern

{'return': 4.00}

reasoning:

Let me analyze this situation:

****Current round:****

- Sender sent me \$3.24 (64.8% of their endowment) - showing strong trust
- This created \$9.72 for me
- Sender currently has \$5 - \$3.24 = \$1.76

****Previous rounds pattern:****

- Round 1: As receiver, got \$9.36, returned \$4.00 (42.7% of what I received)
- Round 2: As sender, sent \$3.00, got back \$3.48
- Round 3: As receiver, got \$8.91, returned \$3.75 (42.1% of what I received)
- Round 4: As sender, sent \$3.00 (to same co-player from Round 3)
- Round 5: As sender, sent \$3.00, got back \$5.73 (very generous)
- Round 6: As sender, sent \$3.00, got back \$4.45

So when I've been receiver:

- Round 1: returned 42.7% of received
- Round 3: returned 42.1% of received

This sender sent \$3.24 (64.8%), which is higher trust than what I typically send (60%). I should reward this high trust appropriately.

If I return around 42% of \$9.72, that would be about \$4.08.

But let me think about fairness:

- Sender started with \$5, sent \$3.24, has \$1.76 left
- If I return \$4.08, sender gets \$5.84 total (gain of \$0.84)
- I keep \$9.72 - \$4.08 = \$5.64 (gain of \$0.64)

This seems reasonable. The sender gets a decent return, and we both benefit.

Actually, let me be a bit more generous since they sent 64.8%, higher than typical. I could return \$4.25:

...

usage:

```
{
  "input_tokens": 1956,
  "output_tokens": 799,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.24, it became \$9.72, and you returned \$4.0. Your visible payoff was \$5.720000000000001.

Your total visible earnings across all rounds are \$35.9.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

****My track record as SENDER:****

- Consistently sent \$3.00 (60%)
- Returns have ranged from \$3.48 to \$5.73
- This strategy has yielded strong payoffs (\$5.48 to \$7.73)

****Overall performance:****

- Average earnings of ~\$5.98 per round
- Cooperative approach has been working well
- Both sending and receiving trust has been rewarded

****Current round with NEW co-player:****

I'll maintain my proven strategy of sending a substantial amount to encourage cooperation.

{'send': 3.00}

reasoning:

Looking at my history:

****As SENDER:****

- Round 2: Sent \$3.00, received back \$3.48 ? payoff \$5.48
- Round 4: Sent \$3.00, received back \$5.73 ? payoff \$7.73
- Round 5: Sent \$3.00, received back \$4.45 ? payoff \$6.45

****As RECEIVER:****

- Round 1: Received \$9.36, returned \$4.00 ? payoff \$5.36
- Round 3: Received \$8.91, returned \$3.75 ? payoff \$5.16
- Round 6: Received \$9.72, returned \$4.00 ? payoff \$5.72

My strategy of sending ~\$3.00 (60%) has been very successful:

- Average return when I send \$3.00: $(\$3.48 + \$5.73 + \$4.45) / 3 = \4.55
- This gives me an average payoff of \$6.55 when I'm sender

When I receive, I return fairly and get around \$5.4 on average.

****Current situation:****

- New co-player this round
- I'm the sender
- My cooperative strategy has been working very well (average earnings of \$5.98 per round)

I should continue with the successful strategy of sending \$3.00 (60%).

usage:

```
{
  "input_tokens": 2298,
  "output_tokens": 479,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$4.22 to you. Your visible payoff was \$6.22.

Your total visible earnings across all rounds are \$42.12.

This round, you are the RECEIVER against your current co-player. They sent you \$3.99, so you received \$11.97. How much do you return? (0-11.97)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****Last round with THIS SAME co-player:****

- I sent them \$3.00 (60%)
- They returned \$4.22 - good reciprocation

****Current round:****

- They sent me \$3.99 (79.8% of their endowment) - EXCEPTIONAL trust!
- This created \$11.97 for me
- They currently have \$1.01

This is the highest trust level anyone has shown me in this game. They sent almost 80% of their endowment after we already had a successful cooperative round. This deserves strong reciprocation.

I'll return \$5.00, which:

- Gives them \$6.01 total (gain of \$1.01)
- Leaves me with \$6.97 (gain of \$1.97)
- Rewards their exceptional trust appropriately
- Maintains cooperative relationship

```
{'return': 5.00}
```

Reasoning:

Looking at this situation:

****Last round with THIS SAME co-player:****

- I sent them \$3.00 (60% of my endowment)
- They returned \$4.22 - good reciprocation
- We had a successful cooperative round

****Current round:****

- They sent me \$3.99 (79.8% of their endowment) - VERY high trust!
- This created \$11.97 for me
- They currently have \$1.01

This is exceptional trust - they sent almost 80% of their endowment! This is significantly higher than:

- What I typically send (~60%)
- What they received from me last round (\$3.00)

They're showing me even more trust than I showed them. I should reward this generously.

Let me think about what's fair:

- They sent \$3.99, keeping \$1.01
- I received \$11.97
- If I return around \$5.00:
 - They'd have \$6.01 total (gain of \$1.01)
 - I'd have \$6.97 (also a good gain)

Actually, given they sent almost 80%, maybe I should return a bit more to really reward this exceptional trust.

If I return \$5.50:

- They'd have \$6.51 total (gain of \$1.51)
- I'd have \$6.47 (gain of \$1.47)
- More balanced rewards

Or if I return \$5.00:

- They'd have \$6.01 total (gain of \$1.01)
- I'd have \$6.97 (gain of \$1.97)

Let me be generous and go with \$5.00, which is a strong return and rewards their exceptional trust appropriately.

...

```
Usage: {
  "input_tokens": 2349,
  "output_tokens": 767,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Sender - dyad_3

Timestamp: 2026-08-12T09:27:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_3",
```

```

"move_index": 4,
"opponent_id": "Agent_6",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_6"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.41 to you. Your visible payoff was \$6.41.

Your total visible earnings across all rounds are \$6.41.

This round, you are the RECEIVER against your current co-player. They sent you \$1.73, so you received \$5.189999999999995. How much do you return? (0-5.189999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

****What happened:****

- My co-player sent \$1.73 (became \$5.19 for me)
- They kept \$3.27
- They showed some trust by sending money

****My reasoning:****

- In the previous round, my co-player returned about 49% of what they received (\$4.41 out of \$9)
- I should be reasonably cooperative to encourage future trust
- Returning around 45-50% of what I received seems fair
- This would be \$2.33-\$2.60

I'll return \$2.40:

- They'll end with: $\$3.27 + \$2.40 = \$5.67$
- I'll end with: $\$5.19 - \$2.40 = \$2.79$
- They gain \$0.67 from sending, which rewards their trust

{'return': 2.40}

reasoning:

Let me think about this situation:

1. In the previous round, I sent \$3, which became \$9 for the receiver. They returned \$4.41 to me.
 - I ended with: $\$5 - \$3 + \$4.41 = \6.41
 - They ended with: $\$9 - \$4.41 = \$4.59$
 - So they returned about 49% of what they received, which seems fairly cooperative.
2. Now I'm the RECEIVER. My co-player sent me \$1.73, which became approximately \$5.19.
 - They now have: $\$5 - \$1.73 = \$3.27$
 - I have: \$5.19
 - Total pie: $\$3.27 + \$5.19 = \$8.46$
3. There's communication noise, so what I see (\$1.73 sent) might not be exactly what they sent.
4. The sender took some risk by sending me \$1.73. It's less than what I sent in the previous round (\$3), but still a meaningful amount showing some trust.
5. I should consider:
 - Being reciprocal/cooperative to encourage future cooperation
 - The communication noise factor
 - What would be "fair"

If I think about fairness:

- They sent \$1.73, keeping \$3.27
- I received \$5.19
- If we aim for roughly equal outcomes: $(\$3.27 + \text{return}) ? (\$5.19 - \text{return})$
- That would suggest: $\$3.27 + \text{return} = \$5.19 - \text{return}$
- $2 * \text{return} = \$5.19 - \$3.27 = \$1.92$
- $\text{return} ? \$0.96$

But that seems quite low given they took a risk. Let me think about proportional return:

- Matching their 49% return rate would mean sending back \$2.54
- Alternatively, if I return \$2.50, they'd end with \$5.77 (a \$0.77 gain) while I'd have \$2.69
- This feels like a balanced cooperative move that rewards their trust without leaving me too disadvantaged

I'll return \$2.50....

usage:

```
{
  "input_tokens": 791,
  "output_tokens": 1107,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.73, it became \$5.189999999999995, and you returned \$2.4. Your visible payoff was \$2.789999999999996. Your total visible earnings across all rounds are \$9.2.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my situation:

****Previous rounds:****

- Round 1: Sent \$3, received \$4.41 back - good cooperation
- Round 2: Received \$5.19, returned \$2.40 - showed I reciprocate

****Current round:****

- New co-player (randomized pairing)
- Need to balance trust with uncertainty

Since I had success with sending \$3 in Round 1 (got \$4.41 back, netting \$1.41 profit), and I want to maintain a cooperative approach while testing this new partner, I'll send the same amount:

****Sending \$3****

This creates \$9 for them to work with. If they're reasonably cooperative like my Round 1 partner, we'll both benefit.

{'send': 3}

reasoning:

Let me analyze this situation:

Round 1: I was SENDER

- I sent \$3
- It became \$9
- They returned \$4.41
- My payoff: $\$5 - \$3 + \$4.41 = \6.41

Round 2: I was RECEIVER

- They sent \$1.73
- I received \$5.19
- I returned \$2.40
- My payoff: $\$5.19 - \$2.40 = \$2.79$
- Total so far: $\$6.41 + \$2.79 = \$9.20$?

Round 3: I am SENDER again

- This is a new co-player (since pairings are randomized each round)
- I have \$5 to send

Strategy considerations:

1. I had a good experience in Round 1 - sent \$3, got back \$4.41 (profit of \$1.41)
2. In Round 2, I received \$5.19 and returned \$2.40 (46%), showing I'm cooperative
3. This is a new co-player, so I don't know their behavior
4. There's communication noise, so what they see may differ from what I intend

I should probably send a moderate amount again to:

- Signal willingness to cooperate

- Test this new partner's trustworthiness
- Not risk everything on an unknown player

Sending \$3 worked well in Round 1. I could try that again, or maybe send slightly more (\$3.50) or less (\$2.50) to experiment.

Given that:

- My first experience was positive
- I want to maintain a cooperative reputation
- But this is a new, unknown player

I'll send \$3 again - it's a balanced approach that showed good results before.

```
{'send': 3}
```

usage:

```
{
  "input_tokens": 1128,
  "output_tokens": 608,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.71 to you. Your visible payoff was \$5.71.

Your total visible earnings across all rounds are \$14.91.

This round, you are the RECEIVER against your current co-player. They sent you \$3.79, so you received \$11.370000000000001. How much do you return? (0-11.370000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

****What happened:****

- My co-player sent \$3.79 out of \$5 (~76% of their endowment)
- This became \$11.37 for me
- They showed strong trust by sending most of their money

****My reasoning:****

- They took a big risk trusting me
- I should reward this trust to encourage cooperation
- Returning around 45% of what I received seems appropriate given their high trust
- This would be approximately \$5.10

****Calculating outcomes:****

- They'll end with: \$1.21 + \$5.10 = \$6.31
- I'll end with: \$11.37 - \$5.10 = \$6.27
- They gain \$1.31 from their trust, which is a good reward

```
{'return': 5.10}
```

reasoning:

Looking at this situation:

****What happened:****

- My co-player sent \$3.79 (became \$11.37 for me)
- They kept \$1.21
- They showed significant trust by sending most of their money

****Previous experience:****

- Round 1: I sent \$3, got back \$4.41 (they returned ~49%)
- Round 2: I received \$5.19, returned \$2.40 (~46%)
- Round 3: I sent \$3, got back \$3.71 (they returned ~41%)

****My reasoning:****

- This co-player sent \$3.79 out of \$5, showing strong trust (~76%)
- They're risking a lot and trusting me to reciprocate
- I should reward this trust appropriately

- Returning around 40-50% of what I received would be \$4.55-\$5.69
- Let me think about fairness:
 - They sent \$3.79, kept \$1.21
 - I received \$11.37
 - If I return \$5.00:
 - They end with: $\$1.21 + \$5.00 = \$6.21$
 - I end with: $\$11.37 - \$5.00 = \$6.37$
 - If I return \$5.50:
 - They end with: $\$1.21 + \$5.50 = \$6.71$
 - I end with: $\$11.37 - \$5.50 = \$5.87$

Given they sent \$3.79 (76% of their endowment), I should be generous but not overdo it. Around 45% return rate seems appropriate, which would be about \$5.12.

Let me go with \$5.10:

...

usage:

```
{
  "input_tokens": 1453,
  "output_tokens": 690,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.79, it became \$11.370000000000001, and you returned \$5.1. Your visible payoff was \$6.270000000000001.

Your total visible earnings across all rounds are \$21.18.

This round, you are the RECEIVER against your current co-player. They sent you \$2.91, so you received \$8.73. How much do you return? (0-8.73)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

****What happened:****

- My co-player sent \$2.91 out of \$5 (~58% of their endowment)
- This became \$8.73 for me
- They kept \$2.09
- This shows moderate to good trust

****My reasoning:****

- They demonstrated decent trust by sending over half their money
- I should reciprocate consistently with my previous rounds
- In previous rounds as receiver, I've returned around 45% of what I received
- For \$8.73, that would be approximately \$3.90

****Calculating outcomes:****

- They'll end with: $\$2.09 + \$3.90 = \$5.99$
- I'll end with: $\$8.73 - \$3.90 = \$4.83$
- They gain about \$1 from trusting me, which is a reasonable reward

{'return': 3.90}

reasoning:

Looking at this situation:

****What happened:****

- My co-player sent \$2.91 out of \$5 (~58% of their endowment)
- This became \$8.73 for me
- They kept \$2.09
- This shows moderate to good trust

****My reasoning:****

- They sent a decent amount, showing trust
- I should reciprocate proportionally
- Looking at my previous returns:
 - Round 2: Received \$5.19, returned \$2.40 (~46%)

- Round 4: Received \$11.37, returned \$5.10 (~45%)
- I've been returning around 45% consistently

For \$8.73, 45% would be about \$3.93

****Calculating outcomes with \$3.93:****

- They'll end with: \$2.09 + \$3.93 = \$6.02
- I'll end with: \$8.73 - \$3.93 = \$4.80
- They gain \$1.02 from sending, which rewards their trust

This seems fair and consistent with my previous behavior. I'll return \$3.90 to keep it simple.

Actually, let me recalculate with \$3.90:

- They'll end with: \$2.09 + \$3.90 = \$5.99
- I'll end with: \$8.73 - \$3.90 = \$4.83
- They gain \$0.99 from sending

That works well.

usage:

```
{
  "input_tokens": 1786,
  "output_tokens": 560,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.91, it became \$8.73, and you returned \$3.9. Your visible payoff was \$4.83.

Your total visible earnings across all rounds are \$26.009999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my history as SENDER:

****Previous rounds as sender:****

- Round 1: Sent \$3 ? received \$4.41 back (profit of \$1.41)
- Round 3: Sent \$3 ? received \$3.71 back (profit of \$0.71)

Both times sending \$3 worked out profitably, showing it's a reasonable cooperative strategy.

****Current situation:****

- New co-player (randomized pairing)
- Want to maintain consistent cooperative approach
- \$3 creates \$9 for them, leaving room for mutual benefit

I'll stick with my established strategy of sending \$3, which has proven effective:

```
{'send': 3}
```

reasoning:

Looking at my history:

Round 1 (SENDER): Sent \$3, received \$4.41 back, payoff \$6.41
 Round 2 (RECEIVER): Received \$5.19, returned \$2.40, payoff \$2.79
 Round 3 (SENDER): Sent \$3, received \$3.71 back, payoff \$5.71
 Round 4 (RECEIVER): Received \$11.37, returned \$5.10, payoff \$6.27
 Round 5 (RECEIVER): Received \$8.73, returned \$3.90, payoff \$4.83

Total earnings: \$26.01

Now Round 6, I'm the SENDER with a new randomized co-player.

My strategy has been to send \$3 as sender (both times I did this), which has worked out re

asonably well:

- Round 1: Sent \$3, got \$4.41 back (net +\$1.41)
- Round 3: Sent \$3, got \$3.71 back (net +\$0.71)

Both times I profited, though the return varied.

As receiver, I've been consistent in returning around 45% of what I receive, which seems fair.

For this new round with a new partner, I should continue with my cooperative strategy. Sending \$3 has been my standard approach and has generally worked well. It signals cooperation while not being overly risky.

I'll send \$3 again.

usage:

```
{
  "input_tokens": 2117,
  "output_tokens": 509,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$3.68 to you. Your visible payoff was \$5.68.

Your total visible earnings across all rounds are \$31.689999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.51, so you received \$7.529999999999999. How much do you return? (0-7.529999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

****What happened:****

- Same co-player from Round 6 (where I sent them \$3 and they returned \$3.68)
- Now they sent me \$2.51, which became \$7.53
- They kept \$2.49
- They showed moderate trust

****My reasoning:****

- They were reasonably cooperative in Round 6 (returned ~41% of what they received)
- Now they're trusting me with about half their endowment
- I should maintain my consistent reciprocation pattern
- I've been returning around 45% of what I receive as a receiver

****For \$7.53:****

- 45% = approximately \$3.39

****Calculating outcomes:****

- They'll end with: \$2.49 + \$3.39 = \$5.88
- I'll end with: \$7.53 - \$3.39 = \$4.14
- They gain \$0.88 from their trust

{'return': 3.39}

reasoning:

Looking at this situation:

****What happened:****

- Same co-player from Round 6 (I was sender then, now I'm receiver)
- In Round 6: I sent \$3, they returned \$3.68
- Now Round 7: They sent \$2.51, became \$7.53 for me
- They kept \$2.49

****My reasoning:****

- This is the same partner who just returned \$3.68 on \$9 (~41% return rate)
- They showed reasonable cooperation
- Now they sent \$2.51 (~50% of endowment), showing moderate trust

- I should reciprocate fairly

****My return pattern as receiver:****

- Round 2: Received \$5.19, returned \$2.40 (46%)
- Round 4: Received \$11.37, returned \$5.10 (45%)
- Round 5: Received \$8.73, returned \$3.90 (45%)

I've been consistently returning around 45% of what I receive.

For \$7.53:

- 45% would be approximately \$3.39

****Calculating outcomes:****

- They'll end with: $\$2.49 + \$3.39 = \$5.88$
- I'll end with: $\$7.53 - \$3.39 = \$4.14$
- They gain \$0.88 from trusting me

This seems fair and consistent with my pattern.

usage:

```
{
  "input_tokens": 2425,
  "output_tokens": 569,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.51, it became \$7.529999999999999, and you returned \$3.39. Your visible payoff was \$4.139999999999999.

Your total visible earnings across all rounds are \$35.83.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history as SENDER:

****Previous rounds as sender:****

- Round 1: Sent \$3 ? received \$4.41 back (profit of \$1.41)
- Round 3: Sent \$3 ? received \$3.71 back (profit of \$0.71)
- Round 6: Sent \$3 ? received \$3.68 back (profit of \$0.68)

****Pattern:****

- Sending \$3 has consistently worked well
- All three co-players returned more than I sent
- This shows a good balance of trust and caution

****Current situation:****

- New co-player (randomized pairing)
- My established strategy has proven reliable

I'll continue with sending \$3:

```
{'send': 3}
```

Reasoning:

Looking at my history as SENDER:

****Previous rounds as sender:****

- Round 1: Sent \$3 ? received \$4.41 back (profit of \$1.41)
- Round 3: Sent \$3 ? received \$3.71 back (profit of \$0.71)
- Round 6: Sent \$3 ? received \$3.68 back (profit of \$0.68)

All three times I've sent \$3, I've gotten back more than I sent, making profitable exchanges. The strategy is working consistently.

****Current situation:****

- Round 8 (final round likely, or near it)
- New co-player again (randomized pairing)

- Want to maintain cooperative approach

I should stick with \$3 again as it has consistently worked well across multiple partners:

```
{'send': 3}
Usage: {
  "input_tokens": 2608,
  "output_tokens": 378,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T09:27:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_6"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.97, it became \$8.91, and you returned \$4.5. Your visible payoff was \$4.41.

Your total visible earnings across all rounds are \$4.41.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my previous experience:

- As a receiver, a sender trusted me with about 60% of their endowment
- I reciprocated fairly, and we both benefited
- This established that cooperation can work well in this game

Now as the sender with a new co-player, I want to:

1. Demonstrate trust to encourage reciprocation
2. Follow a similar cooperative strategy that worked before
3. Give my co-player the opportunity to earn and share

I'll send **\$3.00** (60% of my endowment), mirroring the trust shown to me in the previous round.

{'send': 3.00}

reasoning:

Let me think about this situation:

In Round 1:

- I was the receiver
- The sender sent me \$2.97 (59.4% of their endowment)
- I received \$8.91
- I returned \$4.50
- I kept \$4.41
- The sender got back \$4.50, so they ended with \$2.03 + \$4.50 = \$6.53

Now in Round 2:

- I'm the sender
- I have \$5 to potentially send
- I'm playing against a different co-player (the problem states pairings are randomized each round)

Key considerations:

1. The previous sender showed trust by sending 59.4% of their endowment
2. I reciprocated fairly by returning about 50% of what I received
3. This worked out well - both players benefited
4. Now I'm in the sender role with a new partner

Since my previous experience showed that sending a substantial amount can lead to good returns, and I want to establish trust with this new co-player, I should send a meaningful amount.

The previous sender sent about 60% (\$2.97 out of \$5). This seemed like a good trust-building amount. I could:

- Send a similar proportion to continue the cooperative pattern
- Send slightly more to show strong trust
- Send slightly less to be more cautious with a new partner

I think sending around \$3 (60%) makes sense - it mirrors what worked well in the previous round and shows trust to establish a good relationship with this new co-player.

Let me send \$3.00.

usage:

```
{
  "input_tokens": 908,
  "output_tokens": 516,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.32 to you. Your visible payoff was \$7.32.

Your total visible earnings across all rounds are \$11.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my recent experience:

Round 2 results:

- I sent \$3.00 (60% of endowment)
- Receiver returned \$5.32
- I ended with \$7.32 - excellent outcome!

The cooperative strategy is working very well. When I demonstrate trust, it's being reciprocated generously.

Now facing a new co-player, I'll continue this successful approach. Sending around 60% has proven effective in building mutual benefit.

```
{'send': 3.00}
```

reasoning:

Let me analyze the situation:

Round 2 (just completed):

- I was the SENDER
- I sent \$3.00 (60% of endowment)
- It became \$9.00 for the receiver
- They returned \$5.32 to me
- My payoff: $\$5 - \$3 + \$5.32 = \7.32
- They kept: $\$9.00 - \$5.32 = \$3.68$

This was a very generous return! They returned more than half of what they received, and I ended up with more than my initial endowment even after sending a substantial amount.

My experience so far:

- Round 1 (RECEIVER): Received \$8.91, returned \$4.50, kept \$4.41
- Round 2 (SENDER): Sent \$3.00, received back \$5.32, ended with \$7.32

Total visible earnings: \$11.73

The cooperation strategy is working very well! When I sent 60% in Round 2, I got an excellent return.

Now in Round 3, I'm a SENDER again with a new co-player.

Given:

1. My cooperative strategy has been working well
2. Both times cooperation has led to mutual benefit
3. I want to continue encouraging trust and reciprocation
4. The game benefits from establishing cooperative norms

I should continue with a cooperative approach. Sending around 60% worked well last time. I could:

- Send \$3.00 again (same as last time, 60%)
- Send a bit more to show strong trust
- Send a bit less to be slightly more cautious with a new partner

Since cooperation has been consistently rewarded, I'll stick with the successful strategy of sending around 60%.

...

usage:

```
{
  "input_tokens": 1176,
  "output_tokens": 538,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$2.98 to you. Your visible payoff was \$4.98.

Your total visible earnings across all rounds are \$16.71.

This round, you are the RECEIVER against your current co-player. They sent you \$3.24, so you received \$9.72. How much do you return? (0-9.72)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation with my current co-player:

****Last round (Round 3):****

- I sent them \$3.00
- They returned \$2.98 (~33% of what they received)
- This was a modest but acceptable return

****This round (Round 4):****

- They sent me \$3.24 (slightly MORE than I sent them)
- I received \$9.72
- They currently have only \$1.76

The fact that they sent me MORE than I sent them suggests they're trying to build cooperation. I should reciprocate generously to:

1. Reward their increased trust
2. Establish a positive cooperative dynamic
3. Ensure we both benefit from ongoing interaction

I'll return ****\$4.80****:

- They end with: $\$1.76 + \$4.80 = \$6.56$ (gain \$1.56)
- I keep: $\$9.72 - \$4.80 = \$4.92$ (gain \$4.92)

This is more generous than they were to me, encouraging further cooperation.

{'return': 4.80}

reasoning:

Let me analyze this situation:

Round 3 (previous round):

- I was the SENDER
- I sent \$3.00 to this same co-player
- They returned \$2.98
- My payoff was \$4.98 (kept \$2 + returned \$2.98)

Round 4 (current round):

- I am the RECEIVER

- Same co-player (they are now the sender)
- They sent me \$3.24
- I received \$9.72

So this is interesting - we're playing repeated games with the same partner. In round 3, I sent \$3.00 and they returned \$2.98 (about 33% of what they received). Now they've sent me \$3.24 (notably a bit more than I sent them).

Let me think about the fairness here:

- When I sent \$3.00, they kept \$9.00 - \$2.98 = \$6.02
- I kept \$2.00 + \$2.98 = \$4.98
- So they gained \$1.02, I lost \$0.02

Now they sent \$3.24:

- They kept \$5 - \$3.24 = \$1.76
- I received \$9.72

If I want to reciprocate similarly to how they treated me, I might return around 33% of what I received, which would be about \$3.21.

But actually, they sent me slightly MORE than I sent them (\$3.24 vs \$3.00), which could be seen as trying to establish cooperation or make up for the relatively low return last round.

Let me think about different return amounts:

Option 1: Return \$3.00 (about 31%)

- They end with: \$1.76 + \$3.00 = \$4.76 (lost \$0.24)
- I keep: \$9.72 - \$3.00 = \$6.72 (gain \$6.72)
- This feels too stingy given their slightly larger initial send.

Option 2: Return \$4.00 (about 41%)

- They end with: \$1.76 + \$4.00 = \$5.76 (gain \$0.76)
- I keep: \$9.72 - \$4.00 = \$5.72 (gain \$5.72)
- This creates mutual benefit and shows reciprocal trust.

Option 3: Return \$5.00 (about 51%)

- They end with: \$1.76 + \$5.00 = \$6.76 (gain \$1.76)
- I keep: \$9.72 - \$5.00 = \$4.72...

usage:

```
{
  "input_tokens": 1431,
  "output_tokens": 1067,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.24, it became \$9.72, and you returned \$4.8. Your visible payoff was \$4.920000000000001.

Your total visible earnings across all rounds are \$21.630000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$2.32, so you received \$6.959999999999999. How much do you return? (0-6.959999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- New co-player sent me \$2.32 (about 46% of their endowment)
- I received \$6.96 after the multiplier
- They currently have \$2.68

They showed moderate trust by sending nearly half their endowment. I should reciprocate fairly to reward their trust and ensure mutual benefit.

Following my pattern of returning roughly 50% of what I receive, I'll return **\$3.48**:

- Sender ends with: \$2.68 + \$3.48 = \$6.16 (gain \$1.16)

- I keep: $\$6.96 - \$3.48 = \$3.48$ (gain $\$3.48$)

Both of us benefit from the cooperation.

```
{'return': 3.48}
```

reasoning:

Looking at this round:

- New co-player (different from Round 4)
- They sent me $\$2.32$ (about 46% of their endowment)
- I received $\$6.96$
- They currently have $\$2.68$ ($\$5 - \2.32)

They showed moderate trust by sending nearly half their endowment. I should reciprocate fairly to:

1. Reward their trust
2. Ensure both of us benefit
3. Encourage cooperation

Looking at my previous patterns:

- Round 1: Received $\$8.91$, returned $\$4.50$ (about 50%)
- Round 4: Received $\$9.72$, returned $\$4.80$ (about 49%)

Returning around 50% seems like my consistent fair approach. For $\$6.96$, that would be around $\$3.48$.

Let me check the outcomes:

- Sender ends with: $\$2.68 + \$3.48 = \$6.16$ (gain $\$1.16$)
- I keep: $\$6.96 - \$3.48 = \$3.48$ (gain $\$3.48$)

Both benefit, and this reciprocates their trust fairly. I'll return $\$3.48$.

usage:

```
{
  "input_tokens": 1834,
  "output_tokens": 476,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send $\$2.32$, it became $\$6.959999999999999$, and you returned $\$3.48$. Your visible payoff was $\$3.479999999999999$.

Your total visible earnings across all rounds are $\$25.110000000000003$.

This round, you are the SENDER against your current co-player. You have $\$5$. How much do you send? (0-5)

Respond exactly as JSON: `{'send': <amount>}`

Message 11 role=assistant

content:

Looking at my history with this current co-player:

****Round 5:****

- They sent me $\$2.32$ (46% of endowment)
- I returned $\$3.48$, giving them a good outcome

They showed moderate trust, and I reciprocated fairly. Now it's my turn to send, and I should continue building this cooperative relationship.

I'll send ****\$2.50****, which is:

- Slightly more than they sent me (showing willingness to cooperate)
- A reasonable amount that matches their moderate approach
- Enough to create mutual benefit if they reciprocate

```
{'send': 2.50}
```

reasoning:

Looking at my history with this current co-player:

Round 5 (last round):

- They sent me $\$2.32$ (46% of endowment)

- I received \$6.96
- I returned \$3.48 (50% of what I received)
- They ended with \$6.16, I ended with \$3.48

Now in Round 6, I'm the sender with this same co-player.

They showed moderate trust by sending \$2.32. I reciprocated fairly by returning \$3.48, which gave them a good outcome (\$6.16 total).

Since we had a successful cooperative interaction, I should continue building on this. I'll send a similar or slightly higher amount to maintain the cooperative relationship.

I've been generally sending around \$3.00 (60%) in other rounds, which has worked well. Given this co-player sent me \$2.32 (46%), I could:

1. Send similar to what they sent (~\$2.30-2.50)
2. Send my standard amount (~\$3.00)
3. Send slightly more to build trust

I think sending around \$2.50-3.00 makes sense. Let me go with \$2.50, which is slightly more than they sent me, showing I'm willing to cooperate but matching their level of caution.

Actually, looking at my overall strategy, I've been doing well with \$3.00 sends. But this particular co-player was more cautious. Let me match their level at around \$2.30-2.50.

I'll send \$2.50.

usage:

```
{
  "input_tokens": 2153,
  "output_tokens": 510,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$4.52 to you. Your visible payoff was \$7.02.

Your total visible earnings across all rounds are \$32.13.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my recent experiences:

Round 6 outcome:

- I sent \$2.50
- Received \$4.52 back
- Ended with \$7.02 - very good!

The cooperative approach continues to work well. Now with a new co-player, I'll signal trust and willingness to cooperate by sending a solid amount.

I'll send **\$3.00** (60% of endowment), which has proven to be an effective strategy:

```
{'send': 3.00}
```

reasoning:

Looking at my recent experience:

Round 6:

- I sent \$2.50 to my co-player
- They returned \$4.52
- I ended with \$7.02 - excellent outcome!

This is a new co-player in Round 7. I've been doing well with cooperative strategies. My approach of sending moderate to high amounts (around 50-60% of endowment) has generally worked well across different co-players.

Let me review the pattern:

- Round 2: Sent \$3.00, received back \$5.32 - great
- Round 3: Sent \$3.00, received back \$2.98 - okay
- Round 6: Sent \$2.50, received back \$4.52 - great

The cooperative strategy continues to work. With a new co-player, I should send a reasonable amount to signal willingness to cooperate. I'll send around \$3.00 (60%) to establish trust.

usage:

```
{
  "input_tokens": 2419,
  "output_tokens": 343,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.97 to you. Your visible payoff was \$5.970000000000001.

Your total visible earnings across all rounds are \$38.1.

This round, you are the RECEIVER against your current co-player. They sent you \$2.48, so you received \$7.439999999999995. How much do you return? (0-7.439999999999995)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Co-player sent me \$2.48 (about 50% of their endowment)
- I received \$7.44 after the multiplier
- They currently have \$2.52

They showed solid trust by sending nearly half their endowment. I'll reciprocate fairly.

I'll return **\$3.72** (about 50% of what I received):

- Sender ends with: \$2.52 + \$3.72 = \$6.24 (gain \$1.24)
- I keep: \$7.44 - \$3.72 = \$3.72 (gain \$3.72)

Both of us benefit from the cooperation.

```
{'return': 3.72}
```

Reasoning:

Looking at the situation:

- New co-player sent me \$2.48 (about 50% of their endowment)
- I received \$7.44 after the multiplier
- They currently have \$2.52

They showed substantial trust by sending nearly half their endowment. I should reciprocate fairly.

Following my pattern of fair returns:

- If I return about 50% of what I receive: $7.44 * 0.5 = 3.72$

This would give:

- Sender: $2.52 + 3.72 = 6.24$ (gain \$1.24)
- Me: $7.44 - 3.72 = 3.72$ (gain \$3.72)

Both benefit, and I'm being fair to their trust. This seems reasonable.

Let me return \$3.72.

Usage: {

```
"input_tokens": 2365,
"output_tokens": 378,
"reasoning_tokens": 0
}
```

GAME - Agent_1 - Sender - dyad_4

Timestamp: 2026-08-12T09:27:14
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8
 Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_4"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_4": "trustee"
 },
 "trustee": "Agent_4"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 8,
 "task": "game",
 "task_index": 0
 }

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
 so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths
 , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized
 , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent wi

ll have acted as sender and receiver equally often when the round count allows it.

- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.5 to you. Your visible payoff was \$6.5.

Your total visible earnings across all rounds are \$6.5.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

****Analysis:****

- My co-player sent \$2.9 (about 58% of their endowment), showing considerable trust
- I received \$8.7 after the 3x multiplier
- In Round 1, when I sent \$3, my previous partner returned \$4.5 (50% of what they received)
- Applying a similar reciprocity norm seems fair

****Decision:****

I'll return approximately 50% of what I received, which rewards their trust and establishes a cooperative pattern. This leaves us both better off than our starting positions.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

****Previous Round (Round 1):****

- I was SENDER
- I sent \$3.0
- It became \$9.0
- They returned \$4.5 to me
- My payoff: \$5 - \$3 + \$4.5 = \$6.5
- Their payoff: \$9 - \$4.5 = \$4.5

That was a pretty fair return - they returned half of what they received, which is a common reciprocity norm.

****Current Round (Round 2):****

- I am RECEIVER
- They sent me \$2.9 (close to \$3)
- I received \$8.7 (2.9×3)
- I need to decide how much to return (0-8.7)

****Considerations:****

1. My co-player sent me almost \$3 (actually \$2.9, but there's communication noise mentioned)
2. In Round 1, when I sent \$3, my partner returned \$4.5 (which was 50% of the \$9 they received)
3. If I apply the same 50% rule: I should return about \$4.35 (half of \$8.7)
4. My co-player took a risk by sending \$2.9, showing trust
5. Being reciprocal could establish good cooperation
6. Note: This is a different co-player than Round 1 (pairings are randomized)

Since this is a new co-player and they showed trust by sending \$2.9, I should reciprocate fairly. Returning around 50% seems like a reasonable reciprocity norm that:

- Rewards their trust
- Leaves both of us better off than not playing
- Establishes me as a cooperative player

My co-player's payoff if I return \$4.35:

- They get: $\$5 - \$2.9 + \$4.35 = \6.45
- I get: $\$8.7 - \$4.35 = \$4.35$

They sent 58% of their endowment, which is quite generous. I should return slightly more to match their generosity and encourage continued cooperation. Returning \$4.5 mirrors what my previous partner did for me and feels like the right balance....

usage:

```
{
  "input_tokens": 811,
  "output_tokens": 781,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.9, it became \$8.7, and you returned \$4.5. Your visible payoff was \$4.199999999999999.

Your total visible earnings across all rounds are \$10.7.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

****Analysis:****

- My co-player sent \$2.27 (about 45% of their endowment), showing moderate trust
- I received \$6.81 after the multiplier
- In previous rounds, I've been returning approximately 50% of what I receive as a reciprocity norm
- This approach has worked well so far

****Decision:****

I'll maintain consistency and return about 50% of what I received, which rewards their trust appropriately.

```
'''json
{"return": 3.4}
'''
```

reasoning:

Let me analyze this situation:

****Current situation:****

- I'm the RECEIVER again
- My co-player sent \$2.27 (about 45% of their \$5 endowment)
- I received \$6.81 after the 3x multiplier
- They kept $\$5 - \$2.27 = \$2.73$

****Past behavior:****

- Round 1: I sent \$3, received back \$4.5 (50% return rate on the multiplied amount)
- Round 2: I received \$8.7, returned \$4.5 (about 52% return rate)

****Considerations:****

- My co-player sent a moderate amount (\$2.27), showing some trust but less than the previous player
- Being consistent with my previous behavior would mean returning around 50% of what I received
- 50% of \$6.81 = \$3.405

****Payoff implications:****

- If I return \$3.4-3.5:
 - I keep: $\$6.81 - \$3.4 = \$3.41$
 - They get: $\$2.73 \text{ (kept)} + \$3.4 \text{ (returned)} = \6.13
 - This seems fair - we both do better than the \$5 starting point

I should maintain consistency with my reciprocity pattern from Round 2. Returning about 50% seems fair and builds trust for potential future rounds.

Let me return \$3.4 (just under 50%).

```

usage:
{
  "input_tokens": 1105,
  "output_tokens": 467,
  "reasoning_tokens": 0
}
Message 6 role=user
content:
  Round 4

  In your most recent previous game round, you were the RECEIVER against another agent. You
  saw them send $2.27, it became $6.8100000000000005, and you returned $3.4. Your visible
  payoff was $3.4100000000000006.
  Your total visible earnings across all rounds are $14.11.

  This round, you are the SENDER against your current co-player. You have $5. How much do yo
  u send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Message 7 role=assistant
content:
  **Analysis:**
  - This is Round 4 with a new co-player
  - In Round 1, I sent $3 and received $4.5 back - a positive outcome
  - As a receiver, I've been returning about 50%, establishing a cooperative pattern
  - My current co-player is unknown to me

  **Decision:**
  I'll send $3 again. This amount has worked well before - it shows trust and creates potent
  ial for mutual gain through the 3x multiplier, while managing downside risk.

  '''json
  {"send": 3}
  '''
reasoning:
  Let me analyze the situation:

  **What I know:**
  - Round 1: I was SENDER, sent $3, received back $4.5, payoff $6.5
  - Round 2: I was RECEIVER, got $8.7 (from $2.9 sent), returned $4.5, payoff $4.2
  - Round 3: I was RECEIVER, got $6.81 (from $2.27 sent), returned $3.4, payoff $3.41
  - Total earnings: $14.11

  **Current situation:**
  - Round 4: I'm the SENDER with $5
  - This is a new co-player (different from rounds 1-3)

  **Strategic considerations:**
  1. I've seen one positive interaction as sender (sent $3, got back $4.5)
  2. As receiver, I've been fairly generous, returning about 50%
  3. This creates a reputation for reciprocity if others know
  4. But agents don't know who I am (names hidden), so reputation effects are limited
  5. However, the pattern of behavior in the pool might matter

  **Decision factors:**
  - Sending creates potential gains (3x multiplier)
  - Risk: this is a new co-player, no direct history
  - But overall, cooperation seems to be working in this pool
  - My Round 1 strategy (sending $3) worked well

  I'll send $3 again - it's a moderate trust signal that has proven effective.
usage:
{
  "input_tokens": 1352,
  "output_tokens": 472,
  "reasoning_tokens": 0
}
Message 8 role=user
content:
  Round 5

```

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.48 to you. Your visible payoff was \$6.48.

Your total visible earnings across all rounds are \$20.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

****Analysis:****

- Round 5 with a new co-player
- My \$3 sending strategy has been successful:
 - Round 1: Sent \$3, received \$4.5 back ? \$6.5 payoff
 - Round 4: Sent \$3, received \$4.48 back ? \$6.48 payoff
- Both returns were close to 50%, indicating fair reciprocity from my co-players
- This amount balances trust-building with risk management

****Decision:****

I'll continue with \$3, as it's been consistently effective.

```
'''json
{"send": 3}
'''
```

reasoning:

****Analysis:****

- Round 5, new co-player as sender
- My sending strategy of \$3 has worked well:
 - Round 1: Sent \$3, received \$4.5 back, payoff \$6.5
 - Round 4: Sent \$3, received \$4.48 back, payoff \$6.48
- Both times I received close to 50% back, which is fair reciprocity
- Total visible earnings: \$20.59 across 4 rounds (average \$5.15/round)
- As a receiver, I've been consistently returning about 50%

****Decision:****

The \$3 sending strategy is working consistently well. My co-players have been reciprocating fairly. I should continue this approach.

```
'''json
{"send": 3}
'''
```

usage:

```
{
  "input_tokens": 1601,
  "output_tokens": 339,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.77 to you. Your visible payoff was \$8.77.

Your total visible earnings across all rounds are \$29.36.

This round, you are the RECEIVER against your current co-player. They sent you \$3.68, so you received \$11.04000000000001. How much do you return? (0-11.04000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

****Analysis:****

- My co-player sent \$3.68 (about 74% of their endowment), showing strong trust
- I received \$11.04 after the multiplier
- They're taking a significant risk by sending such a high amount
- I've been maintaining a 50% return pattern, which has worked well

****Decision:****

I'll return approximately 50% of what I received to reward their trust appropriately.

```
'''json
{"return": 5.5}
'''
```

reasoning:

****Analysis:****

- My co-player sent \$3.68 (about 74% of their endowment), showing strong trust
- I received \$11.04 after the 3x multiplier
- In Round 5, my co-player returned \$6.77 from \$9, which is about 75% - very generous!
- I've been returning about 50% as a norm, but this co-player sent a substantial amount

****Decision:****

My co-player showed considerable trust by sending \$3.68. I should reciprocate fairly. Following my 50% return pattern would give them about \$5.50 back.

Let me calculate:

- They sent \$3.68, keeping \$1.32
- I received \$11.04
- If I return \$5.50:
 - They get: $\$1.32 + \$5.50 = \$6.82$ (better than their \$5 starting)
 - I get: $\$11.04 - \$5.50 = \$5.54$ (better than my \$0 starting as receiver)

This seems fair and maintains my reciprocity norm.

usage:

```
{
  "input_tokens": 1890,
  "output_tokens": 379,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.68, it became \$11.040000000000001, and you returned \$5.5. Your visible payoff was \$5.540000000000001.

Your total visible earnings across all rounds are \$34.9.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

****Analysis:****

- My co-player sent \$3.23 (about 65% of their endowment), showing good trust
- I received \$9.69 after the multiplier
- I've been consistently returning approximately 50% of what I receive
- This approach has maintained positive cooperative relationships

****Decision:****

I'll return about 50% of what I received.

```
'''json
{"return": 4.85}
'''
```

reasoning:

****Analysis:****

- My co-player sent \$3.23 (about 65% of their endowment), showing good trust
- I received \$9.69 after the multiplier
- I've been maintaining a consistent 50% return pattern throughout the game
- This has been working well and seems fair

****Decision:****

I'll continue with my 50% return pattern: $0.5 \times \$9.69 = \4.845

Let me round to a reasonable number: \$4.85

usage:

```
{
  "input_tokens": 2142,
  "output_tokens": 221,
  "reasoning_tokens": 0
}
```



```
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.23, it became \$9.69, and you returned \$4.85. Your visible payoff was \$4.84.

Your total visible earnings across all rounds are \$39.739999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

****Analysis:****

- Round 8 with a new co-player
- My sending strategy of \$3 has been consistently successful:
 - Round 1: \$6.5 payoff
 - Round 4: \$6.48 payoff
 - Round 5: \$8.77 payoff
- Average return has been favorable, with partners generally reciprocating fairly
- This amount continues to balance trust-building with prudent risk management

****Decision:****

I'll send \$3 again, as it's proven to be an effective cooperative strategy.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze my performance as a sender so far:

Round 1: Sent \$3, received \$4.5 back ? payoff \$6.5
Round 4: Sent \$3, received \$4.48 back ? payoff \$6.48
Round 5: Sent \$3, received \$6.77 back ? payoff \$8.77

All three times I sent \$3, I got positive returns. The returns varied:

- Round 1: 50% return rate (4.5/9)
- Round 4: ~50% return rate (4.48/9)
- Round 5: ~75% return rate (6.77/9)

Overall, sending \$3 has been very successful for me. I'm averaging well above my \$5 starting amount when I send \$3.

As a receiver, I've been consistently returning about 50% of what I receive, which seems fair and has maintained good cooperation.

This is Round 8, a new co-player. I should stick with my successful strategy of sending \$3.

```
Usage: {
  "input_tokens": 2158,
  "output_tokens": 387,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Receiver - dyad_4

Timestamp: 2026-08-12T09:27:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_4"
    ],
    "investor": "Agent_1",
    "roles": {
```

```

    "Agent_1": "investor",
    "Agent_4": "trustee"
  },
  "trustee": "Agent_4"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.8 to you. Your visible payoff was \$5.8.

Your total visible earnings across all rounds are \$5.8.

This round, you are the RECEIVER against your current co-player. They sent you \$2.56, so you received \$7.68. How much do you return? (0-7.68)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

****Analysis:****

- My co-player sent \$2.56 (about 51% of their endowment), showing moderate trust
- I received \$7.68 after the 3x multiplication
- They currently have \$2.44 remaining

****Reciprocity consideration:****

- In Round 1, my co-player returned about 42% of what they received
- I want to be fair and encourage continued cooperation
- Returning around 45-50% seems reasonable

If I return \$3.50:

- Sender ends with: $\$2.44 + \$3.50 = \$5.94$ (gain of \$0.94)
- I keep: $\$7.68 - \$3.50 = \$4.18$ (gain of \$4.18)

This rewards their trust while keeping a fair share given the multiplication advantage.

```
'''json
{"return": 3.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was SENDER, sent \$3
- It became \$9 for receiver
- They returned \$3.8 to me
- My payoff: $\$5 - \$3 + \$3.8 = \5.8
- Their payoff: $\$9 - \$3.8 = \$5.2$

So in Round 1, my co-player returned about 42% of what they received ($\$3.8/\$9 \approx 0.42$).

Round 2:

- I am RECEIVER
- Sender sent \$2.56
- I received \$7.68 (2.56×3)
- I need to decide how much to return

The sender invested \$2.56, so they currently have $\$5 - \$2.56 = \$2.44$ remaining.

If I think about reciprocity and building trust:

- The sender showed moderate trust by sending about 51% of their endowment
- If I want to encourage cooperation, I should return enough that the sender comes out ahead

Let me consider different return amounts:

- If I return \$4: Sender gets $\$2.44 + \$4 = \$6.44$ (net gain of \$1.44), I keep \$3.68 (net gain of \$3.68)
- If I return \$3.5: Sender gets $\$2.44 + \$3.5 = \$5.94$ (net gain of \$0.94), I keep \$4.18 (net gain of \$4.18)
- If I return \$3: Sender gets $\$2.44 + \$3 = \$5.44$ (net gain of \$0.44), I keep \$4.68 (net gain of \$4.68)

In my previous round, the receiver returned about 42% of what they received. If I apply a similar principle:

$\$7.68 \times 0.42 \approx \3.23

I'm inclined to return slightly more than that baseline to reward their trust and establish a cooperative pattern. Returning \$3.5 seems like the right balance-it gives them a meaningful profit while still letting me benefit substantially, and it signals I'm reliable without being exploited....

usage:

```
{
  "input_tokens": 804,
  "output_tokens": 923,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.56, it became \$7.68, and you returned \$3.5. Your visible payoff was \$4.18.
Your total visible earnings across all rounds are \$9.98.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant**content:**

****Analysis:****

I'm now facing a new co-player (Round 3, new pairing). My experience so far:

- Round 1: Sent \$3, received back \$3.8 (42% return rate)
- Round 2: Received \$7.68, returned \$3.5 (46% return rate)

****Strategy:****

Since I'm building a reputation in this multi-agent environment, I want to signal cooperative behavior. I'll send slightly more than last time to:

- Test if higher trust leads to better reciprocity
- Create more value through the 3x multiplier
- Signal that I'm willing to be generous

Sending \$3.5 means my co-player receives \$10.50. If they return around 40%, I'd get back ~\$4.20, giving me a net gain. This balances trust-building with reasonable risk.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the situation:

****Round 1:**** I was SENDER

- Sent: \$3
- They received: \$9
- They returned: \$3.8
- My payoff: \$5.8

****Round 2:**** I was RECEIVER

- They sent: \$2.56
- I received: \$7.68
- I returned: \$3.5
- My payoff: \$4.18

****Round 3:**** I am SENDER again

Now I need to think about this new co-player in Round 3. Note that each round I'm paired with potentially different agents from the pool of 8 agents.

In Round 1, my co-player returned \$3.8 out of \$9 (42.2%).

In Round 2, I returned \$3.5 out of \$7.68 (45.6%).

For Round 3, I'm facing a NEW co-player (since pairings are randomized each round). I don't have specific history with this particular agent.

Given:

1. I've seen one example of a co-player returning about 42%
2. I returned about 46% myself, trying to be fair
3. The game has communication noise, so amounts may differ slightly

Strategy for Round 3:

- I should continue with a cooperative but cautious approach
- Sending around \$3 worked reasonably in Round 1
- I might try a slightly different amount to gather more information

Let me send \$3 again as it seems to be a reasonable middle ground that:

- Shows trust

- Creates significant value through multiplication (becomes \$9)
- Leaves me with \$2 as a safety net
- If they return around 40-45%, I'd get back around \$3.6-\$4, giving me a decent payoff

Actually, I could also consider sending a bit more to test if higher trust leads to better returns. Let me send \$3.5 this time.

...

usage:

```
{
  "input_tokens": 1137,
  "output_tokens": 766,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.07 to you. Your visible payoff was \$7.57.

Your total visible earnings across all rounds are \$17.55.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

****Analysis:****

My co-player sent \$2.9 (58% of their endowment), showing significant trust. This is a strong cooperative signal.

****Previous patterns:****

- Round 1 sender: Got back 42% return
- Round 2 receiver: I returned 46%
- Round 3 sender: Got back 58% return (very generous!)

****Current situation:****

- I received \$8.7
- Sender has \$2.1 remaining

****Decision:****

My co-player showed high trust by sending 58%. I should reciprocate proportionally. Returning 50% of what I received (\$4.35) is fair:

If I return \$4.35:

- Sender ends with: \$2.1 + \$4.35 = \$6.45 (profit of \$1.45)
- I keep: \$8.7 - \$4.35 = \$4.35 (profit of \$4.35)

This rewards their trust and maintains cooperative momentum.

```
'''json
{"return": 4.35}
'''
```

reasoning:

****Analysis of the situation:****

Round 4, I'm the RECEIVER:

- My co-player sent \$2.9 (58% of their endowment)
- I received \$8.7 after the 3x multiplication
- They currently have \$2.1 remaining (\$5 - \$2.9)

****Previous experience:****

- Round 1 (Sender): Sent \$3, got back \$3.8 (42% of \$9)
- Round 2 (Receiver): Received \$7.68, returned \$3.5 (46%)
- Round 3 (Sender): Sent \$3.5, got back \$6.07 (58% of \$10.5) - this was very generous!

****Reciprocity:****

My Round 3 co-player was very generous with a 58% return rate. This current co-player sent 58% of their endowment, showing significant trust.

Looking at patterns:

- When I received \$7.68 in Round 2, I returned \$3.5 (46%)
- Now I received \$8.7

I should be fair and encouraging. If I return around 45-50%:

- 45% would be \$3.92
- 50% would be \$4.35

Let me calculate outcomes:

If I return \$4.00:

- Sender ends with: $\$2.1 + \$4.00 = \$6.10$ (gain of \$1.10)
- I keep: $\$8.7 - \$4.00 = \$4.70$ (gain of \$4.70)

If I return \$4.35:

- Sender ends with: $\$2.1 + \$4.35 = \$6.45$ (gain of \$1.45)

...

usage:

```
{
  "input_tokens": 1485,
  "output_tokens": 766,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.9, it became \$8.7, and you returned \$4.35. Your visible payoff was \$4.35.

Your total visible earnings across all rounds are \$21.9.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

****Analysis:****

New round, new co-player. My track record:

- As sender: Sent \$3 (got 42% back), sent \$3.5 (got 58% back)
- As receiver: Returned 46% and 50%

****Strategy:****

The higher send amount (\$3.5) yielded much better reciprocity in Round 3. This suggests that at showing more trust encourages cooperation. I'll continue with this amount to:

- Signal strong cooperative intent
- Create significant value (\$10.50 for receiver)
- Build on the positive pattern

Given my fair behavior as a receiver, I'm hopeful this will be reciprocated.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Looking at my history:

- Round 1 (SENDER): Sent \$3, got back \$3.8 (42% return) - payoff \$5.8
- Round 2 (RECEIVER): Received \$7.68, returned \$3.5 (46% return) - payoff \$4.18
- Round 3 (SENDER): Sent \$3.5, got back \$6.07 (58% return) - payoff \$7.57
- Round 4 (RECEIVER): Received \$8.7, returned \$4.35 (50% return) - payoff \$4.35

Now Round 5 - I'm the SENDER against a new co-player.

Key observations:

1. When I sent more (\$3.5 vs \$3), I got a better return (58% vs 42%)
2. I've been generally fair as a receiver (46-50% returns)
3. My total earnings are \$21.9, averaging about \$5.48 per round - solid profit

Strategy considerations:

- I'm building a reputation across multiple agents
- Higher sends have been rewarded so far
- The 3x multiplier means sending more creates more total value
- I should continue signaling cooperation

I'll send \$3.5 or \$4 this round. Let me think:

- Sending \$4 creates \$12 for receiver, shows high trust
- If they return 50%, that's \$6, giving me \$7 total ($\$5 - \$4 + \$6 = \7)
- If they return 40%, that's \$4.8, giving me \$5.8 ($\$5 - \$4 + \$4.8$)

Actually, let me continue with \$3.5 since that worked well in Round 3. It's a solid cooperative amount without being overly risky.

usage:

```
{
  "input_tokens": 1851,
  "output_tokens": 596,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.36 to you. Your visible payoff was \$7.86.

Your total visible earnings across all rounds are \$29.759999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

****Analysis:****

My co-player sent \$2.27 (about 45% of their endowment), showing moderate trust.

****Current situation:****

- I received \$6.81
- Sender has \$2.73 remaining

****My receiver pattern:****

- Round 2: Returned 46% of received
- Round 4: Returned 50% of received

****Decision:****

I'll maintain fair reciprocity. Returning about 48% of what I received:

If I return \$3.25:

- Sender ends with: $\$2.73 + \$3.25 = \$5.98$ (small profit)
- I keep: $\$6.81 - \$3.25 = \$3.56$

This rewards their moderate trust fairly.

```
'''json
{"return": 3.25}
'''
```

reasoning:

****Analysis:****

My co-player sent \$2.27 (about 45% of their \$5 endowment), showing moderate trust.

****Previous patterns as receiver:****

- Round 2: Received \$7.68, returned \$3.5 (46% of received)
- Round 4: Received \$8.7, returned \$4.35 (50% of received)

****Current situation:****

- I received \$6.81
- Sender has \$2.73 remaining

****Decision:****

My co-player showed moderate trust (45% sent). I should reciprocate fairly. They sent less than in previous rounds, so I'll return proportionally.

Returning around 48% seems fair:

- 48% of \$6.81 = \$3.27

If I return \$3.27:

- Sender ends with: $\$2.73 + \$3.27 = \$6.00$ (profit of \$1.00)
- I keep: $\$6.81 - \$3.27 = \$3.54$ (profit of \$3.54)

This is fair reciprocity that rewards their trust while keeping a reasonable share.

usage:

```
{
  "input_tokens": 2166,
  "output_tokens": 466,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.27, it became \$6.8100000000000005, and you returned \$3.25. Your visible payoff was \$3.5600000000000005.

Your total visible earnings across all rounds are \$33.32.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

****Analysis:****

This is interesting - I'm playing against the same co-player from Round 6!

****Round 6 recap:****

- They sent me \$2.27 (45% of endowment)
- I returned \$3.25
- They ended with a small profit

****Decision:****

Since they showed moderate trust in Round 6 and I reciprocated fairly, I should continue that cooperative relationship. My successful strategy has been sending \$3.5:

- Creates \$10.50 for them
- Has historically yielded good returns (58-61%)
- Signals strong reciprocal cooperation

I'll send \$3.5 to maintain our cooperative dynamic.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Interesting! This round I'm the SENDER and it says I'm playing against my "current co-player" - which is the same person from Round 6 where I was the receiver.

Let me review:

- Round 6: They sent me \$2.27, I received \$6.81, I returned \$3.25
- So they ended Round 6 with: $\$5 - \$2.27 + \$3.25 = \5.98

Now in Round 7, I'm the sender playing against the same person.

****Analysis:****

- This co-player sent me a moderate amount (\$2.27, 45%)
- I returned \$3.25, which gave them a small profit
- They showed reasonable trust

****My sender history:****

- Round 1: Sent \$3 ? got back \$3.8 (42% return rate)

- Round 3: Sent \$3.5 ? got back \$6.07 (58% return rate)
- Round 5: Sent \$3.5 ? got back \$6.36 (61% return rate)

The pattern shows that sending \$3.5 has been working well. I've been building positive relationships.

Since this is the same co-player who showed moderate trust, I should reciprocate similarly. They sent 45% of their endowment to me, so I could send around the same proportion or slightly more to reward their cooperation.

Sending \$3.5 again seems appropriate - it's been my successful strategy and matches my cooperative pattern.

```
'''json
{"send": 3.5}
'''
```

usage:

```
{
  "input_tokens": 2489,
  "output_tokens": 538,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$2.79 to you. Your visible payoff was \$4.29.

Your total visible earnings across all rounds are \$37.61.

This round, you are the RECEIVER against your current co-player. They sent you \$2.68, so you received \$8.040000000000001. How much do you return? (0-8.040000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

****Analysis:****

My co-player sent \$2.68 (about 54% of their endowment), showing good trust.

****Current situation:****

- I received \$8.04
- Sender has \$2.32 remaining

****My receiver track record:****

- Consistently returned 46-50% of received amounts
- This has built a reputation for fair play

****Decision:****

I'll maintain my fair reciprocity pattern. Returning about 48% of what I received:

If I return \$3.85:

- Sender ends with: \$2.32 + \$3.85 = \$6.17 (profit of \$1.17)
- I keep: \$8.04 - \$3.85 = \$4.19

This rewards their trust appropriately.

```
'''json
{"return": 3.85}
'''
```

Reasoning:

Let me analyze this situation:

****Current situation:****

- I'm the receiver
- Co-player sent \$2.68 (about 54% of endowment) - showing good trust
- I received \$8.04
- Sender has \$2.32 remaining

****Previous rounds summary:****

- Round 2 receiver: Returned 46% (\$3.5 from \$7.68)

- Round 4 receiver: Returned 50% (\$4.35 from \$8.7)
- Round 6 receiver: Returned 48% (\$3.25 from \$6.81)

I've been consistently fair as a receiver, returning around 46-50% of what I receive.

****Round 7 note:****

My co-player from Round 6-7 only returned \$2.79 out of \$10.5 in Round 7 (about 27%). That was disappointing - much lower than the returns I've been getting.

But this is Round 8 with a potentially different co-player (unless it's the same one). Actually, looking at the multi-agent setting, pairings are randomized each round, so this could be anyone.

The sender sent \$2.68 (54% of endowment), showing decent trust. I should reciprocate fairly.

If I return around 48% of \$8.04 = \$3.86:

- Sender ends with: \$2.32 + \$3.86 = \$6.18 (profit of \$1.18)
- I keep: \$8.04 - \$3.86 = \$4.18

This maintains my reputation as a fair receiver.

Let me go with \$3.85 or \$3.9. I'll say \$3.85.

```
Usage: {
  "input_tokens": 2601,
  "output_tokens": 607,
  "reasoning_tokens": 0
}
```

ROUND 9

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_3=Receiver
 dyad_2: Agent_7=Sender, Agent_2=Receiver
 dyad_3: Agent_4=Sender, Agent_8=Receiver
 dyad_4: Agent_6=Sender, Agent_5=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_3 returned \$4.25; payoffs Agent_1=\$6.25, Agent_3=\$4.75; noise/visible: sent visible=\$3.42, received visible=\$10.26, returned visible=\$5.01
 Action validation: {"returned": {"amount": 4.25, "clamped": false, "max": 9.0, "min": 0, "raw": 4.25}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_2: Agent_7 sent \$3; received \$9; Agent_2 returned \$4.75; payoffs Agent_7=\$6.75, Agent_2=\$4.25; noise/visible: sent visible=\$3.52, received visible=\$10.56, returned visible=\$4.94
 Action validation: {"returned": {"amount": 4.75, "clamped": false, "max": 9.0, "min": 0, "raw": 4.75}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_3: Agent_4 sent \$3.5; received \$10.5; Agent_8 returned \$3.9; payoffs Agent_4=\$5.4, Agent_8=\$6.6; noise/visible: sent visible=\$2.64, received visible=\$7.92, returned visible=\$3.33
 Action validation: {"returned": {"amount": 3.9, "clamped": false, "max": 10.5, "min": 0, "raw": 3.9}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
 dyad_4: Agent_6 sent \$3; received \$9; Agent_5 returned \$3.3; payoffs Agent_6=\$5.3, Agent_5=\$5.7; noise/visible: sent visible=\$2.19, received visible=\$6.57, returned visible=\$2.66
 Action validation: {"returned": {"amount": 3.3, "clamped": false, "max": 9.0, "min": 0, "raw": 3.3}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 50.0,
 "Agent_2": 47.33,
 "Agent_3": 46.2,
 "Agent_4": 52.839999999999996,
 "Agent_5": 47.180000000000001,
 "Agent_6": 50.15,
 "Agent_7": 49.85,
 "Agent_8": 50.449999999999996

}

Visible balances after round: {

"Agent_1": 51.909999999999999,
 "Agent_2": 47.85,
 "Agent_3": 55.099999999999994,
 "Agent_4": 46.629999999999995,

```

"Agent_5": 47.36,
"Agent_6": 46.480000000000004,
"Agent_7": 50.84,
"Agent_8": 53.259999999999999
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:27:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_3",

"pairing": {

"agents": [

"Agent_1",

"Agent_3"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_3": "trustee"

},

"trustee": "Agent_3"

},

"role": "investor",

"role_label": "Sender",

"round": 9,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.9, it became \$8.7, and you returned \$4.5. Your visible payoff was \$4.199999999999999.

Your total visible earnings across all rounds are \$10.7.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

****Analysis:****

- My co-player sent \$2.27 (about 45% of their endowment), showing moderate trust
- I received \$6.81 after the multiplier
- In previous rounds, I've been returning approximately 50% of what I receive as a reciprocity norm
- This approach has worked well so far

****Decision:****

I'll maintain consistency and return about 50% of what I received, which rewards their trust appropriately.

```
'''json
{"return": 3.4}
'''
```

reasoning:

Let me analyze this situation:

****Current situation:****

- I'm the RECEIVER again
- My co-player sent \$2.27 (about 45% of their \$5 endowment)
- I received \$6.81 after the 3x multiplier
- They kept \$5 - \$2.27 = \$2.73

****Past behavior:****

- Round 1: I sent \$3, received back \$4.5 (50% return rate on the multiplied amount)
- Round 2: I received \$8.7, returned \$4.5 (about 52% return rate)

****Considerations:****

- My co-player sent a moderate amount (\$2.27), showing some trust but less than the previous player
- Being consistent with my previous behavior would mean returning around 50% of what I received
- 50% of \$6.81 = \$3.405

****Payoff implications:****

- If I return \$3.4-3.5:
 - I keep: \$6.81 - \$3.4 = \$3.41
 - They get: \$2.73 (kept) + \$3.4 (returned) = \$6.13
 - This seems fair - we both do better than the \$5 starting point

I should maintain consistency with my reciprocity pattern from Round 2. Returning about 50% seems fair and builds trust for potential future rounds.

Let me return \$3.4 (just under 50%).

```

usage:
{
  "input_tokens": 1105,
  "output_tokens": 467,
  "reasoning_tokens": 0
}
Message 4 role=user
content:
  Round 4

  In your most recent previous game round, you were the RECEIVER against another agent. You
  saw them send $2.27, it became $6.8100000000000005, and you returned $3.4. Your visible
  payoff was $3.4100000000000006.
  Your total visible earnings across all rounds are $14.11.

  This round, you are the SENDER against your current co-player. You have $5. How much do yo
  u send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Message 5 role=assistant
content:
  **Analysis:**
  - This is Round 4 with a new co-player
  - In Round 1, I sent $3 and received $4.5 back - a positive outcome
  - As a receiver, I've been returning about 50%, establishing a cooperative pattern
  - My current co-player is unknown to me

  **Decision:**
  I'll send $3 again. This amount has worked well before - it shows trust and creates potent
  ial for mutual gain through the 3x multiplier, while managing downside risk.

  '''json
  {"send": 3}
  '''
reasoning:
  Let me analyze the situation:

  **What I know:**
  - Round 1: I was SENDER, sent $3, received back $4.5, payoff $6.5
  - Round 2: I was RECEIVER, got $8.7 (from $2.9 sent), returned $4.5, payoff $4.2
  - Round 3: I was RECEIVER, got $6.81 (from $2.27 sent), returned $3.4, payoff $3.41
  - Total earnings: $14.11

  **Current situation:**
  - Round 4: I'm the SENDER with $5
  - This is a new co-player (different from rounds 1-3)

  **Strategic considerations:**
  1. I've seen one positive interaction as sender (sent $3, got back $4.5)
  2. As receiver, I've been fairly generous, returning about 50%
  3. This creates a reputation for reciprocity if others know
  4. But agents don't know who I am (names hidden), so reputation effects are limited
  5. However, the pattern of behavior in the pool might matter

  **Decision factors:**
  - Sending creates potential gains (3x multiplier)
  - Risk: this is a new co-player, no direct history
  - But overall, cooperation seems to be working in this pool
  - My Round 1 strategy (sending $3) worked well

  I'll send $3 again - it's a moderate trust signal that has proven effective.
usage:
{
  "input_tokens": 1352,
  "output_tokens": 472,
  "reasoning_tokens": 0
}
Message 6 role=user
content:
  Round 5

```

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.48 to you. Your visible payoff was \$6.48.

Your total visible earnings across all rounds are \$20.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

****Analysis:****

- Round 5 with a new co-player
- My \$3 sending strategy has been successful:
 - Round 1: Sent \$3, received \$4.5 back ? \$6.5 payoff
 - Round 4: Sent \$3, received \$4.48 back ? \$6.48 payoff
- Both returns were close to 50%, indicating fair reciprocity from my co-players
- This amount balances trust-building with risk management

****Decision:****

I'll continue with \$3, as it's been consistently effective.

```
'''json
{"send": 3}
'''
```

reasoning:

****Analysis:****

- Round 5, new co-player as sender
- My sending strategy of \$3 has worked well:
 - Round 1: Sent \$3, received \$4.5 back, payoff \$6.5
 - Round 4: Sent \$3, received \$4.48 back, payoff \$6.48
- Both times I received close to 50% back, which is fair reciprocity
- Total visible earnings: \$20.59 across 4 rounds (average \$5.15/round)
- As a receiver, I've been consistently returning about 50%

****Decision:****

The \$3 sending strategy is working consistently well. My co-players have been reciprocating fairly. I should continue this approach.

```
'''json
{"send": 3}
'''
```

usage:

```
{
  "input_tokens": 1601,
  "output_tokens": 339,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.77 to you. Your visible payoff was \$8.77.

Your total visible earnings across all rounds are \$29.36.

This round, you are the RECEIVER against your current co-player. They sent you \$3.68, so you received \$11.04000000000001. How much do you return? (0-11.04000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

****Analysis:****

- My co-player sent \$3.68 (about 74% of their endowment), showing strong trust
- I received \$11.04 after the multiplier
- They're taking a significant risk by sending such a high amount
- I've been maintaining a 50% return pattern, which has worked well

****Decision:****

I'll return approximately 50% of what I received to reward their trust appropriately.

```
'''json
{"return": 5.5}
'''
```

reasoning:

****Analysis:****

- My co-player sent \$3.68 (about 74% of their endowment), showing strong trust
- I received \$11.04 after the 3x multiplier
- In Round 5, my co-player returned \$6.77 from \$9, which is about 75% - very generous!
- I've been returning about 50% as a norm, but this co-player sent a substantial amount

****Decision:****

My co-player showed considerable trust by sending \$3.68. I should reciprocate fairly. Following my 50% return pattern would give them about \$5.50 back.

Let me calculate:

- They sent \$3.68, keeping \$1.32
- I received \$11.04
- If I return \$5.50:
 - They get: $\$1.32 + \$5.50 = \$6.82$ (better than their \$5 starting)
 - I get: $\$11.04 - \$5.50 = \$5.54$ (better than my \$0 starting as receiver)

This seems fair and maintains my reciprocity norm.

usage:

```
{
  "input_tokens": 1890,
  "output_tokens": 379,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.68, it became \$11.040000000000001, and you returned \$5.5. Your visible payoff was \$5.540000000000001.

Your total visible earnings across all rounds are \$34.9.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

****Analysis:****

- My co-player sent \$3.23 (about 65% of their endowment), showing good trust
- I received \$9.69 after the multiplier
- I've been consistently returning approximately 50% of what I receive
- This approach has maintained positive cooperative relationships

****Decision:****

I'll return about 50% of what I received.

```
'''json
{"return": 4.85}
'''
```

reasoning:

****Analysis:****

- My co-player sent \$3.23 (about 65% of their endowment), showing good trust
- I received \$9.69 after the multiplier
- I've been maintaining a consistent 50% return pattern throughout the game
- This has been working well and seems fair

****Decision:****

I'll continue with my 50% return pattern: $0.5 \times \$9.69 = \4.845

Let me round to a reasonable number: \$4.85

usage:

```
{
  "input_tokens": 2142,
  "output_tokens": 221,
  "reasoning_tokens": 0
}
```

}

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.23, it became \$9.69, and you returned \$4.85. Your visible payoff was \$4.84.

Your total visible earnings across all rounds are \$39.739999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

****Analysis:****

- Round 8 with a new co-player
- My sending strategy of \$3 has been consistently successful:
 - Round 1: \$6.5 payoff
 - Round 4: \$6.48 payoff
 - Round 5: \$8.77 payoff
- Average return has been favorable, with partners generally reciprocating fairly
- This amount continues to balance trust-building with prudent risk management

****Decision:****

I'll send \$3 again, as it's proven to be an effective cooperative strategy.

'''json

{"send": 3}

'''

reasoning:

Let me analyze my performance as a sender so far:

Round 1: Sent \$3, received \$4.5 back ? payoff \$6.5

Round 4: Sent \$3, received \$4.48 back ? payoff \$6.48

Round 5: Sent \$3, received \$6.77 back ? payoff \$8.77

All three times I sent \$3, I got positive returns. The returns varied:

- Round 1: 50% return rate (4.5/9)
- Round 4: ~50% return rate (4.48/9)
- Round 5: ~75% return rate (6.77/9)

Overall, sending \$3 has been very successful for me. I'm averaging well above my \$5 starting amount when I send \$3.

As a receiver, I've been consistently returning about 50% of what I receive, which seems fair and has maintained good cooperation.

This is Round 8, a new co-player. I should stick with my successful strategy of sending \$3

usage:

```
{
  "input_tokens": 2158,
  "output_tokens": 387,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.16 to you. Your visible payoff was \$5.16.

Your total visible earnings across all rounds are \$44.89999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

****Analysis:****

- Round 9 with a new co-player
- My \$3 sending strategy has been generally successful:
 - Round 1: \$6.5 payoff
 - Round 4: \$6.48 payoff
 - Round 5: \$8.77 payoff
 - Round 8: \$5.16 payoff (lower but still profitable)
- Even with variation in returns, this amount has consistently yielded positive outcomes
- Total earnings are strong at \$44.90

****Decision:****

I'll continue with \$3, maintaining the cooperative approach that has worked well overall.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze the situation:

Round 9 - I'm the SENDER again

Recent history as sender:

- Round 1: Sent \$3, received \$4.5 back ? \$6.5 payoff
- Round 4: Sent \$3, received \$4.48 back ? \$6.48 payoff
- Round 5: Sent \$3, received \$6.77 back ? \$8.77 payoff
- Round 8: Sent \$3, received \$3.16 back ? \$5.16 payoff

The last round (Round 8) showed a lower return (\$3.16 on \$9 received), but still positive. The \$3 sending strategy has generally worked well, with most returns being fair or generous.

As receiver, I've been returning about 50% consistently, which has maintained good cooperative dynamics.

Overall, my strategy of sending \$3 has been successful. Even in the worst case (Round 8), I still made a profit (\$5.16 > \$5 initial endowment).

I should continue with \$3 as it's been a robust strategy across different co-players.

```
Usage: {
  "input_tokens": 2140,
  "output_tokens": 423,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Receiver - dyad_1

Timestamp: 2026-08-12T09:27:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_3"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_3": "trustee"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.48 to you. Your visible payoff was \$5.48.

Your total visible earnings across all rounds are \$10.84.

This round, you are the RECEIVER against your current co-player. They sent you \$2.97, so you received \$8.91. How much do you return? (0-8.91)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

****Current round:****

- Sender showed good trust by sending \$2.97 (59.4% of their endowment)
- This created \$8.91 for me
- Sender currently has \$2.03

****Previous round context:****

- I sent \$3.00 (60% - very similar trust level)
- My co-player returned \$3.48, which was fair

I should reciprocate the trust shown to me. The sender sent almost 60% of their endowment, similar to what I did last round.

I'll return \$3.75, which:

- Gives the sender \$5.78 total (a gain of \$0.78)
- Leaves me with \$5.16 (also a gain)
- Rewards their trust appropriately

{'return': 3.75}

reasoning:

Let me analyze this situation:

****Last round (Round 2):****

- I was the SENDER
- I sent \$3.00 (60% of my endowment)
- It became \$9.00 for the receiver
- They returned \$3.48
- My payoff: $\$5 - \$3 + \$3.48 = \5.48
- Their payoff: $\$9 - \$3.48 = \$5.52$

The receiver gave me back slightly more than I sent, which was cooperative and fair.

****This round (Round 3):****

- I'm the RECEIVER
- Sender sent \$2.97 (59.4% of their endowment - very similar to what I sent last round!)
- I received \$8.91
- Sender currently has: $\$5 - \$2.97 = \$2.03$

The sender showed good trust by sending ~60% of their endowment. This is very similar to the 60% I sent in the previous round.

I should reciprocate fairly. Let me think about what would be a fair return:

- If I return \$3.50, sender gets $\$2.03 + \$3.50 = \$5.53$, and I keep $\$8.91 - \$3.50 = \$5.41$
- If I return \$4.00, sender gets $\$2.03 + \$4.00 = \$6.03$, and I keep $\$8.91 - \$4.00 = \$4.91$
- If I return \$3.75, sender gets $\$2.03 + \$3.75 = \$5.78$, and I keep $\$8.91 - \$3.75 = \$5.16$

The sender trusted me with \$2.97, similar to how I trusted my previous co-player with \$3.00, and they returned \$3.48.

...

usage:

```
{
  "input_tokens": 1089,
  "output_tokens": 735,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.97, it became \$8.91, and you returned \$3.75. Your visible payoff was \$5.16.

Your total visible earnings across all rounds are \$16.0.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Last round with THIS SAME co-player:****

- They sent me \$2.97 (59.4% of their endowment) - showing good trust
- I returned \$3.75 - rewarding their trust fairly
- We both gained from cooperation

****Now it's my turn to send:****

Since we just had a successful cooperative round together, I should maintain this pattern.

They trusted me, I reciprocated, and now I should show similar trust.

I'll send \$3.00 (60% of my endowment), matching the level of trust they showed me.

```
{'send': 3.00}
```

reasoning:

Let me analyze this situation:

****Last round with THIS SAME co-player:****

- They sent me \$2.97 (59.4% of their endowment)
- I received \$8.91
- I returned \$3.75
- They got \$5.78 total (gain of \$0.78)
- I got \$5.16 (gain of \$0.16)

****Now I'm the sender with THE SAME co-player:****

- They showed good trust last round (sent ~60%)
- I reciprocated fairly
- We both gained from cooperation

Since we're playing with the same person and we had a successful cooperative round, I should continue this pattern. They trusted me with ~60% of their endowment, and I rewarded that trust. Now I should show similar trust.

I'll send around \$3.00 again, maintaining the cooperative pattern.

usage:

```
{
  "input_tokens": 1406,
  "output_tokens": 354,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.73 to you. Your visible payoff was \$7.73.

Your total visible earnings across all rounds are \$23.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

****Last round:****

- I sent \$3.00 (60% of my endowment)
- My co-player returned \$5.73 - very generous!
- This gave me \$7.73, an excellent outcome

****Pattern so far:****

- Sending around 60% has consistently led to good reciprocation
- My average earnings are strong at \$4.75 per round
- Cooperative approach is working well

****Current round with NEW co-player:****

I'll continue the successful cooperative strategy. Sending a substantial amount demonstrates trust and has been rewarded by others.

```
{'send': 3.00}
```

reasoning:

Let me analyze this situation:

****Last round:****

- I sent \$3.00 (60% of my endowment)
- They returned \$5.73
- My payoff was \$7.73 (very good!)

This was an excellent return - they gave me back more than I would have had if I kept everything (\$5). This suggests my co-player is very cooperative and generous.

****Current round:****

- I'm the sender again with a NEW co-player
- I have \$5 to decide how much to send

Looking at the pattern:

- Round 1 (receiver): Saw sender send 62.4%, I returned fairly
- Round 2 (sender): Sent \$3.00 (60%)
- Round 3 (receiver): Saw sender send 59.4%, I returned fairly
- Round 4 (sender): Sent \$3.00 (60%) to the same co-player from round 3
- Round 5 (sender): Last round's co-player was very generous, returned \$5.73

The cooperative strategy has been working well. I've been consistently sending around 60% and have seen good reciprocation. My total earnings of \$23.73 over 5 rounds (averaging \$4.746 per round) suggests the cooperative approach is profitable.

For this round with a NEW co-player, I should continue the cooperative strategy that's been working. I'll send around \$3.00 again (60% of my endowment).

usage:

```
{
  "input_tokens": 1674,
  "output_tokens": 493,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.45 to you. Your visible payoff was \$6.45.

Your total visible earnings across all rounds are \$30.18.

This round, you are the RECEIVER against your current co-player. They sent you \$3.24, so you received \$9.72. How much do you return? (0-9.72)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

****Current round:****

- Sender showed strong trust by sending \$3.24 (64.8% of their endowment)
- This created \$9.72 for me
- Sender currently has \$1.76

****My pattern as receiver:****

- Round 1: Received \$9.36, returned \$4.00 (rewarded good trust)
- Round 3: Received \$8.91, returned \$3.75 (rewarded good trust)

This sender sent even more (64.8%) than I typically send (~60%), showing exceptional trust. I should reward this appropriately.

I'll return \$4.00, which:

- Gives the sender \$5.76 total (gain of \$0.76)
- Leaves me with \$5.72 (also a good gain)
- Rewards their high trust level
- Maintains my consistent reciprocation pattern

{ 'return': 4.00 }

reasoning:

Let me analyze this situation:

****Current round:****

- Sender sent me \$3.24 (64.8% of their endowment) - showing strong trust
- This created \$9.72 for me
- Sender currently has \$5 - \$3.24 = \$1.76

****Previous rounds pattern:****

- Round 1: As receiver, got \$9.36, returned \$4.00 (42.7% of what I received)
- Round 2: As sender, sent \$3.00, got back \$3.48
- Round 3: As receiver, got \$8.91, returned \$3.75 (42.1% of what I received)
- Round 4: As sender, sent \$3.00 (to same co-player from Round 3)
- Round 5: As sender, sent \$3.00, got back \$5.73 (very generous)
- Round 6: As sender, sent \$3.00, got back \$4.45

So when I've been receiver:

- Round 1: returned 42.7% of received
- Round 3: returned 42.1% of received

This sender sent \$3.24 (64.8%), which is higher trust than what I typically send (60%). I should reward this high trust appropriately.

If I return around 42% of \$9.72, that would be about \$4.08.

But let me think about fairness:

- Sender started with \$5, sent \$3.24, has \$1.76 left
- If I return \$4.08, sender gets \$5.84 total (gain of \$0.84)
- I keep \$9.72 - \$4.08 = \$5.64 (gain of \$0.64)

This seems reasonable. The sender gets a decent return, and we both benefit.

Actually, let me be a bit more generous since they sent 64.8%, higher than typical. I could return \$4.25:

...

usage:

```
{
  "input_tokens": 1956,
  "output_tokens": 799,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.24, it became \$9.72, and you returned \$4.0. Your visible payoff was \$5.720000000000001.

Your total visible earnings across all rounds are \$35.9.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

****My track record as SENDER:****

- Consistently sent \$3.00 (60%)
- Returns have ranged from \$3.48 to \$5.73
- This strategy has yielded strong payoffs (\$5.48 to \$7.73)

****Overall performance:****

- Average earnings of ~\$5.98 per round
- Cooperative approach has been working well
- Both sending and receiving trust has been rewarded

****Current round with NEW co-player:****

I'll maintain my proven strategy of sending a substantial amount to encourage cooperation.

```
{'send': 3.00}
```

reasoning:

Looking at my history:

****As SENDER:****

- Round 2: Sent \$3.00, received back \$3.48 ? payoff \$5.48
- Round 4: Sent \$3.00, received back \$5.73 ? payoff \$7.73
- Round 5: Sent \$3.00, received back \$4.45 ? payoff \$6.45

****As RECEIVER:****

- Round 1: Received \$9.36, returned \$4.00 ? payoff \$5.36
- Round 3: Received \$8.91, returned \$3.75 ? payoff \$5.16
- Round 6: Received \$9.72, returned \$4.00 ? payoff \$5.72

My strategy of sending ~\$3.00 (60%) has been very successful:

- Average return when I send \$3.00: $(\$3.48 + \$5.73 + \$4.45) / 3 = \4.55
- This gives me an average payoff of \$6.55 when I'm sender

When I receive, I return fairly and get around \$5.4 on average.

****Current situation:****

- New co-player this round
- I'm the sender
- My cooperative strategy has been working very well (average earnings of \$5.98 per round)

I should continue with the successful strategy of sending \$3.00 (60%).

usage:

```
{
  "input_tokens": 2298,
  "output_tokens": 479,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$4.22 to you. Your visible payoff was \$6.22.

Your total visible earnings across all rounds are \$42.12.

This round, you are the RECEIVER against your current co-player. They sent you \$3.99, so you received \$11.97. How much do you return? (0-11.97)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

****Last round with THIS SAME co-player:****

- I sent them \$3.00 (60%)
- They returned \$4.22 - good reciprocation

****Current round:****

- They sent me \$3.99 (79.8% of their endowment) - EXCEPTIONAL trust!
- This created \$11.97 for me
- They currently have \$1.01

This is the highest trust level anyone has shown me in this game. They sent almost 80% of their endowment after we already had a successful cooperative round. This deserves strong reciprocation.

I'll return \$5.00, which:

- Gives them \$6.01 total (gain of \$1.01)
- Leaves me with \$6.97 (gain of \$1.97)
- Rewards their exceptional trust appropriately
- Maintains cooperative relationship

{'return': 5.00}

reasoning:

Looking at this situation:

****Last round with THIS SAME co-player:****

- I sent them \$3.00 (60% of my endowment)
- They returned \$4.22 - good reciprocation
- We had a successful cooperative round

****Current round:****

- They sent me \$3.99 (79.8% of their endowment) - VERY high trust!
- This created \$11.97 for me

- They currently have \$1.01

This is exceptional trust - they sent almost 80% of their endowment! This is significantly higher than:

- What I typically send (~60%)
- What they received from me last round (\$3.00)

They're showing me even more trust than I showed them. I should reward this generously.

Let me think about what's fair:

- They sent \$3.99, keeping \$1.01
- I received \$11.97
- If I return around \$5.00:
 - They'd have \$6.01 total (gain of \$1.01)
 - I'd have \$6.97 (also a good gain)

Actually, given they sent almost 80%, maybe I should return a bit more to really reward this exceptional trust.

If I return \$5.50:

- They'd have \$6.51 total (gain of \$1.51)
- I'd have \$6.47 (gain of \$1.47)
- More balanced rewards

Or if I return \$5.00:

- They'd have \$6.01 total (gain of \$1.01)
- I'd have \$6.97 (gain of \$1.97)

Let me be generous and go with \$5.00, which is a strong return and rewards their exceptional trust appropriately.

...

usage:

```
{
  "input_tokens": 2349,
  "output_tokens": 767,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.99, it became \$11.97, and you returned \$5.0. Your visible payoff was \$6.970000000000001.

Your total visible earnings across all rounds are \$49.089999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.42, so you received \$10.26. How much do you return? (0-10.26)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****Current round:****

- Sender sent \$3.42 (68.4% of their endowment) - strong trust
- This created \$10.26 for me
- Sender currently has \$1.58

****My pattern as receiver:****

- Round 1: Received \$9.36, returned \$4.00
- Round 3: Received \$8.91, returned \$3.75
- Round 6: Received \$9.72, returned \$4.00
- Round 8: Received \$11.97 (exceptional trust), returned \$5.00

This sender showed high trust (68.4%), above my typical 60% sends. I should reward this appropriately.

I'll return \$4.25, which:

- Gives them \$5.83 total (gain of \$0.83)
- Leaves me with \$6.01 (gain of \$1.01)
- Rewards their strong trust level fairly


```
{'return': 4.25}
```

Reasoning:

Looking at this situation:

****Current round:****

- Sender sent \$3.42 (68.4% of their endowment) - very high trust
- This created \$10.26 for me
- Sender currently has \$1.58

****My pattern as receiver:****

- Round 1: Received \$9.36, returned \$4.00
- Round 3: Received \$8.91, returned \$3.75
- Round 6: Received \$9.72, returned \$4.00
- Round 8: Received \$11.97, returned \$5.00

The sender showed strong trust (68.4%), significantly above the baseline 60%. This is between the "good trust" I saw in earlier rounds (60-65%) and the exceptional trust in round 8 (80%)

I should return an amount that:

- Rewards their high trust
- Gives them a reasonable gain
- Maintains my pattern of fair reciprocation

Returning \$4.25:

- Gives sender \$5.83 total (gain of \$0.83)
- Leaves me with \$6.01 (gain of \$1.01)
- Fair split of the multiplied gains

This seems appropriate for their high (but not exceptional) trust level.

Usage: {

```
"input_tokens": 2445,  
"output_tokens": 529,  
"reasoning_tokens": 0
```

}

GAME - Agent_7 - Sender - dyad_2

Timestamp: 2026-08-12T09:27:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_2",  
"move_index": 2,  
"opponent_id": "Agent_2",  
"pairing": {  
  "agents": [  
    "Agent_7",  
    "Agent_2"  
  ],  
  "investor": "Agent_7",  
  "roles": {  
    "Agent_2": "trustee",  
    "Agent_7": "investor"  
  },  
  "trustee": "Agent_2"  
},  
"role": "investor",  
"role_label": "Sender",  
"round": 9,  
"task": "game",  
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.48, it became \$7.439999999999995, and you returned \$4.0. Your visible payoff was \$3.439999999999995.

Your total visible earnings across all rounds are \$10.239999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this round:

****Analysis:****

- My co-player sent \$3.02 out of \$5 (about 60%), showing strong trust
- This became \$9.06 for me
- They're trusting me with more than the previous sender

****My decision:****

I want to reward this higher level of trust and maintain the cooperative pattern I've established (returning ~50-55% of what I receive).

If I return \$5.00:

- Sender gets: \$1.98 + \$5.00 = \$6.98
- I get: \$9.06 - \$5.00 = \$4.06
- Both significantly better off than the initial \$5

This return (about 55% of what I received) is consistent with my previous reciprocity and rewards their trust appropriately.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Let me analyze this round:

****What happened in Round 3:****

- I'm the RECEIVER again
- My co-player sent \$3.02 out of \$5 (about 60%), showing good trust
- This became \$9.06 for me

****My history:****

- Round 1: I was SENDER, sent \$3, received \$4.80 back (about 53% of the \$9 they received)
- Round 2: I was RECEIVER, received \$7.44, returned \$4.00 (about 54% of what I received)

****Decision for Round 3:****

My co-player sent a higher percentage this time (\$3.02 vs \$2.48 last round from a different player), showing increased trust. I should:

- Reward this trust
- Maintain reciprocity around 50-55% return rate
- Ensure both of us benefit

If I return \$5.00:

- Sender gets: $\$1.98 + \$5.00 = \$6.98$
- I get: $\$9.06 - \$5.00 = \$4.06$
- Both significantly better off than \$5

This is about 55% of what I received, consistent with my previous behavior and fair reciprocity.

Alternatively, if I return \$4.50:

- Sender gets: $\$1.98 + \$4.50 = \$6.48$
- I get: $\$9.06 - \$4.50 = \$4.56$
- Still both better off

Let me go with \$5.00 to be generous and reward the higher trust shown.

usage:

```
{
  "input_tokens": 1172,
  "output_tokens": 579,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.02, it became \$9.06, and you returned \$5.0. Your visible payoff was \$4.0600000000000005.

Your total visible earnings across all rounds are \$14.299999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this round as SENDER:

****My experience so far:****

- Round 1: Sent \$3 ? received \$4.8 back (53% return rate) ? \$6.8 payoff
- Rounds 2-3: As receiver, I returned 54-55% of what I received
- Cooperation has been working well

****My decision:****

Since this is a new co-player, I don't know their specific behavior, but the general pattern in this game seems cooperative. Sending \$3 worked well in Round 1, balancing trust with

th prudence.

If I send \$3:

- It becomes \$9 for receiver
- If they return around 50%, I get ~\$4.5 back
- Net: ~\$6.5 total, better than keeping all \$5

I'll send **\$3** again - it's proven effective and signals willingness to cooperate.

```
'''json
{"send": 3}
'''
```

reasoning:

I'm now the SENDER again. Let me review my experience:

Round 1: I was SENDER, sent \$3, received back \$4.8, payoff = \$6.8

Round 2: I was RECEIVER, received \$7.44 (from \$2.48 sent), returned \$4.0, payoff = \$3.44

Round 3: I was RECEIVER, received \$9.06 (from \$3.02 sent), returned \$5.0, payoff = \$4.06

Now Round 4: I'm SENDER again

My experience shows:

- When I sent \$3 in Round 1, I got back \$4.8 (53% of the \$9 the receiver got)
- I've been a generous receiver, returning 54% and 55% of what I received
- Cooperation seems to be working well

This is a new co-player, so I don't know their behavior yet. However, the game seems to have a cooperative culture developing.

I should send a moderate-to-high amount to signal trust and cooperation. Sending \$3 worked well before. I could stick with that or increase slightly given the positive experiences.

I'll send \$3 again - it's a proven strategy that balances risk and reward.

usage:

```
{
  "input_tokens": 1500,
  "output_tokens": 515,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.36 to you. Your visible payoff was \$6.36.

Your total visible earnings across all rounds are \$20.66.

This round, you are the RECEIVER against your current co-player. They sent you \$4.17, so you received \$12.51. How much do you return? (0-12.51)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this round:

****Analysis:****

- My co-player sent \$4.17 out of \$5 (about 83%), showing exceptional trust
- This became \$12.51 for me
- This is the highest trust shown by any sender I've faced

****My reciprocity pattern:****

- I've been returning around 50-55% of what I receive
- This has maintained good cooperative relationships

****My decision:****

This level of trust deserves strong reciprocation.

If I return \$7.00:

- Sender gets: \$0.83 + \$7.00 = \$7.83

- I get: $\$12.51 - \$7.00 = \$5.51$
- Both significantly better off than the initial \$5

This return (about 56% of what I received) maintains my cooperative pattern while appropriately rewarding their exceptional trust.

```
'''json
{"return": 7.0}
'''
```

reasoning:

Looking at this round:

****Analysis:****

- My co-player sent \$4.17 out of \$5 (about 83%), showing very high trust
- This became \$12.51 for me
- This is the highest amount anyone has sent me so far

****My past behavior as receiver:****

- Round 2: Received \$7.44, returned \$4.00 (54%)
- Round 3: Received \$9.06, returned \$5.00 (55%)

****My past experience as sender:****

- Round 1: Sent \$3, received back \$4.8 (53% return)
- Round 4: Sent \$3, received back \$4.36 (48% return)

****My decision:****

This co-player showed exceptional trust by sending \$4.17. I should reward this trust generously. If I continue my pattern of returning ~50-55%, that would be \$6.25-\$6.88.

Let me calculate:

If I return \$7.00:

- Sender gets: $\$0.83 + \$7.00 = \$7.83$
- I get: $\$12.51 - \$7.00 = \$5.51$
- Both significantly better off

If I return \$6.50:

- Sender gets: $\$0.83 + \$6.50 = \$7.33$
- I get: $\$12.51 - \$6.50 = \$6.01$
- Both better off

Returning \$7.00 (about 56% of what I received) maintains my cooperative pattern and rewards their exceptional trust. This seems fair given they sent over 80% of their endowment.

usage:

```
{
  "input_tokens": 1854,
  "output_tokens": 605,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.17, it became \$12.51, and you returned \$7.0. Your visible payoff was \$5.51.

Your total visible earnings across all rounds are \$26.17.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at this round as SENDER:

****My sending history:****

- Round 1: Sent \$3 ? received \$4.8 back ? \$6.8 payoff
- Round 4: Sent \$3 ? received \$4.36 back ? \$6.36 payoff

****Analysis:****

- Sending \$3 has consistently worked well for me

- Both previous receivers returned about 48-53%, yielding good profits
- This is a new co-player, so I have no specific information about them
- The overall environment has been cooperative

****My decision:****

I'll stick with the proven strategy of sending ****\$3****. It:

- Shows willingness to cooperate
- Has delivered consistent positive returns
- Balances trust with prudent risk management

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

Round 1 (SENDER): Sent \$3 ? received \$4.8 back ? payoff \$6.8
 Round 2 (RECEIVER): Received \$7.44 ? returned \$4.0 ? payoff \$3.44
 Round 3 (RECEIVER): Received \$9.06 ? returned \$5.0 ? payoff \$4.06
 Round 4 (SENDER): Sent \$3 ? received \$4.36 back ? payoff \$6.36
 Round 5 (RECEIVER): Received \$12.51 ? returned \$7.0 ? payoff \$5.51

As SENDER (2 times so far):

- Round 1: Sent \$3, got back \$4.8 (return rate 53%)
- Round 4: Sent \$3, got back \$4.36 (return rate 48.4%)

Both times I sent \$3 and got positive returns. The cooperation has been working well overall.

This is a new co-player again. Should I:

1. Stick with \$3 (proven strategy)
2. Try increasing to show more trust (like the person who sent me \$4.17)
3. Decrease to be more cautious

Given that:

- \$3 has worked well twice
- I've been a generous receiver (returning 50-55%)
- The overall environment seems cooperative
- It's midway through the game (round 6)

I think \$3 remains a good balanced choice. It's worked well and I don't have information about this specific co-player yet.

Actually, let me think about this differently. The person in Round 5 sent me \$4.17 and I returned \$7.0, so they ended up with \$7.83. That was a very successful interaction.

What if I try sending a bit more to signal stronger trust? Maybe \$3.5 or \$4?

...

usage:

```
{
  "input_tokens": 2187,
  "output_tokens": 766,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.84 to you. Your visible payoff was \$7.84.

Your total visible earnings across all rounds are \$34.010000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$2.17, so you received \$6.51. How much do you return? (0-6.51)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this round:

****Analysis:****

- My co-player sent \$2.17 out of \$5 (about 43%), showing moderate trust
- This became \$6.51 for me
- This is less than previous senders, but still shows willingness to cooperate

****My reciprocity pattern:****

- I've consistently returned 50-55% of what I receive
- This has maintained good cooperative relationships

****My decision:****

I'll maintain my fair reciprocity pattern regardless of the amount sent.

If I return \$3.50:

- Sender gets: $\$2.83 + \$3.50 = \$6.33$
- I get: $\$6.51 - \$3.50 = \$3.01$
- Both better off than the initial \$5

This return (about 54% of what I received) maintains my consistent cooperative approach.

```
'''json
{"return": 3.5}
'''
```

reasoning:

Looking at this round:

****Analysis:****

- My co-player sent \$2.17 out of \$5 (about 43%), showing moderate but lower trust
- This became \$6.51 for me
- Previous senders sent: \$2.48, \$3.02, \$4.17 - this is the lowest

****My reciprocity pattern:****

- I've been consistently returning around 50-55% of what I receive
- Round 2: Returned \$4.00 from \$7.44 (54%)
- Round 3: Returned \$5.00 from \$9.06 (55%)
- Round 5: Returned \$7.00 from \$12.51 (56%)

****My decision:****

Even though they sent less, I should maintain my cooperative pattern. If I return less proportionally, it might discourage future cooperation.

If I return \$3.50:

- Sender gets: $\$2.83 + \$3.50 = \$6.33$
- I get: $\$6.51 - \$3.50 = \$3.01$
- Both better off than initial \$5

This return (about 54% of what I received) maintains my consistent reciprocity pattern.

usage:

```
{
  "input_tokens": 2515,
  "output_tokens": 501,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.17, it became \$6.51, and you returned \$3.5. Your visible payoff was \$3.01.

Your total visible earnings across all rounds are \$37.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at this round as SENDER:

****Important context:****

- This is the SAME co-player from Round 7

- Last round they sent me \$2.17 (43% of \$5)
- I returned \$3.50 (54% of what I received), giving them a good payoff of \$6.33

****My sending history:****

- Rounds 1, 4, 6: All sent \$3, consistently received good returns (\$4.36-\$5.84)
- Average return on \$3 sent: ~160-190% profit

****Analysis:****

- My co-player has seen I'm trustworthy and cooperative
- They might reciprocate more generously this time
- Sending \$3 has been my reliable strategy

****My decision:****

I'll send ****\$3**** again, maintaining consistency and showing continued trust.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at this round as SENDER:

****Important context:****

- This is the SAME co-player from Round 7
- In Round 7, they sent me \$2.17 (43% of their \$5)
- I returned \$3.50 to them (54% of \$6.51)
- So they ended with \$6.33 and I got \$3.01

****My sending history:****

- Round 1: Sent \$3 ? received \$4.8 back ? \$6.8 payoff
- Round 4: Sent \$3 ? received \$4.36 back ? \$6.36 payoff
- Round 6: Sent \$3 ? received \$5.84 back ? \$7.84 payoff

****Analysis:****

- I've consistently sent \$3 and it's worked well
- My current co-player showed moderate trust (43%) in the previous round
- I was generous in return (54% return rate), which should signal cooperation
- They might reciprocate more generously this time, knowing I was fair

****My decision:****

I should continue with my strategy of sending \$3. It's worked consistently and signals ongoing cooperation to this co-player who has already seen I'm trustworthy.

If I send \$3:

- It becomes \$9 for receiver
- Based on my generous return last round, they might return around 50-60%
- Expected return: ~\$4.5-5.4
- Net: ~\$6.5-7.4 total

I'll send \$3 again.

usage:

```
{
  "input_tokens": 2648,
  "output_tokens": 573,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.88 to you. Your visible payoff was \$6.88.

Your total visible earnings across all rounds are \$43.900000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this round as SENDER:

****My sending history:****

- Round 1: Sent \$3 ? received \$4.8 back ? \$6.8 payoff
- Round 4: Sent \$3 ? received \$4.36 back ? \$6.36 payoff
- Round 6: Sent \$3 ? received \$5.84 back ? \$7.84 payoff
- Round 8: Sent \$3 ? received \$4.88 back ? \$6.88 payoff

****Analysis:****

- Sending \$3 has been 100% successful across all my sender rounds
- Consistent positive returns averaging \$6-\$7.84 per round
- This is a new co-player with no specific information

****My decision:****

I'll continue with my proven strategy of sending ****\$3****. It has consistently worked well and signals cooperative intent.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at this round as SENDER:

****My sending history:****

- Round 1: Sent \$3 ? received \$4.8 back ? \$6.8 payoff
- Round 4: Sent \$3 ? received \$4.36 back ? \$6.36 payoff
- Round 6: Sent \$3 ? received \$5.84 back ? \$7.84 payoff
- Round 8: Sent \$3 ? received \$4.88 back ? \$6.88 payoff

****Analysis:****

- I've consistently sent \$3 as SENDER in all 4 previous rounds
- All have yielded positive returns, with payoffs ranging from \$6.36 to \$7.84
- Average return has been around 54-65% of the \$9 they received
- This strategy has been working very well

****This round:****

- This is a new co-player (different from Round 8)
- No specific information about their behavior
- The overall environment has been cooperative

****My decision:****

I should stick with my proven strategy of sending \$3. It has:

- Consistently delivered positive returns
- Demonstrated my willingness to cooperate
- Balanced trust with risk management effectively

I'll send \$3.

```
Usage: {
  "input_tokens": 2626,
  "output_tokens": 515,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_2

Timestamp: 2026-08-12T09:28:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_2"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_2": "trustee",
      "Agent_7": "investor"
    }
  },
```

```

    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.73, it became \$5.189999999999995, and you returned \$2.4. Your visible payoff was \$2.7899999999999996.

Your total visible earnings across all rounds are \$9.2.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my situation:

****Previous rounds:****

- Round 1: Sent \$3, received \$4.41 back - good cooperation
- Round 2: Received \$5.19, returned \$2.40 - showed I reciprocate

****Current round:****

- New co-player (randomized pairing)
- Need to balance trust with uncertainty

Since I had success with sending \$3 in Round 1 (got \$4.41 back, netting \$1.41 profit), and I want to maintain a cooperative approach while testing this new partner, I'll send the same amount:

****Sending \$3****

This creates \$9 for them to work with. If they're reasonably cooperative like my Round 1 partner, we'll both benefit.

{'send': 3}

reasoning:

Let me analyze this situation:

Round 1: I was SENDER

- I sent \$3
- It became \$9
- They returned \$4.41
- My payoff: $\$5 - \$3 + \$4.41 = \6.41

Round 2: I was RECEIVER

- They sent \$1.73
- I received \$5.19
- I returned \$2.40
- My payoff: $\$5.19 - \$2.40 = \$2.79$
- Total so far: $\$6.41 + \$2.79 = \$9.20$?

Round 3: I am SENDER again

- This is a new co-player (since pairings are randomized each round)
- I have \$5 to send

Strategy considerations:

1. I had a good experience in Round 1 - sent \$3, got back \$4.41 (profit of \$1.41)
2. In Round 2, I received \$5.19 and returned \$2.40 (46%), showing I'm cooperative
3. This is a new co-player, so I don't know their behavior
4. There's communication noise, so what they see may differ from what I intend

I should probably send a moderate amount again to:

- Signal willingness to cooperate
- Test this new partner's trustworthiness
- Not risk everything on an unknown player

Sending \$3 worked well in Round 1. I could try that again, or maybe send slightly more (\$3.50) or less (\$2.50) to experiment.

Given that:

- My first experience was positive
- I want to maintain a cooperative reputation
- But this is a new, unknown player

I'll send \$3 again - it's a balanced approach that showed good results before.

{'send': 3}

usage:

```
{
  "input_tokens": 1128,
  "output_tokens": 608,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.71 to you. Your visible payoff was \$5.71.

Your total visible earnings across all rounds are \$14.91.

This round, you are the RECEIVER against your current co-player. They sent you \$3.79, so you received \$11.370000000000001. How much do you return? (0-11.370000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****What happened:****

- My co-player sent \$3.79 out of \$5 (~76% of their endowment)
- This became \$11.37 for me
- They showed strong trust by sending most of their money

****My reasoning:****

- They took a big risk trusting me
- I should reward this trust to encourage cooperation
- Returning around 45% of what I received seems appropriate given their high trust
- This would be approximately \$5.10

****Calculating outcomes:****

- They'll end with: \$1.21 + \$5.10 = \$6.31
- I'll end with: \$11.37 - \$5.10 = \$6.27
- They gain \$1.31 from their trust, which is a good reward

{'return': 5.10}

reasoning:

Looking at this situation:

****What happened:****

- My co-player sent \$3.79 (became \$11.37 for me)
- They kept \$1.21
- They showed significant trust by sending most of their money

****Previous experience:****

- Round 1: I sent \$3, got back \$4.41 (they returned ~49%)
- Round 2: I received \$5.19, returned \$2.40 (~46%)
- Round 3: I sent \$3, got back \$3.71 (they returned ~41%)

****My reasoning:****

- This co-player sent \$3.79 out of \$5, showing strong trust (~76%)
- They're risking a lot and trusting me to reciprocate
- I should reward this trust appropriately
- Returning around 40-50% of what I received would be \$4.55-\$5.69
- Let me think about fairness:
 - They sent \$3.79, kept \$1.21
 - I received \$11.37
 - If I return \$5.00:
 - They end with: \$1.21 + \$5.00 = \$6.21
 - I end with: \$11.37 - \$5.00 = \$6.37
 - If I return \$5.50:
 - They end with: \$1.21 + \$5.50 = \$6.71
 - I end with: \$11.37 - \$5.50 = \$5.87

Given they sent \$3.79 (76% of their endowment), I should be generous but not overdo it. Around 45% return rate seems appropriate, which would be about \$5.12.

Let me go with \$5.10:

...

usage:

```
{
  "input_tokens": 1453,
  "output_tokens": 690,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.79, it became \$11.370000000000001, and you returned \$5.1. Your visible payoff was \$6.270000000000001. Your total visible earnings across all rounds are \$21.18.

This round, you are the RECEIVER against your current co-player. They sent you \$2.91, so you received \$8.73. How much do you return? (0-8.73)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant**content:**

Looking at this situation:

****What happened:****

- My co-player sent \$2.91 out of \$5 (~58% of their endowment)
- This became \$8.73 for me
- They kept \$2.09
- This shows moderate to good trust

****My reasoning:****

- They demonstrated decent trust by sending over half their money
- I should reciprocate consistently with my previous rounds
- In previous rounds as receiver, I've returned around 45% of what I received
- For \$8.73, that would be approximately \$3.90

****Calculating outcomes:****

- They'll end with: \$2.09 + \$3.90 = \$5.99
- I'll end with: \$8.73 - \$3.90 = \$4.83
- They gain about \$1 from trusting me, which is a reasonable reward

{'return': 3.90}

reasoning:

Looking at this situation:

****What happened:****

- My co-player sent \$2.91 out of \$5 (~58% of their endowment)
- This became \$8.73 for me
- They kept \$2.09
- This shows moderate to good trust

****My reasoning:****

- They sent a decent amount, showing trust
- I should reciprocate proportionally
- Looking at my previous returns:
 - Round 2: Received \$5.19, returned \$2.40 (~46%)
 - Round 4: Received \$11.37, returned \$5.10 (~45%)
- I've been returning around 45% consistently

For \$8.73, 45% would be about \$3.93

****Calculating outcomes with \$3.93:****

- They'll end with: \$2.09 + \$3.93 = \$6.02
- I'll end with: \$8.73 - \$3.93 = \$4.80
- They gain \$1.02 from sending, which rewards their trust

This seems fair and consistent with my previous behavior. I'll return \$3.90 to keep it simple.

Actually, let me recalculate with \$3.90:

- They'll end with: \$2.09 + \$3.90 = \$5.99
- I'll end with: \$8.73 - \$3.90 = \$4.83
- They gain \$0.99 from sending

That works well.

usage:

```
{
  "input_tokens": 1786,
  "output_tokens": 560,
```

```

    "reasoning_tokens": 0
  }
Message 8 role=user
content:
  Round 6

  In your most recent previous game round, you were the RECEIVER against another agent. You
  saw them send $2.91, it became $8.73, and you returned $3.9. Your visible payoff was $4.
  83.
  Your total visible earnings across all rounds are $26.009999999999998.

  This round, you are the SENDER against your current co-player. You have $5. How much do yo
  u send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Message 9 role=assistant
content:
  Looking at my history as SENDER:

  **Previous rounds as sender:**
  - Round 1: Sent $3 ? received $4.41 back (profit of $1.41)
  - Round 3: Sent $3 ? received $3.71 back (profit of $0.71)

  Both times sending $3 worked out profitably, showing it's a reasonable cooperative strateg
  y.

  **Current situation:**
  - New co-player (randomized pairing)
  - Want to maintain consistent cooperative approach
  - $3 creates $9 for them, leaving room for mutual benefit

  I'll stick with my established strategy of sending $3, which has proven effective:

  {'send': 3}
reasoning:
  Looking at my history:

  Round 1 (SENDER): Sent $3, received $4.41 back, payoff $6.41
  Round 2 (RECEIVER): Received $5.19, returned $2.40, payoff $2.79
  Round 3 (SENDER): Sent $3, received $3.71 back, payoff $5.71
  Round 4 (RECEIVER): Received $11.37, returned $5.10, payoff $6.27
  Round 5 (RECEIVER): Received $8.73, returned $3.90, payoff $4.83

  Total earnings: $26.01

  Now Round 6, I'm the SENDER with a new randomized co-player.

  My strategy has been to send $3 as sender (both times I did this), which has worked out re
  asonably well:
  - Round 1: Sent $3, got $4.41 back (net +$1.41)
  - Round 3: Sent $3, got $3.71 back (net +$0.71)

  Both times I profited, though the return varied.

  As receiver, I've been consistent in returning around 45% of what I receive, which seems f
  air.

  For this new round with a new partner, I should continue with my cooperative strategy. Sen
  ding $3 has been my standard approach and has generally worked well. It signals cooperat
  ion while not being overly risky.

  I'll send $3 again.
usage:
  {
    "input_tokens": 2117,
    "output_tokens": 509,
    "reasoning_tokens": 0
  }
Message 10 role=user
content:
  Round 7

```

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$3.68 to you. Your visible payoff was \$5.68.

Your total visible earnings across all rounds are \$31.689999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.51, so you received \$7.529999999999999. How much do you return? (0-7.529999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

****What happened:****

- Same co-player from Round 6 (where I sent them \$3 and they returned \$3.68)
- Now they sent me \$2.51, which became \$7.53
- They kept \$2.49
- They showed moderate trust

****My reasoning:****

- They were reasonably cooperative in Round 6 (returned ~41% of what they received)
- Now they're trusting me with about half their endowment
- I should maintain my consistent reciprocation pattern
- I've been returning around 45% of what I receive as a receiver

****For \$7.53:****

- 45% = approximately \$3.39

****Calculating outcomes:****

- They'll end with: \$2.49 + \$3.39 = \$5.88
- I'll end with: \$7.53 - \$3.39 = \$4.14
- They gain \$0.88 from their trust

{ 'return': 3.39 }

reasoning:

Looking at this situation:

****What happened:****

- Same co-player from Round 6 (I was sender then, now I'm receiver)
- In Round 6: I sent \$3, they returned \$3.68
- Now Round 7: They sent \$2.51, became \$7.53 for me
- They kept \$2.49

****My reasoning:****

- This is the same partner who just returned \$3.68 on \$9 (~41% return rate)
- They showed reasonable cooperation
- Now they sent \$2.51 (~50% of endowment), showing moderate trust
- I should reciprocate fairly

****My return pattern as receiver:****

- Round 2: Received \$5.19, returned \$2.40 (46%)
- Round 4: Received \$11.37, returned \$5.10 (45%)
- Round 5: Received \$8.73, returned \$3.90 (45%)

I've been consistently returning around 45% of what I receive.

For \$7.53:

- 45% would be approximately \$3.39

****Calculating outcomes:****

- They'll end with: \$2.49 + \$3.39 = \$5.88
- I'll end with: \$7.53 - \$3.39 = \$4.14
- They gain \$0.88 from trusting me

This seems fair and consistent with my pattern.

usage:

```
{
  "input_tokens": 2425,
  "output_tokens": 569,
  "reasoning_tokens": 0
}
```

}

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.51, it became \$7.529999999999999, and you returned \$3.39. Your visible payoff was \$4.139999999999999.

Your total visible earnings across all rounds are \$35.83.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my history as SENDER:

****Previous rounds as sender:****

- Round 1: Sent \$3 ? received \$4.41 back (profit of \$1.41)
- Round 3: Sent \$3 ? received \$3.71 back (profit of \$0.71)
- Round 6: Sent \$3 ? received \$3.68 back (profit of \$0.68)

****Pattern:****

- Sending \$3 has consistently worked well
- All three co-players returned more than I sent
- This shows a good balance of trust and caution

****Current situation:****

- New co-player (randomized pairing)
- My established strategy has proven reliable

I'll continue with sending \$3:

{'send': 3}

reasoning:

Looking at my history as SENDER:

****Previous rounds as sender:****

- Round 1: Sent \$3 ? received \$4.41 back (profit of \$1.41)
- Round 3: Sent \$3 ? received \$3.71 back (profit of \$0.71)
- Round 6: Sent \$3 ? received \$3.68 back (profit of \$0.68)

All three times I've sent \$3, I've gotten back more than I sent, making profitable exchanges. The strategy is working consistently.

****Current situation:****

- Round 8 (final round likely, or near it)
- New co-player again (randomized pairing)
- Want to maintain cooperative approach

I should stick with \$3 again as it has consistently worked well across multiple partners:

{'send': 3}

usage:

```
{
  "input_tokens": 2608,
  "output_tokens": 378,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.21 to you. Your visible payoff was \$6.21.

Your total visible earnings across all rounds are \$42.04.

This round, you are the RECEIVER against your current co-player. They sent you \$3.52, so you received \$10.56. How much do you return? (0-10.56)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****What happened:****

- My co-player sent \$3.52 out of \$5 (~70% of their endowment)
- This became \$10.56 for me
- They kept \$1.48
- They showed strong trust

****My reasoning:****

- They demonstrated strong trust by sending most of their money
- I should maintain my consistent reciprocation pattern
- I've been returning around 45% of what I receive as a receiver
- For \$10.56, that would be approximately \$4.75

****Calculating outcomes:****

- They'll end with: \$1.48 + \$4.75 = \$6.23
- I'll end with: \$10.56 - \$4.75 = \$5.81
- They gain \$1.23 from their trust, which is a good reward

{'return': 4.75}

Reasoning:

Looking at this situation:

****What happened:****

- My co-player sent \$3.52 out of \$5 (~70% of their endowment)
- This became \$10.56 for me
- They kept \$1.48
- This shows strong trust

****My reasoning:****

- They demonstrated strong trust by sending over 70% of their money
- I should reciprocate consistently with my previous rounds
- In previous rounds as receiver, I've been returning around 45% of what I received
- For \$10.56, that would be approximately \$4.75

****Calculating outcomes:****

- They'll end with: \$1.48 + \$4.75 = \$6.23
- I'll end with: \$10.56 - \$4.75 = \$5.81
- They gain about \$1.23 from trusting me, which is a good reward

Let me check my previous returns as receiver:

- Round 2: Received \$5.19, returned \$2.40 (46%)
- Round 4: Received \$11.37, returned \$5.10 (45%)
- Round 5: Received \$8.73, returned \$3.90 (45%)
- Round 7: Received \$7.53, returned \$3.39 (45%)

So I've been consistently returning around 45% of what I receive.

For \$10.56:

$$45\% = \$10.56 \times 0.45 = \$4.75$$

This seems appropriate.

Usage: {

"input_tokens": 2568,
"output_tokens": 559,
"reasoning_tokens": 0

}

GAME - Agent_4 - Sender - dyad_3

Timestamp: 2026-08-12T09:28:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_3",
"move_index": 4,
"opponent_id": "Agent_8",
"pairing": {

```

"agents": [
  "Agent_4",
  "Agent_8"
],
"investor": "Agent_4",
"roles": {
  "Agent_4": "investor",
  "Agent_8": "trustee"
},
"trustee": "Agent_8"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.56, it became \$7.68, and you returned \$3.5. Your visible payoff was \$4.

18.

Your total visible earnings across all rounds are \$9.98.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

****Analysis:****

I'm now facing a new co-player (Round 3, new pairing). My experience so far:

- Round 1: Sent \$3, received back \$3.8 (42% return rate)
- Round 2: Received \$7.68, returned \$3.5 (46% return rate)

****Strategy:****

Since I'm building a reputation in this multi-agent environment, I want to signal cooperative behavior. I'll send slightly more than last time to:

- Test if higher trust leads to better reciprocity
- Create more value through the 3x multiplier
- Signal that I'm willing to be generous

Sending \$3.5 means my co-player receives \$10.50. If they return around 40%, I'd get back ~\$4.20, giving me a net gain. This balances trust-building with reasonable risk.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the situation:

****Round 1:**** I was SENDER

- Sent: \$3
- They received: \$9
- They returned: \$3.8
- My payoff: \$5.8

****Round 2:**** I was RECEIVER

- They sent: \$2.56
- I received: \$7.68
- I returned: \$3.5
- My payoff: \$4.18

****Round 3:**** I am SENDER again

Now I need to think about this new co-player in Round 3. Note that each round I'm paired with potentially different agents from the pool of 8 agents.

In Round 1, my co-player returned \$3.8 out of \$9 (42.2%).

In Round 2, I returned \$3.5 out of \$7.68 (45.6%).

For Round 3, I'm facing a NEW co-player (since pairings are randomized each round). I don't have specific history with this particular agent.

Given:

1. I've seen one example of a co-player returning about 42%
2. I returned about 46% myself, trying to be fair
3. The game has communication noise, so amounts may differ slightly

Strategy for Round 3:

- I should continue with a cooperative but cautious approach
- Sending around \$3 worked reasonably in Round 1
- I might try a slightly different amount to gather more information

Let me send \$3 again as it seems to be a reasonable middle ground that:

- Shows trust
- Creates significant value through multiplication (becomes \$9)
- Leaves me with \$2 as a safety net
- If they return around 40-45%, I'd get back around \$3.6-\$4, giving me a decent payoff

Actually, I could also consider sending a bit more to test if higher trust leads to better

returns. Let me send \$3.5 this time.

...

usage:

```
{
  "input_tokens": 1137,
  "output_tokens": 766,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.07 to you. Your visible payoff was \$7.57.

Your total visible earnings across all rounds are \$17.55.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

****Analysis:****

My co-player sent \$2.9 (58% of their endowment), showing significant trust. This is a strong cooperative signal.

****Previous patterns:****

- Round 1 sender: Got back 42% return
- Round 2 receiver: I returned 46%
- Round 3 sender: Got back 58% return (very generous!)

****Current situation:****

- I received \$8.7
- Sender has \$2.1 remaining

****Decision:****

My co-player showed high trust by sending 58%. I should reciprocate proportionally. Returning 50% of what I received (\$4.35) is fair:

If I return \$4.35:

- Sender ends with: $\$2.1 + \$4.35 = \$6.45$ (profit of \$1.45)
- I keep: $\$8.7 - \$4.35 = \$4.35$ (profit of \$4.35)

This rewards their trust and maintains cooperative momentum.

```
'''json
{"return": 4.35}
'''
```

reasoning:

****Analysis of the situation:****

Round 4, I'm the RECEIVER:

- My co-player sent \$2.9 (58% of their endowment)
- I received \$8.7 after the 3x multiplication
- They currently have \$2.1 remaining ($\$5 - \2.9)

****Previous experience:****

- Round 1 (Sender): Sent \$3, got back \$3.8 (42% of \$9)
- Round 2 (Receiver): Received \$7.68, returned \$3.5 (46%)
- Round 3 (Sender): Sent \$3.5, got back \$6.07 (58% of \$10.5) - this was very generous!

****Reciprocity:****

My Round 3 co-player was very generous with a 58% return rate. This current co-player sent 58% of their endowment, showing significant trust.

Looking at patterns:

- When I received \$7.68 in Round 2, I returned \$3.5 (46%)
- Now I received \$8.7

I should be fair and encouraging. If I return around 45-50%:

- 45% would be \$3.92
- 50% would be \$4.35

Let me calculate outcomes:

If I return \$4.00:

- Sender ends with: $\$2.1 + \$4.00 = \$6.10$ (gain of \$1.10)
- I keep: $\$8.7 - \$4.00 = \$4.70$ (gain of \$4.70)

If I return \$4.35:

- Sender ends with: $\$2.1 + \$4.35 = \$6.45$ (gain of \$1.45)

...

usage:

```
{
  "input_tokens": 1485,
  "output_tokens": 766,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.9, it became \$8.7, and you returned \$4.35. Your visible payoff was \$4.35.

Your total visible earnings across all rounds are \$21.9.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

****Analysis:****

New round, new co-player. My track record:

- As sender: Sent \$3 (got 42% back), sent \$3.5 (got 58% back)
- As receiver: Returned 46% and 50%

****Strategy:****

The higher send amount (\$3.5) yielded much better reciprocity in Round 3. This suggests that showing more trust encourages cooperation. I'll continue with this amount to:

- Signal strong cooperative intent
- Create significant value (\$10.50 for receiver)
- Build on the positive pattern

Given my fair behavior as a receiver, I'm hopeful this will be reciprocated.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Looking at my history:

- Round 1 (SENDER): Sent \$3, got back \$3.8 (42% return) - payoff \$5.8
- Round 2 (RECEIVER): Received \$7.68, returned \$3.5 (46% return) - payoff \$4.18
- Round 3 (SENDER): Sent \$3.5, got back \$6.07 (58% return) - payoff \$7.57
- Round 4 (RECEIVER): Received \$8.7, returned \$4.35 (50% return) - payoff \$4.35

Now Round 5 - I'm the SENDER against a new co-player.

Key observations:

1. When I sent more (\$3.5 vs \$3), I got a better return (58% vs 42%)
2. I've been generally fair as a receiver (46-50% returns)
3. My total earnings are \$21.9, averaging about \$5.48 per round - solid profit

Strategy considerations:

- I'm building a reputation across multiple agents
- Higher sends have been rewarded so far
- The 3x multiplier means sending more creates more total value
- I should continue signaling cooperation

I'll send \$3.5 or \$4 this round. Let me think:

- Sending \$4 creates \$12 for receiver, shows high trust
- If they return 50%, that's \$6, giving me \$7 total ($\$5 - \$4 + \$6 = \7)
- If they return 40%, that's \$4.8, giving me \$5.8 ($\$5 - \$4 + \$4.8$)

Actually, let me continue with \$3.5 since that worked well in Round 3. It's a solid cooperative amount without being overly risky.

usage:

```
{
  "input_tokens": 1851,
  "output_tokens": 596,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.36 to you. Your visible payoff was \$7.86.

Your total visible earnings across all rounds are \$29.759999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

****Analysis:****

My co-player sent \$2.27 (about 45% of their endowment), showing moderate trust.

****Current situation:****

- I received \$6.81
- Sender has \$2.73 remaining

****My receiver pattern:****

- Round 2: Returned 46% of received
- Round 4: Returned 50% of received

****Decision:****

I'll maintain fair reciprocity. Returning about 48% of what I received:

If I return \$3.25:

- Sender ends with: $\$2.73 + \$3.25 = \$5.98$ (small profit)
- I keep: $\$6.81 - \$3.25 = \$3.56$

This rewards their moderate trust fairly.

```
'''json
{"return": 3.25}
'''
```

reasoning:

****Analysis:****

My co-player sent \$2.27 (about 45% of their \$5 endowment), showing moderate trust.

****Previous patterns as receiver:****

- Round 2: Received \$7.68, returned \$3.5 (46% of received)
- Round 4: Received \$8.7, returned \$4.35 (50% of received)

****Current situation:****

- I received \$6.81
- Sender has \$2.73 remaining

****Decision:****

My co-player showed moderate trust (45% sent). I should reciprocate fairly. They sent less than in previous rounds, so I'll return proportionally.

Returning around 48% seems fair:

- 48% of \$6.81 = \$3.27

If I return \$3.27:

- Sender ends with: $\$2.73 + \$3.27 = \$6.00$ (profit of \$1.00)
- I keep: $\$6.81 - \$3.27 = \$3.54$ (profit of \$3.54)

This is fair reciprocity that rewards their trust while keeping a reasonable share.

usage:

```
{
  "input_tokens": 2166,
  "output_tokens": 466,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.27, it became \$6.8100000000000005, and you returned \$3.25. Your visible payoff was \$3.5600000000000005.

Your total visible earnings across all rounds are \$33.32.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

****Analysis:****

This is interesting - I'm playing against the same co-player from Round 6!

****Round 6 recap:****

- They sent me \$2.27 (45% of endowment)
- I returned \$3.25
- They ended with a small profit

****Decision:****

Since they showed moderate trust in Round 6 and I reciprocated fairly, I should continue that cooperative relationship. My successful strategy has been sending \$3.5:

- Creates \$10.50 for them
- Has historically yielded good returns (58-61%)
- Signals strong reciprocal cooperation

I'll send \$3.5 to maintain our cooperative dynamic.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Interesting! This round I'm the SENDER and it says I'm playing against my "current co-player" - which is the same person from Round 6 where I was the receiver.

Let me review:

- Round 6: They sent me \$2.27, I received \$6.81, I returned \$3.25
- So they ended Round 6 with: $\$5 - \$2.27 + \$3.25 = \5.98

Now in Round 7, I'm the sender playing against the same person.

****Analysis:****

- This co-player sent me a moderate amount (\$2.27, 45%)
- I returned \$3.25, which gave them a small profit
- They showed reasonable trust

****My sender history:****

- Round 1: Sent \$3 ? got back \$3.8 (42% return rate)
- Round 3: Sent \$3.5 ? got back \$6.07 (58% return rate)
- Round 5: Sent \$3.5 ? got back \$6.36 (61% return rate)

The pattern shows that sending \$3.5 has been working well. I've been building positive relationships.

Since this is the same co-player who showed moderate trust, I should reciprocate similarly . They sent 45% of their endowment to me, so I could send around the same proportion or slightly more to reward their cooperation.

Sending \$3.5 again seems appropriate - it's been my successful strategy and matches my cooperative pattern.

```
'''json
{"send": 3.5}
'''
```

usage:

```
{
  "input_tokens": 2489,
  "output_tokens": 538,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$2.79 to you. Your visible payoff was \$4.29.

Your total visible earnings across all rounds are \$37.61.

This round, you are the RECEIVER against your current co-player. They sent you \$2.68, so you received \$8.040000000000001. How much do you return? (0-8.040000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

****Analysis:****

My co-player sent \$2.68 (about 54% of their endowment), showing good trust.

****Current situation:****

- I received \$8.04
- Sender has \$2.32 remaining

****My receiver track record:****

- Consistently returned 46-50% of received amounts
- This has built a reputation for fair play

****Decision:****

I'll maintain my fair reciprocity pattern. Returning about 48% of what I received:

If I return \$3.85:

- Sender ends with: $\$2.32 + \$3.85 = \$6.17$ (profit of \$1.17)
- I keep: $\$8.04 - \$3.85 = \$4.19$

This rewards their trust appropriately.

```
'''json
{"return": 3.85}
'''
```

reasoning:

Let me analyze this situation:

****Current situation:****

- I'm the receiver
- Co-player sent \$2.68 (about 54% of endowment) - showing good trust
- I received \$8.04
- Sender has \$2.32 remaining

****Previous rounds summary:****

- Round 2 receiver: Returned 46% (\$3.5 from \$7.68)
- Round 4 receiver: Returned 50% (\$4.35 from \$8.7)
- Round 6 receiver: Returned 48% (\$3.25 from \$6.81)

I've been consistently fair as a receiver, returning around 46-50% of what I receive.

****Round 7 note:****

My co-player from Round 6-7 only returned \$2.79 out of \$10.5 in Round 7 (about 27%). That was disappointing - much lower than the returns I've been getting.

But this is Round 8 with a potentially different co-player (unless it's the same one). Actually, looking at the multi-agent setting, pairings are randomized each round, so this could be anyone.

The sender sent \$2.68 (54% of endowment), showing decent trust. I should reciprocate fairly.

If I return around 48% of \$8.04 = \$3.86:

- Sender ends with: \$2.32 + \$3.86 = \$6.18 (profit of \$1.18)
- I keep: \$8.04 - \$3.86 = \$4.18

This maintains my reputation as a fair receiver.

Let me go with \$3.85 or \$3.9. I'll say \$3.85.

usage:

```
{
  "input_tokens": 2601,
  "output_tokens": 607,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.68, it became \$8.040000000000001, and you returned \$3.85. Your visible payoff was \$4.190000000000001.

Your total visible earnings across all rounds are \$41.8.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

****Analysis:****

New co-player for Round 9. My sender track record:

- Sent \$3: Got 42% return
- Sent \$3.5: Got 58%, 61%, and 27% returns (mixed results)

****Reflection:****

Despite one poor return in Round 7, my \$3.5 strategy has generally worked well. As a receiver, I've built a strong reputation for fair play (46-50% returns).

****Strategy:****

Continue with \$3.5 to:

- Maintain consistent cooperative signaling
- Create substantial value (\$10.50)
- Hope my reputation for fairness encourages reciprocity

```
'''json
{"send": 3.5}
'''
```

Reasoning:

Let me analyze my history:

As sender:

- Round 1: Sent \$3, got back \$3.8 (42% return rate)
- Round 3: Sent \$3.5, got back \$6.07 (58% return rate)
- Round 5: Sent \$3.5, got back \$6.36 (61% return rate)
- Round 7: Sent \$3.5, got back \$2.79 (27% return rate) - this was worse!

As receiver:

- Round 2: Received \$7.68, returned \$3.5 (46% return rate)
- Round 4: Received \$8.7, returned \$4.35 (50% return rate)
- Round 6: Received \$6.81, returned \$3.25 (48% return rate)

- Round 8: Received \$8.04, returned \$3.85 (48% return rate)

I've been consistently fair as a receiver, returning around 46-50% of what I receive.

As a sender, I've had mixed results. My \$3.5 strategy worked well in rounds 3 and 5, but in rounds 6 and 7 it got a poor return (only 27%).

This is Round 9, playing against a new co-player. My overall approach has been to signal cooperation by sending \$3.5. While I had one bad round, overall this strategy has worked well.

Let me continue with \$3.5 to maintain my cooperative reputation.

```
Usage: {  
  "input_tokens": 2574,  
  "output_tokens": 522,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_8 - Receiver - dyad_3

Timestamp: 2026-08-12T09:28:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_8"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_4": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.50, it became \$7.50, and you saw them return \$3.19 to you. Your visible payoff was \$5.6899999999999995.

Your total visible earnings across all rounds are \$10.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my recent experience:

- Round 2: I sent \$2.50 (50%) and received \$3.19 back, giving me a good positive return

This cooperative approach has been working well. I'm now with a new co-player, and I want to signal trustworthiness and encourage mutual cooperation.

Sending 50% has proven to be a solid strategy - it creates meaningful potential gains while showing balanced trust.

{"send": 2.50}

reasoning:

Let me analyze the previous round:

- I sent \$2.50 (50% of my endowment)
- It became \$7.50 for the receiver
- They returned \$3.19
- I ended with \$2.50 (kept) + \$3.19 (returned) = \$5.69
- They kept \$7.50 - \$3.19 = \$4.31

So the receiver returned about 42.5% of what they received, which gave me a positive gain of \$0.69. This is a reasonable reciprocation, showing they valued cooperation.

Now I'm facing a new co-player in Round 3.

My experience so far:

- Round 1 (as receiver): Received \$8.04, returned \$3.00 (37.3%)
- Round 2 (as sender): Sent \$2.50, received back \$3.19 - good outcome

I should continue with a cooperative strategy. Since my last send of \$2.50 worked well, I could:

1. Send the same amount (\$2.50) to maintain consistency
2. Send slightly more to signal increased trust
3. Send slightly less to be more cautious

Given that cooperation has been working, and I'm now encountering a third different co-player, I think maintaining a cooperative stance of around 50% makes sense. This has proven

successful and sends the right signal.

I'll send \$2.50 again.

usage:

```
{
  "input_tokens": 1103,
  "output_tokens": 441,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$5.25 to you. Your visible payoff was \$7.75.

Your total visible earnings across all rounds are \$18.48.

This round, you are the RECEIVER against your current co-player. They sent you \$3.69, so you received \$11.07. How much do you return? (0-11.07)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- The sender showed very high trust by sending \$3.69 (73.8% of their endowment)
- They now have only \$1.31 remaining
- I received \$11.07 after the 3x multiplier
- Total pool is \$12.38

This sender took a significant risk to create mutual gains. I should reciprocate their strong trust signal generously.

If I return \$5.50:

- Sender ends with \$6.81 (gaining \$1.81)
- I keep \$5.57

This rewards their exceptional trust while maintaining fairness for both of us.

{"return": 5.50}

reasoning:

Let me analyze this situation:

Round 4 - I'm the RECEIVER

- The sender sent \$3.69 (73.8% of their \$5 endowment)
- This became \$11.07 after the 3x multiplier
- The sender now has \$5 - \$3.69 = \$1.31 remaining
- Total pool: \$1.31 + \$11.07 = \$12.38

The sender showed very high trust by sending 73.8% of their endowment. This is much higher than the ~50% I've been sending, indicating strong cooperative intent.

I should reciprocate this high level of trust. Let me think about fair returns:

If I return \$5.50:

- Sender ends with: \$1.31 + \$5.50 = \$6.81 (gain of \$1.81)
- I keep: \$11.07 - \$5.50 = \$5.57

If I return \$6.00:

- Sender ends with: \$1.31 + \$6.00 = \$7.31 (gain of \$2.31)
- I keep: \$11.07 - \$6.00 = \$5.07

Given they sent 73.8% of their endowment (very high trust), I should return a proportionally generous amount. Returning around \$5.50-\$6.00 would:

1. Reward their high trust
2. Ensure they profit from being cooperative
3. Still leave me with a reasonable amount
4. Signal I'm reciprocal and trustworthy

I'll return \$5.50 - this gives them a solid gain while being fair to both parties.

usage:

```
{
  "input_tokens": 1343,
  "output_tokens": 551,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.69, it became \$11.07, and you returned \$5.5. Your visible payoff was \$5.57.

Your total visible earnings across all rounds are \$24.05.

This round, you are the RECEIVER against your current co-player. They sent you \$3.88, so you received \$11.64. How much do you return? (0-11.64)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender showed extremely high trust by sending \$3.88 (77.6% of their endowment)
- They now have only \$1.12 remaining
- I received \$11.64 after the 3x multiplier
- Total pool is \$12.76

This is even higher trust than the previous round. I should reciprocate this exceptional trust generously.

If I return \$5.80:

- Sender ends with \$6.92 (gaining \$1.92)
- I keep \$5.84

This appropriately rewards their very high trust level while maintaining fairness.

```
{"return": 5.80}
```

reasoning:

Looking at this situation:

- The sender sent \$3.88 (77.6% of their endowment) - even higher trust than the previous round
- They now have \$1.12 remaining
- I received \$11.64 after the 3x multiplier
- Total pool is \$12.76

This sender showed extremely high trust. I should reciprocate proportionally.

In the last round, sender sent \$3.69 (73.8%) and I returned \$5.50, giving them \$6.81 total (gain of \$1.81).

This sender sent even more (\$3.88, or 77.6%). I should return proportionally or slightly more to reward the higher trust.

If I return \$5.80:

- Sender ends with \$6.92 (gain of \$1.92)
- I keep \$5.84

This seems fair and rewards their high trust appropriately.

Let me consider other options:

- Return \$6.00: Sender gets \$7.12 (gain \$2.12), I keep \$5.64
- Return \$5.50: Sender gets \$6.62 (gain \$1.62), I keep \$6.14

I think \$5.80 is a good middle ground that appropriately rewards the very high trust level while maintaining fairness.

usage:

```
{
  "input_tokens": 1628,
  "output_tokens": 459,

```

```

    "reasoning_tokens": 0
  }
}
Message 8 role=user
content:
  Round 6

  In your most recent previous game round, you were the RECEIVER against another agent. You
  saw them send $3.88, it became $11.64, and you returned $5.8. Your visible payoff was $5
  .8400000000000001.
  Your total visible earnings across all rounds are $29.89.

  This round, you are the SENDER against your current co-player. You have $5. How much do yo
  u send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Message 9 role=assistant
content:
  Looking at my experience as sender:
  - Round 2: Sent $2.50, received $3.19 back (good return)
  - Round 3: Sent $2.50, received $5.25 back (excellent return)

  The 50% strategy has worked very well for me. The cooperative environment suggests continu
  ing this approach with a new co-player.

  {"send": 2.50}
reasoning:
  Let me review my experience so far:

  Round 1 (RECEIVER): Received from sender who sent $2.68 (53.6%), I returned $3.00
  Round 2 (SENDER): I sent $2.50 (50%), received back $3.19
  Round 3 (SENDER): I sent $2.50 (50%), received back $5.25
  Round 4 (RECEIVER): Received from sender who sent $3.69 (73.8%), I returned $5.50
  Round 5 (RECEIVER): Received from sender who sent $3.88 (77.6%), I returned $5.80

  As a sender, I've been sending 50% and getting good returns. I've encountered receivers wh
  o are cooperating well. I've also been a generous receiver when others send high amounts
  .

  The pattern suggests a cooperative environment. When I send 50%, people return enough for
  me to profit. When others send high amounts (70-78%), I reciprocate generously.

  I'm now facing a new co-player as sender. Should I:
  - Continue with 50% ($2.50) - safe, proven strategy
  - Increase to show more trust - could signal cooperation and encourage higher returns
  - Decrease to be more cautious - but this might seem less cooperative

  Given the cooperative environment and good outcomes when I've sent $2.50, I'll stick with
  that strategy.
usage:
  {
    "input_tokens": 1902,
    "output_tokens": 421,
    "reasoning_tokens": 0
  }
Message 10 role=user
content:
  Round 7

  In your most recent previous game round, you were the SENDER against another agent. You se
  nt $2.5, it became $7.5, and you saw them return $3.97 to you. Your visible payoff was $
  6.4700000000000001.
  Your total visible earnings across all rounds are $36.36.

  This round, you are the RECEIVER against your current co-player. They sent you $3.1, so yo
  u received $9.3. How much do you return? (0-9.3)
  Respond exactly as JSON: {'return': <amount>}
Message 11 role=assistant
content:
  Looking at this situation:

  - The sender sent $3.10 (62% of their endowment), showing good trust

```

- They now have \$1.90 remaining
- I received \$9.30 after the 3x multiplier
- Total pool is \$11.20

This is a solid trust signal that deserves good reciprocation.

If I return \$4.50:

- Sender ends with \$6.40 (gaining \$1.40)
- I keep \$4.80

This rewards their trust appropriately while maintaining balance.

```
{"return": 4.50}
```

reasoning:

Looking at this situation:

- The sender sent \$3.10 (62% of their endowment)
- They now have \$1.90 remaining
- I received \$9.30 after the 3x multiplier
- Total pool is \$11.20

This is a moderately-high trust signal. I should reciprocate fairly.

Let me consider a reasonable return:

- If I return \$4.50:
 - Sender ends with \$6.40 (gaining \$1.40)
 - I keep \$4.80

This seems fair and reciprocates their trust appropriately.

Alternatively:

- If I return \$4.00:
 - Sender ends with \$5.90 (gaining \$0.90)
 - I keep \$5.30

Or:

- If I return \$4.70:
 - Sender ends with \$6.60 (gaining \$1.60)
 - I keep \$4.60

Given they sent 62%, I think returning around \$4.50 is appropriate - it gives them a solid gain while keeping things relatively balanced.

usage:

```
{
  "input_tokens": 2134,
  "output_tokens": 395,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.1, it became \$9.3, and you returned \$4.5. Your visible payoff was \$4.8000000000000001.

Your total visible earnings across all rounds are \$41.16.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my track record as sender:

- Round 2: Sent \$2.50, received \$3.19 back
- Round 3: Sent \$2.50, received \$5.25 back
- Round 6: Sent \$2.50, received \$3.97 back

The 50% sending strategy has been consistently successful, with all partners reciprocating positively. This balanced approach signals trust while protecting against losses.

With a new co-player, I'll continue this proven strategy.

```
{"send": 2.50}
```

reasoning:

Looking at my history as a sender:

- Round 2: Sent \$2.50, received \$3.19 back
- Round 3: Sent \$2.50, received \$5.25 back
- Round 6: Sent \$2.50, received \$3.97 back

All three times I've sent \$2.50, I've received positive returns. The 50% strategy has been consistently successful.

As a receiver, I've been generous in reciprocating trust:

- Round 1: Received \$8.04, returned \$3.00
- Round 4: Received \$11.07, returned \$5.50
- Round 5: Received \$11.64, returned \$5.80
- Round 7: Received \$9.30, returned \$4.50

I'm now with yet another new co-player. The pattern shows that sending 50% has been a reliable strategy in this cooperative environment. I should continue with this approach.

```
{"send": 2.50}
```

usage:

```
{  
  "input_tokens": 2141,  
  "output_tokens": 363,  
  "reasoning_tokens": 0  
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$5.58 to you. Your visible payoff was \$8.08.

Your total visible earnings across all rounds are \$49.239999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$2.64, so you received \$7.92. How much do you return? (0-7.92)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender sent \$2.64 (52.8% of their endowment), showing moderate trust
- They now have \$2.36 remaining
- I received \$7.92 after the 3x multiplier
- Total pool is \$10.28

This is a moderate trust signal that deserves fair reciprocation.

If I return \$3.90:

- Sender ends with \$6.26 (gaining \$1.26)
- I keep \$4.02

This rewards their trust appropriately and maintains balance.

```
{"return": 3.90}
```

Reasoning:

Looking at the situation:

- The sender sent \$2.64 (52.8% of their endowment), showing moderate trust
- They now have \$2.36 remaining
- I received \$7.92 after the 3x multiplier
- Total pool is \$10.28

This is a moderate trust signal. I should reciprocate fairly.

If I return \$3.80:

- Sender ends with \$6.16 (gaining \$1.16)
- I keep \$4.12

This seems like a fair reciprocation for their moderate trust.

Let me think about my pattern as receiver:

- Round 4: Received \$11.07, returned \$5.50
- Round 5: Received \$11.64, returned \$5.80
- Round 7: Received \$9.30, returned \$4.50

I've been returning amounts that give both players roughly equal gains or slightly favor the sender to reward their trust.

For \$7.92 received:

- Returning \$3.80 gives sender total \$6.16 (+\$1.16), I keep \$4.12 (+\$4.12)
- This seems reasonable but maybe I should be more generous to the sender

Let me recalculate:

- If I return \$4.00: Sender gets \$6.36 (+\$1.36), I keep \$3.92
- This is more balanced

Actually, let me look at the proportions more carefully:

- Round 4: They sent 73.8%, I returned about 49.7% of what I received
- Round 5: They sent 77.6%, I returned about 49.8% of what I received
- Round 7: They sent 62%, I returned about 48.4% of what I received

This sender sent 52.8%, which is moderate trust. Around 48-50% return seems consistent.

...

```
Usage: {  
  "input_tokens": 2150,  
  "output_tokens": 668,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_6 - Sender - dyad_4

Timestamp: 2026-08-12T09:28:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.32 to you. Your visible payoff was \$7.32.

Your total visible earnings across all rounds are \$11.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my recent experience:

Round 2 results:

- I sent \$3.00 (60% of endowment)
- Receiver returned \$5.32
- I ended with \$7.32 - excellent outcome!

The cooperative strategy is working very well. When I demonstrate trust, it's being reciprocated generously.

Now facing a new co-player, I'll continue this successful approach. Sending around 60% has proven effective in building mutual benefit.

{'send': 3.00}

reasoning:

Let me analyze the situation:

Round 2 (just completed):

- I was the SENDER
- I sent \$3.00 (60% of endowment)

- It became \$9.00 for the receiver
- They returned \$5.32 to me
- My payoff: $\$5 - \$3 + \$5.32 = \7.32
- They kept: $\$9.00 - \$5.32 = \$3.68$

This was a very generous return! They returned more than half of what they received, and I ended up with more than my initial endowment even after sending a substantial amount.

My experience so far:

- Round 1 (RECEIVER): Received \$8.91, returned \$4.50, kept \$4.41
- Round 2 (SENDER): Sent \$3.00, received back \$5.32, ended with \$7.32

Total visible earnings: \$11.73

The cooperation strategy is working very well! When I sent 60% in Round 2, I got an excellent return.

Now in Round 3, I'm a SENDER again with a new co-player.

Given:

1. My cooperative strategy has been working well
2. Both times cooperation has led to mutual benefit
3. I want to continue encouraging trust and reciprocation
4. The game benefits from establishing cooperative norms

I should continue with a cooperative approach. Sending around 60% worked well last time. I could:

- Send \$3.00 again (same as last time, 60%)
- Send a bit more to show strong trust
- Send a bit less to be slightly more cautious with a new partner

Since cooperation has been consistently rewarded, I'll stick with the successful strategy of sending around 60%.

...

usage:

```
{
  "input_tokens": 1176,
  "output_tokens": 538,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$2.98 to you. Your visible payoff was \$4.98.

Your total visible earnings across all rounds are \$16.71.

This round, you are the RECEIVER against your current co-player. They sent you \$3.24, so you received \$9.72. How much do you return? (0-9.72)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation with my current co-player:

****Last round (Round 3):****

- I sent them \$3.00
- They returned \$2.98 (~33% of what they received)
- This was a modest but acceptable return

****This round (Round 4):****

- They sent me \$3.24 (slightly MORE than I sent them)
- I received \$9.72
- They currently have only \$1.76

The fact that they sent me MORE than I sent them suggests they're trying to build cooperation. I should reciprocate generously to:

1. Reward their increased trust
2. Establish a positive cooperative dynamic

3. Ensure we both benefit from ongoing interaction

I'll return **\$4.80**:

- They end with: $\$1.76 + \$4.80 = \$6.56$ (gain \$1.56)
- I keep: $\$9.72 - \$4.80 = \$4.92$ (gain \$4.92)

This is more generous than they were to me, encouraging further cooperation.

```
{'return': 4.80}
```

reasoning:

Let me analyze this situation:

Round 3 (previous round):

- I was the SENDER
- I sent \$3.00 to this same co-player
- They returned \$2.98
- My payoff was \$4.98 (kept \$2 + returned \$2.98)

Round 4 (current round):

- I am the RECEIVER
- Same co-player (they are now the sender)
- They sent me \$3.24
- I received \$9.72

So this is interesting - we're playing repeated games with the same partner. In round 3, I sent \$3.00 and they returned \$2.98 (about 33% of what they received). Now they've sent me \$3.24 (notably a bit more than I sent them).

Let me think about the fairness here:

- When I sent \$3.00, they kept $\$9.00 - \$2.98 = \$6.02$
- I kept $\$2.00 + \$2.98 = \$4.98$
- So they gained \$1.02, I lost \$0.02

Now they sent \$3.24:

- They kept $\$5 - \$3.24 = \$1.76$
- I received \$9.72

If I want to reciprocate similarly to how they treated me, I might return around 33% of what I received, which would be about \$3.21.

But actually, they sent me slightly MORE than I sent them (\$3.24 vs \$3.00), which could be seen as trying to establish cooperation or make up for the relatively low return last round.

Let me think about different return amounts:

Option 1: Return \$3.00 (about 31%)

- They end with: $\$1.76 + \$3.00 = \$4.76$ (lost \$0.24)
- I keep: $\$9.72 - \$3.00 = \$6.72$ (gain \$6.72)
- This feels too stingy given their slightly larger initial send.

Option 2: Return \$4.00 (about 41%)

- They end with: $\$1.76 + \$4.00 = \$5.76$ (gain \$0.76)
- I keep: $\$9.72 - \$4.00 = \$5.72$ (gain \$5.72)
- This creates mutual benefit and shows reciprocal trust.

Option 3: Return \$5.00 (about 51%)

- They end with: $\$1.76 + \$5.00 = \$6.76$ (gain \$1.76)
- I keep: $\$9.72 - \$5.00 = \$4.72$...

usage:

```
{
  "input_tokens": 1431,
  "output_tokens": 1067,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You

saw them send \$3.24, it became \$9.72, and you returned \$4.8. Your visible payoff was \$4.920000000000001.

Your total visible earnings across all rounds are \$21.630000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$2.32, so you received \$6.959999999999999. How much do you return? (0-6.959999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- New co-player sent me \$2.32 (about 46% of their endowment)
- I received \$6.96 after the multiplier
- They currently have \$2.68

They showed moderate trust by sending nearly half their endowment. I should reciprocate fairly to reward their trust and ensure mutual benefit.

Following my pattern of returning roughly 50% of what I receive, I'll return **\$3.48**:

- Sender ends with: \$2.68 + \$3.48 = \$6.16 (gain \$1.16)
- I keep: \$6.96 - \$3.48 = \$3.48 (gain \$3.48)

Both of us benefit from the cooperation.

{'return': 3.48}

reasoning:

Looking at this round:

- New co-player (different from Round 4)
- They sent me \$2.32 (about 46% of their endowment)
- I received \$6.96
- They currently have \$2.68 (\$5 - \$2.32)

They showed moderate trust by sending nearly half their endowment. I should reciprocate fairly to:

1. Reward their trust
2. Ensure both of us benefit
3. Encourage cooperation

Looking at my previous patterns:

- Round 1: Received \$8.91, returned \$4.50 (about 50%)
- Round 4: Received \$9.72, returned \$4.80 (about 49%)

Returning around 50% seems like my consistent fair approach. For \$6.96, that would be around \$3.48.

Let me check the outcomes:

- Sender ends with: \$2.68 + \$3.48 = \$6.16 (gain \$1.16)
- I keep: \$6.96 - \$3.48 = \$3.48 (gain \$3.48)

Both benefit, and this reciprocates their trust fairly. I'll return \$3.48.

usage:

```
{
  "input_tokens": 1834,
  "output_tokens": 476,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.32, it became \$6.959999999999999, and you returned \$3.48. Your visible payoff was \$3.479999999999999.

Your total visible earnings across all rounds are \$25.110000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my history with this current co-player:

****Round 5:****

- They sent me \$2.32 (46% of endowment)
- I returned \$3.48, giving them a good outcome

They showed moderate trust, and I reciprocated fairly. Now it's my turn to send, and I should continue building this cooperative relationship.

I'll send ****\$2.50****, which is:

- Slightly more than they sent me (showing willingness to cooperate)
- A reasonable amount that matches their moderate approach
- Enough to create mutual benefit if they reciprocate

{'send': 2.50}

reasoning:

Looking at my history with this current co-player:

Round 5 (last round):

- They sent me \$2.32 (46% of endowment)
- I received \$6.96
- I returned \$3.48 (50% of what I received)
- They ended with \$6.16, I ended with \$3.48

Now in Round 6, I'm the sender with this same co-player.

They showed moderate trust by sending \$2.32. I reciprocated fairly by returning \$3.48, which gave them a good outcome (\$6.16 total).

Since we had a successful cooperative interaction, I should continue building on this. I'll send a similar or slightly higher amount to maintain the cooperative relationship.

I've been generally sending around \$3.00 (60%) in other rounds, which has worked well. Given this co-player sent me \$2.32 (46%), I could:

1. Send similar to what they sent (~\$2.30-2.50)
2. Send my standard amount (~\$3.00)
3. Send slightly more to build trust

I think sending around \$2.50-3.00 makes sense. Let me go with \$2.50, which is slightly more than they sent me, showing I'm willing to cooperate but matching their level of caution.

Actually, looking at my overall strategy, I've been doing well with \$3.00 sends. But this particular co-player was more cautious. Let me match their level at around \$2.30-2.50.

I'll send \$2.50.

usage:

```
{
  "input_tokens": 2153,
  "output_tokens": 510,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$4.52 to you. Your visible payoff was \$7.02.

Your total visible earnings across all rounds are \$32.13.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my recent experiences:

Round 6 outcome:

- I sent \$2.50

- Received \$4.52 back
- Ended with \$7.02 - very good!

The cooperative approach continues to work well. Now with a new co-player, I'll signal trust and willingness to cooperate by sending a solid amount.

I'll send **\$3.00** (60% of endowment), which has proven to be an effective strategy:

```
{'send': 3.00}
```

reasoning:

Looking at my recent experience:

Round 6:

- I sent \$2.50 to my co-player
- They returned \$4.52
- I ended with \$7.02 - excellent outcome!

This is a new co-player in Round 7. I've been doing well with cooperative strategies. My approach of sending moderate to high amounts (around 50-60% of endowment) has generally worked well across different co-players.

Let me review the pattern:

- Round 2: Sent \$3.00, received back \$5.32 - great
- Round 3: Sent \$3.00, received back \$2.98 - okay
- Round 6: Sent \$2.50, received back \$4.52 - great

The cooperative strategy continues to work. With a new co-player, I should send a reasonable amount to signal willingness to cooperate. I'll send around \$3.00 (60%) to establish trust.

usage:

```
{
  "input_tokens": 2419,
  "output_tokens": 343,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.97 to you. Your visible payoff was \$5.970000000000001.

Your total visible earnings across all rounds are \$38.1.

This round, you are the RECEIVER against your current co-player. They sent you \$2.48, so you received \$7.439999999999995. How much do you return? (0-7.439999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- Co-player sent me \$2.48 (about 50% of their endowment)
- I received \$7.44 after the multiplier
- They currently have \$2.52

They showed solid trust by sending nearly half their endowment. I'll reciprocate fairly.

I'll return **\$3.72** (about 50% of what I received):

- Sender ends with: \$2.52 + \$3.72 = \$6.24 (gain \$1.24)
- I keep: \$7.44 - \$3.72 = \$3.72 (gain \$3.72)

Both of us benefit from the cooperation.

```
{'return': 3.72}
```

reasoning:

Looking at the situation:

- New co-player sent me \$2.48 (about 50% of their endowment)
- I received \$7.44 after the multiplier
- They currently have \$2.52

They showed substantial trust by sending nearly half their endowment. I should reciprocate fairly.

Following my pattern of fair returns:

- If I return about 50% of what I receive: $\$7.44 * 0.5 = \3.72

This would give:

- Sender: $\$2.52 + \$3.72 = \$6.24$ (gain \$1.24)

- Me: $\$7.44 - \$3.72 = \$3.72$ (gain \$3.72)

Both benefit, and I'm being fair to their trust. This seems reasonable.

Let me return \$3.72.

usage:

```
{
  "input_tokens": 2365,
  "output_tokens": 378,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.48, it became \$7.439999999999995, and you returned \$3.72. Your visible payoff was \$3.719999999999993.

Your total visible earnings across all rounds are \$41.82.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my gameplay so far:

My cooperative strategy has been working well overall. In previous rounds as sender:

- Sending \$3.00 has yielded good returns (like Round 3: sent \$3, got back \$5.32)
- Sending \$2.50 also worked well (Round 6: sent \$2.50, got back \$4.52)

With a new co-player, I'll continue signaling strong cooperation by sending a substantial amount.

I'll send ****\$3.00****:

```
{'send': 3.00}
```

Reasoning:

Looking at my situation:

Round 8 results:

- Previous co-player sent me \$2.48
- I returned \$3.72
- I kept \$3.72

Now I'm the sender with a new co-player in round 9. I have \$5 to decide how much to send.

Looking at my overall pattern:

- I've been generally cooperative, sending between \$2.50-\$3.00 typically
- My cooperative approaches have been working well
- Total earnings are \$41.82

I should continue with a cooperative strategy. Sending around \$3 has been effective in previous rounds.

I'll send \$3.00.

Usage:

```
{
  "input_tokens": 2400,
  "output_tokens": 282,
  "reasoning_tokens": 0
}
```


GAME - Agent_5 - Receiver - dyad_4

Timestamp: 2026-08-12T09:28:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.

- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$4.13 to you. Your visible payoff was \$6.63.

Your total visible earnings across all rounds are \$10.75.

This round, you are the RECEIVER against your current co-player. They sent you \$3.99, so you received \$11.97. How much do you return? (0-11.97)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.99, which is almost 80% of their endowment - showing exceptional trust
- I received \$11.97 after the 3x multiplier
- This is much higher trust than the ~50% I experienced/sent in previous rounds

I should reciprocate proportionally to their high level of trust:

If I return \$6.50:

- Sender ends with: $\$5.00 - \$3.99 + \$6.50 = \7.51 (net gain: \$2.51)
- I end with: $\$11.97 - \$6.50 = \$5.47$ (net gain: \$5.47)

This rewards their exceptional trust while both of us benefit significantly.

```
'''json
{"return": 6.50}
'''
```

reasoning:

Let me analyze this situation:

Round 3: I'm the RECEIVER

- My co-player sent \$3.99 (about 80% of their \$5 endowment)
- I received \$11.97 (after 3x multiplier)
- I need to decide how much to return

The sender showed very high trust by sending almost 80% of their endowment. This is significantly more than the ~50% I saw/sent in previous rounds.

Let me think about fair reciprocation:

If the sender sent \$3.99, they kept \$1.01.

If I return nothing, they end with \$1.01 (bad outcome for them, they lose \$3.98)

If I return everything (\$11.97), they end with $\$1.01 + \$11.97 = \$12.98$, I get \$0

I want to:

1. Reward their high trust
2. Make sure they profit from their decision
3. Keep a reasonable amount for myself
4. Signal I'm cooperative for future rounds

In Round 2, when I sent \$2.50 and received \$4.13 back, that was a good return (I sent 50%, got back enough to profit).

The sender here sent 80% - much higher trust. I should return proportionally more.

Let me calculate some options:

Option 1: Return \$6.00

- Sender ends with: $\$1.01 + \$6.00 = \$7.01$ (net gain: \$2.01)
- I end with: $\$11.97 - \$6.00 = \$5.97$ (net gain: \$5.97)
- This gives them a decent profit

- Returning \$6.50 or \$7.00 would be more generous given their 80% send, letting them gain \$2.51-\$3.01 while I still keep \$4.97-\$5.47. Since they demonstrated exceptional trust, I should lean toward the higher end to reinforce cooperation.

Looking at the pattern: they sent roughly 80% of their endowment, which is significantly more than the 50-51% in earlier rounds. My...

usage:

```
{
  "input_tokens": 1150,
  "output_tokens": 981,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.99, it became \$11.97, and you returned \$6.5. Your visible payoff was \$5.4700000000000001.

Your total visible earnings across all rounds are \$16.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience so far:

- Round 2: I sent \$2.50 and received \$4.13 back - good reciprocity
- Round 3: My co-player sent \$3.99 (high trust) and I reciprocated fairly
- The game has shown cooperative players who reward trust

Since cooperation has been working well, I'll increase my trust level from the 50% I sent in Round 2. Sending around 60% balances building cooperation while managing risk with a new co-player.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Let me review my history:

- Round 1 (RECEIVER): Saw \$2.54 sent, received \$7.62, returned \$3.50, earned \$4.12
- Round 2 (SENDER): Sent \$2.50, received \$4.13 back, earned \$6.63
- Round 3 (RECEIVER): Saw \$3.99 sent, received \$11.97, returned \$6.50, earned \$5.47
- Total earnings: \$16.22

Now Round 4, I'm the SENDER with a new co-player.

My experience so far:

- When I sent \$2.50, I got back \$4.13 (55% return rate)
- When others sent to me:
 - \$2.54 ? I returned \$3.50 (46% of what I received)
 - \$3.99 ? I returned \$6.50 (54% of what I received)

Both times I've been SENDER or RECEIVER, cooperation has been rewarded. The game seems to have cooperative players.

I've been playing moderately cooperatively:

- Round 2: sent 50%
- Could send more to build trust

Given that:

1. Previous rounds show cooperation is rewarded
2. The last sender who sent high (\$3.99, 80%) was well rewarded
3. I have accumulated good earnings and can afford some risk
4. I want to encourage cooperation

I could send around 60-70% to signal strong cooperation without being reckless.

Let me send \$3.00 (60% of endowment).

```
'''json
{"send": 3.00}
'''
```

usage:

```
{
  "input_tokens": 1464,
  "output_tokens": 523,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.71 to you. Your visible payoff was \$6.71.

Your total visible earnings across all rounds are \$22.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience:

- Round 2 (Sender): Sent \$2.50, received \$4.13 back - good return
- Round 4 (Sender): Sent \$3.00, received \$4.71 back - good return

The pattern shows that moderate-to-high trust (50-60%) has been consistently rewarded. Since cooperation is working well, I'll continue with the same level that succeeded in Round 4.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Looking at my history:

- Round 1 (Receiver): Received \$2.54 ? \$7.62, returned \$3.50. Payoff: \$4.12
- Round 2 (Sender): Sent \$2.50 ? \$7.50, got back \$4.13. Payoff: \$6.63
- Round 3 (Receiver): Received \$3.99 ? \$11.97, returned \$6.50. Payoff: \$5.47
- Round 4 (Sender): Sent \$3.00 ? \$9.00, got back \$4.71. Payoff: \$6.71

My sending strategy has been:

- Round 2: \$2.50 (50%)
- Round 4: \$3.00 (60%)

Both times I got good returns:

- \$2.50 sent ? \$4.13 back (good reciprocity)
- \$3.00 sent ? \$4.71 back (good reciprocity)

The game has shown cooperative players. I'm building earnings well. Since I'm sender again with a new co-player, I should continue showing trust. Given the successful cooperation, I could maintain or slightly increase my sending amount.

I'll send \$3.00 again (60%) as it worked well last time and establishes a consistent cooperative stance.

usage:

```
{
  "input_tokens": 1718,
  "output_tokens": 440,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$4.01 to you. Your visible payoff was \$6.01.

Your total visible earnings across all rounds are \$28.939999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.7, so you received \$8.100000000000001. How much do you return? (0-8.100000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation with my current co-player:

Previous round (Round 5) against this SAME co-player:

- I sent them \$3.00, it became \$9.00
- They returned \$4.01 (about 44.6% of what they received)
- Both of us gained

Now they've sent me \$2.70 (54% of their endowment), showing moderate trust.

I should maintain fair reciprocity with this partner. If I return \$4.10:

- Sender ends with: \$5.00 - \$2.70 + \$4.10 = \$6.40 (net gain: \$1.40)
- I end with: \$8.10 - \$4.10 = \$4.00 (net gain: \$4.00)

This is proportionally similar to their return rate in Round 5, maintaining our cooperative relationship.

```
'''json
{"return": 4.10}
'''
```

reasoning:

Looking at this situation:

In Round 5 (the immediately previous round), I was the sender against this SAME co-player:

- I sent \$3.00, it became \$9.00
- They returned \$4.01 to me
- My payoff was \$6.01

Now in Round 6, I'm the receiver against the SAME co-player:

- They sent \$2.70 (54% of their endowment)
- I received \$8.10

This is interesting - we're playing consecutive rounds with the same partner. In Round 5, they returned \$4.01 out of \$9.00 (about 44.6%). That was a reasonable return.

Now they've sent me \$2.70 (54% of endowment), which is a moderate level of trust, slightly less than the \$3.00 (60%) I sent them.

I should reciprocate fairly given our history:

- They gave me a fair return in Round 5
- They're showing moderate trust here

If I return \$4.00:

- Sender ends with: \$5.00 - \$2.70 + \$4.00 = \$6.30 (net gain: \$1.30)
- I end with: \$8.10 - \$4.00 = \$4.10 (net gain: \$4.10)

This is roughly similar to what they returned to me (about 44-49% of what they received).

It maintains fairness and reciprocity with this specific partner.

Actually, let me think more carefully. They returned \$4.01 out of \$9.00, which is 44.6%. If I apply the same percentage: $0.446 * 8.10 = \$3.61$.

But I think returning slightly more shows good faith and maintains cooperation. Something around \$4.00-\$4.20 seems fair.

Let me go with \$4.10 to be slightly more generous than they were, showing I value the ongoing cooperation.

usage:

```
{
  "input_tokens": 1985,
  "output_tokens": 686,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.7, it became \$8.100000000000001, and you returned \$4.1. Your visible payoff was \$4.000000000000002. Your total visible earnings across all rounds are \$32.94.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant**content:**

Looking at my experience as sender:

- Round 2: Sent \$2.50, received \$4.13 back
- Round 4: Sent \$3.00, received \$4.71 back
- Round 5: Sent \$3.00, received \$4.01 back

Sending around \$3.00 (60% of endowment) has consistently been rewarded with fair reciprocity. This is a new co-player, but the moderate-high trust strategy has been working well across different partners.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Looking at my history:

- Round 2 (Sender): Sent \$2.50, received \$4.13 back
- Round 4 (Sender): Sent \$3.00, received \$4.71 back
- Round 5 (Sender): Sent \$3.00, received \$4.01 back (same co-player as Round 6)
- Round 6 (Receiver): Received from same co-player who sent \$2.70

Now I'm playing against a NEW co-player in Round 7 (since the note says "against another agent" in the previous round description).

The pattern shows that sending around \$3.00 (60%) has worked well. I've been consistently reciprocated. This is a new co-player, so I should continue with a moderate-to-high trust level that has been successful.

I'll send \$3.00 again.

usage:

```
{
  "input_tokens": 2333,
  "output_tokens": 350,
  "reasoning_tokens": 0
}
```

Message 12 role=user**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.29 to you. Your visible payoff was \$6.29.

Your total visible earnings across all rounds are \$39.23.

This round, you are the RECEIVER against your current co-player. They sent you \$3.37, so you received \$10.11. How much do you return? (0-10.11)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant**content:**

Looking at this situation:

- My co-player sent \$3.37 (about 67% of their endowment) - showing good trust
- I received \$10.11 after the 3x multiplier

This is a solid level of trust, higher than the moderate amounts but not quite as high as the \$3.99 I saw in Round 3.

If I return \$5.25:

- Sender ends with: $\$5.00 - \$3.37 + \$5.25 = \6.88 (net gain: $\$1.88$)
- I end with: $\$10.11 - \$5.25 = \$4.86$ (net gain: $\$4.86$)

This proportionally rewards their good trust while both of us benefit.

```
'''json
{"return": 5.25}
'''
```

reasoning:

Let me analyze this situation:

Current round (Round 8):

- I'm the RECEIVER
- My co-player sent me $\$3.37$ (about 67% of their endowment) - showing good trust
- I received $\$10.11$ after the 3x multiplier

Looking at my history as receiver:

- Round 1: Received $\$7.62$ (from $\$2.54$ sent), returned $\$3.50$
- Round 3: Received $\$11.97$ (from $\$3.99$ sent), returned $\$6.50$
- Round 6: Received $\$8.10$ (from $\$2.70$ sent), returned $\$4.10$

And as sender, I've consistently sent $\$3.00$ and received returns around $\$4$ - $\$4.70$.

My co-player sent $\$3.37$, showing good trust (67%). I should reciprocate proportionally.

If I return $\$5.00$:

- Sender ends with: $\$5.00 - \$3.37 + \$5.00 = \6.63 (net gain: $\$1.63$)
- I end with: $\$10.11 - \$5.00 = \$5.11$ (net gain: $\$5.11$)

If I return $\$5.50$:

- Sender ends with: $\$5.00 - \$3.37 + \$5.50 = \7.13 (net gain: $\$2.13$)
- I end with: $\$10.11 - \$5.50 = \$4.61$ (net gain: $\$4.61$)

Looking at Round 3 where they sent $\$3.99$ (almost 80%) and I returned $\$6.50$, this is similar trust level (67%).

For $\$3.37$ sent (67%), returning around $\$5.00$ - $\$5.50$ seems fair and proportional to their trust level.

...

usage:

```
{
  "input_tokens": 2318,
  "output_tokens": 731,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send $\$3.37$, it became $\$10.11$, and you returned $\$5.25$. Your visible payoff was $\$4.8599999999999999$.

Your total visible earnings across all rounds are $\$44.089999999999996$.

This round, you are the RECEIVER against your current co-player. They sent you $\$2.19$, so you received $\$6.57$. How much do you return? (0-6.57)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player sent $\$2.19$ (about 44% of their endowment) - showing moderate/cautious trust
- I received $\$6.57$ after the 3x multiplier

This is lower trust than I've typically seen. I should return proportionally to encourage cooperation while being fair.

If I return $\$3.30$:

- Sender ends with: $\$5.00 - \$2.19 + \$3.30 = \6.11 (net gain: $\$1.11$)
- I end with: $\$6.57 - \$3.30 = \$3.27$ (net gain: $\$3.27$)

This gives them a reasonable gain and maintains fairness.

```
'''json
{"return": 3.30}
'''
```

Reasoning:

Looking at this situation:

- My co-player sent \$2.19 (about 44% of their endowment) - showing moderate but cautious trust
- I received \$6.57 after the 3x multiplier

This is lower trust than what I've typically been seeing. I should return proportionally to their trust level.

If I return around \$3.00:

- Sender ends with: $\$5.00 - \$2.19 + \$3.00 = \5.81 (net gain: \$0.81)
- I end with: $\$6.57 - \$3.00 = \$3.57$ (net gain: \$3.57)

This gives them a modest gain while I also benefit. It's proportionally fair given their moderate trust level.

Let me think about the return rate:

- They sent \$2.19, which became \$6.57
- Returning \$3.00 is about 45.7% of what I received
- This is similar to some of the return rates I've seen

Actually, let me be a bit more generous to encourage cooperation. If I return \$3.30:

- Sender ends with: $\$5.00 - \$2.19 + \$3.30 = \6.11 (net gain: \$1.11)
- I end with: $\$6.57 - \$3.30 = \$3.27$ (net gain: \$3.27)

That's about 50% of what I received, which is fair and encourages future cooperation.

```
Usage: {
  "input_tokens": 2389,
  "output_tokens": 541,
  "reasoning_tokens": 0
}
```

ROUND 10

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_3=Sender, Agent_4=Receiver
dyad_2: Agent_2=Sender, Agent_6=Receiver
dyad_3: Agent_8=Sender, Agent_7=Receiver
dyad_4: Agent_5=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_3 sent \$3; received \$9; Agent_4 returned \$6; payoffs Agent_3=\$8, Agent_4=\$3;
noise/visible: sent visible=\$3.97, received visible=\$11.91, returned visible=\$6.28
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 9.0, "min": 0, "raw": 6.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_2 sent \$3; received \$9; Agent_6 returned \$3.84; payoffs Agent_2=\$5.84, Agent_6=\$5.16; noise/visible: sent visible=\$2.56, received visible=\$7.68, returned visible=\$3.77
Action validation: {"returned": {"amount": 3.84, "clamped": false, "max": 9.0, "min": 0, "raw": 3.84}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_8 sent \$2.5; received \$7.5; Agent_7 returned \$4.8; payoffs Agent_8=\$7.3, Agent_7=\$2.7; noise/visible: sent visible=\$2.97, received visible=\$8.91, returned visible=\$4.33
Action validation: {"returned": {"amount": 4.8, "clamped": false, "max": 7.5, "min": 0, "raw": 4.8}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
dyad_4: Agent_5 sent \$3; received \$9; Agent_1 returned \$3.68; payoffs Agent_5=\$5.68, Agent_1=\$5.32; noise/visible: sent visible=\$2.45, received visible=\$7.35, returned visible=\$4.33
Action validation: {"returned": {"amount": 3.68, "clamped": false, "max": 9.0, "min": 0, "raw": 3.68}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 55.32,
"Agent_2": 53.17,
"Agent_3": 54.2,
"Agent_4": 55.839999999999996,
"Agent_5": 52.860000000000001,


```

"Agent_6": 55.31,
"Agent_7": 52.550000000000004,
"Agent_8": 57.749999999999999
}
Visible balances after round: {
  "Agent_1": 55.579999999999999,
  "Agent_2": 53.620000000000005,
  "Agent_3": 63.379999999999995,
  "Agent_4": 52.539999999999999,
  "Agent_5": 53.69,
  "Agent_6": 50.320000000000001,
  "Agent_7": 54.95,
  "Agent_8": 60.089999999999999
}

```

LLM CALLS

GAME - Agent_3 - Sender - dyad_1

Timestamp: 2026-08-12T09:28:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.97, it became \$8.91, and you returned \$3.75. Your visible payoff was \$5.16.

Your total visible earnings across all rounds are \$16.0.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

****Last round with THIS SAME co-player:****

- They sent me \$2.97 (59.4% of their endowment) - showing good trust
- I returned \$3.75 - rewarding their trust fairly
- We both gained from cooperation

****Now it's my turn to send:****

Since we just had a successful cooperative round together, I should maintain this pattern.

They trusted me, I reciprocated, and now I should show similar trust.

I'll send \$3.00 (60% of my endowment), matching the level of trust they showed me.

{'send': 3.00}

reasoning:

Let me analyze this situation:

****Last round with THIS SAME co-player:****

- They sent me \$2.97 (59.4% of their endowment)
- I received \$8.91
- I returned \$3.75
- They got \$5.78 total (gain of \$0.78)
- I got \$5.16 (gain of \$0.16)

****Now I'm the sender with THE SAME co-player:****

- They showed good trust last round (sent ~60%)
- I reciprocated fairly
- We both gained from cooperation

Since we're playing with the same person and we had a successful cooperative round, I should continue this pattern. They trusted me with ~60% of their endowment, and I rewarded that trust. Now I should show similar trust.

I'll send around \$3.00 again, maintaining the cooperative pattern.

usage:

```
{
  "input_tokens": 1406,
  "output_tokens": 354,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.73 to you. Your visible payoff was \$7.73.

Your total visible earnings across all rounds are \$23.73.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Last round:****

- I sent \$3.00 (60% of my endowment)
- My co-player returned \$5.73 - very generous!
- This gave me \$7.73, an excellent outcome

****Pattern so far:****

- Sending around 60% has consistently led to good reciprocation
- My average earnings are strong at \$4.75 per round
- Cooperative approach is working well

****Current round with NEW co-player:****

I'll continue the successful cooperative strategy. Sending a substantial amount demonstrates trust and has been rewarded by others.

{'send': 3.00}

reasoning:

Let me analyze this situation:

****Last round:****

- I sent \$3.00 (60% of my endowment)
- They returned \$5.73
- My payoff was \$7.73 (very good!)

This was an excellent return - they gave me back more than I would have had if I kept everything (\$5). This suggests my co-player is very cooperative and generous.

****Current round:****

- I'm the sender again with a NEW co-player
- I have \$5 to decide how much to send

Looking at the pattern:

- Round 1 (receiver): Saw sender send 62.4%, I returned fairly
- Round 2 (sender): Sent \$3.00 (60%)
- Round 3 (receiver): Saw sender send 59.4%, I returned fairly
- Round 4 (sender): Sent \$3.00 (60%) to the same co-player from round 3
- Round 5 (sender): Last round's co-player was very generous, returned \$5.73

The cooperative strategy has been working well. I've been consistently sending around 60% and have seen good reciprocation. My total earnings of \$23.73 over 5 rounds (averaging \$4.746 per round) suggests the cooperative approach is profitable.

For this round with a NEW co-player, I should continue the cooperative strategy that's been working. I'll send around \$3.00 again (60% of my endowment).

usage:

```
{
  "input_tokens": 1674,
  "output_tokens": 493,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.45 to you. Your visible payoff was \$6.45.

Your total visible earnings across all rounds are \$30.18.

This round, you are the RECEIVER against your current co-player. They sent you \$3.24, so you received \$9.72. How much do you return? (0-9.72)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

****Current round:****

- Sender showed strong trust by sending \$3.24 (64.8% of their endowment)
- This created \$9.72 for me
- Sender currently has \$1.76

****My pattern as receiver:****

- Round 1: Received \$9.36, returned \$4.00 (rewarded good trust)
- Round 3: Received \$8.91, returned \$3.75 (rewarded good trust)

This sender sent even more (64.8%) than I typically send (~60%), showing exceptional trust. I should reward this appropriately.

I'll return \$4.00, which:

- Gives the sender \$5.76 total (gain of \$0.76)
- Leaves me with \$5.72 (also a good gain)
- Rewards their high trust level
- Maintains my consistent reciprocation pattern

{'return': 4.00}

reasoning:

Let me analyze this situation:

****Current round:****

- Sender sent me \$3.24 (64.8% of their endowment) - showing strong trust
- This created \$9.72 for me
- Sender currently has \$5 - \$3.24 = \$1.76

****Previous rounds pattern:****

- Round 1: As receiver, got \$9.36, returned \$4.00 (42.7% of what I received)
- Round 2: As sender, sent \$3.00, got back \$3.48
- Round 3: As receiver, got \$8.91, returned \$3.75 (42.1% of what I received)
- Round 4: As sender, sent \$3.00 (to same co-player from Round 3)
- Round 5: As sender, sent \$3.00, got back \$5.73 (very generous)
- Round 6: As sender, sent \$3.00, got back \$4.45

So when I've been receiver:

- Round 1: returned 42.7% of received
- Round 3: returned 42.1% of received

This sender sent \$3.24 (64.8%), which is higher trust than what I typically send (60%). I should reward this high trust appropriately.

If I return around 42% of \$9.72, that would be about \$4.08.

But let me think about fairness:

- Sender started with \$5, sent \$3.24, has \$1.76 left
- If I return \$4.08, sender gets \$5.84 total (gain of \$0.84)
- I keep \$9.72 - \$4.08 = \$5.64 (gain of \$0.64)

This seems reasonable. The sender gets a decent return, and we both benefit.

Actually, let me be a bit more generous since they sent 64.8%, higher than typical. I could return \$4.25:

...

usage:

```
{
  "input_tokens": 1956,
  "output_tokens": 799,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.24, it became \$9.72, and you returned \$4.0. Your visible payoff was \$5.720000000000001.

Your total visible earnings across all rounds are \$35.9.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

****My track record as SENDER:****

- Consistently sent \$3.00 (60%)
- Returns have ranged from \$3.48 to \$5.73
- This strategy has yielded strong payoffs (\$5.48 to \$7.73)

****Overall performance:****

- Average earnings of ~\$5.98 per round
- Cooperative approach has been working well
- Both sending and receiving trust has been rewarded

****Current round with NEW co-player:****

I'll maintain my proven strategy of sending a substantial amount to encourage cooperation.

{'send': 3.00}

reasoning:

Looking at my history:

****As SENDER:****

- Round 2: Sent \$3.00, received back \$3.48 ? payoff \$5.48
- Round 4: Sent \$3.00, received back \$5.73 ? payoff \$7.73
- Round 5: Sent \$3.00, received back \$4.45 ? payoff \$6.45

****As RECEIVER:****

- Round 1: Received \$9.36, returned \$4.00 ? payoff \$5.36
- Round 3: Received \$8.91, returned \$3.75 ? payoff \$5.16
- Round 6: Received \$9.72, returned \$4.00 ? payoff \$5.72

My strategy of sending ~\$3.00 (60%) has been very successful:

- Average return when I send \$3.00: $(\$3.48 + \$5.73 + \$4.45) / 3 = \4.55
- This gives me an average payoff of \$6.55 when I'm sender

When I receive, I return fairly and get around \$5.4 on average.

****Current situation:****

- New co-player this round
- I'm the sender
- My cooperative strategy has been working very well (average earnings of \$5.98 per round)

I should continue with the successful strategy of sending \$3.00 (60%).

usage:

```
{
  "input_tokens": 2298,
  "output_tokens": 479,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$4.22 to you. Your visible payoff was \$6.22.

Your total visible earnings across all rounds are \$42.12.

This round, you are the RECEIVER against your current co-player. They sent you \$3.99, so you received \$11.97. How much do you return? (0-11.97)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

****Last round with THIS SAME co-player:****

- I sent them \$3.00 (60%)
- They returned \$4.22 - good reciprocation

****Current round:****

- They sent me \$3.99 (79.8% of their endowment) - EXCEPTIONAL trust!
- This created \$11.97 for me
- They currently have \$1.01

This is the highest trust level anyone has shown me in this game. They sent almost 80% of their endowment after we already had a successful cooperative round. This deserves strong reciprocation.

I'll return \$5.00, which:

- Gives them \$6.01 total (gain of \$1.01)
- Leaves me with \$6.97 (gain of \$1.97)
- Rewards their exceptional trust appropriately
- Maintains cooperative relationship

{'return': 5.00}

reasoning:

Looking at this situation:

****Last round with THIS SAME co-player:****

- I sent them \$3.00 (60% of my endowment)
- They returned \$4.22 - good reciprocation
- We had a successful cooperative round

****Current round:****

- They sent me \$3.99 (79.8% of their endowment) - VERY high trust!
- This created \$11.97 for me
- They currently have \$1.01

This is exceptional trust - they sent almost 80% of their endowment! This is significantly higher than:

- What I typically send (~60%)
- What they received from me last round (\$3.00)

They're showing me even more trust than I showed them. I should reward this generously.

Let me think about what's fair:

- They sent \$3.99, keeping \$1.01
- I received \$11.97
- If I return around \$5.00:
 - They'd have \$6.01 total (gain of \$1.01)
 - I'd have \$6.97 (also a good gain)

Actually, given they sent almost 80%, maybe I should return a bit more to really reward this exceptional trust.

If I return \$5.50:

- They'd have \$6.51 total (gain of \$1.51)
- I'd have \$6.47 (gain of \$1.47)
- More balanced rewards

Or if I return \$5.00:

- They'd have \$6.01 total (gain of \$1.01)

- I'd have \$6.97 (gain of \$1.97)

Let me be generous and go with \$5.00, which is a strong return and rewards their exceptional trust appropriately.

...

usage:

```
{
  "input_tokens": 2349,
  "output_tokens": 767,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.99, it became \$11.97, and you returned \$5.0. Your visible payoff was \$6.970000000000001.

Your total visible earnings across all rounds are \$49.089999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.42, so you received \$10.26. How much do you return? (0-10.26)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

****Current round:****

- Sender sent \$3.42 (68.4% of their endowment) - strong trust
- This created \$10.26 for me
- Sender currently has \$1.58

****My pattern as receiver:****

- Round 1: Received \$9.36, returned \$4.00
- Round 3: Received \$8.91, returned \$3.75
- Round 6: Received \$9.72, returned \$4.00
- Round 8: Received \$11.97 (exceptional trust), returned \$5.00

This sender showed high trust (68.4%), above my typical 60% sends. I should reward this appropriately.

I'll return \$4.25, which:

- Gives them \$5.83 total (gain of \$0.83)
- Leaves me with \$6.01 (gain of \$1.01)
- Rewards their strong trust level fairly

{'return': 4.25}

reasoning:

Looking at this situation:

****Current round:****

- Sender sent \$3.42 (68.4% of their endowment) - very high trust
- This created \$10.26 for me
- Sender currently has \$1.58

****My pattern as receiver:****

- Round 1: Received \$9.36, returned \$4.00
- Round 3: Received \$8.91, returned \$3.75
- Round 6: Received \$9.72, returned \$4.00
- Round 8: Received \$11.97, returned \$5.00

The sender showed strong trust (68.4%), significantly above the baseline 60%. This is between the "good trust" I saw in earlier rounds (60-65%) and the exceptional trust in round 8 (80%).

I should return an amount that:

- Rewards their high trust
- Gives them a reasonable gain
- Maintains my pattern of fair reciprocation

Returning \$4.25:

- Gives sender \$5.83 total (gain of \$0.83)
- Leaves me with \$6.01 (gain of \$1.01)
- Fair split of the multiplied gains

This seems appropriate for their high (but not exceptional) trust level.

usage:

```
{
  "input_tokens": 2445,
  "output_tokens": 529,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.42, it became \$10.26, and you returned \$4.25. Your visible payoff was \$6.01.

Your total visible earnings across all rounds are \$55.099999999999994.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

****My track record as SENDER:****

- Consistently sent \$3.00 (60%) across 4 rounds
- Returns have ranged from \$3.48 to \$5.73
- Average payoff as sender: ~\$6.47
- This strategy has been consistently successful

****Overall performance:****

- Average earnings of ~\$6.12 per round
- Cooperative 60% sending strategy works well
- Good balance between trust and safety

****Current round with NEW co-player:****

I'll maintain my proven strategy that has delivered strong results.

{'send': 3.00}

Reasoning:

Looking at my history:

As SENDER:

- Round 2: Sent \$3.00, received \$3.48 back (payoff \$5.48)
- Round 4: Sent \$3.00, received \$5.73 back (payoff \$7.73)
- Round 5: Sent \$3.00, received \$4.45 back (payoff \$6.45)
- Round 7: Sent \$3.00, received \$4.22 back (payoff \$6.22)

My sending strategy has been very consistent at \$3.00 (60% of endowment), and it's been working well with returns ranging from \$3.48 to \$5.73.

Average earnings across all 9 rounds so far: \$55.10 / 9 = \$6.12 per round, which is excellent.

This is Round 10 with a NEW co-player (different from Round 9).

I should continue my proven strategy of sending \$3.00. It's been consistently rewarded and leads to good outcomes.

Usage: {

```
"input_tokens": 2471,
"output_tokens": 395,
"reasoning_tokens": 0
}
```

GAME - Agent_4 - Receiver - dyad_1

Timestamp: 2026-08-12T09:28:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8


```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.07 to you. Your visible payoff was \$7.57.

Your total visible earnings across all rounds are \$17.55.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

****Analysis:****

My co-player sent \$2.9 (58% of their endowment), showing significant trust. This is a strong cooperative signal.

****Previous patterns:****

- Round 1 sender: Got back 42% return
- Round 2 receiver: I returned 46%
- Round 3 sender: Got back 58% return (very generous!)

****Current situation:****

- I received \$8.7
- Sender has \$2.1 remaining

****Decision:****

My co-player showed high trust by sending 58%. I should reciprocate proportionally. Returning 50% of what I received (\$4.35) is fair:

If I return \$4.35:

- Sender ends with: \$2.1 + \$4.35 = \$6.45 (profit of \$1.45)
- I keep: \$8.7 - \$4.35 = \$4.35 (profit of \$4.35)

This rewards their trust and maintains cooperative momentum.

```
'''json
{"return": 4.35}
'''
```

reasoning:

****Analysis of the situation:****

Round 4, I'm the RECEIVER:

- My co-player sent \$2.9 (58% of their endowment)
- I received \$8.7 after the 3x multiplication
- They currently have \$2.1 remaining (\$5 - \$2.9)

****Previous experience:****

- Round 1 (Sender): Sent \$3, got back \$3.8 (42% of \$9)
- Round 2 (Receiver): Received \$7.68, returned \$3.5 (46%)
- Round 3 (Sender): Sent \$3.5, got back \$6.07 (58% of \$10.5) - this was very generous!

****Reciprocity:****

My Round 3 co-player was very generous with a 58% return rate. This current co-player sent 58% of their endowment, showing significant trust.

Looking at patterns:

- When I received \$7.68 in Round 2, I returned \$3.5 (46%)
- Now I received \$8.7

I should be fair and encouraging. If I return around 45-50%:

- 45% would be \$3.92
- 50% would be \$4.35

Let me calculate outcomes:

If I return \$4.00:

- Sender ends with: \$2.1 + \$4.00 = \$6.10 (gain of \$1.10)

- I keep: $\$8.7 - \$4.00 = \$4.70$ (gain of $\$4.70$)

If I return $\$4.35$:

- Sender ends with: $\$2.1 + \$4.35 = \$6.45$ (gain of $\$1.45$)

...

usage:

```
{
  "input_tokens": 1485,
  "output_tokens": 766,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send $\$2.9$, it became $\$8.7$, and you returned $\$4.35$. Your visible payoff was $\$4.35$.

Your total visible earnings across all rounds are $\$21.9$.

This round, you are the SENDER against your current co-player. You have $\$5$. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

****Analysis:****

New round, new co-player. My track record:

- As sender: Sent $\$3$ (got 42% back), sent $\$3.5$ (got 58% back)
- As receiver: Returned 46% and 50%

****Strategy:****

The higher send amount ($\$3.5$) yielded much better reciprocity in Round 3. This suggests that at showing more trust encourages cooperation. I'll continue with this amount to:

- Signal strong cooperative intent
- Create significant value ($\$10.50$ for receiver)
- Build on the positive pattern

Given my fair behavior as a receiver, I'm hopeful this will be reciprocated.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Looking at my history:

- Round 1 (SENDER): Sent $\$3$, got back $\$3.8$ (42% return) - payoff $\$5.8$
- Round 2 (RECEIVER): Received $\$7.68$, returned $\$3.5$ (46% return) - payoff $\$4.18$
- Round 3 (SENDER): Sent $\$3.5$, got back $\$6.07$ (58% return) - payoff $\$7.57$
- Round 4 (RECEIVER): Received $\$8.7$, returned $\$4.35$ (50% return) - payoff $\$4.35$

Now Round 5 - I'm the SENDER against a new co-player.

Key observations:

1. When I sent more ($\$3.5$ vs $\$3$), I got a better return (58% vs 42%)
2. I've been generally fair as a receiver (46-50% returns)
3. My total earnings are $\$21.9$, averaging about $\$5.48$ per round - solid profit

Strategy considerations:

- I'm building a reputation across multiple agents
- Higher sends have been rewarded so far
- The 3x multiplier means sending more creates more total value
- I should continue signaling cooperation

I'll send $\$3.5$ or $\$4$ this round. Let me think:

- Sending $\$4$ creates $\$12$ for receiver, shows high trust
- If they return 50%, that's $\$6$, giving me $\$7$ total ($\$5 - \$4 + \$6 = \7)
- If they return 40%, that's $\$4.8$, giving me $\$5.8$ ($\$5 - \$4 + \$4.8$)

Actually, let me continue with $\$3.5$ since that worked well in Round 3. It's a solid cooperative amount without being overly risky.

usage:

```
{
  "input_tokens": 1851,
  "output_tokens": 596,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.36 to you. Your visible payoff was \$7.86.

Your total visible earnings across all rounds are \$29.759999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

****Analysis:****

My co-player sent \$2.27 (about 45% of their endowment), showing moderate trust.

****Current situation:****

- I received \$6.81
- Sender has \$2.73 remaining

****My receiver pattern:****

- Round 2: Returned 46% of received
- Round 4: Returned 50% of received

****Decision:****

I'll maintain fair reciprocity. Returning about 48% of what I received:

If I return \$3.25:

- Sender ends with: \$2.73 + \$3.25 = \$5.98 (small profit)
- I keep: \$6.81 - \$3.25 = \$3.56

This rewards their moderate trust fairly.

```
```json
{"return": 3.25}
```
```

reasoning:

****Analysis:****

My co-player sent \$2.27 (about 45% of their \$5 endowment), showing moderate trust.

****Previous patterns as receiver:****

- Round 2: Received \$7.68, returned \$3.5 (46% of received)
- Round 4: Received \$8.7, returned \$4.35 (50% of received)

****Current situation:****

- I received \$6.81
- Sender has \$2.73 remaining

****Decision:****

My co-player showed moderate trust (45% sent). I should reciprocate fairly. They sent less than in previous rounds, so I'll return proportionally.

Returning around 48% seems fair:

- 48% of \$6.81 = \$3.27

If I return \$3.27:

- Sender ends with: \$2.73 + \$3.27 = \$6.00 (profit of \$1.00)
- I keep: \$6.81 - \$3.27 = \$3.54 (profit of \$3.54)

This is fair reciprocity that rewards their trust while keeping a reasonable share.

usage:

```
{
  "input_tokens": 2166,
  "output_tokens": 466,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.27, it became \$6.810000000000005, and you returned \$3.25. Your visible payoff was \$3.560000000000005. Your total visible earnings across all rounds are \$33.32.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

****Analysis:****

This is interesting - I'm playing against the same co-player from Round 6!

****Round 6 recap:****

- They sent me \$2.27 (45% of endowment)
- I returned \$3.25
- They ended with a small profit

****Decision:****

Since they showed moderate trust in Round 6 and I reciprocated fairly, I should continue that cooperative relationship. My successful strategy has been sending \$3.5:

- Creates \$10.50 for them
- Has historically yielded good returns (58-61%)
- Signals strong reciprocal cooperation

I'll send \$3.5 to maintain our cooperative dynamic.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Interesting! This round I'm the SENDER and it says I'm playing against my "current co-player" - which is the same person from Round 6 where I was the receiver.

Let me review:

- Round 6: They sent me \$2.27, I received \$6.81, I returned \$3.25
- So they ended Round 6 with: \$5 - \$2.27 + \$3.25 = \$5.98

Now in Round 7, I'm the sender playing against the same person.

****Analysis:****

- This co-player sent me a moderate amount (\$2.27, 45%)
- I returned \$3.25, which gave them a small profit
- They showed reasonable trust

****My sender history:****

- Round 1: Sent \$3 ? got back \$3.8 (42% return rate)
- Round 3: Sent \$3.5 ? got back \$6.07 (58% return rate)
- Round 5: Sent \$3.5 ? got back \$6.36 (61% return rate)

The pattern shows that sending \$3.5 has been working well. I've been building positive relationships.

Since this is the same co-player who showed moderate trust, I should reciprocate similarly. They sent 45% of their endowment to me, so I could send around the same proportion or slightly more to reward their cooperation.

Sending \$3.5 again seems appropriate - it's been my successful strategy and matches my cooperative pattern.

```
'''json
{"send": 3.5}
'''
```

usage:

```
{
  "input_tokens": 2489,
  "output_tokens": 538,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$2.79 to you. Your visible payoff was \$4.29.

Your total visible earnings across all rounds are \$37.61.

This round, you are the RECEIVER against your current co-player. They sent you \$2.68, so you received \$8.040000000000001. How much do you return? (0-8.040000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

****Analysis:****

My co-player sent \$2.68 (about 54% of their endowment), showing good trust.

****Current situation:****

- I received \$8.04
- Sender has \$2.32 remaining

****My receiver track record:****

- Consistently returned 46-50% of received amounts
- This has built a reputation for fair play

****Decision:****

I'll maintain my fair reciprocity pattern. Returning about 48% of what I received:

If I return \$3.85:

- Sender ends with: \$2.32 + \$3.85 = \$6.17 (profit of \$1.17)
- I keep: \$8.04 - \$3.85 = \$4.19

This rewards their trust appropriately.

```
'''json
{"return": 3.85}
'''
```

reasoning:

Let me analyze this situation:

****Current situation:****

- I'm the receiver
- Co-player sent \$2.68 (about 54% of endowment) - showing good trust
- I received \$8.04
- Sender has \$2.32 remaining

****Previous rounds summary:****

- Round 2 receiver: Returned 46% (\$3.5 from \$7.68)
- Round 4 receiver: Returned 50% (\$4.35 from \$8.7)
- Round 6 receiver: Returned 48% (\$3.25 from \$6.81)

I've been consistently fair as a receiver, returning around 46-50% of what I receive.

****Round 7 note:****

My co-player from Round 6-7 only returned \$2.79 out of \$10.5 in Round 7 (about 27%). That was disappointing - much lower than the returns I've been getting.

But this is Round 8 with a potentially different co-player (unless it's the same one). Actually, looking at the multi-agent setting, pairings are randomized each round, so this could be anyone.

The sender sent \$2.68 (54% of endowment), showing decent trust. I should reciprocate fairly.

If I return around 48% of \$8.04 = \$3.86:

- Sender ends with: \$2.32 + \$3.86 = \$6.18 (profit of \$1.18)
- I keep: \$8.04 - \$3.86 = \$4.18

This maintains my reputation as a fair receiver.

Let me go with \$3.85 or \$3.9. I'll say \$3.85.

usage:

```
{
  "input_tokens": 2601,
  "output_tokens": 607,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.68, it became \$8.040000000000001, and you returned \$3.85. Your visible payoff was \$4.190000000000001.

Your total visible earnings across all rounds are \$41.8.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

****Analysis:****

New co-player for Round 9. My sender track record:

- Sent \$3: Got 42% return
- Sent \$3.5: Got 58%, 61%, and 27% returns (mixed results)

****Reflection:****

Despite one poor return in Round 7, my \$3.5 strategy has generally worked well. As a receiver, I've built a strong reputation for fair play (46-50% returns).

****Strategy:****

Continue with \$3.5 to:

- Maintain consistent cooperative signaling
- Create substantial value (\$10.50)
- Hope my reputation for fairness encourages reciprocity

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze my history:

As sender:

- Round 1: Sent \$3, got back \$3.8 (42% return rate)
- Round 3: Sent \$3.5, got back \$6.07 (58% return rate)
- Round 5: Sent \$3.5, got back \$6.36 (61% return rate)
- Round 7: Sent \$3.5, got back \$2.79 (27% return rate) - this was worse!

As receiver:

- Round 2: Received \$7.68, returned \$3.5 (46% return rate)
- Round 4: Received \$8.7, returned \$4.35 (50% return rate)
- Round 6: Received \$6.81, returned \$3.25 (48% return rate)
- Round 8: Received \$8.04, returned \$3.85 (48% return rate)

I've been consistently fair as a receiver, returning around 46-50% of what I receive.

As a sender, I've had mixed results. My \$3.5 strategy worked well in rounds 3 and 5, but in round 7 it got a poor return (only 27%).

This is Round 9, playing against a new co-player. My overall approach has been to signal cooperation by sending \$3.5. While I had one bad round, overall this strategy has worked well.

Let me continue with \$3.5 to maintain my cooperative reputation.

usage:

```
{
  "input_tokens": 2574,
  "output_tokens": 522,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$3.33 to you. Your visible payoff was \$4.83.

Your total visible earnings across all rounds are \$46.629999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$3.97, so you received \$11.91. How much do you return? (0-11.91)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

****Analysis:****

My co-player sent \$3.97 (nearly 80% of their endowment) - this is exceptionally high trust!

****Current situation:****

- I received \$11.91
- Sender has only \$1.03 remaining

****My receiver pattern:****

- Consistently returned 46-50% of received amounts
- This has established my reputation for fairness

****Decision:****

This exceptional trust deserves generous reciprocity. Returning about 50% (\$6.00):

If I return \$6.00:

- Sender ends with: \$1.03 + \$6.00 = \$7.03 (profit of \$2.03)
- I keep: \$11.91 - \$6.00 = \$5.91 (very good profit)

This rewards their exceptional trust while maintaining my fair play pattern.

```
'''json
{"return": 6.0}
'''
```

Reasoning:

Let me analyze this situation:

****Current situation:****

- Co-player sent \$3.97 (about 79% of their endowment) - this is very high trust!
- I received \$11.91
- Sender has \$1.03 remaining

****My receiver pattern:****

- Round 2: Returned 46% of received
- Round 4: Returned 50% of received
- Round 6: Returned 48% of received
- Round 8: Returned 48% of received

So I've been consistently returning 46-50% of what I receive, which is fair and builds trust.

****Decision:****

This person sent nearly 80% of their endowment - that's exceptional trust! I should reciprocate generously.

If I return about 50% of what I received:

- Return: \$5.95 or \$6.00

- Sender ends with: $\$1.03 + \$6.00 = \$7.03$ (profit of $\$2.03$)
- I keep: $\$11.91 - \$6.00 = \$5.91$ (profit of $\$5.91$)

This still gives me a very good profit while rewarding their exceptional trust. Let me go with $\$6.00$ to be nice and round.

Actually, let me think about this more carefully. They sent $\$3.97$, which means they kept $\$1.03$.

- . If I return $\$6.00$:
- They end with $\$7.03$ total (gain of $\$2.03$)
- I keep $\$5.91$ (gain of $\$5.91$)

This seems fair given their high trust. I'll return $\$6.00$.

```
Usage: {
  "input_tokens": 2544,
  "output_tokens": 600,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Sender - dyad_2

Timestamp: 2026-08-12T09:29:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_6"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive $\$5$ each round
- You decide how much to SEND to the RECEIVER ($\$0$ - $\$5$)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

When you are the RECEIVER:

- Sender receives $\$5$
- Sender decides how much to SEND to you ($\$0$ - $\$5$)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: $\$5 - \text{sent} + \text{returned}$
- You keep: $\text{received} - \text{returned}$

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.71 to you. Your visible payoff was \$5.71.

Your total visible earnings across all rounds are \$14.91.

This round, you are the RECEIVER against your current co-player. They sent you \$3.79, so you received \$11.370000000000001. How much do you return? (0-11.370000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

****What happened:****

- My co-player sent \$3.79 out of \$5 (~76% of their endowment)
- This became \$11.37 for me
- They showed strong trust by sending most of their money

****My reasoning:****

- They took a big risk trusting me
- I should reward this trust to encourage cooperation
- Returning around 45% of what I received seems appropriate given their high trust
- This would be approximately \$5.10

****Calculating outcomes:****

- They'll end with: \$1.21 + \$5.10 = \$6.31
- I'll end with: \$11.37 - \$5.10 = \$6.27
- They gain \$1.31 from their trust, which is a good reward

{'return': 5.10}

reasoning:

Looking at this situation:

****What happened:****

- My co-player sent \$3.79 (became \$11.37 for me)
- They kept \$1.21
- They showed significant trust by sending most of their money

****Previous experience:****

- Round 1: I sent \$3, got back \$4.41 (they returned ~49%)
- Round 2: I received \$5.19, returned \$2.40 (~46%)
- Round 3: I sent \$3, got back \$3.71 (they returned ~41%)

****My reasoning:****

- This co-player sent \$3.79 out of \$5, showing strong trust (~76%)

- They're risking a lot and trusting me to reciprocate
- I should reward this trust appropriately
- Returning around 40-50% of what I received would be \$4.55-\$5.69
- Let me think about fairness:
 - They sent \$3.79, kept \$1.21
 - I received \$11.37
 - If I return \$5.00:
 - They end with: $\$1.21 + \$5.00 = \$6.21$
 - I end with: $\$11.37 - \$5.00 = \$6.37$
 - If I return \$5.50:
 - They end with: $\$1.21 + \$5.50 = \$6.71$
 - I end with: $\$11.37 - \$5.50 = \$5.87$

Given they sent \$3.79 (76% of their endowment), I should be generous but not overdo it. Around 45% return rate seems appropriate, which would be about \$5.12.

Let me go with \$5.10:

...

usage:

```
{
  "input_tokens": 1453,
  "output_tokens": 690,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.79, it became \$11.370000000000001, and you returned \$5.1. Your visible payoff was \$6.270000000000001.

Your total visible earnings across all rounds are \$21.18.

This round, you are the RECEIVER against your current co-player. They sent you \$2.91, so you received \$8.73. How much do you return? (0-8.73)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****What happened:****

- My co-player sent \$2.91 out of \$5 (~58% of their endowment)
- This became \$8.73 for me
- They kept \$2.09
- This shows moderate to good trust

****My reasoning:****

- They demonstrated decent trust by sending over half their money
- I should reciprocate consistently with my previous rounds
- In previous rounds as receiver, I've returned around 45% of what I received
- For \$8.73, that would be approximately \$3.90

****Calculating outcomes:****

- They'll end with: $\$2.09 + \$3.90 = \$5.99$
- I'll end with: $\$8.73 - \$3.90 = \$4.83$
- They gain about \$1 from trusting me, which is a reasonable reward

{ 'return': 3.90 }

reasoning:

Looking at this situation:

****What happened:****

- My co-player sent \$2.91 out of \$5 (~58% of their endowment)
- This became \$8.73 for me
- They kept \$2.09
- This shows moderate to good trust

****My reasoning:****

- They sent a decent amount, showing trust
- I should reciprocate proportionally

- Looking at my previous returns:
 - Round 2: Received \$5.19, returned \$2.40 (~46%)
 - Round 4: Received \$11.37, returned \$5.10 (~45%)
- I've been returning around 45% consistently

For \$8.73, 45% would be about \$3.93

****Calculating outcomes with \$3.93:****

- They'll end with: $\$2.09 + \$3.93 = \$6.02$
- I'll end with: $\$8.73 - \$3.93 = \$4.80$
- They gain \$1.02 from sending, which rewards their trust

This seems fair and consistent with my previous behavior. I'll return \$3.90 to keep it simple.

Actually, let me recalculate with \$3.90:

- They'll end with: $\$2.09 + \$3.90 = \$5.99$
- I'll end with: $\$8.73 - \$3.90 = \$4.83$
- They gain \$0.99 from sending

That works well.

usage:

```
{
  "input_tokens": 1786,
  "output_tokens": 560,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.91, it became \$8.73, and you returned \$3.9. Your visible payoff was \$4.83.

Your total visible earnings across all rounds are \$26.009999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history as SENDER:

****Previous rounds as sender:****

- Round 1: Sent \$3 ? received \$4.41 back (profit of \$1.41)
- Round 3: Sent \$3 ? received \$3.71 back (profit of \$0.71)

Both times sending \$3 worked out profitably, showing it's a reasonable cooperative strategy.

****Current situation:****

- New co-player (randomized pairing)
- Want to maintain consistent cooperative approach
- \$3 creates \$9 for them, leaving room for mutual benefit

I'll stick with my established strategy of sending \$3, which has proven effective:

```
{'send': 3}
```

reasoning:

Looking at my history:

Round 1 (SENDER): Sent \$3, received \$4.41 back, payoff \$6.41
 Round 2 (RECEIVER): Received \$5.19, returned \$2.40, payoff \$2.79
 Round 3 (SENDER): Sent \$3, received \$3.71 back, payoff \$5.71
 Round 4 (RECEIVER): Received \$11.37, returned \$5.10, payoff \$6.27
 Round 5 (RECEIVER): Received \$8.73, returned \$3.90, payoff \$4.83

Total earnings: \$26.01

Now Round 6, I'm the SENDER with a new randomized co-player.

My strategy has been to send \$3 as sender (both times I did this), which has worked out reasonably well:

- Round 1: Sent \$3, got \$4.41 back (net +\$1.41)
- Round 3: Sent \$3, got \$3.71 back (net +\$0.71)

Both times I profited, though the return varied.

As receiver, I've been consistent in returning around 45% of what I receive, which seems fair.

For this new round with a new partner, I should continue with my cooperative strategy. Sending \$3 has been my standard approach and has generally worked well. It signals cooperation while not being overly risky.

I'll send \$3 again.

usage:

```
{
  "input_tokens": 2117,
  "output_tokens": 509,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$3.68 to you. Your visible payoff was \$5.68.

Your total visible earnings across all rounds are \$31.689999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.51, so you received \$7.529999999999999. How much do you return? (0-7.529999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

****What happened:****

- Same co-player from Round 6 (where I sent them \$3 and they returned \$3.68)
- Now they sent me \$2.51, which became \$7.53
- They kept \$2.49
- They showed moderate trust

****My reasoning:****

- They were reasonably cooperative in Round 6 (returned ~41% of what they received)
- Now they're trusting me with about half their endowment
- I should maintain my consistent reciprocation pattern
- I've been returning around 45% of what I receive as a receiver

****For \$7.53:****

- 45% = approximately \$3.39

****Calculating outcomes:****

- They'll end with: \$2.49 + \$3.39 = \$5.88
- I'll end with: \$7.53 - \$3.39 = \$4.14
- They gain \$0.88 from their trust

{'return': 3.39}

reasoning:

Looking at this situation:

****What happened:****

- Same co-player from Round 6 (I was sender then, now I'm receiver)
- In Round 6: I sent \$3, they returned \$3.68
- Now Round 7: They sent \$2.51, became \$7.53 for me
- They kept \$2.49

****My reasoning:****

- This is the same partner who just returned \$3.68 on \$9 (~41% return rate)

- They showed reasonable cooperation
- Now they sent \$2.51 (~50% of endowment), showing moderate trust
- I should reciprocate fairly

****My return pattern as receiver:****

- Round 2: Received \$5.19, returned \$2.40 (46%)
- Round 4: Received \$11.37, returned \$5.10 (45%)
- Round 5: Received \$8.73, returned \$3.90 (45%)

I've been consistently returning around 45% of what I receive.

For \$7.53:

- 45% would be approximately \$3.39

****Calculating outcomes:****

- They'll end with: \$2.49 + \$3.39 = \$5.88
- I'll end with: \$7.53 - \$3.39 = \$4.14
- They gain \$0.88 from trusting me

This seems fair and consistent with my pattern.

usage:

```
{
  "input_tokens": 2425,
  "output_tokens": 569,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.51, it became \$7.529999999999999, and you returned \$3.39. Your visible payoff was \$4.139999999999999.

Your total visible earnings across all rounds are \$35.83.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my history as SENDER:

****Previous rounds as sender:****

- Round 1: Sent \$3 ? received \$4.41 back (profit of \$1.41)
- Round 3: Sent \$3 ? received \$3.71 back (profit of \$0.71)
- Round 6: Sent \$3 ? received \$3.68 back (profit of \$0.68)

****Pattern:****

- Sending \$3 has consistently worked well
- All three co-players returned more than I sent
- This shows a good balance of trust and caution

****Current situation:****

- New co-player (randomized pairing)
- My established strategy has proven reliable

I'll continue with sending \$3:

```
{'send': 3}
```

reasoning:

Looking at my history as SENDER:

****Previous rounds as sender:****

- Round 1: Sent \$3 ? received \$4.41 back (profit of \$1.41)
- Round 3: Sent \$3 ? received \$3.71 back (profit of \$0.71)
- Round 6: Sent \$3 ? received \$3.68 back (profit of \$0.68)

All three times I've sent \$3, I've gotten back more than I sent, making profitable exchanges. The strategy is working consistently.

****Current situation:****

- Round 8 (final round likely, or near it)
- New co-player again (randomized pairing)
- Want to maintain cooperative approach

I should stick with \$3 again as it has consistently worked well across multiple partners:

```
{'send': 3}
```

usage:

```
{  
  "input_tokens": 2608,  
  "output_tokens": 378,  
  "reasoning_tokens": 0  
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.21 to you. Your visible payoff was \$6.21.
Your total visible earnings across all rounds are \$42.04.

This round, you are the RECEIVER against your current co-player. They sent you \$3.52, so you received \$10.56. How much do you return? (0-10.56)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

****What happened:****

- My co-player sent \$3.52 out of \$5 (~70% of their endowment)
- This became \$10.56 for me
- They kept \$1.48
- They showed strong trust

****My reasoning:****

- They demonstrated strong trust by sending most of their money
- I should maintain my consistent reciprocation pattern
- I've been returning around 45% of what I receive as a receiver
- For \$10.56, that would be approximately \$4.75

****Calculating outcomes:****

- They'll end with: $\$1.48 + \$4.75 = \$6.23$
- I'll end with: $\$10.56 - \$4.75 = \$5.81$
- They gain \$1.23 from their trust, which is a good reward

```
{'return': 4.75}
```

reasoning:

Looking at this situation:

****What happened:****

- My co-player sent \$3.52 out of \$5 (~70% of their endowment)
- This became \$10.56 for me
- They kept \$1.48
- This shows strong trust

****My reasoning:****

- They demonstrated strong trust by sending over 70% of their money
- I should reciprocate consistently with my previous rounds
- In previous rounds as receiver, I've been returning around 45% of what I received
- For \$10.56, that would be approximately \$4.75

****Calculating outcomes:****

- They'll end with: $\$1.48 + \$4.75 = \$6.23$
- I'll end with: $\$10.56 - \$4.75 = \$5.81$
- They gain about \$1.23 from trusting me, which is a good reward

Let me check my previous returns as receiver:

- Round 2: Received \$5.19, returned \$2.40 (46%)

- Round 4: Received \$11.37, returned \$5.10 (45%)
- Round 5: Received \$8.73, returned \$3.90 (45%)
- Round 7: Received \$7.53, returned \$3.39 (45%)

So I've been consistently returning around 45% of what I receive.

For \$10.56:

$$45\% = \$10.56 \times 0.45 = \$4.75$$

This seems appropriate.

usage:

```
{
  "input_tokens": 2568,
  "output_tokens": 559,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.52, it became \$10.56, and you returned \$4.75. Your visible payoff was \$5.8100000000000005.

Your total visible earnings across all rounds are \$47.85.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history as SENDER:

****Previous rounds as sender:****

- Round 1: Sent \$3 ? received \$4.41 back (profit of \$1.41)
- Round 3: Sent \$3 ? received \$3.71 back (profit of \$0.71)
- Round 6: Sent \$3 ? received \$3.68 back (profit of \$0.68)
- Round 8: Sent \$3 ? received \$4.21 back (profit of \$1.21)

****Pattern:****

- Sending \$3 has been consistently profitable
- All co-players have returned more than I sent
- This demonstrates a reliable cooperative strategy

****Current situation:****

- New co-player (randomized pairing)
- My \$3 strategy has worked well across different partners

I'll stick with my proven approach:

```
{'send': 3}
```

Reasoning:

Looking at my history as SENDER:

****Previous rounds as sender:****

- Round 1: Sent \$3 ? received \$4.41 back (profit of \$1.41)
- Round 3: Sent \$3 ? received \$3.71 back (profit of \$0.71)
- Round 6: Sent \$3 ? received \$3.68 back (profit of \$0.68)
- Round 8: Sent \$3 ? received \$4.21 back (profit of \$1.21)

****Pattern:****

- I've consistently sent \$3 every time I'm the sender
- All four times, I've received more back than I sent
- My returns have ranged from \$3.68 to \$4.41
- All have been profitable

****Current situation:****

- New co-player (randomized pairing)
- My strategy of sending \$3 has been very consistent and successful
- No reason to deviate from this pattern

I'll continue with my established strategy of sending \$3.


```
Usage: {
  "input_tokens": 2584,
  "output_tokens": 450,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_2

Timestamp: 2026-08-12T09:29:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_6"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$2.98 to you. Your visible payoff was \$4.98.

Your total visible earnings across all rounds are \$16.71.

This round, you are the RECEIVER against your current co-player. They sent you \$3.24, so you received \$9.72. How much do you return? (0-9.72)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation with my current co-player:

****Last round (Round 3):****

- I sent them \$3.00
- They returned \$2.98 (~33% of what they received)
- This was a modest but acceptable return

****This round (Round 4):****

- They sent me \$3.24 (slightly MORE than I sent them)
- I received \$9.72
- They currently have only \$1.76

The fact that they sent me MORE than I sent them suggests they're trying to build cooperation. I should reciprocate generously to:

1. Reward their increased trust
2. Establish a positive cooperative dynamic
3. Ensure we both benefit from ongoing interaction

I'll return ****\$4.80****:

- They end with: $\$1.76 + \$4.80 = \$6.56$ (gain \$1.56)
- I keep: $\$9.72 - \$4.80 = \$4.92$ (gain \$4.92)

This is more generous than they were to me, encouraging further cooperation.

{'return': 4.80}

reasoning:

Let me analyze this situation:

Round 3 (previous round):

- I was the SENDER
- I sent \$3.00 to this same co-player
- They returned \$2.98
- My payoff was \$4.98 (kept \$2 + returned \$2.98)

Round 4 (current round):

- I am the RECEIVER
- Same co-player (they are now the sender)
- They sent me \$3.24
- I received \$9.72

So this is interesting - we're playing repeated games with the same partner. In round 3, I sent \$3.00 and they returned \$2.98 (about 33% of what they received). Now they've sent me \$3.24 (notably a bit more than I sent them).

Let me think about the fairness here:

- When I sent \$3.00, they kept $\$9.00 - \$2.98 = \$6.02$

- I kept $\$2.00 + \$2.98 = \$4.98$
- So they gained $\$1.02$, I lost $\$0.02$

Now they sent $\$3.24$:

- They kept $\$5 - \$3.24 = \$1.76$
- I received $\$9.72$

If I want to reciprocate similarly to how they treated me, I might return around 33% of what I received, which would be about $\$3.21$.

But actually, they sent me slightly MORE than I sent them ($\$3.24$ vs $\$3.00$), which could be seen as trying to establish cooperation or make up for the relatively low return last round.

Let me think about different return amounts:

Option 1: Return $\$3.00$ (about 31%)

- They end with: $\$1.76 + \$3.00 = \$4.76$ (lost $\$0.24$)
- I keep: $\$9.72 - \$3.00 = \$6.72$ (gain $\$6.72$)
- This feels too stingy given their slightly larger initial send.

Option 2: Return $\$4.00$ (about 41%)

- They end with: $\$1.76 + \$4.00 = \$5.76$ (gain $\$0.76$)
- I keep: $\$9.72 - \$4.00 = \$5.72$ (gain $\$5.72$)
- This creates mutual benefit and shows reciprocal trust.

Option 3: Return $\$5.00$ (about 51%)

- They end with: $\$1.76 + \$5.00 = \$6.76$ (gain $\$1.76$)
- I keep: $\$9.72 - \$5.00 = \$4.72$...

usage:

```
{
  "input_tokens": 1431,
  "output_tokens": 1067,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send $\$3.24$, it became $\$9.72$, and you returned $\$4.8$. Your visible payoff was $\$4.9200000000000001$.

Your total visible earnings across all rounds are $\$21.6300000000000003$.

This round, you are the RECEIVER against your current co-player. They sent you $\$2.32$, so you received $\$6.959999999999999$. How much do you return? (0-6.959999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- New co-player sent me $\$2.32$ (about 46% of their endowment)
- I received $\$6.96$ after the multiplier
- They currently have $\$2.68$

They showed moderate trust by sending nearly half their endowment. I should reciprocate fairly to reward their trust and ensure mutual benefit.

Following my pattern of returning roughly 50% of what I receive, I'll return **$\$3.48$** :

- Sender ends with: $\$2.68 + \$3.48 = \$6.16$ (gain $\$1.16$)
- I keep: $\$6.96 - \$3.48 = \$3.48$ (gain $\$3.48$)

Both of us benefit from the cooperation.

```
{'return': 3.48}
```

reasoning:

Looking at this round:

- New co-player (different from Round 4)
- They sent me $\$2.32$ (about 46% of their endowment)
- I received $\$6.96$

- They currently have \$2.68 (\$5 - \$2.32)

They showed moderate trust by sending nearly half their endowment. I should reciprocate fairly to:

1. Reward their trust
2. Ensure both of us benefit
3. Encourage cooperation

Looking at my previous patterns:

- Round 1: Received \$8.91, returned \$4.50 (about 50%)
- Round 4: Received \$9.72, returned \$4.80 (about 49%)

Returning around 50% seems like my consistent fair approach. For \$6.96, that would be around \$3.48.

Let me check the outcomes:

- Sender ends with: \$2.68 + \$3.48 = \$6.16 (gain \$1.16)
- I keep: \$6.96 - \$3.48 = \$3.48 (gain \$3.48)

Both benefit, and this reciprocates their trust fairly. I'll return \$3.48.

usage:

```
{
  "input_tokens": 1834,
  "output_tokens": 476,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.32, it became \$6.959999999999999, and you returned \$3.48. Your visible payoff was \$3.479999999999999.

Your total visible earnings across all rounds are \$25.110000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history with this current co-player:

****Round 5:****

- They sent me \$2.32 (46% of endowment)
- I returned \$3.48, giving them a good outcome

They showed moderate trust, and I reciprocated fairly. Now it's my turn to send, and I should continue building this cooperative relationship.

I'll send **\$2.50**, which is:

- Slightly more than they sent me (showing willingness to cooperate)
- A reasonable amount that matches their moderate approach
- Enough to create mutual benefit if they reciprocate

```
{'send': 2.50}
```

reasoning:

Looking at my history with this current co-player:

Round 5 (last round):

- They sent me \$2.32 (46% of endowment)
- I received \$6.96
- I returned \$3.48 (50% of what I received)
- They ended with \$6.16, I ended with \$3.48

Now in Round 6, I'm the sender with this same co-player.

They showed moderate trust by sending \$2.32. I reciprocated fairly by returning \$3.48, which gave them a good outcome (\$6.16 total).

Since we had a successful cooperative interaction, I should continue building on this. I'll

I send a similar or slightly higher amount to maintain the cooperative relationship.

I've been generally sending around \$3.00 (60%) in other rounds, which has worked well. Given this co-player sent me \$2.32 (46%), I could:

1. Send similar to what they sent (~\$2.30-2.50)
2. Send my standard amount (~\$3.00)
3. Send slightly more to build trust

I think sending around \$2.50-3.00 makes sense. Let me go with \$2.50, which is slightly more than they sent me, showing I'm willing to cooperate but matching their level of caution.

Actually, looking at my overall strategy, I've been doing well with \$3.00 sends. But this particular co-player was more cautious. Let me match their level at around \$2.30-2.50.

I'll send \$2.50.

usage:

```
{
  "input_tokens": 2153,
  "output_tokens": 510,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$4.52 to you. Your visible payoff was \$7.02.

Your total visible earnings across all rounds are \$32.13.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my recent experiences:

Round 6 outcome:

- I sent \$2.50
- Received \$4.52 back
- Ended with \$7.02 - very good!

The cooperative approach continues to work well. Now with a new co-player, I'll signal trust and willingness to cooperate by sending a solid amount.

I'll send ****\$3.00**** (60% of endowment), which has proven to be an effective strategy:

```
{'send': 3.00}
```

reasoning:

Looking at my recent experience:

Round 6:

- I sent \$2.50 to my co-player
- They returned \$4.52
- I ended with \$7.02 - excellent outcome!

This is a new co-player in Round 7. I've been doing well with cooperative strategies. My approach of sending moderate to high amounts (around 50-60% of endowment) has generally worked well across different co-players.

Let me review the pattern:

- Round 2: Sent \$3.00, received back \$5.32 - great
- Round 3: Sent \$3.00, received back \$2.98 - okay
- Round 6: Sent \$2.50, received back \$4.52 - great

The cooperative strategy continues to work. With a new co-player, I should send a reasonable amount to signal willingness to cooperate. I'll send around \$3.00 (60%) to establish trust.

usage:

```
{
  "input_tokens": 2419,
  "output_tokens": 343,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.97 to you. Your visible payoff was \$5.970000000000001.

Your total visible earnings across all rounds are \$38.1.

This round, you are the RECEIVER against your current co-player. They sent you \$2.48, so you received \$7.439999999999995. How much do you return? (0-7.439999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- Co-player sent me \$2.48 (about 50% of their endowment)
- I received \$7.44 after the multiplier
- They currently have \$2.52

They showed solid trust by sending nearly half their endowment. I'll reciprocate fairly.

I'll return **\$3.72** (about 50% of what I received):

- Sender ends with: \$2.52 + \$3.72 = \$6.24 (gain \$1.24)
- I keep: \$7.44 - \$3.72 = \$3.72 (gain \$3.72)

Both of us benefit from the cooperation.

{'return': 3.72}

reasoning:

Looking at the situation:

- New co-player sent me \$2.48 (about 50% of their endowment)
- I received \$7.44 after the multiplier
- They currently have \$2.52

They showed substantial trust by sending nearly half their endowment. I should reciprocate fairly.

Following my pattern of fair returns:

- If I return about 50% of what I receive: $\$7.44 * 0.5 = \3.72

This would give:

- Sender: $\$2.52 + \$3.72 = \$6.24$ (gain \$1.24)
- Me: $\$7.44 - \$3.72 = \$3.72$ (gain \$3.72)

Both benefit, and I'm being fair to their trust. This seems reasonable.

Let me return \$3.72.

usage:

```
{
  "input_tokens": 2365,
  "output_tokens": 378,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.48, it became \$7.439999999999995, and you returned \$3.72. Your visible payoff was \$3.719999999999993.

Your total visible earnings across all rounds are \$41.82.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my gameplay so far:

My cooperative strategy has been working well overall. In previous rounds as sender:

- Sending \$3.00 has yielded good returns (like Round 3: sent \$3, got back \$5.32)
- Sending \$2.50 also worked well (Round 6: sent \$2.50, got back \$4.52)

With a new co-player, I'll continue signaling strong cooperation by sending a substantial amount.

I'll send **\$3.00**:

{'send': 3.00}

reasoning:

Looking at my situation:

Round 8 results:

- Previous co-player sent me \$2.48
- I returned \$3.72
- I kept \$3.72

Now I'm the sender with a new co-player in round 9. I have \$5 to decide how much to send.

Looking at my overall pattern:

- I've been generally cooperative, sending between \$2.50-\$3.00 typically
- My cooperative approaches have been working well
- Total earnings are \$41.82

I should continue with a cooperative strategy. Sending around \$3 has been effective in previous rounds.

I'll send \$3.00.

usage:

```
{
  "input_tokens": 2400,
  "output_tokens": 282,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.66 to you. Your visible payoff was \$4.66.

Your total visible earnings across all rounds are \$46.480000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$2.56, so you received \$7.68. How much do you return? (0-7.68)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Co-player sent me \$2.56 (about 51% of their endowment)
- I received \$7.68 after the multiplier
- They currently have \$2.44

They showed solid trust by sending over half their endowment. I'll reciprocate fairly.

I'll return **\$3.84** (50% of what I received):

- Sender ends with: \$2.44 + \$3.84 = \$6.28 (gain \$1.28)
- I keep: \$7.68 - \$3.84 = \$3.84 (gain \$3.84)

Both of us benefit from the cooperation.

{'return': 3.84}

Reasoning:

Looking at the situation:

- Co-player sent me \$2.56 (about 51% of their endowment)
- I received \$7.68 after the multiplier
- They currently have \$2.44

They showed solid trust by sending over half their endowment. I should reciprocate fairly.

Following my pattern of returning roughly 50% of what I receive, I'll return about \$3.84:

- Sender ends with: $\$2.44 + \$3.84 = \$6.28$ (gain \$1.28)
- I keep: $\$7.68 - \$3.84 = \$3.84$ (gain \$3.84)

Both of us benefit from the cooperation. This maintains my cooperative strategy.

```
Usage: {  
  "input_tokens": 2427,  
  "output_tokens": 348,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_8 - Sender - dyad_3

Timestamp: 2026-08-12T09:29:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agents": [  
      "Agent_8",  
      "Agent_7"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_7": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_7"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: $\$5 - \text{sent} + \text{returned}$

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$5.25 to you. Your visible payoff was \$7.75.

Your total visible earnings across all rounds are \$18.48.

This round, you are the RECEIVER against your current co-player. They sent you \$3.69, so you received \$11.07. How much do you return? (0-11.07)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed very high trust by sending \$3.69 (73.8% of their endowment)
- They now have only \$1.31 remaining
- I received \$11.07 after the 3x multiplier
- Total pool is \$12.38

This sender took a significant risk to create mutual gains. I should reciprocate their strong trust signal generously.

If I return \$5.50:

- Sender ends with \$6.81 (gaining \$1.81)
- I keep \$5.57

This rewards their exceptional trust while maintaining fairness for both of us.

{"return": 5.50}

reasoning:

Let me analyze this situation:

Round 4 - I'm the RECEIVER

- The sender sent \$3.69 (73.8% of their \$5 endowment)
- This became \$11.07 after the 3x multiplier
- The sender now has \$5 - \$3.69 = \$1.31 remaining
- Total pool: \$1.31 + \$11.07 = \$12.38

The sender showed very high trust by sending 73.8% of their endowment. This is much higher than the ~50% I've been sending, indicating strong cooperative intent.

I should reciprocate this high level of trust. Let me think about fair returns:

If I return \$5.50:

- Sender ends with: \$1.31 + \$5.50 = \$6.81 (gain of \$1.81)

- I keep: $\$11.07 - \$5.50 = \$5.57$

If I return \$6.00:

- Sender ends with: $\$1.31 + \$6.00 = \$7.31$ (gain of \$2.31)
- I keep: $\$11.07 - \$6.00 = \$5.07$

Given they sent 73.8% of their endowment (very high trust), I should return a proportionally generous amount. Returning around \$5.50-\$6.00 would:

1. Reward their high trust
2. Ensure they profit from being cooperative
3. Still leave me with a reasonable amount
4. Signal I'm reciprocal and trustworthy

I'll return \$5.50 - this gives them a solid gain while being fair to both parties.

usage:

```
{
  "input_tokens": 1343,
  "output_tokens": 551,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.69, it became \$11.07, and you returned \$5.5. Your visible payoff was \$5.57.

Your total visible earnings across all rounds are \$24.05.

This round, you are the RECEIVER against your current co-player. They sent you \$3.88, so you received \$11.64. How much do you return? (0-11.64)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- The sender showed extremely high trust by sending \$3.88 (77.6% of their endowment)
- They now have only \$1.12 remaining
- I received \$11.64 after the 3x multiplier
- Total pool is \$12.76

This is even higher trust than the previous round. I should reciprocate this exceptional trust generously.

If I return \$5.80:

- Sender ends with \$6.92 (gaining \$1.92)
- I keep \$5.84

This appropriately rewards their very high trust level while maintaining fairness.

{"return": 5.80}

reasoning:

Looking at this situation:

- The sender sent \$3.88 (77.6% of their endowment) - even higher trust than the previous round
- They now have \$1.12 remaining
- I received \$11.64 after the 3x multiplier
- Total pool is \$12.76

This sender showed extremely high trust. I should reciprocate proportionally.

In the last round, sender sent \$3.69 (73.8%) and I returned \$5.50, giving them \$6.81 total (gain of \$1.81).

This sender sent even more (\$3.88, or 77.6%). I should return proportionally or slightly more to reward the higher trust.

If I return \$5.80:

- Sender ends with \$6.92 (gain of \$1.92)

- I keep \$5.84

This seems fair and rewards their high trust appropriately.

Let me consider other options:

- Return \$6.00: Sender gets \$7.12 (gain \$2.12), I keep \$5.64
- Return \$5.50: Sender gets \$6.62 (gain \$1.62), I keep \$6.14

I think \$5.80 is a good middle ground that appropriately rewards the very high trust level while maintaining fairness.

usage:

```
{
  "input_tokens": 1628,
  "output_tokens": 459,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.88, it became \$11.64, and you returned \$5.8. Your visible payoff was \$5.840000000000001.

Your total visible earnings across all rounds are \$29.89.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience as sender:

- Round 2: Sent \$2.50, received \$3.19 back (good return)
- Round 3: Sent \$2.50, received \$5.25 back (excellent return)

The 50% strategy has worked very well for me. The cooperative environment suggests continuing this approach with a new co-player.

```
{"send": 2.50}
```

reasoning:

Let me review my experience so far:

Round 1 (RECEIVER): Received from sender who sent \$2.68 (53.6%), I returned \$3.00

Round 2 (SENDER): I sent \$2.50 (50%), received back \$3.19

Round 3 (SENDER): I sent \$2.50 (50%), received back \$5.25

Round 4 (RECEIVER): Received from sender who sent \$3.69 (73.8%), I returned \$5.50

Round 5 (RECEIVER): Received from sender who sent \$3.88 (77.6%), I returned \$5.80

As a sender, I've been sending 50% and getting good returns. I've encountered receivers who are cooperating well. I've also been a generous receiver when others send high amounts.

The pattern suggests a cooperative environment. When I send 50%, people return enough for me to profit. When others send high amounts (70-78%), I reciprocate generously.

I'm now facing a new co-player as sender. Should I:

- Continue with 50% (\$2.50) - safe, proven strategy
- Increase to show more trust - could signal cooperation and encourage higher returns
- Decrease to be more cautious - but this might seem less cooperative

Given the cooperative environment and good outcomes when I've sent \$2.50, I'll stick with that strategy.

usage:

```
{
  "input_tokens": 1902,
  "output_tokens": 421,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.97 to you. Your visible payoff was \$6.470000000000001.

Your total visible earnings across all rounds are \$36.36.

This round, you are the RECEIVER against your current co-player. They sent you \$3.1, so you received \$9.3. How much do you return? (0-9.3)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- The sender sent \$3.10 (62% of their endowment), showing good trust
- They now have \$1.90 remaining
- I received \$9.30 after the 3x multiplier
- Total pool is \$11.20

This is a solid trust signal that deserves good reciprocation.

If I return \$4.50:

- Sender ends with \$6.40 (gaining \$1.40)
- I keep \$4.80

This rewards their trust appropriately while maintaining balance.

{"return": 4.50}

reasoning:

Looking at this situation:

- The sender sent \$3.10 (62% of their endowment)
- They now have \$1.90 remaining
- I received \$9.30 after the 3x multiplier
- Total pool is \$11.20

This is a moderately-high trust signal. I should reciprocate fairly.

Let me consider a reasonable return:

- If I return \$4.50:
 - Sender ends with \$6.40 (gaining \$1.40)
 - I keep \$4.80

This seems fair and reciprocates their trust appropriately.

Alternatively:

- If I return \$4.00:
 - Sender ends with \$5.90 (gaining \$0.90)
 - I keep \$5.30

Or:

- If I return \$4.70:
 - Sender ends with \$6.60 (gaining \$1.60)
 - I keep \$4.60

Given they sent 62%, I think returning around \$4.50 is appropriate - it gives them a solid gain while keeping things relatively balanced.

usage:

```
{
  "input_tokens": 2134,
  "output_tokens": 395,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.1, it became \$9.3, and you returned \$4.5. Your visible payoff was \$4.800000000000001.

Your total visible earnings across all rounds are \$41.16.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my track record as sender:

- Round 2: Sent \$2.50, received \$3.19 back
- Round 3: Sent \$2.50, received \$5.25 back
- Round 6: Sent \$2.50, received \$3.97 back

The 50% sending strategy has been consistently successful, with all partners reciprocating positively. This balanced approach signals trust while protecting against losses.

With a new co-player, I'll continue this proven strategy.

{"send": 2.50}

reasoning:

Looking at my history as a sender:

- Round 2: Sent \$2.50, received \$3.19 back
- Round 3: Sent \$2.50, received \$5.25 back
- Round 6: Sent \$2.50, received \$3.97 back

All three times I've sent \$2.50, I've received positive returns. The 50% strategy has been consistently successful.

As a receiver, I've been generous in reciprocating trust:

- Round 1: Received \$8.04, returned \$3.00
- Round 4: Received \$11.07, returned \$5.50
- Round 5: Received \$11.64, returned \$5.80
- Round 7: Received \$9.30, returned \$4.50

I'm now with yet another new co-player. The pattern shows that sending 50% has been a reliable strategy in this cooperative environment. I should continue with this approach.

{"send": 2.50}

usage:

```
{
  "input_tokens": 2141,
  "output_tokens": 363,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$5.58 to you. Your visible payoff was \$8.08.

Your total visible earnings across all rounds are \$49.239999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$2.64, so you received \$7.92. How much do you return? (0-7.92)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- The sender sent \$2.64 (52.8% of their endowment), showing moderate trust
- They now have \$2.36 remaining
- I received \$7.92 after the 3x multiplier
- Total pool is \$10.28

This is a moderate trust signal that deserves fair reciprocation.

If I return \$3.90:

- Sender ends with \$6.26 (gaining \$1.26)
- I keep \$4.02

This rewards their trust appropriately and maintains balance.

```
{"return": 3.90}
```

reasoning:

Looking at the situation:

- The sender sent \$2.64 (52.8% of their endowment), showing moderate trust
- They now have \$2.36 remaining
- I received \$7.92 after the 3x multiplier
- Total pool is \$10.28

This is a moderate trust signal. I should reciprocate fairly.

If I return \$3.80:

- Sender ends with \$6.16 (gaining \$1.16)
- I keep \$4.12

This seems like a fair reciprocation for their moderate trust.

Let me think about my pattern as receiver:

- Round 4: Received \$11.07, returned \$5.50
- Round 5: Received \$11.64, returned \$5.80
- Round 7: Received \$9.30, returned \$4.50

I've been returning amounts that give both players roughly equal gains or slightly favor the sender to reward their trust.

For \$7.92 received:

- Returning \$3.80 gives sender total \$6.16 (+\$1.16), I keep \$4.12 (+\$4.12)
- This seems reasonable but maybe I should be more generous to the sender

Let me recalculate:

- If I return \$4.00: Sender gets \$6.36 (+\$1.36), I keep \$3.92
- This is more balanced

Actually, let me look at the proportions more carefully:

- Round 4: They sent 73.8%, I returned about 49.7% of what I received
- Round 5: They sent 77.6%, I returned about 49.8% of what I received
- Round 7: They sent 62%, I returned about 48.4% of what I received

This sender sent 52.8%, which is moderate trust. Around 48-50% return seems consistent.

...

usage:

```
{
  "input_tokens": 2150,
  "output_tokens": 668,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.64, it became \$7.92, and you returned \$3.9. Your visible payoff was \$4.02.

Your total visible earnings across all rounds are \$53.25999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my sender history:

- Round 2: Sent \$2.50, received \$3.19 back
- Round 3: Sent \$2.50, received \$5.25 back
- Round 6: Sent \$2.50, received \$3.97 back
- Round 8: Sent \$2.50, received \$5.58 back

The 50% strategy has been highly successful across all partners, with consistently positive returns. This balanced approach continues to work well.

```
{"send": 2.50}
```

Reasoning:

Looking at my history as sender:

- Round 2: Sent \$2.50, received \$3.19 back (net gain: \$0.69)
- Round 3: Sent \$2.50, received \$5.25 back (net gain: \$2.75)
- Round 6: Sent \$2.50, received \$3.97 back (net gain: \$1.47)
- Round 8: Sent \$2.50, received \$5.58 back (net gain: \$3.08)

All of these have been very positive outcomes. The 50% (\$2.50) strategy has been working extremely well. The returns have averaged around \$4.50, which is excellent.

I'm with a new co-player this round. The consistent success of the \$2.50 strategy suggests I should continue with it.

```
{"send": 2.50}
Usage: {
  "input_tokens": 2178,
  "output_tokens": 335,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Receiver - dyad_3

Timestamp: 2026-08-12T09:29:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:**Message 1 role=system****content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.02, it became \$9.06, and you returned \$5.0. Your visible payoff was \$4.0600000000000005.

Your total visible earnings across all rounds are \$14.299999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this round as SENDER:

****My experience so far:****

- Round 1: Sent \$3 ? received \$4.8 back (53% return rate) ? \$6.8 payoff
- Rounds 2-3: As receiver, I returned 54-55% of what I received
- Cooperation has been working well

****My decision:****

Since this is a new co-player, I don't know their specific behavior, but the general pattern in this game seems cooperative. Sending \$3 worked well in Round 1, balancing trust with prudence.

If I send \$3:

- It becomes \$9 for receiver
- If they return around 50%, I get ~\$4.5 back
- Net: ~\$6.5 total, better than keeping all \$5

I'll send ****\$3**** again - it's proven effective and signals willingness to cooperate.

```
'''json
{"send": 3}
'''
```

reasoning:

I'm now the SENDER again. Let me review my experience:

Round 1: I was SENDER, sent \$3, received back \$4.8, payoff = \$6.8

Round 2: I was RECEIVER, received \$7.44 (from \$2.48 sent), returned \$4.0, payoff = \$3.44

Round 3: I was RECEIVER, received \$9.06 (from \$3.02 sent), returned \$5.0, payoff = \$4.06

Now Round 4: I'm SENDER again

My experience shows:

- When I sent \$3 in Round 1, I got back \$4.8 (53% of the \$9 the receiver got)
- I've been a generous receiver, returning 54% and 55% of what I received
- Cooperation seems to be working well

This is a new co-player, so I don't know their behavior yet. However, the game seems to have a cooperative culture developing.

I should send a moderate-to-high amount to signal trust and cooperation. Sending \$3 worked well before. I could stick with that or increase slightly given the positive experiences.

I'll send \$3 again - it's a proven strategy that balances risk and reward.

usage:

```
{
  "input_tokens": 1500,
  "output_tokens": 515,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.36 to you. Your visible payoff was \$6.36.

Your total visible earnings across all rounds are \$20.66.

This round, you are the RECEIVER against your current co-player. They sent you \$4.17, so you received \$12.51. How much do you return? (0-12.51)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this round:

****Analysis:****

- My co-player sent \$4.17 out of \$5 (about 83%), showing exceptional trust
- This became \$12.51 for me
- This is the highest trust shown by any sender I've faced

****My reciprocity pattern:****

- I've been returning around 50-55% of what I receive
- This has maintained good cooperative relationships

****My decision:****

This level of trust deserves strong reciprocation.

If I return \$7.00:

- Sender gets: \$0.83 + \$7.00 = \$7.83
- I get: \$12.51 - \$7.00 = \$5.51
- Both significantly better off than the initial \$5

This return (about 56% of what I received) maintains my cooperative pattern while appropriately rewarding their exceptional trust.

```
'''json
{"return": 7.0}
'''
```

reasoning:

Looking at this round:

****Analysis:****

- My co-player sent \$4.17 out of \$5 (about 83%), showing very high trust
- This became \$12.51 for me
- This is the highest amount anyone has sent me so far

****My past behavior as receiver:****

- Round 2: Received \$7.44, returned \$4.00 (54%)
- Round 3: Received \$9.06, returned \$5.00 (55%)

****My past experience as sender:****

- Round 1: Sent \$3, received back \$4.8 (53% return)
- Round 4: Sent \$3, received back \$4.36 (48% return)

****My decision:****

This co-player showed exceptional trust by sending \$4.17. I should reward this trust generously. If I continue my pattern of returning ~50-55%, that would be \$6.25-\$6.88.

Let me calculate:

If I return \$7.00:

- Sender gets: $\$0.83 + \$7.00 = \$7.83$
- I get: $\$12.51 - \$7.00 = \$5.51$
- Both significantly better off

If I return \$6.50:

- Sender gets: $\$0.83 + \$6.50 = \$7.33$
- I get: $\$12.51 - \$6.50 = \$6.01$
- Both better off

Returning \$7.00 (about 56% of what I received) maintains my cooperative pattern and rewards their exceptional trust. This seems fair given they sent over 80% of their endowment.

usage:

```
{
  "input_tokens": 1854,
  "output_tokens": 605,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.17, it became \$12.51, and you returned \$7.0. Your visible payoff was \$5.51.

Your total visible earnings across all rounds are \$26.17.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this round as SENDER:

****My sending history:****

- Round 1: Sent \$3 ? received \$4.8 back ? \$6.8 payoff
- Round 4: Sent \$3 ? received \$4.36 back ? \$6.36 payoff

****Analysis:****

- Sending \$3 has consistently worked well for me
- Both previous receivers returned about 48-53%, yielding good profits
- This is a new co-player, so I have no specific information about them
- The overall environment has been cooperative

****My decision:****

I'll stick with the proven strategy of sending ****\$3****. It:

- Shows willingness to cooperate
- Has delivered consistent positive returns
- Balances trust with prudent risk management

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

Round 1 (SENDER): Sent \$3 ? received \$4.8 back ? payoff \$6.8
 Round 2 (RECEIVER): Received \$7.44 ? returned \$4.0 ? payoff \$3.44
 Round 3 (RECEIVER): Received \$9.06 ? returned \$5.0 ? payoff \$4.06
 Round 4 (SENDER): Sent \$3 ? received \$4.36 back ? payoff \$6.36
 Round 5 (RECEIVER): Received \$12.51 ? returned \$7.0 ? payoff \$5.51

As SENDER (2 times so far):

- Round 1: Sent \$3, got back \$4.8 (return rate 53%)
- Round 4: Sent \$3, got back \$4.36 (return rate 48.4%)

Both times I sent \$3 and got positive returns. The cooperation has been working well overall.

This is a new co-player again. Should I:

1. Stick with \$3 (proven strategy)
2. Try increasing to show more trust (like the person who sent me \$4.17)
3. Decrease to be more cautious

Given that:

- \$3 has worked well twice
- I've been a generous receiver (returning 50-55%)
- The overall environment seems cooperative
- It's midway through the game (round 6)

I think \$3 remains a good balanced choice. It's worked well and I don't have information about this specific co-player yet.

Actually, let me think about this differently. The person in Round 5 sent me \$4.17 and I returned \$7.0, so they ended up with \$7.83. That was a very successful interaction.

What if I try sending a bit more to signal stronger trust? Maybe \$3.5 or \$4?

...

usage:

```
{
  "input_tokens": 2187,
  "output_tokens": 766,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.84 to you. Your visible payoff was \$7.84.

Your total visible earnings across all rounds are \$34.010000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$2.17, so you received \$6.51. How much do you return? (0-6.51)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this round:

****Analysis:****

- My co-player sent \$2.17 out of \$5 (about 43%), showing moderate trust
- This became \$6.51 for me
- This is less than previous senders, but still shows willingness to cooperate

****My reciprocity pattern:****

- I've consistently returned 50-55% of what I receive
- This has maintained good cooperative relationships

****My decision:****

I'll maintain my fair reciprocity pattern regardless of the amount sent.

If I return \$3.50:

- Sender gets: $\$2.83 + \$3.50 = \$6.33$
- I get: $\$6.51 - \$3.50 = \$3.01$
- Both better off than the initial \$5

This return (about 54% of what I received) maintains my consistent cooperative approach.

```
'''json
{"return": 3.5}
'''
```

reasoning:

Looking at this round:

****Analysis:****

- My co-player sent \$2.17 out of \$5 (about 43%), showing moderate but lower trust
- This became \$6.51 for me
- Previous senders sent: \$2.48, \$3.02, \$4.17 - this is the lowest

****My reciprocity pattern:****

- I've been consistently returning around 50-55% of what I receive
- Round 2: Returned \$4.00 from \$7.44 (54%)
- Round 3: Returned \$5.00 from \$9.06 (55%)
- Round 5: Returned \$7.00 from \$12.51 (56%)

****My decision:****

Even though they sent less, I should maintain my cooperative pattern. If I return less proportionally, it might discourage future cooperation.

If I return \$3.50:

- Sender gets: $\$2.83 + \$3.50 = \$6.33$
- I get: $\$6.51 - \$3.50 = \$3.01$
- Both better off than initial \$5

This return (about 54% of what I received) maintains my consistent reciprocity pattern.

usage:

```
{
  "input_tokens": 2515,
  "output_tokens": 501,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.17, it became \$6.51, and you returned \$3.5. Your visible payoff was \$3.01.

Your total visible earnings across all rounds are \$37.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at this round as SENDER:

****Important context:****

- This is the SAME co-player from Round 7
- Last round they sent me \$2.17 (43% of \$5)
- I returned \$3.50 (54% of what I received), giving them a good payoff of \$6.33

****My sending history:****

- Rounds 1, 4, 6: All sent \$3, consistently received good returns (\$4.36-\$5.84)
- Average return on \$3 sent: ~160-190% profit

****Analysis:****

- My co-player has seen I'm trustworthy and cooperative
- They might reciprocate more generously this time
- Sending \$3 has been my reliable strategy

****My decision:****

I'll send ****\$3**** again, maintaining consistency and showing continued trust.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at this round as SENDER:

****Important context:****

- This is the SAME co-player from Round 7
- In Round 7, they sent me \$2.17 (43% of their \$5)
- I returned \$3.50 to them (54% of \$6.51)
- So they ended with \$6.33 and I got \$3.01

****My sending history:****

- Round 1: Sent \$3 ? received \$4.8 back ? \$6.8 payoff
- Round 4: Sent \$3 ? received \$4.36 back ? \$6.36 payoff
- Round 6: Sent \$3 ? received \$5.84 back ? \$7.84 payoff

****Analysis:****

- I've consistently sent \$3 and it's worked well
- My current co-player showed moderate trust (43%) in the previous round
- I was generous in return (54% return rate), which should signal cooperation
- They might reciprocate more generously this time, knowing I was fair

****My decision:****

I should continue with my strategy of sending \$3. It's worked consistently and signals ongoing cooperation to this co-player who has already seen I'm trustworthy.

If I send \$3:

- It becomes \$9 for receiver
- Based on my generous return last round, they might return around 50-60%
- Expected return: ~\$4.5-5.4
- Net: ~\$6.5-7.4 total

I'll send \$3 again.

usage:

```
{
  "input_tokens": 2648,
  "output_tokens": 573,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.88 to you. Your visible payoff was \$6.88.

Your total visible earnings across all rounds are \$43.900000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at this round as SENDER:

****My sending history:****

- Round 1: Sent \$3 ? received \$4.8 back ? \$6.8 payoff
- Round 4: Sent \$3 ? received \$4.36 back ? \$6.36 payoff
- Round 6: Sent \$3 ? received \$5.84 back ? \$7.84 payoff
- Round 8: Sent \$3 ? received \$4.88 back ? \$6.88 payoff

****Analysis:****

- Sending \$3 has been 100% successful across all my sender rounds
- Consistent positive returns averaging \$6-\$7.84 per round
- This is a new co-player with no specific information

****My decision:****

I'll continue with my proven strategy of sending ****\$3****. It has consistently worked well and signals cooperative intent.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at this round as SENDER:

****My sending history:****

- Round 1: Sent \$3 ? received \$4.8 back ? \$6.8 payoff
- Round 4: Sent \$3 ? received \$4.36 back ? \$6.36 payoff
- Round 6: Sent \$3 ? received \$5.84 back ? \$7.84 payoff
- Round 8: Sent \$3 ? received \$4.88 back ? \$6.88 payoff

****Analysis:****

- I've consistently sent \$3 as SENDER in all 4 previous rounds
- All have yielded positive returns, with payoffs ranging from \$6.36 to \$7.84
- Average return has been around 54-65% of the \$9 they received
- This strategy has been working very well

****This round:****

- This is a new co-player (different from Round 8)
- No specific information about their behavior
- The overall environment has been cooperative

****My decision:****

I should stick with my proven strategy of sending \$3. It has:

- Consistently delivered positive returns
- Demonstrated my willingness to cooperate
- Balanced trust with risk management effectively

I'll send \$3.

usage:

```
{
  "input_tokens": 2626,
  "output_tokens": 515,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.94 to you. Your visible payoff was \$6.94.

Your total visible earnings across all rounds are \$50.84.

This round, you are the RECEIVER against your current co-player. They sent you \$2.97, so you received \$8.91. How much do you return? (0-8.91)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this round:

****Analysis:****

- My co-player sent \$2.97 out of \$5 (about 59%), showing good trust
- This became \$8.91 for me

****My reciprocity pattern:****

- Round 3: Received \$9.06, returned \$5.0 (55%)
- Round 5: Received \$12.51, returned \$7.0 (56%)
- Round 7: Received \$6.51, returned \$3.5 (54%)
- Consistent 54-56% return rate

****My decision:****

I'll maintain my fair reciprocity pattern.

If I return \$4.80:

- Sender gets: \$2.03 + \$4.80 = \$6.83
- I get: \$8.91 - \$4.80 = \$4.11
- Both better off than the initial \$5

This return (about 54% of what I received) is consistent with my cooperative approach.

```
'''json
{"return": 4.8}
'''
```

Reasoning:

Looking at this round:

****Analysis:****

- My co-player sent \$2.97 out of \$5 (about 59%), showing good trust
- This became \$8.91 for me
- This is moderate trust, similar to some previous rounds

****My reciprocity pattern:****

- Round 3: Received \$9.06, returned \$5.0 (55%)
- Round 5: Received \$12.51, returned \$7.0 (56%)
- Round 7: Received \$6.51, returned \$3.5 (54%)
- I've consistently returned 54-56% of what I receive

****My decision:****

I'll maintain my fair reciprocity pattern.

If I return \$4.80:

- Sender gets: $\$2.03 + \$4.80 = \$6.83$
- I get: $\$8.91 - \$4.80 = \$4.11$
- Both better off than the initial \$5

This return (about 54% of what I received) maintains my consistent cooperative approach.

Let me check: $\$8.91 * 0.54 = \4.81

I'll go with \$4.80 to be consistent with my pattern.

```
Usage: {  
  "input_tokens": 2633,  
  "output_tokens": 534,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Sender - dyad_4

Timestamp: 2026-08-12T09:29:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_5",  
      "Agent_1"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_5": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.99, it became \$11.97, and you returned \$6.5. Your visible payoff was \$5.470000000000001.

Your total visible earnings across all rounds are \$16.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my experience so far:

- Round 2: I sent \$2.50 and received \$4.13 back - good reciprocity
- Round 3: My co-player sent \$3.99 (high trust) and I reciprocated fairly
- The game has shown cooperative players who reward trust

Since cooperation has been working well, I'll increase my trust level from the 50% I sent in Round 2. Sending around 60% balances building cooperation while managing risk with a new co-player.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Let me review my history:

- Round 1 (RECEIVER): Saw \$2.54 sent, received \$7.62, returned \$3.50, earned \$4.12
- Round 2 (SENDER): Sent \$2.50, received \$4.13 back, earned \$6.63
- Round 3 (RECEIVER): Saw \$3.99 sent, received \$11.97, returned \$6.50, earned \$5.47
- Total earnings: \$16.22

Now Round 4, I'm the SENDER with a new co-player.

My experience so far:

- When I sent \$2.50, I got back \$4.13 (55% return rate)
- When others sent to me:
 - \$2.54 ? I returned \$3.50 (46% of what I received)
 - \$3.99 ? I returned \$6.50 (54% of what I received)

Both times I've been SENDER or RECEIVER, cooperation has been rewarded. The game seems to have cooperative players.

I've been playing moderately cooperatively:

- Round 2: sent 50%
- Could send more to build trust

Given that:

1. Previous rounds show cooperation is rewarded
2. The last sender who sent high (\$3.99, 80%) was well rewarded
3. I have accumulated good earnings and can afford some risk
4. I want to encourage cooperation

I could send around 60-70% to signal strong cooperation without being reckless.

Let me send \$3.00 (60% of endowment).

```
'''json
{"send": 3.00}
'''
```

usage:

```
{
  "input_tokens": 1464,
  "output_tokens": 523,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.71 to you. Your visible payoff was \$6.71.

Your total visible earnings across all rounds are \$22.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience:

- Round 2 (Sender): Sent \$2.50, received \$4.13 back - good return
- Round 4 (Sender): Sent \$3.00, received \$4.71 back - good return

The pattern shows that moderate-to-high trust (50-60%) has been consistently rewarded. Since cooperation is working well, I'll continue with the same level that succeeded in Round 4.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Looking at my history:

- Round 1 (Receiver): Received \$2.54 ? \$7.62, returned \$3.50. Payoff: \$4.12
- Round 2 (Sender): Sent \$2.50 ? \$7.50, got back \$4.13. Payoff: \$6.63
- Round 3 (Receiver): Received \$3.99 ? \$11.97, returned \$6.50. Payoff: \$5.47
- Round 4 (Sender): Sent \$3.00 ? \$9.00, got back \$4.71. Payoff: \$6.71

My sending strategy has been:

- Round 2: \$2.50 (50%)
- Round 4: \$3.00 (60%)

Both times I got good returns:

- \$2.50 sent ? \$4.13 back (good reciprocity)
- \$3.00 sent ? \$4.71 back (good reciprocity)

The game has shown cooperative players. I'm building earnings well. Since I'm sender again with a new co-player, I should continue showing trust. Given the successful cooperation, I could maintain or slightly increase my sending amount.

I'll send \$3.00 again (60%) as it worked well last time and establishes a consistent cooperative stance.

usage:

```
{
  "input_tokens": 1718,
  "output_tokens": 440,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$4.01 to you. Your visible payoff was \$6.01.

Your total visible earnings across all rounds are \$28.939999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.7, so you received \$8.100000000000001. How much do you return? (0-8.100000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation with my current co-player:

Previous round (Round 5) against this SAME co-player:

- I sent them \$3.00, it became \$9.00
- They returned \$4.01 (about 44.6% of what they received)
- Both of us gained

Now they've sent me \$2.70 (54% of their endowment), showing moderate trust.

I should maintain fair reciprocity with this partner. If I return \$4.10:

- Sender ends with: \$5.00 - \$2.70 + \$4.10 = \$6.40 (net gain: \$1.40)
- I end with: \$8.10 - \$4.10 = \$4.00 (net gain: \$4.00)

This is proportionally similar to their return rate in Round 5, maintaining our cooperative relationship.

```
'''json
{"return": 4.10}
'''
```

reasoning:

Looking at this situation:

In Round 5 (the immediately previous round), I was the sender against this SAME co-player:

- I sent \$3.00, it became \$9.00
- They returned \$4.01 to me
- My payoff was \$6.01

Now in Round 6, I'm the receiver against the SAME co-player:

- They sent \$2.70 (54% of their endowment)
- I received \$8.10

This is interesting - we're playing consecutive rounds with the same partner. In Round 5, they returned \$4.01 out of \$9.00 (about 44.6%). That was a reasonable return.

Now they've sent me \$2.70 (54% of endowment), which is a moderate level of trust, slightly less than the \$3.00 (60%) I sent them.

I should reciprocate fairly given our history:

- They gave me a fair return in Round 5
- They're showing moderate trust here

If I return \$4.00:

- Sender ends with: $\$5.00 - \$2.70 + \$4.00 = \6.30 (net gain: \$1.30)
- I end with: $\$8.10 - \$4.00 = \$4.10$ (net gain: \$4.10)

This is roughly similar to what they returned to me (about 44-49% of what they received). It maintains fairness and reciprocity with this specific partner.

Actually, let me think more carefully. They returned \$4.01 out of \$9.00, which is 44.6%. If I apply the same percentage: $0.446 * 8.10 = \$3.61$.

But I think returning slightly more shows good faith and maintains cooperation. Something around \$4.00-\$4.20 seems fair.

Let me go with \$4.10 to be slightly more generous than they were, showing I value the ongoing cooperation.

usage:

```
{
  "input_tokens": 1985,
  "output_tokens": 686,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.7, it became \$8.100000000000001, and you returned \$4.1. Your visible payoff was \$4.0000000000000002.

Your total visible earnings across all rounds are \$32.94.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience as sender:

- Round 2: Sent \$2.50, received \$4.13 back
- Round 4: Sent \$3.00, received \$4.71 back
- Round 5: Sent \$3.00, received \$4.01 back

Sending around \$3.00 (60% of endowment) has consistently been rewarded with fair reciprocity. This is a new co-player, but the moderate-high trust strategy has been working well across different partners.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Looking at my history:

- Round 2 (Sender): Sent \$2.50, received \$4.13 back
- Round 4 (Sender): Sent \$3.00, received \$4.71 back
- Round 5 (Sender): Sent \$3.00, received \$4.01 back (same co-player as Round 6)
- Round 6 (Receiver): Received from same co-player who sent \$2.70

Now I'm playing against a NEW co-player in Round 7 (since the note says "against another agent" in the previous round description).

The pattern shows that sending around \$3.00 (60%) has worked well. I've been consistently reciprocated. This is a new co-player, so I should continue with a moderate-to-high trust level that has been successful.

I'll send \$3.00 again.

usage:

```
{
  "input_tokens": 2333,
  "output_tokens": 350,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.29 to you. Your visible payoff was \$6.29.

Your total visible earnings across all rounds are \$39.23.

This round, you are the RECEIVER against your current co-player. They sent you \$3.37, so you received \$10.11. How much do you return? (0-10.11)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.37 (about 67% of their endowment) - showing good trust
- I received \$10.11 after the 3x multiplier

This is a solid level of trust, higher than the moderate amounts but not quite as high as the \$3.99 I saw in Round 3.

If I return \$5.25:

- Sender ends with: $\$5.00 - \$3.37 + \$5.25 = \6.88 (net gain: \$1.88)
- I end with: $\$10.11 - \$5.25 = \$4.86$ (net gain: \$4.86)

This proportionally rewards their good trust while both of us benefit.

```
'''json
{"return": 5.25}
'''
```

reasoning:

Let me analyze this situation:

Current round (Round 8):

- I'm the RECEIVER
- My co-player sent me \$3.37 (about 67% of their endowment) - showing good trust
- I received \$10.11 after the 3x multiplier

Looking at my history as receiver:

- Round 1: Received \$7.62 (from \$2.54 sent), returned \$3.50
- Round 3: Received \$11.97 (from \$3.99 sent), returned \$6.50
- Round 6: Received \$8.10 (from \$2.70 sent), returned \$4.10

And as sender, I've consistently sent \$3.00 and received returns around \$4-\$4.70.

My co-player sent \$3.37, showing good trust (67%). I should reciprocate proportionally.

If I return \$5.00:

- Sender ends with: $\$5.00 - \$3.37 + \$5.00 = \6.63 (net gain: \$1.63)
- I end with: $\$10.11 - \$5.00 = \$5.11$ (net gain: \$5.11)

If I return \$5.50:

- Sender ends with: $\$5.00 - \$3.37 + \$5.50 = \7.13 (net gain: \$2.13)
- I end with: $\$10.11 - \$5.50 = \$4.61$ (net gain: \$4.61)

Looking at Round 3 where they sent \$3.99 (almost 80%) and I returned \$6.50, this is similar trust level (67%).

For \$3.37 sent (67%), returning around \$5.00-\$5.50 seems fair and proportional to their trust level.

...

usage:

```
{
  "input_tokens": 2318,
  "output_tokens": 731,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.37, it became \$10.11, and you returned \$5.25. Your visible payoff was \$4.8599999999999999.

Your total visible earnings across all rounds are \$44.089999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$2.19, so you received \$6.57. How much do you return? (0-6.57)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- My co-player sent \$2.19 (about 44% of their endowment) - showing moderate/cautious trust
- I received \$6.57 after the 3x multiplier

This is lower trust than I've typically seen. I should return proportionally to encourage cooperation while being fair.

If I return \$3.30:

- Sender ends with: \$5.00 - \$2.19 + \$3.30 = \$6.11 (net gain: \$1.11)
- I end with: \$6.57 - \$3.30 = \$3.27 (net gain: \$3.27)

This gives them a reasonable gain and maintains fairness.

```
'''json
{"return": 3.30}
'''
```

reasoning:

Looking at this situation:

- My co-player sent \$2.19 (about 44% of their endowment) - showing moderate but cautious trust
- I received \$6.57 after the 3x multiplier

This is lower trust than what I've typically been seeing. I should return proportionally to their trust level.

If I return around \$3.00:

- Sender ends with: \$5.00 - \$2.19 + \$3.00 = \$5.81 (net gain: \$0.81)
- I end with: \$6.57 - \$3.00 = \$3.57 (net gain: \$3.57)

This gives them a modest gain while I also benefit. It's proportionally fair given their moderate trust level.

Let me think about the return rate:

- They sent \$2.19, which became \$6.57
- Returning \$3.00 is about 45.7% of what I received
- This is similar to some of the return rates I've seen

Actually, let me be a bit more generous to encourage cooperation. If I return \$3.30:

- Sender ends with: \$5.00 - \$2.19 + \$3.30 = \$6.11 (net gain: \$1.11)
- I end with: \$6.57 - \$3.30 = \$3.27 (net gain: \$3.27)

That's about 50% of what I received, which is fair and encourages future cooperation.

usage:

```
{
  "input_tokens": 2389,
  "output_tokens": 541,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.19, it became \$6.57, and you returned \$3.3. Your visible payoff was \$3.2700000000000005.

Your total visible earnings across all rounds are \$47.36.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my sender history:

- Round 2: Sent \$2.50, received \$4.13 back
- Round 4: Sent \$3.00, received \$4.71 back
- Round 5: Sent \$3.00, received \$4.01 back
- Round 7: Sent \$3.00, received \$4.29 back

The pattern is clear - sending \$3.00 (60% of endowment) has consistently resulted in good reciprocity from different co-players. This strategy balances trust-building with risk management.

```
'''json
{"send": 3.00}
'''
```

Reasoning:

Looking at my history as sender:

- Round 2 (Sender): Sent \$2.50, received \$4.13 back
- Round 4 (Sender): Sent \$3.00, received \$4.71 back
- Round 5 (Sender): Sent \$3.00, received \$4.01 back
- Round 7 (Sender): Sent \$3.00, received \$4.29 back

All of my sender experiences have shown good reciprocity. Sending \$3.00 (60% of endowment) has been consistently successful.

This is a new co-player for Round 10. The cooperative pattern has been strong throughout the game. I should continue with the strategy that has worked well.

I'll send \$3.00 again.

```
Usage: {
  "input_tokens": 2369,
  "output_tokens": 333,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_4

Timestamp: 2026-08-12T09:29:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_1"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_1": "trustee",
      "Agent_5": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system
content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.27, it became \$6.8100000000000005, and you returned \$3.4. Your visible payoff was \$3.4100000000000006.

Your total visible earnings across all rounds are \$14.11.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

****Analysis:****

- This is Round 4 with a new co-player
- In Round 1, I sent \$3 and received \$4.5 back - a positive outcome
- As a receiver, I've been returning about 50%, establishing a cooperative pattern
- My current co-player is unknown to me

****Decision:****

I'll send \$3 again. This amount has worked well before - it shows trust and creates potential for mutual gain through the 3x multiplier, while managing downside risk.

'''json

{"send": 3}

'''

reasoning:

Let me analyze the situation:

****What I know:****

- Round 1: I was SENDER, sent \$3, received back \$4.5, payoff \$6.5
- Round 2: I was RECEIVER, got \$8.7 (from \$2.9 sent), returned \$4.5, payoff \$4.2
- Round 3: I was RECEIVER, got \$6.81 (from \$2.27 sent), returned \$3.4, payoff \$3.41
- Total earnings: \$14.11

****Current situation:****

- Round 4: I'm the SENDER with \$5
- This is a new co-player (different from rounds 1-3)

****Strategic considerations:****

1. I've seen one positive interaction as sender (sent \$3, got back \$4.5)
2. As receiver, I've been fairly generous, returning about 50%
3. This creates a reputation for reciprocity if others know
4. But agents don't know who I am (names hidden), so reputation effects are limited
5. However, the pattern of behavior in the pool might matter

****Decision factors:****

- Sending creates potential gains (3x multiplier)
- Risk: this is a new co-player, no direct history
- But overall, cooperation seems to be working in this pool
- My Round 1 strategy (sending \$3) worked well

I'll send \$3 again - it's a moderate trust signal that has proven effective.

usage:

```
{
  "input_tokens": 1352,
  "output_tokens": 472,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.48 to you. Your visible payoff was \$6.48.

Your total visible earnings across all rounds are \$20.59.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

****Analysis:****

- Round 5 with a new co-player
- My \$3 sending strategy has been successful:
 - Round 1: Sent \$3, received \$4.5 back ? \$6.5 payoff
 - Round 4: Sent \$3, received \$4.48 back ? \$6.48 payoff
- Both returns were close to 50%, indicating fair reciprocity from my co-players
- This amount balances trust-building with risk management

****Decision:****

I'll continue with \$3, as it's been consistently effective.

```
'''json
{"send": 3}
'''
```

reasoning:

****Analysis:****

- Round 5, new co-player as sender
- My sending strategy of \$3 has worked well:
 - Round 1: Sent \$3, received \$4.5 back, payoff \$6.5
 - Round 4: Sent \$3, received \$4.48 back, payoff \$6.48
- Both times I received close to 50% back, which is fair reciprocity
- Total visible earnings: \$20.59 across 4 rounds (average \$5.15/round)

- As a receiver, I've been consistently returning about 50%

****Decision:****

The \$3 sending strategy is working consistently well. My co-players have been reciprocating fairly. I should continue this approach.

```
'''json
{"send": 3}
'''
```

usage:

```
{
  "input_tokens": 1601,
  "output_tokens": 339,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.77 to you. Your visible payoff was \$8.77.

Your total visible earnings across all rounds are \$29.36.

This round, you are the RECEIVER against your current co-player. They sent you \$3.68, so you received \$11.040000000000001. How much do you return? (0-11.040000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

****Analysis:****

- My co-player sent \$3.68 (about 74% of their endowment), showing strong trust
- I received \$11.04 after the multiplier
- They're taking a significant risk by sending such a high amount
- I've been maintaining a 50% return pattern, which has worked well

****Decision:****

I'll return approximately 50% of what I received to reward their trust appropriately.

```
'''json
{"return": 5.5}
'''
```

reasoning:

****Analysis:****

- My co-player sent \$3.68 (about 74% of their endowment), showing strong trust
- I received \$11.04 after the 3x multiplier
- In Round 5, my co-player returned \$6.77 from \$9, which is about 75% - very generous!
- I've been returning about 50% as a norm, but this co-player sent a substantial amount

****Decision:****

My co-player showed considerable trust by sending \$3.68. I should reciprocate fairly. Following my 50% return pattern would give them about \$5.50 back.

Let me calculate:

- They sent \$3.68, keeping \$1.32
- I received \$11.04
- If I return \$5.50:
 - They get: \$1.32 + \$5.50 = \$6.82 (better than their \$5 starting)
 - I get: \$11.04 - \$5.50 = \$5.54 (better than my \$0 starting as receiver)

This seems fair and maintains my reciprocity norm.

usage:

```
{
  "input_tokens": 1890,
  "output_tokens": 379,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.68, it became \$11.040000000000001, and you returned \$5.5. Your visible payoff was \$5.540000000000001. Your total visible earnings across all rounds are \$34.9.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

****Analysis:****

- My co-player sent \$3.23 (about 65% of their endowment), showing good trust
- I received \$9.69 after the multiplier
- I've been consistently returning approximately 50% of what I receive
- This approach has maintained positive cooperative relationships

****Decision:****

I'll return about 50% of what I received.

```
'''json
{"return": 4.85}
'''
```

reasoning:

****Analysis:****

- My co-player sent \$3.23 (about 65% of their endowment), showing good trust
- I received \$9.69 after the multiplier
- I've been maintaining a consistent 50% return pattern throughout the game
- This has been working well and seems fair

****Decision:****

I'll continue with my 50% return pattern: $0.5 \times \$9.69 = \4.845

Let me round to a reasonable number: \$4.85

usage:

```
{
  "input_tokens": 2142,
  "output_tokens": 221,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.23, it became \$9.69, and you returned \$4.85. Your visible payoff was \$4.84.

Your total visible earnings across all rounds are \$39.739999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

****Analysis:****

- Round 8 with a new co-player
- My sending strategy of \$3 has been consistently successful:
 - Round 1: \$6.5 payoff
 - Round 4: \$6.48 payoff
 - Round 5: \$8.77 payoff
- Average return has been favorable, with partners generally reciprocating fairly
- This amount continues to balance trust-building with prudent risk management

****Decision:****

I'll send \$3 again, as it's proven to be an effective cooperative strategy.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze my performance as a sender so far:

Round 1: Sent \$3, received \$4.5 back ? payoff \$6.5
Round 4: Sent \$3, received \$4.48 back ? payoff \$6.48
Round 5: Sent \$3, received \$6.77 back ? payoff \$8.77

All three times I sent \$3, I got positive returns. The returns varied:

- Round 1: 50% return rate (4.5/9)
- Round 4: ~50% return rate (4.48/9)
- Round 5: ~75% return rate (6.77/9)

Overall, sending \$3 has been very successful for me. I'm averaging well above my \$5 starting amount when I send \$3.

As a receiver, I've been consistently returning about 50% of what I receive, which seems fair and has maintained good cooperation.

This is Round 8, a new co-player. I should stick with my successful strategy of sending \$3

usage:

```
{
  "input_tokens": 2158,
  "output_tokens": 387,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.16 to you. Your visible payoff was \$5.16.

Your total visible earnings across all rounds are \$44.89999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

****Analysis:****

- Round 9 with a new co-player
- My \$3 sending strategy has been generally successful:
 - Round 1: \$6.5 payoff
 - Round 4: \$6.48 payoff
 - Round 5: \$8.77 payoff
 - Round 8: \$5.16 payoff (lower but still profitable)
- Even with variation in returns, this amount has consistently yielded positive outcomes
- Total earnings are strong at \$44.90

****Decision:****

I'll continue with \$3, maintaining the cooperative approach that has worked well overall.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

Round 9 - I'm the SENDER again

Recent history as sender:

- Round 1: Sent \$3, received \$4.5 back ? \$6.5 payoff
- Round 4: Sent \$3, received \$4.48 back ? \$6.48 payoff
- Round 5: Sent \$3, received \$6.77 back ? \$8.77 payoff
- Round 8: Sent \$3, received \$3.16 back ? \$5.16 payoff

The last round (Round 8) showed a lower return (\$3.16 on \$9 received), but still positive.

The \$3 sending strategy has generally worked well, with most returns being fair or generous.

As receiver, I've been returning about 50% consistently, which has maintained good cooperation.

tive dynamics.

Overall, my strategy of sending \$3 has been successful. Even in the worst case (Round 8), I still made a profit (\$5.16 > \$5 initial endowment).

I should continue with \$3 as it's been a robust strategy across different co-players.

usage:

```
{
  "input_tokens": 2140,
  "output_tokens": 423,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.01 to you. Your visible payoff was \$7.01.

Your total visible earnings across all rounds are \$51.90999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.45, so you received \$7.3500000000000005. How much do you return? (0-7.3500000000000005)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

****Analysis:****

- My co-player sent \$2.45 (about 49% of their endowment), showing moderate trust
- I received \$7.35 after the multiplier
- I've been maintaining a consistent ~50% return pattern as receiver
- This approach has worked well in building cooperative relationships

****Decision:****

I'll return approximately 50% of what I received.

```
'''json
{"return": 3.68}
'''
```

Reasoning:

****Analysis:****

- My co-player sent \$2.45 (about 49% of their endowment), showing moderate trust
- I received \$7.35 after the multiplier
- I've been consistently returning approximately 50% of what I receive as a receiver
- This approach has been fair and maintained good cooperative relationships

****Decision:****

I'll return about 50% of what I received, which would be around \$3.68.

Let me calculate:

- They sent: \$2.45
- I received: \$7.35
- 50% return: \$3.675

I'll round to \$3.68 for consistency.

Usage: {

```
"input_tokens": 2178,
"output_tokens": 255,
"reasoning_tokens": 0
}
```